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| Inventor(s)          | Ostertag; Eric M. et al. |

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### BCMA specific VCAR compositions and methods for use

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#### Abstract

Disclosed are VHH chimeric antigen receptors (VCARs), VCAR transposons encoding VCARs of the disclosure, cells modified to express VCARs of the disclosure, as well as methods of making and methods of using the same for adoptive cell therapy.

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**Field of Classification Search**

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## Background/Summary

RELATED APPLICATIONS (1) This application is a U.S. National Phase Application, filed under 35 U.S.C. § 371 of International Patent Application No. PCT/US2018/066936, filed Dec. 20, 2018, which claims the benefit of provisional applications U.S. Ser. No. 62/608,571, filed Dec. 20, 2017 and U.S. Ser. No. 62/608,894, filed Dec. 21, 2017, the contents of which are herein incorporated by reference in their entirety.

## FIELD OF THE DISCLOSURE

(1) The disclosure is directed to molecular biology, and more, specifically, to chimeric antigen receptors, and to transposons containing one or more VCARs, as well as methods of making and using

the same.

## INCORPORATION OF SEQUENCE LISTING

(2) The contents of the text file named “POTH-034\_N01US\_SegListing\_ST25\_R.txt”, which was created on Jun. 10, 2020, and is 54 MB in size are hereby incorporated by reference in their entirety.

## BACKGROUND

(3) There has been a long-felt but unmet need in the art for a method of directing the specificity of an immune cell without using traditional antibody sequences or fragments thereof. The disclosure provides a superior chimeric antigen receptor.

## SUMMARY

(4) The disclosure provides a chimeric antigen receptor (CAR) comprising: (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one single domain antibody; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the single domain antibody comprises a human or humanized sequence. In some embodiments, the single domain antibody comprises a non-naturally occurring sequence. In some embodiments, the single domain antibody comprises a recombinant or chimeric sequence. In some embodiments, the single domain antibody comprises a VHH or a sequence encoding a VHH. In some embodiments, the single domain antibody comprises a VH or a sequence encoding a VH. In some embodiments, the single domain antibody comprises a VH of the disclosure or a sequence encoding a VH of the disclosure.

(5) The disclosure provides a chimeric antigen receptor (CAR) comprising: (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VHH; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. As used throughout the disclosure, a CAR comprising a VHH is referred to as a VCAR. In certain embodiments, the antigen recognition region may comprise two VHHs to produce a bi-specific or tandem VCAR. In certain embodiments, the antigen recognition region may comprise three VHHs to produce a tri-specific VCAR.

(6) In certain embodiments of the VCARs of the disclosure, the ectodomain may further comprise a signal peptide. Alternatively, or in addition, in certain embodiments, the ectodomain may further comprise a hinge between the antigen recognition region and the transmembrane domain. In certain embodiments, the ectodomain may further comprise a signal peptide. Alternatively, or in addition, in certain embodiments, the ectodomain may further comprise a hinge between the antigen recognition region and the transmembrane domain.

(7) In certain embodiments of the VCARs of the disclosure, the VHH comprises or consists of the amino acid sequence

(8) TABLE-US-00001 (VH-A; SEQ ID NO: 18000)

malpvtalllplalllhaarpevqllesggglvqpqgslrlscaa sgftssyamnwvrqapgkglewvagiigsggstyyadsvkgrfs  
isrdnskntldlqmnsraedtavyycvkdwnntmitergqgtlv tvssttppaprpptpaptiasqplslrpeacrpaaggavhtrgld  
facdiyiwaplagtcgvllslvtlyckrgrkkllyifkqpfmr pvqttqeedgcscrpfpeeeeggcelrvkfsrsadapaykqgqnql  
ynelnlgrreeydvldkrrgrdpemggkprknpqeglynelqkd kmaeayseigmkgenrgkghdglyqglstatkdydalhmqalp  
pr

or the VHH comprises or consists of the nucleic acid sequence

(9) TABLE-US-00002 (VH-A; SEQ ID NO: 18001)

atggctctgcctgtgacagctctgctgctgctctggctctgctt cttcatgcggcgccgacctgaagttcagctgctgaatctggcgga  
ggcctgggtcaacctggcggtatctgagactgagctgtgccgcc agcggcttcaccttagcagctacgccatgaactgggtccgacag  
gccctggcaaaggactggaatgggtggccggaatcatcggcagc ggccggcagcacatattacgccgattctgtgaagggccgcttcagc  
atcagccgggacaacagcaagaacacctggacctgcagatgaac agcctgagagccgaggataccgccgtgtactactgctgaaggat  
tggaacaccaccatgatcaccgagagaggccagggcacactggtc accgtgtcctctacaacaacaccggcgctcggcctccaacacca  
gctcctacaatcgcgagtcagcccctgtctctcagacccgaagcc ttagacctgctgctggcgagctgtgcataccagaggactggat  
ttcgctgcgacatctacatctgggtcctctggctggcacatgc ggagtttctgctgagcctggcatcacctgtactgtaagaga  
ggcaggaagaagctgctgtatatcttcaagcagccctcatgaga cccgtgcagaccacacaggaggaggacggctgctctgtaggttc  
ccagaggaggaggaggaggatgcgagctgcgcgtgaagtttagc cgggtccgccgatgcacctgcatacaagcaggagacagaaccagctg  
tataacgagctgaatctgggcccggagagaggagtacgacgtgctg gataagaggcggggccgggaccccagatgggaggcaagcctcgg

agaaagaaccacagggagggcctgtacaatgagctgcaaaaggac aagatggccgaggcctattctgagatcggcacatgaaggagagagg  
cgccggggcaaggacacgatggcctgtaccagggcctgagcacc gccacaaaggacacctatgatgcctgcacatgcaggccctgccc  
cctagatga.

(10) In certain embodiments of the VCARS of the disclosure, the VHH comprises or consists of the amino acid sequence

(11) TABLE-US-00003 (VH-B; SEQ ID NO: 18002)

malpvtalllplalllhaarpevqllesggglvqpggsltlscasgftfsnyamnwwrqapgkglewvsgiiigsgattyyadsvkgrft  
isrdnskntlnlqmnsraedtaiyyvcvkdwnntmitergqgtlv tvssttpaprpptpaptiasqplslrpeacrpaaggavhtrgld  
facdiyiwaplagtcgvlllslvitlyckrgrkkllyifkqpfmr pvqttqeedgcscrpfpeeeeggcelrvkfsrsadapaykqgqnql  
ynelnlgrreedydvldkirgrdpemggkprkrnpqeglynelqkd kmaeyseigmkgerrrgkghdglyqglstatkdydalhmqlp  
pr

or the VHH comprises or consists of the nucleic acid sequence

(12) TABLE-US-00004 (VH-B; SEQ ID NO: 18003)

atggctctgcctgtgacagctctgctgctgcctctggctctgctt cttcatgcggcgccctgaagttcagctgcttgaatctggcgga  
ggcctggttcaacctggcgcatctctgacactgagctgtgccgcc agcggcttcaccttcagcaactacgccatgaactgggtccgacag  
gccctggcaaaggccttgaatgggtgtccggcatcattggctct ggccgaccacactactacgccgattctgtgaaggcgagattcacc  
atcagccgggacaacagcaagaacaccctgaacctgcagatgaac agcctgagagccgaggacaccgccatctactactgcgtgaaggac  
tggaacaccacatgatcaccgagagaggccagggcacactggctc accgtgtcctctacaacaacaccggcgctcggcctccaacacca  
gctcctacaatcgcgagtcagccccgtctctcagacccgaagcc ttagacctgctgctggcgagctgtgcataccagaggactggat  
ttcgctgcgacatctacatctgggctcctctggctggcacatgc ggagtttctgctgagcctggctcatcacctgtactgtaagaga  
ggcaggaagaagctgctgtatatcttcaagcagccctcatgaga cccgtgcagaccacacaggaggaggacggctgctctttaggttc  
ccagaggaggaggaggaggatgcgagctgcgctgaagtttagc cggctccgccgatgcacctgcatacaagcaggacagaaccagctg  
tataacgagctgaatctgggccggagagaggagtacgacgtgctg gataagaggcgggggccgggaccccgagatgggaggcaagcctcgg  
agaaagaacccacaggaggggcctgtacaatgagctgcaaaaggac aagatggccgaggcctattctgagatcgccatgaaggagagagg  
cgccggggcaaggacacgatggcctgtaccagggcctgagcacc gccacaaaggacacctatgatgcctgcacatgcaggccctgccc  
cctagatga.

(13) In certain embodiments of the VCARS of the disclosure, the VHH comprises or consists of the amino acid sequence

(14) TABLE-US-00005 (VH-C; SEQ ID NO: 18004)

malpvtalllplalllhaarpevqllesggglvqpgeslrscasgftfsnyamnwwrqapgkglewvsgivggggtsyyadsvrgrft  
isrdnskntlylqmnsraedtavyyvcvkdwnntmitergqgtlv tvssttpaprpptpaptiasqplslrpeacrpaaggavhtrgld  
facdiyiwaplagtcgvlllslvitlyckrgrkkllyifkqpfmr pvqttqeedgcscrpfpeeeeggcelrvkfsrsadapaykqgqnql  
ynelnlgrreedydvldkrrgrdpemggkprkrnpqeglynelqkd kmaeyseigmkgerrrgkghdglyqglstatkdydalhmqlp  
pr

or the VHH comprises or consists of the nucleic acid sequence

(15) TABLE-US-00006 (VH-C; SEQ ID NO: 18005)

atggctctgcctgtgacagctctgctgctgcctctggctctgct tcttcatgcggcgccctgaagttcagctgcttgaatctggcg  
gaggcctggttcagcctggcgaatctctgagactgagctgtgcc gccagcggcttcaccttcagcaactacgccatgaactgggtccg  
acaggccccctggcaaaggccttgaatgggtgtccggaatcgttg gcggcggaggcacaagctactacgccgattctgtgcggggcaga  
ttcaccatcagccgggacaacagcaagaacaccctgtacctgca gatgaacagcctgagagccgaggacaccgccgtgtactactgcg  
tgaaggactggaacaccacatgatcaccgagagaggccagggc aactggctaccgtgtcctctacaacaacaccggcgctcggcc  
tccaacaccagctcctacaatcgcgagtcagccccgtctctca gaccgaagcctgtagacctgctgctggcgaggctgtgcatacc  
agaggactggatttcgctgcgacatctacatctgggctcctct ggctggcacatgcggagtttctgctgagcctggctcatcccc  
tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag cagcccttcatgagacccgtgcagaccacacaggaggaggacgg  
ctgctctttaggttcccagaggaggaggaggaggtatcgagc tgcgcgtgaagtttagccggtccgccgatgcacctgcatacaag  
caggacagaaccagctgtataacgagctgaatctgggccggag agaggagtacgacgtgctggataagaggcgggggccgggaccccg  
agatgggagggaagcctcgagaaagaacccacaggaggggcctg tacaatgagctgcaaaaggacaagatggccgaggcctattctga  
gatcgccatgaaggagagaggcgccggggcaaggacacgatg gcctgtaccagggcctgagcaccgccacaaaggacacctatgat  
gcctgcacatgcaggccctgccccctagatga.

(16) In certain embodiments of the VCARS of the disclosure, the VHH comprises or consists of the amino acid sequence

(17) TABLE-US-00007 (VH-D; SEQ ID NO: 18006)

malpvtalllplalllhaarpevqllesggglvqpqgslrlsca asgftfsnyamtwirqapgkglewvsgitdggstfyadsvkgr  
ftisrdnsntlylqmnsraedtavyycvkdwnntmitergqg tlvtvssttppaprpptpaptiasqplslrpeacrpaaggavht  
rgldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifk qpfmrpvqttqeedgcscrpfpeeeeggcelrvkfsrsadapayk  
qgqnqlynelnlgrreedydldkrrrdpemmkgkprknpqegl ynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdyd  
alhmqalppr

or the VHH comprises or consists of the nucleic acid sequence

(18) TABLE-US-00008 (VH-D; SEQ ID NO: 18007)

atggctctgcctgtgacagctctgctgctgctctggctctgctt tcatgcggcgcgccctgaagttcagctgcttgaatctggcggagg  
cctggttcaacctggcggatctctgagactgagctgtgccgccagc ggcttcaccttcagcaattacgccatgacctggatcagacaggccc  
ctggcaaaggcctggaatgggtgtccggaattacaggcgacggcgg cagcaccttttacgccgattctgtgaagggcagattcaccatcagc  
cgggacaacagcaagaacaccctgtacctgcagatgaacagcctga gagccgaggacaccgccgtgtactactgcgtgaaggactggaacac  
caccatgatcaccgagagaggccagggcacactggcaccgtgtcc tctacaacaacaccggcgccctggcctccaacaccagctcctacaa  
tcgcgagtcagcccctgtctctcagacccgaagcctgtagacctgc tgctggcggagctgtgcataccagaggactggatttcgctgcgac  
atctacatctgggctcctctggctggcacatgcggagtttctgctc tgacctggctcatcacctgtactgtaagagaggcaggaagaagct  
gctgtatatcttaagcagcccttcatgagaccctgcagaccaca caggaggaggacggctgtctttaggttcccagaggaggaggagg  
gaggatgcgagctgcgcgtgaagtttagccggtccgccgatgcacc tgcatacaagcaggggacagaaccagctgtataacgagctgaatctg  
ggccggagagaggagtacgacgtgctggataagaggcgggggccggg  
accccgagatgggagggaagcctcgagaaagaacccacaggaggg cctgtacaatgagctgcaaaaggacaagatggccgaggcctattct  
gagatcggcctgaaggagagagaggcgccgggggcaagggaacacgatg gcctgtaccagggcctgagcaccgccacaaaggacacctatgatg  
cctgcacatgcaggccctgccccctagatga.

(19) In certain embodiments of the VCARs of the disclosure, the VHH comprises or consists of the amino acid sequence

(20) TABLE-US-00009 (VH-E; SEQ ID NO: 18008)

malpvtalllplalllhaarpevqllesggglvqpqgslrlsca sgftfsnyamtwirqapgkglewvsgisgsggstyyadsvkgrft  
isrdnsntvylqmnsraedtavyycvkdwnntmitergqgtlv tvssttppaprpptpaptiasqplslrpeacrpaaggavhtrgld  
facdiyiwaplagtcgvllslvitlyckrgrkkllyifkqpfmr pvqttqeedgcscrpfpeeeeggcelrvkfsrsadapaykqgqnql  
ynelnlgrreedydldkrrrdpemmkgkprknpqeglynelqkd kmaeayseigmkgerrrgkghdglyqglstatkdydalhmqalp  
pr

or the VHH comprises or consists of the nucleic acid sequence

(21) TABLE-US-00010 (VH-E; SEQ ID NO: 18009)

atggcactgcctgtgacagccctgctgctgctctggccctgct gctgcacgcagcacggcccagggtgcagctgctggagtccggag  
gaggcctggcccagcctggcggcagcctgaggctgtcctgcgcc gcctctggcttcaccttagctcctacgccatgaactggatcag  
acaggcccctggcaagggcctggagtgggtgtccggcatctccg gctctggaggctctacatactatgccgacagcgtgaagggccgg  
ttcaccatcagcagagataactccaagaataccgtgtacctcca gatgaactctctgcggggccgaggacaccgccgtgtactattgcg  
tgaaggattggaataaccacaatgatcacagagagggggccagggc accctggtgacagtgtctagcaccacaaccctgccccagacc  
tccacacccgcccctaccatcgcgagtcagccactgtccctgc ggctgaggcctgccggcccgcggcgaggcagtgacaca  
cggggcctggactttgctgtgacatctacatatgggcaccact ggaggaaacctgcggcgtgctgctgctgagcctggatcaccc  
tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag cagcccttcatgagacccgtgcagaccacacaggaggaggacgg  
ctgctctttaggttcccagaggaggaggaggaggatgcgagc tgcgcgtgaagtttagccggtccgccgatgcacctgcatacaag  
cagggaacagaaccagctgtataacgagctgaatctgggccggag agaggagtacgacgtgctggataagaggcgggggccgggaccccg  
agatgggagggaagcctcgagaaagaacccacaggaggggcctg tacaatgagctgcaaaaggacaagatggccgaggcctattctga  
gatcggcctgaaggagagagaggcgccgggggcaagggaacacgatg gcctgtaccagggcctgagcaccgccacaaaggacacctatgat  
gcctgcacatgcaggccctgccccctagatga.

(22) In certain embodiments of the VCARs of the disclosure, the VHH comprises or consists of the amino acid sequence

(23) TABLE-US-00011 (VH-F; SEQ ID NO: 18010)

malpvtalllplalllhaarpevqllesggglvqpqgslrlsca asgftfsnyamtwirqapgkglewvsgisggggstyyadsvkgr  
ftisrdnsntlylqmnsraedtavyycvkdwnntmitergqg tlvtvssttppaprpptpaptiasqplslrpeacrpaaggavht  
rgldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifk qpfmrpvqttqeedgcscrpfpeeeeggcelrvkfsrsadapayk  
qgqnqlynelnlgrreedydldkrrrdpemmkgkprknpqegl ynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdyd  
alhmqalppr

or the VHH comprises or consists of the nucleic acid sequence

(24) TABLE-US-00012 (VH-F; SEQ ID NO: 18011)

atggcactgcctgtgacagccctgctgctgctctggccctgct gctgcacgcagcacggcccagggtgcagctgctggagctctggag  
gaggcctggtgcagccccggccggtccctgagactgtcttgcgcc gccagcgggttcacctttacaaactacgccatgaattgggtgcg  
gcaggcccctggcaagggcctggagtggtgtctggcatcagcg gaggaggaggcagcacctactatgcagactccgtgaagggcagg  
ttcaccatctcccgcgataactctaagaatacactgtacctcca gatgaacagcctgagggcagaggacaccgccgtgtactattgcg  
tgaaggattggaataaccacaatgatcacagagaggggacagggc accctggtgaccgtgagcagcaccacaacccctgccccagacc  
tcccacacccgcccctaccatcgcgagtcagccactgtccctgc ggcctgaggcctgccggcccggccggcgaggcagtgacaca  
cggggcctggactttgctgtgacatctacatatgggcaccact ggcaggaacctgcggcgtgctgctgctgagcctggtcatcacc  
tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag cagcccttcagtagaccctgagaccacacaggaggaggacgg  
ctgctctttaggttcccagaggaggaggaggaggtatgcgagc tgcgcgtgaagtttagccggtccgccgatgcacctgcatacaag  
caggggacagaaccagctgtataacgagctgaatctgggcccggag agaggagtacgacgtgctggataagaggcgggggccgggacccg  
agatgggagggaagcctcgagaaagaacccacaggaggggcctg tacaatgagctgcaaaaggacaagatggccgaggcctattctga  
gatcggcatgaaggagagaggcgccgggggcaagggacacgatg gcctgtaccagggcctgagcaccgccacaaaggacacctatgat  
gccctgcacatgcaggccctgccccctagatga.

(25) The disclosure provides a chimeric antigen receptor (CAR) comprising: (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. As used throughout the disclosure, a CAR comprising a VH is referred to as a VCAR. In certain embodiments, the antigen recognition region may comprise two VHs to produce a bi-specific or tandem VCAR. In certain embodiments, the antigen recognition region may comprise three VHs to produce a tri-specific VCAR. In certain embodiments, the ectodomain may further comprise a signal peptide. Alternatively, or in addition, in certain embodiments, the ectodomain may further comprise a hinge between the antigen recognition region and the transmembrane domain. In certain embodiments, the ectodomain may further comprise a signal peptide. Alternatively, or in addition, in certain embodiments, the ectodomain may further comprise a hinge between the antigen recognition region and the transmembrane domain.

(26) In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH comprises a human or a humanized sequence.

(27) In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH comprises a non-naturally occurring sequence.

(28) In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH is not naturally occurring.

(29) In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH comprises a recombinant or chimeric sequence.

(30) In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH is produced by an in vitro procedure of affinity selection and recombination.

(31) In certain embodiments of the VCARs of the disclosure, the VH comprises or consists of the amino acid sequence

(32) TABLE-US-00013 (VH-A; SEQ ID NO: 18000)

malpvtalllplalllhaarpevqllesggglvqpggsrlsca asgftfssyamnwvrqapgkglewvagiigsggstyyadsvkgr  
fsisrdnsknldlqmnsraedtavyycvkdwnntmitergqg tlvtvssttpaprpptpaptiasqplslrpeacrpaaggavht  
rgldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifk qpfmrpvqtteedgcscrpfeeeeggcelrvkfsrsadapayk  
qgqnqlynelnlgrreedydvldkrrgrdpemggkprkrnpqegl ynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdyd  
alhmqalppr

or the VH comprises or consists of the nucleic acid sequence

(33) TABLE-US-00014 (VH-A; SEQ ID NO: 18001)

atggctctgcctgtgacagctctgctgctgctctggctctgct tcttcatgcggcgcgcctgaagttcagctgcttgaatctggcg

gaggcctggttcaacctggcgatctctgagactgagctgtgcc gccagcggcttcaccttttagcagctacgcatgaactgggtccg  
acaggccccctggcaaaggactggaatgggtggccggaatcatcg gcagcggcggcagcacatattacgccgattctgtgaagggccgc  
ttcagcatcagccgggacaacagcaagaacaccctggacctgca gatgaacagcctgagagccgaggataaccgccgtgtactactgcg  
tgaaggattggaacaccaccatgatcaccgagagaggccagggc aactggtcaccgtgtcctctacaacaacaccggcgctcggcc  
tccaacaccagctcctacaatcgcgagtcagcccctgtctctca gaccgaagcctgtagacctgctgctggcggagctgtgcatacc  
agaggactggatttcgctgcgacatctacatctgggctcctct ggctggcacatgcggagttttgctgctgagcctgggtcatcacc  
tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag cagccctcatgagacccgtgcagaccacacaggaggaggacgg  
ctgctctgtaggttcccagaggaggaggaggaggtatgcgagc tgcgcgtgaagttagccgggtccgccgatgcacctgcatacaag  
cagggacagaaccagctgtataacgagctgaatctgggcccggag agaggagtacgacgtgctggataagaggcggggccgggaccccg  
agatgggagggaagcctcggagaaagaacccacaggagggcctg tacaatgagctgcaaaaggacaagatggccgaggcctattctga  
gatcgcatgaaggagagaggcgccgggggcaaggacacgatg gcctgtaccagggcctgagcaccgccacaaaggacacctatgat  
gcctgcacatgcaggcctgccccctagatga.

(34) In certain embodiments of the VCARs of the disclosure, the VH comprises or consists of the amino acid sequence

(35) TABLE-US-00015 (VH-B; SEQ ID NO: 18002)

malpvtalllplalllhaarpevqllesggglvqpgsltsca asgttsnyamnwwrqapgkglewvsgngsgattyyadsvkgrt  
tisrdnsntlnlqmnsraedtaiyycvkdwnntmitergqgt lvtvssttpaprpptpaptiasqplsrpeacrpaaggavhtr  
glldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifkq pfmrpvqttqeedgcscrpfeeeeggcelrvkfsrsadapaykq  
gqnqlynelnlgrreedyvldkrrgrdpemggkprrknpqegly nelqkdkmaeayseigmkgerrrgkghdglyqglstatkdyda  
lhmqlppr

or the VH comprises or consists of the nucleic acid sequence

(36) TABLE-US-00016 (VH-B; SEQ ID NO: 18003)

atggctctgcctgtgacagctctgctgctgcctctggctctgct tcttcatgcggcgccctgaagttcagctgcttgaatctggcg  
gaggcctggttcaacctggcgatctctgacactgagctgtgcc gccagcggcttcacctcagcaactacgccatgaactgggtccg  
acaggccccctggcaaaggccttgaatgggtgtccggcatcattg gctctggcgccaccacactactacgccgattctgtgaagggcaga  
ttcaccatcagccgggacaacagcaagaacaccctgaacctgca gatgaacagcctgagagccgaggacaccgccatctactactgcg  
tgaaggactggaacaccaccatgatcaccgagagaggccagggc aactggtcaccgtgtcctctacaacaacaccggcgctcggcc  
tccaacaccagctcctacaatcgcgagtcagcccctgtctctca gaccgaagcctgtagacctgctgctggcggagctgtgcatacc  
agaggactggatttcgctgcgacatctacatctgggctcctct ggctggcacatgcggagttttgctgctgagcctgggtcatcacc  
tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag cagccctcatgagacccgtgcagaccacacaggaggaggacgg  
ctgctctgtaggttcccagaggaggaggaggaggtatgcgagc tgcgcgtgaagttagccgggtccgccgatgcacctgcatacaag  
cagggacagaaccagctgtataacgagctgaatctgggcccggag agaggagtacgacgtgctggataagaggcggggccgggaccccg  
agatgggagggaagcctcggagaaagaacccacaggagggcctg tacaatgagctgcaaaaggacaagatggccgaggcctattctga  
gatcgcatgaaggagagaggcgccgggggcaaggacacgatg gcctgtaccagggcctgagcaccgccacaaaggacacctatgat  
gcctgcacatgcaggcctgccccctagatga.

(37) In certain embodiments of the VCARs of the disclosure, the VH comprises or consists of the amino acid sequence

(38) TABLE-US-00017 (VH-C; SEQ ID NO: 18004)

malpvtalllplalllhaarpevqllesggglvqpgeslrscasgftfsnyamnwwrqapgkglewvsgivgggtsyyadsvrgr  
ftisrdnsntlylqmnsraedtavyycvkdwnntmitergqg tlvtvssttpaprpptpaptiasqplsrpeacrpaaggavht  
rgldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifk qpfmrpvqttqeedgcscrpfeeeeggcelrvkfsrsadapayk  
qgqnqlynelnlgrreedyvldkrrgrdpemggkprrknpqegl ynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdyd  
alhmqlppr

or the VH comprises or consists of the nucleic acid sequence

(39) TABLE-US-00018 (VH-C; SEQ ID NO: 18005)

atggctctgcctgtgacagctctgctgctgcctctggctctgct tcttcatgcggcgccctgaagttcagctgcttgaatctggcg  
gaggcctggttcagcctggcgaatctctgagactgagctgtgcc gccagcggcttcacctcagcaactacgccatgaactgggtccg  
acaggccccctggcaaaggccttgaatgggtgtccggaatcgttg gcggcggaggcacaagctactacgccgattctgtgcggggcaga  
ttcaccatcagccgggacaacagcaagaacaccctgtacctgca gatgaacagcctgagagccgaggacaccgccgtgtactactgcg  
tgaaggactggaacaccaccatgatcaccgagagaggccagggc aactggtcaccgtgtcctctacaacaacaccggcgctcggcc  
tccaacaccagctcctacaatcgcgagtcagcccctgtctctca gaccgaagcctgtagacctgctgctggcggagctgtgcatacc  
agaggactggatttcgctgcgacatctacatctgggctcctct ggctggcacatgcggagttttgctgctgagcctgggtcatcacc



tgtagtgaagagaggcaggaagaagctgtgtatcttcaag cagcccttcagagaccctgtgcagaccacacaggaggaggacgg  
ctgctctttagaggtccagaggaggaggaggaggatgcgagc tgcgcgtgaagtttagccggtccgccgatgcacctgcatacaag  
caggacagaaccagctgtataacgagctgaatctgggcccggag agaggagtacgacgtgctggataagaggcgggggccgggaccccg  
agatgggaggcaagcctcgagaaagaacccacaggaggggcctg tacaatgagctgcaaaaggacaagatggccgaggcctattctga  
gatcggcatagaaggagagaggcgccggggcaaggacacgatg gcctgtaccagggcctgagcaccgccacaaaggacacctatgat  
gccctgcacatgcaggccctgccccctagatga.

(40) In certain embodiments of the VCARS of the disclosure, the VH comprises or consists of the amino acid sequence

(41) TABLE-US-00019 (VH-D; SEQ ID NO: 18006)

malpvtalllplalllhaarpevqllesggglvqpqgsrlrsc asgftfsnyamtwirqapgkglewvsgitdggstfyadsvkgr  
ftisrdnsntlylqmnsraedtavyycvkdwnntmitergqg tlvtvssttpaprpptpaptiasqplslrpeacrpaaggavht  
rgldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifk qpfmrpvqttqeedgcscrpfeeeeggcelrvkfsrsadapayk  
qgqnqlynelnlgrreeydvldkrrrdpemmkgkprknpegl ynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdyd  
alhmqlppr

or the VH comprises or consists of the nucleic acid sequence

(42) TABLE-US-00020 (VH-D; SEQ ID NO: 18007)

atggctctgcctgtgacagctctgctgctgcctctggctctgc ttctcatgcggcgccctgaagttcagctgcttgaatctgg  
cggaggcctggttcaacctggcgcatctctgagactgagctgt gccgccagcggcttcaccttcagcaattacccatgacctgga  
tcagacaggccccctggcaaaggcctggaatgggtgtccggaat tacaggcgacggcggcagcaccttttacgccgattctgtgaag  
ggcagattcaccatcagccgggacaacagcaagaacaccctgt acctgcagatgaacagcctgagagccgaggacaccgccgtgta  
ctactgcgtgaaggactggaacaccacatgatcaccgagaga ggccagggcacactggtcaccgtgtcctctacaacaacaccgg  
cgctcggcctccaacaccagctctacaatcgcgagtcagcc cctgtctctcagaccgaagcctgtagacctgctgctggcgga  
gctgtgcataccagaggactggattcgctgcacatctaca tctgggctcctctggctggcacatgcggagttttgctgctgag  
cctggtcatcacctgtactgtaagagaggcaggaagaagctg ctgtatatcttaagcagcccttcagacccgtgcagacca  
cacaggaggaggacggctgctctttaggttcccagaggagga ggaggaggagatgcgagctgcgcgtgaagtttagccgggtccgcc  
gatgcacctgcatacaagcaggacagaaccagctgtataacg agctgaatctgggcccggagagaggagtacgacgtgctggataa  
gaggcgggggccgggaccccgagatgggaggcaagcctcgga aagaaccacagaggaggcctgtacaatgagctgcaaaaggaca  
agatggccgaggcctattctgagatcgcatgaaggagagag gcgccggggcaaggacacgatggcctgtaccagggcctgagc  
accgccacaaaggacacctatgatgcctgcacatgcaggccc tgccccctagatga.

(43) In certain embodiments of the VCARS of the disclosure, the VH comprises or consists of the amino acid sequence

(44) TABLE-US-00021 (VH-F; SEQ ID NO: 18010)

malpvtalllplalllhaarpevqllesggglvqpgrslrsc aasgftfnyamnwwrqpapgkglewvsgisggggstyyadsvk  
grftisrdnsntlylqmnsraedtavyycvkdwnntmiter gqgtlvtvssttpaprpptpaptiasqplslrpeacrpaagg  
avhtrgldfacdiyiwaplagtcgvllslvitlyckrgrkkl lyifkqpfmrpvqttqeedgcscrpfeeeeggcelrvkfsrsa  
dapaykqgqnqlynelnlgrreeydvldkrrrdpemmkgkpr knppeglynelqkdkmaeayseigmkgerrrgkghdglyqgls  
tatkdtydalhmqlppr

or the VH comprises or consists of the nucleic acid sequence

(45) TABLE-US-00022 (VH-E; SEQ ID NO: 18009)

atggcactgcctgtgacagccctgctgctgcctctggccctg ctgctgcacgcagcacggccccgaggtgcagctgctggagtcc  
ggaggaggcctggcccagcctggcgagcctgaggctgtcc tgcgccgctctggcttcaccttagctcctacgccatgaac  
tggatcagacaggccccctggcaaaggcctggagtgggtgtcc ggcatctccggctctggaggctctacatactatgccgacagc  
gtgaaggggccggttcaccatcagcagagataactccaagaat accgtgtacctccagatgaactctctgcgggcccaggacacc  
gccgtgtactattgcgtgaaggattggaataccacaatgatc acagagagggggccagggcaccctgggtgacagtgtctagcacc  
acaaccctgccccagacctccacacccgccccctaccatc gcgagtcagccactgtccctgcggcctgaggcctgccggccc  
gccgccggcgaggagcagtcacacacggggcctggactttgcc tgtgacatctacatatgggcaccactggcaggaacctgcggc  
gtgctgctgctgagcctggctcatcacctgtactgtaagaga ggaggaagaagctgctgtatatcttaagcagcccttcag  
agaccctgcagaccacacaggaggaggacggctgctcttgt aggttcccagaggaggaggaggaggatgcgagctgcgcgtg  
aagtttagccggtccgccgatgcacctgcatacaagcaggga cagaaccagctgtataacgagctgaatctgggcccggagagag  
gagtacgacgtgctggataagaggcgggggccgggacccccgag atgggaggcaagcctcggaagaagaacccacaggagggcctg  
tacaatgagctgcaaaaggacaagatggccgaggcctattct gagatcgcatgaaggagagaggcgccggggcaaggacac  
gatggcctgtaccagggcctgagcaccgccacaaaggacacc tatgatgcctgcacatgcaggccctgccccctagatga.

(46) In certain embodiments of the VCARS of the disclosure, the VH comprises or consists of the amino acid sequence

(47) TABLE-US-00023 (VH-F; SEQ ID NO: 18010)

malpvtalllplalllhaarpevqllesggglyqpgrsldscaasgftft  
nyamnwyraqapgkglewysgisggggstyyadsvkgrftisrdnskntly  
lqmnsiraedtavyycvkdwnttmitergqgtlvtvssttpaprpptpa  
ptiasqplslrpeacrpaaggavhtrglfacdiyiwaplagtcgvllls  
lvitlyckrgrkkllyifkqpfmrpvqttqeedgcscrpfeeeeggcelr  
vkfsrsadapaykqgqnqlynelnlgrreedydldkrrgrdpemggkpr  
knpqeglynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdy dalhmqalppr

or the VH comprises or consists of the nucleic acid sequence

(48) TABLE-US-00024 (VH-F; SEQ ID NO: 18011)

atggcactgcctgtgacagccctgctgctgcctctggccctgctgctgca  
cgagcagcagggccgaggtgcagctgctggagtctggaggaggcctggtgc  
agcccggccggtccctgagactgtcttgcgccgagcggcttcacctt  
acaaactacgccatgaattgggtgcggcaggcccctggcaagggcctgga  
gtgggtgtctggcatcagcggaggaggaggcagcacctactatgcagact  
ccgtgaagggcaggttcacatctcccgcgataactctaagaatacactg  
tacctccagatgaacagcctgagggcagaggacaccgccgtgtactattg  
cgtgaaggattggaataaccacaatgatcacagagaggggacagggcaccc  
tggtgaccgtgagcagcaccacaacccctgccccagacctccacaccc  
gcccctaccatcgcgagtcagccactgtccctgcggcctgaggcctgccg  
gcccgcgcggcgaggagcagtcacacacggggcctggactttgcctgtg  
acatctacatatgggcaccactggcaggaacctgcggcgtgctgctgctg  
agcctggtcatcacctgtactgtaagagaggcaggaagaagctgctgta  
tatcttcaagcagcccttcagagacccgtgcagaccacacaggaggagg  
acggctgctctttaggttcccagaggaggaggaggaggatgcgagctg  
cgctgaagttagccggtccgcccgatgcacctgcatacaagcagggaca  
gaaccagctgtataacgagctgaatctgggcccggagagaggagtacgacg  
tgctggataagaggcggggcccgggaccccagatgggaggcaagcctcgg  
agaaagaacccacaggaggggcctgtacaatgagctgcaaaaggacaagat  
ggccgaggcctattctgagatcggcagatgaaggagagaggcgccggggca  
agggacacgatggcctgtaccagggcctgagcaccgccacaaaggacacc tatgatgcctgcacatgcaggccctgccccctagatga.

(49) In certain embodiments of the VCARS of the disclosure, the VCAR comprises a single domain antibody, VHH, VH or a combination thereof. In some embodiments, the single domain antibody, VHH or VH comprises or consists of a recombinant sequence and/or a chimeric sequence. In some embodiments, the single domain antibody, VHH or VH comprises or consists of a human sequence and/or a humanized sequence.

(50) In certain embodiments of the VCARS of the disclosure, the VCAR comprises a single domain antibody. In some embodiments, the single domain antibody is a VHH or a VH antibody. In some embodiments, the VH antibody is a UniDab antibody. In some embodiments, VH antibody is not a fragment of a naturally occurring monoclonal antibody.

(51) In certain embodiments of the VCARS of the disclosure, the signal peptide may comprise a sequence encoding a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR signal peptide. In certain embodiments of the VCARS of the disclosure, the signal peptide may comprise a sequence encoding a human CD8α signal peptide. The human CD8α signal peptide may comprise an amino acid sequence comprising MALPVTALLLPLALLLHAARP (SEQ ID NO: 18012). The human CD8α signal peptide may comprise an amino acid sequence comprising MALPVTALLLPLALLLHAARP (SEQ ID NO: 18012) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the an amino acid sequence comprising MALPVTALLLPLALLLHAARP (SEQ ID NO: 18012). The human CD8α signal peptide may be encoded by a nucleic acid sequence comprising atggcactgccagtcaccgccctgctgctgcctctggctctgctgctgcacgcagctagacca (SEQ ID NO: 18013).

(52) In certain embodiments of the VCARs of the disclosure, the transmembrane domain may comprise a sequence encoding a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR transmembrane domain. In certain embodiments of the VCARs of the disclosure, the transmembrane domain may comprise a sequence encoding a human CD8α transmembrane domain. The CD8α transmembrane domain may comprise an amino acid sequence comprising IYIWAPLAGTCGVLLLSLVITLYC (SEQ ID NO: 18014) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising IYIWAPLAGTCGVLLLSLVITLYC (SEQ ID NO: 18014). The CD8α transmembrane domain may be encoded by the nucleic acid sequence comprising atctacatttgggcaccactggccgggacctgtggagtgtgtgtgtgagcctgggtcatcacactgtactgc (SEQ ID NO: 18015).

(53) In certain embodiments of the VCARs of the disclosure, the endodomain may comprise a human CD3ζ endodomain.

(54) In certain embodiments of the VCARs of the disclosure, the at least one costimulatory domain may comprise a human 4-1BB, CD28, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof. In certain embodiments of the VCARs of the disclosure, the at least one costimulatory domain may comprise a CD28 and/or a 4-1BB costimulatory domain. The CD3zeta costimulatory domain may comprise an amino acid sequence comprising RVKFSRSADAPAYKQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGL YNELQKDKMAEAYSEIGMKGERRRGKGGHDGLYQGLSTATKDTYDALHMQALPPR (SEQ ID NO: 18016) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising

RVKFSRSADAPAYKQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGL YNELQKDKMAEAYSEIGMKGERRRGKGGHDGLYQGLSTATKDTYDALHMQALPPR (SEQ ID NO: 18016). The CD3zeta costimulatory domain may be encoded by the nucleic acid sequence comprising

cgcgtgaagtttagtcgatcagcagatgccccagcttacaacagggacagaaccagctgtataacgagctgaatctgggcccgcgagag gaatatgacgtgctggataagcggagaggacgcgaccccgaaatgggaggcaagcccaggcgcaaaaccctcaggaaggcctgtat aacgagctgcagaaggacaaaatggcagaagcctattctgagatcggcatgaagggggagcgcgagaggcgaaggggcacgatgg gctgtaccagggactgagcaccgccacaaaggacacctatgatgctctgcatatgcaggcactgcctccaagg (SEQ ID NO: 18017).

The 4-1BB costimulatory domain may comprise an amino acid sequence comprising KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCEL (SEQ ID NO: 18018) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCEL (SEQ ID NO: 18018). The 4-1BB costimulatory domain may be encoded by the nucleic acid sequence comprising

aagagaggcaggaagaaactgctgtatatatttcaaacagcccttcagtcgccccgtgcagactaccagagggaagacgggtgctcctgt cgattccctgaggaagaggaaggcgggtgtgagctg (SEQ ID NO: 18019). The 4-1BB costimulatory domain may be located between the transmembrane domain and the CD28 costimulatory domain.

(55) In certain embodiments of the VCARs of the disclosure, the hinge may comprise a sequence derived from a human CD8α, IgG4, and/or CD4 sequence. In certain embodiments of the VCARs of the disclosure, the hinge may comprise a sequence derived from a human CD8α sequence. The hinge may comprise a human CD8α amino acid sequence comprising

TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACD (SEQ ID NO: 18020) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACD (SEQ ID NO: 18020). The human CD8α hinge amino acid sequence may be encoded by the nucleic acid sequence comprising

(56) TABLE-US-00025 (SEQ ID NO: 18021)

actaccacaccagcacctagaccaccaactccagctccaaccatcgcgag

tcagcccctgagtctgagacctgaggcctgcaggccagctgcaggaggag ctgtgcacaccaggggcctggacttcgcctgcgac.

(57) VHHs and/or VCARs of the disclosure may bind an antigen with at least one affinity selected from a K.sub.D of less than or equal to 10.sup.-9M, less than or equal to 10.sup.-10M, less than or equal to 10.sup.-11M, less than or equal to 10.sup.-12M, less than or equal to 10.sup.-13M, less than or equal to 10.sup.-14M, and less than or equal to 10.sup.-15M. The K.sub.D may be determined by surface

plasmon resonance.

(58) The disclosure provides an anti-BCMA VCAR. The disclosure provides a composition comprising the VCAR of the disclosure and at least one pharmaceutically acceptable carrier.

(59) The disclosure provides a transposon comprising the VCAR of the disclosure.

(60) Transposons of the disclosure may comprise a selection gene for identification, enrichment and/or isolation of cells that express the transposon. Exemplary selection genes encode any gene product (e.g., transcript, protein, enzyme) essential for cell viability and survival. Exemplary selection genes encode any gene product (e.g., transcript, protein, enzyme) essential for conferring resistance to a drug challenge against which the cell is sensitive (or which could be lethal to the cell) in the absence of the gene product encoded by the selection gene. Exemplary selection genes encode any gene product (e.g., transcript, protein, enzyme) essential for viability and/or survival in a cell media lacking one or more nutrients essential for cell viability and/or survival in the absence of the selection gene. Exemplary selection genes include, but are not limited to, neo (conferring resistance to neomycin), DHFR (encoding Dihydrofolate Reductase and conferring resistance to Methotrexate), TYMS (encoding Thymidylate Synthetase), MGMT (encoding O(6)-methylguanine-DNA methyltransferase), multidrug resistance gene (MDR1), ALDH1 (encoding Aldehyde dehydrogenase 1 family, member A1), FRNCF, RAD51C (encoding RAD51 Paralog C), GCS (encoding glucosylceramide synthase), and NKX2.2 (encoding NK2 Homeobox 2).

(61) Transposons of the disclosure may comprise an inducible proapoptotic polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. In certain embodiments, the ligand binding region may be a multimeric ligand binding region. Inducible proapoptotic polypeptides of the disclosure may also be referred to as an “iC9 safety switch”. In certain embodiments, transposons of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, transposons of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, transposons of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand-binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide. In certain embodiments, the amino acid sequence of the ligand binding region that comprise a FK506 binding protein 12 (FKBP12) polypeptide may comprise a modification at position 36 of the sequence. The modification may be a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In certain embodiments, the FKBP12 polypeptide is encoded by an amino acid sequence comprising

GVQVETISPGDGRTPKRGQTCVVHYTGMLLEDGKKVDSSRDRNKPFFKMLGKQEVIRG  
WEEGVAQMSVGQRAKLTISPDYAYGATGHPGIIPPHATLVFDVELLKLE (SEQ ID NO: 18022). In certain embodiments, the FKBP12 polypeptide is encoded by a nucleic acid sequence comprising  
GGGGTCCAGGTCGAGACTATTTACACAGGGGATGGGCGAACATTTCCAAAAAGGGG  
CCAGACTTGCGTCGTGCATTACACCGGGATGCTGGAGGACGGGAAGAAAGTGGACA  
GCTCCAGGGATCGCAACAAGCCCTTCAAGTTCATGCTGGGAAAGCAGGAAGTGATC  
CGAGGATGGGAGGAAGGCGTGGCACAGATGTCAGTCGGCCAGCGGGCCAAACTGA  
CCATTAGCCCTGACTACGCTTATGGAGCAACAGGCCACCCAGGGATCATTCCCCCTC  
ATGCCACCCTGGTCTTCGAT GTGGAAGTCTGAAGCTGGAG (SEQ ID NO: 18023). In certain embodiments, the induction agent specific for the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V) comprises AP20187 and/or AP1903, both synthetic drugs.

(62) In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides

or truncated caspase 9 polypeptides of the disclosure, the linker region is encoded by an amino acid comprising GGGGS (SEQ ID NO: 18024) or a nucleic acid sequence comprising GGAGGAGGAGGATCC (SEQ ID NO: 18025). In certain embodiments, the nucleic acid sequence encoding the linker does not comprise a restriction site.

(63) In certain embodiments of the truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an arginine (R) at position 87 of the sequence. Alternatively, or in addition, in certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an alanine (A) at position 282 the sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid comprising

(64) TABLE-US-00026 (SEQ ID NO: 18026)

GFGDVGALESLRGNADLAYILSMEPCGHCLIINNPNFCRESGLRTRTGSN  
IDCEKLRRRFSSLHFMVEVKGDLTAKKMVLALLELAQQDHGALDCCVVVI  
LSHGCQASHLQFPGAVYGTGCPVSVEKIVNIFNGTSCPSLGGKPKLFFI  
QACGGEQKDHGFEVASTPEDESPGSNPEPDATPFQEGLRTFDQLDAISS  
LPTPSDIFVSYSTFPGFVSWRDPKSGSWYVETLDDIFEQWAHSEDLQSL  
LRVANAVSVKGIYKQMPGCFNFLRKKLFFKTS

a nucleic acid sequence comprising

(65) TABLE-US-00027 (SEQ ID NO: 18027)

TTTGGGGACGTGGGGGCCCTGGAGTCTCTGCGAGGAAATGCCGATCTGGC  
TTACATCCTGAGCATGGAACCCTGCGGCCACTGTCTGATCATTAAACAATG  
TGAACCTCTGCAGAGAAAGCGGACTGCGAACACGGACTGGCTCCAATATT  
GACTGTGAGAAGCTGCGGAGAAGGTTCTCTAGTCTGCACTTTATGGTCGA  
AGTGAAAGGGGATCTGACCGCCAAGAAAATGGTGCTGGCCCTGCTGGAGC  
TGGCTCAGCAGGACCATGGAGCTCTGGATTGCTGCGTGGTCGTGATCCTG  
TCCCACGGGTGCCAGGCTTCTCATCTGCAGTTCCCCGGAGCAGTGTACGG  
AACAGACGGCTGTCCTGTCAGCGTGGAGAAGATCGTCAACATCTTCAACG  
GCACTTCTTGCCCTAGTCTGGGGGGAAAGCCAAAAGTGTCTTTATCCAG  
GCCTGTGGCGGGGAACAGAAAGATCACGGCTTCGAGGTGGCCAGCACCAG  
CCCTGAGGACGAATCACCAGGGAGCAACCCTGAACCAGATGCAACTCCAT  
TCCAGGAGGGACTGAGGACCTTTGACCAGCTGGATGCTATCTCAAGCCTG  
CCCCTCCTAGTGACATTTTCGTGTCTTACAGTACCTTCCCAGGCTTTGT  
CTCATGGCGCGATCCCAAGTCAGGGAGCTGGTACGTGGAGACACTGGACG  
ACATCTTTGAACAGTGGGCCCATTCAGAGGACCTGCAGAGCCTGCTGCTG  
CGAGTGGCAAACGCTGTCTCTGTGAAGGGCATCTACAAACAGATGCCCGG  
GTGCTTCAATTTTCTGAGAAAGAAAGTGTCTTTAAGACTTCC.

(66) In certain embodiments of the inducible proapoptotic polypeptides, wherein the polypeptide comprises a truncated caspase 9 polypeptide, the inducible proapoptotic polypeptide is encoded by an amino acid sequence comprising

(67) TABLE-US-00028 (SEQ ID NO: 18028)

GVQVETISPGDGRTPKRGQTCVVYHYTGMLEDGKKVDSSRDRNKPFFKM  
LGKQEVIRGWEEGVAQMSVGQRAKLTISPDYAYGATGHPGIIPPHATLVF  
DVELLKLEGGGGSGFGDVGALESLRGNADLAYILSMEPCGHCLIINNPNF  
CRESGLRTRTGSNIDCEKLRRRFSSLHFMVEVKGDLTAKKMVLALLELAQ  
QDHGALDCCVVVILSHGCQASHLQFPGAVYGTGCPVSVEKIVNIFNGTS  
CPSLGGKPKLFFIQACGGEQKDHGFEVASTPEDESPGSNPEPDATPFQE  
GLRTFDQLDAISSLPTPSDIFVSYSTFPGFVSWRDPKSGSWYVETLDDIF  
EQWAHSEDLQSLLLRVANAVSVKGIYKQMPGCFNFLRKKLFFKTS

or the nucleic acid sequence comprising

(68) TABLE-US-00029 (SEQ ID NO: 18029)

gggggtcagggtcgagactatttcaccagggggatgggcgaacatttccaaa  
aaggggcccagacttgctgcgtgcattacaccgggatgctggaggacggga  
agaaagtggacagctccagggtatcgcaacaagcccttcaagttcatgctg  
ggaaagcaggaaagtgatccgaggatgggaggaaggcgtggcacagatgct  
agtcgggccagcgggccaactgaccattagccctgactacgcttatggag  
caacaggccaccagggtatcattccccctcatgccaccctggtcttcgat  
gtggaactgctgaagctggaggaggaggaggatccggatttggggacgt  
ggggggccctggagtctctgcgaggaaatgccgatctggcttacatcctga  
gcatggaaccctgcggccactgtctgatcattaacaatgtgaacttctgc  
agagaaagcggactgcgaacacggactggctccaatattgactgtgagaa  
gctgcggagaagggttcttagtctgcactttatggtcgaagtgaagggg  
atctgaccgcaagaaaatgggtgctggccctgctggagctggctcagcag  
gacatggagctctggattgctgcgtggctgctgacctgtcccacgggtg  
ccaggcttctcatctgcagttccccggagcaggtacggaacagacggct  
gtcctgtcagcgtggagaagatcgtaacatcttcaacggcacttctgc  
cctagtctggggggaaagccaaaactgttctttatccaggcctgtggcgg  
ggaacagaaagatcacggcttcgaggtggccagcaccagccctgaggacg  
aatcaccaggggagcaaccctgaaccagatgcaactccattccaggaggga  
ctgaggacctttgaccagctggatgctatctcaagcctgccactcctag tgacatttctgtgtcttacagtaccttccaggccttgtctcatggcgcg  
atcccaagtcaggggagctggtagctggagacactggacgacatctttgaa  
cagtggggccattcagaggacctgcagagcctgctgctgcgagtggcaaa  
cgctgtctctgtgaagggtcatctacaacagatgcccggtgcttcaatt ttctgagaaagaaactgttctttaagacttcc.

(69) Transposons of the disclosure may comprise at least one self-cleaving peptide(s) located, for example, between one or more VHH(s) or VCAR(s) of the disclosure and a selection gene of the disclosure. Transposons of the disclosure may comprise at least one self-cleaving peptide(s) located, for example, between one or more VHH(s) or VCAR(s) of the disclosure and an inducible proapoptotic polypeptide of the disclosure. Transposons of the disclosure may comprise at least two self-cleaving peptide(s), a first self-cleaving peptide located, for example, upstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure and a second first self-cleaving peptide located, for example, downstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure.

(70) The at least one self-cleaving peptide may comprise, for example, a T2A peptide, GSG-T2A peptide, an E2A peptide, a GSG-E2A peptide, an F2A peptide, a GSG-F2A peptide, a P2A peptide, or a GSG-P2A peptide. A T2A peptide may comprise an amino acid sequence comprising EGRGSLLTCGDVEENPGP (SEQ ID NO: 18030) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising EGRGSLLTCGDVEENPGP (SEQ ID NO: 18030). A GSG-T2A peptide may comprise an amino acid sequence comprising GSGEGRGSLLTCGDVEENPGP (SEQ ID NO: 18031) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGEGRGSLLTCGDVEENPGP (SEQ ID NO: 18031). A GSG-T2A peptide may comprise a nucleic acid sequence comprising ggatctggagaggggaagggaagcctgctgacctgtggagacgtggaggaaaaccaggacca (SEQ ID NO: 18032). An E2A peptide may comprise an amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033). A GSG-E2A peptide may comprise an amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034). An F2A peptide may comprise an amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035). A GSG-F2A peptide may comprise an amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising

GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036). A P2A peptide may comprise an amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037). A GSG-P2A peptide may comprise an amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038).

(71) Transposons of the disclosure may comprise a first and a second self-cleaving peptide, the first self-cleaving peptide located, for example, upstream of one or more VHH(s) or VCAR(s) of the disclosure the second self-cleaving peptide located, for example, downstream of the one or more VHH(s) or VCAR(s) of the disclosure. The first and/or the second self-cleaving peptide may comprise, for example, a T2A peptide, GSG-T2A peptide, an E2A peptide, a GSG-E2A peptide, an F2A peptide, a GSG-F2A peptide, a P2A peptide, or a GSG-P2A peptide. A T2A peptide may comprise an amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030). A GSG-T2A peptide may comprise an amino acid sequence comprising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031). A GSG-T2A peptide may comprise a nucleic acid sequence comprising ggatctggagagggaagggaagcctgtgacctgtggagacgtggaggaaaaccaggacca (SEQ ID NO: 18032). An E2A peptide may comprise an amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033). A GSG-E2A peptide may comprise an amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034). An F2A peptide may comprise an amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035). A GSG-F2A peptide may comprise an amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036). A P2A peptide may comprise an amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037). A GSG-P2A peptide may comprise an amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038).

(72) The disclosure provides a composition comprising the transposon the disclosure. In certain embodiments, the composition may further comprise a plasmid comprising a sequence encoding a transposase enzyme. The sequence encoding a transposase enzyme may be an mRNA sequence.

(73) Transposons of the disclosure may comprise piggyBac transposons. In certain embodiments of this method, the transposon is a plasmid DNA transposon with a sequence encoding the chimeric antigen receptor flanked by two cis-regulatory insulator elements. In certain embodiments, the transposon is a piggyBac transposon. Transposase enzymes of the disclosure may include piggyBac transposases or compatible enzymes. In certain embodiments, and, in particular, those embodiments wherein the transposon is a piggyBac transposon, the transposase is a piggyBac or a Super piggyBac (SPB) transposase. In certain embodiments, and, in particular, those embodiments wherein the transposase is a Super piggyBac (SPB) transposase, the sequence encoding the transposase is an mRNA sequence.

(74) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac (PB) transposase enzyme. The piggyBac (PB) transposase enzyme may comprise or consist of an amino

acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(75) TABLE-US-00030 (SEQ ID NO: 14487) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEI SDHVSEDDVQ SDTEEFIDE VHEVQPTSSG 61 SEILDEQNV  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTGATFRD  
TNEDEIYAFF 181 GILVMTAVRK DNHMSTDDL FDRSLSMVYVS VMSRDRFDFL  
IRCLRMDDKS IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ  
LLGFRGRCPF RMYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN  
GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT  
VRSNKREIPE VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421  
DEDASINEST GKPQMVMYYN QTKGGVDTL QMCSVMTCR KTNRWPMALL  
YGMINIACIN 481 SFIIYSHNVS SKGEKVQSRK KFMRLNLYMSL TSSFMRKRLE  
APTLKRYLRD NISNILPNEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA  
NASCKKCKKV ICREHNIDMC QSCF.

(76) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at one or more of positions 30, 165, 282, or 538 of the sequence:

(77) TABLE-US-00031 (SEQ ID NO: 14487) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEI SDHVSEDDVQ SDTEEFIDE VHEVQPTSSG 61 SEILDEQNV  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTGATFRD  
TNEDEIYAFF 181 GILVMTAVRK DNHMSTDDL FDRSLSMVYVS VMSRDRFDFL  
IRCLRMDDKS IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ  
LLGFRGRCPF RMYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN  
GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT  
VRSNKREIPE VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421  
DEDASINEST GKPQMVMYYN QTKGGVDTL QMCSVMTCR KTNRWPMALL  
YGMINIACIN 481 SFIIYSHNVS SKGEKVQSRK KFMRLNLYMSL TSSFMRKRLE  
APTLKRYLRD NISNILPNEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA  
NASCKKCKKV ICREHNIDMC QSCF.

(78) In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at two or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at three or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at each of the following positions 30, 165, 282, and 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the amino acid substitution at position 30 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 165 of the sequence of SEQ ID NO: 14487 is a substitution of a serine (S) for a glycine (G). In certain embodiments, the amino acid substitution at position 282 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 538 of the sequence of SEQ ID NO: 14487 is a substitution of a lysine (K) for an asparagine (N).

(79) In certain embodiments of the methods of the disclosure, the transposase enzyme is a Super piggyBac (SPB) transposase enzyme. In certain embodiments, the Super piggyBac (SPB) transposase enzymes of the disclosure may comprise or consist of the amino acid sequence of the sequence of SEQ ID NO: 14487 wherein the amino acid substitution at position 30 is a substitution of a valine (V) for an isoleucine (I), the amino acid substitution at position 165 is a substitution of a serine (S) for a glycine (G), the amino acid substitution at position 282 is a substitution of a valine (V) for a methionine (M), and the amino acid substitution at position 538 is a substitution of a lysine (K) for an asparagine (N). In



certain embodiments, the Super piggyBac (SPB) transposase enzyme may comprise or consist of an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(80) TABLE-US-00032 (SEQ ID NO: 14484) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEV SDHVSEDDVQ SDTEEFIDE VHEVQPTSSG 61 SEILDEQNVI  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIIEIVKW TNAEISLKRR ESMTSATFRD  
TNEDEIYAFF 181 GILVMTAVRK DNHMSTDDLF DRSLSMVYVS VMSRDRFDFL  
IRCLRMDDKS IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ  
LLGFRGRCPF RVYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN  
GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTVGT  
VRSNKREIPE VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421  
DEDASINEST GKPQMVMYYN QTKGGVDTLQ QMCSVMTCSR KTNRWPMALL  
YGMINIACIN 481 SFIIYSHNVS SKGEKVQSRK KFMRLNLYMSL TSSFMRKRLE  
APTLKRYLRD NISNILPKEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRKKA  
NASCKKCKKV ICREHNIDMC QSCF.

(81) The disclosure provides a vector comprising the VCAR of the disclosure. In certain embodiments, the vector is a viral vector. The vector may be a recombinant vector.

(82) Viral vectors of the disclosure may comprise a sequence isolated or derived from a retrovirus, a lentivirus, an adenovirus, an adeno-associated virus or any combination thereof. The viral vector may comprise a sequence isolated or derived from an adeno-associated virus (AAV). The viral vector may comprise a recombinant AAV (rAAV). Exemplary adeno-associated viruses and recombinant adeno-associated viruses of the disclosure comprise two or more inverted terminal repeat (ITR) sequences located in cis next to a sequence encoding a VHH or VCAR of the disclosure. Exemplary adeno-associated viruses and recombinant adeno-associated viruses of the disclosure include, but are not limited to all serotypes (e.g., AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, and AAV9). Exemplary adeno-associated viruses and recombinant adeno-associated viruses of the disclosure include, but are not limited to, self-complementary AAV (scAAV) and AAV hybrids containing the genome of one serotype and the capsid of another serotype (e.g., AAV2/5, AAV-DJ and AAV-DJ8). Exemplary adeno-associated viruses and recombinant adeno-associated viruses of the disclosure include, but are not limited to, rAAV-LK03.

(83) Viral vectors of the disclosure may comprise a selection gene. The selection gene may encode a gene product essential for cell viability and survival. The selection gene may encode a gene product essential for cell viability and survival when challenged by selective cell culture conditions. Selective cell culture conditions may comprise a compound harmful to cell viability or survival and wherein the gene product confers resistance to the compound. Exemplary selection genes of the disclosure may include, but are not limited to, neo (conferring resistance to neomycin), DHFR (encoding Dihydrofolate Reductase and conferring resistance to Methotrexate), TYMS (encoding Thymidylate Synthetase), MGMT (encoding O(6)-methylguanine-DNA methyltransferase), multidrug resistance gene (MDR1), ALDH1 (encoding Aldehyde dehydrogenase 1 family, member A1), FRNCF, RAD51C (encoding RAD51 Paralog C), GCS (encoding glucosylceramide synthase), NKX2.2 (encoding NK2 Homeobox 2) or any combination thereof.

(84) Viral vectors of the disclosure may comprise an inducible proapoptotic polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. In certain embodiments, the ligand binding region may be a multimeric ligand binding region. Inducible proapoptotic polypeptides of the disclosure may also be referred to as an “iC9 safety switch”. In certain embodiments, viral vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, viral vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein

the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, viral vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide. In certain embodiments, the amino acid sequence of the ligand binding region that comprise a FK506 binding protein 12 (FKBP12) polypeptide may comprise a modification at position 36 of the sequence. The modification may be a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In certain embodiments, the FKBP12 polypeptide is encoded by an amino acid sequence comprising

GVQVETISPGDGRFTFPRGQTCVVHYTGMLEDGKKVDSSRDRNKPFFKFMLGKQEVIRG  
WEEGVAQMSVGQRAKLTISPDYAYGATGHPGIIPPHATLVFDVELLKLE (SEQ ID NO: 18022). In certain embodiments, the FKBP12 polypeptide is encoded by a nucleic acid sequence comprising  
GGGGTCCAGGTCGAGACTATTTACACAGGGGATGGGCGAACATTTCCAAAAAGGGG  
CCAGACTTGCGTCGTGCATTACACCGGGATGCTGGAGGACGGGAAGAAAGTGGACA  
GCTCCAGGGATCGCAACAAGCCCTTCAAGTTCATGCTGGGAAAGCAGGAAGTGATC  
CGAGGATGGGAGGAAGGCGTGGCACAGATGTCAGTCGGCCAGCGGGCCAAACTGA  
CCATTAGCCCTGACTACGCTTATGGAGCAACAGGCCACCCAGGGATCATTCCCCCTC  
ATGCCACCCTGGTCTTCGAT GTGGAAGTCTGAAGCTGGAG (SEQ ID NO: 18023). In certain embodiments, the induction agent specific for the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V) comprises AP20187 and/or AP1903, both synthetic drugs.

(85) In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the linker region is encoded by an amino acid comprising GGGGS (SEQ ID NO: 18024) or a nucleic acid sequence comprising GGAGGAGGAGGATCC (SEQ ID NO: 18025). In certain embodiments, the nucleic acid sequence encoding the linker does not comprise a restriction site.

(86) In certain embodiments of the truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an arginine (R) at position 87 of the sequence. Alternatively, or in addition, in certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an alanine (A) at position 282 the sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid comprising

(87) TABLE-US-00033 (SEQ ID NO: 18026)

GFGDVGALESLRGNADLAYILSMEPCGHCLIINN VNFCRESGLRTRTGSN  
IDCEKLRRRFSSLHFMVEVKGDLTAKKMVLALLELAQQDHGALDCCVVVI  
LSHGCQASHLQFPGAVYGTGCPVSVEKIVNIFNGTSCPSLGGKPKLFFI  
QACGGEQKDHGFEVASTSPEDESPGSNPEPDATPFQEGLRTFDQLDAISS  
LPTPSDIFVSYSTFPGFVSWRDPKSGSWYVETLDDIFEQWAHSEDLQSL  
LRVANAVSVKGIYKQMPGCFNFLRKKLFFKTS

or a nucleic acid sequence comprising

(88) TABLE-US-00034 (SEQ ID NO: 18027)

TTTGGGGACGTGGGGGCCCTGGAGTCTCTGCGAGGAAATGCCGATCTGGC  
TTACATCCTGAGCATGGAACCCTGCGGCCACTGTCTGATCATTAACAATG  
TGAAGTCTGCGAGAGAAAGCGGACTGCGAACACGGACTGGCTCCAATATT  
GACTGTGAGAAGCTGCGGAGAAGGTTCTCTAGTCTGCACTTTATGGTCTGA  
AGTGAAAGGGGATCTGACCGCCAAGAAAATGGTGCTGGCCCTGCTGGAGC  
TGGCTCAGCAGGACCATGGAGCTCTGGATTGCTGCGTGGTCGTGATCCTG  
TCCCACGGGTGCCAGGCTTCTCATCTGCAGTTCCCCGGAGCAGTGTACGG

(92) Viral vectors of the disclosure may comprise at least one self-cleaving peptide. In some embodiments, the vector may comprise at least one self-cleaving peptide and wherein a self-cleaving peptide is located between a CAR and a selection gene. In some embodiments, the vector may comprise at least one self-cleaving peptide and wherein a first self-cleaving peptide is located upstream of a CAR and a second self-cleaving peptide is located downstream of a CAR. Viral vectors of the disclosure may comprise at least one self-cleaving peptide(s) located, for example, between one or more of a VCAR,

VCAR or VCAR and an inducible proapoptotic polypeptide of the disclosure. Viral vectors of the disclosure may comprise at least two self-cleaving peptide(s), a first self-cleaving peptide located, for example, upstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure and a second first self-cleaving peptide located, for example, downstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure. The self-cleaving peptide may comprise, for example, a T2A peptide, GSG-T2A peptide, an E2A peptide, a GSG-E2A peptide, an F2A peptide, a GSG-F2A peptide, a P2A peptide, or a GSG-P2A peptide. A T2A peptide may comprise an amino acid sequence comprising EGRGSLTCTGCDVEENPGP (SEQ ID NO: 18030) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising EGRGSLTCTGCDVEENPGP (SEQ ID NO: 18030). A GSG-T2A peptide may comprise an amino acid sequence comprising GSGEGRGSLTCTGCDVEENPGP (SEQ ID NO: 18031) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGEGRGSLTCTGCDVEENPGP (SEQ ID NO: 18031). A GSG-T2A peptide may comprise a nucleic acid sequence comprising ggatctggagaggggaaggggaagcctgctgacctgtggagacgtggaggaaaaccaggacca (SEQ ID NO: 18032). An E2A peptide may comprise an amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033). A GSG-E2A peptide may comprise an amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034). An F2A peptide may comprise an amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035). A GSG-F2A peptide may comprise an amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036). A P2A peptide may comprise an amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037). A GSG-P2A peptide may comprise an amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038).

(93) The disclosure provides a vector comprising the VCAR of the disclosure. In certain embodiments, the vector is a nanoparticle. Exemplary nanoparticle vectors of the disclosure include, but are not limited to, nucleic acids (e.g., RNA, DNA, synthetic nucleotides, modified nucleotides or any combination thereof), amino acids (L-amino acids, D-amino acids, synthetic amino acids, modified amino acids, or any combination thereof), polymers (e.g., polymersomes), micelles, lipids (e.g., liposomes), organic molecules (e.g., carbon atoms, sheets, fibers, tubes), inorganic molecules (e.g., calcium phosphate or gold) or any combination thereof. A nanoparticle vector may be passively or actively transported across a cell membrane.

(94) Nanoparticle vectors of the disclosure may comprise a selection gene. The selection gene may encode a gene product essential for cell viability and survival. The selection gene may encode a gene product essential for cell viability and survival when challenged by selective cell culture conditions. Selective cell culture conditions may comprise a compound harmful to cell viability or survival and wherein the gene product confers resistance to the compound. Exemplary selection genes of the disclosure may include, but are not limited to, neo (conferring resistance to neomycin), DHFR (encoding Dihydrofolate Reductase and conferring resistance to Methotrexate), TYMS (encoding Thymidylate Synthetase), MGMT (encoding O(6)-methylguanine-DNA methyltransferase), multidrug resistance gene (MDR1), ALDH1 (encoding Aldehyde dehydrogenase 1 family, member A1), FRNCF, RAD51C (encoding RAD51 Paralog C), GCS (encoding glucosylceramide synthase), NKX2.2 (encoding NK2 Homeobox 2) or any combination thereof.

(95) Nanoparticle vectors of the disclosure may comprise an inducible proapoptotic polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. In certain embodiments, the ligand binding region may be a multimeric ligand binding region. Inducible proapoptotic polypeptides of the disclosure may also be referred to as an “iC9 safety switch”. In certain embodiments, nanoparticle vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, nanoparticle vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, nanoparticle vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide. In certain embodiments, the amino acid sequence of the ligand binding region that comprise a FK506 binding protein 12 (FKBP12) polypeptide may comprise a modification at position 36 of the sequence. The modification may be a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In certain embodiments, the FKBP12 polypeptide is encoded by an amino acid sequence comprising

GVQVETISPGDGRFTFPRGQTCVVHYTGMLLEDGKKVDSSRDRNKPFFKFMLGKQEVIRG  
WEEGVAQMSVGQRAKLITSPDYAYGATGHPGIIPPHATLVFDVELLKLE (SEQ ID NO: 18022). In certain embodiments, the FKBP12 polypeptide is encoded by a nucleic acid sequence comprising  
GGGGTCCAGGTCGAGACTATTTACCAGGGGATGGGCGAACATTTCCAAAAAGGGG  
CCAGACTTGCGTCGTGCATTACACCGGGATGCTGGAGGACGGGAAGAAAGTGGACA  
GCTCCAGGGATCGCAACAAGCCCTTCAAGTTCATGCTGGGAAAGCAGGAAGTGATC  
CGAGGATGGGAGGAAGGCGTGGCACAGATGTCAGTCGGCCAGCGGGCCAAACTGA  
CCATTAGCCCTGACTACGCTTATGGAGCAACAGGCCACCCAGGGATCATTCCCCCTC  
ATGCCACCCTGGTCTTCGAT GTGGAAGTCTGAAGCTGGAG (SEQ ID NO: 18023). In certain embodiments, the induction agent specific for the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V) comprises AP20187 and/or AP1903, both synthetic drugs.

(96) In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the linker region is encoded by an amino acid comprising GGGGS (SEQ ID NO: 18024) or a nucleic acid sequence comprising  
GGAGGAGGAGGATCC (SEQ ID NO: 18025). In certain embodiments, the nucleic acid sequence encoding the linker does not comprise a restriction site.

(97) In certain embodiments of the truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an arginine (R) at position 87 of the sequence. Alternatively, or in addition, in certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an alanine (A) at position 282 the sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid comprising

(98) TABLE-US-00037 (SEQ ID NO: 18026)

GFGDVGALSLRGNADLAYILSMEPCGHCLINNVNFCRESGLRTRTGSN  
IDCEKLRRRFSSLHFMVEVKGDLTAKKMVLALLELAQQDHGALDCCVVVI  
LSHGCGQASHLQFPGAVYGTGCPVSVEKIVNIFNGTSCPSLGGKPKLFFI  
QACGGEQKDHGFEVASTSPEDESPGSNPEPDATPFQEGLRTFDQLDAISS  
LTPSDIFVSYSTFPGFVSWRDPKSGSWYVETLDDIFEQWAHSEDLQSL

LRVANVSKMGIYKQMPGCFNFLRKKLFFKTS

or a nucleic acid sequence comprising

(99) TABLE-US-00038 (SEQ ID NO: 18027)

TTTGGGGACGTGGGGGCCCTGGAGTCTCTGCGAGGAAATGCCGATCTGGC  
TTACATCCTGAGCATGGAACCCTGCGGCCACTGTCTGATCATTAACAATG  
TGAAC TTCTGCAGAGAAAGCGGACTGCGAACACGGACTGGCTCCAATATT  
GACTGTGAGAAGCTGCGGAGAAGGTTCTCTAGTCTGCACTTTATGGTCTGA  
AGTGAAGAGGGGATCTGACCGCCAAGAAAATGGTGCTGGCCCTGCTGGAGC  
TGGCTCAGCAGGACCATGGAGCTCTGGATTGCTGCGTGGTCGTGATCCTG  
TCCCACGGGTGCCAGGCTTCTCATCTGCAGTTCCCCGGAGCAGTGTACGG  
AACAGACGGCTGTCCTGTCAGCGTGGAGAAGATCGTCAACATCTTCAACG  
GCACTTCTTGCCCTAGTCTGGGGGGAAAGCCAAAACCTGTTCTTTATCCAG  
GCCTGTGGCGGGGAACAGAAAGATCACGGCTTCGAGGTGGCCAGCACCAG  
CCCTGAGGACGAATCACCAGGGAGCAACCCTGAACCAGATGCAACTCCAT  
TCCAGGAGGGACTGAGGACCTTTGACCAGCTGGATGCTATCTCAAGCCTG  
CCCCTCCTAGTGACATTTTCGTGTCTTACAGTACCTTCCCAGGCTTTGT  
CTCATGGCGCGATCCCAAGTCAGGGAGCTGGTACGTGGAGACACTGGACG  
ACATCTTTGAACAGTGGGCCCATTTCAGAGGACCTGCAGAGCCTGCTGCTG  
CGAGTGGCAAACGCTGTCTCTGTGAAGGGCATCTACAAACAGATGCCCCG  
GTGCTTCAATTTTCTGAGAAAGAAACTGTTCTTTAAGACTTCC.

(100) In certain embodiments of the inducible proapoptotic polypeptides, wherein the polypeptide comprises a truncated caspase 9 polypeptide, the inducible proapoptotic polypeptide is encoded by an amino acid sequence comprising

(101) TABLE-US-00039 (SEQ ID NO: 18028)

GVQVETISPGDGRTPKRGQTCVVHYTGMLEDGKKVDSSRDRNKPFFKML  
GKQEVIRGWEEGVAQMSVGQRAKLTISPDYAYGATGHPGIIPPHATLVFD  
VELLKLEGGGGSGFGDVGALSLRGNADLAYILSMEPCGHCLIINNPNFC  
RESGLRTRTGSNIDCEKLRRRFFSLHFMVEVKGDLTAKKMVLALLELAQQ  
DHGALDCCVVVILSHGCQASHLQFPGAVYGTGCPVSVEKIVNIFNGTSC  
PSLGGKPKLFFIQACGGEQKDHGFEVASTSPEDESPGSNPEPDATPFQEG  
LRTFDQLDAISSLPTPSDIFVSYSTFPGFVSWRDPKSGSWYVETLDDIFE  
QWAHSEDLQSLLLRVANAVSVKGIYKQMPGCFNFLRKKLFFKTS

or the nucleic acid sequence comprising

(102) TABLE-US-00040 (SEQ ID NO: 18029)

gggggtccaggtcgagactatttcaccaggggatgggcgaacatttcaaa  
aaggggccagacttgcgtcgtgcattacaccgggatgctggaggacggga  
agaaagtggacagctccagggatcgcaacaagcccttcaagttcatgctg  
ggaaagcaggaagtgatccgaggtgggaggaaggcgtggcacagatgct  
agtcggccagcgggcaaaactgaccattagccctgactacgcttatggag  
caacaggccacccagggatcattccccctcatgccaccctggtcttcgat  
gtggaactgctgaagctggagggaggaggaggatccggatttggggacgt  
ggggggccctggagtctctgcgaggaaatgccgatctggcttacatcctga  
gcatggaaccctgcggccactgtctgatcattaacaatgtgaacttctgc  
agagaaaagcggactgcgaacacggactggctccaatattgactgtgagaa  
gctgcggagaaggttctctagctgcactttatggcgaagtgaagggg  
atctgaccgccaagaaaatgggtgctggccctgctggagctggctcagcag  
gaccatggagctctggattgctgcgtggctgctgatcctgtcccacgggtg  
ccaggcttctcatctgcagttccccggagcagtgtagcgaacagacggct  
gtcctgtcagcgtggagaagatcgtaacatcttcaacggcacttcttc  
cctagtctggggggaaagccaaaactgttcttatccaggcctgtggcgg  
ggaacagaaagatcacggcttcgaggtggccagcaccagccctgaggacg  
aatcaccagggagcaaccctgaaccagatgcaactccattccaggaggga

ctgaggacatttgagcagctggtatctctcagcctgcccagcttag tgacattttctgtgtcttacagtaccttcccaggttctgtctcatggcgcg  
atcccaagtcagggagctggtagctggagacactggacgacatcttgaa  
cagtggggccattcagaggacctgcagagcctgctgctgcgagtgga  
cgctgtctctgtgaagggcatctacaaacagatgcccgggtgcttcaatt ttctgagaaagaaactgttctttaagacttcc.

(103) Nanoparticle vectors of the disclosure may comprise at least one self-cleaving peptide. In some embodiments, the nanoparticle vector may comprise at least one self-cleaving peptide and wherein a self-cleaving peptide is located between a VCAR and the nanoparticle. In some embodiments, the nanoparticle vector may comprise at least one self-cleaving peptide and wherein a first self-cleaving peptide is located upstream of a VCAR and a second self-cleaving peptide is located downstream of a VCAR. In some embodiments, the nanoparticle vector may comprise at least one self-cleaving peptide and wherein a first self-cleaving peptide is located between a VCAR and the nanoparticle and a second self-cleaving peptide is located downstream of the VCAR. In some embodiments, the nanoparticle vector may comprise at least one self-cleaving peptide and wherein a first self-cleaving peptide is located between a VCAR and the nanoparticle and a second self-cleaving peptide is located downstream of the VCAR, for example, between the VCAR and a selection gene.

(104) Nanoparticle vectors of the disclosure may comprise at least one self-cleaving peptide(s) located, for example, between one or more VHH(s) or VCAR(s) of the disclosure and an inducible proapoptotic polypeptide of the disclosure. Nanoparticle vectors of the disclosure may comprise at least two self-cleaving peptide(s), a first self-cleaving peptide located, for example, upstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure and a second first self-cleaving peptide located, for example, downstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure. The self-cleaving peptide may comprise, for example, a T2A peptide, GSG-T2A peptide, an E2A peptide, a GSG-E2A peptide, an F2A peptide, a GSG-F2A peptide, a P2A peptide, or a GSG-P2A peptide. A T2A peptide may comprise an amino acid sequence comprising EGRGSLLTCGDVEENPGP (SEQ ID NO: 18030) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising EGRGSLLTCGDVEENPGP (SEQ ID NO: 18030). A GSG-T2A peptide may comprise an amino acid sequence comprising GSGEGRGSLLTCGDVEENPGP (SEQ ID NO: 18031) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGEGRGSLLTCGDVEENPGP (SEQ ID NO: 18031). A GSG-T2A peptide may comprise a nucleic acid sequence comprising ggatctggagaggggaaggggaagcctgctgacgtgtggagacgtggaggaaaacccaggacca (SEQ ID NO: 18032). An E2A peptide may comprise an amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033). A GSG-E2A peptide may comprise an amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034). An F2A peptide may comprise an amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035). A GSG-F2A peptide may comprise an amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036). A P2A peptide may comprise an amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037). A GSG-P2A peptide may comprise an amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038).

(105) The disclosure provides a composition comprising a vector of the disclosure.

(106) The disclosure provides a cell comprising a VCAR of the disclosure. The disclosure provides a cell comprising a transposon of the disclosure. In certain embodiments, the cell comprising a VCAR, a

transposon, or a vector of the disclosure may express a VCAR on the cell surface. The cell may be any type of cell. Preferably, the cell is an immune cell. The immune cell may be a T-cell, a Natural Killer (NK) cell, a Natural Killer (NK)-like cell, a Cytokine Induced Killer (CIK) cell, a hematopoietic progenitor cell, a peripheral blood (PB) derived T cell or an umbilical cord blood (UCB) derived T-cell. Preferably, the immune cell is a T-cell. The T-cell may be an early memory cell, a stem-like T-cell, a T.sub.SCM-like cell, a T.sub.SCM or a T.sub.CM. The T-cell may be a T.sub.SCM. The cell may be an artificial antigen presenting cell, which, optionally, may be used to stimulate and expand a modified immune cell or T cell of the disclosure. The cell may be a tumor cell, which, optionally, may be used as an artificial or modified antigen presenting cell.

(107) Modified cells of the disclosure that may be used for adoptive therapy may be autologous or allogeneic.

(108) The disclosure provides a method for expressing a VCAR on the surface of a cell, comprising: (a) obtaining a cell population; (b) contacting the cell population to a composition comprising a VCAR of the disclosure or a sequence encoding the VCAR, under conditions sufficient to transfer the VCAR across a cell membrane of at least one cell in the cell population, thereby generating a modified cell population; (c) culturing the modified cell population under conditions suitable for integration of the transposon; and (d) expanding and/or selecting at least one cell from the modified cell population that express the VCAR on the cell surface.

(109) In certain embodiments of this method of expressing a VCAR, the cell population may comprise leukocytes and/or CD4<sup>+</sup> and CD8<sup>+</sup> leukocytes. The cell population may comprise CD4<sup>+</sup> and CD8<sup>+</sup> leukocytes in an optimized ratio. The optimized ratio of CD4<sup>+</sup> to CD8<sup>+</sup> leukocytes does not naturally occur in vivo. The cell population may comprise a tumor cell.

(110) In certain embodiments of this method of expressing a VCAR, a transposon or vector comprises the VCAR or the sequence encoding the VCAR.

(111) In certain embodiments of this method of expressing a VCAR, the conditions sufficient to transfer the sequence encoding the VCAR across a cell membrane of at least one cell in the cell population comprise nucleofection.

(112) In certain embodiments of this method of expressing a VCAR, wherein the conditions sufficient to transfer the sequence encoding the VCAR across a cell membrane of at least one cell in the cell population comprise at least one of an application of one or more pulses of electricity at a specified voltage, a buffer, and one or more supplemental factor(s). In certain embodiments, the buffer may comprise PBS, HBSS, OptiMEM, BTXpress, Amaxa Nucleofector, Human T cell nucleofection buffer or any combination thereof. In certain embodiments, the one or more supplemental factor(s) may comprise (a) a recombinant human cytokine, a chemokine, an interleukin or any combination thereof; (b) a salt, a mineral, a metabolite or any combination thereof; (c) a cell medium; (d) an inhibitor of cellular DNA sensing, metabolism, differentiation, signal transduction, one or more apoptotic pathway(s) or combinations thereof, and (e) a reagent that modifies or stabilizes one or more nucleic acids. The recombinant human cytokine, the chemokine, the interleukin or any combination thereof may comprise IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN-gamma, IL-1 alpha/IL-1F1, IL-1 beta/IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 beta, IL-32 gamma, IL-33, LAP (TGF-beta 1), Lymphotoxin-alpha/TNF-beta, TGF-beta, TNF-alpha, TRANCE/TNFSF11/RANK L or any combination thereof. The salt, the mineral, the metabolite or any combination thereof may comprise HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl<sub>2</sub>, Na<sub>2</sub>HPO<sub>4</sub>, NaH<sub>2</sub>PO<sub>4</sub>, Sodium lactobionate, Mannitol, Sodium succinate, Sodium Chloride, CINA, Glucose, Ca(NO<sub>3</sub>)<sub>2</sub>, Tris/HCl, K<sub>2</sub>HPO<sub>4</sub>, KH<sub>2</sub>PO<sub>4</sub>, Polyethylenimine, Poly-ethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5, or any combination thereof. The cell medium may comprise PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-



VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium or any combination thereof. The inhibitor of cellular DNA sensing, metabolism, differentiation, signal transduction, one or more apoptotic pathway(s) or combinations thereof comprise inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-1, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1B, PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3 $\beta$  (GSK-3 $\beta$ ) (e.g., TWS119), Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK or any combination thereof. The reagent that modifies or stabilizes one or more nucleic acids comprises a pH modifier, a DNA-binding protein, a lipid, a phospholipid, CaPO<sub>4</sub>, a net neutral charge DNA binding peptide with or without a NLS sequence, a TREX1 enzyme or any combination thereof.

(113) In certain embodiments of this method of expressing a VCAR, the conditions suitable for integration of the VCAR or a sequence encoding the VCAR of the disclosure comprise at least one of a buffer and one or more supplemental factor(s). In certain embodiments, a transposon or vector of the disclosure comprise the VCAR or a sequence encoding the VCAR of the disclosure. In certain embodiments, the buffer may comprise PBS, HBSS, OptiMEM, BTXpress, Amaxa Nucleofector, Human T cell nucleofection buffer or any combination thereof. In certain embodiments, the one or more supplemental factor(s) may comprise (a) a recombinant human cytokine, a chemokine, an interleukin or any combination thereof; (b) a salt, a mineral, a metabolite or any combination thereof; (c) a cell medium; (d) an inhibitor of cellular DNA sensing, metabolism, differentiation, signal transduction, one or more apoptotic pathway(s) or combinations thereof, and (e) a reagent that modifies or stabilizes one or more nucleic acids. The recombinant human cytokine, the chemokine, the interleukin or any combination thereof may comprise IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN-gamma, IL-1 alpha/IL-1F1, IL-1 beta/IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 beta, IL-32 gamma, IL-33, LAP (TGF-beta 1), Lymphotoxin-alpha/TNF-beta, TGF-beta, TNF-alpha, TRANCE/TNFSF11/RANK L or any combination thereof. The salt, the mineral, the metabolite or any combination thereof may comprise HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl<sub>2</sub>, Na<sub>2</sub>HPO<sub>4</sub>, NaH<sub>2</sub>PO<sub>4</sub>, Sodium lactobionate, Mannitol, Sodium succinate, Sodium Chloride, CINa, Glucose, Ca(NO<sub>3</sub>)<sub>2</sub>, Tris/HCl, K<sub>2</sub>HPO<sub>4</sub>, KH<sub>2</sub>PO<sub>4</sub>, Polyethylenimine, Polyethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5, or any combination thereof. The cell medium may comprise PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium or any combination thereof. The inhibitor of cellular DNA sensing, metabolism, differentiation, signal transduction, one or more apoptotic pathway(s) or combinations thereof comprise inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-1, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1B, PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3 $\beta$  (GSK-3 $\beta$ ) (e.g., TWS119), Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK or any combination thereof. The reagent that modifies or stabilizes one or more nucleic acids comprises a pH modifier, a DNA-binding protein, a lipid, a phospholipid, CaPO<sub>4</sub>, a net neutral charge DNA binding peptide with or without a NLS sequence, a TREX1 enzyme or any combination thereof.

(114) In certain embodiments of this method of expressing a VCAR, the expansion and selection steps occur sequentially. The expansion may occur prior to selection. The expansion may occur following selection, and, optionally, a further (i.e. second) selection may occur following expansion.

(115) In certain embodiments of this method of expressing a VCAR, the expansion and selection steps

may occur simultaneously.

(116) In certain embodiments of this method of expressing a VCAR, the expansion may comprise contacting at least one cell of the modified cell population with an antigen to stimulate the at least one cell through the VCAR, thereby generating an expanded cell population. The antigen may be presented on the surface of a substrate. The substrate may have any form, including, but not limited to a surface, a well, a bead or a plurality thereof, and a matrix. The substrate may further comprise a paramagnetic or magnetic component. In certain embodiments of this method of expressing a VCAR, the antigen may be presented on the surface of a substrate, wherein the substrate is a magnetic bead, and wherein a magnet may be used to remove or separate the magnetic beads from the modified and expanded cell population. The antigen may be presented on the surface of a cell or an artificial antigen presenting cell. Artificial antigen presenting cells of the disclosure may include, but are not limited to, tumor cells and stem cells.

(117) In certain embodiments of this method of expressing a VCAR, wherein the transposon or vector comprises a selection gene and wherein the selection step comprises contacting at least one cell of the modified cell population with a compound to which the selection gene confers resistance, thereby identifying a cell expressing the selection gene as surviving the selection and identifying a cell failing to express the selection gene as failing to survive the selection step.

(118) In certain embodiments of this method of expressing a VCAR, the expansion and/or selection steps may proceed for a period of 10 to 14 days, inclusive of the endpoints.

(119) The disclosure provides a composition comprising the modified, expanded and selected cell population of the methods of the disclosure.

(120) The disclosure provides a method of treating cancer in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on a tumor cell. In certain embodiments, the tumor cell may be a malignant tumor cell. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic.

(121) The disclosure provides a method of treating an autoimmune condition in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on an autoimmune cell of the subject. In certain embodiments, the autoimmune cell may be a lymphocyte that specifically binds to a self-antigen on a target cell of the subject. In certain embodiments, the autoimmune cell may be a B lymphocyte (i.e. a B cell). In certain embodiments, the autoimmune cell may be a T lymphocyte (i.e. a T cell). In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic.

(122) The disclosure provides a method of treating an infection in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on a cell comprising an infectious agent, a cell in communication with an infectious agent or a cell exposed to an infectious agent. In some embodiments, a cell in communication with an infectious agent may be in air communication (e.g., the infectious agent is airborne or inhaled) or fluid communication (e.g., the infectious agent is carried in an aqueous or a biological fluid) with the infectious agent. The infectious agent causing the infection of the host cell may be a bacterium, a virus, a yeast, or a microbe. The infectious agent may induce in the cell or the cell's host organism (the subject), exemplary conditions including, but not limited to, a viral infection, an immunodeficiency condition, an inflammatory condition and a proliferative disorder. In certain embodiments, the infection causes tuberculosis, microencephaly, neurodegeneration or malaria. In certain embodiments, the infection causes microencephaly in a fetus of the subject. In certain embodiments, including those wherein the infection causes microencephaly in a fetus of the subject, the infectious agent is a virus and wherein the virus is a Zika virus. In certain embodiments, the immunodeficiency condition is acquired immune deficiency syndrome (AIDS). In certain embodiments, the proliferative disorder is a cancer. In

cervical cancer is a human papilloma virus (HPV). In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic.

(123) The disclosure provides a method of treating a mast cell disease in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on a mast cell. In certain embodiments, the VCAR specifically binds to an antigen on a mast cell of the subject. In certain embodiments, the mast cell disease may include, but is not limited to, disorders associated with an excessive proliferation of mast cells, disorders associated with mast cells having abnormal activity, and disorders associated with both abnormal numbers of mast cells and abnormal mast cell activity. Exemplary disorders associated with an excessive proliferation of mast cells include, but are not limited to, mastocytosis, cutaneous mastocytosis (e.g., urticaria pigmentosa or maculopapular cutaneous mastocytosis), systemic mastocytosis (including mast cell leukaemia), and localized mast cell proliferations. Exemplary disorders associated with mast cells having abnormal activity, include, but are not limited to, mast cell activation syndrome (MCAS) or mast cell activation disorder (MCAD), allergic disease (including anaphylaxis), asthma, inflammatory disease (including autoimmune related inflammation of, for example, joint tissues, arthritis, etc.), or any combination thereof. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic. The disclosure provides a method of treating a degenerative disease in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on a deleterious cell or an aged cell. In certain embodiments, the VCAR specifically binds to an antigen on a deleterious cell or an aged cell of the subject. In certain embodiments, the degenerative disease may include, but is not limited to, a neurodegenerative disorder, a metabolic disorder, a vascular disorder and aging. Exemplary neurodegenerative disorders include, but are not limited to, disorders associated with a loss of a function or efficacy of one or more of a neuron, a glial cell or a microglia. Exemplary neurodegenerative disorders include, but are not limited to, disorders associated with an accumulation of one or more of a signaling molecule, a protein, or a prion that interferes with a function or decreases an efficacy of one or more of a neuron, a glial cell or a microglia. Exemplary metabolic disorders include, but are not limited to, disorders associated with mitochondrial disorders, interruptions of the electron transport chain, interruptions of cellular production of ATP, a loss of a function or a decreased efficacy of one or more mitochondria of one or more of a neuron, a glial cell or a microglia. Exemplary metabolic disorders include, but are not limited to, disorders associated with a loss of circulating blood flow or a decreased blood flow to a neuron, a glial cell or a microglia (e.g., a stroke); a transient or permanent state of hypoxia in a neuron, a glial cell or a microglia (for example, sufficient to release free radicals in a cell); a loss of circulating CNS or a decreased CNS flow to a neuron, a glial cell or a microglia during a sleep state of the subject sufficient to decrease efficacy of removal of a waste product of a neuron, a glial cell or a microglia during that sleep state. Exemplary aging disorders include, but are not limited to, disorders associated with an increased shortened or shortened telomeres on one or more chromosomes of a neuron, a glial cell or a microglia; a loss of a function or a decreased efficacy of telomerase in a neuron, a glial cell or a microglia; or a loss of a function or a decreased efficacy of a DNA repair mechanism in a neuron, a glial cell or a microglia. In certain embodiments, the deleterious cell or the aged cell interferes with a function or decreases an efficacy of another cell in a network comprising the deleterious cell or the aged cell and the targeted removal of the deleterious cell or the aged cell improves or restores a function or increases an efficacy of the network. In certain embodiments, the deleterious cell or the aged cell may transform the function or efficacy of a second cell and the targeted removal of the deleterious cell or the aged cell prevents the transformation of the second cell. In certain embodiments, the degenerative disease is a neurodegenerative disorder and the

deleterious cell or the aged cell is a stem cell, an immune cell, a neuron, a glia or a microglia. In certain embodiments, the degenerative disease is a metabolic disorder and the deleterious cell or the aged cell is a stem cell, a somatic cell, a neuron, a glia or a microglia. In certain embodiments, the degenerative disease is a vascular disorder and the deleterious cell or the aged cell is a stem cell, a somatic cell, an immune cell, an endothelial cell, a neuron, a glia or a microglia. In certain embodiments, the degenerative disease is aging and the deleterious cell or the aged cell is an oocyte, a sperm, a stem cell, a somatic cell, an immune cell, an endothelial cell, a neuron, a glia or a microglia. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic.

(124) The disclosure provides a method of modifying a cell therapy in a subject in need thereof, comprising administering to the subject a composition comprising a cell comprising a transposon or vector of the composition comprising an inducible proapoptotic polypeptide, wherein apoptosis may be selectively induced in the cell by contacting the cell with an induction agent. In certain embodiments, the cell is autologous. In certain embodiments, the cell is allogeneic. In certain embodiments of this method, the cell therapy is an adoptive cell therapy. In certain embodiments of this method, modifying the cell therapy comprises a termination of the cell therapy. In certain embodiments of this method, modifying the cell therapy comprises a depletion of a portion of the cells provided in the cell therapy. In certain embodiments, the method further comprises the step of administering an inhibitor of the induction agent to inhibit modification of the cell therapy, thereby restoring the function and/or efficacy of the cell therapy.

(125) Methods of modifying a cell therapy of the disclosure may be used to terminate or dampen a therapy in response to, for example, a sign of recovery or a sign of decreasing disease severity/progression, a sign of disease remission/cessation, and/or the occurrence of an adverse event. Cell therapies of the disclosure may be resumed by inhibiting the induction agent should a sign or symptom of the disease reappear or increase in severity and/or an adverse event is resolved.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a plot showing weight change versus days post treatment for VCARs of the disclosure.
- (2) FIG. 2 is a plot showing tumor burden versus days post treatment in for VCARs of the disclosure.
- (3) FIG. 3 is a plot showing percent survival versus days post treatment for VCARs of the disclosure.
- (4) FIG. 4 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-A (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).
- (5) FIG. 5 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-B (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).
- (6) FIG. 6 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-C (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).
- (7) FIG. 7 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-D (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).
- (8) FIG. 8 a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-E (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).
- (9) FIG. 9 a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-F (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).
- (10) FIG. 10A is a table providing consensus sequences of framework and CDR sequences for exemplary VHs of the disclosure.
- (11) FIG. 10B is an alignment of exemplary VHs of the disclosure. From top to bottom, the sequences correspond with the consensus sequences of the frameworks VH-B (SEQ ID NO: 18050), VH-D (SEQ ID NO: 18051), VH-A (SEQ ID NO: 18052), VH-E (SEQ ID NO: 18053), VH-F (SEQ ID NO: 18054) and VH-C (SEQ ID NO: 18055).

(12) FIG. **11** is a bar graph depicting the knock out efficiency of checkpoint signaling proteins on armored T-cells. Primary human T-cells are typically in resting state when isolated from normal healthy donors. Cas-CLOVER was used to knockout the checkpoint receptors, PD-1, TGFB2, LAG-3, TIM-3 and CTLA-4. Percent knock-out is shown on the y-axis. Gene editing resulted in 30-70% loss of protein expression at the cell surface as measured by flow cytometry.

(13) FIG. **12** are schematic diagrams of wildtype, null and switch receptors and their effects on intracellular signaling, either inhibitory or stimulatory, in primary T-cells. Binding of the wildtype inhibitory receptor expressed endogenously on a T-cell with its endogenous ligand results in transmission of an inhibitory signal which, in part, reduces T-cell effector function. However, mutation (Mutated null) or deletion (Truncated null) of the intracellular domain (ICD) of a checkpoint receptor protein, such as PD1 (top panel) or TGFBRII (bottom panel), reduces or eliminates its signaling capability when cognate ligand(s) is bound. Thus, expression of engineered mutated or truncated null receptors on the surface of modified T cells results in a competition with endogenously-expressed wildtype receptors for binding of the free endogenous ligand(s), effectively reducing or eliminating delivery of inhibitory signals by endogenously-expressed wildtype receptors. Specifically, any binding by a mutated or null receptor sequesters the endogenous ligand(s) from binding the wildtype receptor and results in dilution of the overall level of checkpoint signaling effectively delivered to the modified T-cell, thereby reducing or blocking checkpoint inhibition and functional exhaustion of the modified T cells. A switch receptor is created by replacement of the wildtype ICD with an ICD from either a co-stimulatory molecule (such as CD3z, CD28, 4-1B) or a different inhibitory molecule (such as CTLA4, PD1, Lag3). In the former case, binding of the endogenous ligand(s) by the modified switch receptor results in the delivery of a positive signal to the T-cells, thereby helping to enhance stimulation of the modified T cell and potentially enhance target tumor cell killing. In the latter case, binding of the endogenous ligand(s) by the modified switch receptor results in the delivery of a negative signal to the T-cells, thereby eliminating stimulation of the modified T cell and potentially reducing target tumor cell killing. The signal peptide (purple arrow), extracellular domain (ECD) (bright green), transmembrane domain (yellow), intracellular signaling domain (ICD)(orange), and replacement ICD (green) are displayed in the receptor diagrams. “\*” indicates a mutated ICD. “+” indicates the presence of a checkpoint signal. “-” indicates the absence of a checkpoint signal.

(14) FIG. **13** is a schematic diagram showing the design of PD1 and TGFBRII null receptors. The signal peptide domain (SP), transmembrane domain (TM) and extracellular domain (ECD) of truncated null receptors for PD1 (top panel) and TGFBRII (bottom panel) are shown. The first of the top four molecules is the wildtype PD-1 receptor, which encodes the wildtype PD-1 SP and TM. For the PD1 null receptor, replacement of PD1 wildtype SP or TM domain (green; light green) with the SP or TM domain of a human T cell CD8 $\alpha$  receptor (red) is depicted. The second molecule encodes the CD8 $\alpha$  SP along with the native PD-1 TM, the third encodes the wildtype PD-1 SP and the alternative CD8 $\alpha$  TM, and the fourth encodes both the alternative CD8 $\alpha$  SP and TM. Similarly, for the null receptor of TGFBRII, replacement of the wildtype TGFBRII SP (pink) with a SP domain of a human T cell CD8 $\alpha$  receptor (red). The names of the constructs and the amino acid lengths (aa) of each construct protein is listed on the left of the diagram.

(15) FIG. **14** is a series of histograms depicting the expression of the PD1 and TGFBRII null Receptors on the surface of primary human T cells determined by flow cytometry. Each of the six truncated null constructs from FIG. **13** were expressed on the surface of primary human T cells. T cells were stained with either anti-PD1 (top; blue histograms) or anti-TGFBRII (bottom; blue histograms), or isotype control or secondary only (gray histograms). Cells staining positive for PD-1 or TGF $\beta$ RII expression were gated (frequency shown above gate) and mean fluorescence intensity (MFI) value is displayed above each positive histogram. The names of the null receptor constructs are depicted above each plot. Both null receptor gene strategies, replacement of the wildtype SP with the alternative CD8 $\alpha$  were successfully expressed. 02.8aSP-PD-1 and 02.8aSP-TGF $\beta$ RII resulted in the highest level of expression at the T-cell surface. 02.8aSP-PD-1 null receptor exhibited an MFI of 43,680, which is 177-fold higher than endogenous T cell PD-1 expression and 2.8-fold higher than the wildtype PD-1 null receptor. 02.8aSP-TGF $\beta$ RII null receptor exhibited an MFI of 13,809, which is 102-fold higher than endogenous

T cell TGF $\beta$ RII expression and 1.8-fold higher than the wildtype TGF $\beta$ RII null receptor. Replacement of wildtype SP with the alternative CD8 $\alpha$  SP for both PD1 and TGRBR II results in enhanced surface expression of the null or Switch receptor, which helps to maximize checkpoint inhibition or co-stimulation, respectively, upon binding of the endogenous ligand(s).

(16) FIG. 15A-B is a pair of schematic diagrams depicting NF-KB inducible vectors for expression in T-cells. Two T cell activation NF-KB inducible vectors were developed; one with the gene expression system (GES) in the forward orientation (A) and the other in the complementary direction (B), both preceding the constitutive EF1a promoter. These vectors also direct expression of a CAR molecule and a DHFR selection gene, separated by a T2A sequence. Both the conditional NF-KB inducible system and the EF1a directed genes are a part of a piggyBac transposon that can be permanently integrated into T cells using electroporation (EP). Once integrated into the genome, the T cells will constitutively express the CAR on the membrane surface and the DHFR within the cell, while expression of the NF-KB inducible gene, GFP, will be expressed to the highest level only upon T cell activation.

(17) FIG. 16 is a pair of graphs depicting NF-KB inducible expression of GFP in activated T cells. T cells were nucleofected with a piggyBac vector expressing an anti-BCMA CAR and a DHFR mutein gene under control of an EF1a promoter along with the absence (No GES control) or presence of an NF-KB inducible expression system driving GFP expression in either the forward (pNFKB-GFP forward) or reverse orientation (pNFKB-GFP reverse). Cells were cultured in the presence of methotrexate selection until the cells were almost completely resting (Day 19) and GFP expression was assessed at Day 5 and Day 19. At Day 5, all T cells are proliferating and highly stimulated, with cells harboring the NF-KB inducible expression cassette producing high levels of GFP due to strong NF $\kappa$ B activity. The No GES control cells did not express detectable levels of GFP. By Day 19, the GES T cells were almost fully resting and GFP expression was significantly lower than Day 5 ( $\sim 1/8$  MFI), since NF $\kappa$ B activity is lower. GFP expression is still observed at Day 19, which may be due to the long half-life of GFP protein ( $\sim 30$  hr), or, basal level of NF $\kappa$ B activity through, for example, a TCR, a CAR, a cytokine receptor, or a growth factor receptor signal.

(18) FIG. 17 is a series of graphs depicting anti-BCMA CAR-mediated activation of NF-KB inducible expression of GFP in presence of BCMA<sup>+</sup> tumor cells. T cells were either unmodified (Mock T cells) or nucleofected with a piggyBac vector expressing an anti-BCMA CAR and a DHFR mutein gene under control of an EF1a promoter along with the absence (No GES control) or presence of an NF-KB inducible expression system driving GFP expression in either the forward (pNFKB-GFP forward) or reverse orientation (pNFKB-GFP reverse). All cells were cultured for 22 days, either with or without methotrexate selection (Mock T cells), until the cells were almost completely resting. Cells were then stimulated for 3 days in the absence (No stimulation) or presence of BCMA<sup>-</sup> (K562), BCMA<sup>+</sup>(RPMI 8226), or positive control anti-CD3 anti-CD28 activation reagent (CD3/28 stimulation). GFP expression was undetectable under all conditions with the No GES control or Mock T cells. However, while pNFKB-GFP forward- and reverse-transposed cells exhibited little GFP expression over the No stimulation control when cultured with BCMA<sup>-</sup> K562 cells, they both demonstrated dramatic upregulation of gene expression either in the presence of BCMA<sup>+</sup> tumor cells or under positive control conditions. Little difference in GFP expression was observed between the pNFKB-GFP forward- and reverse-transposed cells that were cocultured with BCMA<sup>+</sup> tumor cells.

(19) FIG. 18 is a series of graphs demonstrating that the Expression level of inducible gene can be regulated by number of response elements preceding the promoter T cells were nucleofected with a piggyBac vector encoding an anti-BCMA CARTyrin followed by a selection gene, both under control of a human EF1a promoter. Further, vectors either additionally encoded the conditional NF-KB inducible gene expression system driving expression of a truncated CD19 protein (dCD19) and included a number of NF $\kappa$ B response elements (RE) varying from 0-5, no GES (No GES), or received an electroporation pulse but no piggyBac nucleic acid (Mock). Data are shown for only the GES in the reverse (opposite) direction/orientation. All cells were cultured for 18 days and included selection for piggyBac-modified T cells using methotrexate addition. Cells were then stimulated for 3 days using anti-CD3 anti-CD28 bead activation reagent and dCD19 surface expression was assessed by FACS at Days 0, 3 and 18, and data are shown as FACS histograms and MFI of target protein staining. Surface

dCD19 expression was detected at low levels at Day 0 in all T cells transposed with vectors encoding the GES. At 3 days post-stimulation, dramatic upregulation of dCD19 expression was observed for all T cells expressing the GES, with a greater fold increase in surface expression in those with higher numbers of REs. Thus, surface dCD19 expression was directly proportional with the number of REs encoded in the GES. No dCD19 was detected on the surface of T cells that did not harbor the GES: No GES and Mock controls.

(20) FIG. 19 is a schematic depiction of the Csy4-T2A-Clo051-G4Slinker-dCas9 construct map (Embodiment 2).

(21) FIG. 20 is a schematic depiction of the pRT1-Clo051-dCas9 Double NLS construct map (Embodiment 1).

(22) FIG. 21 is a pair of graphs comparing the efficacy of knocking out expression of either B2M in Pan T-cells (left) or the  $\alpha$ -chain of the T-cell Receptor in Jurkat cells (right) for either Embodiment 1 (pRT1-Clo051-dCas9 Double NLS, as shown in FIG. 20) or Embodiment 2 (Csy4-T2A-Clo051-G4Slinker-dCas9, as shown in FIG. 19) of a Cas-Clover fusion protein of the disclosure. For the right-hand graph, the fusion protein is provided at either 10  $\mu$ g or 20  $\mu$ g, as indicated.

(23) FIG. 22 is a photograph of a gel electrophoresis analysis of mRNA in the presence of each of Embodiment 1 (pRT1-Clo051-dCas9 Double NLS, as shown in FIG. 20) and Embodiment 2 (Csy4-T2A-Clo051-G4Slinker-dCas9, as shown in FIG. 19). As shown, both are effective at knocking down mRNA expression.

(24) FIG. 23 is a diagram of the UniRat™ Human Heavy Chain Antibody production platform.

(25) FIG. 24 is a diagram of the pipeline of immunization, B-cell isolation, mRNA purification, Next Generation Sequencing (NGS), bioinformatics analysis, high-throughput vector assembly and high-throughput expression and screening used to identify the full antigen-specific repertoire of heavy-chain antibodies generated by the UniRat™ after immunization. Unique gene assembly methods convert the antibody repertoire sequence information into large collections of fully human heavy chain antibodies that can be screened for a variety of functions.

(26) FIG. 25 is a pair of heat maps showing how next generation sequencing (NGS) analysis reveals expanded VH sequence lineages. The degree of red (high) or blue (low) in the heatmap indicates expanded VH sequence families due to immunization. Some highly ranked VH sequences are unique to a single animal. Other highly ranked VH sequences appear in more than one animal suggesting convergent selection of those sequence families.

(27) FIG. 26 is a series of flow cytometry plots showing that a CAR-T population expressing P-PSMA-101 and comprising a mixture of T.sub.SCM/T.sub.CM, give rise to CAR+T.sub.CM, T.sub.EM, and Teff to attack solid tumor. After solid tumor elimination, a population of CAR-T+T.sub.SCM persists. Although demonstrated with a CARTyrin, this principle shown here applies to VCAR+T.sub.SCM and VCAR+T.sub.CM populations of the disclosure. Specifically, a murine xenograft model using a luciferase-expressing LNCaP cell line (LNCaP.luc) injected subcutaneously (SC) into NSG mice was utilized to assess in vivo anti-tumor efficacy of a CAR (P-PSMA5-101 and P-PSMA8-101) at a 'stress' dose ( $4 \times 10^6$ ) total CAR-T cells. For these in vivo studies, all CAR-T cells were produced using PB delivery of either the P-PSMA5-101 or P-PSMA8-101 plasmid using the Poseida manufacturing process. Mice were injected in the axilla with LNCaP and treated when tumors were established (100-300 mm.sup.3 by caliper measurement). Mice were treated with a 'stress' dose ( $4 \times 10^6$ ) of P-PSMA-101 by IV injection in order to tease out any possible differences in efficacy between the PSMA5 and the PSMA8 CARs. Anti-tumor activity was evaluated by survival, CD8+ T cell expansion and detection in the blood, tumor volume assessment by caliper measurement, and bioluminescence of LNCaP tumor. P-PSMA5-101 and P-PSMA8-101 at a 'stress' dose demonstrated significantly enhanced anti-tumor efficacy and survival in comparison to the T cells (no CAR) control mice against established SC LNCaP.luc solid tumors in NSG mice. Specifically, there was no survival in T cells (no CAR) control animals, 25% survival in the P-BCMA-101 treated group, 75% survival in the P-PSMA5-101 treated group, and 100% survival in animals treated with a 'stress' dose of P-PSMA8-101. In the peripheral blood, P-PSMA5-101 and P-PSMA8-101 expanded and gave rise to differentiated effector CARTyrin+ T-cells that were concomitant with a decrease in tumor burden

below detectable caliper and bioluminescent imaging limits. These cells then contracted, yet persisted in the peripheral blood.

#### DETAILED DESCRIPTION

(28) The disclosure provides chimeric antigen receptors (CARs) comprising at least one VHH (VCAR). Chimeric antigen receptors of the disclosure may comprise more than one VHH. For example, a bi-specific VCAR may comprise two VHHs that specifically bind two distinct antigens.

(29) VHH proteins of the disclosure specifically bind to an antigen. Chimeric antigen receptors of the disclosure comprising one or more VHHs that specifically bind an antigen may be used to direct the specificity of a cell, (e.g., a cytotoxic immune cell) towards the specific antigen.

(30) The disclosure provides chimeric antigen receptors (CARs) an antigen recognition region comprising a single domain antibody (VCARs). In some embodiments, the single domain antibody is a VHH antibody. In some embodiments, the single domain antibody is a VH antibody.

(31) Chimeric antigen receptors of the disclosure may comprise a signal peptide of human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR. A hinge/spacer domain of the disclosure may comprise a hinge/spacer/stalk of human CD8α, IgG4, and/or CD4. An intracellular domain or endodomain of the disclosure may comprise an intracellular signaling domain of human CD3ζ and may further comprise human 4-1BB, CD28, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof. Exemplary transmembrane domains include, but are not limited to a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR transmembrane domain.

(32) The disclosure provides genetically modified cells, such as T cells, NK cells, hematopoietic progenitor cells, peripheral blood (PB) derived T cells (including T cells from G-CSF-mobilized peripheral blood), umbilical cord blood (UCB) derived T cells rendered specific for one or more antigens by introducing to these cells a VCAR of the disclosure. Cells of the disclosure may be modified by electrotransfer of a transposon encoding a VCAR of the disclosure and a plasmid comprising a sequence encoding a transposase of the disclosure (preferably, the sequence encoding a transposase of the disclosure is an mRNA sequence).

(33) VHs of the Disclosure

(34) The disclosure provides chimeric antigen receptors (CARs) comprising a single domain antibody (VCARs). In some embodiments, the single domain antibody comprises a VH. In some embodiments, the VH is isolated or derived from a human sequence. In some embodiments, VH comprises a human CDR sequence and/or a human framework sequence and a non-human or humanized sequence (e.g., a rat Fe domain). In some embodiments, the VH is a fully humanized VH. In some embodiments, the VH is neither a naturally occurring antibody nor a fragment of a naturally occurring antibody. In some embodiments, the VH is not a fragment of a monoclonal antibody. In some embodiments, the VH is a UniDab™ antibody (TeneoBio).

(35) In some embodiments, the VH is fully engineered using the UniRat™ (TeneoBio) system and “NGS-based Discovery” to produce the VH. Using this method, the specific VH are not naturally-occurring and are generated using fully engineered systems. The VH are not derived from naturally-occurring monoclonal antibodies (mAbs) that were either isolated directly from the host (for example, a mouse, rat or human) or directly from a single clone of cells or cell line (hybridoma). These VHs were not subsequently cloned from said cell lines. Instead, VH sequences are fully-engineered using the UniRat™ system as transgenes that comprise human variable regions (VH domains) with a rat Fc domain, and are thus human/rat chimeras without a light chain and are unlike the standard mAb format. The native rat genes are knocked out and the only antibodies expressed in the rat are from transgenes with VH domains linked to a Rat Fc (UniAbs). These are the exclusive Abs expressed in the UniRat. Next generation sequencing (NGS) and bioinformatics are used to identify the full antigen-specific repertoire of the heavy-chain antibodies generated by UniRat™ after immunization. Then, a unique gene assembly method is used to convert the antibody repertoire sequence information into large collections of fully-human heavy-chain antibodies that can be screened in vitro for a variety of functions. In some embodiments, fully humanized VH are generated by fusing the human VH domains with human Fcs in vitro (to generate a non-naturally occurring recombinant VH antibody). In some



embodiments, the VH are fully humanized, but they are expressed in vivo as human/rat chimera (human VH, rat Fc) without a light chain. Fully humanized VHs are expressed in vivo as human/rat chimera (human VH, rat Fc) without a light chain are about 80 kDa (vs 150 kDa).

(36) VCARs of the disclosure may comprise at least one VH of the disclosure. In some embodiments, the VH of the disclosure may be modified to remove an Fc domain or a portion thereof. In some embodiments, a framework sequence of the VH of the disclosure may be modified to, for example, improve expression, decrease immunogenicity or to improve function.

(37) Exemplary VCARs of the Disclosure

(38) In some embodiments of the VCARs of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

(39) TABLE-US-00041 VH-A: (SEQ ID NO: 18000)

malpvtalllplalllhaarpevqllesggglvqppgslrlscaasgftf  
ssyamnwvrqapgkglewvagiigsggstyyadsvkgrfsisrdnsknt  
dlqmnsraedtavyycvkdwnntmitergqgtlvtvssttppaprpptp  
aptiasqplslrpeacrpaaggavhtrglfacdiyiwaplagtcgvlll  
slvitlyckrgrkkllyifkqpfmrpvqttqeedgcscrpfpeeeeggcel  
rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr  
rknppqeglynelqkdmaeayseigmkgerrrgkghdglyqglstatkdt ydalhmqalppr.

(40) In some embodiments of the VCARs of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

(41) TABLE-US-00042 VH-A: (SEQ ID NO: 18001)

atggctctgcctgtgacagctctgctgctgctctggctctgcttctca tgcggcgcgccctgaagttcagctgcttgaatctggcggaggcctgggtc  
aacctggcgggatctctgagactgagctgtgccgccagcggcttcacctt  
agcagctacgccatgaactgggtccgacaggccccctggcaaaggactgga  
atgggtggccggaatcatcggcagcggcggcagcacatattacgccgatt  
ctgtgaagggccgcttcagcatcagccgggacaacagcaagaacaccctg  
gacctgcagatgaacagcctgagagccgaggataccgccgtgtactactg  
cgtgaaggattggaacaccaccatgatcaccgagagaggccagggcacac  
tggtcaccgtgtcctctacaacaacaccggcgctcggcctccaacacca  
gctcctacaatcgcgagtcagcccctgtctctcagacccgaagcctgtag  
acctgctgctggcggagctgtgcataccagaggactggatttcgcctgcg  
acatctacatctgggctcctctggctggcacatgcggagttttgctgctg  
agcctggctcatccctgtactgtaagagaggcaggaagaagctgctgta  
tatcttcaagcagcccttcatgagacccgtgcagaccacacaggaggagg  
acggctgctctttaggttcccagaggaggaggaggaggatgcgagctg  
cgctgaagtttagccggtccgccgatgcacctgcatacaagcagggaca  
gaaccagctgtataacgagctgaatctgggcccggagagaggagtacgacg  
tgctggataagaggcggggccgggacccccgagatgggaggcaagcctcgg  
agaaaagaaccacaggagggcctgtacaatgagctgcaaaaggacaagat  
ggccgaggcctattctgagatcgccatgaagggagagaggcgccggggca  
agggacacgatggcctgtaccagggcctgagcaccgccacaaaggacacc tatgatgccctgcacatgcaggccctgccccctagatga.

(42) In some embodiments of the VCARs of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

(43) TABLE-US-00043 VH-B: (SEQ ID NO: 18002)

malpvtalllplalllhaarpevqllesggglyqppgslrlscaasgftf  
snyamnwvrqapgkglewysgiigsgattyyadsvkgrftisrdnsknt  
nlqmnsraedtaiyycvkdwnntmitergqgtlvtvssttppaprpptp  
aptiasqplslrpeacrpaaggavhtrglfacdiyiwaplagtcgvlll  
slvitlyckrgrkkllyifkqpfmrpvqttqeedgcscrpfpeeeeggcel

rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr  
rknppqeglynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdt ydalhmqalppr.

(44) In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

(45) TABLE-US-00044 VH-B (SEQ ID NO: 18003)

atggctctgcctgtgacagctctgctgtgcctctggctctgcttctca tgcggcgcgccctgaagttcagctgcttgaatctggcggaggcctgggtc  
aacctggcggatctctgacactgagctgtgccgccagcggcttcaccttc  
agcaactacgcatgaactgggtccgacaggcccctggcaaaggccttga  
atgggtgtccggcatcattggctctggcgccaccacactactacgccgatt  
ctgtgaagggcagattcaccatcagccgggacaacagcaagaacaccctg  
aacctgcagatgaacagcctgagagccgaggacaccgccatctactactg  
cgtgaaggactggaacaccacatgatcaccgagagaggccagggcacac  
tggtcaccgtgtcctctacaacaacaccggcgccctcggtccaacacca  
gctcctacaatcgcgagtcagcccctgtctctcagacccgaagcctgtag  
acctgctgctggcggagctgtgcataccagaggactggatttcgctgctg  
acatctacatctgggctcctctggctggcacatgcggagttttgctgctg  
agcctggctacaccctgtactgtaagagaggcaggaagaagctgctgta  
tatcttcaagcagccatcatgagaccctgcagaccacacaggaggagga  
cggctgctctttaggttcccagaggaggaggaggaggatgcgagctgc  
gcgtgaagtttagccgggtccgccgatgcacctgcatacaagcagggacag  
aaccagctgtataacgagctgaatctgggcccggagagaggagtacgacgt  
gctggataagaggcggggccgggaccccagatgggaggcaagcctcgga  
gaaagaacccacaggaggcctgtacaatgagctgcaaaaggacaagatg  
gccgaggcctattctgagatcggtcatgaaggagagaggcgccggggcaa  
gggacacgatggcctgtaccaggcctgagcaccgccacaaaggacacct atgatgccctgcacatgcaggcctgccccctagatgac.

(46) In some embodiments of the VCARS of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

(47) TABLE-US-00045 VH-C: (SEQ ID NO: 18004)

malpvtalllplalllhaarpevqllesggglvqpgeslrscasgftf  
snyamnwvrqapgkglewvsgivgggtsyyadsvrgrftisrnsntnl  
ylqmnsraedtavyycvkdwnttmitergqgtlvtvssttppaprpptp  
aptiasqplslrpeacrpaaggavhtrglfacdiyiwaplagtcgvlll  
slvitlyckrgrkkllyifkqpfmrpvqttqeedgcscrfeeeeeggcel  
rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr  
rknppqeglynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdt ydalhmqalppr.

(48) In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of

(49) TABLE-US-00046 VH-C: (SEQ ID NO: 18005)

atggctctgcctgtgacagctctgctgtgcctctggctctgcttctca tgcggcgcgccctgaagttcagctgcttgaatctggcggaggcctgggtc  
agcctggcgaatctctgagactgagctgtgccgccagcggatcaccttca  
gcaactacgcatgaactgggtccgacaggcccctggcaaaggccttgaa  
tgggtgtccggaatcgttggcggcgaggacacaagctactacgccgattc  
tgtgcggggcagattcaccatcagccgggacaacagcaagaacaccctgt  
acctgcagatgaacagcctgagagccgaggacaccgccgtgtactactgc  
gtgaaggactggaacaccacatgatcaccgagagaggccagggcacact  
ggtcaccgtgtcctctacaacaacaccggcgccctcggtccaacaccag  
ctcctacaatcgcgagtcagcccctgtctctcagacccgaagcctgtaga  
cctgctgctggcggagctgtgcataccagaggactggatttcgctgctga  
catctacatctgggctcctctggctggcacatgcggagttttgctgctga

gcctgggtcatcctgtactgtaagagaggcaggaagaagctgctgtat  
atcttcaagcagccatcatgagacccgtgcagaccacacaggaggaggac  
ggctgctctttaggttcccagaggaggaggaggaggatgagctgagctgag  
cgtgaagtttagccggtccgcccgtgacacctgcatacaagcagggacaga  
accagctgtataacgagctgaatctgggcccggagagaggagtacgacgtg  
ctggataagaggcggggccgggaccccgagatgggaggcaagcctcgag  
aaagaacccacaggaggcctgtacaatgagctgcaaaaggacaagatgg  
ccgaggcctattctgagatcggcatgaaggagagaggcgccggggcaag  
ggacacgatggcctgtaccaggcctgagcaccgccacaaaggacaccta tgatgccctgcacatgcaggccctgccccctagatga.

(50) In some embodiments of the VCARS of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

(51) TABLE-US-00047 VH-D: (SEQ ID NO: 18006)

malpvtalllplalllhaarpevqllesggglvqpggslrlscaasgftf  
snyamtwirqapgkglewvsgitdggstfyadsvkgrftisrdnsknt  
ylqmnsiraedtavyycvkdwnntmitergqgtlvtssttppaprpptp  
aptiasqplslrpeacrpaaggavhtrglfacdiyiwaplagtcgvlll  
slvitlyckrgrkkllyifkqpfmrpvqttqeedgcsrfpeeeeggcel  
rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr  
rknppqeglynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdt ydalhmqalprr.

(52) In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

(53) TABLE-US-00048 VH-D (SEQ ID NO: 18007)

atggctctgcctgtgacagctctgctgctgcctctggctctgcttctca tgcggcgcgccctgaagttcagctgcttgaatctggcggaggcctgggtc  
aacctggcggatctctgagactgagctgtgccgccagcggcttcaccttc  
agcaattacgccatgacctggatcagacaggcccctggcaaaggcctgga  
atgggtgtccggaattacaggcgacggcggcagcaccttttacgccgatt  
ctgtgaaggggcagattcaccatcagccgggacaacagcaagaacaccctg  
tacctgcagatgaacagcctgagagccgaggacaccgccgtgtactactg  
cgtgaaggactggaacaccaccatgatcaccgagagaggccagggcacac  
tggtcaccgtgtcctctacaacaacaccggcgccctggcctccaacacca  
gctcctacaatcgcgagtcagcccctgtctctcagacccgaagcctgtag  
acctgctgctggcggagctgtgcataccagaggactggatttcgcctgcg  
acatctacatctgggctcctctggctggcacatgcggagttttgctgctg  
agcctgggtcatcaccctgtactgtaagagaggcaggaagaagctgctgta  
tatcttcaagcagccatcatgagaccgtgcagaccacacaggaggagga  
cggctgctctttaggttcccagaggaggaggaggaggatgagctgagctgc  
gcgtgaagtttagccggtccgcccgtgacacctgcatacaagcagggacag  
aaccagctgtataacgagctgaatctgggcccggagagaggagtacgacgt  
gctggataagaggcggggccgggaccccgagatgggaggcaagcctcgga  
gaaagaacccacaggaggcctgtacaatgagctgcaaaaggacaagatg  
gccgaggcctattctgagatcggcatgaaggagagaggcgccggggcaa  
gggacacgatggcctgtaccaggcctgagcaccgccacaaaggacacct atgatgccctgcacatgcaggccctgccccctagatga.

(54) In some embodiments of the VCARS of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

(55) TABLE-US-00049 VH-E: (SEQ ID NO: 18008)

malpvtalllplalllhaarpevqllesggglvqpggslrlscaasgftf  
ssyamnwirqapgkglewvsgisgsggstyyadsvkgrftisrdnskntv  
ylqmnsiraedtavyycvkdwnntmitergqgtlvtssttppaprpptp  
aptiasqplslrpeacrpaaggavhtrglfacdiyiwaplagtcgvlll

slvitlyckrgrkllyifkqpfmrpvqttqeedgcscrfeeeggcel  
rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr  
rknpeglynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdt ydalhmqalppr.

(56) In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

(57) TABLE-US-00050 VH-E (SEQ ID NO: 18009)

atggcactgcctgtgacagccctgtgctgcctctggccctgctgctgca  
cgagcagcggcccgaggtgcagctgctggagtcggaggaggcctggccc  
agcctggcggcagcctgaggctgtcctgcgccgctctggcttcacctt  
agctcctacgccatgaactggatcagacaggccctggcaagggcctgga  
gtgggtgtccggcatctccggctctggaggtctacatactatgccgaca  
gcgtgaaggggccggttcacatcagcagagataactccaagaataccgtg  
tacctccagatgaactctctgcgggcccaggacaccgccgtgtactattg  
cgtgaaggattggaataaccacaatgatcacagagagggggccagggcaccc  
tggtgacagtgtctagaccacaacccctgccccagacctccacaccc  
gcccctaccatcgcgagtcagccactgtccctgcggcctgaggcctgccg  
gcccggccggcgaggagtcacacacggggcctggactttgcctgtg  
acatctacatatgggcaccactggcaggaacctgcggcgtgctgctgctg  
agcctggtcatcaccctgtactgtaagagaggcaggaagaagctgagtat  
atcttcaagcagccctcatgagaccgtgcagaccacacaggaggagga  
cggctgacttgtaggtcccagaggaggaggaggaggatgcgagagcgc  
gtgaagtttagccgggtccgccgatgcacctgcatacaagcagggacagaa  
ccagagtataacgagctgaatctgggcccggagagaggagtacgacgtgct  
ggataagaggcgggggccgggaccccgagatgggaggcaagcctcgagaa  
agaaccacaggaggggcctgtacaatgagagcaaaaggacaagatggccg  
aggcctattctgagatcggtatgaaggagagaggcgccggggcaaggga  
cacgatggcctgtaccagggcctgagcaccgccacaaaggacacctatga tgccagcacatgcaggccagccccctagatga.

(58) In some embodiments of the VCARS of the disclosure, the VCAR at least one of comprises an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

(59) TABLE-US-00051 VH-F: (SEQ ID NO: 18010)

malpvtalllplalllhaarpevqllesggglvqpgrslrlscaasgftf  
tnyamnwvrqapgkglewvsgisggggstyyadsvkgrftisrdnsnt  
ylqmnsraedtavyycvkdwnntmitergqgtlvtvssttpprptp  
aptiasqplslrpeacrpaaggavhtrglfacdiyiwaplagtcgvlll  
slvitlyckrgrkllyifkqpfmrpvqttqeedgcscrfeeeggcel  
rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr  
rknpeglynelqkdkmaeayseigmkgerrrgkghdglyqglstatkdt ydalhmqalppr.

(60) In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

(61) TABLE-US-00052 VH-F (SEQ ID NO: 18011)

atggcactgcctgtgacagccctgtgctgcctctggccctgctgctgca  
cgagcagcggcccgaggtgcagctgctggagtcggaggaggcctgggtgc  
agccccggccggtcctgagactgtcttgcgccgcccagcggcttcacctt  
acaaactacgccatgaattgggtgcggcaggcccctggcaagggcctgga  
gtgggtgtctggcatcagcggaggaggaggcagcacctactatgcagact  
ccgtgaaggggcaggttcacatctcccgcgataactctaagaatacactg  
tacctccagatgaacagcctgagggcagaggacaccgccgtgtactattg  
cgtgaaggattggaataaccacaatgatcacagagagggggacagggcaccc  
tggtgaccgtgagcagcaccacaacccctgccccagacctccacaccc

gccccctacatcgcgagtcagccactgtccctgcggcctgaggcctgccg  
 gccccgccgcccgggagcagtgacacacggggcctggactttgcctgtg  
 acatctacatatgggcaccactggcaggaacctgcggcgtgctgctgctg  
 agcctggtcatcacctgtactgtaagagaggcaggaagaagctgctgta  
 tatcttcaagcagcccttcatgagacccgtgcagaccacacaggaggagg  
 acggctgctctttagaggtcccagaggaggaggaggaggatgcgagctg  
 cgcggtgaagtttagccggtccgccgatgcacctgcatacaagcagggaca  
 gaaccagctgtataacgagctgaatctgggcccggagagaggagtacgacg  
 tgctggataagaggcggggccgggacccccgagatgggagggaagcctcgg  
 agaaagaacccacaggaggggcctgtacaatgagctgcaaaaggacaagat  
 ggccgaggcctattctgagatcgccatgaagggagagaggcgccggggca  
 agggacacgatggcctgtaccaggcctgagcaccgccacaaaggacacc tatgatgccagcacatgcaggccagccccctagatga.

(62) In some embodiments of the VCARs of the disclosure, the VCAR comprises a sequence encoding VH-A, VH-B, VH-C, VH-D, VH-E, or VH-F. In some embodiments of the VCARs of the disclosure, the VCAR comprises two sequences encoding en VH-A, VH-B, VH-C, VH-D, VH-E, or VH-F.

(63) Immune and Immune Precursor Cells

(64) In certain embodiments, immune cells of the disclosure comprise lymphoid progenitor cells, natural killer (NK) cells, T lymphocytes (T-cell), stem memory T cells (T.sub.SCM cells), central memory T cells (T.sub.CM), stem cell-like T cells, B lymphocytes (B-cells), myeloid progenitor cells, neutrophils, basophils, eosinophils, monocytes, macrophages, platelets, erythrocytes, red blood cells (RBCs), megakaryocytes or osteoclasts.

(65) In certain embodiments, immune precursor cells comprise any cells which can differentiate into one or more types of immune cells. In certain embodiments, immune precursor cells comprise multipotent stem cells that can self renew and develop into immune cells. In certain embodiments, immune precursor cells comprise hematopoietic stem cells (HSCs) or descendants thereof. In certain embodiments, immune precursor cells comprise precursor cells that can develop into immune cells. In certain embodiments, the immune precursor cells comprise hematopoietic progenitor cells (HPCs).

(66) Hematopoietic Stem Cells (HSCs)

(67) Hematopoietic stem cells (HSCs) are multipotent, self-renewing cells. All differentiated blood cells from the lymphoid and myeloid lineages arise from HSCs. HSCs can be found in adult bone marrow, peripheral blood, mobilized peripheral blood, peritoneal dialysis effluent and umbilical cord blood.

(68) HSCs of the disclosure may be isolated or derived from a primary or cultured stem cell. HSCs of the disclosure may be isolated or derived from an embryonic stem cell, a multipotent stem cell, a pluripotent stem cell, an adult stem cell, or an induced pluripotent stem cell (iPSC).

(69) Immune precursor cells of the disclosure may comprise an HSC or an HSC descendent cell. Exemplary HSC descendent cells of the disclosure include, but are not limited to, multipotent stem cells, lymphoid progenitor cells, natural killer (NK) cells, T lymphocyte cells (T-cells), B lymphocyte cells (B-cells), myeloid progenitor cells, neutrophils, basophils, eosinophils, monocytes, and macrophages.

(70) HSCs produced by the methods of the disclosure may retain features of “primitive” stem cells that, while isolated or derived from an adult stem cell and while committed to a single lineage, share characteristics of embryonic stem cells. For example, the “primitive” HSCs produced by the methods of the disclosure retain their “stemness” following division and do not differentiate. Consequently, as an adoptive cell therapy, the “primitive” HSCs produced by the methods of the disclosure not only replenish their numbers, but expand in vivo. “Primitive” HSCs produced by the methods of the disclosure may be therapeutically-effective when administered as a single dose. In some embodiments, primitive HSCs of the disclosure are CD34+. In some embodiments, primitive HSCs of the disclosure are CD34+ and CD38-. In some embodiments, primitive HSCs of the disclosure are CD34+, CD38- and CD90+. In some embodiments, primitive HSCs of the disclosure are CD34+, CD38-, CD90+ and CD45RA-. In some embodiments, primitive HSCs of the disclosure are CD34+, CD38-, CD90+, CD45RA-, and CD49f+. In some embodiments, the most primitive HSCs of the disclosure are CD34+, CD38-, CD90+, CD45RA-, and CD49f+.

(71) In some embodiments of the disclosure, primitive HSCs, HSCs, and/or HSC descendent cells may be modified according to the methods of the disclosure to express an exogenous sequence (e.g., a chimeric antigen receptor or therapeutic protein). In some embodiments of the disclosure, modified primitive HSCs, modified HSCs, and/or modified HSC descendent cells may be forward differentiated to produce a modified immune cell including, but not limited to, a modified T cell, a modified natural killer cell and/or a modified B-cell of the disclosure.

## (72) T Cells

(73) Modified T cells of the disclosure may be derived from modified hematopoietic stem and progenitor cells (HSPCs) or modified HSCs.

(74) Unlike traditional biologics and chemotherapeutics, modified-T cells of the disclosure possess the capacity to rapidly reproduce upon antigen recognition, thereby potentially obviating the need for repeat treatments. To achieve this, in some embodiments, modified-T cells of the disclosure not only drive an initial response, but also persist in the patient as a stable population of viable memory T cells to prevent potential relapses. Alternatively, in some embodiments, when it is not desired, modified-T cells of the disclosure do not persist in the patient.

(75) Intensive efforts have been focused on the development of antigen receptor molecules that do not cause T cell exhaustion through antigen-independent (tonic) signaling, as well as of a modified-T cell product containing early memory T cells, especially stem cell memory (T.sub.SCM) or stem cell-like T cells. Stem cell-like modified-T cells of the disclosure exhibit the greatest capacity for self-renewal and multipotent capacity to derive central memory (T.sub.CM) T cells or T.sub.CM like cells, effector memory (T.sub.EM) and effector T cells (T.sub.E), thereby producing better tumor eradication and long-term modified-T cell engraftment. A linear pathway of differentiation may be responsible for generating these cells: Naïve T cells (T.sub.N)>T.sub.SCM>T.sub.CM>T.sub.EM>T.sub.E>T.sub.TE, whereby T.sub.N is the parent precursor cell that directly gives rise to T.sub.SCM, which then, in turn, directly gives rise to T.sub.CM, etc. Compositions of T cells of the disclosure may comprise one or more of each parental T cell subset with T.sub.SCM or T.sub.CM cells being the most abundant (e.g., T.sub.SCM>T.sub.CM>T.sub.EM>T.sub.E>T.sub.TE).

(76) In some embodiments of the methods of the disclosure, the immune cell precursor is differentiated into or is capable of differentiating into an early memory T cell, a stem cell like T-cell, a Naïve T cells (T.sub.N), a T.sub.SCM, a T.sub.CM, a T.sub.EM, a T.sub.E, or a Tm. In some embodiments, the immune cell precursor is a primitive HSC, an HSC, or a HSC descendent cell of the disclosure.

(77) In some embodiments of the methods of the disclosure, the immune cell is an early memory T cell, a stem cell like T-cell, a Naïve T cells (T.sub.N), a T.sub.SCM, a T.sub.CM, a T.sub.EM, a T.sub.E, or a T.sub.TE.

(78) In some embodiments of the methods of the disclosure, the immune cell is an early memory T cell.

(79) In some embodiments of the methods of the disclosure, the immune cell is a stem cell like T-cell.

(80) In some embodiments of the methods of the disclosure, the immune cell is a T.sub.SCM.

(81) In some embodiments of the methods of the disclosure, the immune cell is a T.sub.CM.

(82) In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of an early memory T cell. In certain embodiments, the plurality of modified early memory T cells comprises at least one modified stem cell-like T cell. In certain embodiments, the plurality of modified early memory T cells comprises at least one modified T.sub.SCM. In certain embodiments, the plurality of modified early memory T cells comprises at least one modified T.sub.CM.

(83) In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a stem cell-like T cell. In certain embodiments, the plurality of modified stem cell-like T cells comprises at least one modified T.sub.SCM. In certain embodiments, the plurality of modified stem cell-like T cells comprises at least

one modified T.sub.CM

(84) In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a stem memory T cell (T.sub.SCM). In certain embodiments, the cell-surface markers comprise CD62L and CD45RA. In certain embodiments, the cell-surface markers comprise one or more of CD62L, CD45RA, CD28, CCR7, CD127, CD45RO, CD95, CD95 and IL-2RP. In certain embodiments, the cell-surface markers comprise one or more of CD45RA, CD95, IL-2R $\beta$ , CCR7, and CD62L.

(85) In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a central memory T cell (T.sub.CM). In certain embodiments, the cell-surface markers comprise one or more of CD45RO, CD95, IL-2RP, CCR7, and CD62L.

(86) In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a naïve T cell (T.sub.N). In certain embodiments, the cell-surface markers comprise one or more of CD45RA, CCR7 and CD62L.

(87) In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of an effector T-cell (modified TEFF). In certain embodiments, the cell-surface markers comprise one or more of CD45RA, CD95, and IL-2RP.

(88) In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a stem cell-like T cell, a stem memory T cell (T.sub.SCM) or a central memory T cell (T.sub.CM).

(89) In some embodiments of the methods of the disclosure, a buffer comprises the immune cell or precursor thereof. The buffer maintains or enhances a level of cell viability and/or a stem-like phenotype of the immune cell or precursor thereof, including T-cells. In certain embodiments, the buffer maintains or enhances a level of cell viability and/or a stem-like phenotype of the primary human T cells prior to the nucleofection. In certain embodiments, the buffer maintains or enhances a level of cell viability and/or a stem-like phenotype of the primary human T cells during the nucleofection. In certain embodiments, the buffer maintains or enhances a level of cell viability and/or a stem-like phenotype of the primary human T cells following the nucleofection. In certain embodiments, the buffer comprises one or more of KCl, MgCl.sub.2, ClNa, Glucose and Ca(NO.sub.3).sub.2 in any absolute or relative abundance or concentration, and, optionally, the buffer further comprises a supplement selected from the group consisting of HEPES, Tris/HCl, and a phosphate buffer. In certain embodiments, the buffer comprises 5 mM KCl, 15 mM MgCl.sub.2, 90 mM ClNa, 10 mM Glucose and 0.4 mM Ca(NO.sub.3).sub.2. In certain embodiments, the buffer comprises 5 mM KCl, 15 mM MgCl.sub.2, 90 mM ClNa, 10 mM Glucose and 0.4 mM Ca(NO.sub.3).sub.2 and a supplement comprising 20 mM HEPES and 75 mM Tris/HCl. In certain embodiments, the buffer comprises 5 mM KCl, 15 mM MgCl.sub.2, 90 mM ClNa, 10 mM Glucose and 0.4 mM Ca(NO.sub.3).sub.2 and a supplement comprising 40 mM Na.sub.2HPO.sub.4/NaH.sub.2PO.sub.4 at pH 7.2. In certain embodiments, the composition comprising primary human T cells comprises 100  $\mu$ l of the buffer and between 5 $\times$ 10<sup>6</sup> and 25 $\times$ 10<sup>6</sup> cells. In certain embodiments, the composition comprises a scalable ratio of 250 $\times$ 10<sup>6</sup> primary human T cells per milliliter of buffer or other media during the introduction step.

(90) In some embodiments of the methods of the disclosure, the methods comprise contacting an immune cell of the disclosure, including a T cell of the disclosure, and a T-cell expansion composition. In some embodiments of the methods of the disclosure, the step of introducing a transposon and/or transposase of the disclosure into an immune cell of the disclosure may further comprise contacting the immune cell and a T-cell expansion composition. In some embodiments, including those in which the introducing step of the methods comprises an electroporation or a nucleofection step, the electroporation or a nucleofection step may be performed with the immune cell contacting T-cell expansion composition of the disclosure.

(91) In some embodiments of the methods of the disclosure, the T-cell expansion composition comprises, consists essentially of or consists of phosphorus; one or more of an octanoic acid, a palmitic acid, a linoleic acid, and an oleic acid; a sterol; and an alkane.

(92) In certain embodiments of the methods of producing a modified T cell of the disclosure, the expansion supplement comprises one or more cytokine(s). The one or more cytokine(s) may comprise any cytokine, including but not limited to, lymphokines. Exemplary lymphokines include, but are not limited to, interleukin-2 (IL-2), interleukin-3 (IL-3), interleukin-4 (IL-4), interleukin-5 (IL-5), interleukin-6 (IL-6), interleukin-7 (IL-7), interleukin-15 (IL-15), interleukin-21 (IL-21), granulocyte-macrophage colony-stimulating factor (GM-CSF) and interferon-gamma (INF). The one or more cytokine(s) may comprise IL-2.

(93) In some embodiments of the methods of the disclosure, the T-cell expansion composition comprises human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid, nicotinamide, 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD), diisopropyl adipate (DIPA), n-butyl-benzenesulfonamide, 1,2-benzenedicarboxylic acid, bis(2-methylpropyl) ester, palmitic acid, linoleic acid, oleic acid, stearic acid hydrazide, oleamide, a sterol and an alkane. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid, palmitic acid, linoleic acid, oleic acid and a sterol. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid at a concentration of between 0.9 mg/kg to 90 mg/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; oleic acid at a concentration of 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; and a sterol at a concentration of about 0.1 mg/kg to 10 mg/kg, inclusive of the endpoints. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid at a concentration of about 9 mg/kg, palmitic acid at a concentration of about 2 mg/kg, linoleic acid at a concentration of about 2 mg/kg, oleic acid at a concentration of about 2 mg/kg and a sterol at a concentration of about 1 mg/kg. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid at a concentration of between 6.4  $\mu$ mol/kg and 640  $\mu$ mol/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.7  $\mu$ mol/kg and 70  $\mu$ mol/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.75  $\mu$ mol/kg and 75  $\mu$ mol/kg, inclusive of the endpoints; oleic acid at a concentration of between 0.75  $\mu$ mol/kg and 75  $\mu$ mol/kg, inclusive of the endpoints; and a sterol at a concentration of between 0.25  $\mu$ mol/kg and 25  $\mu$ mol/kg, inclusive of the endpoints. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid at a concentration of about 64  $\mu$ mol/kg, palmitic acid at a concentration of about 7  $\mu$ mol/kg, linoleic acid at a concentration of about 7.5  $\mu$ mol/kg, oleic acid at a concentration of about 7.5  $\mu$ mol/kg and a sterol at a concentration of about 2.5  $\mu$ mol/kg.

(94) In certain embodiments, the T-cell expansion composition comprises one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement to produce a plurality of expanded modified T-cells, wherein at least 2% of the plurality of modified T-cells expresses one or more cell-surface marker(s) of an early memory T cell, a stem cell-like T cell, a stem memory T cell (TscM) and/or a central memory T cell (TcM). In certain embodiments, the T-cell expansion composition comprises or further comprises one or more of octanoic acid, nicotinamide, 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD), diisopropyl adipate (DIPA), n-butyl-



benzenesulfonamide, 1,2-benzenedicarboxylic acid, bis(2-methylpropyl) ester, palmitic acid, linoleic acid, oleic acid, stearic acid hydrazide, oleamide, a sterol and an alkane. In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid, palmitic acid, linoleic acid, oleic acid and a sterol (e.g., cholesterol). In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of between 0.9 mg/kg to 90 mg/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; oleic acid at a concentration of 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; and a sterol at a concentration of about 0.1 mg/kg to 10 mg/kg, inclusive of the endpoints (wherein mg/kg=parts per million). In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of about 9 mg/kg, palmitic acid at a concentration of about 2 mg/kg, linoleic acid at a concentration of about 2 mg/kg, oleic acid at a concentration of about 2 mg/kg, and a sterol at a concentration of about 1 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of 9.19 mg/kg, palmitic acid at a concentration of 1.86 mg/kg, linoleic acid at a concentration of about 2.12 mg/kg, oleic acid at a concentration of about 2.13 mg/kg, and a sterol at a concentration of about 1.01 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the T-cell expansion composition comprises octanoic acid at a concentration of 9.19 mg/kg, palmitic acid at a concentration of 1.86 mg/kg, linoleic acid at a concentration of 2.12 mg/kg, oleic acid at a concentration of about 2.13 mg/kg, and a sterol at a concentration of 1.01 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of between 6.4  $\mu$ mol/kg and 640  $\mu$ mol/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.7  $\mu$ mol/kg and 70  $\mu$ mol/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.75  $\mu$ mol/kg and 75  $\mu$ mol/kg, inclusive of the endpoints; oleic acid at a concentration of between 0.75  $\mu$ mol/kg and 75  $\mu$ mol/kg, inclusive of the endpoints; and a sterol at a concentration of between 0.25  $\mu$ mol/kg and 25  $\mu$ mol/kg, inclusive of the endpoints. In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of about 64  $\mu$ mol/kg, palmitic acid at a concentration of about 7  $\mu$ mol/kg, linoleic acid at a concentration of about 7.5  $\mu$ mol/kg, oleic acid at a concentration of about 7.5  $\mu$ mol/kg and a sterol at a concentration of about 2.5  $\mu$ mol/kg. In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of about 63.75  $\mu$ mol/kg, palmitic acid at a concentration of about 7.27  $\mu$ mol/kg, linoleic acid at a concentration of about 7.57  $\mu$ mol/kg, oleic acid at a concentration of about 7.56  $\mu$ mol/kg and a sterol at a concentration of about 2.61  $\mu$ mol/kg. In certain embodiments, the T-cell expansion composition comprises octanoic acid at a concentration of about 63.75  $\mu$ mol/kg, palmitic acid at a concentration of about 7.27  $\mu$ mol/kg, linoleic acid at a concentration of about 7.57  $\mu$ mol/kg, oleic acid at a concentration of 7.56  $\mu$ mol/kg and a sterol at a concentration of 2.61  $\mu$ mol/kg.

(95) As used herein, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement at 37° C. Alternatively, or in addition, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of phosphorus, an octanoic fatty acid, a palmitic fatty acid, a linoleic fatty acid and an oleic acid. In certain embodiments, the media comprises an amount of phosphorus that is 10-fold higher than may be found in, for example, Iscove's Modified Dulbecco's Medium ((IMDM); available at ThermoFisher Scientific as Catalog number 12440053).

(96) As used herein, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, Iscove's MDM, and an expansion supplement at 37° C. Alternatively, or in addition, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of the following elements: boron, sodium, magnesium, phosphorus, potassium, and

calcium. In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of the following elements present in the corresponding average concentrations: boron at 3.7 mg/L, sodium at 3000 mg/L, magnesium at 18 mg/L, phosphorus at 29 mg/L, potassium at 15 mg/L and calcium at 4 mg/L.

(97) As used herein, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement at 37° C. Alternatively, or in addition, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of the following components: octanoic acid (CAS No. 124-07-2), nicotinamide (CAS No. 98-92-0), 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD) (CAS No. 126-86-3), diisopropyl adipate (DIPA) (CAS No. 6938-94-9), n-butyl-benzenesulfonamide (CAS No. 3622-84-2), 1,2-benzenedicarboxylic acid, bis(2-methylpropyl) ester (CAS No. 84-69-5), palmitic acid (CAS No. 57-10-3), linoleic acid (CAS No. 60-33-3), oleic acid (CAS No. 112-80-1), stearic acid hydrazide (CAS No. 4130-54-5), oleamide (CAS No. 3322-62-1), sterol (e.g., cholesterol) (CAS No. 57-88-5), and alkanes (e.g., nonadecane) (CAS No. 629-92-5). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of the following components: octanoic acid (CAS No. 124-07-2), nicotinamide (CAS No. 98-92-0), 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD) (CAS No. 126-86-3), diisopropyl adipate (DIPA) (CAS No. 6938-94-9), n-butyl-benzenesulfonamide (CAS No. 3622-84-2), 1,2-benzenedicarboxylic acid, bis(2-methylpropyl) ester (CAS No. 84-69-5), palmitic acid (CAS No. 57-10-3), linoleic acid (CAS No. 60-33-3), oleic acid (CAS No. 112-80-1), stearic acid hydrazide (CAS No. 4130-54-5), oleamide (CAS No. 3322-62-1), sterol (e.g., cholesterol) (CAS No. 57-88-5), alkanes (e.g., nonadecane) (CAS No. 629-92-5), and phenol red (CAS No. 143-74-8). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of the following components: octanoic acid (CAS No. 124-07-2), nicotinamide (CAS No. 98-92-0), 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD) (CAS No. 126-86-3), diisopropyl adipate (DIPA) (CAS No. 6938-94-9), n-butyl-benzenesulfonamide (CAS No. 3622-84-2), 1,2-benzenedicarboxylic acid, bis(2-methylpropyl) ester (CAS No. 84-69-5), palmitic acid (CAS No. 57-10-3), linoleic acid (CAS No. 60-33-3), oleic acid (CAS No. 112-80-1), stearic acid hydrazide (CAS No. 4130-54-5), oleamide (CAS No. 3322-62-1), phenol red (CAS No. 143-74-8) and lanolin alcohol.

(98) In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement at 37° C. Alternatively, or in addition, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of the following ions: sodium, ammonium, potassium, magnesium, calcium, chloride, sulfate and phosphate.

(99) As used herein, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement at 37° C. Alternatively, or in addition, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of the following free amino acids: histidine, asparagine, serine, glutamate, arginine, glycine, aspartic acid, glutamic acid, threonine, alanine, proline, cysteine, lysine, tyrosine, methionine, valine, isoleucine, leucine, phenylalanine and tryptophan. In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of the following free amino acids in the corresponding average mole percentages: histidine (about 1%), asparagine (about 0.5%), serine (about 1.5%), glutamine (about 67%), arginine (about 1.5%), glycine (about 1.5%), aspartic acid (about 1%),

glutamic acid (about 2%), threonine (about 2%), alanine (about 1%), proline (about 1.5%), cysteine (about 1.5%), lysine (about 3%), tyrosine (about 1.5%), methionine (about 1%), valine (about 3.5%), isoleucine (about 3%), leucine (about 3.5%), phenylalanine (about 1.5%) and tryptophan (about 0.5%). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of the following free amino acids in the corresponding average mole percentages: histidine (about 0.78%), asparagine (about 0.4%), serine (about 1.6%), glutamine (about 67.01%), arginine (about 1.67%), glycine (about 1.72%), aspartic acid (about 1.00%), glutamic acid (about 1.93%), threonine (about 2.38%), alanine (about 1.11%), proline (about 1.49%), cysteine (about 1.65%), lysine (about 2.84%), tyrosine (about 1.62%), methionine (about 0.85%), valine (about 3.45%), isoleucine (about 3.14%), leucine (about 3.3%), phenylalanine (about 1.64%) and tryptophan (about 0.37%).

(100) As used herein, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, Iscove's MDM, and an expansion supplement at 37° C. Alternatively, or in addition, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of phosphorus, an octanoic fatty acid, a palmitic fatty acid, a linoleic fatty acid and an oleic acid. In certain embodiments, the media comprises an amount of phosphorus that is 10-fold higher than may be found in, for example, Iscove's Modified Dulbecco's Medium ((IMDM); available at ThermoFisher Scientific as Catalog number 12440053).

(101) In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid, palmitic acid, linoleic acid, oleic acid and a sterol (e.g., cholesterol). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of between 0.9 mg/kg to 90 mg/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; oleic acid at a concentration of 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; and a sterol at a concentration of about 0.1 mg/kg to 10 mg/kg, inclusive of the endpoints (wherein mg/kg=parts per million). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of about 9 mg/kg, palmitic acid at a concentration of about 2 mg/kg, linoleic acid at a concentration of about 2 mg/kg, oleic acid at a concentration of about 2 mg/kg, and a sterol at a concentration of about 1 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of 9.19 mg/kg, palmitic acid at a concentration of 1.86 mg/kg, linoleic acid at a concentration of about 2.12 mg/kg, oleic acid at a concentration of about 2.13 mg/kg, and a sterol at a concentration of about 1.01 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of 9.19 mg/kg, palmitic acid at a concentration of 1.86 mg/kg, linoleic acid at a concentration of 2.12 mg/kg, oleic acid at a concentration of about 2.13 mg/kg, and a sterol at a concentration of 1.01 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of between 6.4 μmol/kg and 640 μmol/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.7 μmol/kg and 70 μmol/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.75 μmol/kg and 75 mol/kg, inclusive of the endpoints; oleic acid at a concentration of between 0.75 mol/kg and 75 mol/kg, inclusive of the endpoints; and a sterol at a concentration of between 0.25 mol/kg and 25 mol/kg, inclusive of the endpoints. In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic

acid at a concentration of about 64 mol/kg, palmitic acid at a concentration of about 7  $\mu$ mol/kg, linoleic acid at a concentration of about 7.5  $\mu$ mol/kg, oleic acid at a concentration of about 7.5  $\mu$ mol/kg and a sterol at a concentration of about 2.5  $\mu$ mol/kg.

(102) In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of about 63.75  $\mu$ mol/kg, palmitic acid at a concentration of about 7.27  $\mu$ mol/kg, linoleic acid at a concentration of about 7.57  $\mu$ mol/kg, oleic acid at a concentration of about 7.56 mol/kg and a sterol at a concentration of about 2.61  $\mu$ mol/kg. In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of about 63.75  $\mu$ mol/kg, palmitic acid at a concentration of about 7.27  $\mu$ mol/kg, linoleic acid at a concentration of about 7.57  $\mu$ mol/kg, oleic acid at a concentration of 7.56 mol/kg and a sterol at a concentration of 2.61  $\mu$ mol/kg.

(103) In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T.sub.SCM and/or a T.sub.CM) of the disclosure, the method comprises contacting a modified T cell and an inhibitor of the P13K-Akt-mTOR pathway. Modified T-cells of the disclosure, including modified stem cell-like T cells, T.sub.SCM and/or T.sub.CM of the disclosure, may be incubated, cultured, grown, stored, or otherwise, combined at any step in the methods of the procedure with a growth medium comprising one or more inhibitors a component of a PI3K pathway. Exemplary inhibitors a component of a PI3K pathway include, but are not limited to, an inhibitor of GSK3 $\beta$  such as TWS119 (also known as GSK 3B inhibitor XII; CAS Number 601514-19-6 having a chemical formula C.sub.18H.sub.14N.sub.4O.sub.2). Exemplary inhibitors of a component of a PI3K pathway include, but are not limited to, bb007 (BLUEBIRDBIO™). Additional Exemplary inhibitors of a component of a PI3K pathway include, but are not limited to, an allosteric Akt inhibitor VIII (also referred to as Akti-1/2 having Compound number 10196499), ATP competitive inhibitors (Orthosteric inhibitors targeting the ATP-binding pocket of the protein kinase B (Akt)), Isoquinoline-5-sulfonamides (H-8, H-89, and NL-71-101), Azepane derivatives (A series of structures derived from (-)-balanol), Aminofurazans (GSK690693), Heterocyclic rings (7-azaindole, 6-phenylpurine derivatives, pyrrolo[2,3-d]pyrimidine derivatives, CCT128930, 3-aminopyrrolidine, anilinothiazole derivatives, spiroindoline derivatives, AZD5363, ipatasertib (GDC-0068, RG7440), A-674563, and A-443654), Phenylpyrazole derivatives (AT7867 and AT13148), Thiophenecarboxamide derivatives (Afuresertib (GSK2110183), 2-pyrimidyl-5-amidothiophene derivative (DC120), uprosertib (GSK2141795)), Allosteric inhibitors (Superior to orthosteric inhibitors providing greater specificity, reduced side-effects and less toxicity), 2,3-diphenylquinoxaline analogues (2,3-diphenylquinoxaline derivatives, triazolo[3,4-f][1,6]naphthyridin-3(2H)-one derivative (MK-2206)), Alkylphospholipids (Edelfosine (1-O-octadecyl-2-O-methyl-rac-glycero-3-phosphocholine, ET-18-OCH.sub.3) ilmofosine (BM 41.440), miltefosine (hexadecylphosphocholine, HePC), perifosine (D-21266), erucylphosphocholine (ErPC), erufosine (ErPC3, erucylphosphohomocholine), Indole-3-carbinol analogues (Indole-3-carbinol, 3-chloroacetylindole, diindolylmethane, diethyl 6-methoxy-5,7-dihydroindolo [2,3-b]carbazole-2,10-dicarboxylate (SR13668), OSU-A9), Sulfonamide derivatives (PH-316 and PHT-427), Thiourea derivatives (PIT-1, PIT-2, DM-PIT-1, N-[(1-methyl-1H-pyrazol-4-yl)carbonyl]-N'-(3-bromophenyl)-thiourea), Purine derivatives (Triciribine (TCN, NSC 154020), triciribine mono-phosphate active analogue (TCN-P), 4-amino-pyrido[2,3-d]pyrimidine derivative API-1, 3-phenyl-3H-imidazo[4,5-b]pyridine derivatives, ARQ 092), BAY 1125976, 3-methyl-xanthine, quinoline-4-carboxamide and 2-[4-(cyclohexa-1,3-dien-1-yl)-1H-pyrazol-3-yl]phenol, 3-oxo-tirucallic acid, 3 $\alpha$ - and 3 $\beta$ -acetoxy-tirucallic acids, acetoxy-tirucallic acid, and irreversible inhibitors (antibiotics, Lactoquinomycin, Frenolicin B, kalafungin, medermycin, Boc-Phe-vinyl ketone, 4-hydroxynonenal (4-HNE), 1,6-naphthyridinone derivatives, and imidazo-1,2-pyridine derivatives).

(104) In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T.sub.SCM and/or a T.sub.CM) of the disclosure, the method comprises contacting a modified T cell and an inhibitor of T cell effector differentiation. Exemplary inhibitors of T cell effector differentiation include, but are not limited to, a BET inhibitor (e.g., JQ1, a hienotriazolodiazepine)

and/or an inhibitor of the BET family of proteins (e.g., BRD2, BRD3, BRD4, and BRDT).

(105) In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T.sub.SCM and/or a T.sub.CM) of the disclosure, the method comprises contacting a modified T cell and an agent that reduces nucleocytoplasmic Acetyl-CoA. Exemplary agents that reduce nucleocytoplasmic Acetyl-CoA include, but are not limited to, 2-hydroxy-citrate (2-HC) as well as agents that increase expression of Acss1.

(106) In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T.sub.SCM and/or a T.sub.CM) of the disclosure, the method comprises contacting a modified T cell and a composition comprising a histone deacetylase (HDAC) inhibitor. In some embodiments, the composition comprising an HDAC inhibitor comprises or consists of valproic acid, Sodium Phenylbutyrate (NaPB) or a combination thereof. In some embodiments, the composition comprising an HDAC inhibitor comprises or consists of valproic acid. In some embodiments, the composition comprising an HDAC inhibitor comprises or consists of Sodium Phenylbutyrate (NaPB).

(107) In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T.sub.SCM and/or a T.sub.CM) of the disclosure, the activation supplement may comprise one or more cytokine(s). The one or more cytokine(s) may comprise any cytokine, including but not limited to, lymphokines. Exemplary lymphokines include, but are not limited to, interleukin-2 (IL-2), interleukin-3 (IL-3), interleukin-4 (IL-4), interleukin-5 (IL-5), interleukin-6 (IL-6), interleukin-7 (IL-7), interleukin-15 (IL-15), interleukin-21 (IL-21), granulocyte-macrophage colony-stimulating factor (GM-CSF) and interferon-gamma (INF). The one or more cytokine(s) may comprise IL-2.

(108) In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T.sub.SCM and/or a T.sub.CM) of the disclosure, the activation supplement may comprise one or more activator complexes. Exemplary and nonlimiting activator complexes may comprise a monomeric, dimeric, trimeric or tetrameric antibody complex that binds one or more of CD3, CD28, and CD2. In some embodiments, the activation supplement comprises or consists of an activator complex that comprises a human, a humanized or a recombinant or a chimeric antibody. In some embodiments, the activation supplement comprises or consists of an activator complex that binds CD3 and CD28. In some embodiments, the activation supplement comprises or consists of an activator complex that binds CD3, CD28 and CD2.

(109) Natural Killer (NK) Cells

(110) In certain embodiments, the modified immune or immune precursor cells of the disclosure are natural killer (NK) cells. In certain embodiments, NK cells are cytotoxic lymphocytes that differentiate from lymphoid progenitor cells.

(111) Modified NK cells of the disclosure may be derived from modified hematopoietic stem and progenitor cells (HSPCs) or modified HSCs.

(112) In certain embodiments, non-activated NK cells are derived from CD3-depleted leukopheresis (containing CD14/CD19/CD56+ cells).

(113) In certain embodiments, NK cells are electroporated using a Lonza 4D nucleofector or BTX ECM 830 (500V, 700 usec pulse length, 0.2 mm electrode gap, one pulse). All Lonza 4D nucleofector programs are contemplated as within the scope of the methods of the disclosure.

(114) In certain embodiments,  $5 \times 10^6$  cells were electroporated per electroporation in 100  $\mu$ L P3 buffer in cuvettes. However, this ratio of cells per volume is scalable for commercial manufacturing methods.

(115) In certain embodiments, NK cells were stimulated by co-culture with an additional cell line. In certain embodiments, the additional cell line comprises artificial antigen presenting cells (aAPCs). In certain embodiments, stimulation occurs at day 1, 2, 3, 4, 5, 6, or 7 following electroporation. In certain embodiments, stimulation occurs at day 2 following electroporation.

(116) In certain embodiments, NK cells express CD56.

(117) B Cells

(118) In certain embodiments, the modified immune or immune precursor cells of the disclosure are B cells. B cells are a type of lymphocyte that express B cell receptors on the cell surface. B cell receptors bind to specific antigens.

(119) Modified B cells of the disclosure may be derived from modified hematopoietic stem and

progenitor cells (HSPCs) or modified HSCs.

(120) In certain embodiments, HSPCs are modified using the methods of the disclosure, and then primed for B cell differentiation in presence of human IL-3, Flt3L, TPO, SCF, and G-CSF for at least 3 days, at least 4 days, at least 5 days, at least 6 days or at least 7 days. In certain embodiments, HSPCs are modified using the methods of the disclosure, and then primed for B cell differentiation in presence of human IL-3, Flt3L, TPO, SCF, and G-CSF for 5 days.

(121) In certain embodiments, following priming, modified HSPC cells are transferred to a layer of feeder cells and fed bi-weekly, along with transfer to a fresh layer of feeders once per week. In certain embodiments, the feeder cells are MS-5 feeder cells.

(122) In certain embodiments, modified HSPC cells are cultured with MS-5 feeder cells for at least 7, 14, 21, 28, 30, 33, 35, 42 or 48 days. In certain embodiments, modified HSPC cells were cultured with MS-5 feeder cells for 33 days.

(123) Transposition Systems

(124) Exemplary transposon/transposase systems of the disclosure include, but are not limited to, piggyBac transposons and transposases, piggyBac-like transposons and transposases, Sleeping Beauty transposons and transposases, Helraiser transposons and transposases and Tol2 transposons and transposases.

(125) The piggyBac transposase recognizes transposon-specific inverted terminal repeat sequences (ITRs) on the ends of the transposon, and moves the contents between the ITRs into TTA chromosomal sites. The piggyBac transposon system has no payload limit for the genes of interest that can be included between the ITRs. In certain embodiments, and, in particular, those embodiments wherein the transposon is a piggyBac transposon, the transposase is a piggyBac or a Super piggyBac (SPB) transposase. In certain embodiments, and, in particular, those embodiments wherein the transposase is a Super piggyBac (SPB) transposase, the sequence encoding the transposase is an mRNA sequence.

(126) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac (PB) transposase enzyme. The piggyBac (PB) transposase enzyme may comprise or consist of an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(127) TABLE-US-00053 (SEQ ID NO: 14487) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEI SDHVSEDDVQ SDTEEAFIDE VHEVQPTSSG 61 SEILDEQNV  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTGATFRD TNEDEIYAFF  
181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRFDFL IRCLRMDDKS  
IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF  
RMYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN GVPLGEYYVK  
ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT VRSNKREIPE  
VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421 DEDASINEST  
GKPQMVMYYN QTKGGVDTL D QMCSVMTC SR KTNRWPMALL YGMINIACIN 481  
SFIYSHNV S SKGEKVQSRK KFM RNLYMSL TSSFMRKRLE APTLKRYLRD  
NISNILPNEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA NASCKKCKKV  
ICREHNIDMC QSCF.

(128) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at one or more of positions 30, 165, 282, or 538 of the sequence:

(129) TABLE-US-00054 (SEQ ID NO: 14487) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEI SDHVSEDDVQ SDTEEAFIDE VHEVQPTSSG 61 SEILDEQNV  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTGATFRD TNEDEIYAFF  
181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRFDFL IRCLRMDDKS  
IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF  
RMYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN GVPLGEYYVK  
ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT VRSNKREIPE

VLKNSRSPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421 DEDASINEST  
GKPQMVMYYN QTKGGVDTLD QMCSVMTCR KTNRWPMALL YGMINIACIN 481  
SFIIYSHNVS SKGEKVQSRK KFMRLNLYMSL TSSFMRKRLE APTLKRYLRD  
NISNILPNEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA NASCKKCKKV  
ICREHNIDMC QSCF.

(130) In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at two or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at three or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at each of the following positions 30, 165, 282, and 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the amino acid substitution at position 30 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 165 of the sequence of SEQ ID NO: 14487 is a substitution of a serine (S) for a glycine (G). In certain embodiments, the amino acid substitution at position 282 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 538 of the sequence of SEQ ID NO: 14487 is a substitution of a lysine (K) for an asparagine (N).

(131) In certain embodiments of the methods of the disclosure, the transposase enzyme is a Super piggyBac (SPB) transposase enzyme. In certain embodiments, the Super piggyBac (SPB) transposase enzymes of the disclosure may comprise or consist of the amino acid sequence of the sequence of SEQ ID NO: 14487 wherein the amino acid substitution at position 30 is a substitution of a valine (V) for an isoleucine (I), the amino acid substitution at position 165 is a substitution of a serine (S) for a glycine (G), the amino acid substitution at position 282 is a substitution of a valine (V) for a methionine (M), and the amino acid substitution at position 538 is a substitution of a lysine (K) for an asparagine (N). In certain embodiments, the Super piggyBac (SPB) transposase enzyme may comprise or consist of an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(132) TABLE-US-00055 (SEQ ID NO: 14484) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEV SDHVSEDDVQ SDTEEFIDE VHEVQPTSSG 61 SEILDEQNVI  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTSATFRD TNEDEIYAFF  
181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRFDFL IRCLRMDDKS  
IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF  
RVYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN GVPLGEYYVK  
ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT VRSNKREIPE  
VLKNSRSPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421 DEDASINEST  
GKPQMVMYYN QTKGGVDTLD QMCSVMTCR KTNRWPMALL YGMINIACIN 481  
SFIIYSHNVS SKGEKVQSRK KFMRLNLYMSL TSSFMRKRLE APTLKRYLRD  
NISNILPKEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA NASCKKCKKV  
ICREHNIDMC QSCF.

(133) In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or Super piggyBac transposase enzyme may further comprise an amino acid substitution at one or more of positions 3, 46, 82, 103, 119, 125, 177, 180, 185, 187, 200, 207, 209, 226, 235, 240, 241, 243, 258, 296, 298, 311, 315, 319, 327, 328, 340, 421, 436, 456, 470, 486, 503, 552, 570 and 591 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or Super piggyBac transposase enzyme may further comprise an amino acid substitution at one or more of positions 46, 119, 125, 177, 180, 185, 187, 200, 207, 209, 226, 235,

240, 241, 243, 296, 298, 311, 315, 319, 327, 328, 340, 421, 436, 456, 470, 485, 503, 552 and 570. In certain embodiments, the amino acid substitution at position 3 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an asparagine (N) for a serine (S). In certain embodiments, the amino acid substitution at position 46 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an alanine (A). In certain embodiments, the amino acid substitution at position 46 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a threonine (T) for an alanine (A). In certain embodiments, the amino acid substitution at position 82 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 103 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a serine (S). In certain embodiments, the amino acid substitution at position 119 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for an arginine (R). In certain embodiments, the amino acid substitution at position 125 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) a cysteine (C). In certain embodiments, the amino acid substitution at position 125 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a cysteine (C). In certain embodiments, the amino acid substitution at position 177 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 177 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a histidine (H) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 185 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 187 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for an alanine (A). In certain embodiments, the amino acid substitution at position 200 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 207 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a valine (V). In certain embodiments, the amino acid substitution at position 209 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a valine (V). In certain embodiments, the amino acid substitution at position 226 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a methionine (M). In certain embodiments, the amino acid substitution at position 235 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a leucine (L). In certain embodiments, the amino acid substitution at position 240 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a valine (V). In certain embodiments, the amino acid substitution at position 241 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 243 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a proline (P). In certain embodiments, the amino acid substitution at position 258 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an asparagine (N). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for a leucine (L). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tyrosine (Y) for a leucine (L). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a leucine (L). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for a methionine (M). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 311 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an



isoleucine (I) for a proline (P). In certain embodiments, the amino acid substitution at position 311 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine for a proline (P). In certain embodiments, the amino acid substitution at position 315 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for an arginine (R). In certain embodiments, the amino acid substitution at position 319 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a threonine (T). In certain embodiments, the amino acid substitution at position 327 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 328 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 340 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a cysteine (C). In certain embodiments, the amino acid substitution at position 340 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a cysteine (C). In certain embodiments, the amino acid substitution at position 421 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a histidine (H) for the aspartic acid (D). In certain embodiments, the amino acid substitution at position 436 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a valine (V). In certain embodiments, the amino acid substitution at position 456 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tyrosine (Y) for a methionine (M). In certain embodiments, the amino acid substitution at position 470 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a leucine (L). In certain embodiments, the amino acid substitution at position 485 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a serine (S). In certain embodiments, the amino acid substitution at position 503 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 503 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a methionine (M). In certain embodiments, the amino acid substitution at position 552 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a valine (V). In certain embodiments, the amino acid substitution at position 570 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a threonine (T) for an alanine (A). In certain embodiments, the amino acid substitution at position 591 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a glutamine (Q). In certain embodiments, the amino acid substitution at position 591 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a glutamine (Q).

(134) In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac transposase enzyme may comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at one or more of positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac transposase enzyme may comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at two, three, four, five, six or more of positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac transposase enzyme may comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, the amino acid substitution at position 103 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a serine (S). In certain embodiments, the amino acid substitution at position 194 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 372 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for an arginine (R). In certain embodiments, the amino acid substitution at position 375 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for a lysine (K). In certain embodiments, the amino acid substitution at position 450 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an asparagine (N) for an aspartic acid (D). In certain embodiments, the amino acid

at substitution at position of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a serine (S). In certain embodiments, the amino acid substitution at position 570 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an asparagine (N). In certain embodiments, the piggyBac transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487. In certain embodiments, including those embodiments wherein the piggyBac transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, the piggyBac transposase enzyme may further comprise an amino acid substitution at positions 372, 375 and 450 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, the piggyBac transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, a substitution of an alanine (A) for an arginine (R) at position 372 of SEQ ID NO: 14487, and a substitution of an alanine (A) for a lysine (K) at position 375 of SEQ ID NO: 14487. In certain embodiments, the piggyBac transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, a substitution of an alanine (A) for an arginine (R) at position 372 of SEQ ID NO: 14487, a substitution of an alanine (A) for a lysine (K) at position 375 of SEQ ID NO: 14487 and a substitution of an asparagine (N) for an aspartic acid (D) at position 450 of SEQ ID NO: 14487.

(135) The sleeping beauty transposon is transposed into the target genome by the Sleeping Beauty transposase that recognizes ITRs, and moves the contents between the ITRs into TA chromosomal sites. In various embodiments, SB transposon-mediated gene transfer, or gene transfer using any of a number of similar transposons, may be used in the compositions and methods of the disclosure.

(136) In certain embodiments, and, in particular, those embodiments wherein the transposon is a Sleeping Beauty transposon, the transposase is a Sleeping Beauty transposase or a hyperactive Sleeping Beauty transposase (SB100X).

(137) In certain embodiments of the methods of the disclosure, the Sleeping Beauty transposase enzyme comprises an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(138) TABLE-US-00056 (SEQ ID NO: 14485) 1 MGKSKEISQD LRKKIVDLHK  
SGSSLGAISK RLKVPRSSVQ TIVRKYKHHG TTQPSYRSGR 61 RRVLSRDER  
TLVRKVQINP RTTAKDLVKM LEETGTKVSI STVKRVLYRH NLKGRSARKK 121  
PLLQNRHKKA RLRFATAHGD KDRTFWRNVL WSDETKIELF GHNDHRYVWR  
KKGEACKPKN 181 TIPTVKHGGG SIMLWGCFAA GGTGALHKID GIMRKENYVD  
ILKQHLKTSV RKLKLGRKWV 241 FQMDNDPKHT SKVVAKWLKD NKVKVLEWPS  
QSPDLNPIEN LWAEKKRVR ARRPTNLTQL 301 HQLCQEEWAK IHPTYCGKLV  
EGYPKRLTQV KQFKGNATKY.

(139) In certain embodiments of the methods of the disclosure, the hyperactive Sleeping Beauty (SB100X) transposase enzyme comprises an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(140) TABLE-US-00057 (SEQ ID NO: 14486) 1 MGKSKEISQD LRKRIVDLHK  
SGSSLGAISK RLAVPRSSVQ TIVRKYKHHG TTQPSYRSGR 61 RRVLSRDER  
TLVRKVQINP RTTAKDLVKM LEETGTKVSI STVKRVLYRH NLKGHSARKK 121  
PLLQNRHKKA RLRFATAHGD KDRTFWRNVL WSDETKIELF GHNDHRYVWR  
KKGEACKPKN 181 TIPTVKHGGG SIMLWGCFAA GGTGALHKID GIMDAVQYVD  
ILKQHLKTSV RKLKLGRKWV 241 FQHDNDPKHT SKVVAKWLKD NKVKVLEWPS  
QSPDLNPIEN LWAEKKRVR ARRPTNLTQL 301 HQLCQEEWAK IHPNYCGKLV  
EGYPKRLTQV KQFKGNATKY.

(141) The Helraiser transposon is transposed by the Helitron transposase. Helitron transposases mobilize the Helraiser transposon, an ancient element from the bat genome that was active about 30 to 36 million years ago. An exemplary Helraiser transposon of the disclosure includes Helibat1, which comprises a nucleic acid sequence comprising:

(142) TABLE-US-00058 (SEQ ID NO: 17006) 1 TCCTATATAA TAAAAGAGAA  
ACATGCAAAT TGACCATCCC TCCGCTACGC TCAAGCCACG 61 CCCACCAGCC

AATACGAAAGT GACTATGCAA ATTACCAACCA CAAAGATGGC AGTTAAATTT 121  
 GCATACGCAG GTGTCAAGCG CCCCAGGAGG CAACGGCGGC CGCGGGCTCC  
 CAGGACCTTC 181 GCTGGCCCCG GGAGGCGAGG CCGGCCGCGC CTAGCCACAC  
 CCGCGGGCTC CCGGGACCTT 241 CGCCAGCAGA GAGCAGAGCG GGAGAGCGGG  
 CGGAGAGCGG GAGGTTTGA GACTTTGGCA 301 GAGCAGGAGG CCGCTGGACA  
 TAGAGCAGAG CGAGAGAGAG GGTGGCTTGG AGGGCGTGGC 361 TCCCTCTGTC  
 ACCCCAGCTT CCTCATCACA GCTGTGAAA CTGACAGCAG GGAGGAGGAA 421  
 GTCCCACCCC CACAGAATCA GCCAGAATCA GCCGTTGGTC AGACAGCTCT  
 CAGCGGCCTG 481 ACAGCCAGGA CTCTCATTCA CCTGCATCTC AGACCGTGAC  
 AGTAGAGAGG TGGGACTATG 541 TCTAAAGAAC AACTGTTGAT ACAACGTAGC  
 TCTGCAGCCG AAAGATGCCG GCGTTATCGA 601 CAGAAAATGT CTGCAGAGCA  
 ACGTGCGTCT GATCTTGAAA GAAGGCGGCG CCTGCAACAG 661 AATGTATCTG  
 AAGAGCAGCT ACTGGAAAAA CGTCGCTCTG AAGCCGAAAA ACAGCGGCGT 721  
 CATCGACAGA AAATGTCTAA AGACCAACGT GCCTTTGAAG TTGAAAGAAG  
 GCGGTGGCGA 781 CGACAGAATA TGTCTAGAGA ACAGTCATCA ACAAGTACTA  
 CCAATACCGG TAGGAAGTGC 841 CTTCTCAGCA AAAATGGAGT ACATGAGGAT  
 GCAATTCTCG AACATAGTTG TGGTGGAATG 901 ACTGTTCGAT GTGAATTTTG  
 CCTATCACTA AATTTCTCTG ATGAAAAACC ATCCGATGGG 961 AAATTTACTC  
 GATGTTGTAG CAAAGGGAAA GTCTGTCCAA ATGATATACA TTTTCCAGAT 1021  
 TACCCGGCAT ATTTAAAAAG ATTAATGACA AACGAAGATT CTGACAGTAA  
 AAATTTTCATG 1081 GAAAATATTC GTTCCATAAA TAGTTCTTTT GCTTTTGCTT  
 CCATGGGTGC AAATATTGCA 1141 TCGCCATCAG GATATGGGCC ATACTGTTTT  
 AGAATACACG GACAAGTTTA TCACCGTACT 1201 GGAACCTTAC ATCCTTCGGA  
 TGGTGTTTCT CGGAAGTTTG CTCAACTCTA TATTTTGGAT 1261 ACAGCCGAAG  
 CTACAAGTAA AAGATTAGCA ATGCCAGAAA ACCAGGGCTG CTCAGAAAGA 1321  
 CTCATGATCA ACATCAACAA CCTCATGCAT GAAATAAATG AATTAACAAA  
 ATCGTACAAG 1381 ATGCTACATG AGGTAGAAAA GGAAGCCCAA TCTGAAGCAG  
 CAGCAAAAGG TATTGCTCCC 1441 ACAGAAGTAA CAATGGCGAT TAAATACGAT  
 CGTAACAGTG ACCCAGGTAG ATATAATTCT 1501 CCCC GTGTAA CCGAGGTTGC  
 TGTCATATTC AGAAACGAAG ATGGAGAACC TCCTTTTGAA 1561 AGGGACTTGC  
 TCATTCATTG TAAACCAGAT CCCAATAATC CAAATGCCAC TAAATGAAA 1621  
 CAAATCAGTA TCCTGTTTCC TACATTAGAT GCAATGACAT ATCCTATTCT  
 TTTTCCACAT 1681 GGTGAAAAAG GCTGGGGAAC AGATATTGCA TTAAGACTCA  
 GAGACAACAG TGTAATCGAC 1741 AATAATACTA GACAAAATGT AAGGACACGA  
 GTCACACAAA TGCAGTATTA TGGATTTTCAT 1801 CTCTCTGTGC GGGACACGTT  
 CAATCCTATT TTAAATGCAG GAAAATTAAC TCAACAGTTT 1861 ATTGTGGATT  
 CATATTCAA AATGGAGGCC AATCGGATAA ATTTTCATCA AGCAAACCAA 1921  
 TCTAAGTTGA GAGTTGAAAA ATATAGTGGT TTGATGGATT ATCTCAAATC  
 TAGATCTGAA 1981 AATGACAATG TGCCGATTGG TAAAATGATA ATACTTCCAT  
 CATCTTTTGA GGGTAGTCCC 2041 AGAAATATGC AGCAGCGATA TCAGGATGCT  
 ATGGCAATTG TAACGAAGTA TGGCAAGCCC 2101 GATTTATTCA TAACCATGAC  
 ATGCAACCCC AAATGGGCAG ATATTACAAA CAATTTACAA 2161 CGCTGGCAAA  
 AAGTTGAAAA CAGACCTGAC TTGGTAGCCA GAGTTTTTAA TATTAAGCTG 2221  
 AATGCTCTTT TAAATGATAT ATGTAAATTC CATTATTG GCAAAGTAAT  
 AGCTAAAATT 2281 CATGTCATTG AATTTAGAAA ACGCGGACTG CCTCACGCTC  
 ACATATTATT GATATTAGAT 2341 AGTGAGTCCA AATTACGTTT AGAAGATGAC  
 ATTGACCGTA TAGTTAAGGC AGAAATTCCA 2401 GATGAAGACC AGTGTCTCTG  
 ACTTTTTCAA ATTGTAAAAT CAAATATGGT ACATGGACCA 2461 TGTGGAATAC  
 AAAATCCAAA TAGTCCATGT ATGGAAAATG GAAAATGTTC AAAGGGATAT 2521  
 CCAAAAGAAT TTCAAATATGC GACCATTGGA AATATTGATG GATATCCCAA  
 ATACAAACGA 2581 AGATCTGGTA GCACCATGTC TATTGGAAAT AAAGTTGTCTG  
 ATAACACTTG GATTGTCCCT 2641 TATAACCCGT ATTTGTGCCT TAAATATAAC

TGTACATATA 2701 ATGTTAAGTG TCAAAATATT  
 ATTTAAATAC ATCTATAAAG GGCACGATTG TGCAAATATT 2761 CAAATTTCTG  
 AAAAAAATAT TATCAATCAT GACGAAGTAC AGGACTTCAT TGACTCCAGG 2821  
 TATGTGAGCG CTCCTGAGGC TGTTTGGAGA CTTTTTGCAA TGCGAATGCA  
 TGACCAATCT 2881 CATGCAATCA CAAGATTAGC TATTCATTG CCAAATGATC  
 AGAATTTGTA TTTTCATACC 2941 GATGATTTTG CTGAAGTTTT AGATAGGGCT  
 AAAAGGCATA ACTCGACTTT GATGGCTTGG 3001 TTCTTATTGA ATAGAGAAGA  
 TTCTGATGCA CGTAATTATT ATTATTGGGA GATTCCACAG 3061 CATTATGTGT  
 TTAATAATTC TTTGTGGACA AAACGCCGAA AGGGTGGGAA TAAAGTATTA 3121  
 GGTAGACTGT TCACTGTGAG CTTTAGAGAA CCAGAACGAT ATTACCTTAG  
 ACTTTTGCTT 3181 CTGCATGTAA AAGGTGCGAT AAGTTTTGAG GATCTGCGAA  
 CTGTAGGAGG TGTAACCTTAT 3241 GATACATTTC ATGAAGCTGC TAAACACCGA  
 GGATTATTAC TTGATGACAC TATCTGGAAA 3301 GATACGATTG ACGATGCAAT  
 CATCCTTAAT ATGCCCAAAC AACTACGGCA ACTTTTTGCA 3361 TATATATGTG  
 TGTTTGGATG TCCTTCTGCT GCAGACAAAT TATGGGATGA GAATAAATCT 3421  
 CATTTTATTG AAGATTTCTG TTGGAAATTA CACCGAAGAG AAGGTGCCTG  
 TGTGAACTGT 3481 GAAATGCATG CCCTTAACGA AATTCAGGAG GTATTCACAT  
 TGCATGGAAT GAAATGTTCA 3541 CATTTCAAAC TTCCGGACTA TCCTTTATTA  
 ATGAATGCAA ATACATGTGA TCAATTGTAC 3601 GAGCAACAAC AGGCAGAGGT  
 TTTGATAAAT TCTCTGAATG ATGAACAGTT GGCAGCCTTT 3661 CAGACTATAA  
 CTTCAGCCAT CGAAGATCAA ACTGTACACC CCAAATGCTT TTTCTTGGAT 3721  
 GGTCCAGGTG GTAGTGGAAA AACATATCTG TATAAAGTTT TAACACATTA  
 TATTAGAGGT 3781 CGTGGTGGTA CTGTTTTACC CACAGCATCT ACAGGAATTG  
 CTGCAAATTT ACTTCTTGGT 3841 GGAAGAACCT TTCATTCCCA ATATAAATTA  
 CCAATTCCAT TAAATGAAAC TTCAATTTCT 3901 AGACTCGATA TAAAGAGTGA  
 AGTTGCTAAA ACCATTAAAA AGGCCCAACT TCTCATTATT 3961 GATGAATGCA  
 CCATGGCATC CAGTCATGCT ATAAACGCCA TAGATAGATT ACTAAGAGAA 4021  
 ATTATGAATT TGAATGTTGC ATTTGGTGGG AAAGTTCTCC TTCTCGGAGG  
 GGATTTTCGA 4081 CAATGTCTCA GTATTGTACC ACATGCTATG CGATCGGCCA  
 TAGTACAAAC GAGTTTAAAG 4141 TACTGTAATG TTTGGGGATG TTTCAGAAAG  
 TTGTCTCTTA AAACAAATAT GAGATCAGAG 4201 GATTCTGCTT ATAGTGAATG  
 GTTAGTAAAA CTTGGAGATG GCAAACCTGA TAGCAGTTTT 4261 CATTTAGGAA  
 TGGATATTAT TGAAATCCCC CATGAAATGA TTTGTAACGG ATCTATTATT 4321  
 GAAGCTACCT TTGGAAATAG TATATCTATA GATAATATTA AAAATATATC  
 TAAACGTGCA 4381 ATTCTTTGTC CAAAAAATGA GCATGTTCAA AAATTAAATG  
 AAGAAATTTT GGATATACTT 4441 GATGGAGATT TTCACACATA TTTGAGTGAT  
 GATTCCATTG ATTCAACAGA TGATGCTGAA 4501 AAGGAAAATT TTCCCATCGA  
 ATTTCTTAAT AGTATTACTC CTTCGGGAAT GCCGTGTCAT 4561 AAATTAAAAAT  
 TGAAAGTGGG TGCAATCATC ATGCTATTGA GAAATCTTAA TAGTAAATGG 4621  
 GGTCTTTGTA ATGGTACTAG ATTTATTATC AAAAGATTAC GACCTAACAT  
 TATCGAAGCT 4681 GAAGTATTAA CAGGATCTGC AGAGGGAGAG GTTGTTCTGA  
 TTCCAAGAAT TGATTTGTCC 4741 CCATCTGACA CTGGCCTCCC ATTTAAATTA  
 ATTCGAAGAC AGTTTCCCGT GATGCCAGCA 4801 TTTGCGATGA CTATTAATAA  
 ATCACAAGGA CAAACTCTAG ACAGAGTAGG AATATTCCTA 4861 CCTGAACCCG  
 TTTTCGCACA TGGTCAGTTA TATGTTGCTT TCTCTCGAGT TCGAAGAGCA 4921  
 TGTGACGTTA AAGTTAAAGT TGTAATAACT TCATCACAAG GGAAATTAGT  
 CAAGCACTCT 4981 GAAAGTGTTT TTAATCACTT 5041 TATCAGTCAT TGTTTGCATC AATGTTGTTT  
 TTATATCATG TTTTTGTTGT TTTTATATCA 5101 TGTCTTTGTT GTTGTTATAT  
 CATGTTGTTA TTGTTTATTT ATTAATAAAT TTATGTATTA 5161 TTTTCATATA  
 CATTTTACTC ATTTCTTTTC ATCTCTCACA CTTCTATTAT AGAGAAAGGG 5221  
 CAAATAGCAA TATTAAAATA TTTCTCTAA TTAATTCCCT TTCAATGTGC

ACGATTTCG 5281 TGCACCGGGC CACTAG.

(143) Unlike other transposases, the Helitron transposase does not contain an RNase-H like catalytic domain, but instead comprises a RepHel motif made up of a replication initiator domain (Rep) and a DNA helicase domain. The Rep domain is a nuclease domain of the HUH superfamily of nucleases.

(144) An exemplary Helitron transposase of the disclosure comprises an amino acid sequence comprising:

(145) TABLE-US-00059 (SEQ ID NO: 14501) 1 MSKEQLLIQR SSAAERCRRY  
RQKMSAEQRA SDLERRRRLQ QNVSEEQLLE KRRSEAEKQR 61 RHRQKMSKDQ  
RAFEVERRRW RRQNMSREQS STSTNTGRN CLLSKNGVHE DAILEHSCGG 121  
MTVRCEFCLS LNFSDKPSD GKFTRCCKSG KVCPCNDIHF DYPAYLKRLM  
TNEDSDSKNF 181 MENIRSINSS FAFASMGANI ASPSGYGPYC FRIHGQVYHR  
TGTLHPSDGV SRKFAQLYIL 241 DTAEATSKRL AMPENQGCSE RLMINNNLM  
HEINELTKSY KMLHEVEKEA QSEAAAKGIA 301 PTEVIMAIKY DRNSDPGRYN  
SPRVTEVAVI FRNEDGEPPF ERDLLIHCKP DPNPNATKM 361 KQISILFPTL  
DAMTYPILFP HGEKGWGTDI ALRLRDNSVI DNNTRQNVRT RVTQMYYGF 421  
HLSVRDTFNP ILNAGKLTQQ FIVDSYSKME ANRINFIKAN QSKLRVEKYS  
GLMDYLKSRS 481 ENDNVPIGKM IILPSSFEGS PRNMQQRYQD AMAIVTKYK  
PDLFITMTCN PKWADITNNL 541 QRWQKVENRP DLVARVFNIK LNALLNDICK  
FHLFGKVIK IHVIEFQKRG LPHAHILLIL 601 DSESKLRSED DIDRIVKAEI  
PDEDQCPRLF QIVKSNMVHG PCGIQNPNSP CMENGKCSKG 661 YPKEFQNATI  
GNIDGYPKYK RRSSTMSIG NKVVDNTWIV PYNPYLCLKY NCHINVEVCA 721  
SIKSVKYLK YIYKGHDCAN IQISEKNIIN HDEVQDFIDS RYVSAPEAVW  
RLFAMRMHDQ 781 SHAITRLAIH LPNDQNLVYH TDDFAEVLDR AKRHNSTLMA  
WFLNREDSD ARNYYYWEIP 841 QHYVFNNSLW TKRRKGGNKV LGRLFTVSFR  
EPERYYLRL LLHVKGAI SF EDLRTVGGVT 901 YDTFHEAAKH RGLLLDDTIW  
KDTIDDAIIL NMPKQLRQLF AYICVFGCPS AADKLWDENK 961 SHFIEDFCWK  
LHRREGACVN CEMHALNEIQ EVFTLHGMKC SHFKLPDYPL LMNANTCDQL 1021  
YEQQQAEVLI NSLNDEQLAA FQTITSAIED QTVHPKCFFL DPGGSGSKTY  
LYKVLTHYIR 1081 GRGGTVLPTA STGIAANLLL GGRTFHSQYK LPIPLNETSI  
SRLDIKSEVA KTIKKAQLLI 1141 IDECTMASSH AINAIDRLR EIMNLNVAFG  
GKVLLLGDF RQCLSIVPHA MRSIVQTS 1201 KYCNVWGCFR KLSLKTNMRS  
EDSAYSEWLK LGDGKLDSS FHLGMDIIEI PHEMICNGSI 1261 IEATFGNSIS  
IDNIKNISKR AILCPKNEHV QKLNEEILDI LDGDFHTYLS DDSIDSTDDA 1321  
EKENFPIEFL NSITPSGMPC HKLKLKVGAI IMLLRNLNSK WGLCNGTRFI IKRLRPNIIE  
1381 AEVLTGSAEG EVVLIPRIDL SPSDTGLPFK LIRRQFPVMP AFAMTINKSQ  
GQTLDRVGIF 1441 LPEPVFAHGQ LYVAFSRVRR ACDVKVKVNV TSSQGKLVKH  
SESVFTLNVV YREILE.

(146) In Helitron transpositions, a hairpin close to the 3' end of the transposon functions as a terminator. However, this hairpin can be bypassed by the transposase, resulting in the transduction of flanking sequences. In addition, Helraiser transposition generates covalently closed circular intermediates. Furthermore, Helitron transpositions can lack target site duplications. In the Helraiser sequence, the transposase is flanked by left and right terminal sequences termed LTS (5' terminal sequence) and RTS (3' terminal sequence). These sequences terminate with a conserved 5'-TC/CTAG-3' motif. A 19 bp palindromic sequence with the potential to form the hairpin termination structure is located 11 nucleotides upstream of the RTS and consists of the sequence

GTGCACGAATTCGTGCACCGGGCCACTAG (SEQ ID NO: 14500).

(147) Tol2 transposons may be isolated or derived from the genome of the medaka fish, and may be similar to transposons of the hAT family. Exemplary Tol2 transposons of the disclosure are encoded by a sequence comprising about 4.7 kilobases and contain a gene encoding the Tol2 transposase, which contains four exons. An exemplary Tol2 transposase of the disclosure comprises an amino acid sequence comprising the following:

(148) TABLE-US-00060 (SEQ ID NO: 14502) 1 MEEVCDSSAA ASSTVQNQPQ

DQEHFYPYLR EFFSGLVNLK DSFKMKCVLC LPLNKEISAF 61 KSSPSNLRKH  
IERMHPNYLK NYSKLTAQKR KIGTSTHASS SKQLKVDSVF PVKHVSPVTV 121  
NKAILRYIIQ GLHPFSTVDL PSFKELISTL QPGISVITRP TLRSKIAEAA LIMKQKVTA  
181 MSEVEWIATT TDCWTARRKS FIGVTAHWIN PGSLEHSA LACKRLMGSH  
TFEVLASAMN 241 DIHSEYEIRD KVVCTTTDSG SNFMKAFRVF GVENNDIETE  
ARRCESDDTD SEGCGEGSDG 301 VEFQDASRVL DQDDGFEFQL PKHQKCACHL  
LNLVSSVDAQ KALSNEHYKK LYRSVFGKCQ 361 ALWNKSSRSA LAAEAVESES  
RLQLLRPNQT RWNSTFMAVD RILQICKEAG EGALRNICTS 421 LEVPMFNPAE  
MLFLTEWANT MRPVAKVLDI LQAETNTQLG WLLPSVHQLS LKLQRLHHSL 481  
RYCDPLVDAL QQGIQTRFKH MFEDPEITAA AILLPKFRTS WTNDETIHKR  
GMDYIRVHLE 541 PLDHKKELAN SSSDDEDEFFA SLKPTTHEAS KELDGYLACV  
SDTRESLLTF PAICSLSIKT 601 NTPLPASAAC ERLFSTAGLL FSPKRARLDT  
NNFENQLLLK LNLRFYNFE.

(149) An exemplary Tol2 transposon of the disclosure, including inverted repeats, subterminal sequences and the Tol2 transposase, is encoded by a nucleic acid sequence comprising the following:

(150) TABLE-US-00061 (SEQ ID NO: 17007) 1 CAGAGGTGTA AAGTACTTGA  
GTAATTTTAC TTGATTACTG TACTTAAGTA TTATTTTGG 61 GGATTTTAC  
TTTACTTGAG TACAATTAAA AATCAATACT TTTACTTTTA CTTAATTACA 121  
TTTTTTTAGA AAAAAAAGTA CTTTTTACTC CTTACAATTT TATTACAGT  
CAAAAAGTAC 181 TTATTTTTTTG GAGATCACTT CATTCTATTT TCCCTTGCTA  
TTACCAAACC AATTGAATTG 241 CGCTGATGCC CAGTTTAATT TAAATGTTAT  
TTATTCTGCC TATGAAAATC GTTTTCACAT 301 TATATGAAAT TGGTCAGACA  
TGTTCAATTGG TCCTTTGGAA GTGACGTCAT GTCACATCTA 361 TTACCACAAT  
GCACAGCACC TTGACCTGGA AATTAGGGAA ATTATAACAG TCAATCAGTG 421  
GAAGAAAATG GAGGAAGTAT GTGATTCATC AGCAGCTGCG AGCAGCACAG  
TCCAAAATCA 481 GCCACAGGAT CAAGAGCACC CGTGGCCGTA TCTTCGCGAA  
TTCTTTTCTT TAAGTGGTGT 541 AAATAAAGAT TCATTCAAGA TGAAATGTGT  
CCTCTGTCTC CCGCTTAATA AAGAAATATC 601 GGCCTTCAAA AGTTCGCCAT  
CAAACCTAAG GAAGCATATT GAGGTAAGTA CATTAAAGTAT 661 TTTGTTTTAC  
TGATAGTTTT TTTTTTTTTT **TTTTTTTTTT** TTTTGGGTG TGCATGTTTT 721  
GACGTTGATG GCGCGCCTTT TATATGTGTA GTAGGCCTAT TTCACTAAT  
GCATGCGATT 781 GACAATATAA GGCTCACGTA ATAAAATGCT AAAATGCATT  
TGTAATTGGT AACGTTAGGT 841 CCACGGGAAA TTTGGCGCCT ATTGCAGCTT  
TGAATAATCA TTATCATTCC GTGCTCTCAT 901 TGTGTTTGAA TTCATGCAAA  
ACACAAGAAA ACCAAGCGAG AAATTTTTTT CCAAACATGT 961 TGTATTGTCA  
AAACGGTAAC ACTTTACAAT GAGGTTGATT AGTTCATGTA TTAATAACA 1021  
TTAAATAACC ATGAGCAATA CATTTGTTAC TGTATCTGTT AATCTTTGTT  
AACGTTAGTT 1081 AATAGAAATA CAGATGTTCA TTGTTTGTTT ATGTTAGTTC  
ACAGTGCATT AACTAATGTT 1141 AACAAGATAT AAAGTATTAG TAAATGTTGA  
AATTAACATG TATACGTGCA GTTCATTATT 1201 AGTTCATGTT AACTAATGTA  
GTAACTAAC GAACCTTATT GTAAAAGTGT TACCATCAAA 1261 ACTAATGTAA  
TGAAATCAAT TCACCCTGTC ATGTCAGCCT TACAGTCCTG TGTTTTTGTC 1321  
AATATAATCA GAAATAAAAT TAATGTTTGA TTGTCACTAA ATGCTACTGT  
ATTTCTAAAA 1381 TCAACAAGTA TTAAACATTA TAAAGTGTGC AATTGGCTGC  
AAATGTCAGT TTTATTAAAG 1441 GGTTAGTTCA CCCAAAATG AAAATAATGT  
CATTAAATGAC TCGCCCTCAT GTCGTTCCAA 1501 GCCCGTAAGA CCTCCGTTCA  
TCTTCAGAAC ACAGTTTAAG ATATTTTAGA TTTAGTCCGA 1561 GAGCTTTCTG  
TGCCTCCATT GAGAATGTAT GTACGGTATA CTGTCCATGT CCAGAAAGGT 1621  
AATAAAAACA TCAAAGTAGT CCATGTGACA TCAGTGGGTT AGTTAGAATT  
TTTTGAAGCA 1681 TCGAATACAT TTTGGTCCAA AAATAACAAA ACCTACGACT  
TTATTCGGCA TTGTATTCTC 1741 TTCCGGGTCT GTTGTCATC CGCGTTCACG  
ACTTCGCAGT GACGCTACAA TGCTGAATAA 1801 AGTCGTAGGT TTTGTTATTT

TTGACCAAA ATGATTTTC GATGCTTCAA ATAATCTTAC 1861 CTAACCCACT  
 GATGTCACAT GGACTACTTT GATGTTTTTA TTACCTTTCT GGACATGGAC 1921  
 AGTATACCGT ACATACATTT TCAGTGGAGG GACAGAAAGC TCTCGGACTA  
 AATCTAAAAT 1981 ATCTTAAACT GTGTTCCGAA GATGAACGGA GGTGTTACGG  
 GCTTGGAACG ACATGAGGGT 2041 GAGTCATTAA TGACATCTTT TCATTTTTTG  
 GTGAACTAAC CCTTTAATGC TGTAATCAGA 2101 GAGTGTATGT GTAATTGTTA  
 CATTTATTGC ATACAATATA AATATTTATT TGTTGTTTTT 2161 ACAGAGAATG  
 CACCCAAATT ACCTCAAAAA CTACTCTAAA TTGACAGCAC AGAAGAGAAA 2221  
 GATCGGGACC TCCACCCATG CTTCCAGCAG TAAGCAACTG AAAGTTGACT  
 CAGTTTTCCC 2281 AGTCAAACAT GTGTCTCCAG TCACTGTGAA CAAAGCTATA  
 TTAAGGTACA TCATTCAAGG 2341 ACTTCATCCT TTCAGCACTG TTGATCTGCC  
 ATCATTTAAA GAGCTGATTA GTACACTGCA 2401 GCCTGGCATT TCTGTCATTA  
 CAAGGCCTAC TTTACGCTCC AAGATAGCTG AAGCTGCTCT 2461 GATCATGAAA  
 CAGAAAGTGA CTGCTGCCAT GAGTGAAGTT GAATGGATTG CAACCACAAC 2521  
 GGATTGTTGG ACTGCACGTA GAAAGTCATT CATTGGTGTA ACTGCTCACT  
 GGATCAACCC 2581 TGGAAGTCTT GAAAGACATT CCGCTGCACT TGCCTGCAAA  
 AGATTAATGG GCTCTCATACT 2641 TTTTGAGGTA CTGGCCAGTG CCATGAATGA  
 TATCCACTCA GAGTATGAAA TACGTGACAA 2701 GGTTGTTTGC ACAACCACAG  
 ACAGTGGTTC CAACTTTATG AAGGCTTTCA GAGTTTTTTGG 2761 TGTGGAAAAC  
 AATGATATCG AGACTGAGGC AAGAAGGTGT GAAAGTGATG ACACTGATTC 2821  
 TGAAGGCTGT GGTGAGGGAA GTGATGGTGT GGAATTCCAA GATGCCTCAC  
 GAGTCCTGGA 2881 CCAAGACGAT GGCTTCGAAT TCCAGCTACC AAAACATCAA  
 AAGTGTGCCT GTCACTTACT 2941 TAACCTAGTC TCAAGCGTTG ATGCCCAAAA  
 AGCTCTCTCA AATGAACACT ACAAGAACT 3001 CTACAGATCT GTCTTTGGCA  
 AATGCCAAGC TTTATGGAAT AAAAGCAGCC GATCGGCTCT 3061 AGCAGCTGAA  
 GCTGTTGAAT CAGAAAGCCG GCTTCAGCTT TTAAGGCCAA ACCAAACGCG 3121  
 GTGGAATTCA ACTTTTATGG CTGTTGACAG AATTCTTCAA ATTTGCAAAG  
 AAGCAGGAGA 3181 AGGCGCACTT CGGAATATAT GCACCTCTCT TGAGGTTCCA  
 ATGTAAGTGT TTTTCCCCTC 3241 TATCGATGTA AACAAATGTG GGTTGTTTTT  
 GTTTAATACT CTTTGATTAT GCTGATTCT 3301 CCTGTAGGTT TAATCCAGCA  
 GAAATGCTGT TCTTGACAGA GTGGGCCAAC ACAATGCGTC 3361 CAGTTGCAAA  
 AGTACTCGAC ATCTTGCAAG CGGAAACGAA TACACAGCTG GGGTGGCTGC 3421  
 TGCCTAGTGT CCATCAGTTA AGCTTGAAAC TTCAGCGACT CCACCATTCT  
 CTCAGGTACT 3481 GTGACCCACT TGTGGATGCC CTACAACAAG GAATCCAAAC  
 ACGATTCAAG CATATGTTTG 3541 AAGATCCTGA GATCATAGCA GCTGCCATCC  
 TTCTCCCTAA ATTTCCGACC TCTTGACAA 3601 ATGATGAAAC CATCATAAAA  
 CGAGGTAAAT GAATGCAAGC AACATACACT TGACGAATTC 3661 TAATCTGGGC  
 AACCTTTGAG CCATACCAAA ATTATTCTTT TATTTATTTA TTTTGGCACT 3721  
 TTTTAGGAAT GTTATATCCC ATCTTTGGCT GTGATCTCAA TATGAATATT  
 GATGTAAAGT 3781 ATTCTTGACAG CAGGTTGTAG TTATCCCTCA GTGTTTCTTG  
 AAACCAAAT CATATGTATC 3841 ATATGTGGTT TGGAATGCA GTTAGATTTT  
 ATGCTAAAAT AAGGGATTTG CATGATTTTA 3901 GATGTAGATG ACTGCACGTA  
 AATGTAGTTA ATGACAAAAT CCATAAAATT TGTTCCAGT 3961 CAGAAGCCCC  
 TCAACCAAAC TTTTCTTTGT GTCTGCTCAC TGTGCTTGTA GGCATGGACT 4021  
 ACATCAGAGT GCATCTGGAG CCTTTGGACC ACAAGAAGGA ATTGGCCAAC  
 AGTTCATCTG 4081 ATGATGAAGA TTTTTTCGCT TCTTTGAAAC CGACAACACA  
 TGAAGCCAGC AAAGAGTTGG 4141 ATGGATATCT GGCCTGTGTT TCAGACACCA  
 GGGAGTCTCT GCTCACGTTT CCTGCTATTT 4201 GCAGCCTCTC TATCAAGACT  
 AATACACCTC TTCCCGCATC GGCTGCCTGT GAGAGGCTTT 4261 TCAGCACTGC  
 AGGATTGCTT TTCAGCCCCA AAAGAGCTAG GCTTGACACT AACAATTTTG 4321  
 AGAATCAGCT TCTACTGAAG TTAAATCTGA GGTTTTACAA CTTTGAGTAG  
 CGTGTACTGG 4381 CATTAGATTG TCTGTCTTAT AGTTTGATAA TTAAATACAA

ACAGTCTTAA AGCAGGATAA 4441 AACCTTGTAT GCATTTCATT TAATGTTTTT  
TGAGATTAAA AGCTTAAACA AGAATCTCTA 4501 GTTTTCTTTC TTGCTTTTAC  
TTTTACTTCC TTAATACTCA AGTACAATT TAATGGAGTA 4561 CTTTTTTACT  
TTTACTCAAG TAAGATTCTA GCCAGATACT TTTACTTTTA ATTGAGTAAA 4621  
ATTTTCCCTA AGTACTTGTA CTTTCACTTG AGTAAAATT TTGAGTACTT  
TTTACACCTC 4681 TG.

(151) Exemplary transposon/transposase systems of the disclosure include, but are not limited to, piggyBac and piggyBac-like transposons and transposases.

(152) PiggyBac and piggyBac-like transposases recognizes transposon-specific inverted terminal repeat sequences (ITRs) on the ends of the transposon, and moves the contents between the ITRs into TTAA or TTAT chromosomal sites. The piggyBac or piggyBac-like transposon system has no payload limit for the genes of interest that can be included between the ITRs.

(153) In certain embodiments, and, in particular, those embodiments wherein the transposon is a piggyBac transposon, the transposase is a piggyBac, Super piggyBac (SPB) transposase. In certain embodiments, and, in particular, those embodiments wherein the transposase is a piggyBac, Super piggyBac (SPB), the sequence encoding the transposase is an mRNA sequence.

(154) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme.

(155) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or a piggyBac-like transposase enzyme. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(156) TABLE-US-00062 (SEQ ID NO: 14487) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEI SDHVSEDDVQ SDTEEAFIDE VHEVQPTSSG 61 SEILDEQNVI  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTGATFRD TNEDEIYAFF  
181 GILVMTAVRK DNHMSTDDL F DRLSMVYVS VMSRDRFDFL IRCLRMDDKS  
IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF  
RMYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN GVPLGEYYVK  
ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT VRSNKREIPE  
VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421 DEDASINEST  
GKPQMVMYYN QTKGGVDTLD QMCSVMTCR KTNRWPMALL YGMINIACIN 481  
SFIYSHNVS SKGEKVQSRK KFMRNLYMSL TSSFMRKRLE APTLKRYLRD  
NISNILPNEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA NASCKKCKKV  
ICREHNIDMC QSCF.

(157) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at one or more of positions 30, 165, 282, or 538 of the sequence:

(158) TABLE-US-00063 (SEQ ID NO: 14487) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEI SDHVSEDDVQ SDTEEAFIDE VHEVQPTSSG 61 SEILDEQNVI  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTGATFRD TNEDEIYAFF  
181 GILVMTAVRK DNHMSTDDL F DRLSMVYVS VMSRDRFDFL IRCLRMDDKS  
IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF  
RMYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN GVPLGEYYVK  
ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT VRSNKREIPE  
VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421 DEDASINEST  
GKPQMVMYYN QTKGGVDTLD QMCSVMTCR KTNRWPMALL YGMINIACIN 481  
SFIYSHNVS SKGEKVQSRK KFMRNLYMSL TSSFMRKRLE APTLKRYLRD  
NISNILPNEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA NASCKKCKKV  
ICREHNIDMC QSCF.



(159) In certain embodiments, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at two or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at three or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at each of the following positions 30, 165, 282, and 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the amino acid substitution at position 30 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 165 of the sequence of SEQ ID NO: 14487 is a substitution of a serine (S) for a glycine (G). In certain embodiments, the amino acid substitution at position 282 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 538 of the sequence of SEQ ID NO: 14487 is a substitution of a lysine (K) for an asparagine (N).

(160) In certain embodiments of the methods of the disclosure, the transposase enzyme is a Super piggyBac (SPB) or piggyBac-like transposase enzyme. In certain embodiments, the Super piggyBac (SPB) or piggyBac-like transposase enzyme of the disclosure may comprise or consist of the amino acid sequence of the sequence of SEQ ID NO: 14487 wherein the amino acid substitution at position 30 is a substitution of a valine (V) for an isoleucine (I), the amino acid substitution at position 165 is a substitution of a serine (S) for a glycine (G), the amino acid substitution at position 282 is a substitution of a valine (V) for a methionine (M), and the amino acid substitution at position 538 is a substitution of a lysine (K) for an asparagine (N). In certain embodiments, the Super piggyBac (SPB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(161) TABLE-US-00064 (SEQ ID NO: 14484) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEV SDHVSEDDVQ SDTEEFIDE VHEVQPTSSG 61 SEILDEQNVI  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTSATFRD TNEDEIYAFF  
181 GILVMTAVRK DNHMSTDDL FDRSLSMVYVS VMSRDRFDFL IRCLRMDDKS  
IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF  
RVYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN GVPLGEYYVK  
ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT VRSNKREIPE  
VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421 DEDASINEST  
GKPQMVMYYN QTKGGVDTL DQMCSVMTC SR KTNRWPMALL YGMINIACIN 481  
SFIIYSHNVS SKGEKVQSRK KFMRNLYMSL TSSFMRKRLE APTLKRYLRD  
NISNILPKEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA NASCKKCKKV  
ICREHNIDMC QSCF.

(162) In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac, Super piggyBac or piggyBac-like transposase enzyme may further comprise an amino acid substitution at one or more of positions 3, 46, 82, 103, 119, 125, 177, 180, 185, 187, 200, 207, 209, 226, 235, 240, 241, 243, 258, 296, 298, 311, 315, 319, 327, 328, 340, 421, 436, 456, 470, 486, 503, 552, 570 and 591 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac, Super piggyBac or piggyBac-like transposase enzyme may further comprise an amino acid substitution at one or more of positions 46, 119, 125, 177, 180, 185, 187, 200, 207, 209, 226, 235, 240, 241, 243, 296, 298, 311, 315, 319, 327, 328, 340, 421, 436, 456, 470, 485, 503, 552 and 570. In certain embodiments, the amino acid substitution at position 3 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an asparagine (N) for a serine (S). In certain embodiments, the amino acid substitution at position 46 of SEQ ID NO: 14487 or SEQ ID NO: 14484

is a substitution of a serine (S) for an alanine (A). In certain embodiments, the amino acid substitution at position 46 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a threonine (T) for an alanine (A). In certain embodiments, the amino acid substitution at position 82 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 103 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a serine (S). In certain embodiments, the amino acid substitution at position 119 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for an arginine (R). In certain embodiments, the amino acid substitution at position 125 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) a cysteine (C). In certain embodiments, the amino acid substitution at position 125 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a cysteine (C). In certain embodiments, the amino acid substitution at position 177 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 177 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a histidine (H) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 185 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 187 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for an alanine (A). In certain embodiments, the amino acid substitution at position 200 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 207 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a valine (V). In certain embodiments, the amino acid substitution at position 209 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a valine (V). In certain embodiments, the amino acid substitution at position 226 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a methionine (M). In certain embodiments, the amino acid substitution at position 235 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a leucine (L). In certain embodiments, the amino acid substitution at position 240 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a valine (V). In certain embodiments, the amino acid substitution at position 241 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 243 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a proline (P). In certain embodiments, the amino acid substitution at position 258 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an asparagine (N). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for a leucine (L). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tyrosine (Y) for a leucine (L). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a leucine (L). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for a methionine (M). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 311 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a proline (P). In certain embodiments, the amino acid substitution at position 311 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine for a proline (P). In certain embodiments, the amino acid substitution at position 315 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for an arginine (R). In certain embodiments, the

amino acid substitution at position 319 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a threonine (T). In certain embodiments, the amino acid substitution at position 327 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 328 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 340 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a cysteine (C). In certain embodiments, the amino acid substitution at position 340 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a cysteine (C). In certain embodiments, the amino acid substitution at position 421 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a histidine (H) for the aspartic acid (D). In certain embodiments, the amino acid substitution at position 436 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a valine (V). In certain embodiments, the amino acid substitution at position 456 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tyrosine (Y) for a methionine (M). In certain embodiments, the amino acid substitution at position 470 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a leucine (L). In certain embodiments, the amino acid substitution at position 485 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a serine (S). In certain embodiments, the amino acid substitution at position 503 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 503 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a methionine (M). In certain embodiments, the amino acid substitution at position 552 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a valine (V). In certain embodiments, the amino acid substitution at position 570 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a threonine (T) for an alanine (A). In certain embodiments, the amino acid substitution at position 591 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a glutamine (Q). In certain embodiments, the amino acid substitution at position 591 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a glutamine (Q).

(163) In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or piggyBac-like transposase enzyme or may comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at one or more of positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or piggyBac-like transposase enzyme may comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at two, three, four, five, six or more of positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or piggyBac-like transposase enzyme may comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, the amino acid substitution at position 103 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a serine (S). In certain embodiments, the amino acid substitution at position 194 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 372 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for an arginine (R). In certain embodiments, the amino acid substitution at position 375 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for a lysine (K). In certain embodiments, the amino acid substitution at position 450 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an asparagine (N) for an aspartic acid (D). In certain embodiments, the amino acid substitution at position 509 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a serine (S). In certain embodiments, the amino acid substitution at position 570 of SEQ ID NO: 14487 or SEQ ID NO:

14484 is a substitution of a serine (S) for an asparagine (N). In certain embodiments, the piggyBac or piggyBac-like transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487. In certain embodiments, including those embodiments wherein the piggyBac or piggyBac-like transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, the piggyBac or piggyBac-like transposase enzyme may further comprise an amino acid substitution at positions 372, 375 and 450 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, a substitution of an alanine (A) for an arginine (R) at position 372 of SEQ ID NO: 14487, and a substitution of an alanine (A) for a lysine (K) at position 375 of SEQ ID NO: 14487. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, a substitution of an alanine (A) for an arginine (R) at position 372 of SEQ ID NO: 14487, a substitution of an alanine (A) for a lysine (K) at position 375 of SEQ ID NO: 14487 and a substitution of an asparagine (N) for an aspartic acid (D) at position 450 of SEQ ID NO: 14487.

(164) In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from an insect. In certain embodiments, the insect is *Trichoplusia ni* (GenBank Accession No. AAA87375; SEQ ID NO: 16796), *Argyrogramma agnata* (GenBank Accession No. GU477713; SEQ ID NO: 14534, SEQ ID NO: 16797), *Anopheles gambiae* (GenBank Accession No. XP\_312615 (SEQ ID NO: 16798); GenBank Accession No. XP\_320414 (SEQ ID NO: 16799); GenBank Accession No. XP\_310729 (SEQ ID NO: 16800)), *Aphis gossypii* (GenBank Accession No. GU329918; SEQ ID NO: 16801, SEQ ID NO: 16802), *Acyrtosiphon pisum* (GenBank Accession No. XP\_001948139; SEQ ID NO: 16803), *Agrotis ipsilon* (GenBank Accession No. GU477714; SEQ ID NO: 14537, SEQ ID NO: 16804), *Bombyx mori* (GenBank Accession No. BAD11135; SEQ ID NO: 14505), *Chilo suppressalis* (GenBank Accession No. JX294476; SEQ ID NO: 16805, SEQ ID NO: 16806), *Drosophila melanogaster* (GenBank Accession No. AAL39784; SEQ ID NO: 16807), *Helicoverpa armigera* (GenBank Accession No. ABS18391; SEQ ID NO: 14525), *Heliothis virescens* (GenBank Accession No. ABD76335; SEQ ID NO: 16808), *Macdunnoughia crassisigna* (GenBank Accession No. EU287451; SEQ ID NO: 16809, SEQ ID NO: 16810), *Pectinophora gossypiella* (GenBank Accession No. GU270322; SEQ ID NO: 14530, SEQ ID NO: 16811), *Tribolium castaneum* (GenBank Accession No. XP\_001814566; SEQ ID NO: 16812), *Ctenoplia agnata* (also called *Argyrogramma agnata*), *Messour bouvieri*, *Megachile rotundata*, *Bombus impatiens*, *Mamestra brassicae*, *Mayetiola destructor* or *Apis mellifera*.

(165) In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from an insect. In certain embodiments, the insect is *Trichoplusia ni* (AAA87375).

(166) In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from an insect. In certain embodiments, the insect is *Bombyx mori* (BAD11135).

(167) In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from a crustacean. In certain embodiments, the crustacean is *Daphnia pulex* (AAM76342, SEQ ID NO: 16813).

(168) In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from a vertebrate. In certain embodiments, the vertebrate is *Xenopus tropicalis* (GenBank Accession No. BAF82026; SEQ ID NO: 14518), *Homo sapiens* (GenBank Accession No. NP\_689808; SEQ ID NO: 16814), *Mus musculus* (GenBank Accession No. NP\_741958; SEQ ID NO: 16815), *Macaca fascicularis* (GenBank Accession No. AB179012; SEQ ID NO: 16816, SEQ ID NO: 16817), *Rattus norvegicus* (GenBank Accession No. XP\_220453; SEQ ID NO: 16818) or *Myotis lucifugus*.

(169) In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from a urochordate. In certain embodiments, the urochordate is *Ciona intestinalis* (GenBank Accession No. XP\_002123602; SEQ ID NO: 16819).

(170) In certain embodiments, the piggyBac or piggyBac-like transposase inserts a transposon at the sequence 5'-TTAT-3' within a chromosomal site (a TTAT target sequence).

(171) In certain embodiments, the piggyBac or piggyBac-like transposase inserts a transposon at the

sequence 5'-TTAA-3' within a chromosomal site (a TTAA target sequence).

(172) In certain embodiments, the target sequence of the piggyBac or piggyBac-like transposon comprises or consists of 5'-CTAA-3', 5'-TTAG-3', 5'-ATAA-3', 5'-TCAA-3', 5'-AGTT-3', 5'-ATTA-3', 5'-GTTA-3', 5'-TTGA-3', 5'-TTTA-3', 5'-TTAC-3', 5'-ACTA-3', 5'-AGGG-3', 5'-CTAG-3', 5'-TGAA-3', 5'-AGGT-3', 5'-ATCA-3', 5'-CTCC-3', 5'-TAAA-3', 5'-TCTC-3', 5'-TGAA-3', 5'-AAAT-3', 5'-AATC-3', 5'-ACAA-3', 5'-ACAT-3', 5'-ACTC-3', 5'-AGTG-3', 5'-ATAG-3', 5'-CAAA-3', 5'-CACA-3', 5'-CATA-3', 5'-CCAG-3', 5'-CCCA-3', 5'-CGTA-3', 5'-GTCC-3', 5'-TAAG-3', 5'-TCTA-3', 5'-TGAG-3', 5'-TGTT-3', 5'-TTCA-3', 5'-TTCT-3' and 5'-TTTT-3'.

(173) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Bombyx mori*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(174) TABLE-US-00065 (SEQ ID NO: 14504) 1 MDIERQEERI RAMLEEELSD  
YSESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61 EANAIIANES  
DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE 121  
NIVTEQAQVK NIARDASTEY ECWNIFVTSD MLQEILHTN SSIRHRQTKT  
AAENSSAETS 181 FYMQETTLCE LKALIALLYL AGLIKSNRQS LKDLWRDGT  
GVDIFRTTMS LQRFQFLQNN 241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ  
CCQNAYSPSE FLTIDEMLLS FRGRCLFRVY 301 IPNKPAKYGI KILALVDAKN  
FDVVNLEVYA GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG  
YELMLHLLNE YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN STKAGVDVVD  
ELSANYNVS 481 NSKRWPMTLF YGVLNMAAIN ACIIYRANKN VTIKRTEFIR  
SLGLSMIYEH LHSRNKKKNI 541 PTYLRQRIEK QLGEPSPRHV NVPGRYVRCQ  
DCPYKKDRKT KHSCNACAKP ICMEHAKFLC 601 ENCAELDSSL.

(175) The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(176) TABLE-US-00066 (SEQ ID NO: 14505) 1 MDIERQEERI RAMLEEELSD  
YSESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61 EANAIIANES  
DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE 121  
NIVTEQAQVK NIARDASTEY ECWNIFVTSD MLQEILHTN SSIRHRQTKT  
AAENSSAETS 181 FYMQETTLCE LKALIALLYL AGLIKSNRQS LKDLWRDGT  
GVDIFRTTMS LQRFQFLQNN 241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ  
CCQNAYSPSE FLTIDEMLLS FRGRCLFRVY 301 IPNKPAKYGI KILALVDAKN  
FYVVNLEVYA GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG  
YELMLHLLNE YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN STKAGVDVVD  
ELCANYNVS 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN VTIKRTEFIR  
SLGLSMIYEH LHSRNKKKNI 541 PTYLRQRIEK QLGEPSPRHV NVPGRYVRCQ  
DCPYKKDRKT KRSCNACAKP ICMEHAKFLC 601 ENCAELDSSL.

(177) In certain embodiments, the piggyBac or piggyBac-like transposase is fused to a nuclear localization signal. In certain embodiments, the amino acid sequence of the piggyBac or piggyBac-like transposase fused to a nuclear localization signal is encoded by a polynucleotide sequence comprising:

(178) TABLE-US-00067 (SEQ ID NO: 14629) 1 atggcaccca aaaagaaacg taaagtgatg  
gacattgaaa gacaggaaga aagaatcagg 61 gcgatgctcg aagaagaact gagcgactac tccgacgaat  
cgatcatcaga ggaatgaaacc 121 gaccactgta gcgagcatga ggtaactac gacaccgagg aggagagaat  
cgactctgtg 181 gatgtgccct ccaactcacg ccaagaagag gccaatgcaa ttatcgcaaa cgaatcggac 241  
agcgatccag acgatgatct gccactgtcc ctctgtgcgc agcggggccag cgcttcgaga 301 caagtgtcag  
gtccattcta cacttcgaag gacggcacta agtgggtacaa gaattgccag 361 cgacctaacg tcagactccg ctccgagaat

atcgtgaccg aacagggtca ggtcaagaat 421 atgcgccgag acgcctcgac tgagtacgag tgttggaata  
tcttcgtgac ttcggacatg 481 ctgcaagaaa ttctgacgca caccaacagc tcgattaggc atcgccagac caagactgca  
541 gcggagaact catcggccga aacctccttc tatatgcaag agactactct gtgcgaactg 601 aaggcgctga  
ttgactgct gtacttggcc ggcctcatca aatcaaatag gcagagcctc 661 aaagatctct ggagaacgga tggaactgga  
gtggatatct ttcggacgac tatgagcttg 721 cagcggttcc agtttctgca aaacaataac agattcgacg acaagtccac  
ccgggacgaa 781 aggaaacaga ctgacaacat ggctgcgttc cggtaaatat tcgatcagtt tgtgcagtgc 841  
tgccaaaacg cttatagccc atcggaattc ctgaccatcg acgaaatgct tctctccttc 901 cggggggcgct gcctgttccg  
agtgtacatc ccgaacaagc cggctaaata cggaatcaaa 961 atcctggccc tggaggacgc caagaatttc tacgtcgtga  
atctcgaagt gtacgcagga 1021 aagcaaccgt cgggaccgta cgctgtttcg aaccgcccgt ttgaagtcgt  
cgagcggctt 1081 attcagccgg tggccagatc ccaccgcaat gttaccttcg acaattggtt caccggctac 1141  
gagctgatgc ttcaccttct gaacgagtag cggctcacta gcgtggggac tgtcaggaag 1201 aacaagcggc  
agatcccaga atccttcacg cgcaccgacc gccagcctaa ctctgccgtg 1261 ttcggatttc aaaaggatat cacgcttgc  
tcgtacgccc ccaagaaaaa caaggctcgtg 1321 gtcgtgatga gcaccatgca tcacgacaac agcatcgacg  
agtcaaccgg agaaaagcaa 1381 aagcccgaga tgatcacctt ctacaattca actaaggccg gcgtcgacgt  
cgtggatgaa 1441 ctgtgcgcga actataacgt gtcccggaa tctaagcggg ggcctatgac tctcttctac 1501  
ggagtgtcga atatggccgc aatcaacgcg tgcattatct accgcaccaa caagaacgtg 1561 accatcaagc  
gcaccgagtt catcagatcg ctgggtttga gcatgatcta cgagcacctc 1621 cattcacgga acaagaagaa gaatatcct  
acttacctga ggcagcgtat cgagaagcag 1681 ttgggagaac caagcccgcg ccacgtgaac gtgccggggc  
gctacgtgcg gtgccaagat 1741 tgcccgtaca aaaaggaccg caaaaccaa agatcgtgta acgcgtgcgc  
caaactatc 1801 tgcattggagc atgcaaaatt tctgtgtgaa aattgtgctg aactcgattc ctccctg.

(179) In certain embodiments, the piggyBac or piggyBac-like transposase is hyperactive. A hyperactive piggyBac or piggyBac-like transposase is a transposase that is more active than the naturally occurring variant from which it is derived. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Bombyx mori*. In certain embodiments, the piggyBac or piggyBac-like transposase is a hyperactive variant of SEQ ID NO: 14505. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to:

(180) TABLE-US-00068 (SEQ ID NO: 14576) 1 MDIERQEERI RAMLEEELSD  
YDESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61  
EANAIIANES DSDPDDDLPL SLVRQRASAS RQMSGPHYTS KDGTKWYKNC  
QRPNVRLRSE 121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD  
MLQEILHTN SSIRWRQTKT AAENSSASTS 181 FYMQETTLCE LKALIGLLYI  
AGLIKSNRQS LKDLWRTDGT GVDIFRTTMS LQRFQFLQNN 241  
IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ SCQNAYSPSE FLTIDEMLLS  
FRGRCLFRVY 301 IPNKPAYGI KILALVDAKN FYVKNLEVYA  
GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG YELMLHLLNE  
YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN  
STKAGVDVVD ELCANYNVSR 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN  
VTIKRTEFIR SLGLSMIYEH LHSRNKKKNI 541 PTYLRQRIEK  
QLGEPSPRHV NVPGRYVRCQ DCPYKKDRKT KRSCNACAKP ICMEHAKFLC  
601 ENCAELDSHL.

(181) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14576. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(182) TABLE-US-00069 (SEQ ID NO: 14630) 1 MDIERQEERI RAMLEEELSD  
YDESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61  
EANAIIANES DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC  
QRPNVRLRSE 121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD  
MLQEILHTN SSIRWRQTKT AAENSSAETS 181 FYMQETTLCE LKALIGLLYI  
AGLIKSNRQS LKDLWRTDGT GVDIFRTTMS LQRFQFLNN 241  
IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ SCQNAYSPSE FLTIDEMLLS

FRGRCLFRVY 301 IPNKPAKYGI KILALVDAKN FYVKNLEVYA  
 GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG YEVMLHLLNE  
 YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN  
 STKAGVDVVD ELCANYNVSR 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN  
 VTIKRTEFIR SLGLSMIYEH LHSRNKKKNI 541 PTYLRQRIEK  
 QLGEPSPRHV NVPGRYVRCQ DCPYKKDRKT KRSCNACAKP ICMEHAKFLC  
 601 ENCAHLDS.

(183) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(184) TABLE-US-00070 (SEQ ID NO: 14631) 1 MDIERQEERI RAMLEEELSD  
 YSDESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61  
 EANAIIANES DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC  
 QRPNVRLRSE 121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD  
 MLQEILHTN SSIRWRQTKT AAENSSASTS 181 FYMQETTLCE LKALIGLLYI  
 AGLIKSNRQS LKDLWRTDGT GVDIFRTTMS LQRFQFLLNN 241  
 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ SCQNAYSPSE FLTIDEMLLS  
 FRGRCLFRVY 301 IPNKPAKYGI KILALVDAKN FYVKNLEVYA  
 GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG YELMLHLLNE  
 YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN  
 STKAGVDVVD ELCANYNVSR 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN  
 VTIKRTEFIR SLGLSMIYEH LHSRNKKKNI 541 PTYLRQRIAM  
 QLGEPSPRHV NVPGRYVRCQ DCPYKKDRKT KRSCNACAKP ICMEHAKFLC  
 601 ENCAELDSSL.

(185) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(186) TABLE-US-00071 (SEQ ID NO: 14632) 1 MDIERQEERI RAMLEEELSD  
 YSDESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61  
 EANAIIANES DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC  
 QRPNVRLRSE 121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD  
 MLQEILHTN SSIRWRQTKT AAENSSAETS 181 FYMQETTLCE LKALIGLLYI  
 AGLIKSNRQS LKDLWRTDGT GVDIFRTTMS LQRFQFLLNN 241  
 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ SCQNAYSPSE FLTIDEMLLS  
 FRGRCLFRVY 301 IPNKPAKYGI KILALVDAKN FYVKNLEVYA  
 GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG YELMLHLLNE  
 YRLTSVGTVR KNKTQIPENF IRTDRQPNSS VFGFQKDITL 421  
 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN  
 STKAGVDVVD ELQANYNVSR 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN  
 VTIKRTEFIR SLGLSMIYEH LHSRNKKKNI 541 PTYLRQRIEK  
 QLGEPSPRHV NVPGRYVRCQ DCPYKKDRKT KRSCNACAKP ICMEHAKFLC  
 601 ENCAELDSSL.

(187) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(188) TABLE-US-00072 (SEQ ID NO: 14633) 1 MDIERQEERI RAMLEEELSD  
 YSDESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61  
 EANAIIANES DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC  
 QRPNVRLRSE 121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD  
 MLQEILHTN SSIRWRQTKT AAENSSAETS 181 FYMQETTLCE LKALIGLLYI  
 AGLIKSNRQS LKDLWRTDGT GVDIFRTTMS LQRFQFLQNN 241  
 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ SCQNAYSPSE FLTIDEMLLS  
 FRGRCLFRVY 301 IPNKPAKYGI KILALVDAKN FYVKNLEVYA

GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG YELMLHLLNE  
 YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN  
 STKAGVDVVD ELCANYNVSR 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN  
 VTIKRTEFIR SLGLSMIYEH LHSRNKKKNI 541 PTYLRQRIEK  
 QLGEPSPRHV NVPGRYVRCQ DCPYKKDRKT KRSCNACAKP ICMEHAKFLC  
 601 ENCAELDSSL.

(189) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(190) TABLE-US-00073 (SEQ ID NO: 14634) 1 MDIRQEERI RAMLEEELSD  
 YSDESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61  
 EANAIIANES DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC  
 QRPNVRLRSE 121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD  
 MLQEILHTN SSIRHRQTKT AAENSSAETS 181 FYMQETTLCE LKALIALLYL  
 AGLIKSNRQS LKDLWRTDGT GVDIFRTTMS LQRFQFLQNN 241  
 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ CCQNAYSPSE FLTIDEMLLS  
 FRGRCLFRVY 301 IPNPKAKYGI KILALVDAKN DYVVNLEVYA  
 GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG YELMLHLLNE  
 YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN  
 STKAGVDVVD ELCANYNVSR 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN  
 VTIKRTEFIR SLGLSMIYEH LHSRNKKKNI 541 PTYLRQRIEK  
 QLGEPSRHV NVKGRYVRCQ DCPYKKDRKT KRSCNACAKP ICMEHAKFLC  
 601 ENCAELDSSL.

(191) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase is more active than the transposase of SEQ ID NO: 14505. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase is at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or any percentage in between identical to SEQ ID NO: 14505.

(192) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution at a position selected from 92, 93, 96, 97, 165, 178, 189, 196, 200, 201, 211, 215, 235, 238, 246, 253, 258, 261, 263, 271, 303, 321, 324, 330, 373, 389, 399, 402, 403, 404, 448, 473, 484, 507, 523, 527, 528, 543, 549, 550, 557, 601, 605, 607, 609, 610 or a combination thereof (relative to SEQ ID NO: 14505). In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution of Q92A, V93L, V93M, P96G, F97H, F97C, H165E, H165W, E178S, E178H, C189P, A196G, L200I, A201Q, L211A, W215Y, G219S, Q235Y, Q235G, Q238L, K246I, K253V, M258V, F261L, S263K, C271S, N303R, F321W, F321D, V324K, V324H, A330V, L373C, L373V, V389L, S399N, R402K, T403L, D404Q, D404S, D404M, N441R, G448W, E449A, V469T, C473Q, R484K T507C, G523A, I527M, Y528K Y543I, E549A, K550M, P557S, E601V, E605H, E605W, D607H, S609H, L610I or any combination thereof. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution of Q92A, V93L, V93M, P96G, F97H, F97C, H165E, H165W, E178S, E178H, C189P, A196G, L200I, A201Q, L211A, W215Y, G219S, Q235Y, Q235G, Q238L, K246I, K253V, M258V, F261L, S263K, C271S, N303R, F321W, F321D, V324K, V324H, A330V, L373C, L373V, V389L, S399N, R402K, T403L, D404Q, D404S, D404M, N441R, G448W, E449A, V469T, C473Q, R484K T507C, G523A, I527M, Y528K Y543I, E549A, K550M, P557S, E601V, E605H, E605W, D607H, S609H and L610I.

(193) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises one or more substitutions of an amino acid that is not wild type, wherein the one or more substitutions for a wild type amino acid comprises a substitution of E4X, A12X, M13X, L14X, E15X, D20X, E24X, S25X, S26X, S27X, D32X, H33X, E36X, E44X, E45X, E46X, I48X, D49X, R58X, A62X, N63X, A64X, I65X, I66X, N68X, E69X, D71X, S72X, D76X, P79X, R84X, Q85X, A87X, S88X, Q92X, V93X, S94X, G95X, P96X, F97X, Y98X, T99X, I145X, S149X, D150X, L152X, E154X, T157X, N160X, S161X, S162X, H165X, R166X, T168X, K169X, T170X, A171X, E173X, S175X, S176X,



T178X, T179X, M183X, Q184X, T186X, T187X, L188X, C189X, L194X, I195X, A196X, L198X, L200X, A201X, L203X, I204X, K205X, A206X, N207X, Q209X, S210X, L211X, K212X, D213X, L214X, W215X, R216X, T217X, G219X, V222X, D223X, I224X, T227X, M229X, Q235X, L237X, Q238X, N239X, N240X, P302X, N303X, P305X, A306X, K307X, Y308X, I310X, K311X, I312X, L313X, A314X, L315X, V316X, D317X, A318X, K319X, N320X, F321X, Y322X, V323X, V324X, L326X, E327X, V328X, A330X, Q333X, P334X, S335X, G336X, P337X, A339X, V340X, S341X, N342X, R343X, P344X, F345X, E346X, V347X, E349X, I352X, Q353X, V355X, A356X, R357X, N361X, D365X, W367X, T369X, G370X, L373X, M374X, L375X, H376X, N379X, E380X, R382X, V386X, V389X, N392X, R394X, Q395X, S399X, F400X, I401X, R402XT403X, D404X, R405X, Q406X, P407X, N408X, S409X, S410X, V411X, F412X, F414X, Q415X, I418X, T419X, L420X, N428XV432X, M434X, D440X, N441X, S442X, I443X, D444X, E445X, G448X, E449X, Q451X, K452X, M455X, I456X, T457X, F458X, S461X, A464X, V466X, Q468X, V469X, E471X, L472X, C473X, A474X, K483X, W485X, T488X, L489X, Y491X, G492X, V493X, M496X, I499X, C502X, I503X, T507X, K509X, N510X, V511X, T512X, I513X, R515X, E517X, S521X, G523X, L524X, S525X, I527X, Y528X, E529X, H532X, S533X, N535X, K536X, K537X, N539X, I540X, T542X, Y543X, Q546X, E549X, K550X, Q551X, G553X, E554X, P555X, S556X, P557X, R558X, H559X, V560X, N561X, V562X, P563X, G564X, R565X, Y566X, V567X, Q570X, D571X, P573X, Y574X, K576X, K581X, S583X, A586X, A588X, E594X, F598X, L599X, E601X, N602X, C603X, A604X, E605X, L606X, D607X, S608X, S609X or L610X (relative to SEQ ID NO: 14505). A list of hyperactive amino acid substitutions can be found in U.S. Pat. No. 10,041,077, the contents of which are incorporated herein by reference in their entirety.

(194) In certain embodiments, the piggyBac or piggyBac-like transposase is integration deficient. In certain embodiments, an integration deficient piggyBac or piggyBac-like transposase is a transposase that can excise its corresponding transposon, but that integrates the excised transposon at a lower frequency than a corresponding wild type transposase. In certain embodiments, the piggyBac or piggyBac-like transposase is an integration deficient variant of SEQ ID NO: 14505.

(195) In certain embodiments, the excision competent, integration deficient piggyBac or piggyBac-like transposase comprises one or more substitutions of an amino acid that is not wild type, wherein the one or more substitutions a for wild type amino acid comprises a substitution of R9X, A12X, M13X, D20X, Y21K, D23X, E24X, S25X, S26X, S27X, E28X, E30X, D32X, H33X, E36X, H37X, A39X, Y41X, D42X, T43X, E44X, E45X, E46X, R47X, D49X, S50X, S55X, A62X, N63X, A64X, I66X, A67X, N68X, E69X, D70X, D71X, S72X, D73X, P74X, D75X, D76X, D77X, I78X, S81X, V83X, R84X, Q85X, A87X, S88X, A89X, S90X, R91X, Q92X, V93X, S94X, G95X, P96X, F97X, Y98X, T99X, W012X, G103X, Y107X, K108X, L117X, I122X, Q128X, I312X, D135X, S137X, E139X, Y140X, I145X, S149X, D150X, Q153X, E154X, T157X, S161X, S162X, R164X, H165X, R166X, Q167X, T168X, K169X, T170X, A171X, A172X, E173X, R174X, S175X, S176X, A177X, E178X, T179X, S180X, Y182X, Q184X, E185X, T187X, L188X, C189X, L194X, I195X, A196X, L198X, L200X, A201X, L203X, I204X, K205X, N207X, Q209X, L211X, D213X, L214X, W215X, R216X, T217X, G219X, T220X, V222X, D223X, I224X, T227X, T228X, F234X, Q235X, L237X, Q238X, N239X, N240X, N303X, K304X, I310X, I312X, L313X, A314X, L315X, V316X, D317X, A318X, K319X, N320X, F321X, Y322X, V323X, V324X, N325X, L326X, E327X, V328X, A330X, G331X, K332X, Q333X, S335X, P337X, P344X, F345X, E349X, H359X, N361X, V362X, D365X, F368X, Y371X, E372X, L373X, H376X, E380X, R382X, R382X, V386X, G387X, T388X, V389X, K391X, N392X, R394X, Q395X, E398X, S399X, F400X, I401X, R402XT403X, D404X, R405X, Q406X, P407X, N408X, S409X, S410X, Q415X, K416X, A424X, K426X, N428X, V430X, V432X, V433X, M434X, D436X, D440X, N441X, S442X, I443X, D444X, E445X, S446X, T447X, G448X, E449X, K450X, Q451X, E454X, M455X, I456X, T457X, F458X, S461X, A464X, V466X, Q468X, V469X, C473X, A474X, N475X, N477X, K483X, R484X, P486X, T488X, L489X, G492X, V493X, M496X, I499X, I503X, Y505X, T507X, N510X, V511X, T512X, I513X, K514X, T516X, E517X, S521X, G523X, L524X, S525X, I527X, Y528X, L531X, H532X, S533X, N535X, I540X, T542X, Y543X, R545X, Q546X, E549X, L552X, G553X, E554X, P555X, S556X, P557X, R558X, H559X, V560X, N561X, V562X, P563X, G564X, V567X, Q570X, D571X, P573X, Y574X, K575X, K576X, N585X, A586X,

M593X, E601X, N602X, A604X, E605X, L606X, D607X, S608X, S609X or L610X (relative to SEQ ID NO: 14505). A list of integration deficient amino acid substitutions can be found in U.S. Pat. No. 10,041,077, the contents of which are incorporated by reference in their entirety.

(196) In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence of:

(197) TABLE-US-00074 (SEQ ID NO: 14606) 1 MDIERQEERI RAMLEEELSD  
YSESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61  
EANAIIANES DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC  
QRPNVRLRSE 121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD  
MLQEILHTN SSIRHRQTKT AAENSSAETS 181 FYMQETTLCE LKALIALLYL  
AGLIKSNRQS LKDLWRKDGT GVDIFRTTMS LQRFQFLLNN 241  
IRFDDISTRD ERKQTDNMAA FRSIFDQFVQ CCQNAYSPSE FLTIDEMLLS  
FRGRCLFRVY 301 IPNKPAKYGI KILALVDAKN FYVVNLEVYA  
GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG YELMLHLLNE  
YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN  
STKAGVDVVD ELCANYNVSR 481 NSKKWPMTLF YGVLNMAAIN ACIIYRTNKN  
VTIKRTEFIR SLGLSMMYEH LHSRNKKKNI 541 PTYLRQRIEK  
QLGEPVPRHV NVPGRYVRCQ DCPYKKDRKT KRSCNACAKP ICMEHAKFLC  
601 ENCAELDSSL.

In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence of:

(198) TABLE-US-00075 (SEQ ID NO: 14607) 1 MDIERQEERI RAMLEEELSD  
YSESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61  
EANAIIANES DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC  
QRPNVRLRSE 121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD  
MLQEILHTN SSIRHRQTKT AAENSSAETS 181 FYMQETTLCE LKALIGLLYL  
AGLIKSNRQS LKDLWRDGT GVDIFRTTMS LQRFYFLQNN 241  
IRFDDKSTLD ERKQTDNMAA FRSIFDQFVQ SCQNAYSPSE FLTIDEMLLS  
FRGRCLFRVY 301 IPNKPAKYGI KILALVDAKN FYVVNLEVYA  
GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG YELMLHLLNE  
YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN  
STKAGVDVVD ELCANYNVSR 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN  
VTIKRTEFIR SLGLSMIYEH LHSRNKKKNI 541 PTYLRQRIEK  
QLGEPSPRHV NYPGRYVRCQ DCPYKKDRKT KRSCNACAKP ICMEHAKFLC  
601 VNCAELDSSL.

In certain embodiments, the piggyBac or piggyBac-like transposase that is integration deficient comprises a sequence of:

(199) TABLE-US-00076 (SEQ ID NO: 14608) 1 MDIERQEERI RAMLEEELSD  
YSESSSEDE TDHCSEHEVN YDTEERIDS VDVPSNSRQE 61  
EANAIIANES DSDPDDDLPL SLVRQRASAS RQVSGPFYTS KDGTKWYKNC  
QRPNVRLRSE 121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD  
MLQEILHTN SSIRHRQTKT AAENSSAETS 181 FYMQETTLCE LKALIALLYL  
AGLIKSNRQS LKDLWRKDGT GVDIFRTTMS LQRFQFLLNN 241  
IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ CCQNAYSPSE FLTIDEMLLS  
FRGRCLFRVY 301 IPNKPAKYGI KILALVDAKN DYVVNLEVYA  
GKQPSGPYAV SNRPFEVVER LIQPVARSHR 361 NVTFDNWFTG YECMLHLLNE  
YRLTSVGTVR KNKRQIPESF IRTDRQPNSS VFGFQKDITL 421  
VSYAPKKNKV VVVMSTMHHD NSIDESTGEK QKPEMITFYN  
STKAGVDVVD ELCANYNVSR 481 NSKKWPMTLF YGVLNMAAIN ACIIYRTNKN  
VTIKRTEFIR SLGLSMIKEH LHSRNKKKNI 541 PTYLRQRIEK

In certain embodiments, the integration deficient transposase comprises a sequence that is at least 90% identical to SEQ ID NO: 14608.

(200) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Bombyx mori*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(201) TABLE-US-00077 (SEQ ID NO: 14506) 1 ttatcccggc gagcatgagg cagggtatct  
cataccctgg taaaatttta aagttgtgta 61 tttataaaa tttcgtctg acaacactag  
cgcgctcagt agctggaggc aggagcgtgc 121 gggaggggat agtggcgtga tcgcagtgtg  
gcacgggaca ccggcgagat attcgtgtgc 181 aaacctgtt cgggtatgtt ataccctgcc tcattgttga  
cgtatttttt ttatgtaatt 241 ttccgatta ttaattcaa ctgtttatt ggtatttta tgttatccat tgttctttt  
301 ttatgattta ctgtatcggg tgtctttcgt tccttagtt gagtttttt ttattattt 361 cagttttga  
tcaaa .

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(202) TABLE-US-00078 (SEQ ID NO: 14507) 1 tcatattttt agtttaaaaa aataattata  
tgttttataa tgaaaagaat ctcatatct 61 ttcagtatta ggttgattta tattccaaag  
aataatattt ttgttaaatt gttgattttt 121 gtaaacctct aaatgtttgt tgctaaaatt actgtgttta  
agaaaaagat taataaataa 181 taataattc ataataaaa acttctttca ttgaatgcca ttaaataaac  
cattatttta 241 caaaataaga tcaacataat tgagtaaata ataataagaa caatattata gtacaacaaa  
301 atatgggtat gtcataacct gccacattct tgatgtaact tttttcacc tcatgctcgc 361 cgggttat.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(203) TABLE-US-00079 (SEQ ID NO: 14508) 1 ttatcccggc gagcatgagg cagggtatct  
cataccctgg taaaatttta aagttgtgta 61 tttataaaa tttcgtctg acaacactag  
cgcgctcagt agctggaggc aggagcgtgc 121 gggaggggat agtggcgtga tcgcagtgtg  
gcacgggaca ccggcgagat attcgtgtgc 181 aaacctgtt cgggtatgtt ataccctgcc tcat.  
In certain embodiments, the piggyBac (PB) or piggyBac-like transposon comprises a sequence of:

(204) TABLE-US-00080 (SEQ ID NO: 14509) 1 taaataataa taatttcata attaaaaact  
tctttcattg aatgccatta aataaacat 61 tattttacaa aataagatca acataattga  
gtaaataata ataagaacaa tattatagta 121 caacaaaata tgggtatgtc ataccctgcc acattcttga  
tgtaactttt tttcacctca 181 tgctcgccgg gttat.

(205) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 5' sequence corresponding to SEQ ID NO: 14506 and a 3' sequence corresponding to SEQ ID NO: 14507. In certain embodiments, one piggyBac or piggyBac-like transposon end is at least 85%, at least 90%, at least 95%, at least 98%, at least 99% identical or any percentage in between identical to SEQ ID NO: 14506 and the other piggyBac or piggyBac-like transposon end is at least 85%, at least 90%, at least 95%, at least 98%, at least 99% or any percentage in between identical to SEQ ID NO: 14507. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14506 and SEQ ID NO: 14507 or SEQ ID NO: 14509. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14508 and SEQ ID NO: 14507 or SEQ ID NO: 14509. In certain embodiments, the 5' and 3' transposon ends share a 16 bp repeat sequence at their ends of CCCGGCGAGCATGAGG (SEQ ID NO: 14510) immediately adjacent to the 5'-TTAT-3 target insertion site, which is inverted in the orientation in the two ends. In certain embodiments, 5' transposon end begins with a sequence comprising 5'-TTATCCCGGCGAGCATGAGG-3 (SEQ ID NO: 14511), and the 3' transposon ends with a sequence comprising the reverse complement of this sequence: 5'-CCTCATGCTCGCCGGGTTAT-3' (SEQ ID NO: 14512).

(206) In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end comprising at least 14, 16, 18, 20, 30 or 40 contiguous nucleotides of SEQ ID NO: 14506 or SEQ ID NO: 14508. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end comprising at least 14, 16, 18, 20, 30 or 40 contiguous nucleotides of SEQ ID NO: 14507 or SEQ ID NO: 14509. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end with at least 90% identity to SEQ ID NO: 14506 or SEQ ID NO: 14508. In certain embodiments, the piggyBac or piggyBac-like

transposon comprises one end with at least 90% identity to SEQ ID NO: 14507 or SEQ ID NO: 14509.

(207) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(208) TABLE-US-00081 (SEQ ID NO: 14515) 1 ttaacccggc gagcatgagg cagggtatct

cataccctgg taaaatttta aagttgtgta 61 tttataaaa tttcgtctg acaacactag

cgcgctcagt agctggaggc aggagcgtgc 121 gggaggggat agtggcgtga tcgcagtgtg

gcacgggaca ccggcgagat attcgtgtgc 181 aaacctgtt cgggtatgtt ataccctgcc tcattgttga

cgtattttt ttatgtaatt 241 ttccgatta ttaatttcaa ctgttttatt ggtattttta tgttatccat tgttctttt

301 ttatgattta ctgtatcggg tgtcttctgt tccttagtt gagtttttt ttattatttt 361 cagtttttga tcaaa.

(209) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(210) TABLE-US-00082 (SEQ ID NO: 14516) 1 tcatatttt agtttaaaaa aataattata

tgttttataa tgaaaagaat ctcatatct 61 ttcagtatta ggttgattta tattccaaag

aataatattt ttgttaaatt gttgatttt 121 gtaaacctct aaatgtttgt tgctaaaatt actgtgttta

agaaaaagat taataaataa 181 taataatttc ataattaaaa acttctttca ttgaatgcca ttaaataatt

cattatttta 241 caaaataaga tcaacataat tgagtaaata ataataagaa caatattata gtacaacaaa

301 atatgggtat gtcataccct tttttttt tttttttt tttttcggg tagaggggccg 361 aacctctac

gaggtccccg cgcaaaaggg gcgcgcgggg tatgtgagac tcaacgatct 421 gcatggtgtt

gtgagcagac cgcgggccca aggattttag agcccacca ctaaagcact 481 cctctgcact

cttacacccg acgtccgatc ccctccgagg tcagaacccg gatgaggtag 541 gggggctacc

gcggtcaaca ctacaaccag acggcgcggc tcacccaag gacgcccagc 601 cgacggagcc

ttcgaggcga atcgaaggct ctgaaacgtc ggccgtctcg gtacggcagc 661 ccgtcgggcc

gcccagacgg tgccgctggt gtcccggaat acccgctgg accagaacca 721 gcctgccggg

tcgggacgcg atacaccgtc gaccggtcgc tctaactact ccacggcagc 781 gcgctagagt gctggta.

(211) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of CCCGGCGAGCATGAGG (SEQ ID NO: 14510). In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of SEQ ID NO: 14510. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TTATCCCGGCGAGCATGAGG (SEQ ID NO: 14511). In certain embodiments, the piggyBac or piggyBac-like transposon comprises at least 16 contiguous nucleotides from SEQ ID NO: 14511. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of CCTCATGCTCGCCGGGTTAT (SEQ ID NO: 14512). In certain embodiments, the piggyBac or piggyBac-like transposon comprises at least 16 contiguous nucleotides from SEQ ID NO: 14512. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end comprising at least 16 contiguous nucleotides from SEQ ID NO: 14511 and one end comprising at least 16 contiguous nucleotides from SEQ ID NO: 14512. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14511 and SEQ ID NO: 14512. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TTAACCCGGCGAGCATGAGG (SEQ ID NO: 14513). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of

(212) TABLE-US-00083 (SEQ ID NO: 14514) CCTCATGCTCGCCGGGTTAA.

(213) In certain embodiments, the piggyBac or piggyBac-like transposon may have ends comprising SEQ ID NO: 14506 and SEQ ID NO: 14507, or a variant of either or both of these having at least 90% sequence identity to SEQ ID NO: 14506 or SEQ ID NO: 14507, and the piggyBac or piggyBac-like transposase has the sequence of SEQ ID NO: 14504 or SEQ ID NO: 14505, or a sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identity to SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a heterologous polynucleotide inserted between a pair of inverted repeats, where the transposon is capable of transposition by a piggyBac or piggyBac-like transposase having at least 5% 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identity to SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments, the transposon comprises two transposon ends, each of which comprises SEQ ID NO: 14510 in inverted orientations in the two transposon ends. In certain embodiments, each inverted terminal repeat (ITR) is at least 90% identical to SEQ ID NO: 14510.

(214) In certain embodiments, the piggyBac or piggyBac-like transposon is capable of insertion by a piggyBac or piggyBac-like transposase at the sequence 5'-TTAT-3' within a target nucleic acid. In certain embodiments, one end of the piggyBac or piggyBac-like transposon comprises at least 16 contiguous nucleotides from SEQ ID NO: 14506 and the other transposon end comprises at least 16 contiguous nucleotides from SEQ ID NO: 14507. In certain embodiments, one end of the piggyBac or piggyBac-like transposon comprises at least 17, at least 18, at least 19, at least 20, at least 22, at least 25, at least 30 contiguous nucleotides from SEQ ID NO: 14506 and the other transposon end comprises at least 17, at least 18, at least 19, at least 20, at least 22, at least 25, at least 30 contiguous nucleotides from SEQ ID NO: 14507.

(215) In certain embodiments, the piggyBac or piggyBac-like transposon comprises transposon ends (each end comprising an ITR) corresponding to SEQ ID NO: 14506 and SEQ ID NO: 14507, and has a target sequence corresponding to 5'-TTAT3'. In certain embodiments, the piggyBac or piggyBac-like transposon also comprises a sequence encoding a transposase (e.g. SEQ ID NO: 14505). In certain embodiments, the piggyBac or piggyBac-like transposon comprises one transposon end corresponding to SEQ ID NO: 14506 and a second transposon end corresponding to SEQ ID NO: 14516. SEQ ID NO: 14516 is very similar to SEQ ID NO: 14507, but has a large insertion shortly before the ITR. Although the ITR sequences for the two transposon ends are identical (they are both identical to SEQ ID NO: 14510), they have different target sequences: the second transposon has a target sequence corresponding to 5'-TTAA-3', providing evidence that no change in ITR sequence is necessary to modify the target sequence specificity. The piggyBac or piggyBac-like transposase (SEQ ID NO: 14504), which is associated with the 5'-TTAA-3' target site differs from the 5'-TTAT-3'-associated transposase (SEQ ID NO: 14505) by only 4 amino acid changes (D322Y, S473C, A507T, H582R). In certain embodiments, the piggyBac or piggyBac-like transposase (SEQ ID NO: 14504), which is associated with the 5'-TTAA-3' target site is less active than the 5'-TTAT-3'-associated piggyBac or piggyBac-like transposase (SEQ ID NO: 14505) on the transposon with 5'-TTAT-3' ends. In certain embodiments, piggyBac or piggyBac-like transposons with 5'-TTAA-3' target sites can be converted to piggyBac or piggyBac-like transposases with 5'-TTAT-3' target sites by replacing 5'-TTAA-3' target sites with 5'-TTAT-3'. Such transposons can be used either with a piggyBac or piggyBac-like transposase such as SEQ ID NO: 14504 which recognizes the 5'-TTAT-3' target sequence, or with a variant of a transposase originally associated with the 5'-TTAA-3' transposon. In certain embodiments, the high similarity between the 5'-TTAA-3' and 5'-TTAT-3' piggyBac or piggyBac-like transposases demonstrates that very few changes to the amino acid sequence of a piggyBac or piggyBac-like transposase alter target sequence specificity. In certain embodiments, modification of any piggyBac or piggyBac-like transposon-transposase gene transfer system, in which 5'-TTAA-3' target sequences are replaced with 5'-TTAT-3'-target sequences, the ITRs remain the same, and the transposase is the original piggyBac or piggyBac-like transposase or a variant thereof resulting from using a low-level mutagenesis to introduce mutations into the transposase. In certain embodiments, piggyBac or piggyBac-like transposon transposase transfer systems can be formed by the modification of a 5'-TTAT-3'-active piggyBac or piggyBac-like transposon-transposase gene transfer systems in which 5'-TTAT-3' target sequences are replaced with 5'-TTAA-3'-target sequences, the ITRs remain the same, and the piggyBac or piggyBac-like transposase is the original transposase or a variant thereof.

(216) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Bombyx mori*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(217) TABLE-US-00084 (SEQ ID NO: 14577) 1 cccggcgagc atgaggcagg gtatctcata  
ccctggtaaa attttaaagt tgtgtatttt 61 ataaaatttt cgtctgacaa cactagcgcg ctcatgtagct  
ggaggcagga gcggtcgagg 121 ggggatagtg gcggtgatcgc agtgtggcac gggacaccgg cgagatattc  
gtgtgcaaac 181 ctgtttcggg tatgttatac cctgcctcat tgttgacgta t.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(218) TABLE-US-00085 (SEQ ID NO: 14578) 1 tttaagaaaa agattaataa ataataataa  
tttcataatt aaaaacttct ttcatgaat 61 gccattaaat aaaccattat ttacaaaat aagatcaaca taattgagta  
aataataata 121 agaacaatat tatagtacaa caaatatgg gtatgtcata ccctgccaca ttcttgatgt 181 aactttttt

cacctcatgc tcgccggg.

In certain embodiments, the transposon comprises at least 16 contiguous bases from SEQ ID NO: 14577 and at least 16 contiguous bases from SEQ ID NO: 14578, and inverted terminal repeats that are at least 87% identical to CCCGGCGAGCATGAGG (SEQ ID NO: 14510). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(219) TABLE-US-00086 (SEQ ID NO: 14595) 1 cccggcgagc atgaggcagg gtatctcata  
ccctggtaaa attttaaagt tgtgtatgtt 61 ataaaatttt cgtctgacaa cactagcgcg ctcatagtagct  
ggaggcagga gcggtgcggga 121 ggggatagtg gcggtgatcgc agtgtggcac gggacaccgg cgagatattc  
gtgtgcaaac 181 ctgtttcggg tatgttatac cctgcctcat tgttgacgta tttttttat gtaattttc 241 cgattattaa  
tttcaactgt tttattggta ttttatgtt atccattgtt cttttttat 301 gatttactgt atcggttgc tttcgttctt  
ttagttgagt tttttttat tatttcagtt 361 ttttgatcaa a.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(220) TABLE-US-00087 (SEQ ID NO: 14596) 1 tcatatttt agtttaaaaa aataattata  
tgttttataa tgaaaagaat ctcatatct 61 ttcagtatta ggttgattta tattccaaag aataatattt ttgttaaatt  
gttgattttt 121 gtaaacctct aaatgtttgt tgctaaaatt actgtgttta agaaaaagat taataaataa 181 taataatttc  
ataattaaaa acttctttca ttgaatgcca ttaaataaac cattatttta 241 caaaataaga tcaacataat tgagtaaata  
ataataagaa caatattata gtacaacaaa 301 atatgggtat gtcataacct gccacattct tgatgtaact tttttcacc  
tcatgctcgc 361 cggg.

(221) In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14595 and SEQ ID NO: 14596, and is transposed by the piggyBac or piggyBac-like transposase of SEQ ID NO: 14505. In certain embodiments, the ITRs of SEQ ID NO: 14595 and SEQ ID: 14596 are not flanked by a 5'-TTAA-3' sequence. In certain embodiments, the ITRs of SEQ ID NO: 14595 and SEQ ID: 14596 are flanked by a 5'-TTAT-3' sequence.

(222) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(223) TABLE-US-00088 (SEQ ID NO: 14597) 1 cccggcgagc atgaggcagg gtatctcata  
ccctggtaaa attttaaagt tgtgtatgtt 61 ataaaatttt cgtctgacaa cactagcgcg ctcatagtagct  
ggaggcagga gcggtgcggga 121 ggggatagtg gcggtgatcgc agtgtggcac gggacaccgg cgagatattc  
gtgtgcaaac 181 ctgtttcggg tatgttatac cctgcctcat tgttgacgta tttttttat gtaattttc 241 cgattattaa  
tttcaactgt tttattggta ttttatgtt atccattgtt cttttttat 301 g.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(224) TABLE-US-00089 (SEQ ID NO: 14598) 1 cagggtatct cataacctgg taaaatttta  
aagttgtgta ttttataaaa ttttcgtctg 61 acaacactag cgcgctcagt agctggaggc aggagcgtgc  
gggaggggat agtggcgtga 121 tcgcagtgtg gcacgggaca ccggcgagat attcgtgtgc aaacctgttt  
cgggtatgtt 181 atacctgcc tcattgttga cgtattttt ttatgtaatt ttccgatta ttaatttcaa 241 ctgtttttat  
ggattttta ttttatccat tttctttt ttatg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(225) TABLE-US-00090 (SEQ ID NO: 14599) 1 cagggtatct cataacctgg taaaatttta  
aagttgtgta ttttataaaa ttttcgtctg 61 acaacactag cgcgctcagt agctggaggc aggagcgtgc  
gggaggggat agtggcgtga 121 tcgcagtgtg gcacgggaca ccggcgagat attcgtgtgc aaacctgttt  
cgggtatgtt 181 atacctgcc tcattgttga cgtat.

In certain embodiments, the 5' end of the piggyBac or piggyBac-like transposon comprises a sequence of SEQ ID NO: 14577, SEQ ID NO: 14595, or SEQ ID NOs: 14597-14599. In certain embodiments, the 5' end of the piggyBac or piggyBac-like transposon is preceded by a 5' target sequence.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(226) TABLE-US-00091 (SEQ ID NO: 14600) 1 tcatatttt agtttaaaaa aataattata  
tgttttataa tgaaaagaat ctcatatct 61 ttcagtatta ggttgattta tattccaaag aataatattt ttgttaaatt  
gttgattttt 121 gtaaacctct aaatgtttgt tgctaaaatt actgtgttta agaaaaagat taataaataa 181 taataatttc  
ataattaaaa acttctttca ttgaatgcca ttaaataaac cattatttta 241 caaaataaga tcaacataat tgagtaaata  
ataataagaa caatattata gtacaacaaa 301 atatgggtat gtcataacct gccacattct tgatgtaact tttttcacc  
tcatgctcgc 361 cggg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(227) TABLE-US-00092 (SEQ ID NO: 14601) 1 ttaagaaaa agattaataa ataataataa

tttcataatt aaaaacttct ttcattgaat 61 gccattaat ttacaaaat aagatcaaca taattgagta  
aataataata 121 agaacaatat tatagtacaa caaaatatgg gtatgtcata ccctgccaca ttctgatgt 181 aactttttt  
ca.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(228) TABLE-US-00093 (SEQ ID NO: 14602) 1 cccggcgagc atgaggcagg gtatctcata  
ccctggtaaa attttaaagt tgtgtatttt 61 ataaaatttt cgtctgacaa cactagcgcg ctcatagtagt  
ggaggcagga gcggtcgggga 121 ggggatagtg gcggtgatcgc agtgtggcac gggacaccgg cgagatattc  
gtgtgcaaac 181 ctgttcggg tatgttatac cctgcctcat tgttgacgta tttttttat gtaattttc 241 cgattattaa  
tttcaactgt ttattggta ttttatgtt atccattgtt cttttttat 301 gatttactgt atcggtgtc ttcgttctc  
ttagttgagt tttttttat tattttcagt 361 ttttgatcaa a.

(229) In certain embodiments, the 3' end of the piggyBac or piggyBac-like transposon comprises a sequence of SEQ ID NO: 14578, SEQ ID NO: 14596, or SEQ ID NOs: 14600-14601. In certain embodiments, the 3' end of the piggyBac or piggyBac-like transposon is followed by a 3' target sequence. In certain embodiments, the transposon is transposed by the transposase of SEQ ID NO: 14505. In certain embodiments, the 5' and 3' ends of the piggyBac or piggyBac-like transposon share a 16 bp repeat sequence of SEQ ID NO: 14510 in inverted orientation and immediately adjacent to the target sequence. In certain embodiments, the 5' transposon end begins with SEQ ID NO: 14510, and the 3' transposon end ends with the reverse complement of SEQ ID NO: 14510, 5'-

CCTCATGCTCGCCGGG-3' (SEQ ID NO: 14603). In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR with at least 93%, at least 87%, or at least 81% or any percentage in between identity to SEQ ID NO: 14510 or SEQ ID NO: 14603. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a target sequence followed by a 5' transposon end comprising a sequence selected from SEQ ID NOs: 88, 105 or 107 and a 3' transposon end comprising SEQ ID NO: 14578 or 106 followed by a target sequence. In certain embodiments, the piggyBac or piggyBac like transposon comprises one end that comprises a sequence that is at least 90%, at least 95% or at least 99% or any percentage in between identical to SEQ ID NO: 14577 and one end that comprises a sequence that is at least 90%, at least 95% or at least 99% or any percentage in between identical to SEQ ID NO: 14578. In certain embodiments, one transposon end comprises at least 14, at least 16, at least 18 or at least 20 contiguous bases from SEQ ID NO: 14577 and one transposon end comprises at least 14, at least 16, at least 18 or at least 20 contiguous bases from SEQ ID NO: 14578. (230) In certain embodiments, the piggyBac or piggyBac-like transposon comprises two transposon ends wherein each transposon ends comprises a sequence that is at least 81% identical, at least 87% identical or at least 93% identical or any percentage in between identical to SEQ ID NO: 14510 in inverted orientation in the two transposon ends. One end may further comprise at least 14, at least 16, at least 18 or at least 20 contiguous bases from SEQ ID NO: 14599, and the other end may further comprise at least 14, at least 16, at least 18 or at least 20 contiguous bases from SEQ ID NO: 14601. The piggyBac or piggyBac-like transposon may be transposed by the transposase of SEQ ID NO: 14505, and the transposase may optionally be fused to a nuclear localization signal.

(231) In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14595 and SEQ ID NO: 14596 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14597 and SEQ ID NO: 14596 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14595 and SEQ ID NO: 14578 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14602 and SEQ ID NO: 14600 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14504 or SEQ ID NO: 14505.

(232) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 5' end comprising 1, 2, 3, 4, 5, 6, or 7 sequences selected from ATGAGGCAGGGTAT (SEQ ID NO: 14614), ATACCCTGCCTCAT (SEQ ID NO: 14615), GGCAGGGTAT (SEQ ID NO: 14616), ATACCCTGCC (SEQ ID NO: 14617), TAAAATTTTA (SEQ ID NO: 14618), ATTTTATAAAAT (SEQ ID NO: 14619), TCATACCCTG (SEQ ID NO: 14620) and TAAATAATAATAA (SEQ ID NO: 14621). In certain

embodiments, the piggyBac or piggyBac-like transposon comprises a 3' end comprising 1, 2 or 3 sequences selected from SEQ ID NO: 14617, SEQ ID NO: 14620 and SEQ ID NO: 14621.

(233) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Xenopus tropicalis*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(234) TABLE-US-00094 (SEQ ID NO: 14517) 1 MAKRFYSAEE AAAHCMASST  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLPRY ARAHAWHPTD IAEMKRFBVGL  
TLAMGLIKAN 181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL RFLHFNNNAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSEFRAAVY TPCQNICIDE SLLLFKGRLLQ  
FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT 361 PACGTINRNR  
KGLPRALLDK KLNREGTYAL RKNELLAIFK FDKKNVFMILT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKVGIY  
LIQMALARNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP  
SDNVARLIGK HFIDTLPPPT 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCFKPCF EIYHTQLHY.

(235) In some embodiments, the piggyBac or piggyBac-like transposase is a hyperactive variant of SEQ ID NO: 14517. In certain embodiments, the piggyBac or piggyBac-like transposase is an integration defective variant of SEQ ID NO: 14517. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(236) TABLE-US-00095 (SEQ ID NO: 14518) 1 MAKRFYSAEE AAAHCMAPSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLPRY ARAHAWHPTD IAEMKRFBVGL  
TLAMGLIKAN 181 SLESYWNTTT VLSIPVFSAT MSRNRYQLLL RFLHFNNNAT  
AVPPDQPDHD RLHKLRLPLID 241 SLSEFRAAVY TPCQNICIDE SLLLFKGRLLR  
FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT 361 PACGTINRTR  
KGLPRALLDK KLNREGTYAL RKNELLAIFK FDKKNVFMILT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT SAWYKKVGIY  
LIQMALARNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMLP  
SDNVARLIGK HFIDTLPPPT 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCFKPCF EIYHTQLHY.

(237) In certain embodiments, the piggyBac or piggyBac-like transposase is isolated or derived from *Xenopus tropicalis*. In certain embodiments, the piggyBac or piggyBac-like transposase is a hyperactive piggyBac or piggyBac-like transposase. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence at least 90% identical to:

(238) TABLE-US-00096 (SEQ ID NO: 14572) 1 MAKRFYSAEE AAAHCSASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLTRG ARAHAWHPTD IAEMKRFBVGL  
TLAMGLIKAN 181 SIESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSEFANVY TPCQNICIDE SLMLFKGRLLQ  
FRQYIPSKRA RYGIKFYKLC ESSTGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT 361 PACGTINRNR



KGLPRALLDK KLNRGETYAL RKNELLAIFK FDKKNVFMLT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY  
LIQMALRNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPD  
SDNVARLIGK HFIDTLPPTP 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCRKPCF EIYHTQLHY.

(239) In certain embodiments, piggyBac or piggyBac-like transposase is a hyperactive piggyBac or piggyBac-like transposase. A hyperactive piggyBac or piggyBac-like transposase is a transposase that is more active than the naturally occurring variant from which it is derived. In certain embodiments, a hyperactive piggyBac or piggyBac-like transposase is more active than the transposase of SEQ ID NO: 14517. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(240) TABLE-US-00097 (SEQ ID NO: 14572) 1 MAKRFYSAEE AAAHCSASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLTRG ARAHAWHPTD IAEMKRFBVGL  
TLAMGLIKAN 181 SIESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSERFANVY TPCQNICIDE SLMLFKGRLQ  
FRQYIPSKRA RYGIKFYKLC ESSTGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT 361 PACGTINRNR  
KGLPRALLDK KLNRGETYAL RKNELLAIFK FDKKNVFMLT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY  
LIQMALRNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPD  
SDNVARLIGK HFIDTLPPTP 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCRKPCF EIYHTQLHY.

(241) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(242) TABLE-US-00098 (SEQ ID NO: 14624) 1 MAKRFYSAEE AAAHCMASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLTRY ARAHAWHPTD IAEMKRFBVGL  
TLAMGLIKAN 181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL RFLHFNNNAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSERFAAVY TPCQNICIDE SLLLFBKRLQ  
FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT 361 PACGTINRNR  
KGLPRALLDK KLNRGETYAL RKNELLAIFK FDKKNVFMLT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY  
LIQMALRNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPD  
SDNVARLIGK HFIDTLPPTP 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCRKPCF EIYHTQLHY.

(243) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(244) TABLE-US-00099 (SEQ ID NO: 14625) 1 MAKRFYSAEE AAAHCMASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLPRY ARAHAWHPTD IAEMKRFBVGL  
TLAMGLIKAN 181 SLESYWDTTT VLKIPVFSAT MSRNRYQLLL RFLHFNNNAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSERFAAVY TPCQNICIDE SLLLFBKRLQ  
FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT 361 PACGTINRNR  
KGLPRALLDK KLNRGETYAL RKNELLAIFK FDKKNVFMLT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY  
LIQMALRNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPD

SDNVARLIGK HFIDTLPPTP 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCFKPCF EIYHTQLHY.

(245) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(246) TABLE-US-00100 (SEQ ID NO: 14627) 1 MAKRFYSAEE AAAHCMASSS  
EQTS GSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLTRY ARAHAWHPTD IAEMKR FVGL  
TLAMGLIKAN 181 SIESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSERFANVY TPCQNICIDE SLLLFKGRLQ  
FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT 361 PACGTINRNR  
KGLPRALLDK KLNREGETYAL RKNELLAIFK FDKKNVFMLT SIHDESVIRE 421  
QRVGRKPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY  
LIQMALRNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMP  
SDNVARLIGK HFIDTLPPTP 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCRKPCF EIYHTQLHY.

(247) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(248) TABLE-US-00101 (SEQ ID NO: 14628) 1 MAKRFYSAEE AAAHCSASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLTRG ARAHAWHPTD IAEMKR FVGL  
TLAMGLIKAN 181 SLESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSERFANVY TPCQNICIDE SLMLFKGRLQ  
FRQYIPSKRA RYGIKFYKLC ESSTGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT 361 PACGTINRNR  
KGLPRALLDK KLNREGETYAL RKNELLAIFK FDKKNVFMLT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY  
LIQMALRNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMP  
SDNVARLIGK HFIDTLPPTP 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCRKPCF EIYHTQLHY.

(249) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(250) TABLE-US-00102 (SEQ ID NO: 16820) 1 MAKRFYSAEE AAAHCMASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLTRY ARAHAWHPTD IAEMKR FVGL  
TLAMGLIKAN 181 SLESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSERFANVY TPCQNICIDE SLLLFKGRLQ  
FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT 361 PACGTINRNR  
KGLPRALLDK KLNREGETYAL RKNELLAIFK FDKKNVFMLT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY  
LIQMALRNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMP  
SDNVARLIGK HFIDTLPPTP 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCRKPCF EIYHTQLHY.

(251) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution at a position selected from amino acid 6, 7, 16, 19, 20, 21, 22, 23, 24, 26, 28, 31, 34, 67, 73, 76, 77, 88, 91, 141, 145, 146, 148, 150, 157, 162, 179, 182, 189, 192, 193, 196, 198, 200, 210, 212, 218, 248, 263, 270, 294, 297, 308, 310, 333, 336, 354, 357, 358, 359, 377, 423, 426, 428, 438, 447, 450, 462, 469, 472, 498, 502, 517, 520, 523, 533, 534, 576, 577, 582, 583 or 587

(relative to SEQ ID NO: 14517). In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution of Y6C, S7G, M16S, S19G, S20Q, S20G, S20D, E21D, E22Q, F23T, F23P, S24Y, S26V, S28Q, V31K, A34E, L67A, G73H, A76V, D77N, P88A, N91D, Y141Q, Y141A, N145E, N145V, P146T, P146V, P146K, P148T, P148H, Y150G, Y150S, Y150C, H157Y, A162C, A179K, L182I, L182V, T189G, L192H, S193N, S193K, V196I, S198G, T200W, L210H, F212N, N218E, A248N, L263M, Q270L, S294T, T297M, S308R, L310R, L333M, Q336M, A354H, C357V, L358F, D359N, L377I, V 423H, P426K, K428R, S438A, T447G, T447A, L450V, A462H, A462Q, I469V, I472L, Q498M, L502V, E517I, P520D, P520G, N523S, I533E, D534A, F576R, F576E, K577, I582R, Y583F, L587Y or L587W, or any combination thereof including at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or all of these mutations (relative to SEQ ID NO: 14517).

(252) In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises one or more substitutions of an amino acid that is not wild type, wherein the one or more substitutions for a wild type amino acid comprises a substitution of A2X, K3X, R4X, F5X, Y6X, S7X, A11X, A13X, C15X, M16X, A17X, S18X, S19X, S20X, E21X, E22X, F23X, S24X, G25X, 26X, D27X, S28X, E29X, E42X, E43X, S44X, C46X, S47X, S48X, S49X, T50X, V51X, S52X, A53X, L54X, E55X, E56X, P57X, M58X, E59X, E62X, D63X, V64X, D65X, D66X, L67X, E68X, D69X, Q70X, E71X, A72X, G73X, D74X, R75X, A76X, D77X, A78X, A79X, A80X, G81X, G82X, E83X, P84X, A85X, W86X, G87X, P88X, P89X, C90X, N91X, F92X, P93X, E95X, I96X, P97X, P98X, F99X, T100X, T101X, P103X, G104X, V105X, K106X, V107X, D108X, T109X, N111X, P114X, I115X, N116X, F117X, F118X, Q119X, M122X, T123X, E124X, A125X, I126X, L127X, Q128X, D129X, M130X, L132X, Y133X, V126X, Y127X, A138X, E139X, Q140X, Y141X, L142X, Q144X, N145X, P146X, L147X, P148X, Y150X, A151X, A155X, H157X, P158X, I161X, A162X, V168X, T171X, L172X, A173X, M174X, I177X, A179X, L182X, D187X, T188X, T189X, T190X, L192X, S193X, I194X, P195X, V196X, S198X, A199X, T200X, S202X, L208X, L209X, L210X, R211X, F212X, F215X, N217X, N218X, A219X, T220X, A221X, V222X, P224X, D225X, Q226X, P227X, H229X, R231X, H233X, L235X, P237X, I239X, D240X, L242X, S243X, E244X, R244X, F246X, A247X, A248X, V249X, Y250X, T251X, P252X, C253X, Q254X, I256X, C257X, I258X, D259X, E260X, S261X, L262X, L263X, L264X, F265X, K266X, G267X, R268X, L269X, Q270X, F271X, R272X, Q273X, Y274X, I275X, P276X, S277X, K278X, R279X, A280X, R281X, Y282X, G283X, I284X, K285X, F286X, Y287X, K288X, L289X, C290X, E291X, S292X, S293XS294X, G295X, Y296X, T297X, S298X, Y299X, F300X, E304X, L310X, P313X, G314X, P316X, P317X, D318X, L319X, T320X, V321X, K324X, E328X, I330X, S331X, P332X, L333X, L334X, G335X, Q336X, F338X, L340X, D343X, N344X, F345X, Y346X, S347X, L351X, F352X, A354X, L355X, Y356X, C357X, L358X, D359X, T360X, R422X, Y423X, G424X, P426X, K428X, N429X, K430X, P431X, L432X, S434X, K435X, E436X, S438X, K439X, Y440X, G443X, R446X, T447X, L450X, Q451X, N455X, T460X, R461X, A462X, K465X, V467X, G468X, I469X, Y470X, L471X, I472X, M474X, A475X, L476X, R477X, S479X, Y480X, V482XY483X, K484X, A485X, A486X, V487X, P488X, P490X, K491X, S493X, Y494X, Y495X, K496X, Y497T, Q498X, L499X, Q500X, I501X, L502X, P503X, A504X, L505X, L506X, F507X, G508X, G509X, V510X, E511X, E512X, Q513X, T514X, V515X, E517X, M518X, P519X, P520X, S521X, D522X, N523X, V524X, A525X, L527X, I528X, K530X, H531X, F532X, I533X, D534X, T535X, L536X, T539X, P540X, Q546X, K550X, R553X, K554X, R555X, G556X, I557X, R558X, R559X, D560X, T561X, Y564X, P566X, K567X, P569X, R570X, N571X, L574X, C575X, F576X, K577X, P578X, F580X, E581X, I582X, Y583X, T585X, Q586X, L587X, H588X or Y589X (relative to SEQ ID NO: 14517). A list of hyperactive amino acid substitutions can be found in U.S. Pat. No. 10,041,077, the contents of which are incorporated by reference in their entirety.

(253) In certain embodiments, the piggyBac or piggyBac-like transposase is integration deficient. In certain embodiments, an integration deficient piggyBac or piggyBac-like transposase is a transposase that can excise its corresponding transposon, but that integrates the excised transposon at a lower frequency than a corresponding naturally occurring transposase. In certain embodiments, the piggyBac or piggyBac-like transposase is an integration deficient variant of SEQ ID NO: 14517. In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase is deficient relative to SEQ ID NO: 14517.

(254) In certain embodiments, the piggyBac or piggyBac-like transposase is active for excision but deficient in integration. In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of

(255) TABLE-US-00103 (SEQ ID NO: 14605) 1 MAKRFYSAEE AAAHCMASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRVDAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNPLPRY ARAHAWHPTD IAEMKRFBVGL  
TLAMGLIKAN 181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL KFLHFNNEAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSEFAAVY TPCQNICIDE SLLLFKGRLQ  
FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT 361 PACGTINRNR  
KGLPRALLDK KLNREGTYAL RKNELLAIF FDKKNVFMILT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKVGIY  
LIQMALARNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP  
SDNVARLIGK HFIDTLPPPT 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCFKPCF EIYHTQLHYG RR.

(256) In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of:

(257) TABLE-US-00104 (SEQ ID NO: 14604) 1 MAKRFYSAEE AAAHCMASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQVPLPRY ARAHAWHPTD IAEMKRFBVGL  
TLAMGLIKAN 181 SLESYWDTTT VLNIPVFSAT MSRNRYQLLL RFLEFNNEAT  
AVPPDQPGHD RLHKLRLPLID 241 SLSEFAAVY TPCQNICIDE SLLLFKGRLQ  
FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT 361 PACGTINRNR  
KGLPRALLDK KLNREGTYAL RKNELLAIF FDKKNVFMILT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKVGIY  
LIQMALARNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP  
SDNVARLIGK HFIDTLPPPT 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCFKPCF EIYHTQLHY.

(258) In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of:

(259) TABLE-US-00105 (SEQ ID NO: 14611) 1 MAKRFYSAEE AAAHCMASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
FMTEAILQDM VLYTNVYAEQ YLTQNVLPY ARAHAWHPTD IAEMKRFBVGL  
TLAMGLIKAN 181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL RFLHFNN DAT  
AVPPDQPGHD RLHKLRLPLID 241 SLTERFAAVY TPCQNICIDE SLLLFKGRLQ  
FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT 361 PACGTINRNR  
KGLPRALLDK KLNREGTYAL RKNELLAIF FDKKNVFMILT SIHDESVIRE 421  
QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKVGIY  
LIQMALARNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP  
SDNVARLIGK HFIDTLPPPT 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
NPGLCFKPCF EIYHTQLHYG RR.

(260) In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14611. In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of:

(261) TABLE-US-00106 (SEQ ID NO: 14612) 1 MAKRFYSAEE AAAHCMASSS  
EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
EAGDRADAAP GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121

FMTEAILQDM VLYTNVYAEQ YLTQVPLPRY ARAHAWHPTD IAEMKRFVGL  
 TLAMGLIKAN 181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL RFLHFNNEAT  
 AVPPDQPGHD RLHKLRLPLID 241 SLSEFAAVY TPCQNICIDE SLLLFKGRLQ  
 FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
 VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT 361 PACGTINRNR  
 KGLPRALLDK KLNREGTYAL RKNELLAIF FDKKNVFMILT SIHDESVIRE 421  
 QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKVGIY  
 LIQMALRNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP  
 SDNVARLIGK HFIDTLPPPT 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
 NPGLCFKPCF EIYHTQLHYG RR.

(262) In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14612. In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of:

(263) TABLE-US-00107 (SEQ ID NO: 14613) 1 MAKRFYSAEE AAAHCMASSS  
 EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV 61 DEDVDDLEDQ  
 EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL 121  
 FMTEAILQDM VLYTNVYAEQ YLTQVPLPRY ARAHAWHPTD IAEMKRFVGL  
 TLAMGLIKAN 181 SLESYWDTTT VLNIPVFSAT MSRNRYQLLL RFLEFNNNAT  
 AVPPDQPGHD RLHKLRLPLID 241 SLSEFAAVY TPCQNICIDE SLLLFKGRLQ  
 FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF 301 LIYEGKDSKL DPPGCPPDLT  
 VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT 361 PACGTINRNR  
 KGLPRALLDK KLNREGTYAL RKNELLAIF FDKKNVFMILT SIHDESVIRE 421  
 QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKVGIY  
 LIQMALRNSY 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP  
 SDNVARLIGK HFIDTLPPPT 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR  
 NPGLCFKPCF EIYHTQLHYG RR.

(264) In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14613. In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises an amino acid substitution wherein the Asn at position 218 is replaced by a Glu or an Asp (N218D or N218E) (relative to SEQ ID NO: 14517).

(265) In certain embodiments, the excision competent, integration deficient piggyBac or piggyBac-like transposase comprises one or more substitutions of an amino acid that is not wild type, wherein the one or more substitutions a for wild type amino acid comprises a substitution of A2X, K3X, R4X, F5X, Y6X, S7X, A8X, E9X, E10X, A11X, A12X, A13X, H14X, C15X, M16X, A17X, S18X, S19X, S20X, E21X, E22X, F23X, S24X, G25X, 26X, D27X, S28X, E29X, V31X, P32X, P33X, A34X, S35X, E36X, S37X, D38X, S39X, S40X, T41X, E42X, E43X, S44X, W45X, C46X, S47X, S48X, S49X, T50X, V51X, S52X, A53X, L54X, E55X, E56X, P57X, M58X, E59X, V60X, M122X, T123X, E124X, A125X, L127X, Q128X, D129X, L132X, Y133X, V126X, Y127X, E139X, Q140X, Y141X, L142X, T143X, Q144X, N145X, P146X, L147X, P148X, R149X, Y150X, A151X, H154X, H157X, P158X, T159X, D160X, I161X, A162X, E163X, M164X, K165X, R166X, F167X, V168X, G169X, L170X, T171X, L172X, A173X, M174X, G175X, L176X, I177X, K178X, A179X, N180X, S181X, L182X, S184X, Y185X, D187X, T188X, T189X, T190X, V191X, L192X, S193X, I194X, P195X, V196X, F197X, S198X, A199X, T200X, M201X, S202X, R203X, N204X, R205X, Y206X, Q207X, L208X, L209X, L210X, R211X, F212X, L213X, H241X, F215X, N216X, N217X, N218X, A219X, T220X, A221X, V222X, P223X, P224X, D225X, Q226X, P227X, G228X, H229X, D230X, R231X, H233X, K234X, L235X, R236X, L238X, I239X, D240X, L242X, S243X, E244X, R244X, F246X, A247X, A248X, V249X, Y250X, T251X, P252X, C253X, Q254X, N255X, I256X, C257X, I258X, D259X, E260X, S261X, L262X, L263X, L264X, F265X, K266X, G267X, R268X, L269X, Q270X, F271X, R272X, Q273X, Y274X, I275X, P276X, S277X, K278X, R279X, A280X, R281X, Y282X, G283X, I284X, K285X, F286X, Y287X, K288X, L289X, C290X, E291X, S292X, S293X, S294X, G295X, Y296X, T297X, S298X, Y299X, F300X, I302X, E304X, G305X, K306X, D307X, S308X, K309X, L310X, D311X, P312X, P313X, G314X, C315X, P316X, P317X, D318X, L319X, T320X, V321X,

S322X, G323X, I324X, V326X, W327X, E328X, L329X, I330X, S331X, P332X, L333X, L334X, G335X, Q336X, F338X, H339X, L340X, V342X, N344X, F345X, Y346X, S347X, S348X, I349X, L351X, T353X, A354X, Y356X, C357X, L358X, D359X, T360X, P361X, A362X, C363X, G364X, I366X, N367X, R368X, D369X, K371X, G372X, L373X, R375X, A376X, L377X, L378X, D379X, K380X, K381X, L382X, N383X, R384XG385X, T387X, Y388X, A389X, L390X, K392X, N393X, E394X, A397X, K399X, F400X, F401X, D402X, N405X, L406X, L409X, R422X, Y423X, G424X, E425X, P426X, K428X, N429X, K430X, P431X, L432X, S434X, K435X, E436X, S438X, K439X, Y440X, G442X, G443X, V444X, R446X, T447X, L450X, Q451X, H452X, N455X, T457X, R458X, T460X, R461X, A462X, Y464X, K465X, V467X, G468X, I469X, L471X, I472X, Q473X, M474X, L476X, R477X, N478X, S479X, Y480X, V482XY483X, K484X, A485X, A486X, V487X, P488X, G489X, P490X, K491X, L492X, S493X, Y494X, Y495X, K496X, Q498X, L499X, Q500X, I501X, L502X, P503X, A504X, L505X, L506X, F507X, G508X, G509X, V510X, E511X, E512X, Q513X, T514X, V515X, E517X, M518X, P519X, P520X, S521X, D522X, N523X, V524X, A525X, L527X, I528X, G529X, K530X, F532X, I533X, D534X, T535X, L536X, P537X, P538X, T539X, P540X, G541X, F542X, Q543X, R544X, P545X, Q546X, K547X, G548X, C549X, K550X, V551X, C552X, R553X, K554X, R555X, G556X, I557X, R558X, R559X, D560X, T561X, R562X, Y563X, Y564X, C565X, P566X, K567X, C568X, P569X, R570X, N571X, P572X, G573X, L574X, C575X, F576X, K577X, P578X, C579X, F580X, E581X, I582X, Y583X, H584X, T585X, Q586X, L587X, H588X or Y589X (relative to SEQ ID NO: 14517). A list of excision competent, integration deficient amino acid substitutions can be found in U.S. Pat. No. 10,041,077, the contents of which are incorporated by reference in their entirety.

(266) In certain embodiments, the piggyBac or piggyBac-like transposase is fused to a nuclear localization signal. In certain embodiments, SEQ ID NO: 14517 or SEQ ID NO: 14518 is fused to a nuclear localization signal. In certain embodiments, the amino acid sequence of the piggyBac or piggyBac like transposase fused to a nuclear localization signal is encoded by a polynucleotide sequence comprising:

(267) TABLE-US-00108 (SEQ ID NO: 14626)                      1 atggcaccca    aaaagaaacg    taaagtgatg  
gccaaaagat    ttacagcgc    cgaagaagca    61 gcagcacatt    gcatggcatc    gtcattccgaa    gaatttctcg  
ggagcgattc    cgaatatgtc    121 ccaccggcct    cggaaagcga    ttcgagcact    gaggagtcgt    ggtgttctc  
ctcaactgtc    181 tcggctcttg    aggagccgat    ggaagtggat    gaggatgtgg    acgacttgga    ggaccaggaa    241  
gccggagaca    gggccgacgc    tgccgcggga    ggggagccgg    cgtggggacc    tccatgcaat    301 tttctcccc  
aaatcccacc    gttcactact    gtgccgggag    tgaaggtcga    cacgtccaac    361 ttcgaaccga    tcaatttctt  
tcaactcttc    atgactgaag    cgatctgca    agatatggtg    421 ctctacacta    atgtgtacgc    cgagcagtac  
ctgactcaaa    acccgctgcc    tcgctacgcg    481 agagcgcattg    cgtggcaccc    gaccgatatc    gcggagatga  
agcggttcgt    gggactgacc    541 ctgcgaatgg    gcctgatcaa    ggccaacagc    ctcgagtcac    actgggatac  
cacgactgtg    601 cttagcattc    cggtgttctc    cgctaccatg    tcccgttaacc    gctaccaact    cctgctgcgg    661  
ttctccact    tcaacaacaa    tgcgaccgct    gtgccacctg    accagccagg    acacgacaga    721 ctccacaagc  
tgccggccatt    gatcgactcg    ctgagcgagc    gattcgccgc    ggtgtacacc    781 ccttgccaaa    acatttgcac  
cgacgagtcg    ctctgctgt    ttaaaggccg    gcttcagttc    841 cgccagtaca    tccatcgaa    gcgcgctcgc  
tatggtatca    aattctacaa    actctgcgag    901 tcgtccagcg    gctacacgtc    atacttcttg    atctacgagg  
ggaaggactc    taagctggac    961 ccaccggggg    gtccaccgga    tcttactgtc    tccggaaaaa    tcgtgtggga  
actcatctca    1021 cctctcctcg    gacaaggctt    tcatctctac    gtcgacaatt    tctactcatc    gatccctctg    1081  
ttaccgccc    tctactgcct    ggatactcca    gcctgtggga    ccattaacag    aaaccggaag    1141 ggtctgccga  
gagcactgct    ggataagaag    ttgaacaggg    gagagactta    cgcgctgaga    1201 aagaacgaac    tcctcgccat  
caaattcttc    gacaagaaaa    atgtgtttat    gctcacctcc    1261 atccacgacg    aatccgtcat    ccgggagcag  
cgcgtgggca    ggccgcccga    aaacaagccg    1321 ctgtgctcta    aggaatactc    caagtacatg    ggggggtgtc  
accggaccga    tcagctgcag    1381 cattactaca    acgccactag    aaagaccggg    gcctggatca    agaaagtcgg  
catctacctg    1441 atccaaatgg    cactgaggaa    ttcgtatatt    gtctacaagg    ctgccgttcc    gggcccga    1501  
ctgtcactact    acaagtacca    gttcaaactc    ctgccggcgc    tgctgttcgg    tggagtggaa    1561 gaacagactg  
tgcccagat    gccgccatcc    gacaacgtgg    cccggttgat    cggaaagcac    1621 ttattgata    ccctgcctcc  
gacgcctgga    aagcagcggc    cacagaaggg    atgcaaagtt    1681 tgccgcaagc    gcggaatacg    gcgcgatacc  
cgctactatt    gcccgaaagt    ccccgcaat    1741 cccggactgt    gtttcaagcc    ctgtttgaa    atctaccaca

ttacgtttgca ttac.

(268) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Xenopus tropicalis*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(269) TABLE-US-00109 (SEQ ID NO: 14519) 1 ttaacctttt tactgccaat gacgcatggg  
atacgtcgtg gcagtaaaag ggcttaaatg 61 ccaacgacgc gtcccatagc ttgttgcat ttaagtctt  
ctctctgcag cggcagcatg 121 tgccgccgct gcagagagtt tctagcgatg acagcccctc tgggcaacga  
gccggggggg 181 ctgtc.

(270) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(271) TABLE-US-00110 (SEQ ID NO: 14520) 1 ttgcatttt tagacattta gaagcctata  
tctgtttaca gaattggaat tacacaaaaa 61 ttctaccata tttgaaagc ttaggttgtt ctgaaaaaaa caatatattg  
tttctctggg 121 taaactaaaa gtcccctcga ggaaaggccc ctaaagttaa acagtgcaaa acgttcaaaa 181  
actgtctggc aatacaagtt ccactttgac caaaacggct ggcagtaaaa ggggttaa.

(272) In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14519 and SEQ ID NO: 14520. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(273) TABLE-US-00111 (SEQ ID NO: 14521) 1 ttaacctttt gcctgccaat cacgcatggg  
atacgtcgtg gcagtaaaag ggcttaaatg 61 ccaacgacgc gtcccatagc ttgttgcat ttaagtctt  
ctctctgcag cggcagcatg 121 tgccgccgct gcagagagtt tctagcgatg acagcccctc tgggcaacga  
gccggggggg 181 ctgtc.

(274) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(275) TABLE-US-00112 (SEQ ID NO: 14522) 1 ttgcatttt tagacattta gaagcctata  
tctgtttaca gaattggaat tacacaaaaa 61 ttctaccata tttgaaagc ttaggttgtt ctgaaaaaaa caatatattg  
tttctctggg 121 taaactaaaa gtcccctcga ggaaaggccc ctaaagttaa acagtgcaaa acgttcaaaa 181  
actgtctggc aatacaagtt ccactttggg acaaatcggc tggcagttaa aggggttaa.

(276) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(277) TABLE-US-00113 (SEQ ID NO: 14523) 1 ttaacctttt tactgccaat gacgcatggg  
atacgtcgtg gcagtaaaag ggcttaaatg 61 ccaacgacgc gtcccatagc ttgttgcat  
ttaaattctt ctctctgcag cggcagcatg 121 tgccgccgct gcagagagtt tctagcgatg acagcccctc  
tgggcaacga gccggggggg 181 ctgtc.

(278) In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14520 and SEQ ID NO: 14519, SEQ ID NO: 14521 or SEQ ID NO: 14523. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14522 and SEQ ID NO: 14519, SEQ ID NO: 14521 or SEQ ID NO: 14523. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end comprising at least 14, 16, 18, 20, 30 or 40 contiguous nucleotides from SEQ ID NO: 14519, SEQ ID NO: 14521 or SEQ ID NO: 14523. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end comprising at least 14, 16, 18, 20, 30 or 40 contiguous nucleotides from SEQ ID NO: 14520 or SEQ ID NO: 14522. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end with at least 90% identity to SEQ ID NO: 14519, SEQ ID NO: 14521 or SEQ ID NO: 14523. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end with at least 90% identity to SEQ ID NO: 14520 or SEQ ID NO: 14522. In one embodiment, one transposon end is at least 90% identical to SEQ ID NO: 14519 and the other transposon end is at least 90% identical to SEQ ID NO: 14520.

(279) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TTAACCTTTTACTGCCA (SEQ ID NO: 14524). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TTAACCCTTTGCCTGCCA (SEQ ID NO: 14526). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TTAACCYTTTACTGCCA (SEQ ID NO: 14527). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TGGCAGTAAAAGGGTTAA (SEQ ID NO: 14529). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TGGCAGTGAAAGGGTTAA (SEQ ID NO: 14531). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TTAACCYTTTKMCTGCCA (SEQ ID NO: 14533).

In certain embodiments, one end of the piggyBac or piggyBac-like transposon comprises a sequence selected from SEQ ID NO: 14524, SEQ ID NO: 14526 and SEQ ID NO: 14527. In certain embodiments, one end of the piggyBac (PB) or piggyBac-like transposon comprises a sequence selected from SEQ ID NO: 14529 and SEQ ID NO: 14531. In certain embodiments, each inverted terminal repeat of the piggyBac or piggyBac-like transposon comprises a sequence of ITR sequence of CCYTTTKMCTGCCA (SEQ ID NO: 14563). In certain embodiments, each end of the piggyBac (PB) or piggyBac-like transposon comprises SEQ ID NO: 14563 in inverted orientations. In certain embodiments, one ITR of the piggyBac or piggyBac-like transposon comprises a sequence selected from SEQ ID NO: 14524, SEQ ID NO: 14526 and SEQ ID NO: 14527. In certain embodiments, one ITR of the piggyBac or piggyBac-like transposon comprises a sequence selected from SEQ ID NO: 14529 and SEQ ID NO: 14531. In certain embodiments, the piggyBac or piggyBac like transposon comprises SEQ ID NO: 14533 in inverted orientation in the two transposon ends.

(280) In certain embodiments, The piggyBac or piggyBac-like transposon may have ends comprising SEQ ID NO: 14519 and SEQ ID NO: 14520 or a variant of either or both of these having at least 90% sequence identity to SEQ ID NO: 14519 or SEQ ID NO: 14520, and the piggyBac or piggyBac-like transposase has the sequence of SEQ ID NO: 14517 or a variant showing at least %, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between sequence identity to SEQ ID NO: 14517 or SEQ ID NO: 14518. In certain embodiments, one piggyBac or piggyBac-like transposon end comprises at least 14 contiguous nucleotides from SEQ ID NO: 14519, SEQ ID NO: 14521 or SEQ ID NO: 14523, and the other transposon end comprises at least 14 contiguous nucleotides from SEQ ID NO: 14520 or SEQ ID NO: 14522. In certain embodiments, one transposon end comprises at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 22, at least 25, at least 30 contiguous nucleotides from SEQ ID NO: 14519, SEQ ID NO: 14521 or SEQ ID NO: 14523, and the other transposon end comprises at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 22, at least 25 or at least 30 contiguous nucleotides from SEQ ID NO: 14520 or SEQ ID NO: 14522.

(281) In certain embodiments, the piggyBac or piggyBac-like transposase recognizes a transposon end with a 5' sequence corresponding to SEQ ID NO: 14519, and a 3' sequence corresponding to SEQ ID NO: 14520. It will excise the transposon from one DNA molecule by cutting the DNA at the 5'-TTAA-3' sequence at the 5' end of one transposon end to the 5'-TTAA-3' at the 3' end of the second transposon end, including any heterologous DNA that is placed between them, and insert the excised sequence into a second DNA molecule. In certain embodiments, truncated and modified versions of the 5' and 3' transposon ends will also function as part of a transposon that can be transposed by the piggyBac or piggyBac-like transposase. For example, the 5' transposon end can be replaced by a sequence corresponding to SEQ ID NO: 14521 or SEQ ID NO: 14523, the 3' transposon end can be replaced by a shorter sequence corresponding to SEQ ID NO: 14522. In certain embodiments, the 5' and 3' transposon ends share an 18 bp almost perfectly repeated sequence at their ends (5'-TTAACCYTTTKMCTGCCA: SEQ ID NO: 14533) that includes the 5'-TTAA-3' insertion site, which sequence is inverted in the orientation in the two ends. That is in (SEQ ID NO: 14519) and SEQ ID NO: 14523 the 5' transposon end begins with the sequence 5'-TTAACCTTTTTACTGCCA-3' (SEQ ID NO: 14524), or in (SEQ ID NO: 14521) the 5' transposon end begins with the sequence 5'-TTAACCTTTGCCTGCCA-3' (SEQ ID NO: 14526); the 3' transposon ends with approximately the reverse complement of this sequence: in SEQ ID NO: 14520 it ends 5' TGGCAGTAAAAGGGTTAA-3' (SEQ ID NO: 14529), in (SEQ ID NO: 14522) it ends 5'-TGGCAGTGAAAGGGTTAA-3' (SEQ ID NO: 14531.) One embodiment of the invention is a transposon that comprises a heterologous polynucleotide inserted between two transposon ends each comprising SEQ ID NO: 14533 in inverted orientations in the two transposon ends. In certain embodiments, one transposon end comprises a sequence selected from SEQ ID NOS: 14524, SEQ ID NO: 14526 and SEQ ID NO: 14527. In some embodiments, one transposon end comprises a sequence selected from SEQ ID NO: 14529 and SEQ ID NO: 14531.

(282) In certain embodiments, the piggyBac (PB) or piggyBac-like transposon is isolated or derived from *Xenopus tropicalis*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:



(283) TABLE-US-00114 (SEQ ID NO: 14573) 1 ccctttgcct gccaatcacg catgggatac  
gtcgtggcag taaaagggct taaatgccaa 61 cgacgcgtcc catacgtt.

(284) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(285) TABLE-US-00115 (SEQ ID NO: 14574) 1 cctgggtaaa ctaaaagtcc cctcgaggaa  
aggcccctaa agtgaacag tgcaaaacgt 61 tcaaaaactg tctggcaata caagttccac  
tttgggacaa atcggctggc agtgaaaggg.

(286) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at least 16 contiguous bases from SEQ ID NO: 14573 or SEQ ID NO: 14574, and inverted terminal repeat of CCYTTTBMCTGCCA (SEQ ID NO: 14575).

(287) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(288) TABLE-US-00116 (SEQ ID NO: 14579) 1 ccctttgcct gccaatcacg catgggatac  
gtcgtggcag taaaagggct taaatgccaa 61 cgacgcgtcc catacgttgt tggcatttta  
agtcttctct ctgcagcggc agcatgtgcc 121 gccgctgcag agagtttcta gcgatgacag cccctctggg  
caacgagccg ggggggctgt 181 c.

(289) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(290) TABLE-US-00117 (SEQ ID NO: 14580) 1 cctttttact gccaatgacg catgggatac  
gtcgtggcag taaaagggct taaatgccaa 61 cgacgcgtcc catacgttgt tggcatttta  
attcttctct ctgcagcggc agcatgtgcc 121 gccgctgcag agagtttcta gcgatgacag cccctctggg  
caacgagccg ggggggctgt 181 c.

(291) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(292) TABLE-US-00118 (SEQ ID NO: 14581) 1 cctttttact gccaatgacg catgggatac  
gtcgtggcag taaaagggct taaatgccaa 61 cgacgcgtcc catacgttgt tggcatttta  
agtcttctct ctgcagcggc agcatgtgcc 121 gccgctgcag agagtttcta gcgatgacag cccctctggg  
caacgagccg ggggggctgt 181 c .

(293) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(294) TABLE-US-00119 (SEQ ID NO: 14582) 1 cctttttact gccaatgacg catgggatac  
gtcgtggcag taaaagggct taaatgccaa 61 cgacgcgtcc catacgttgt tggcatttta  
agtcttctct ctgcagcggc agcatgtgcc 121 gccgctgcag agag.

(295) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(296) TABLE-US-00120 (SEQ ID NO: 14583) 1 cctttttact gccaatgacg catgggatac  
gtcgtggcag taaaagggct taaatgccaa 61 cgacgcgtcc catacgttgt tggcatttta agtctt.

(297) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(298) TABLE-US-00121 (SEQ ID NO: 14584) 1 ccctttgcct gccaatcacg catgggatac  
gtcgtggcag taaaagggct taaatgccaa 61 cgacgcgtcc catacgttgt tggcatttta agtctt.

(299) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(300) TABLE-US-00122 (SEQ ID NO: 14585) 1 ttatcctttt tactgccaat gacgcatggg  
atacgtcgtg gcagtaaaag ggcttaaatg 61 ccaacgacgc gtcccatagc ttgttggcat  
tttaagtctt ctctctgcag cggcagcatg 121 tgccgccgct gcagagagtt tctagcatg acagcccctc  
tgggcaacga gccggggggg 181 ctgtc.

(301) In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(302) TABLE-US-00123 (SEQ ID NO: 14586) 1 tttgcatttt tagacattta gaagcctata  
tcttgttaca gaattggaat tacacaaaaa 61 ttctaccata ttttgaaagc ttaggttgtt  
ctgaaaaaaa caatatattg ttttctggg 121 taaactaaaa gtcccctcga ggaaaggccc ctaaagtga  
acagtgcata acgttcaaaa 181 actgtctggc aatacaagtt ccactttggg acaaatcggc tggcagtga  
aggg.

(303) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 5' transposon end sequence selected from SEQ ID NO: 14573 and SEQ ID NOs: 14579-14585. In certain embodiments, the 5' transposon end sequence is preceded by a 5' target sequence. In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(304) TABLE-US-00124 (SEQ ID NO: 14587) 1 tttgcatttt tagacattta gaagcctata  
tcttgttaca gaattggaat tacacaaaaa 61 ttctaccata ttttgaaagc ttaggttgtt  
ctgaaaaaaa caatatattg ttttctggg 121 taaactaaaa gtcccctcga ggaaaggccc ctaaagtga

acagtgcacaa acgttcaaaa181 actgtctggc aatacaagtt ccactttgac caaacggct ggagtaaaa  
ggg .

(305) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(306) TABLE-US-00125 (SEQ ID NO: 14588) 1 ttgtctgaa aaaaacaata tattgtttc  
ctgggtaaac taaaagtccc ctcgaggaaa 61 ggcccctaaa gtgaaacagt gcaaacggt  
caaaaactgt ctggcaatac aagttccact 121 ttgacaaaaa cggctggcag taaaagg.

(307) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(308) TABLE-US-00126 (SEQ ID NO: 14589) 1 ttgcatttt tagacattta gaagcctata  
tctgtttaca gaattggaat tacacaaaaa 61 ttctaccata tttgaaagc ttaggttgtt  
ctgaaaaaaa caatatattg ttttctggg 121 taaactaaaa gtcccctcga ggaaaggccc ctaaagtga  
acagtgcacaa acgttcaaaa 181 actgtctggc aatacaagtt ccactttgac caaacggct ggagtaaaa  
gggttat.

(309) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(310) TABLE-US-00127 (SEQ ID NO: 14590) 1 ttgtctgaa aaaaacaata tattgtttc  
ctgggtaaac taaaagtccc ctcgaggaaa 61 ggcccctaaa gtgaaacagt gcaaacggt  
caaaaactgt ctggcaatac aagttccact 121 ttgggacaaa tcggctggca gtgaaagg.

(311) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 3' transposon end sequence selected from SEQ ID NO: 14574 and SEQ ID NOs: 14587-14590. In certain embodiments, the 3' transposon end sequence is followed by a 3' target sequence. In certain embodiments, the 5' and 3' transposon ends share a 14 repeated sequence inverted in orientation in the two ends (SEQ ID NO: 14575) adjacent to the target sequence. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 5' transposon end comprising a target sequence and a sequence that is selected from SEQ ID NOs: 14582-14584 and 14573, and a 3' transposon end comprising a sequence selected from SEQ ID NOs: 14588-14590 and 14574 followed by a 3' target sequence.

(312) In certain embodiments, the 5' transposon end of the piggyBac or piggyBac-like transposon comprises

(313) TABLE-US-00128 1 atcacgcatg ggatacgtcg tggcagtaaa agggcttaaa tgccaacgac  
gcgtcccata 61 cggt

(SEQ ID NO: 14591), and an ITR. In certain embodiments, the 5' transposon end comprises

(314) TABLE-US-00129 1 atgacgcatg ggatacgtcg tggcagtaaa agggcttaaa tgccaacgac  
gcgtcccata 61 cggtgtggc atttaagtc tt

(SEQ ID NO: 14592) and an ITR. In certain embodiments, the 3' transposon end of the piggyBac or piggyBac-like transposon comprises

(315) TABLE-US-00130 1 cctgggtaaa ctaaaagtcc cctcaggaaa agggccctaa agtgaacag  
tgcaaacgt 61 tcaaaaactg tctggcaata caagtccac ttgggacaa atcggc

(SEQ ID NO: 14593) and an ITR. In certain embodiments, the 3' transposon end comprises

(316) TABLE-US-00131 1 ttgtctgaa aaaaacaata tattgtttc ctgggtaaac taaaagtccc  
ctcaggagaaa 61 ggcccctaaa gtgaaacagt gcaaacggt caaaaactgt ctggcaatac aagttccact  
121 ttgacaaaaa cggc

(SEQ ID NO: 14594) and an ITR.

(317) In certain embodiments, one transposon end comprises a sequence that is at least 90%, at least 95%, at least 99% or any percentage in between identical to SEQ ID NO: 14573 and the other transposon end comprises a sequence that is at least 90%, at least 95%, at least 99% or any percentage in between identical to SEQ ID NO: 14574. In certain embodiments, one transposon end comprises at least 14, at least 16, at least 18, at least 20 or at least 25 contiguous nucleotides from SEQ ID NO: 14573 and one transposon end comprises at least 14, at least 16, at least 18, at least 20 or at least 25 contiguous nucleotides from SEQ ID NO: 14574. In certain embodiments, one transposon end comprises at least 14, at least 16, at least 18, at least 20 from SEQ ID NO: 14591, and the other end comprises at least 14, at least 16, at least 18, at least 20 from SEQ ID NO: 14593. In certain embodiments, each transposon end comprises SEQ ID NO: 14575 in inverted orientations.

(318) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence selected from of SEQ ID NO: 14573, SEQ ID NO: 14579, SEQ ID NO: 14581, SEQ ID NO: 14582, SEQ ID

NO: 14583, and SEQ ID NO: 14588, and a sequence selected from SEQ ID NO: 14587, SEQ ID NO: 14588, SEQ ID NO: 14589 and SEQ ID NO: 14586 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14517 or SEQ ID NO: 14518.

(319) In certain embodiments, the piggyBac or piggyBac-like transposon comprises ITRs of CCCTTTGCCTGCCA (SEQ ID NO: 14622) (5' ITR) and TGGCAGTGAAAGGG (SEQ ID NO: 14623) (3' ITR) adjacent to the target sequences.

(320) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Helicoverpa armigera*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(321) TABLE-US-00132 (SEQ ID NO: 14525) 1 MASRQRLNHD EIATILENDD  
DYSPLDSESE KEDCVVEDDV WSDNEDAIVD FVEDTSAQED 61  
PDNNIASRES PNLEVTSLTS HRIITLPQRS IRGKNNHVWS TTKGRTTGRT  
SAINIIRTNR 121 GPTRMCRNIV DPLLCFQLFI TDEIHEIVK WTNVEIIVKR  
QNLKDISASY RDINTMEIWA 181 LVGILTTLTAV MKDNHLSTDE LFDATFSGTR  
YVSVMSRERF EFLIRCIRMD DKTLRPTLRS 241 DDAFLPVRKI WEIFINQCRQ  
NHVPGSNLTV DEQLLGRFRGR CPFRMYIPNK PDKYGIKFBPM 301  
MCAAATKYMI DAIPYLKST KTNGLPLGEF YVKDLTKTVH GTNRNITCDN  
WFTSIPLAKN 361 MLQAPYNLTI VGTIRSNKRE MPEEIKNSRS  
RPVGSSMFCF DGPLTLVSYK PKPSKMVFL 421 SSCDENAVIN ESNGKPDMIL  
FYNQTKGGVD SFDQMCKSMS ANRKTNRWPM AVFYGMLNMA 481  
FVNSYIIYCH NKINKQEKPI SRKEFMKKLS IQLTTPWMQE RLQAPTLKRT  
LRDNITNVLK 541 NVVPASSENI SNEPEPKRR YCGVCSYKKR  
RMTKAQCKC KKAICGEHNI DVCQDCI.

(322) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Helicoverpa armigera*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(323) TABLE-US-00133 (SEQ ID NO: 14570) 1 ttaaccctag aagcccaatc tacgtaaatt  
tgacgtatac cgcggcgaaa tatctctgtc 61 tctttcatgt ttaccgtcgg atcgccgcta  
acttctgaac caactcagta gccattggga 121 cctcgcagga cacagttgcg tcatctcgggt aagtgcgccc  
atthttgtgt actctctatt 181 acaacacacg tcacgtcacg tcgttgacg tcattttgac gtataattgg  
gctttgtgta 241 acttttgaat ttgtttcaaa tttttatgt ttgtgattta ttgagttaa tcgtattgtt 301  
tcgttacatt ttcatataa taataatatt ttcaggttga gtacaaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(324) TABLE-US-00134 (SEQ ID NO: 14528) 1 agactgtttt tttctaagag acttctaaaa  
tattattacg agttgattta attttatgaa 61 aacatttaaa actagttgat ttttttata  
attacataat ttaagaaaa agtgtagag 121 gcttgatttt ttgttgatt tttctaaga ttgattaaa  
gtgccataat agtattaata 181 aagagtattt ttaacttaa aatgtatttt atttattaat taaaacttca  
attatgataa 241 ctcatgcaaa aatatagttc attaacagaa aaaaatagga aaactttgaa gttttgttt 301  
tacacgtcat ttttacgtat gattgggctt tatagctagt taaatatgat tgggcttcta 361 ggggttaa.

(325) in certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Pectinophora gossypiella*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(326) TABLE-US-00135 (SEQ ID NO: 14530) 1 MDLRKQDEKI RQWLEQDIEE  
DSKGESDNSS SETEDIVEME VHKNSTSESE VSSES DYEPV 61  
CPSKRQRTQI IESEESDNSE SIRPSRRQTS RVIDSDETDE DVMSSTPQNI  
PRNPVNIQPS 121 SRFLYGKNKH KWSSAAKPSS VRTSRRNIH FIPGPKERAR

EVSEPIDIFS LFISEDMLQQ 181 VVFTNAEML IRKNKYKTET FTVSPTNLEE  
IRALLGLLFN AAAMKSNHLP TRMLFNTHRS 241 GTIFKACMSA ERLNFLIKCL  
RFDDKLTRNV RQRDDRFAPL RDLWQALISN FQKWYTPGSY 301  
ITVDEQLVGF RGRCSFRMYI PNKPNKYGIK LVMAADVNSK YIVNAIPYLG  
KGTDPQNQPL 361 ATFFIKEITS TLHGTRNIT MDNWFTSVPL  
ANELLMAPYN LTLVGTLRSN KREIPEKLN 421 SKSRAIGTSM FCYDGDKTLV  
SYKAKSNKVV FILSTIHDQP DINQETGKPE MIHFYNSTKG 481  
AVDTVDMQCS SISTNRKTQR WPLCVFYNML NLSIINAYVV YVYNNVRNNK  
KPMSSRRDFVI 541 KLGDQLMEPW LRQLQTVTL REDIKVMIQD  
ILGESSDLEA PVPSVSNVRK IYYLCPSKAR 601 RMTKHRCIKC KQAICGPHNI  
DICSRCE.

(327) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Pectinophora gossypiella*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(328) TABLE-US-00136 (SEQ ID NO: 14532) 1 ttaaccctag ataactaac attcgccgc  
tcgacgacgc gctatgccgc gaaattgaag 61 ttacctatt attccgcgtc ccccgcccc  
gccgctttt ctagcttct gatttgcaaa 121 atagtgcac gcgtgacacg ctcgaggta caccagaatt  
aggctgaaag ttacaggaat 181 ttcgtcgcc gctcgacgaa agtttagtaa ttacgtaagt ttggcaaagg  
taagtgaatg 241 aagtatttt ttataattat ttttaattc tttatagtga taacgtaagg tttattaaa 301  
tttattactt ttatagttat ttagccaatt gttataaatt ccttggtatt gctgaaaaat 361 ttgcctgtt  
tagtcaaat ttattaactt ttcgatcgtt ttttag.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(329) TABLE-US-00137 (SEQ ID NO: 14571) 1 tttcactaag taatttggtt cctatttagt  
agataagtaa cacataatta ttgtgatatt 61 caaaacttaa gaggtttaat aaataataat  
aaaaaaaaa tgggttttat ttcgtagtct 121 gctcgacgaa tgtttagtta ttacgtaacc gtgaatatag  
tttagtagtc tagggttaa.

(330) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Ctenoplosia agnata*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(331) TABLE-US-00138 (SEQ ID NO: 14534) 1 MASRQHLYQD EIAAILENED  
DYSPHDTDSE MEDCVTQDDV RSDVEDEMVD NINGTSPAS 61  
RHEDPETPDP SSEASNLEVT LSSHRIILP QRSIREKNNH IWSTTKGQSS  
GRTAAINIVR 121 TNRGPTRMCR NIVDPLLCFQ LFIKEEIVVE  
IVKWTNVEMV QKRVNLKDIS ASYRDTNEME 181 IWAHSMMLTL SAVMKDNHLS  
TDELFNVSYG TRYVSVMSRE RFEFLRLLR MGDKLLRPNL 241  
RQEDAFTPVR KIWEIFINQC RLNYVPGTNL TVDEQLLGFR GRCPFRMYIP  
NKPDKYGIKF 301 PMVCDAATKY MVDAIPYLGK STKTQGLPLG  
EFYVKELTQT VHGTNRNVTC DNWFTSVPLA 361 KSLNLPYNL TLVGTIRSNK  
REIPEEVKNS RSRQVGSSMF CFDGPLTLVS YKPKPSKMVF 421  
LLSSCNEDAV VNQSNKPKDM ILFYNQTKGG VDSFDQMCS  
MSTNRKTNRW PMAVFYGMLN 481 MAFVNSYIY CHNMLAKKEK PLSRKDFMKK  
LSTDLTTPSM QKRLEAPTLK RSLRDNITNV 541 LKIVPQAAID  
TSFDEPEPKK RRYCGFCSYK KKRMTKTQCF KCKKPVCGEH NIDVCQDCI.

(332) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Ctenoplosia agnata*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(333) TABLE-US-00139 (SEQ ID NO: 14535) 1 ttaaccctag aagcccaatc tacgtcattc  
tgacgtgat gtcgccgaaa atactctgtc 61 tctttctct gcacgatcgg attgccgcga  
acgctcgatt caaccagtt ggcgccgaga 121 tctattggag gactgcggcg ttgattcggg aagtcgcc

attttgcat agtaacagta 181 ttgcacgtca gcttgacgta tatttgggct ttgtgttatt tttgtaaatt  
ttcaacgta 241 gtttattatt gcatctttt gttacattac tggtttattt gcatgtatta ctcaaatt 301  
atttttatt tagcgtagaa aataca.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(334) TABLE-US-00140 (SEQ ID NO: 14536) 1 agactgtttt tttgtattt gcattatata  
ttatattcta aagttgattt aattctaaga 61 aaaacattaa aataagtttc ttttgtaaa  
atttaattaa ttataagaaa aagtttaagt 121 tgatctcatt tttataaaa atttgcaatg tttccaaagt  
tattattgta aaagaataaa 181 taaaagtaaa ctgagtttta attgatgttt tattatatca ttatactata  
tattacttaa 241 ataaaacaat aactgaatgt atttctaaaa ggaatcacta gaaaatatag tgatcaaaaa  
301 tttacacgtc attttgcgt atgattgggc tttataggtt ctaaaaatat gattgggcct 361 ctagggttaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCTAGAAGCCCAATC (SEQ ID NO: 14564).

(335) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Agrotis ipsilon*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(336) TABLE-US-00141 (SEQ ID NO: 14537) 1 MESRQRLNQD EIATILENDD  
DYSPLDSDSE AEDRVVEDDV WSDNEDAMID YVEDTSRQED 61  
PDNNIASQES ANLEVTS LTS HRIISLPQRS ICGKNNHVWS TTKGRTTGRT  
SAINIIRTNR 121 GPTRMCRNIV DPLLCFQLFI TDEIHEIVK WTNVEMIVKR  
QNLIDISASY RDTNTMEMWA 181 LVGILTAV MKDNHLSTDE LFDATFSGTR  
YVSVMSRERF EFLIRCMRMD DKTLRPTLRS 241 DDAFIPVRKL  
WEIFINQCRL NYVPGGNLTV DEQLLGFRGR CPFRMYIPNK PDKYGIRFPM  
301 MCDAATKYMI DAIPYLKST KTNGPLGFEF YVKELTKTVH  
GTNRNVTC DN WFTSIPLAKN 361 MLQAPYNLTI VGTIRSNKRE IPEEIKNSRS  
RPVGSSMFCF DGPLTLVSYK PKPSRMVFL 421 SSCDENAVIN ESNGKPDMIL  
FYNQTKGGVD SFDQMCKSMS ANRKTNRWPM AVFYGMLNMA 481  
FVNSYIIYCH NKINKQKKPI NRKEFMKNLS TDLTPWMQE RLKAPTLKRT  
LRDNITNVLK 541 NVVPPSPANN SEEPGPKKRS YCGFCSYKKR  
RMTKTQFYKC KKAICGEHNI DVCQDCV.

(337) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Agrotis ipsilon*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(338) TABLE-US-00142 (SEQ ID NO: 14538) 1 ttaaccctag aagcccaatc tacgtaaatt  
tgacgtatac cgcggcgaaa tatatctgtc 61 tctttcacgt ttaccgtcgg attcccgcta  
acttcggaac caactcagta gccattgaga 121 actcccagga cacagttgcg tcatctcggg aagtgccgcc  
attttgtgt aatagacagg 181 ttgcacgtca tttgacgta taattgggct ttgtgtaact tttgaaatta  
ttataaatt 241 ttattgatgt gatttatttg agttaatcgt attgttcgt tacattttc atatgatatt 301  
aatattttca gattgaatat aaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(339) TABLE-US-00143 (SEQ ID NO: 14539) 1 agactgtttt ttttaaaagg cttataaagt attactattg  
cgtgatttaa tttataaaa 61 atatttaaaa ccagttgatt ttttaataa ttacctaatt ttaagaaaaa atgtagaag 121  
cttgatattt ttgttgattt ttttctaaga tttgattaaa aggccataat tgtattaata 181 aagagtattt ttaactcaa  
attatttta tttattaatt aaaactcaa ttatgataat 241 acatgcaaaa atatagttca tcaacagaaa aatataggaa  
aactctaata gttttattt 301 tacacgtcat ttttacgtat gattgggctt tatagctagt caaatatgat tgggcttcta 361  
gggttaa.

(340) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Megachile rotundata*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%10%, 15%, 20%,

25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(341) TABLE-US-00144 (SEQ ID NO: 14540) 1 MNGKDSLGEF YLDDLSDCLD  
CRSASSTDDE SDSSNIAIRK RCPIPLIYSD SEDEDMNNNV 61 EDNNHFVKES  
NRYHYQIVEK YKITSKTKKW KDVTVTEMKK FLGLIILMGQ VKKDVLYDYW 121  
STDPSIETPF FSKVMSRNR LQIMQSWHFY NNNDISPNSH RLVKIQPVID  
YFKEKFNNVY 181 KSDQQLSLDE CLIPWRGRLS IKTYNPAKIT KYGILVRVLS  
EARTGYVSNF CVYAADGKKI 241 EETVLSVIGP YKNMWHHVYQ DNYYNNSVNIA  
KIFLKNKLRV CGTIRKNRSL PQILQTVKLS 301 RGQHQLRNG HTLLEVWNNG  
KRVNVMISTI HSAQMAESRN RSRTSDCPIQ KPISIIDYNK 361 YMKGVD RADQ  
YLSYYSIFRK TKKWTKRVVM FFINCALFNS FKVYTTLNGQ KITYKNFLHK 421  
AALSLIEDCG TEEQGTDLPN SEPTTTRTTS RVDHPGRLEN FGKHKLVNIV  
TSGQCKKPLR 481 QCRVCASKKK LSRTGFACKY CNVPLHKGDC FERYHSLKKY.

(342) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Megachile rotundata*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(343) TABLE-US-00145 (SEQ ID NO: 14541) 1 ttaaataatg cccactctag atgaacttaa cactttaccg  
accggccgctc gattattcga 61 cgtttgctcc ccagcgctta ccgaccggcc atcgattatt cgacgtttgc ttccagcgc 121  
ttaccgaccg gtcatcgact ttgatcttt ccgttagatt tggtaggtc agattgacaa 181 gtagcaagca ttgcgcatc  
ttattcaaa taatcgggtgc tttttctaa gcttagccc 241 ttagaa.

(344) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(345) TABLE-US-00146 (SEQ ID NO: 14542) 1 acaacttctt ttttcaacaa atattgttat atggattatt  
tattatttta tttatttatg 61 gtatatttta tgtttattta tttatggta ttatggtata tttatgtaa ataataaact 121  
gaaaacgatt gtaatatag aaataaatat tgttttaaca ctaatatat taaagtaaaa 181 gattttaata aatttcgta  
ccctacaata acacgaagcg tacaatttta ccagagtta 241 ttaa.

(346) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Bombus impatiens*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(347) TABLE-US-00147 (SEQ ID NO: 14543) 1 MNEKNGIGEF YLDDLSDCPD  
SYSRSNSGDE SDGSDTIIRK RGSVLPPRYS DSEDDEINNV 61 EDNANNVENN  
DDIWSTNDEA IILEPFEGSP GLKIMPSSAE SVTDNVNLFF GDDFFEHLVR 121  
ESNRYHYQVM EKYKIPSKAK KWTDITVPEM KKFLGLIVLM GQIKKDVLYD  
YWSTDPSIET 181 PFFSQVMSRN RFVQIMQSWH FCNNDNIPHD SHRLAKIQPV  
IDYFRRKFND VYKPCQQLSL 241 DESIIPWRGR LSIKTYNPAK ITKYGILVRV  
LSEAVTGYVC NFDVYAADGK KLEDTAVIEP 301 YKNIWHQIYQ DNYYN SVKMA  
RILLKNKVRV CGTIRKNRGL PRSLKTIQLS RGQYEFRRNH 361 QILLEVWNNG  
RRNVNMISTI HSAQLMESRS KSKRSDVPIQ KPNSIIDYNK YMKGVD RADQ 421  
YLAYYSIFRK TKKWTKRVVM FFINCALFNS FRVYTILNGK NITYKNFLHK  
VAVSWIEDGE 481 TNCTEQDDNL PNSEPTRRAP RLDHPGRLSN YGKHKLINIV  
TSGRSLKPQR QCRVCAVQKK 541 RSRTCFCVCKF CNVPLHKGDC FERYHTLKKY.

(348) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Bombus impatiens*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(349) TABLE-US-00148 (SEQ ID NO: 14544) 1 ttaattttt aacattttac cgaccgatag ccgattaatc  
gggtttttgc cgctgacgct 61 taccgaccga taacctatta atcggctttt tgcgtcgaa gcttaccac ctatagccta 121  
cctatagtta atcggttgcc atggcgataa acaatcttc tcatatatg agcagtaatt 181 tgttatttag tactaaggta  
ccttgctcag ttgcgtcagt tgcgttgctt tgtaagctcc 241 cacagttta taccaattcg aaaaacttac cgttcgcg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(350) TABLE-US-00149 (SEQ ID NO: 14545) 1 actatttcac atttgaacta aaaaccgttg taatagataa

aataaatata ttaggtatt 61aatattatgg aaacaaaaga ttttattcaa tttattatc ctatagtaac aaaagcggc 121  
caattttatc tgagcatatc aaaagcacag atactccgc cgcacagtct aaaccgaaac 181 agagccggcg  
ccagggagaa tctgcgctg agcagccggt cggacgtgcg tttgctgttg 241 aaccgctagt ggtcagtaaa  
ccagaaccag tcagtaagcc agtaactgat cagttaacta 301 gattgtatag ttcaaattga acttaatat gttttaagc  
gtttgaatgt tgtctaactt 361 cgttatatat tatattcttt ttaa.

(351) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Mamestra brassicae*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(352) TABLE-US-00150 (SEQ ID NO: 14546) 1 MFSFVPNKEQ TRTVLIFCFH  
LKTAAESHR PLVEAFGEQV PTVKTCERWF QRFKSGDFDV 61 DDKEHGKPPK  
RYEDAELQAL LDEDDAQTQK QLAEQLEVSQ QAVSNRLREG GKIQKVGRWV 121  
PHELNERQRE RRKNTCEILL SRYKRKSFLH RIVTGEEKWI FFVNPKRKKS  
YVDPGQPATS 181 TARP NRFGKK TRLCVWWDQS GVIYYELLKP GETVNTARYQ  
QQLINLNRAL QRKRPEYQKR 241 QHRVIFLHDN APSHTARAVR DTLETNLWEV  
LPHAAAYSPDL APSDYHLFAS MGHALAEQRF 301 DSYESVEEWL DEWFAAKDDE  
FYWRGIHKLP ERWDNVCVASD GKYFE.

(353) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Mamestra brassicae*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(354) TABLE-US-00151 (SEQ ID NO: 14547) 1 ttattgggtt gcccaaaaag taattgcgga ttttcatat  
acctgtcttt taaacgtaca 61 tagggatcga actcagtaaa actttgacct tgtgaaataa caaacttgac tgtccaacca 121  
ccatagtgtg gcgcgaattg agcgtcataa ttgtttgac ttttgcagt caac.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(355) TABLE-US-00152 (SEQ ID NO: 14548) 1 atgatttttt ctttttaaac caattttaat tagttaattg  
atataaaaat ccgcaattac 61 ttttgggca acccaataa.

(356) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Mayetiola destructor*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5% 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(357) TABLE-US-00153 (SEQ ID NO: 14549) 1 MENFENWRKR RHLREVLLGH  
FFAKKTAAES HRLLEVEYGE HALAKTQCFE WFQRFKSGDF 61 DTEDKERPGQ  
PKKFEDEELE ALLDEDCCQT QEELAKSLGV TQQAISKRLK AAGYIQKQGN 121  
WVPHCLKPRD VERRFCMSEM LLQRHKKKSF LSRIITGDEK WIHYDNSKRK  
KSYVKRGGRA 181 KSTPKSNLHG AKVMLCIWWD QRGVLYYELL EPGQTITGDL  
YRTQLIRLKQ ALAEKRPEYA 241 KRHGAVIFHH DNARPHVALP VKNYLENSGW  
EVLPHPPYSP DLAPSDYHLF RSMQNDLAGK 301 RFTSEQGIRK WLDSFLAAKP  
AKFFEKGIHE LSERWEKVIA SDGQYFE.

(358) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Mayetiola destructor*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(359) TABLE-US-00154 (SEQ ID NO: 14550) 1 taagacttcc aaaattcca cccgaacttt accttccccg  
cgcattatgt ctctcttttc 61 accctctgat ccctgtgatt gttgtcgagc acgatttata ttgggtgtac aacttaaaaa 121  
ccggaattgg acgctagatg tccacactaa cgaatagtgt aaaagcacia atttcatata 181 tacgtcattt tgaaggtaca  
tttgacagct atcaaaatca gtcaataaaa ctattctatc 241 tgtgtgcatc atattttttt attact.

(360) In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(361) TABLE-US-00155 (SEQ ID NO: 14551) 1 tgcattcatt cattttgtta tcgaaataaa gcattaattt  
tcactaaaaa attccgggtt 61 ttaagttgta cacccaatat catccttagt gacaattttc aaatggcttt cccattgagc 121

tgaacccgtg gctctagtaa gaaaaacgcc caaccgctca tcatatgcct ttttttctc 181 aacatccg.

(362) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Apis mellifera*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(363) TABLE-US-00156 (SEQ ID NO: 14552) 1 MENQKEHYRH ILLFYFRKKGK  
NASQAHKKLC AVYGDEALKE RQCQNWDFKF RSGDFSLKDE 61 KRSGRPVEVD  
DDLKAIIDS DRHSTTREIA EKLHVSHTCI ENHLKQLGYV QKLDTWVPHE 121  
LKEKHLTQRI NSCDLLKKRN ENDPFLKRLI TGDEKWVVYN NIKRKRSWSR  
PREPAQTTSK 181 AGIHRKKVLL SVWWDYKGIV YFELLPPNRT INSVVYIEQL  
TKLNNAVEEK RPELTNRKGV 241 VFHHDNARPH TSLVTRQKLL ELGWDVLPHP  
PYSPDLAPSD YFLFRSLQNS LNGKNFNND 301 DIKSYLIQFF ANKNQKFYER  
GIMMLPERWQ KVIDQNGQHI TE.

(364) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Apis mellifera*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(365) TABLE-US-00157 (SEQ ID NO: 14553) 1 ttgggttggc aactaagtaa ttgcggattt cactcataga  
tggcttcagt tgaatttta 61 ggtttgctgg cgtagtccaa atgtaaaaca cattttgtta ttgatagtt ggcaattcag 121  
ctgtcaatca gtaaaaaaag tttttgatc gggtgcgtag ttttcgttg gcgttcggtg 181 aaaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(366) TABLE-US-00158 (SEQ ID NO: 14554) 1 agttatttag ttcatgaaa aaattgtctt  
tgattttcta aaaaaaatcc gcaattactt 61 agttgccaat ccaa.

(367) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Messor bouvieri*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(368) TABLE-US-00159 (SEQ ID NO: 14555) 1 MSSFVPENVH LRHALLFLFH  
QKKRAAESHR LLVETYGEHA PTIRTCETWF RQFKCGDFNV 61 QDKERPGRPK  
TFEDAELQEL LDEDSTQTQK QLAEKLNVSR VAICERLQAM GKIQKMGRWV 121  
PHELNDRQME NRKIVSEMLL QRYERKSFLH RIVTGDEKWI YFENPKRKKK  
WLSPGEAGPS 181 TARPNNRFGRK TMLCVWWDQI GVVYYELLKP GETVNTDRYR  
QQMINLNCAL IEKRPQYAQR 241 HDKVILQHDN APSHTAKPVK EMLKSLGWEV  
LSHPPYSPDL APSDYHLFAS MGHALAEQHF 301 ADFEEVKKWL DEWFSSKEKL  
FFWNGIHKLS ERWTKCIESN GQYFE.

(369) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Messor bouvieri*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(370) TABLE-US-00160 (SEQ ID NO: 14556) 1 agtcagaaat gacacctega tcgacgacta  
atcgacgtct aatcgacgtc gattttatgt 61 caacatgtta ccaggtgtgt cggttaattcc ttccggttt ttccggcaga  
tgtcactagc 121 cataagtatg aaatgttatg atttgataca tatgtcattt tattctactg acattaacct 181 taaaactaca  
caagttacgt tccgcaaaa taacagcgtt atagatttat aatttttga 241 aa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(371) TABLE-US-00161 (SEQ ID NO: 14557) 1 ataaatttga actatccatt ctaagtaacg  
tgttttcttt aacgaaaaaa ccggaaaaga 61 attaccgaca ctctggtat gtcaacatgt tattttcgac attgaatcgc  
gtcgattcga 121 agtcgatcga ggtgtcattt ctgact.

(372) In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Trichoplusia ni*. The piggyBac (PB) or piggyBac-like transposase



enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(373) TABLE-US-00162 (SEQ ID NO: 14558) 1 MGSSLDDEHI LSALLQSDDE  
LVGEDSDSEV SDHVSEDDVQ SDTEEFIDE VHEVQPTSSG 61 SEILDEQNVI  
EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
PTRMCRNIYD PLLCFKLFFT DEIIEIVKW TNAEISLKRR ESMTSATFRD  
TNEDEIYAFF 181 GILVMTAVRK DNHMSTDDLF DRSLSMVYVS VMSRDRFDFL  
IRCLRMDDKS IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ  
LLGFRGRCPF RVYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN  
GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT  
VRSNKREIPE VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421  
DEDASINEST GKPQMVMYYN QTKGGVDTLQ QMCSVMTCSR KTNRWPMALL  
YGMINIACIN 481 SFIIYSHNVS SKGEKVQSRK KFMRNLYMSL TSSFMRKRLE  
APTLKRYLRD NISNILPKEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA  
NASCKKCKKV ICREHNIDMC QSCF.

(374) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Trichoplusia ni*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(375) TABLE-US-00163 (SEQ ID NO: 14559) 1 ttaaccctag aaagatagtc tgcgtaaaat  
tgacgcatgc attcttgaaa tattgctctc 61 tctttctaaa tagcgcgaaat ccgctcgctgt gcatttagga catctcagtc  
gccgcttgga 121 gctcccgtag ggctgcttg tcaatgcggt aagtgtcact gattttgaac tataacgacc 181  
gcgtgagtc aaatgacgca tgattatctt ttacgtgact ttaagattt aactcatagc 241 ataattatat tgttattca  
tgttctactt acgtgataac ttattatata tatattttct 301 tgttatagat atc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(376) TABLE-US-00164 (SEQ ID NO: 14560) 1 ttgttactt tatagaagaa atttgagtt  
ttgtttttt tttaataaat aaataaacat 61 aaataaattg ttgttgaat ttattattag tatgtaagtg taaatataat  
aaaacttaat 121 atctattcaa attaataaat aaacctcgat atacagaccg ataaaacaca tgcgtcaatt 181  
ttacgcatga ttatctttaa cgtacgtcac aatatgatta tctttctagg gtaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(377) TABLE-US-00165 (SEQ ID NO: 14561) 1 ccctagaaag atagtctgcg taaaattgac  
gcatgcattc ttgaaatatt gctctctctt 61 tctaaatagc gcgaatccgt cgctgtgcat ttaggacatc tcagtcgccg  
cttgagctc 121 ccgtgaggcg tgcttgtaa tgcggtaagt gtcactgatt ttgaactata acgaccgcgt 181  
gagtcaaaat gacgcatgat tatcttttac gtgactttta agatttaact catagcataa 241 ttatattggt atttcatggt  
ctacttacgt gataacttat tatatatata ttttctggt 301 atagatatc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(378) TABLE-US-00166 (SEQ ID NO: 14562) 1 ttgttactt tatagaagaa atttgagtt  
ttgtttttt tttaataaat aaataaacat 61 aaataaattg ttgttgaat ttattattag tatgtaagtg taaatataat  
aaaacttaat 121 atctattcaa attaataaat aaacctcgat atacagaccg ataaaacaca tgcgtcaatt 181  
ttacgcatga ttatctttaa cgtacgtcac aatatgatta tctttctagg g.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(379) TABLE-US-00167 (SEQ ID NO: 14609) 1 tctaaatagc gcgaatccgt cgctgtgcat  
ttaggacatc tcagtcgccg cttgagctc 61 ccgtgaggcg tgcttgtaa tgcggtaagt gtcactgatt  
ttgaactata acgaccgcgt 121 gagtcaaaat gacgcatgat tatcttttac gtgactttta agatttaact catagcataa 181  
ttatattggt atttcatggt ctacttacgt gataacttat tatatatata ttttctggt 241 atagatatc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(380) TABLE-US-00168 (SEQ ID NO: 14610) 1 ttgttactt tatagaagaa atttgagtt  
ttgtttttt tttaataaat aaataaacat 61 aaataaattg ttgttgaat ttattattag tatgtaagtg taaatataat  
aaaacttaat 121 atctattcaa attaataaat aaacctcgat atacagaccg ataaaacaca tgcgtcaatt 181  
ttacgcatga ttatctttaa cgtacgtcac aatatgatta tctttctagg g.

(381) In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14561 and SEQ ID NO: 14562, and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14558.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14609 and SEQ ID NO: 14610, and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14558.

(382) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Aphis gossypii*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCTTCCAGCGGGCGCGC (SEQ ID NO: 14565).

(383) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Chilo suppressalis*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCAGATTAGCCT (SEQ ID NO: 14566).

(384) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Heliothis virescens*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCTTAATTACTCGCG (SEQ ID NO: 14567).

(385) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Pectinophora gossypiella*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCTAGATAACTAAAC (SEQ ID NO: 14568).

(386) In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Anopheles stephensi*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCTAGAAAGATA (SEQ ID NO: 14569).

(387) Non-Transposition Based Methods of Genetic Modification

(388) In some embodiments of the methods of the disclosure, a modified HSC or modified HSC descendent cell of the disclosure may be produced by introducing a transgene into an HSC or an HSC descendent cell of the disclosure. The introducing step may comprise delivery of a nucleic acid sequence and/or a genomic editing construct via a non-transposition delivery system.

(389) In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ comprises one or more of topical delivery, adsorption, absorption, electroporation, spin-fectation, co-culture, transfection, mechanical delivery, sonic delivery, vibrational delivery, magnetofection or by nanoparticle-mediated delivery. In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ comprises liposomal transfection, calcium phosphate transfection, fugene transfection, and dendrimer-mediated transfection. In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ by mechanical transfection comprises cell squeezing, cell bombardment, or gene gun techniques. In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ by nanoparticle-mediated transfection comprises liposomal delivery, delivery by micelles, and delivery by polymerosomes.

(390) In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ comprises a non-viral vector. In some embodiments, the non-viral vector comprises a nucleic acid. In some embodiments, the non-viral vector comprises plasmid DNA, linear double-stranded DNA (dsDNA), linear single-stranded DNA (ssDNA), DoggyBone™ DNA, nanoplasmids, minicircle DNA, single-stranded oligodeoxynucleotides (ssODN), DDNA oligonucleotides, single-stranded mRNA (ssRNA), and double-stranded mRNA (dsRNA). In some embodiments, the non-viral vector comprises a transposon of the disclosure.

(391) In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ comprises a viral vector. In some embodiments, the viral vector is a non-integrating non-chromosomal vector. Exemplary non-integrating non-chromosomal vectors include, but are not limited to, adeno-associated virus (AAV), adenovirus, and herpes viruses. In some embodiments, the viral vector is an integrating chromosomal vector. Integrating chromosomal vectors include, but are not limited to, adeno-associated vectors (AAV), Lentiviruses, and gamma-retroviruses.

(392) In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence

and/or a genomic editing construct into an HSC or HSC descendant cell ex vivo, in vivo, in vitro or in situ comprises a combination of vectors. Exemplary, non-limiting vector combinations include: viral and non-viral vectors, a plurality of non-viral vectors, or a plurality of viral vectors. Exemplary but non-limiting vectors combinations include: a combination of a DNA-derived and an RNA-derived vector, a combination of an RNA and a reverse transcriptase, a combination of a transposon and a transposase, a combination of a non-viral vector and an endonuclease, and a combination of a viral vector and an endonuclease.

(393) In some embodiments of the methods of the disclosure, genome modification comprising introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendant cell ex vivo, in vivo, in vitro or in situ stably integrates a nucleic acid sequence, transiently integrates a nucleic acid sequence, produces site-specific integration a nucleic acid sequence, or produces a biased integration of a nucleic acid sequence. In some embodiments, the nucleic acid sequence is a transgene.

(394) In some embodiments of the methods of the disclosure, genome modification comprising introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendant cell ex vivo, in vivo, in vitro or in situ stably integrates a nucleic acid sequence. In some embodiments, the stable chromosomal integration can be a random integration, a site-specific integration, or a biased integration. In some embodiments, the site-specific integration can be non-assisted or assisted. In some embodiments, the assisted site-specific integration is co-delivered with a site-directed nuclease. In some embodiments, the site-directed nuclease comprises a transgene with 5' and 3' nucleotide sequence extensions that contain a percentage homology to upstream and downstream regions of the site of genomic integration. In some embodiments, the transgene with homologous nucleotide extensions enable genomic integration by homologous recombination, microhomology-mediated end joining, or nonhomologous end-joining. In some embodiments the site-specific integration occurs at a safe harbor site. Genomic safe harbor sites are able to accommodate the integration of new genetic material in a manner that ensures that the newly inserted genetic elements function reliably (for example, are expressed at a therapeutically effective level of expression) and do not cause deleterious alterations to the host genome that cause a risk to the host organism. Potential genomic safe harbors include, but are not limited to, intronic sequences of the human albumin gene, the adeno-associated virus site 1 (AAVS1), a naturally occurring site of integration of AAV virus on chromosome 19, the site of the chemokine (C—C motif) receptor 5 (CCR5) gene and the site of the human ortholog of the mouse Rosa26 locus.

(395) In some embodiments, the site-specific transgene integration occurs at a site that disrupts expression of a target gene. In some embodiments, disruption of target gene expression occurs by site-specific integration at introns, exons, promoters, genetic elements, enhancers, suppressors, start codons, stop codons, and response elements. In some embodiments, exemplary target genes targeted by site-specific integration include but are not limited to TRAC, TRAB, PDI, any immunosuppressive gene, and genes involved in allo-rejection.

(396) In some embodiments, the site-specific transgene integration occurs at a site that results in enhanced expression of a target gene. In some embodiments, enhancement of target gene expression occurs by site-specific integration at introns, exons, promoters, genetic elements, enhancers, suppressors, start codons, stop codons, and response elements.

(397) In some embodiments of the methods of the disclosure, enzymes may be used to create strand breaks in the host genome to facilitate delivery or integration of the transgene. In some embodiments, enzymes create single-strand breaks. In some embodiments, enzymes create double-strand breaks. In some embodiments, examples of break-inducing enzymes include but are not limited to: transposases, integrases, endonucleases, CRISPR-Cas9, transcription activator-like effector nucleases (TALEN), zinc finger nucleases (ZFN), Cas-CLOVER™, and Cpf1. In some embodiments, break-inducing enzymes can be delivered to the cell encoded in DNA, encoded in mRNA, as a protein, as a nucleoprotein complex with a guide RNA (gRNA).

(398) In some embodiments of the methods of the disclosure, the site-specific transgene integration is controlled by a vector-mediated integration site bias. In some embodiments vector-mediated integration site bias is controlled by the chosen lentiviral vector. In some embodiments vector-mediated integration

site bias is controlled by the chosen gamma-retroviral vector.

(399) In some embodiments of the methods of the disclosure, the site-specific transgene integration site is a non-stable chromosomal insertion. In some embodiments, the integrated transgene may become silenced, removed, excised, or further modified.

(400) In some embodiments of the methods of the disclosure, the genome modification is a non-stable integration of a transgene. In some embodiments, the non-stable integration can be a transient non-chromosomal integration, a semi-stable non chromosomal integration, a semi-persistent non-chromosomal insertion, or a non-stable chromosomal insertion. In some embodiments, the transient non-chromosomal insertion can be epi-chromosomal or cytoplasmic.

(401) In some embodiments, the transient non-chromosomal insertion of a transgene does not integrate into a chromosome and the modified genetic material is not replicated during cell division.

(402) In some embodiments of the methods of the disclosure, the genome modification is a semi-stable or persistent non-chromosomal integration of a transgene. In some embodiments, a DNA vector encodes a Scaffold/matrix attachment region (S-MAR) module that binds to nuclear matrix proteins for episomal retention of a non-viral vector allowing for autonomous replication in the nucleus of dividing cells.

(403) In some embodiments of the methods of the disclosure, the genome modification is a non-stable chromosomal integration of a transgene. In some embodiments, the integrated transgene may become silenced, removed, excised, or further modified.

(404) In some embodiments of the methods of the disclosure, the modification to the genome by transgene insertion can occur via host cell-directed double-strand breakage repair (homology-directed repair) by homologous recombination (HR), microhomology-mediated end joining (MMEJ), nonhomologous end joining (NHEJ), transposase enzyme-mediated modification, integrase enzyme-mediated modification, endonuclease enzyme-mediated modification, or recombinant enzyme-mediated modification. In some embodiments, the modification to the genome by transgene insertion can occur via CRISPR-Cas9, TALEN, ZFNs, Cas-CLOVER, and Cpf1.

(405) In gene editing systems that involve inserting new or existing nucleotides/nucleic acids, insertion tools (e.g., DNA template vectors, transposable elements (transposons or retrotransposons) must be delivered to the cell in addition to the cutting enzyme (e.g., a nuclease, recombinase, integrase or transposase). Examples of such insertion tools for a recombinase may include a DNA vector. Other gene editing systems require the delivery of an integrase along with an insertion vector, a transposase along with a transposon/retrotransposon, etc. In some embodiments, an example recombinase that may be used as a cutting enzyme is the CRE recombinase. In various embodiments, example integrases that may be used in insertion tools include viral based enzymes taken from any of a number of viruses including, but not limited to, AAV, gamma retrovirus, and lentivirus. Example transposons/retrotransposons that may be used in insertion tools include, but are not limited to, the piggyBac transposon, Sleeping Beauty transposon, and the L1 retrotransposon.

(406) In certain embodiments of the methods of the disclosure, the transgene is delivered in vivo. In certain embodiments of the methods of the disclosure, in vivo transgene delivery can occur by: topical delivery, adsorption, absorption, electroporation, spin-fecton, co-culture, transfection, mechanical delivery, sonic delivery, vibrational delivery, magnetofection or by nanoparticle-mediated delivery. In certain embodiments of the methods of the disclosure, in vivo transgene delivery by transfection can occur by liposomal transfection, calcium phosphate transfection, fugene transfection, and dendrimer-mediated transfection. In certain embodiments of the methods of the disclosure, in vivo mechanical transgene delivery can occur by cell squeezing, bombardment, and gene gun. In certain embodiments of the methods of the disclosure, in vivo nanoparticle-mediated transgene delivery can occur by liposomal delivery, delivery by micelles, and delivery by polymerosomes. In various embodiments, nucleases that may be used as cutting enzymes include, but are not limited to, Cas9, transcription activator-like effector nucleases (TALENs) and zinc finger nucleases.

(407) In certain embodiments of the methods of the disclosure, non-viral vectors are used for transgene delivery. In certain embodiments, the non-viral vector is a nucleic acid. In certain embodiments, the nucleic acid non-viral vector is plasmid DNA, linear double-stranded DNA (dsDNA), linear single-stranded DNA (ssDNA), DoggyBone™ DNA, nanoplastids, minicircle DNA, single-stranded

oligodeoxynucleotides (ssODN), DDNA oligonucleotides, single-stranded mRNA (ssRNA), and double-stranded mRNA (dsRNA). In certain embodiments, the non-viral vector is a transposon. In certain embodiments, the transposon is piggyBac.

(408) In certain embodiments of the methods of the disclosure, transgene delivery can occur via viral vector. In certain embodiments, the viral vector is a non-integrating non-chromosomal vectors. Non-integrating non-chromosomal vectors can include adeno-associated virus (AAV), adenovirus, and herpes viruses. In certain embodiments, the viral vector is an integrating chromosomal vectors. Integrating chromosomal vectors can include adeno-associated vectors (AAV), Lentiviruses, and gamma-retroviruses.

(409) In certain embodiments of the methods of the disclosure, transgene delivery can occur by a combination of vectors. Exemplary but non-limiting vector combinations can include: viral plus non-viral vectors, more than one non-viral vector, or more than one viral vector. Exemplary but non-limiting vectors combinations can include: DNA-derived plus RNA-derived vectors, RNA plus reverse transcriptase, a transposon and a transposase, a non-viral vectors plus an endonuclease, and a viral vector plus an endonuclease.

(410) In certain embodiments of the methods of the disclosure, the genome modification can be a stable integration of a transgene, a transient integration of a transgene, a site-specific integration of a transgene, or a biased integration of a transgene.

(411) In certain embodiments of the methods of the disclosure, the genome modification can be a stable chromosomal integration of a transgene. In certain embodiments, the stable chromosomal integration can be a random integration, a site-specific integration, or a biased integration. In certain embodiments, the site-specific integration can be non-assisted or assisted. In certain embodiments, the assisted site-specific integration is co-delivered with a site-directed nuclease. In certain embodiments, the site-directed nuclease comprises a transgene with 5' and 3' nucleotide sequence extensions that contain homology to upstream and downstream regions of the site of genomic integration. In certain embodiments, the transgene with homologous nucleotide extensions enable genomic integration by homologous recombination, microhomology-mediated end joining, or nonhomologous end-joining. In certain embodiments the site-specific integration occurs at a safe harbor site. Genomic safe harbor sites are able to accommodate the integration of new genetic material in a manner that ensures that the newly inserted genetic elements function reliably (for example, are expressed at a therapeutically effective level of expression) and do not cause deleterious alterations to the host genome that cause a risk to the host organism. Potential genomic safe harbors include, but are not limited to, intronic sequences of the human albumin gene, the adeno-associated virus site 1 (AAVS1), a naturally occurring site of integration of AAV virus on chromosome 19, the site of the chemokine (C—C motif) receptor 5 (CCR5) gene and the site of the human ortholog of the mouse Rosa26 locus.

(412) In certain embodiments, the site-specific transgene integration occurs at a site that disrupts expression of a target gene. In certain embodiments, disruption of target gene expression occurs by site-specific integration at introns, exons, promoters, genetic elements, enhancers, suppressors, start codons, stop codons, and response elements. In certain embodiments, exemplary target genes targeted by site-specific integration include but are not limited to TRAC, TRAB, PDI, any immunosuppressive gene, and genes involved in allo-rejection.

(413) In certain embodiments, the site-specific transgene integration occurs at a site that results in enhanced expression of a target gene. In certain embodiments, enhancement of target gene expression occurs by site-specific integration at introns, exons, promoters, genetic elements, enhancers, suppressors, start codons, stop codons, and response elements.

(414) In certain embodiments of the methods of the disclosure, enzymes may be used to create strand breaks in the host genome to facilitate delivery or integration of the transgene. In certain embodiments, enzymes create single-strand breaks. In certain embodiments, enzymes create double-strand breaks. In certain embodiments, examples of break-inducing enzymes include but are not limited to: transposases, integrases, endonucleases, CRISPR-Cas9, transcription activator-like effector nucleases (TALEN), zinc finger nucleases (ZFN), Cas-CLOVER™, and Cpf1. In certain embodiments, break-inducing enzymes can be delivered to the cell encoded in DNA, encoded in mRNA, as a protein, as a nucleoprotein

complex with a guide RNA (gRNA).

(415) In certain embodiments of the methods of the disclosure, the site-specific transgene integration is controlled by a vector-mediated integration site bias. In certain embodiments vector-mediated integration site bias is controlled by the chosen lentiviral vector. In certain embodiments vector-mediated integration site bias is controlled by the chosen gamma-retroviral vector.

(416) In certain embodiments of the methods of the disclosure, the site-specific transgene integration site is a non-stable chromosomal insertion. In certain embodiments, the integrated transgene may become silenced, removed, excised, or further modified. In certain embodiments of the methods of the disclosure, the genome modification is a non-stable integration of a transgene. In certain embodiments, the non-stable integration can be a transient non-chromosomal integration, a semi-stable non chromosomal integration, a semi-persistent non-chromosomal insertion, or a non-stable chromosomal insertion. In certain embodiments, the transient non-chromosomal insertion can be epi-chromosomal or cytoplasmic. In certain embodiments, the transient non-chromosomal insertion of a transgene does not integrate into a chromosome and the modified genetic material is not replicated during cell division.

(417) In certain embodiments of the methods of the disclosure, the genome modification is a semi-stable or persistent non-chromosomal integration of a transgene. In certain embodiments, a DNA vector encodes a Scaffold/matrix attachment region (S-MAR) module that binds to nuclear matrix proteins for episomal retention of a non-viral vector allowing for autonomous replication in the nucleus of dividing cells.

(418) In certain embodiments of the methods of the disclosure, the genome modification is a non-stable chromosomal integration of a transgene. In certain embodiments, the integrated transgene may become silenced, removed, excised, or further modified.

(419) In certain embodiments of the methods of the disclosure, the modification to the genome by transgene insertion can occur via host cell-directed double-strand breakage repair (homology-directed repair) by homologous recombination (HR), microhomology-mediated end joining (MMEJ), nonhomologous end joining (NHEJ), transposase enzyme-mediated modification, integrase enzyme-mediated modification, endonuclease enzyme-mediated modification, or recombinant enzyme-mediated modification. In certain embodiments, the modification to the genome by transgene insertion can occur via CRISPR-Cas9, TALEN, ZFNs, Cas-CLOVER, and Cpf1.

(420) In certain embodiments of the methods of the disclosure, a cell with an in vivo or ex vivo genomic modification can be a germline cell or a somatic cell. In certain embodiments the modified cell can be a human, non-human, mammalian, rat, mouse, or dog cell. In certain embodiments, the modified cell can be differentiated, undifferentiated, or immortalized. In certain embodiments, the modified undifferentiated cell can be a stem cell. In certain embodiments, the modified cell can be differentiated, undifferentiated, or immortalized. In certain embodiments, the modified undifferentiated cell can be an induced pluripotent stem cell. In certain embodiments, the modified cell can be a T cell, a hematopoietic stem cell, a natural killer cell, a macrophage, a dendritic cell, a monocyte, a megakaryocyte, or an osteoclast. In certain embodiments, the modified cell can be modified while the cell is quiescent, in an activated state, resting, in interphase, in prophase, in metaphase, in anaphase, or in telophase. In certain embodiments, the modified cell can be fresh, cryopreserved, bulk, sorted into sub-populations, from whole blood, from leukapheresis, or from an immortalized cell line.

(421) Production and Generation of VCARS

(422) At least one VHH protein or VCAR of the disclosure can be optionally produced by a cell line, a mixed cell line, an immortalized cell or clonal population of immortalized cells, as well known in the art. See, e.g., Ausubel, et al., ed., *Current Protocols in Molecular Biology*, John Wiley & Sons, Inc., NY, N.Y. (1987-2001); Sambrook, et al., *Molecular Cloning: A Laboratory Manual*, 2nd Edition, Cold Spring Harbor, N.Y. (1989); Harlow and Lane, *Antibodies, a Laboratory Manual*, Cold Spring Harbor, N.Y. (1989); Colligan, et al., eds., *Current Protocols in Immunology*, John Wiley & Sons, Inc., NY (1994-2001); Colligan et al., *Current Protocols in Protein Science*, John Wiley & Sons, NY, N.Y., (1997-2001).

(423) Amino acids from a VHH protein can be altered, added and/or deleted to reduce immunogenicity or reduce, enhance or modify binding, affinity, on-rate, off-rate, avidity, specificity, half-life, stability,

solubility or any other suitable characteristic, as known in the art.

(424) Optionally, VHH proteins can be engineered with retention of high affinity for the antigen and other favorable biological properties. To achieve this goal, the VHH proteins can be optionally prepared by a process of analysis of the parental sequences and various conceptual engineered products using three-dimensional models of the parental and engineered sequences. Three-dimensional models are commonly available and are familiar to those skilled in the art. Computer programs are available which illustrate and display probable three-dimensional conformational structures of selected candidate sequences and can measure possible immunogenicity (e.g., Immunofilter program of Xencor, Inc. of Monrovia, Calif.). Inspection of these displays permits analysis of the likely role of the residues in the functioning of the candidate sequence, i.e., the analysis of residues that influence the ability of the candidate VHH protein to bind its antigen. In this way, residues can be selected and combined from the parent and reference sequences so that the desired characteristic, such as affinity for the target antigen(s), is achieved. Alternatively, or in addition to, the above procedures, other suitable methods of engineering can be used.

#### (425) Screening of VHH Proteins

(426) Screening VHH for specific binding to similar proteins or fragments can be conveniently achieved using nucleotide (DNA or RNA display) or peptide display libraries, for example, in vitro display. This method involves the screening of large collections of peptides for individual members having the desired function or structure. The displayed nucleotide or peptide sequences can be from 3 to 5000 or more nucleotides or amino acids in length, frequently from 5-100 amino acids long, and often from about 8 to 25 amino acids long. In addition to direct chemical synthetic methods for generating peptide libraries, several recombinant DNA methods have been described. One type involves the display of a peptide sequence on the surface of a bacteriophage or cell. Each bacteriophage or cell contains the nucleotide sequence encoding the particular displayed peptide sequence. The VHH proteins of the disclosure can bind human or other mammalian proteins with a wide range of affinities (KD). In a preferred embodiment, at least one VHH of the present invention can optionally bind to a target protein with high affinity, for example, with a KD equal to or less than about  $10 \times 10^{-7}$  M, such as but not limited to,  $0.1 \times 10^{-9}$  (or any range or value therein) to  $10 \times 10^{-11}$ ,  $10 \times 10^{-12}$ ,  $10 \times 10^{-13}$ ,  $10 \times 10^{-14}$ ,  $10 \times 10^{-15}$  or any range or value therein, as determined by surface plasmon resonance or the Kinexa method, as practiced by those of skill in the art.

(427) The affinity or avidity of a VHH or a VCAR for an antigen can be determined experimentally using any suitable method. (See, for example, Berzofsky, et al., "Antibody-Antigen Interactions," In Fundamental Immunology, Paul, W. E., Ed., Raven Press: New York, N.Y. (1984); Kuby, Janis Immunology, W.H. Freeman and Company: New York, N.Y. (1992); and methods described herein). The measured affinity of a particular VHH-antigen or VCAR-antigen interaction can vary if measured under different conditions (e.g., salt concentration, pH). Thus, measurements of affinity and other antigen-binding parameters (e.g., KD, Kon, Koff) are preferably made with standardized solutions of VHH or VCAR and antigen, and a standardized buffer, such as the buffer described herein.

(428) Competitive assays can be performed with the VHH or VCAR of the disclosure in order to determine what proteins, antibodies, and other antagonists compete for binding to a target protein with the VHH or VCAR of the present invention and/or share the epitope region. These assays as readily known to those of ordinary skill in the art evaluate competition between antagonists or ligands for a limited number of binding sites on a protein. The protein and/or antibody is immobilized or insolubilized before or after the competition and the sample bound to the target protein is separated from the unbound sample, for example, by decanting (where the protein/antibody was preinsolubilized) or by centrifuging (where the protein/antibody was precipitated after the competitive reaction). Also, the competitive binding may be determined by whether function is altered by the binding or lack of binding of the VHH or VCAR to the target protein, e.g., whether the VCAR molecule inhibits or potentiates the enzymatic activity of, for example, a label. ELISA and other functional assays may be used, as well known in the art.

#### (429) Nucleic Acid Molecules

(430) Nucleic acid molecules of the disclosure encoding VHH proteins or VCARs can be in the form of

RNA, such as mRNA, hnRNA, tRNA or any other form, or in the form of DNA, including, but not limited to, cDNA and genomic DNA obtained by cloning or produced synthetically, or any combinations thereof. The DNA can be triple-stranded, double-stranded or single-stranded, or any combination thereof. Any portion of at least one strand of the DNA or RNA can be the coding strand, also known as the sense strand, or it can be the non-coding strand, also referred to as the anti-sense strand.

(431) Isolated nucleic acid molecules of the disclosure can include nucleic acid molecules comprising an open reading frame (ORF), optionally, with one or more introns, e.g., but not limited to, at least one specified portion of at least one VCAR; nucleic acid molecules comprising the coding sequence for a VCAR; and nucleic acid molecules which comprise a nucleotide sequence substantially different from those described above but which, due to the degeneracy of the genetic code, still encode the VCAR as described herein and/or as known in the art. Of course, the genetic code is well known in the art. Thus, it would be routine for one skilled in the art to generate such degenerate nucleic acid variants that code for specific VCARS of the present invention. See, e.g., Ausubel, et al., supra, and such nucleic acid variants are included in the present invention.

(432) As indicated herein, nucleic acid molecules of the disclosure which comprise a nucleic acid encoding a VCAR can include, but are not limited to, those encoding the amino acid sequence of a VHH fragment, by itself, the coding sequence for the entire VCAR or a portion thereof; the coding sequence for a VHH, fragment or portion, as well as additional sequences, such as the coding sequence of at least one signal leader or fusion peptide, with or without the aforementioned additional coding sequences, such as at least one intron, together with additional, non-coding sequences, including but not limited to, non-coding 5' and 3' sequences, such as the transcribed, non-translated sequences that play a role in transcription, mRNA processing, including splicing and polyadenylation signals (for example, ribosome binding and stability of mRNA); an additional coding sequence that codes for additional amino acids, such as those that provide additional functionalities. Thus, the sequence encoding a VCAR can be fused to a marker sequence, such as a sequence encoding a peptide that facilitates purification of the fused VCAR comprising a VHH fragment or portion.

(433) Polynucleotides Selectively Hybridizing to a Polynucleotide as Described Herein

(434) The disclosure provides isolated nucleic acids that hybridize under selective hybridization conditions to a polynucleotide disclosed herein. Thus, the polynucleotides of this embodiment can be used for isolating, detecting, and/or quantifying nucleic acids comprising such polynucleotides. For example, polynucleotides of the present invention can be used to identify, isolate, or amplify partial or full-length clones in a deposited library. In some embodiments, the polynucleotides are genomic or cDNA sequences isolated, or otherwise complementary to, a cDNA from a human or mammalian nucleic acid library.

(435) Preferably, the cDNA library comprises at least 80% full-length sequences, preferably, at least 85% or 90% full-length sequences, and, more preferably, at least 95% full-length sequences. The cDNA libraries can be normalized to increase the representation of rare sequences. Low or moderate stringency hybridization conditions are typically, but not exclusively, employed with sequences having a reduced sequence identity relative to complementary sequences. Moderate and high stringency conditions can optionally be employed for sequences of greater identity. Low stringency conditions allow selective hybridization of sequences having about 70% sequence identity and can be employed to identify orthologous or paralogous sequences.

(436) Optionally, polynucleotides of this invention will encode at least a portion of a VCAR encoded by the polynucleotides described herein. The polynucleotides of this invention embrace nucleic acid sequences that can be employed for selective hybridization to a polynucleotide encoding a VCAR of the present invention. See, e.g., Ausubel, supra; Colligan, supra, each entirely incorporated herein by reference.

(437) Construction of Nucleic Acids

(438) The isolated nucleic acids of the disclosure can be made using (a) recombinant methods, (b) synthetic techniques, (c) purification techniques, and/or (d) combinations thereof, as well-known in the art.



(439) The nucleic acids can conveniently comprise sequences in addition to a polynucleotide of the present invention. For example, a multi-cloning site comprising one or more endonuclease restriction sites can be inserted into the nucleic acid to aid in isolation of the polynucleotide. Also, translatable sequences can be inserted to aid in the isolation of the translated polynucleotide of the disclosure. For example, a hexa-histidine marker sequence provides a convenient means to purify the proteins of the disclosure. The nucleic acid of the disclosure, excluding the coding sequence, is optionally a vector, adapter, or linker for cloning and/or expression of a polynucleotide of the disclosure.

(440) Additional sequences can be added to such cloning and/or expression sequences to optimize their function in cloning and/or expression, to aid in isolation of the polynucleotide, or to improve the introduction of the polynucleotide into a cell. Use of cloning vectors, expression vectors, adapters, and linkers is well known in the art. (See, e.g., Ausubel, supra; or Sambrook, supra).

(441) Recombinant Methods for Constructing Nucleic Acid

(442) The isolated nucleic acid compositions of this disclosure, such as RNA, cDNA, genomic DNA, or any combination thereof, can be obtained from biological sources using any number of cloning methodologies known to those of skill in the art. In some embodiments, oligonucleotide probes that selectively hybridize, under stringent conditions, to the polynucleotides of the present invention are used to identify the desired sequence in a cDNA or genomic DNA library. The isolation of RNA, and construction of cDNA and genomic libraries are well known to those of ordinary skill in the art. (See, e.g., Ausubel, supra; or Sambrook, supra).

(443) Nucleic Acid Screening and Isolation Methods

(444) A cDNA or genomic library can be screened using a probe based upon the sequence of a polynucleotide of the disclosure. Probes can be used to hybridize with genomic DNA or cDNA sequences to isolate homologous genes in the same or different organisms. Those of skill in the art will appreciate that various degrees of stringency of hybridization can be employed in the assay; and either the hybridization or the wash medium can be stringent. As the conditions for hybridization become more stringent, there must be a greater degree of complementarity between the probe and the target for duplex formation to occur. The degree of stringency can be controlled by one or more of temperature, ionic strength, pH and the presence of a partially denaturing solvent, such as formamide. For example, the stringency of hybridization is conveniently varied by changing the polarity of the reactant solution through, for example, manipulation of the concentration of formamide within the range of 0% to 50%. The degree of complementarity (sequence identity) required for detectable binding will vary in accordance with the stringency of the hybridization medium and/or wash medium. The degree of complementarity will optimally be 100%, or 70-100%, or any range or value therein. However, it should be understood that minor sequence variations in the probes and primers can be compensated for by reducing the stringency of the hybridization and/or wash medium.

(445) Methods of amplification of RNA or DNA are well known in the art and can be used according to the disclosure without undue experimentation, based on the teaching and guidance presented herein.

(446) Known methods of DNA or RNA amplification include, but are not limited to, polymerase chain reaction (PCR) and related amplification processes (see, e.g., U.S. Pat. Nos. 4,683,195, 4,683,202, 4,800,159, 4,965,188, to Mullis, et al.; U.S. Pat. Nos. 4,795,699 and 4,921,794 to Tabor, et al; U.S. Pat. No. 5,142,033 to Innis; U.S. Pat. No. 5,122,464 to Wilson, et al.; U.S. Pat. No. 5,091,310 to Innis; U.S. Pat. No. 5,066,584 to Gyllensten, et al; U.S. Pat. No. 4,889,818 to Gelfand, et al; U.S. Pat. No. 4,994,370 to Silver, et al; U.S. Pat. No. 4,766,067 to Biswas; U.S. Pat. No. 4,656,134 to Ringold) and RNA mediated amplification that uses anti-sense RNA to the target sequence as a template for double-stranded DNA synthesis (U.S. Pat. No. 5,130,238 to Malek, et al, with the tradename NASBA), the entire contents of which references are incorporated herein by reference. (See, e.g., Ausubel, supra; or Sambrook, supra.)

(447) For instance, polymerase chain reaction (PCR) technology can be used to amplify the sequences of polynucleotides of the disclosure and related genes directly from genomic DNA or cDNA libraries. PCR and other in vitro amplification methods can also be useful, for example, to clone nucleic acid sequences that code for proteins to be expressed, to make nucleic acids to use as probes for detecting the presence of the desired mRNA in samples, for nucleic acid sequencing, or for other purposes.

Examples of techniques sufficient to direct persons of skill through in vitro amplification methods are found in Berger, *supra*, Sambrook, *supra*, and Ausubel, *supra*, as well as Mullis, et al., U.S. Pat. No. 4,683,202 (1987); and Innis, et al., *PCR Protocols A Guide to Methods and Applications*, Eds., Academic Press Inc., San Diego, Calif. (1990). Commercially available kits for genomic PCR amplification are known in the art. See, e.g., Advantage-GC Genomic PCR Kit (Clontech). Additionally, e.g., the T4 gene 32 protein (Boehringer Mannheim) can be used to improve yield of long PCR products.

(448) Synthetic Methods for Constructing Nucleic Acids

(449) The isolated nucleic acids of the disclosure can also be prepared by direct chemical synthesis by known methods (see, e.g., Ausubel, et al., *supra*). Chemical synthesis generally produces a single-stranded oligonucleotide, which can be converted into double-stranded DNA by hybridization with a complementary sequence, or by polymerization with a DNA polymerase using the single strand as a template. One of skill in the art will recognize that while chemical synthesis of DNA can be limited to sequences of about 100 or more bases, longer sequences can be obtained by the ligation of shorter sequences.

(450) Recombinant Expression Cassettes

(451) The disclosure further provides recombinant expression cassettes comprising a nucleic acid of the disclosure. A nucleic acid sequence of the disclosure, for example, a cDNA or a genomic sequence encoding a VCAR of the disclosure, can be used to construct a recombinant expression cassette that can be introduced into at least one desired host cell. A recombinant expression cassette will typically comprise a polynucleotide of the disclosure operably linked to transcriptional initiation regulatory sequences that will direct the transcription of the polynucleotide in the intended host cell. Both heterologous and non-heterologous (i.e., endogenous) promoters can be employed to direct expression of the nucleic acids of the disclosure.

(452) In some embodiments, isolated nucleic acids that serve as promoter, enhancer, or other elements can be introduced in the appropriate position (upstream, downstream or in the intron) of a non-heterologous form of a polynucleotide of the disclosure so as to up or down regulate expression of a polynucleotide of the disclosure. For example, endogenous promoters can be altered *in vivo* or *in vitro* by mutation, deletion and/or substitution.

(453) Nanotransposons

(454) The disclosure provides a nanotransposon comprising: (a) a sequence encoding a transposon insert, comprising a sequence encoding a first inverted terminal repeat (ITR), a sequence encoding a second inverted terminal repeat (ITR), and an intra-ITR sequence; (b) a sequence encoding a backbone, wherein the sequence encoding the backbone comprises a sequence encoding an origin of replication having between 1 and 450 nucleotides, inclusive of the endpoints, and a sequence encoding a selectable marker having between 1 and 200 nucleotides, inclusive of the endpoints, and (c) an inter-ITR sequence. In some embodiments, the inter-ITR sequence of (c) comprises the sequence of (b). In some embodiments, the intra-ITR sequence of (a) comprises the sequence of (b).

(455) In some embodiments of the nanotransposons of the disclosure, the sequence encoding the backbone comprises between 1 and 600 nucleotides, inclusive of the endpoints. In some embodiments, the sequence encoding the backbone consists of between 1 and 50 nucleotides, between 50 and 100 nucleotides, between 100 and 150 nucleotides, between 150 and 200 nucleotides, between 200 and 250 nucleotides, between 250 and 300 nucleotides, between 300 and 350 nucleotides, between 350 and 400 nucleotides, between 400 and 450 nucleotides, between 450 and 500 nucleotides, between 500 and 550 nucleotides, between 550 and 600 nucleotides, each range inclusive of the endpoints.

(456) In some embodiments of the nanotransposons of the disclosure, the inter-ITR sequence comprises between 1 and 1000 nucleotides, inclusive of the endpoints. In some embodiments, the inter-ITR sequence consists of between 1 and 50 nucleotides, between 50 and 100 nucleotides, between 100 and 150 nucleotides, between 150 and 200 nucleotides, between 200 and 250 nucleotides, between 250 and 300 nucleotides, between 300 and 350 nucleotides, between 350 and 400 nucleotides, between 400 and 450 nucleotides, between 450 and 500 nucleotides, between 500 and 550 nucleotides, between 550 and 600 nucleotides, between 600 and 650 nucleotides, between 650 and 700 nucleotides, between 700 and

750 nucleotides, between 750 and 800 nucleotides, between 800 and 850 nucleotides, between 850 and 900 nucleotides, between 900 and 950 nucleotides, or between 950 and 1000 nucleotides, each range inclusive of the endpoints.

(457) In some embodiments of the nanotransposons of the disclosure, including the short nanotransposons (SNTs) of the disclosure, the inter-ITR sequence comprises between 1 and 200 nucleotides, inclusive of the endpoints. In some embodiments, the inter-ITR sequence consists of between 1 and 10 nucleotides, between 10 and 20 nucleotides, between 20 and 30 nucleotides, between 30 and 40 nucleotides, between 40 and 50 nucleotides, between 50 and 60 nucleotides, between 60 and 70 nucleotides, between 70 and 80 nucleotides, between 80 and 90 nucleotides, or between 90 and 100 nucleotides, each range inclusive of the endpoints.

(458) In some embodiments of the nanotransposons of the disclosure, the selectable marker having between 1 and 200 nucleotides, inclusive of the endpoints, comprises a sequence encoding a sucrose-selectable marker. In some embodiments, the sequence encoding a sucrose-selectable marker comprises a sequence encoding an RNA-OUT sequence. In some embodiments, the sequence encoding an RNA-OUT sequence comprises or consists of 137 base pairs (bp). In some embodiments, the selectable marker having between 1 and 200 nucleotides, inclusive of the endpoints, comprises a sequence encoding a fluorescent marker. In some embodiments, the selectable marker having between 1 and 200 nucleotides, inclusive of the endpoints, comprises a sequence encoding a cell surface marker.

(459) In some embodiments of the nanotransposons of the disclosure, the sequence encoding an origin of replication having between 1 and 450 nucleotides, inclusive of the endpoints, comprises a sequence encoding a mini origin of replication. In some embodiments, the sequence encoding an origin of replication having between 1 and 450 nucleotides, inclusive of the endpoints, comprises a sequence encoding an R6K origin of replication. In some embodiments, the R6K origin of replication comprises an R6K gamma origin of replication. In some embodiments, the R6K origin of replication comprises an R6K mini origin of replication. In some embodiments, the R6K origin of replication comprises an R6K gamma mini origin of replication. In some embodiments, the R6K gamma mini origin of replication comprises or consists of 281 base pairs (bp).

(460) In some embodiments of the nanotransposons of the disclosure, the sequence encoding the backbone does not comprise a recombination site, an excision site, a ligation site or a combination thereof. In some embodiments, neither the nanotransposon nor the sequence encoding the backbone comprises a product of a recombination site, an excision site, a ligation site or a combination thereof. In some embodiments, neither the nanotransposon nor the sequence encoding the backbone is derived from a recombination site, an excision site, a ligation site or a combination thereof.

(461) In some embodiments of the nanotransposons of the disclosure, a recombination site comprises a sequence resulting from a recombination event. In some embodiments, a recombination site comprises a sequence that is a product of a recombination event. In some embodiments, the recombination event comprises an activity of a recombinase (e.g., a recombinase site).

(462) In some embodiments of the nanotransposons of the disclosure, the sequence encoding the backbone does not further comprise a sequence encoding foreign DNA.

(463) In some embodiments of the nanotransposons of the disclosure, the inter-ITR sequence does not comprise a recombination site, an excision site, a ligation site or a combination thereof. In some embodiments, the inter-ITR sequence does not comprise a product of a recombination event, an excision event, a ligation event or a combination thereof. In some embodiments, the inter-ITR sequence is not derived from a recombination event, an excision event, a ligation event or a combination thereof.

(464) In some embodiments of the nanotransposons of the disclosure, the inter-ITR sequence comprises a sequence encoding foreign DNA.

(465) In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises at least one sequence encoding an insulator and a sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell. In some embodiments, the mammalian cell is a human cell.

(466) In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises a first sequence encoding an insulator, a sequence encoding a promoter capable of expressing an

exogenous sequence in a mammalian cell and a second sequence encoding an insulator.

(467) In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises a first sequence encoding an insulator, a sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell, a polyadenosine (polyA) sequence and a second sequence encoding an insulator.

(468) In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises a first sequence encoding an insulator, a sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell, at least one exogenous sequence, a polyadenosine (polyA) sequence and a second sequence encoding an insulator.

(469) In some embodiments of the nanotransposons of the disclosure, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell is capable of expressing an exogenous sequence in a human cell. In some embodiments, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell comprises a sequence encoding a constitutive promoter. In some embodiments, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell comprises a sequence encoding an inducible promoter. In some embodiments, the intra-ITR sequence comprises a first sequence encoding a first promoter capable of expressing an exogenous sequence in a mammalian cell and a second sequence encoding a second promoter capable of expressing an exogenous sequence in mammalian cell, wherein the first promoter is a constitutive promoter, wherein the second promoter is an inducible promoter, and wherein the first sequence encoding the first promoter and the second sequence encoding the second promoter are oriented in opposite directions. In some embodiments, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell comprises a sequence encoding a cell-type or tissue-type specific promoter. In some embodiments, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell comprises a sequence encoding an EF1a promoter, a sequence encoding a CMV promoter, a sequence encoding an MND promoter, a sequence encoding an SV40 promoter, a sequence encoding a PGK1 promoter, a sequence encoding a Ubc promoter, a sequence encoding a CAG promoter, a sequence encoding an H1 promoter, or a sequence encoding a U6 promoter.

(470) In some embodiments of the nanotransposons of the disclosure, the polyadenosine (polyA) sequence is isolated or derived from a viral polyA sequence. In some embodiments, the polyadenosine (polyA) sequence is isolated or derived from an (SV40) polyA sequence.

(471) In some embodiments of the nanotransposons of the disclosure, the at least one exogenous sequence comprises an inducible proapoptotic polypeptide. In some embodiments, the inducible caspase polypeptide comprises (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In some embodiments, the inducible caspase polypeptide comprises (a) a ligand binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence.

(472) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide, the ligand binding region comprises a FK506 binding protein 12 (FKBP12) polypeptide. In some embodiments, the amino acid sequence of the ligand binding region comprises a FK506 binding protein 12 (FKBP12) polypeptide. In some embodiments, the FK506 binding protein 12 (FKBP12) polypeptide comprises a modification at position 36 of the sequence. In some embodiments, the modification comprises a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In some embodiments, the FKBP12 polypeptide is encoded by an amino acid sequence comprising

(473) TABLE-US-00169 (SEQ ID NO: 14635)

GVQVETISPGDGRTPKRGQTCVVHYTGMLEDGKKVDSSRDRNKPFFK

FMLGKQEVIRGWEEGVAQM SVGQRAKLTISPDYAYGATGHPGIIPPH ATLVFDVELLKLE.

In some embodiments, the FKBP12 polypeptide is encoded by a nucleic acid sequence comprising

(474) TABLE-US-00170 (SEQ ID NO: 14636)

GGGGTCCAGGTCGAGACTATTTCACCAGGGGATGGGCGAACATTTC

AAAGGAGCCAGAGCTTCGCTCGTATTACACCGGGATGCTGGAGG  
ACGGGAAGAAAGTGGACAGCTCCAGGGATCGCAACAAGCCCTTCAAG  
TTCATGCTGGGAAAGCAGGAAGTGATCCGAGGATGGGAGGAAGGCGT  
GGCACAGATGTCAGTCGGCCAGCGGGCCAAACTGACCATTAGCCCTG  
ACTACGCTTATGGAGCAACAGGCCACCCAGGGATCATTCCCCCTCAT  
GCCACCCTGGTCTTCGATGTGGAAGCTGCTGAAGCTGGAG.

(475) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide, the linker region is encoded by an amino acid comprising GGGGS (SEQ ID NO: 14637) or a nucleic acid sequence comprising GGAGGAGGAGGATCC (SEQ ID NO: 14638). In some embodiments, the nucleic acid sequence encoding the linker does not comprise a restriction site.

(476) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an arginine (R) at position 87 of the sequence. In some embodiments, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an alanine (A) at position 282 the sequence. In some embodiments, the truncated caspase 9 polypeptide is encoded by an amino acid comprising

(477) TABLE-US-00171 (SEQ ID NO: 14639)

GFGDVGALESRLGNADLAYILSMEPCGHCLIINNVNFCRESGLRTRT  
GSNIDCEKLRRRFSSLHFMVEVKGDLTAKKMVLLALLELAQQDHGALD  
CCVVVILSHGCQASHLQFPGAVYGTGCPVSVEKIVNIFNGTSCPSL  
GGKPKLFFIQACGGEQKDHGFEVASTPEDESPGSNPEPDATPFQEG  
LRTFDQLDAISSLPTPSDIFVSYSTFPGFVSWRDPKSGSWYVETLDD  
IFEQWAHSEDLQSLLLRVANAVSVKGIYKQMPGCFNFLRKKLFFKTS.

In some embodiments, the truncated caspase 9 polypeptide is encoded by a nucleic acid sequence comprising

(478) TABLE-US-00172 (SEQ ID NO: 14640)

GGATTTGGGGACGTGGGGGCCCTGGAGTCTCTGCGAGGAAATGCCGA  
TCTGGCTTACATCCTGAGCATGGAACCCTGCGGCCACTGTCTGATCA  
TTAACAATGTGAACTTCTGCAGAGAAAGCGGACTGCGAACACGGACT  
GGCTCCAATATTGACTGTGAGAAGCTGCGGAGAAGGTTCTCTAGTCT  
GCACTTTATGGTCGAAGTGAAAGGGGATCTGACCGCCAAGAAAATGG  
TGCTGGCCCTGCTGGAGCTGGCTCAGCAGGACCATGGAGCTCTGGAT  
TGCTGCGTGGTCGTGATCCTGTCCCACGGGTGCCAGGCTTCTCATCT  
GCAGTTCCCCGGAGCAGTGTACGGAACAGACGGCTGTCCTGTCAGCG  
TGGAGAAGATCGTCAACATCTTCAACGGCACTTCTTGCCCTAGTCTG  
GGGGGAAAGCCAAAAGTGTCTTTATCCAGGCCTGTGGCGGGGAACA  
GAAAGATCACGGCTTCGAGGTGGCCAGCACCAGCCCTGAGGACGAAT  
CACCAGGGAGCAACCCTGAACCAGATGCAACTCCATTCCAGGAGGGA  
CTGAGGACCTTTGACCAGCTGGATGCTATCTCAAGCCTGCCCACTCC  
TAGTGACATTTTCGTGTCTTACAGTACCTTCCCAGGCTTTGTCTCAT  
GGCGCGATCCCAAGTCAGGGAGCTGGTACGTGGAGACACTGGACGAC  
ATCTTTGAACAGTGGGCCCATTTCAGAGGACCTGCAGAGCCTGCTGCT  
GCGAGTGGCAAACGCTGTCTCTGTGAAGGGCATCTACAAACAGATGC  
CCGGGTGCTTCAATTTTCTGAGAAAGAAACTGTTCTTTAAGACTTCC.

(479) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide, the inducible proapoptotic polypeptide is encoded by an amino acid sequence comprising

(480) TABLE-US-00173 (SEQ ID NO: 14641)

GVQVETISPGDGRTPKRGQTCVVHYTGMLEDGKKVDSSRDRNKPFFKML  
GKQEVIRGWEEGVAQMSVGQRAKLTISPDYAYGATGHPGIIPPHATLVFD  
VELLKLEGGGSGFGDVGALESRLGNADLAYILSMEPCGHCLIINNVNFC

RESGLRTRTGSLNIDCEKLRRRFFSSLHFMEVTKGDLTAKKMLALLELAQQ  
DHGALDCCVVVILSHGCQASHLQFPGAVYGTGCPVSVEKIVNIFNGTSC  
PSLGGKPKLFFIQACGGEQKDHGFEVASTSPEDESPGSNPEPDATPFQEG  
LRTFDQLDAISSLPTPSDIFVSYSTFPGFVSWRDPKSGSWYVETLDDIFE  
QWAHSEDLQSLLLRVANAVSVKGIYKQMPGCFNFLRKKLFFKTS.

In some embodiments, the inducible proapoptotic polypeptide is encoded by a nucleic acid sequence comprising

(481) TABLE-US-00174 (SEQ ID NO: 14642)

gggggtccagggtcgagactatttcaccaggggatgggcgaacatttccaaa  
aaggggccagacttgctgctgcattacaccgggatgctggaggacggga  
agaaagtggacagctccagggatcgcaacaagcccttcaagttcatgctg  
ggaaagcaggaaagtgatccgaggatgggaggaaggcgtggcacagatgct  
agtcggccagcgggccaactgaccattagccctgactacgcttatggag  
caacaggccacccagggatcattccccctcatgccaccctggcttcgat  
gtggaactgctgaagctggaggaggaggaggatccggatttggggacgt  
ggggggccttgagtgctctgaggaatgccgatctggcttacatcctga  
gcatggaaccctgeggccactgtctgacattaacaatgtgaacttctgc  
agagaaagcggactgcaacacggactggctccaatattgactgtgagaa  
gctgcggagaaggttctctagctgcactttatggtcgaagtgaagggg  
atctgaccgcaagaaaatggtgctggcctgctggagctggctcagcag  
gacatggagctctggattgctgctggctgctgacctgtccacgggtg  
ccaggcttctcatctgcagttccccggagcagtgtacggaacagacggct  
gtcctgtcagcgtggagaagatcgtaacatcttcaacggcacttettgc  
cctagtctgggggggaaagccaaaactgttctttatccaggcctgtggcgg  
ggaacagaaagatcacggcttcgaggtggccagcaccagccctgaggacg  
aatcaccaggaggagcaaccctgaaccagatgcaactccattccaggaggga  
ctgaggacctttgaccagctggatgctatctcaagcctgccactectag tgacatttctgttettacagtaccttccaggccttctcatggcgcg  
atcccaagtcaggagctggtacgtggagacactggacgacatctttgaa  
cagtgggcccattcagaggacctgcagagcctgctgctgcgagtggcaaa  
cgctgtctctgtgaagggcattacaacagatgcccggtgcttcaatt ttctgagaaagaaactgttctttaagacttcc.

(482) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide, the exogenous sequence further comprises a sequence encoding a selectable marker. In some embodiments, the sequence encoding the selectable marker comprises a sequence encoding a detectable marker. In some embodiments, the detectable marker comprises a fluorescent marker or a cell-surface marker. In some embodiments, the sequence encoding the selectable marker comprises a sequence encoding a protein that is active in dividing cells and not active in non-dividing cells. In some embodiments, the sequence encoding the selectable marker comprises a sequence encoding a metabolic marker. In some embodiments, the sequence encoding the selectable marker comprises a sequence encoding a dihydrofolate reductase (DHFR) mutein enzyme. In some embodiments, the DHFR mutein enzyme comprises or consists of the amino acid sequence of:

(483) TABLE-US-00175 (SEQ ID NO: 17012) 1 MVGSLNCIVA VSQNMGIGKN  
GDFWPWPLRN ESRYFQRMTT TSSVEGKQNL 61 VIMGKKTWFS IPEKNRPLKG  
RINLVLSREL KEPPQGAHFL SRSLLDDALKL 121 TEQPELANKV DMVWIVGGSS  
VYKEAMNHPG HLKLFVTRIM QDFESDTFFP 181 EIDLEKYKLL PEYPGVLSDV  
QEEKGIKYKF EVYEKND.

In some embodiments, the DHFR mutein enzyme is encoded by a the nucleic acid sequence comprising or consisting of

(484) TABLE-US-00176 (SEQ ID NO: 17095)

atggtcgggtctctgaattgtatcgctcgccgtgagtcagaacatgggcat  
tggaagaatggcgatttcccatggccacctctgcgcaacgagtcccgat  
actttcagcggatgacaactacctcctctgtggaagggaacagaatctg

gtcatcatgggaaagaacttgggttcagcattccagagaagaaccggcc  
cctgaaaggcagaatcaatctgggtgctgtcccagaaactgaaggagccac  
cacagggagctcactttctgagccgggtccctggacgatgcactgaagctg  
acagaacagcctgagctggccaacaaagtcgatatgggtggtgatcgtcgg  
gggaagttcagtgataaggaggccatgaatcaccccgccatctgaaac tgttcgtcacacggatcatgcaggactttgagagcgatactttcttct  
gaaattgacctggagaagtacaaactgctgcccgaatatcctggcgtgct  
gtccgatgtccaggaagagaaaggcatcaaatacaagttcgaggtctatg agaagaatgac.

In some embodiments, the amino acid sequence of the DHFR mutein enzyme further comprises a mutation at one or more of positions 80, 113, or 153. In some embodiments, the amino acid sequence of the DHFR mutein enzyme comprises one or more of a substitution of a Phenylalanine (F) or a Leucine (L) at position 80, a substitution of a Leucine (L) or a Valine (V) at position 113, and a substitution of a Valine (V) or an Aspartic Acid (D) at position 153.

(485) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide and/or the exogenous sequence comprises a sequence encoding a selectable marker, the exogenous sequence further comprises a sequence encoding a non-naturally occurring antigen receptor, and/or a sequence encoding a therapeutic polypeptide. In some embodiments, the non-naturally occurring antigen receptor comprises a T cell Receptor (TCR). In some embodiments, a sequence encoding the TCR comprises one or more of an insertion, a deletion, a substitution, an inversion, a transposition or a frameshift compared to a corresponding wild type sequence. In some embodiments, a sequence encoding the TCR comprises a chimeric or recombinant sequence. In some embodiments, the non-naturally occurring antigen receptor comprises a chimeric antigen receptor (CAR). In some embodiments, the CAR comprises: (a) an ectodomain comprising an antigen recognition region, (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the ectodomain of (a) of the CAR further comprises a signal peptide. In some embodiments, the ectodomain of (a) of the CAR further comprises a hinge between the antigen recognition region and the transmembrane domain. In some embodiments, the endodomain comprises a human CD3 $\zeta$  endodomain. In some embodiments, the at least one costimulatory domain comprises a human 4-1BB, CD28, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof. In some embodiments, the at least one costimulatory domain comprises a human CD28 and/or a 4-1BB costimulatory domain. In some embodiments, the antigen recognition region comprises one or more of a scFv, a VHH, a VH, and a Centyrin.

(486) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide and/or the exogenous sequence comprises a sequence encoding a selectable marker, the exogenous sequence further comprises a sequence encoding a transposase.

(487) In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises a sequence encoding a selectable marker, an exogenous sequence, a sequence encoding an inducible caspase polypeptide, and at least one sequence encoding a self-cleaving peptide. In some embodiments, the at least one sequence encoding a self-cleaving peptide is positioned between one or more of: (a) the sequence encoding a selectable marker and the exogenous sequence, (b) the sequence encoding a selectable marker and the inducible caspase polypeptide, and (c) the exogenous sequence and the inducible caspase polypeptide. In some embodiments, a first sequence encoding a self-cleaving peptide is positioned between the sequence encoding a selectable marker and the exogenous sequence and a second sequence encoding a self-cleaving peptide is positioned between the exogenous sequence and the inducible caspase polypeptide.

(488) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a piggyBac transposase or a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a piggyBac transposase. In some embodiments, the sequence

encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprise a TTAA, a TTAT or a TTAX recognition sequence. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprise a TTAA, a TTAT or a TTAX recognition sequence and a sequence having at least 50% identity to a sequence isolated or derived from a piggyBac transposase or a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprise at least 2 nucleotides (nts), 3 nts, 4 nts, 5 nts, 6 nts, 7 nts, 8 nts, 9 nts, 10 nts, 11 nts, 12 nts, 13 nts, 14 nts, 15 nts, 16 nts, 17 nts, 18 nts, 19 nts, or 20 nts.

(489) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a piggyBac transposase or a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of

CCCTAGAAAGATAGTCTGCGTAAAATTGACGCATG (SEQ ID NO: 17096) or a sequence having at least 70% identity to the sequence of CCCTAGAAAGATAGTCTGCGTAAAATTGACGCATG (SEQ ID NO: 17096). In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of CCCTAGAAAGATAATCATATTGTGACGTACGTAAAGATAATCATGCGTAAAATTGACGCATG (SEQ ID NO: 17097). In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of CCCTAGAAAGATAGTCTGCGTAAAATTGACGCATG (SEQ ID NO: 17096) and comprises the sequence of

CCCTAGAAAGATAATCATATTGTGACGTACGTAAAGATAATCATGCGTAAAATTGACGCATG (SEQ ID NO: 17097). In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of CCCTAGAAAGATAGTCTGCGTAAAATTGACGCATG (SEQ ID NO: 17096) and comprises the sequence of

CCCTAGAAAGATAATCATATTGTGACGTACGTAAAGATAATCATGTGTAAAATTGACGCATG (SEQ ID NO: 17098). In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of CCCTAGAAAGATAGTCTGCGTAAAATTGACGCATG (SEQ ID NO: 17096) and comprises the sequence of

(490) TABLE-US-00177 (SEQ ID NO: 17099)

TTAACCTAGAAAGATAATCATATTGTGACGTACGTAAAGATAATCATG  
TGTAATAATTGACGCATGTGTTTTATCGGTCTGTATATCGAGGTTTATTTA  
TTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACATACTAAT  
AATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAA  
CAAAAACCTCAAAATTTCTTCTATAAAGTAACAAAACCTTTTA.

(491) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a piggyBac transposase or a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) is recognized by a piggyBac transposase having an amino acid sequence of at least 20% identity to the amino acid sequence of

(492) TABLE-US-00178 (SEQ ID NO: 14487) 1 MGSSLDDEHI LSALLQSDDE



LVGEDSDSEI SDHVSEDDVQ SDTEEAFIDE VHEVQPTSSG 61 SEILDEQNVI  
 EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTGATFRD  
 TNEDEIYAFF 181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRFDFL  
 IRCLRMDDKS IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ  
 LLGFRGRCPF RMYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN  
 GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT  
 VRSNKREIPE VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421  
 DEDASINEST GKPQMVMYYN QTKGGVDTLD QMCSVMTCR KTNRWPMALL  
 YGMINIACIN 481 SFIYSHNVS SKGEKVQSRK KFMRNLYMSL TSSFMRKRLE  
 APTLKRYLRD NISNILPNEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA  
 NASCKKCKKV ICREHNIDMC QSCF.

In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) is recognized by a piggyBac transposase having the amino acid sequence of

(493) TABLE-US-00179 (SEQ ID NO: 14487) 1 MGSSLDDEHI LSALLQSDDE  
 LVGEDSDSEI SDHVSEDDVQ SDTEEAFIDE VHEVQPTSSG 61 SEILDEQNVI  
 EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTGATFRD  
 TNEDEIYAFF 181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRFDFL  
 IRCLRMDDKS IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ  
 LLGFRGRCPF RMYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN  
 GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT  
 VRSNKREIPE VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421  
 DEDASINEST GKPQMVMYYN QTKGGVDTLD QMCSVMTCR KTNRWPMALL  
 YGMINIACIN 481 SFIYSHNVS SKGEKVQSRK KFMRNLYMSL TSSFMRKRLE  
 APTLKRYLRD NISNILPNEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA  
 NASCKKCKKV ICREHNIDMC QSCF.

In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) is recognized by a piggyBac transposase having an amino acid sequence of at least 20% identity to the amino acid sequence of

(494) TABLE-US-00180 (SEQ ID NO: 14484) 1 MGSSLDDEHI LSALLQSDDE  
 LVGEDSDSEV SDHVSEDDVQ SDTEEAFIDE VHEVQPTSSG 61 SEILDEQNVI  
 EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTSATFRD  
 TNEDEIYAFF 181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRFDFL  
 IRCLRMDDKS IRPTLRENDV 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ  
 LLGFRGRCPF RVIYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN  
 GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT  
 VRSNKREIPE VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421  
 DEDASINEST GKPQMVMYYN QTKGGVDTLD QMCSVMTCR KTNRWPMALL  
 YGMINIACIN 481 SFIYSHNVS SKGEKVQSRK KFMRNLYMSL TSSFMRKRLE  
 APTLKRYLRD NISNILPKEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA  
 NASCKKCKKV ICREHNIDMC QSCF.

In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) is recognized by a piggyBac transposase having the amino acid sequence of

(495) TABLE-US-00181 (SEQ ID NO: 14484) 1 MGSSLDDEHI LSALLQSDDE  
 LVGEDSDSEV SDHVSEDDVQ SDTEEAFIDE VHEVQPTSSG 61 SEILDEQNVI  
 EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRRSRVS ALNIVRSQRG 121  
 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTSATFRD  
 TNEDEIYAFF 181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRFDFL

IRCLRMDDDKS IR241FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ  
 LLGFRGRCPF RVYIPNKPSK YGIKILMMCD 301 SGTKYMINGM PYLGRGTQTN  
 GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ 361 EPYKLTIVGT  
 VRSNKREIPE VLKNSRSRPV GTSMFCFDGP LTLVSYKPKP AKMVYLLSSC 421  
 DEDASINEST GKPQMVMYYN QTKGGVDTLD QMCSVMTCSR KTNRWPMALL  
 YGMINIACIN 481 SFIYSHNVS SKGEKVQSRK KFMRNLYMSL TSSFMRKRLE  
 APTLKRYLRD NISNILPKEV 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRKA  
 NASCKKCKKV ICREHNIDMC QSCF.

(496) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a Sleeping Beauty transposase. In some embodiments, the Sleeping Beauty transposase is a hyperactive Sleeping Beauty transposase (SB100X).

(497) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a Helitron transposase.

(498) In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a Tol2 transposase.

(499) The disclosure provides a cell comprising a nanotransposon of the disclosure. In some embodiments, the cell further comprises a transposase composition. In some embodiments, the transposase composition comprises a transposase or a sequence encoding the transposase that is capable of recognizing the first ITR or the second ITR of the nanotransposon. In some embodiments, the transposase composition comprises a nanotransposon comprising the sequence encoding the transposase. In some embodiments, the cell comprises a first nanotransposon comprising an exogenous sequence and a second nanotransposon comprising a sequence encoding a transposase. In some embodiments, the cell is an allogeneic cell.

(500) The disclosure provides a composition comprising the nanotransposon of the disclosure.

(501) The disclosure provides a composition comprising the cell of the disclosure. In some embodiments, the cell comprises a nanotransposon of the disclosure. In some embodiments, the cell is not further modified. In some embodiments, the cell is allogeneic.

(502) The disclosure provides a composition comprising the cell of the disclosure. In some embodiments, the cell comprises a nanotransposon of the disclosure. In some embodiments, the cell is not further modified. In some embodiments, the cell is autologous.

(503) The disclosure provides a composition comprising a plurality of cells of the disclosure. In some embodiments, at least one cell of the plurality of cells comprises a nanotransposon of the disclosure. In some embodiments, a portion of the plurality of cells comprises a nanotransposon of the disclosure. In some embodiments, the portion comprises at least 1%, 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 99% or any percentage in between of the plurality of cells. In some embodiments, each cell of the plurality of cells comprises a nanotransposon of the disclosure. In some embodiments, the plurality of cells does not comprise a modified cell of the disclosure. In some embodiments, at least one cell of the plurality of cells is not further modified. In some embodiments, none of the plurality of cells is not further modified. In some embodiments, plurality of cells is allogeneic. In some embodiments, an allogeneic plurality of cells are produced according to the methods of the disclosure. In some embodiments, plurality of cells is autologous. In some embodiments, an autologous plurality of cells are produced according to the methods of the disclosure.

(504) The disclosure provides a modified cell comprising: (a) a nanotransposon of the disclosure; (b) a sequence encoding an inducible proapoptotic polypeptide; and wherein the cell is a T cell, (c) a modification of an endogenous sequence encoding a T cell Receptor (TCR), wherein the modification reduces or eliminates a level of expression or activity of the TCR. In some embodiments, the cell further comprises: (d) a non-naturally occurring sequence comprising an HLA class I histocompatibility antigen, alpha chain E (HLA-E), and (e) a modification of an endogenous sequence encoding Beta-2-Microglobulin (B2M), wherein the modification reduces or eliminates a level of expression or activity of a major histocompatibility complex (MHC) class I (MHC-I).

(505) The disclosure provides a modified cell comprising: (a) a nanotransposon of the disclosure; (b) a sequence encoding an inducible proapoptotic polypeptide; (c) a non-naturally occurring sequence comprising an HLA class I histocompatibility antigen, alpha chain E (HLA-E), and (e) a modification of an endogenous sequence encoding Beta-2-Microglobulin (B2M), wherein the modification reduces or eliminates a level of expression or activity of a major histocompatibility complex (MHC) class I (MHC-I).

(506) In some embodiments of the modified cells of the disclosure, the non-naturally occurring sequence comprising a HLA-E further comprises a sequence encoding a B2M signal peptide. In some embodiments, the non-naturally occurring sequence comprising an HLA-E further comprises a linker, wherein the linker is positioned between the sequence encoding the sequence encoding a B2M polypeptide and the sequence encoding the HLA-E. In some embodiments, the non-naturally occurring sequence comprising an HLA-E further comprises a sequence encoding a peptide and a sequence encoding a B2M polypeptide. In some embodiments, the non-naturally occurring sequence comprising an HLA-E further comprises a first linker positioned between the sequence encoding the B2M signal peptide and the sequence encoding the peptide, and a second linker positioned between the sequence encoding the B2M polypeptide and the sequence encoding the HLA-E.

(507) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a mammalian cell.

(508) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a human cell.

(509) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a stem cell.

(510) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a differentiated cell.

(511) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a somatic cell.

(512) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is an immune cell or an immune cell precursor. In some embodiments, the immune cell is a lymphoid progenitor cell, a natural killer (NK) cell, a cytokine induced killer (CIK) cell, a T lymphocyte (T cell), a B lymphocyte (B-cell) or an antigen presenting cell (APC). In some embodiments, the immune cell is a T cell, an early memory T cell, a stem cell-like T cell, a stem memory T cell (Tscm), or a central memory T cell (Tcm). In some embodiments, the immune cell precursor is a hematopoietic stem cell (HSC). In some embodiments, the cell is an antigen presenting cell (APC).

(513) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell further comprises a gene editing composition. In some embodiments, the gene editing composition comprises a sequence encoding a DNA binding domain and a sequence encoding a nuclease protein or a nuclease domain thereof. In some embodiments, the gene editing composition comprises a sequence encoding a nuclease protein or a sequence encoding a nuclease domain thereof. In some embodiments, the sequence encoding a nuclease protein or the sequence encoding a nuclease domain thereof comprises a DNA sequence, an RNA sequence, or a combination thereof. In some embodiments, the nuclease or the nuclease domain thereof comprises one or more of a CRISPR/Cas protein, a Transcription Activator-Like Effector Nuclease (TALEN), a Zinc Finger Nuclease (ZFN), and an endonuclease. In some embodiments, the CRISPR/Cas protein comprises a nuclease-inactivated Cas (dCas) protein.

(514) In some embodiments of the cells, modified cells of the disclosure, the cell further comprises a gene editing composition. In some embodiments, the gene editing composition comprises a sequence encoding a DNA binding domain and a sequence encoding a nuclease protein or a nuclease domain thereof. In some embodiments, the nuclease or the nuclease domain thereof comprises a nuclease-inactivated Cas (dCas) protein and an endonuclease. In some embodiments, the endonuclease comprises a Clo051 nuclease or a nuclease domain thereof. In some embodiments, the gene editing composition comprises a fusion protein. In some embodiments, the fusion protein comprises a nuclease-inactivated Cas9 (dCas9) protein and a Clo051 nuclease or a Clo051 nuclease domain. In some embodiments, the gene editing composition further comprises a guide sequence. In some embodiments, the guide sequence comprises an RNA sequence. In some embodiments, the fusion protein comprises or consists of the amino acid sequence:

(515) TABLE-US-00182 (SEQ ID NO: 17013)

MAPKKKRKVEGIKSNISLLKDELRGQISHISHEYLSLIDLAFDSKQNRFLF  
EMKVLELLVNEYGFKGRHLGGSRKPDGIVYSTTLEDNFGIIVDTKAYSEG  
YSLPISQADEMERYVRENSNRDEEVNPNKWWENFSEEVKKYYFVFISGSF  
KGKFEEQLRRLSMTTG VNGSAVNVNLLLGA EKIRSGEMTIEELERAMFN  
NSEFILKYGGGGSDKKYSIGLAIGTNSVGWAVITDEYKVPSKKEKVLGNT  
DRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSN  
EMAKVDDSSFFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHL  
RKKLVDSTDKADLRLIYLA LAHMIKFRGHFLIEGDLNPDNSDVKLFIQL  
VQTYNQLFEE NPINASGVDAKAILSARLSKSRRLENLIAQLPGEKKNGLF  
GNLIALSLGLTPNEKSNEDLAEDAKLQLSKDTYDDDLDNLLAQIGDQYAD  
LFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKAL  
VRQQLPKEYKEIFFDQSKNGYAGYIDGGASQEEFYKFIPKILEKMDGTTEE  
LLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYPPFLKDNRE  
KIEKILTFRIPYYVGPLARGNSRFAWMTRKSEETITPWNFEEVVDKGASA  
QSFIERMTNEDKNLPNEKVLPKHSLLYEYFTVYNELTKVKYVTEGMRKPA  
ELSGEQKKAIVDLLKTNRKVIVKQLKEDYFKKIECEDSVESGVEDRFN  
ASLGTYHDLLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMIEERLKT  
YAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGF  
ANRNFMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKKGIL  
QTVKVVDLVKVMGRHKPENIVIAMARENQTTQKGQKNSRERMKRIEEGI  
KELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQELDINRLSDYDV  
DAIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMKNYWRQLLN  
AKLITQRKEDNLIKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSR  
MNTKYDENDKLIREVKVITLKS KLVSDERKDFQFYKVREINNYHHAHDAY  
LNAVVG TALIKKYPKLESEFVYGDYKVYDVRKMIKSEQEIGKATAKYFF  
YSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLS  
MPQVNIVKKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPT  
VAYSVLVVAKVEKGKSKKLKSVKELLGITIMERSSEFEKNPIDFLEAKGYK  
EVKCDLIKLPKYSLFELENGRKRMLASAGELQKGNELALPSKYVNFLYL  
ASHYEKLKGS PEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVLADANLD  
KVLSAYNKH RDKPIREQAENIIHLFTLTNLGAPAAF KYFDTTIDRKRYTS  
TKEVL DATLIHQ SITGLYETRIDLSQLGGDGSPKKKRKVSS

or a nucleic acid comprising or consisting of the sequence:

(516) TABLE-US-00183 (SEQ ID NO: 17014) 1 atggcaccaa agaagaaaag aaaagtggag  
ggcatcaagt caaacatcag cctgctgaaa 61 gacgaactgc ggggacagat tagtcacatc agtcacgagt  
acctgtcact gattgatctg 121 gccttcgaca gcaagcagaa tagactgttt gagatgaaag tgctggaact  
gctggtcaac 181 gagtatggct tcaagggcag acatctgggc ggggtctagga aacctgacgg catcgtgtac 241  
agtaccacac tggaagacaa cttcgggaatc attgtcgata ccaaggctta ttccgagggc 301 tactctctgc  
caattagtcagg ggcagatgag atggaaaggt acgtgcgcga aaactcaaat 361 agggacgagg aagtcaaccc  
caataagtgg tgggagaatt tcagcgagga agtgaagaaa 421 tactacttcg tctttatctc aggcagcttc aaagggaagt

ttgaggaaac gctgcggaga 481 ctgtcatga ctacgggggt gaacggatct gctgtcaacg tggctcaatct  
 gctgctgggc 541 gcagaaaaga tcaggtccgg ggagatgaca attgaggaac tggaacgcgc catgttcaac 601  
 aattctgagt ttatctgaa gtatggaggc gggggaagcg ataagaaata ctccatcgga 661 ctggccattg  
 gcaccaatc cgtgggctgg gctgtcatca cagacgagta caaggtgcca 721 agcaagaagt tcaaggtcct  
 ggggaacacc gatcgccaca gtatcaagaa aaatctgatt 781 ggagccctgc tgttcgactc aggcgagact  
 gctgaagcaa cccgactgaa gcggactgct 841 aggcgccgat ataccgagg aaaaaatcgg atctgctacc  
 tgcaggaaat tttcagcaac 901 gagatggcca aggtggacga tagtttctt caccgcctgg aggaatcatt  
 cctggtggag 961 gaagataaga aacacgagcg gcatcccatc ttggcaaca ttgtggacga agtcgcttat 1021  
 cacgagaagt accctactat ctatcatctg aggaagaaac tgggtggactc caccgataag 1081 gcagacctgc  
 gcctgatcta tctggccctg gctcacatga tcaagtccg ggggcatttt 1141 ctgatcgagg gagatctgaa  
 ccctgacaat tctgatgtgg acaagctgtt catccagctg 1201 gtccagacat acaatcagct gtttgaggaa  
 aacccaatta atgcctcagg cgtggacgca 1261 aaggccatcc tgagcgccag actgtccaaa ttaggcgcc  
 tggaaaacct gatcgctcag 1321 ctgccaggag agaagaaaaa cggcctgtt gggaatctga tgcactgtc  
 cctgggcctg 1381 acaccaact tcaagtctaa tttgatctg gccgaggacg ctaagctgca gctgtccaaa 1441  
 gacacttatg acgatgacct ggataacctg ctggctcaga tcggcgatca gtacgcagac 1501 ctgttctctg  
 ccgctaagaa tctgagtgac gccatctgc tgcagatat tctgcgcgtg 1561 aacacagaga ttactaaggc  
 cccactgagt gttcaatga taaaagata tgacgagcac 1621 catcaggatc tgaccctgct gaaggctctg  
 gtgaggcagc agtgcccga gaaatacaag 1681 gaaatcttct ttgatcagag caagaatgga tacgccggct  
 atattgacgg cggggcttcc 1741 caggaggagt tctacaagt catcaagccc attctggaaa agatggacgg  
 caccgaggaa 1801 ctgctggtga agctgaatcg ggaggacctg ctgagaaaac agaggacatt tgataacgga 1861  
 agcatccctc accagattca tctgggcgaa ctgcacgcca tctgcgacg gcaggaggac 1921 ttctacccat  
 ttctgaagga taaccgcgag aaaatcgaag agatcctgac ctccagaatc 1981 ccctactatg tggggcctct  
 ggcacgggga aatagtagat ttgcttgat gacaagaaag 2041 tcagaggaaa ctatcccc ctggaactc  
 gaggaagtgg tcgataaagg cgctagcgca 2101 cagtccttca ttgaaaggat gacaaattt gacaagaacc  
 tgccaaatga gaaggtgctg 2161 ccaaacaca gcctgctgta cgaatattc acagtgtata acgagctgac  
 taaagtgaag 2221 tacgtcaccg aagggatcg caagcccgca ttctgtccg gagagcagaa gaaagccatc 2281  
 gtggacctgc tgttaagac aaatcggaag gtgactgtca aacagctgaa ggaagactat 2341 ttcaagaaaa  
 ttgagtgtt cgattcagt gaaatcagcg gcgtcgagga caggtttaac 2401 gcctccctgg ggacctacca  
 cgatctgctg aagatcatca aggataagga ctctctggac 2461 aacgaggaaa atgaggacat cctggaggac  
 attgtgctga cactgactct gtttgaggat 2521 cgcgaaatga tcgaggaacg actgaagact tatgccatc  
 tgttcgatga caaagtgatg 2581 aagcagctga aaagaaggcg ctacaccgga tggggacgcc tgagccgaaa  
 actgatcaat 2641 gggattagag acaagcagag cggaaaaact atcctggact ttctgaagtc cgatggcttc 2701  
 gccaacagga acttcatgca gctgattcac gatgactctc tgaccttcaa ggaggacatc 2761 cagaaagcac  
 aggtgtctgg ccagggggac agtctgcacg agcatatcgc aaacctggcc 2821 ggcagccccg ccatcaagaa  
 agggattctg cagaccgtga aggtggtgga cgaactggc 2881 aaggtcatgg gacgacaaa acctgagaac  
 atcgtgattg agatggcccg cgaaaatcag 2941 acaactcaga agggccagaa aaacagtcga gaacggatga  
 agagaatcga ggaaggcatc 3001 aaggagctgg ggtcacagat cctgaaggag catcctgtgg aaaacactca  
 gctgcagaat 3061 gagaaactgt atctgtacta tctgcagaat ggacgggata tgtacgtgga ccaggagctg 3121  
 gatattaaca gactgagtga ttatgacgtg gatgccatcg tcctcagag ctctctgaag 3181 gatgactcca  
 ttgacaacaa ggtgctgacc aggtccgaca agaaccgcgg caaatcagat 3241 aatgtgccaa gcgaggaagt  
 ggtcaagaaa atgaagaact actggaggca gctgctgaat 3301 gccaaactga tcacacagcg gaaattgat  
 aacctgacta aggcagaaaag aggggcctg 3361 tctgagctgg acaaggccgg ctcatcaag cggcagctgg  
 tggagacaag acagatcact 3421 aagcacgtcg ctgagattct ggatagcaga atgaacaaa agtacgatga  
 aaacgacaag 3481 ctgatcaggg aggtgaaagt cattactctg aaatccaagc tgggtgtctga cttagaaaag 3541  
 gatttccagt ttataaagt caggagatc aacaactacc accatgtca tgacgcatc 3601 ctgaacgcag  
 tggtcgggac cgccctgatt aagaaatacc ccaagctgga gtccgagttc 3661 gtgtacggag actataaagt  
 gtacgatgtc cggaagatga tcgccaatc tgagcaggaa 3721 attggcaagg ccaccgtaa gtatttctt  
 tacagtaaca tcatgaattt cttaagacc 3781 gaaatcacac tggcaaatgg ggagatcaga aaaaggcctc  
 tgattgagac caacggggag 3841 acaggagaaa tcgtgtggga caagggaagg gattttgcta ccgtgcgcaa  
 agtctgtctc 3901 atgccccag tgaatttgt caagaaaact gaagtgcaga ccgggggatt cttaaggag 3961  
 agtattctgc ctaagcgaag ctctgataaa ctgatcgccc ggaagaaaga ctgggacccc 4021 aagaagtatg  
 gcgggttcga ctctccaaca gtggcttaca gtgtcctggt ggtcgcaaag 4081 gtggaaaagg ggaagtcaa

gaaactgaag tctgtcaaaag agctgctggg aatcactatt 4141 atggaacgca gctccttcga gaagaatcct  
atcgattttc tggaagccaa gggctataaa 4201 gaggtgaaga aagacctgat cattaagctg ccaaaatact  
cactgtttga gctggaaaac 4261 ggacgaaagc gaatgctggc aagcgccgga gaactgcaga agggcaatga  
gctggccctg 4321 ccttccaaat acgtgaactt cctgtatctg gctagccact acgagaaact gaaggggtcc 4381  
cctgaggata acgaacagaa gcagctgttt gtggagcagc acaaacatta tctggacgag 4441 atcattgaac  
agatttcaga gttcagcaag agagtgatcc tggctgacgc aaatctggat 4501 aaagtctga gcgcatacaa  
caagcaccga gacaaaccaa tccgggagca ggccgaaaat 4561 atcattcatc tgttcaccct gacaaacctg  
ggcgcccctg cagccttcaa gtattttgac 4621 accacaatcg atcggaagag atacacttct accaaagagg  
tgctggatgc taccctgatc 4681 caccagagta ttaccggcct gtatgagaca cgcacgacc tgtcacagct  
gggaggcgat 4741 gggagcccca agaaaaagcg gaaggtgtct agttaa.

In some embodiments, the fusion protein comprises or consists of the amino acid sequence:

(517) TABLE-US-00184 ((SEQ ID NO: 17058) 1 MPKKKRKVEG IKSNISLLKD  
ELRGQISHIS HEYLSLIDLA FDSKQNRLE MKVLELLVNE 61 YGFKGRHLGG  
SRKPDGIVYS TTLEDNFGII VDTKAYSEGY SLPISQADEM ERYVRENSNR 121  
DEEVNPNKWW ENFSEEVKKY YFVFISGSFK GKFEQLRRL SMTTGVNGSA  
VNVVNLLLGA 181 EKIRSGEMTI EELERAMFNN SEFILKYGGG GSDKKYSIGL  
AIGTNSVGWA VITDEYKVPS 241 KKFKVLGNTD RSHIKKNLIG ALLFDSGETA  
EATRLKRTAR RRYTRRKNRI CYLQEIFSNE 301 MAKVDDSFFH RLEESFLVEE  
DKKHERHPIF GNIVDEVAYH EKYPTIYHLR KKLVDSTDKA 361 DLRLIYLA  
HMIKFRGHFL IEGDLNPDNS DVDKLFQILV QTYNQLFEEN PINASGVDAK 421  
AILSARLSKS RRLENLIAQL PGEKKNGLFG NLIALSLGLT PNFKSNFDLA  
EDAKLQLSKD 481 TYDDDLNLL AQIGDQYADL FLAAKNLSA ILLSDILRVN  
TEITKAPLSA SMIKRYDEHH 541 QDLTLLKALV RQQLPEKYKE IFFDQSKNGY  
AGYIDGGASQ EEFYKFIKPI LEKMDGTEEL 601 LVKLNREDLL RKQRTFDNGS  
IPHQIHLGEL HAILRRQEDF YPFLKDNREK IEKILTFRIP 661 YYVGPLARGN  
SRFAWMTRKS EETITPWNFE EVVDKGASAQ SFIERMTNFD KNLPNEKVLP 721  
KHSLLYEYFT VYNELTKVKY VTEGMRKPAF LSGEQKKAIV DLLFKTNRKV  
TVKQLKEDYF 781 KKIECFDSVE ISGVEDRFNA SLGTYHDLK IIKDKDFLDN  
EENEDILEDI VLTTLTFEDR 841 EMIEERLKTY AHLFDDKVMK QLKRRRYTGW  
GRLSRKLING IRDKQSGKTI LDFLKSDGFA 901 NRNFMQLIHD DSLTFKEDIQ  
KAQVSGQGDS LHEHIANLAG SPAIKKGILQ TVKVVDLVK 961 VMGRHKPENI  
VIEMARENQT TQKGQKNSRE RMKRIEELIK ELGSQILKEH PVENTQLQNE 1021  
KLYLYYLQNG RDMYVDQELD INRLSDYDVD AIVPQSFLKD DSIDNKVLTR  
SDKNRGKSDN 1081 VPSEEVVKKM KNYWRQLLNA KLITQRKFDN LTKAERGGLS  
ELDKAGFIKR QLVETRQITK 1141 HVAQILDSRM NTKYDENDKL IREVKVITLK  
SKLVSDFRKD FQFYKVREIN NYHHAHDAYL 1201 NAVVGTA  
LIK KYPKLESEFV  
YGDYKVYDVR KMIKSEQEIGKATAKYFFY SNIMNFFKTE 1261 ITLANGEIRK  
RPLIETNGET GEIVWDKGRD FATVRKVLSPQVNIVKKTE VQTGGFSKES 1321  
ILPKRNSDKL IARKKDWDPK KYGGFDSPTV AYSVLVAKV EKGKSKKLKS  
VKELLGITIM 1381 ERSSFENPI DFLEAKGYKE VKKDLIIKLP KYSLEFELNG  
RKRMLASAGE LQKGNELALP 1441 SKYVNFLYLA SHYEKLKGSP EDNEQKQLFV  
EQHKHYLDEI IEQISEFSKR VILADANLDK 1501 VLSAYNKH  
RD KPIREQAENI  
IHLFTLINLG APAAFKYFDT TIDRKRYTST KEVLDTLIH 1561 QSITGLYETR  
IDLSQLGGDG SPKKKRKV

or a nucleic acid comprising or consisting of the sequence:

(518) TABLE-US-00185 (SEQ ID NO: 17059) 1 atgcctaaga agaagcggaa ggtggaaggc  
atcaaaagca acatctcct cctgaaagac 61 gaactccggg ggcagattag ccacattagt cacgaatacc tctcctcat  
cgacctggct 121 ttcgatagca agcagaacag gctctttgag atgaaagtgc tggaactgct cgtcaatgag 181  
tacgggttca agggctgaca cctcggcggg tctaggaaac cagacggcat cgtgtatagt 241 accacactgg  
aagacaactt tgggatcatt gtggatacca aggcatactc tgagggttat 301 agtctgcca ttcacaggc  
cgacgagatg gaacggtacg tgcgcgagaa ctcaaataga 361 gatgaggaag tcaaccctaa caagtgggtg  
gagaacttct ctgaggaagt gaagaaatac 421 tacttcgtct ttatcagcgg gtccttaag ggtaaatttg aggaacagct

caggagactg 481 agcatgacta ccggcgtgaa tggcagcgcc gtcaacgtgg tcaatctgct cctgggcgct 541  
gaaaagattc ggagcggaga gatgaccatc gaagagctgg agaggggcaat gtttaataat 601 agcagagtta  
tcctgaaata cgggtggcggg ggatccgata aaaagtattc tattggttta 661 gccatcggca ctaattccgt  
tggatgggct gtcataaccg atgaatacaa agtaccttca 721 aagaaattta aggtgttggg gaacacagac cgtcattcga  
ttaaaaagaa tcttatcggt 781 gccctcctat tcgatagtgg cgaaacggca gaggcgactc gcctgaaacg  
aaccgctcgg 841 agaaggtata cacgtcgcaa gaaccgaata tgttacttac aagaaattht tagcaatgag 901  
atggccaaag ttgacgattc tttctttcac cgtttggaag agtccttctt tgtcgaagag 961 gacaagaaac  
atgaacggca ccccatcttt ggaaacatag tagatgaggt ggcataatcat 1021 gaaaagtacc caacgattta  
tcacctcaga aaaaagctag ttgactcaac tgataaagcg 1081 gacctgaggt taatctactt ggctcttgcc catatgataa  
agttccgtgg gcactttctc 1141 attgaggggtg atctaaatcc ggacaactcg gatgtcgaca aactgttcat  
ccagtttagta 1201 caaacctata atcagttgtt tgaagagaac cctataaatg caagtggcgt ggatgcgaag 1261  
gctattctta gcgcccgcct ctctaaatcc cgacggctag aaaacctgat cgcacaatta 1321 cccggagaga  
agaaaaatgg gttgttcggg aaccttatag cgctctcact aggcctgaca 1381 ccaaatttta agtcgaactt  
cgacttagct gaagatgcca aattgcagct tagtaaggac 1441 acgtacgatg acgatctcga caatctactg  
gcacaaattg gagatcagta tgcggactta 1501 ttttggtgctg caaaaaacct tagcgatgca atcctcctat ctgacatact  
gagagttaat 1561 actgagatta ccaaggcgcc gttatccgct tcaatgatca aaaggtacga tgaacatcac 1621  
caagacttga cacttctcaa ggccctagtc cgtcagcaac tgcctgagaa atataaggaa 1681 atattctttg  
atcagtcgaa aaacgggtac gcaggttata ttgacggcgg agcgagtcaa 1741 gaggaattct acaagtttat  
caaaccata ttagagaaga tggatgggac ggaagagttg 1801 ctgtgaaaac tcaatcgcgga agatctactg  
cgaaagcagc ggactttcga caacggtagc 1861 attccacatc aaatccactt aggcgaattg catgctatac  
ttagaaggca ggaggatttt 1921 tatccgttcc tcaaagacaa tcgtgaaaag attgagaaaa tctaacctt  
tcgcatacct 1981 tactatgtgg gaccctggc ccgaggggaa tctcgggtcg catggatgac aagaaagtc 2041  
gaagaaacga ttactccatg gaattttgag gaagttgtcg ataaaggtgc gtcagctcaa 2101 tcgttcacg  
agaggatgac caactttgac aagaatttac cgaacgaaaa agtattgcct 2161 aagcacagtt tactttacga  
gtatttcaca gtgtacaatg aactcacgaa agttaagtat 2221 gtcactgagg gcatgcgtaa accgccttt  
ctaagcggag aacagaagaa agcaatagta 2281 gatctgttat tcaagaccaa ccgcaaagt acagttaagc  
aattgaaaga ggactacttt 2341 aagaaaattg aatgcttcga ttctgtcgag atctccgggg tagaagatcg  
atttaatgcg 2401 tcacttggtg cgtatcatga cctcctaaag ataattaaag ataaggactt cctggataac 2461  
gaagagaatg aagatatctt agaagatata gtgttgactc ttaccctctt tgaagatcgg 2521 gaaatgattg  
aggaaagact aaaaacatac gtcacctgt tcgacgataa ggttatgaaa 2581 cagttaaaga ggctgtcgta  
tacgggctgg ggacgattgt cgcggaaact tatcaacggg 2641 ataagagaca agcaaagtgg taaaactatt  
ctcgattttc taaagagcga cggcttcgcc 2701 aataggaact ttatgcagct gatccatgat gactctttaa ccttcaaaga  
ggatatacaa 2761 aaggcacagg tttccggaca aggggactca ttgcacgaac atattgcgaa tcttgctggg 2821  
tcgccagcca tcaaaaaggg catactccag acagtcaaag tagtggatga gctagttaag 2881 gtcattgggac  
gtcaciaaac ggaaaacatt gtaatcgaga tggcacgcga aaatcaaacg 2941 actcagaagg ggcaaaaaa  
cagtcgagag cggatgaaga gaatagaaga gggattataa 3001 gaactgggca gccagatctt aaaggagcat  
cctgtggaaa ataccaatt gcagaacgag 3061 aaactttacc tctattacct acaaaatgga agggacatgt  
atgttgatca ggaactggac 3121 ataaaccgtt tatctgatta cgacgtcgat gccattgtac ccaatcctt  
tttgaaggac 3181 gattcaatcg acaataaagt gcttacacgc tcggataaga accgagggaa aagtgacaat 3241  
gttccaagcg aggaagtcgt aaagaaaatg aagaactatt ggccggcagct cctaaatgcg 3301 aaactgataa  
cgcaaaagaa gttcgataac ttaactaaag ctgagagggg tggctgtct 3361 gaacttgaca aggccggatt  
tattaaacgt cagctcgtgg aaaccgcca aatcacaaaag 3421 catgttgac agatactaga ttcccgaatg  
aatacgaaat acgacgagaa cgataagctg 3481 attcgggaag tcaaagtaac cactttaaag tcaaaattgg  
tgtcggactt cagaaaggat 3541 ttcaattct ataaagttag ggagataaat aactaccacc atgcgcacga  
cgcttatctt 3601 aatgccgtcg tagggaccgc actcattaag aaatacccga agctagaaag tgagttgtg 3661  
tatgggtgatt acaaaagtta tgacgtccgt aagatgatcg cgaaaagcga acaggagata 3721 ggcaaggcta  
cagccaaata cttcttttat tctaacatta tgaattctt taagacggaa 3781 atcactctgg caaacggaga  
gatacgcaaa cgacctttaa ttgaaaccaa tggggagaca 3841 ggtgaaatcg tatgggataa gggccgggac  
ttcgcgacgg tgagaaaagt tttgtccatg 3901 ccccaagtca acatagtaaa gaaaactgag gtgcagaccg  
gagggttttc aaaggaatcg 3961 attcttcaa aaaggaatag tgataagctc atcgctcgta aaaaggactg  
ggaccgaaa 4021 aagtacgggtg gcttcgatag ccctacagtt gcctattctg tctagtagt ggcaaaagt 4081  
gagaagggaa aatccaagaa actgaagtca gtcaaagaat tattggggat aacgattatg 4141 gagcgtcgt

cttttgaaaa gaaccccatc gacttccttg aggcgaaagg ttacaaggaa 4201 gtaaaaaagg attcataat  
 taaactacca aagtatagtc tgtttgagtt agaaaatggc 4261 cgaaaacgga tgttggctag cgccggagag  
 ctcaaaagg ggaacgaact cgcactaccg 4321 tctaaatacg tgaatttcct gtatttagcg tcccattacg  
 agaagttgaa aggttcacct 4381 gaagataacg aacagaagca acttttgtt gagcagcaca aacattatct  
 cgacgaaatc 4441 atagagcaaa ttcggaatt cagtaagaga gtcctcctag ctgatgccaa tctggacaaa 4501  
 gtattaagcg catacaaaa gcacagggat aaaccatac gtgagcaggc ggaaaatatt 4561 atccattgt  
 ttactcttac caacctcggc gctccagccg cattcaagta tttgacaca 4621 acgatagatc gcaaacgata  
 cactctacc aaggaggtgc tagacgcgac actgattcac 4681 caatccatca cgggattata tgaaactcgg  
 atagatttgt cacagcttgg ggggtgacgga 4741 tcccccaaga agaagaggaa agtctga.

(519) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, a nanotransposon comprises the gene editing composition comprising a guide sequence and a sequence encoding a fusion protein comprising a sequence encoding an inactivated Cas9 (dCas9) and a sequence encoding a Clo051 nuclease or a nuclease domain thereof.

(520) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell expresses the gene editing composition transiently.

(521) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a T cell and the guide RNA comprises a sequence complementary to a target sequence encoding an endogenous TCR. In some embodiments, the guide RNA comprises a sequence complementary to a target sequence encoding a B2M polypeptide.

(522) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the guide RNA comprises a sequence complementary to a target sequence within a safe harbor site of a genomic DNA sequence.

(523) In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the Clo051 nuclease or a nuclease domain thereof induces a single or double strand break in a target sequence. In some embodiments, a donor sequence, a donor plasmid, or a donor nanotransposon intra-ITR sequence integrated at a position of single or double strand break and/or at a position of cellular repair within a target sequence.

(524) The disclosure provides a composition comprising a modified cell according to the disclosure. In some embodiments, the composition further comprises a pharmaceutically-acceptable carrier.

(525) The disclosure provides a composition comprising a plurality of modified cells according to the disclosure. In some embodiments, the composition further comprises a pharmaceutically-acceptable carrier.

(526) The disclosure provides a composition of the disclosure for use in the treatment of a disease or disorder.

(527) The disclosure provides the use of a composition of the disclosure for the treatment of a disease or disorder.

(528) The disclosure provides a method of treating a disease or disorder comprising administering to a subject in need thereof a therapeutically-effective amount of a composition of the disclosure. In some embodiments, the subject does not develop graft vs. host (GvH) and/or host vs. graft (HvG) following administration of the composition. In some embodiments, the administration is systemic. In some embodiments, the composition is administered by an intravenous route. In some embodiments, the composition is administered by an intravenous injection or an intravenous infusion.

(529) The disclosure provides a method of treating a disease or disorder comprising administering to a subject in need thereof a therapeutically-effective amount of a composition of the disclosure. In some embodiments, the subject does not develop graft vs. host (GvH) and/or host vs. graft (HvG) following administration of the composition. In some embodiments, the administration is local. In some embodiments, the composition is administered by an intra-tumoral route, an intraspinal route, an intracerebroventricular route, an intraocular route or an intraosseous route. In some embodiments, the composition is administered by an intra-tumoral injection or infusion, an intraspinal injection or infusion, an intracerebroventricular injection or infusion, an intraocular injection or infusion or an intraosseous injection or infusion.

(530) In some embodiments of the methods of treating a disease or disorder of the disclosure, the



therapeutically effective dose is a single dose and wherein the allogeneic cells of the composition engraft and/or persist for a sufficient time to treat the disease or disorder. In some embodiments, the single dose is one of at least 2, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100 or any number of doses in between that are manufactured simultaneously.

(531) In some embodiments of the methods of treating a disease or disorder of the disclosure, the therapeutically effective dose is a single dose and wherein the autologous cells of the composition engraft and/or persist for a sufficient time to treat the disease or disorder. In some embodiments, the single dose is one of at least 2, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100 or any number of doses in between that are manufactured simultaneously.

(532) In some embodiments of the composition and methods of the disclosure, allogeneic cells are stem cells. In some embodiments, allogeneic cells are derived from stem cells. Exemplary stem cells include, but are not limited to, embryonic stem cells, adult stem cells, induced pluripotent stem cells (iPSCs), multipotent stem cells, pluripotent stem cells, and hematopoietic stem cells (HSCs).

(533) In some embodiments of the composition and methods of the disclosure, allogeneic cells are differentiated somatic cells.

(534) In some embodiments of the composition and methods of the disclosure, allogeneic cells are immune cells. In some embodiments, allogeneic cells are T lymphocytes (T cells). In some embodiments, allogeneic cells are T cells that do not express one or more components of a naturally-occurring T-cell Receptor (TCR). In some embodiments, allogeneic cells are T cells that express a non-naturally occurring antigen receptor. Alternatively, or in addition, in some embodiments, allogeneic cells are T cells that express a non-naturally occurring Chimeric Stimulatory Receptor (CSR). In some embodiments, the non-naturally occurring CSR comprises or consists of a switch receptor. In some embodiments, the switch receptor comprises an extracellular domain, a transmembrane domain, and an intracellular domain. In some embodiments, the extracellular domain of the switch receptor binds to a TCR co-stimulatory molecule and transduces a signal to the intracellular space of the allogeneic cell that recapitulates TCR signaling or TCR co-stimulatory signaling.

(535) Chimeric Stimulatory Receptors (CSRs)

(536) Adoptive cell compositions that are “universally” safe for administration to any patient requires a significant reduction or elimination of alloreactivity.

(537) Towards this end, allogeneic cells of the disclosure are modified to interrupt expression or function of a T-cell Receptor (TCR) and/or a class of Major Histocompatibility Complex (MHC). The TCR mediates graft vs host (GvH) reactions whereas the MHC mediates host vs graft (HvG) reactions. In preferred embodiments, any expression and/or function of the TCR is eliminated in allogeneic cells of the disclosure to prevent T-cell mediated GvH that could cause death to the subject. Thus, in particularly preferred embodiments, the disclosure provides a pure TCR-negative allogeneic T-cell composition (e.g. each cell of the composition expresses at a level so low as to either be undetectable or non-existent).

(538) In preferred embodiments, expression and/or function of MHC class I (MHC-I, specifically, HLA-A, HLA-B, and HLA-C) is reduced or eliminated in allogeneic cells of the disclosure to prevent HvG and, consequently, to improve engraftment of allogeneic cells of the disclosure in a subject. Improved engraftment of the allogeneic cells of the disclosure results in longer persistence of the cells, and, therefore, a larger therapeutic window for the subject. Specifically, in the allogeneic cells of the disclosure, expression and/or function of a structural element of MHC-I, Beta-2-Microglobulin (B2M), is reduced or eliminated in allogeneic cells of the disclosure.

(539) The above strategies for generating an allogeneic cell of the disclosure induce further challenges. T Cell Receptor (TCR) knockout (KO) in T cells results in loss of expression of CD3-zeta (CD3z or CD3ζ), which is part of the TCR complex. The loss of CD3ζ in TCR-KO T-cells dramatically reduces the ability of optimally activating and expanding these cells using standard stimulation/activation reagents, including, but not limited to, agonist anti-CD3 mAb. When the expression or function of any one component of the TCR complex is interrupted, all components of the complex are lost, including TCR-alpha (TCRα), TCR-beta (TCRβ), CD3-gamma (CD3γ), CD3-epsilon (CD3ε), CD3-delta (CD3δ), and CD3-zeta (CD3ζ). Both CD3ε and CD3ζ are required for T cell activation and expansion. Agonist

anti-CD3 mAbs typically recognize CD3ε and possibly another protein within the complex which, in turn, signals to CD3ζ. CD3ζ provides the primary stimulus for T cell activation (along with a secondary co-stimulatory signal) for optimal activation and expansion. Under normal conditions, full T-cell activation depends on the engagement of the TCR in conjunction with a second signal mediated by one or more co-stimulatory receptors (e.g. CD28, CD2, 4-1BBL, etc. . . . ) that boost the immune response. However, when the TCR is not present, T cell expansion is severely reduced when stimulated using standard activation/stimulation reagents, including agonist anti-CD3 mAb. In fact, T cell expansion is reduced to only 20-40% of the normal level of expansion when stimulated using standard activation/stimulation reagents, including agonist anti-CD3 mAb.

(540) The disclosure provides a Chimeric Stimulatory Receptor (CSR) to deliver CD3ζ primary stimulation to allogeneic T cells in the absence of an endogenous TCR (and, consequently, an endogenous CD3ζ) when stimulated using standard activation/stimulation reagents, including agonist anti-CD3 mAb.

(541) In the absence of an endogenous TCR, Chimeric Stimulatory Receptors (CSRs) of the disclosure provide a CD3ζ stimulus to enhance activation and expansion of allogeneic T cells. In other words, in the absence of an endogenous TCR, Chimeric Stimulatory Receptors (CSRs) of the disclosure rescue the allogeneic cell from an activation-based disadvantage when compared to non-allogeneic T-cells that express an endogenous TCR. In some embodiments, CSRs of the disclosure comprise an agonist mAb epitope extracellularly and a CD3ζ stimulatory domain intracellularly and, functionally, convert an anti-CD28 or anti-CD2 binding event on the surface into a CD3ζ signaling event in an allogeneic T cell modified to express the CSR. In some embodiments, a CSR comprises a wild type CD28 or CD2 protein and a CD3ζ intracellular stimulation domain, to produce CD28ζ CSR and CD2ζ CSR, respectively. In preferred embodiments, CD28ζ CSR and/or CD2ζ CSR further express a non-naturally occurring antigen receptor and/or a therapeutic protein. In preferred embodiments, the non-naturally occurring antigen receptor comprises a Chimeric Antigen Receptor.

(542) The data provided herein demonstrate that modified allogeneic T cells of the disclosure comprising/expressing a CSR of the disclosure improve or rescue, the expansion of allogeneic T cells that no longer express endogenous TCR when compared to those cells that do not comprise/express a CSR of the disclosure.

(543) Endogenous TCR Knock-Out

(544) Gene editing compositions of the disclosure, including but not limited to, RNA-guided fusion proteins comprising dCas9-Clo051, may be used to target and decrease or eliminate expression of an endogenous T-cell receptor of an allogeneic cell of the disclosure. In preferred embodiments, the gene editing compositions of the disclosure target and delete a gene, a portion of a gene, or a regulatory element of a gene (such as a promoter) encoding an endogenous T-cell receptor of an allogeneic cell of the disclosure.

(545) Nonlimiting examples of primers (including a T7 promoter, genome target sequence, and gRNA scaffold) for the generation of guide RNA (gRNA) templates for targeting and deleting TCR-alpha (TCR-α) are provided in Table 10.

(546) TABLE-US-00186 TABLE 10 Target sequences underlined SEQ ID Name Sequence NO:

| TCRa-                   | TAATACGACTCACTATA         | 16821                    | gRNA-                     | <u>GCTGGTACACGGCAGGG</u> | WT1                      |
|-------------------------|---------------------------|--------------------------|---------------------------|--------------------------|--------------------------|
| TCAGTTTTAGAGCTAGAA      | AATAG                     | TCRa-                    | TAATACGACTCACTATA         | 16822                    | gRNA-                    |
| <u>GAGAATCAA</u>        | <u>AATCGGTG</u>           | WT2                      | <u>AAT</u>                | TCRa-                    | 16823                    |
| gRNA--                  | TAATACGACTCACTATA         | WT4                      | <u>GTGCTAGACATGAGGTC</u>  | TA                       | TCRa-                    |
| TAATACGACTCACTATA       | 16824                     | gRNA-                    | <u>GGCTGGTACACGGCAGG</u>  | WT1-2G                   | <u>GTCA</u>              |
| TCRa-                   | TAATACGACTCACTATA         | 16825                    | gRNA-                     | <u>GAGAATCAA</u>         | <u>AATCGGTG</u>          |
| WT2                     | <u>AATGTTTTAGAGCTAGAA</u> | AATAG                    | TCRa-                     | gRNA-                    |                          |
| TAATACGACTCACTATA       | WT3                       | <u>GGATTTAGAGTCTCTCA</u> | 16826                     | <u>GCGTTTTAGAGCTAGAA</u> |                          |
| ATAG                    | TCRa-                     | TAATACGACTCACTATA        | 16827                     | gRNA-                    | <u>GTGCTAGACATGAGGTC</u> |
| WT4                     | <u>TAGTTTTAGAGCTAGAA</u>  | ATAG                     | TCRa-                     | TAATACGACTCACTATA        | 16828                    |
| gRNA-                   | <u>GACACCTTCTTCCCCAG</u>  | WT5                      | <u>CCCGTTTTAGAGCTAGAA</u> | AATAG                    | TCRa-                    |
| TAATACGACTCACTATA       | 16829                     | gRNA-                    | <u>gtggaataatgctgttg</u>  | NG1-                     | <u>ttgaGTTTTAGAGCTAG</u> |
| L                       | AAATAG                    | TCRa-                    | TAATACGACTCACTATA         | 16830                    | gRNA-                    |
| <u>gcatcacaggaacttc</u> |                           | NG2-                     |                           |                          |                          |

taaaGTTTATAG L TAATACGACTCACTATA 16831 gRNA-  
gtaaaaccaagaggcca NG3- cagGTTTTAGAGCTAGA L AAATAG TCRA- TAATACGACTCACTATA  
 16832 gRNA- gacccggccactttcag NG4- gaggGTTTTAGAGCTAG L AAATAG TCRA-  
 TAATACGACTCACTATA 16833 gRNA- gattaaaccggccact NGS- ttcGTTTTAGAGCTAGA L  
 AAATAG TCRA- TAATACGACTCACTATA 16834 gRNA- gagcccaggtaagggca NG1-  
gcttGTTTTAGAGCTAG R AAATAG TCRA- TAATACGACTCACTATA 16835 gRNA-  
gagctttgaaacaggta NG2- agacGTTTTAGAGCTAG 1-R AAATAG TCRA- TAATACGACTCACTATA  
 16836 gRNA- gctttgaaacaggtaag NG2- acaGTTTTAGAGCTAGA 2-R AAATAG TCRA-  
 TAATACGACTCACTATA 16837 gRNA- gtttcaaaacctgtcag NG3-R tgatGTTTTAGAGCTAG  
 AAATAG TCRA- TAATACGACTCACTATA 16838 gRNA- gctgcggctgtggtcca NG4-R  
gctgGTTTTAGAGCTAG AAATAG TCRA- TAATACGACTCACTATA 16839 gRNA-  
gctgtggtccagctgag NG5- gtgGTTTTAGAGCTAGA 1-R AAATAG TCRA- TAATACGACTCACTATA  
 16840 gRNA- gctgtggtccagctgag NG5- gtgaGTTTTAGAGCTAG 2-R AAATAG TCRA-  
 TAATACGACTCACTATA 16841 gRNA- gtgtggtccagctgagg NG5- tgagGTTTTAGAGCTAG 3-R  
 AAATAG TCRA- TAATACGACTCACTATA 16842 gRNA- gtgtggtccagctgagg NG5-  
tgagGTTTTAGAGCTAG 3-Rb AAATAG

(547) Nonlimiting examples of primers for the generation of guide RNA (gRNA) templates for targeting and deleting TCR-beta (TCR-β) are provided in Table 11.

(548) TABLE-US-00187 TABLE 11 Target sequences underlined SEQ ID Name Sequence NO:  
 TCRb- TAATACGACTCACTATAG 16843 gRNA- GCTGCTCCTTGAGGGGCT WT1  
GGTTTTAGAGCTAGAAAT AG TCRb- TAATACGACTCACTATAG 16844 gRNA-  
GCAGTATCTGGAGTCATT WT2 GGTTTTAGAGCTAGAAAT AG TCRb-  
 TAATACGACTCACTATAG 16845 gRNA- GCCTCGGCGCTGACGATC WT3 T TCRb-  
 TAATACGACTCACTATAG 16846 gRNA- GCTCTCGGAGAATGACGA WT5 G TCRb-  
 TAATACGACTCACTATAG 16847 gRNA- GCCTCGGCGCTGACGATC WT3  
TGTTTTAGAGCTAGAAAT AG TCRb- TAATACGACTCACTATAG 16848 gRNA-  
GAGAATGACGAGTGGACC WT4 CGTTTTAGAGCTAGAAAT AG TCRb-  
 TAATACGACTCACTATAG 16849 gRNA- GCTCTCGGAGAATGACGA WT5  
GGTTTTAGAGCTAGAAAT AG TCRb- TAATACGACTCACTATAG 16850 gRNA-  
CAAACACAGCGACCTCGG NG1-L GTGTTTTAGAGCTAGAAA TAG TCRb-  
 TAATACGACTCACTATAG 16851 gRNA- TGGCTCAAACACAGCGAC NG2-L  
CTGTTTTAGAGCTAGAAA TAG TCRb- TAATACGACTCACTATAG 16852 gRNA-  
AGGGCGGGCTGCTCCTTG NG3-L AGGTTTTAGAGCTAGAAA TAG TCRb-  
 TAATACGACTCACTATAG 16853 gRNA- TATCTGGAGTCATTGAGG NG4-L  
GGTTTTAGAGCTAGAAAT AG TCRb- TAATACGACTCACTATAG 16854 gRNA-  
ACTGGACTTGACAGCGGA NGS-L AGGTTTTAGAGCTAGAAA TAG TCRb-  
 TAATACGACTCACTATAG 16855 gRNA- AGAGATCTCCCACACCCA NG1-R  
AAGTTTTAGAGCTAGAAA TAG TCRb- TAATACGACTCACTATAG 16856 gRNA-  
CCACACCCAAAAGGCCAC NG2-R ACGTTTTAGAGCTAGAAA TAG TCRb-  
 TAATACGACTCACTATAG 16857 gRNA- ACTGCCTGAGCAGCCGCC NG3-R  
TGGTTTTAGAGCTAGAAA TAG TCRb- TAATACGACTCACTATAG 16858 gRNA-  
TGAGGGTCTCGGCCACCT NG4-R TCGTTTTAGAGCTAGAAA TAG TCRb-  
 TAATACGACTCACTATAG 16859 gRNA- ATGACGAGTGGACCCAGG NGS-R  
ATGTTTTAGAGCTAGAAA TAG TCRb- TAATACGACTCACTATAG 16860 gRNA-  
TGGCTCAAACACAGCGAC NG6-L CTGTTTTAGAGCTAGAAA TAG TCRb-  
 TAATACGACTCACTATAG 16861 gRNA- CCACACCCAAAAGGCCAC NG6-R  
ACGTTTTAGAGCTAGAAA TAG

(549) Nonlimiting examples of primers for the generation of guide RNA (gRNA) templates for targeting and deleting beta-2-microglobulin (β2M) are provided in Table 12.

(550) TABLE-US-00188 TABLE 12 Target sequences underlined Primer No. Name Sequence  
 SEQ ID NO: 1 B2- TAATACGACTCACTATAG 16862 Prom- AGACAGGTGACGGTCCCTGC  
 NG1-R GTTTTAGAGCTAGAAATAG 2 B2- TAATACGACTCACTATA 16863 Prom-

(554) TABLE-US-00190 TABLE 14 SEQ SEQ SEQ Exon or NG-Left Target ID NG-Right  
Target ID Target sequence ID Gene Reagent/Type region Sequence NO: Sequence NO: (if WT  
crispr) NO Beta2- B2-Promoter- promoter GCAGTGCCAGGTTAG 16889 AGACAGGTGACGGTC  
16913 MG NG1 AGAGA CCTGC B2-Promoter- promoter CAAGCCAGCGACGCA 16890  
CCTGCGGGCCTTGTC 16914 NG2 GTGCC CTGAT B2-Promoter- promoter  
CCAATCAGGACAAGG 16891 TATAAGTGAGGCGT 16915 NG3 CCCGC CGCGC B2-Ex2-NG  
exon 2 ACCCAGACACATAGC 16892 GAAGTTGACTTACTG 16916 AATTC AAGAA B2-Ex2-  
NG2 exon 2 **gatattcctcagGTA** 16893 **TCACGTCATCCAGCA** 16917 **CTCCA GAGAA** B2-Ex1-  
NG exon 1 GGCCACGGAGCGAGA 16894 ACTCTCTCTTTCTGG 16918 CATCT CCTGG WT-  
B2MG-exon2 GGAGAGAGAATTGAAAAAG 16937 WT-B2MG- cuts in promoter-4 promoter  
GGGCCTTGTCCTGATTGGC 16938 region Y WT-B2MG- cuts in promoter-5 promoter  
GGCACTGCGTCGCTGGCT 16939 region C2TA C2TA exon 4 exon 4 CATCGCTGTTAAGAA  
16895 CTACCACTTCTATGA 16919 NG GCTCC CCAGA C2TA exon4 exon 4  
**GGTCCATCTGGTCAT** 16896 **AGATTGAGCTCTACT** 16920 NG2 **AGAAG CAGGT** C2TA  
exon6 exon6 GGCCCTCCAGCTGGG 16897 CAGTAAGTTTGTGGT 16921 NG AGTCC GGGTG  
C2TA exon4- exon 4 GGTCCATCTGGTCATAGAAG 16940 WT C2TA exon6- exon6  
GGAGTCCTGGAAGACATAC 16941 WT C2TA exon6 exon6 **CCTTGCTCAGGCCCT** 16898  
**TGTGGTGGGTGGGGA** 16922 NG2 **CCAGC GGTCT** RFXANK RFXANK exon1 exon 1  
GGGTCTGCTGGGTCT 16899 GGACCCTGAAGACCC 16923 NG1 GGATG CGGAG RFXANK

exon1 exon 1 GTTCTGAGGCAGGG 16900 CCCGGAGAGGAGGCT 16924 NG2 TCTGC  
 GCAGA RFXANK exon2 exon 2 TGAGAGTGGTGGAGT 16901 GAACGAGGTGTCAGC  
 16925 NG1 GCTTC TCTGC RFXANK Exon2 exon 2 CTCGTTCCCTCGCTG 16902  
**GGCCACCCTAGACTG** 16926 NG2 **CCGGT TGAGT** RFXANK-WT- exon 1  
 GGTCCCCAAGTTCTGAGGC 16942 exon1-3 RFXANK-WT- exon1 GGCAGGGGTCTGCTGGGTC  
 16943 exon1-4 RFXAP RFXAP Exon 1 exon 1 CCCGCCCAACGCTG 16903  
 CTGTGCGAAGGGGCC 16927 NG1 CCCCC GGGGA RFXAP Exon 1 exon 1  
 CCTTCGCACAGGTAC 16904 AGAGGAGGCTGGGGA 16928 NG2 CTAAC GGACG RFXAP  
 Exon1 exon 1 **CAGCCGGGGCTAGGG** 16905 **CTTGGCGCCAGCCTC** 16929 NG3 **CCGCG**  
**GGTGG** RFXAP Exon1 exon 1 **GCCGCGGCCGCCACC** 16906 **CTAGTGATGCAACCC**  
 16930 NG4 **GAGGC TGTGC** RFXAP Exon1 exon 1 **GCCGCGCTCTCGCCT** 16907  
**GAGGACGAGGAGACT** 16931 NG5 **CCCC CACTC** WT- RFXAP- exon 1  
 GGCCCCCGGGGGCAGCGTT 16944 ex1-3 WT- RFXAP- exon 1  
 GGTACCTGTGCGAAGGGGC 16945 ex1-4 RFX5 RFX5 exon1 exon 1 GTCTTGGGGCTCTTA  
 16908 CCCAGGTGGTGCTGA 16932 NG1 GCATC GGCTG RFX5 exon2 exon 2  
 ACGGCCTTGCTGTGG 16909 GGGATCCTGTGAAGT 16933 NG2 GGAAG GTGTT RFX5 exon5  
 exon5 TCTGATGATCTTGCC 16910 ATCAAAGCTCGAAGG 16934 NG3 AAAGT CTTGG RFX5  
 exon1 **GTCTTGGGGCTCTTA** 16911 **CCCCAGGTGGTGCTG** 16935 NG2 **GCATC AGGCT**  
 RFX5 exon1 **AGGCTCATCTTCTGC** 16912 **ACTGGGGGAAGGGCC** 16936 NG3 **CATCC**  
**CCCC** WT-RFX5- GGAAGGGCCCCCCCCAGG 16946 ex1-4 Exon1 WT-RFX5-  
 GCCTTCGAGCTTTGATGTC 16947 ex5-5 Exon 5

#### Engineered HLA-E Compositions

(555) MHCI knockout (KO) renders cells resistant to killing by T cells, but also makes them susceptible to natural killer (NK) cell-mediated cytotoxicity (“Missing-self hypothesis”) (see FIG. 23). It is hypothesized that NK rejection would reduce the in vivo efficacy and/or persistence of these KO cells in a therapeutic setting, such as allogeneic (allo) CAR-T therapy. Retention of MHCI on the surface of allo CAR-T cells would render them susceptible to killing by host T cells, as observed in the classic mixed lymphocyte reaction (MLR) experiment. It is estimated that up to 10% of a person's T cells are specific to foreign MHC, which would mediate the rejection of foreign cells and tissues. A targeted KO of MHCI, specifically HLA-A, B and C, which can be achieved by targeted KO of B2M, results in a loss of additional HLA molecules including HLA-E. Loss of HLA-E, for example, renders the KO cells more susceptible to NK cell-mediated cytotoxicity due to the “Missing-self Hypothesis”. NK-mediated cytotoxicity against missing-self cells is a defense mechanism against pathogens that downregulate MHC on the surface of infected cells to evade detection and killing by cells of the adaptive immune system.

(556) Two strategies are contemplated by the disclosure for engineering allo (MHCI-neg) T cells (including CAR-T cells) more resistant to NK cell-mediated cytotoxicity. In some embodiments, a sequence encoding a molecule (such as single-chain HLA-E) that reduces or prevents NK killing is introduced or delivered to an allogeneic cell. Alternatively, or in addition, gene editing methods of the disclosure retain certain endogenous HLA molecules (such as endogenous HLA-E). For example, the first approach involves piggyBac (PB) delivery of a single-chain (sc)HLA-E molecule to B2M KO T cells.

(557) The second approach uses a gene editing composition with guide RNAs selective for HLA-A, HLA-B and HLA-C, but not, for example, HLA-E or other molecules that are protective against natural-killer cell mediated cytotoxicity for MHCI KO cells.

(558) Alternative or additional molecules to HLA-E that are protective against NK cell-mediated cytotoxicity include, but are not limited to, interferon alpha/beta receptor 1 (IFNAR1), human IFNAR1, interferon alpha/beta receptor 2 (IFNAR2), human IFNAR2, HLA-G1, HLA-G2, HLA-G3, HLA-G4, HLA-G5, HLA-G6, HLA-G7, human carcino embryonic antigen-related cell adhesion molecule 1 (CEACAM1), viral hemoagglutinins, CD48, LIT1 (also referred to as C-type lectin domain family 2 member (CLC2D)), ULBP2, ULBP3, and sMICA or a variant thereof.

(559) An exemplary INFAR1 protein of the disclosure comprises or consists of the amino acid sequence

of (Signal peptide, Extracellular, TM, Cytoplasmic):

(560) TABLE-US-00191 (SEQ ID NO: 17017)

MMVVLLGATTLVLVAVAPWVLSAAAGGKNLKSPQKVEVDIIDDNFILRWNRSDESVGNVT  
FSFDYQKTGMDNWIKLSGCQNITSTKCNFSSLKLNVEEIKLRIRAEKENTSSWYEVDSF  
TPFRKAQIGPPEVHLEADKAIVIHISPGTKDSVMWALDGLSFTYSLVIWKNSSGVEERI  
ENIYSRHKIYKLSPETTYCLKVKAALLTSWKIGVYSPVHCIKTTVENELPPPENIEVSVQ  
NQNYVLKWDYTYANMTFQVQWLHAFLKRNPGNHLYKWKQIPDCENVKTTQCCVFPQNVFQK  
GIYLLRVQASDGNNTSFWSEEIKFDTEIQAFLLPPVFNIRSLSDSFHIYIGAPKQSGNTP  
VIQDYPLIYEIIFWENTSNAERKIIEKKTDVTPNLKPLTVYCVKARAHTMDEKLNKSSV  
FSDAVCEKTKPGNTSK KVFLRCINYVFFPSLKPSSSIDE  
YFSEQPLKNLLSTSEEQIEKCFIENISTIATVEETNQTDEDEHKKYSSQTSQDSGNYSN  
EDESESKTSEELQQDFV.

(561) An exemplary INFAR2 protein of the disclosure comprises or consists of the amino acid sequence of (Signal peptide, Extracellular, TM, Cytoplasmic):

(562) TABLE-US-00192 (SEQ ID NO: 17018)

MLLSQNAFIFRSLNLVLMVYISLVFGISYDSPDYTDESCTFKISLRNFRSILSWELKNHS  
IVPTHYTLTYTIMSKPEDLKVVKNCANTIRSFCDLTDEWRSTHEAYVTVLEGFSGNITLF  
SCSHNFWLAIDMSFEPPEFEIVGFTNHINVMVKFPSIVEEELQFDLSLVIEEQSEGIVKK  
HKPEIKGNMSGNFTYIIDKLIPNTNYCVSVYLEHSDEQAVIKSPLKCTLLPPGQESESAE SAK  
 KWIGYICLRNSLPKVLNFHNFLAWPFPNLPPLEAMD  
MVEVIYNRKKKVWDYNYDDESSDTEAAPRTSGGGYTMHGLTVRPLGQASATSTESQLI  
DPESEEPDLPEVDVELPTMPKDSPQQLELLSGPCERRKSPLQDPFPEEDYSSTEGSGGR  
ITFNVDLNSVFLRVLDDEDSDDLEAPMLSSHLEEMVDPEDPDNVQSNHLLASGEGTOPT  
FPSPSSEGLWSEDAPSDQSDISESDVDLGDGYIMR.

(563) An exemplary HLA-G1 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3):

(564) TABLE-US-00193 (SEQ ID NO: 17019)

MVVMAPRTLFLLLSGALTTLTETWAGSHSMRYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS  
DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEASSHTLQWMIG  
CDLGSDGRLLRGYEQYAYDGKDYALNEDLRSWTAADTAAOISKRKCEANVAEQRRAYLEG  
TCVEWLHRYLENGKEMLORADPPKTHVTHHPVFDYEATLRCWALGFYPAEIILTWQRDGEDQ  
TQDVELVETRPAGDGTFQKWAAVVPSGEEQRYTCHVQHEGLPEPLMLRWKQSSLPTIPIMG  
IVAGLVVLAAVVTGAAAVAAVLWRKKSSD.

(565) An exemplary HLA-G2 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3):

(566) TABLE-US-00194 (SEQ ID NO: 17020)

MVVMAPRTLFLLLSGALTTLTETWAGSHSMRYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS  
DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEADPPKTHVTH  
PVFDYEATLRCWALGFYPAEIILTWQRDGEDQTQDVELVETRPAGDGTFQKWAAVVPSGEE  
QRYTCHVQHEGLPEPLMLRWKQSSLPTIPIMGIVAGLVVLAAVVTGAAAVAAVLWRKKSSD.

(567) An exemplary HLA-G3 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3):

MVVMAPRTLFLLLSGALTTLTETWAGSHSMRYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS  
DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEAKQSSLPTIPI  
MGIVAGLVVLAAVVTGAAAVAAVLWRKKSSD (SEQ ID NO: 17021).

(568) An exemplary HLA-G4 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3):

(569) TABLE-US-00195 (SEQ ID NO: 17022)

MVVMAPRTLFLLLSGALTTLTETWAGSHSMRYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS  
DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEASSHTLQWMIG  
CDLGSDGRLLRGYEQYAYDGKDKLALNEDLRSWTAADTAAQISKRKCEANVAEORRAYLEG  
TCVEWLHRYLENGKEMLQRAKQSSLPTIPIMGIVAGLVVLAAVVTGAAAVAAVLWRKKSSD.

(570) An exemplary HLA-G5 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3, intron 4):

(571) TABLE-US-00196 (SEQ ID NO: 17023)

**MVVMAPRTLFLLLSGALTLIETWAGSHSMRYFSAAVSRPGRGEPREIAMGYVDDTQFVREDS**  
**DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEASSHTLQWMIG**  
**CDLGSDGRLLRGYEQYAYDGKDKLALNEDLRSWTAADTAAQISKRKCEANVAEORRAYLEG**  
**TCVEWLHRYLENGKEMLQRADPPKTHVTHHPVFDYEATLRCWALGFYPAEIILTQQRDGEDQ**  
**TQDVELVETRPAGDGTQKWA**AVVPSGEEQRYTCHVQHEGLPEPLMLRWcustom character  
custom character .

(572) An exemplary HLA-G5 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3, intron 4):

(573) TABLE-US-00197 (SEQ ID NO: 17024)

**MVVMAPRTLFLLLSGALTLIETWAGSHSMRYFSAAVSRPGRGEPREFIAMGYVDDTQFVRFD**  
**DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEADPPKTHVTHH**  
**PVFDYEATLRCWALGFYPAEIILTQQRDGEDQTQDVELVETRPAGDGTQKWA**AVVPSGEE  
QRYTCHVQHEGLPEPLMLRWcustom character .

(574) An exemplary HLA-G5 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3, intron 2):

(575) TABLE-US-00198 (SEQ ID NO: 17025)

**MVVMAPGALTLTETWAGSHSMRYFSAAVSRPGRGEPREFIAMGYVDDTQFV**  
**RFDS****DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGY YNQSEA**  
custom character .

(576) An exemplary CEACAM1 protein of the disclosure comprises or consists of the amino acid sequence of (Extracellular, TM, Cytoplasmic):

(577) TABLE-US-00199 (SEQ ID NO: 17026)

**MGHLSAPLHRVRVPWQGLLLTASLLTFWNPPTTAQLTTESMPFNVAEGKE**  
**VLLLVHNLPPQLFGYSWYKGERVDGNRQIVGYAIGTQQATPGPANSRET**  
**IYPNASLLIQNVTQNDTGFTLQVIKSDLVNEEATGQFHVPELPKPSIS**  
**SNNSNPVEDKDAVAFTCEPETQDTTYLWWINNQSLPVSPRLQLSNGNRTL**  
**TLLSVTRNDTGPYECEIQNPVSANRSDPVTLNVTYGPDTPTISPSDTYYR**  
**PGANLSLSCYAASNPPAQYSWLINGTFQQSTQELFIPNITVNNSGSYCH**  
**ANNSVTGCNRRTTVKTIIVTELSPVAKPQIKASKTTVTGDKDSVNLTCST**  
**NDTGISIRWFFKNQSLPSSERMKLSQGNNTLSINPVKREDAGTYWCEVN**  
**PISKNQSDPIMLVNYNALPQENGLSPGAIAGIVIGVVALVALIAVALAC**  
**FLHFGKTGRASDQRDLTEHKPSVSNHTQDHSNDPPNKMNEVTYSTLNFEA**  
**QQPTQPTSASPSLTATEIYSEVKKQ.**

(578) An exemplary viral hemagglutinin protein of the disclosure comprises or consists of the amino acid sequence of (HAforInfluenzaAvirus(A/NewCaledonia/20/1999(H1N1); TM):

(579) TABLE-US-00200 (SEQ ID NO: 17027)

**MKAKLLVLLCTFTATYADTICIGYHANNSTDVDTVLEKNVTVTHSVNL**  
**EDSHNGKLCCLKGIAPLQLGNCSVAGWILGNPECELLISKESWSYIVETP**  
**NPENGTCYPGYFADYEELREQLSSVSSFERFEIFPKESSWPNHTVTGVSA**  
**SCSHNGKSSFYRNLLWLTGKNGLYPNLSKSYVNNKEKEVLVLWGVHHPN**  
**IGNQRALYHTENAYVSVVSSHYSRRFTPEIAKRPKVRDQEGRINYYWILL**  
**EPGDTIIFEANGNLIAPWYAFALSRGFGSGIITSNAPMDECDKACQTPQG**  
**AINSSLPFQNVHPVTIGECPKYVRSACLKRMVTGLRNIPSIQSRGLFGAIA**  
**GFIEGGWTGMVDGWYGYHHQNEQSGYAADQKSTQNAINGITNKVNSVIE**  
**KMNTQFTAVGKEFNKLERRMENLNKKVDDGFLDIWTYNAELLVLENERT**  
**LDFHDSNVKNLYEKVKSQLKNNAKEIGNGCFEFYHKCNNECMESVKNGTY**  
**DYPKYSEESKLNREKIDGVKLESMGVYQILAIYSTVASSLVLLVSLGAIS**  
**FWMCSNGSLQCRICI.**

(580) An exemplary CD48 protein of the disclosure comprises or consists of the amino acid sequence of



(Signal peptide, Chain, Pro peptide removed in mature form):

(581) TABLE-US-00201 (SEQ ID NO: 17028)

**MCSRGWDSCLALELLLLPLSLLVTSIQGHLVHMTVVSGSNVTLNISESLP  
ENYKQLTWFYTEDQKIVEWDSRKSKEYFESKFKGRVRLDPQSGALYISKVQ  
KEDNSTYIMRVLKKTGNEQEWKIKLQVLDPVPKPKVIEKIEDMDDNCYL  
KLSCVIPGESVNYTWYGDKRPFPELQNSVLETTLMPHNYSRCYTCQVSN  
SVSSKNGTVCLSPCTLARSFGVEWIASWLVTPTILGLLLT.**

(582) An exemplary LLT1 protein of the disclosure comprises or consists of the amino acid sequence of (Cytoplasmic, TM, Extracellular):

(583) TABLE-US-00202 (SEQ ID NO: 17029)

**MHDSNNVEKDITPSEL PANPGCLHSKEHSIKATLIWRLFFLIMFLTIIVC  
GMVAALSAIRANCHQEPSVCLQAACPESWIGFQRKCFYFSDDTKNWTSSQ  
RFCDSQDADLAQVESFQELNFLRLRYKGPSDHWIGLSREQGPWKWINGTE  
WTRQFPILGAGECAYLNDKGASSARHYTERKWICKSKSDIHV.**

(584) An exemplary ULBP2 protein of the disclosure comprises or consists of the amino acid sequence of (also known as NKG2D ligand; Genbank ACCESSION No. AAQ89028):

(585) TABLE-US-00203 (SEQ ID NO: 17030) 1 maaaaatkil lclplllls gwsragradp hslcyditvi  
pkfrpgprwc avqgqvdekt 61 flhydcgntk vtpvsplgkk lntttawkaq npvrevvdi lteqlrdiql enytpkeplt  
121 lqarmsceqk aeghssgswq fsfdgqifll fdsekrmwtt vhpqarkmke kwendkvvam 181 sfhyfsmgdc  
igwledflmg mdstlepszag aplamssgtt qlratattli lclliilpc 241 filpgi.

(586) An exemplary ULBP3 protein of the disclosure comprises or consists of the amino acid sequence of (also known as NKG2D ligand; Genbank ACCESSION No. NP\_078794):

(587) TABLE-US-00204 (SEQ ID NO: 17031) 1 maaaaspail prlailpyll fdwsgtgrad  
ahslwynfti ihlprhgqqw cevqsqvdk 61 nflsydcgsd kvlsmgglee qlyatdawgk qlemlrevvg  
rlrllelacte ledftpsgpl 121 tlqvrmscec eadgyirgsw qfsfdgrkfl lfdsnrkwt vvahagarmk ekwekdsglt  
181 tffkmvsmrd ckswirdflm hrkkrlpta pptmapglag pkaiattlsp wsfliilcfi 241 lpgi.

(588) An exemplary sMICA protein of the disclosure comprises or consists of the amino acid sequence of (Signal Peptide, Portion of Extracellular domain, TM and cytoplasmic domain) (Genbank Accession No. Q29983):

(589) TABLE-US-00205 (SEQ ID NO: 17032) 1 mglgpvfill agifpfappg aaaphslry  
nlvlswdgs vqsgfltevh ldgqpfllrcd 61 rqlcrakpqq qwaedvlgmk twdretrdlt gngkdlrmtl  
ahikdqkegl hslqeirvce 121 ihednstrss qhfyydgelf lsqnetkew tmpqssraqt **lamnvrnflk  
edamktkthy** 181 hamhadclqe lrrylksgvv lrrtvppmvn vtrseasegn itvtcrasgf ypwnitlswr 241  
qdgvsishdt qqwgdvlpdg ngtyqtwwat ricqgeeqrf tcymehsgnh sthpvpvsgkv 301 lvlqshwqtf  
hvsavaaaai fviiiifyrc ckkktsaaeg pelvslqvld qhpvgtsdhr 361 datqlgfqpl msdlgstgst ega.

(590) An exemplary sMICA protein of the disclosure comprises or consists of the amino acid sequence of (Alpha-1, Alpha-2, Alpha-3):

(591) TABLE-US-00206 (SEQ ID NO: 17033) 1 mglgpvfill agifpfappg aaaphslry  
nlvlswdgs vqsgfltevh ldgqpfllrcd 61 rqlcrakpqq qwaedvlgmk twdretrdlt gngkdlrmtl **ahikdqkegl  
hslqeirvce** 121 **ihednstrss qhfyydgelf lsqnetkew tmpassraqt lamnvrnflk edamktkthy** 181  
**hamhadclqe lrrylksgvv lrrtvppmvn vtrseasegn itvtcrasgf ypwnitlswr** 241 **qdgvsishdt  
qqwgdvlpdg ngtyqtwwat ricqgeeqrf tcymehsgnh sthpvpvsgkv** 301 **lvlqshwqtf** hvsavaaaai  
fviiiifyrc ckkktsaaeg pelvslqvld qhpvgtsdhr 361 datqlgfqpl msdlgstgst ega.

(592) An exemplary sMICA protein of the disclosure comprises or consists of the amino acid sequence of (Signal peptide; Alpha-, Alpha-2, Alpha-3):

(593) TABLE-US-00207 (SEQ ID NO: 17034) custom character custom character ephslry  
nlvlswdgs vqsgfltevh ldgqpfllrcd 61 rqlcrakpqq qwaedvlgmk twdretrdlt gngkdlrmtl  
**ahikdqkegl hslqeirvce** 121 **ihednstrss qhfyydgelf lsqnetkew tmpassraqt**  
**I thy** 181 **hamhadclqe lrrylksgvv lrrtvppmvn**  
**vtrseasegn itvtcrasgf ypwnitlswr** 241 **qdgvsishdt qqwgdvlpdg ngtyqtwwat ricqgeeqrf**  
**tcymehsgnh sthpvpvsgkv** 301 **lvlqshw.**

(594) An exemplary sMICA protein of the disclosure comprises or consists of the amino acid sequence



of (Signal peptide):

(595) TABLE-US-00208 (SEQ ID NO: 17035)    
EPHSLRYNLTVLSWDGSGVQSGFLT

EVHLDGQPFLRCDRQKCRAKPQGQWAEDVLGNKTWDRETRDLTGNGKLDL  
RMTLAHIKLDQKEGLHSLQEIRVCEIHEDNSTRSSQHFYYNGELFLSQNL  
ETKEWTMPQSSRAQTLTHYHAMHADCLQELRRYLKSGVVLRRTPPPMVDV  
TRSEASEGNITVTCRASGFYPWNITLSWRQDGVSLSHDTQQWGDVLPDGN  
GTYQTWVATRICQGEEQRFTCYMEHSGNHSTHPVPSGKVLVLQSHW.

(596) An exemplary bGBE Trimer (20G and 484S) protein of the disclosure comprises or consists of the amino acid sequence of:

(597) TABLE-US-00209 (SEQ ID NO: 16972)

MSRSVALAVLALLSLSGLEAVMAPRILILGGGGSGGGGSGGGGSIQRT  
KIQVYSRHPAENGKSNFLNCYVSGFHPDIEVDLLKNGERIEKVEHSDL  
SFSKDWSFYLLYYTEFTPTEKDEYACRVNHVILSQPKIVKWDRDMGGGG  
SGGGGSGGGGSGGGGSGSHSLKYFHTSVSRPGRGEPRFISVGYVDDTQF  
VRFDNDAASPRMVPRAPWMEQEGSEYWDRETRSARDTAQIFRVNLRTLR  
GYYNQSEAGSHTLQWMHGCGLGPDGRFLRGYEQFAYDGKDYLTNEDLR  
SWTAVDTAAQISEQKSNDASEAEHQRAYLEDTCVEWLHKYLEKGKETLL  
HLEPPKTHVTHHPISDHEATLRCWALGFYPAEITLTWQQDGEHTQDTE  
LVETRPAGDGTGQKWA AVVVPSGEEQRYTCHVQHEGLPEPVTLRWKPAS  
QPTIPIVGIIAGLVLLGSVVSGAVVA AVIWRKKSSGGKGGSY  KAEWSD  
SAQGSSEHSL\*.

(598) An exemplary bGBE Trimer (270G and 484S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(599) TABLE-US-00210 (SEQ ID NO: 16973)

atgtctcgagcgtggccctggccgtgctggccctgctgtccctgtctgg  
cctggaggccgtgatggccccggaccctgatcctgggaggaggaggca  
gcggcgaggagggtccggaggcgggcgtctatccagcgcacacctaag  
atccaggtgtattctcggcacccagccgagaacggcaagagcaacttct  
gaattgctacgtgagcggcttcacccttcgacatcgaggtggatctgc  
tgaagaatggcgagagaatcgagaaggtggagcactccgacctgagcttc tccaaggattggctcttttatctgctgtactataccgagtttaccctac  
agagaaggacgagtagcctgtcgctgaaccacgtgacactgtcccagc  
caaagatcgtgaagtgggaccgggatatgggcggcgggcgtctggcggc  
ggcggcgagcgggcgggcgggctccggaggaggcggtctggcagccactc  
cctgaagtatttccacacctgtgtgagccggccaggcagaggagagccac  
ggttcatctctgtgggtacgtggacgatacacagttcgtgaggtttgac  
aatgatgccgccagcccaagaatggtgcctagggccccatggatggagca  
ggagggcagcgagtattgggacagggagacccggagcgccagagacacag  
cacagatttccgggtgaacctgagaacctgaggggctactataatcag  
tccgaggccggctctcacacactccagtggatgcacggatgcgagctggg  
accagatggccgcttctgcggggctacgagcagtttgcctatgacggca  
aggattacctgacctgaacgaggacctgagatcctggaccgccgtggat  
acagccgcccagatcagcgagcagaagtccaatgacgcatctgaggcaga  
gcaccagagggcatatctggaggatacctgcgtggagtggtgcacaagt  
acctggagaagggaaggagacactgctgcacctggagccccctaagacc  
cacgtgacacaccaccaatcagcgaccacgaggccacctgaggtgttg  
ggcactgggcttctatcccgccgagatcacctgacatggcagcaggacg  
gagagggacacacccaggatacagagctggtggagaccaggcccgccggc  
gatggcacatttcagaagtgggcccgtggtggtgccttccggagagga  
gcagagatacactgtcacgtgcagcacgaggactgccagagccagtga  
ccctgaggtggaagcctgccagccagcccacaatccctatcgtgggaatc  
atcgaggcctggtgctgctgggctctgtggtgagcggagcagtggtggc

cgccgtgatctggcggaagaagagcagcggagggaaggagctcctact  custom character

caaggcagagtgagcgactccgcccagggctctgagagccactccctg tga.

(600) An exemplary bGBE Trimer (270R and 484S) protein of the disclosure comprises or consists of the amino acid sequence of:

(601) TABLE-US-00211 (SEQ ID NO: 16974)

MSRSVALAVLALLSLSGLEAVMAPRTLILGGGSGGGGSGGGGSIQRTP  
KIQVYSRHPAENGKSNFLNCYVSGFHPSDIEVDLLKNGERIEKVEHSDL  
SFSKDWSFYLLYYTEFTPTTEKDEYACRVNHVILSQPKIVKWDRDMGGGG  
SGGGGSGGGGSGGGGSGSHSLKYFHTSVSRPGRGEPRFISVGYVDDTQF  
VRFDNDAASPRMVPRAPWMEQEGSEYWDRETRSARDTAQIFRVNLRTLR  
GYYNQSEAGSHTLQWMHGCELGPDRRFLRGYEQFAYDGGKDYLTNLNEDLR  
SWTAVDTAAQISEQKSNDASEAEHQRAYLEDTCVEWLHKYLEKGKETLL  
HLEPPKTHVTHHPISDHEATLRCWALGFYPAEITLTWQQDGEGHTQDTE  
LVETRPAGDGTGFKWA~~AV~~VVPSGEEQRYTCHVQHEGLPEPVTLRWKPAS  
QPTIPIVGIIAGLVLLGSVVS~~GA~~VVA~~AV~~IWRKKSSGGKGGSY  custom character KAEWSD  
SAQGSESHSL\*.

(602) An exemplary bGBE Trimer (270R and 484S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(603) TABLE-US-00212 (SEQ ID NO: 16975)

atgtctcgagcgtggccctggccgtgctggccctgctgtccctgtctgg  
cctggaggccgtgatggcccccgaccctgatcctgggaggaggaggca  
gcggcgaggagggtccggaggcgggctctatccagcgcacacctaag  
atccaggtgtattctcggcacccagccgagaacggcaagagcaacttcct  
gaattgctacgtgagcggcttcacccttcgacatcgaggtggatctgc  
tgaagaatggcgagagaatcgagaaggtggagcactccgacctgagcttc tccaaggattggctcttttatctgctgtactataccgagtttaccctac  
agagaaggacgagtacgcctgtcgctgaaccacgtgacactgtcccagc  
caaagatcgtgaagtgggaccgggatatgggcggcgggctctggcggc  
ggcggcagcgggcgggcgggctccggaggaggcggtctggcagccactc  
cctgaagtatttcacacctctgtgagccggccaggcagaggagagccac  
ggttcatctctgtgggtacgtggacgatacacagttcgtgaggtttgac  
aatgatgccgccagcccaagaatgggtgcctagggcccatggatggagca  
ggaggggcagcgagtattgggacagggagacccggagcgccagagacacag  
cacagatttccgggtgaacctgagaacctgaggggctactataatcag  
tccgaggccggctctcacacactccagtggatgcacggatgcgagctggg  
accagatcgccgcttctgcggggctacgagcagtttgcctatgacggca  
aggattacctgacctgaacgaggacctgagatcctggaccgccgtggat  
acagccgcccagatcagcgagcagaagtccaatgacgcatctgaggcaga  
gcaccagagggcatatctggaggatacctgcgtggagtggctgcacaagt  
acctggagaagggcaaggagacactgctgcacctggagccccctaagacc  
cacgtgacacaccaccaatcagcgaccacgaggccaccctgaggtgttg  
ggcactgggcttctatcccgcgagatcacctgacatggcagcaggacg  
gagagggacacacccaggatacagagctgggtggagaccaggccccgggc  
gatggcacatttcagaagtgggcccgcgtgggtggcttccggagagga  
gcagagatacactgtcacgtgcagcacgagggactgccagagccagtga  
ccctgaggtggaagcctgccagccagcccacaatccctatcgtgggaatc  
atcgagggcctggctgctgggctctgtggtgagcggagcagtggtggc  
cgccgtgatctggcggaagaagagcagcggaggcaaggaggctcctact  custom character  
caaggcagagtgagcgactccgcccagggctctgagagccactccctg tga.

(604) An exemplary gBE Dimer (Rand S) protein of the disclosure comprises or consists of the amino acid sequence of:

(605) TABLE-US-00213 (SEQ ID NO: 16976)

MSRSVALAVLALLSLSGLEAIQRTPKIQVYSRHPAENGKSNFLNCYVSGF

HPSDILLKNGERIEKVEHSDLSFSKDWSFYLLYYTEFTPTTEKDEYAC  
RVNHVTL SQPKIVKWDRDMGGGGSGGGGSGGGGSGGGGSGSHSLKYFHTS  
VSRPGRGEPRFISVG YVDDTQFVRFDNDAASPRMVPRAPWMEQEGSEYWD  
RETRSARDTAQIFRVNLRTL RGYYNQSEAGSHTLQWMHGCELGPD~~RR~~FLR  
GYEQFAYDGDYLTNLNEDLRSWTAVDTAAQISEQKSNDA SEAEHQRAYLE  
DTCVEWLHKYLEKGKETLLHLEPPKTHVTHHPISDHEATLRCWALGFYPA  
EITLTWQQDGE GHTQDTEL VETRPAGDGT FQKWA AVVVP SGEEQRYTCHV  
QHEGLPEPVTLRWKPASQPTIPIVGIIAGLVLLGSVVSGAVVA AVIWRKK SSGGKGGSY  
 KAEWSDSAQGSESHSL.

(606) An exemplary gBE Dimer (R and S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(607) TABLE-US-00214 (SEQ ID NO: 16977)

ATGAGCAGATCTGTGGCCCTGGCTGTTCTGGCTCTGCTGTCTCTGTCTGG  
CCTGGAAGCCATCCAGCGGACCCCTAAGATCCAGGTGTACAGCAGACACC  
CCGCCGAGAACGGCAAGAGCAACTTCCTGAAGTGTACGTGTCCGGCTTT  
CACCCCAGCGACATTGAGGTGGACCTGCTGAAGAACGGCGAGCGGATCGA  
GAAGGTGGAACACAGCGATCTGAGCTTCAGCAAGGACTGGTCCTTCTACC  
TGCTGTACTACACCGAGTTCACCCCTACCGAGAAGGACGAGTACGCCTGC  
AGAGTGAACCACGTGACACTGAGCCAGCCTAAGATCGTGAAGTGGGACAG  
AGATATGGGCGGAGGCGGATCTGGTGGCGGAGGAAGTGGCGGCGGAGGAT  
CTGGCGGTGGTGGTTCTGGATCTCACAGCCTGAAGTACTTTCACACCTCC  
GTGTCCAGACCTGGCAGAGGCGAGCCTAGATTTCATCAGCGTGGGCTACGT  
GGACGACACCCAGTTCGTCAGATTTCGACAACGACGCCGCTCTCCTCGGA  
TGGTTCCTAGAGCACCTGGATGGAACAAGAGGGCAGCGAGTACTGGGAT  
CGCGAGACAAGAAGCGCCAGAGACACAGCCCAGATCTTCCGCGTGAACCT  
GAGAACCTGCGGGGCTACTACAATCAGTCTGAGGCCGGCTCTCACACCC  
TGCAGTGGATGCATGGATGTGAACTGGGCCCGACAGAC~~CGG~~TTCCTGAGA  
GGCTATGAGCAGTTCGCCTACGACGGCAAGGACTACCTGACACTGAACGA  
GGACCTGAGAAGCTGGACCGCCGTGGATACAGCCGCTCAGATCAGCGAGC  
AGAAGTCTAACGACGCCAGCGAGGCCGAACACCAGAGAGCCTATCTGGAA  
GATACCTGCGTGGAAATGGCTGCACAAGTACCTGGAAAAGGGCAAAGAGAC  
ACTGCTGCACCTGGAACCTCCAAAGACACATGTGACCCACCATCCTATCA  
GCGACCACGAGGCCACACTGAGATGTTGGGCCCTGGGCTTTTACCCTGCC  
GAGATCACACTGACATGGCAGCAGGATGGCGAGGGCCACACACAGGATAC  
AGAGCTGGTGGAAACAAGACCTGCCGGCGACGGCACCTTCCAGAAATGGG  
CTGCTGTGGTTGTGCCAGCGGCGAGGAACAGAGATACACCTGTCACGTG  
CAGCACGAGGGACTGCCTGAACCTGTGACTCTGAGATGGAAGCCTGCCAG  
CCAGCCAACAATCCCCATCGTGGGAATCATTGCCGGCCTGGTGTCTGG  
GATCTGTGGTTTCTGGTGTCTGTGGTGGCCGCCGTGATTTGGAGAAAGAAG  
TCCTCTGGCGGCAAAGGCGGCTCCTAC~~custom character~~ AAGGCCGAGTGGAGCGATTC  
TGCCCAGGGCTCTGAAAGCCACAGCCT~~GT~~AGATAA.

(608) An exemplary gBE Dimer (G and S) protein of the disclosure comprises or consists of the amino acid sequence of:

(609) TABLE-US-00215 (SEQ ID NO: 16978)

DLLKNGERIEKVEHSDLSFSKDWSFYLLYYTEFTPTTEKDEYACRVNHVTL  
SQPKIVKWDRDMGGGGSGGGGSGGGGSGGGGSGSHSLKYFHTSVSRPGRG  
EPRFISVG YVDDTQFVRFDNDAASPRMVPRAPWMEQEGSEYWDRETRSAR  
DTAQIFRVNLRTL RGYYNQSEAGSHTLQWMHGCELGPD~~G~~RFLRGYEQFAY  
DGKDYLTLNEDLRSWTAVDTAAQISEQKSNDA SEAEHQRAYLEDTCVEWL  
HKYLEKGKETLLHLEPPKTHVTHHPISDHEATLRCWALGFYPAEITLTWQ  
QDGE GHTQDTEL VETRPAGDGT FQKWA AVVVP SGEEQRYTCHVQHEGLPE  
PVTLRWKPASQPTIPIVGIIAGLVLLGSVVSGAVVA AVIWRKKSSSGKGG SY~~custom character~~

KAEWSDSAQGSSESHSL

(610) An exemplary gBE Dimer (G and S) protein of the disclosure comprises or consists of the amino acid sequence of:

(611) TABLE-US-00216 (SEQ ID NO: 16979)

ATGAGCAGATCTGTGGCCCTGGCTGTTCTGGCTCTGCTGTCTCTGTCTGG  
CCTGGAAGCCATCCAGCGGACCCCTAAGATCCAGGTGTACAGCAGACACC  
CCGCCGAGAACGGCAAGAGCAACTTCCTGAACTGCTACGTGTCCGGCTTT  
CACCCCAGCGACATTGAGGTGGACCTGCTGAAGAACGGCGAGCGGATCGA  
GAAGGTGGAACACAGCGATCTGAGCTTCAGCAAGGACTGGTCCTTCTACC  
TGCTGTACTACACCGAGTTCACCCCTACCGAGAAGGACGAGTACGCCTGC  
AGAGTGAACCACGTGACACTGAGCCAGCCTAAGATCGTGAAGTGGGACAG  
AGATATGGGCGGAGGCGGATCTGGTGGCGGAGGAAGTGGCGGCGGAGGAT  
CTGGCGGTGGTGGTTCTGGATCTCACAGCCTGAAGTACTTTCACACCTCC  
GTGTCCAGACCTGGCAGAGGCGAGCCTAGATTTCATCAGCGTGGGCTACGT  
GGACGACACCCAGTTCGTCAGATTTCGACAACGACGCCGCTCTCCTCGGA  
TGGTTCCTAGAGCACCTGGATGGAACAAGAGGGGCAGCGAGTACTGGGAT  
CGCGAGACAAGAAGCGCCAGAGACACAGCCCAGATCTTCCGCGTGAACCT  
GAGAACCCTGCGGGGCTACTACAATCAGTCTGAGGCCGGCTCTCACACCC  
TGCAGTGGATGCATGGATGTGAACTGGGCCCCGACAGACAGTTTCCTGAGA  
GGCTATGAGCAGTTCGCCTACGACGGCAAGGACTACCTGACACTGAACGA  
GGACCTGAGAAGCTGGACCGCCGTGGATACAGCCGCTCAGATCAGCGAGC  
AGAAGTCTAACGACGCCAGCGAGGCCGAACACCAGAGAGCCTATCTGGAA  
GATACCTGCGTGGAATGGCTGCACAAGTACCTGGAAAAGGGCAAAGAGAC  
ACTGCTGCACCTGGAACCTCCAAAGACACATGTGACCCACCATCCTATCA  
GCGACCACGAGGCCACACTGAGATGTTGGGCCCTGGGCTTTTACCCTGCC  
GAGATCACACTGACATGGCAGCAGGATGGCGAGGGCCACACACAGGATAC  
AGAGCTGGTGGAAACAAGACCTGCCGGCGACGGCACCTTCCAGAAATGGG  
CTGCTGTGGTTGTGCCAGCGGCGAGGAACAGAGATACACCTGTCACGTG  
CAGCACGAGGGACTGCCTGAACCTGTGACTCTGAGATGGAAGCCTGCCAG  
CCAGCCAACAATCCCCATCGTGGGAATCATTGCCGGCCTGGTGTCTGG  
GATCTGTGGTTTCTGGTGTCTGTGGTGGCCGCCGTGATTTGGAGAAAGAAG  
TCCTCTGGCGGCAAAGGCGGCTCCTACcustom character AGGCCGAGTGGAGCGATTC  
TGCCCAGGGCTCTGAAAGCCACAGCCTGTAGATAA

(612) An exemplary WT HLA-E Monomer (Rand S) protein of the disclosure comprises or consists of the amino acid sequence of:

(613) TABLE-US-00217 (SEQ ID NO: 16980)

MSRSVALAVLALLSLSGLEAGSHSLKYFHTSVSRPGRGEPFISVGYVDD  
TQFVRFDNDAASPRMVPRAPWMEQEGSEYWDRETRSARDTAQIFRVNLRT  
LRGYYNQSEAGSHTLQWMHGCCELGPDRRFLRGYEQFAYDGKDYLTNLNEDL  
RSWTAVDTAQAQISEQKSNDASEAEHQRAYLEDTCVEWLHKYLEKGKETLL  
HLEPPKTHVTHHPISDHEATLRCWALGFYPAEITLTWQQDGEGHTQDTEL  
VETRPAGDGTFFQKWAAVVPSGEEQRYTCHVQHEGLPEPVTLRWKPASQP  
TIPIVGIIAGLVLLGSVVSGAVVAAVIWRKKSSGGKGGSYcustom character KAEWSDSAQ  
GSSESHSL

(614) An exemplary WT HLA-E Monomer (Rand S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(615) TABLE-US-00218 (SEQ ID NO: 16981)

ATGAGCAGATCTGTGGCCCTGGCTGTTCTGGCTCTGCTGTCTCTGTCTGG  
ACTGGAAGCCGGCAGCCACAGCCTGAAGTACTTTCACACCAGCGTGTCCA  
GACCTGGCAGAGGCGAGCCTAGATTTCATCAGCGTGGGCTACGTGGACGAC  
ACCCAGTTCGTCAGATTCGACAACGACGCCGCTCTCCTCGGATGGTTCC  
TAGAGCACCCCTGGATGGAACAAGAGGGGCAGCGAGTACTGGGACAGAGAGA

CAAGACCGCCAGACAGCTACCGTCTTTCAGAGTGAACCTGCGGACC  
CTGCGGGGCTACTACAATCAGTCTGAAGCCGGCTCTCACACCCTGCAGTG  
GATGCACGGATGTGAACTGGGCCCCGACAGAAAGATTCCTGAGAGGCTACG  
AGCAGTTCGCCTACGACGGCAAGGACTACCTGACACTGAACGAGGACCTG  
AGAAGCTGGACCGCCGTGGATACAGCCGCTCAGATCAGCGAGCAGAAGTC  
TAACGACGCCTCTGAGGCCGAACACCAGAGAGCCTACCTGGAAGATACCT  
GCGTGGAATGGCTGCACAAGTACCTGGAAAAGGGCAAAGAGACACTGCTG  
CACCTGGAACCTCCAAAGACACACGTGACCCACCATCCTATCAGCGACCA  
CGAGGCCACACTGAGATGTTGGGCCCTGGGCTTTTACCCCGCCGAGATCA  
CACTGACATGGCAGCAGGATGGCGAGGGCCACACACAGGATACAGAGCTG  
GTGGAACAAGACCTGCCGGCGACGGCACCTTCCAGAAATGGGCTGCTGT  
GGTGGTTCCAGCGGCGAGGAACAGAGATACACCTGTCACGTGCAGCACG  
AGGGACTGCCTGAACCTGTGACACTGAGGTGGAAGCCTGCCAGCCAGCCT  
ACAATCCCCATCGTGGGAATCATTGCCGGCCTGGTGCTGCTGGGATCTGT  
GGTTTCTGGTGCAGTGGTGGCCGCCGTGATCTGGCGGAAAAAAAGCTCAG  
GCGGCAAAGGCGGCTCCTACcustom character AAAGCCGAGTGGAGCGATTCTGCCCAG  
GGCTCTGAAAGCCACTCTCTGTAGATAA.

(616) An exemplary WT HLA-E Monomer (G and S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(617) TABLE-US-00219 (SEQ ID NO: 16982)

MSRSVALAVLALLSLSGLEAGSHSLKYFHTSVSRPGRGEPFISVGYVDD  
TQFVRFDNDAASPRMVPRAPWMEQEGSEYWDRETRSARDTAQIFRVNLRT  
LRGYYNQSEAGSHTLQWMHGCCELGPDGRFLRGYEQFAYDGKDYLTNEDL  
RSWTAVDTAQAISEQKSNDAEAEHQRAYLEDTCVEWLHKYLEKGKETLL  
HLEPPKTHVTHHPISDHEATLRCWALGFYPAEITLTWQQDGEGHTQDTEL  
VETRPAGDGTQKWAAVVVPSGEEQRYTCHVQHEGLPEPVTLRWKPASQP  
TIPIVGIIAGLVLLGSVVSGAVVAAVIWRKKSSGGKGGSYcustom character KAEWSDSAQ  
GSESHSL.

(618) An exemplary WT HLA-E Monomer (Q and S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(619) TABLE-US-00220 (SEQ ID NO: 16983)

ATGAGCAGATCTGTGGCCCTGGCTGTTCTGGCTCTGCTGTCTCTGTCTGG  
ACTGGAAGCCGGCAGCCACAGCCTGAAGTACTTTCACACCAGCGTGTCCA  
GACCTGGCAGAGGCGAGCCTAGATTCATCAGCGTGGGCTACGTGGACGAC  
ACCCAGTTCGTCAGATTCGACAACGACGCCGCCTCTCCTCGGATGGTTCC  
TAGAGCACCCCTGGATGGAACAAGAGGGCAGCGAGTACTGGGACAGAGAGA  
CAAGAAGCGCCAGAGACACAGCCCAGATCTTCAGAGTGAACCTGCGGACC  
CTGCGGGGCTACTACAATCAGTCTGAAGCCGGCTCTCACACCCTGCAGTG  
GATGCACGGATGTGAACTGGGCCCCGACGGAAAGATTCCTGAGAGGCTACG  
AGCAGTTCGCCTACGACGGCAAGGACTACCTGACACTGAACGAGGACCTG  
AGAAGCTGGACCGCCGTGGATACAGCCGCTCAGATCAGCGAGCAGAAGTC  
TAACGACGCCTCTGAGGCCGAACACCAGAGAGCCTACCTGGAAGATACCT  
GCGTGGAATGGCTGCACAAGTACCTGGAAAAGGGCAAAGAGACACTGCTG  
CACCTGGAACCTCCAAAGACACACGTGACCCACCATCCTATCAGCGACCA  
CGAGGCCACACTGAGATGTTGGGCCCTGGGCTTTTACCCCGCCGAGATCA  
CACTGACATGGCAGCAGGATGGCGAGGGCCACACACAGGATACAGAGCTG  
GTGGAACAAGACCTGCCGGCGACGGCACCTTCCAGAAATGGGCTGCTGT  
GGTGGTTCCAGCGGCGAGGAACAGAGATACACCTGTCACGTGCAGCACG  
AGGGACTGCCTGAACCTGTGACACTGAGGTGGAAGCCTGCCAGCCAGCCT  
ACAATCCCCATCGTGGGAATCATTGCCGGCCTGGTGCTGCTGGGATCTGT  
GGTTTCTGGTGCAGTGGTGGCCGCCGTGATCTGGCGGAAAAAAAGCTCAG  
GCGGCAAAGGCGGCTCCTACcustom character AAAGCCGAGTGGAGCGATTCTGCCCAG

GGCTCTGAAAGCCACTCTCTGTAGATAA.

## Vectors and Host Cells

(620) The disclosure also relates to vectors that include isolated nucleic acid molecules of the disclosure, host cells that are genetically engineered with the recombinant vectors, and the production of at least one VHH or VCAR by recombinant techniques, as is well known in the art. See, e.g., Sambrook, et al., supra; Ausubel, et al., supra, each entirely incorporated herein by reference.

(621) For example, the PB-EF1a vector may be used. The vector comprises the following nucleotide sequence:

(622) TABLE-US-00221 (SEQ ID NO: 17502) tgtacatagattaaccctagaaagataatcatattgtgacgtacgttaaa  
gataatcatgcgtaaaattgacgcatgtgtttatcggctgtatatcga ggtttatttattaattgaatagatattaagttttattatatttacactt  
acatactaataataaattcaacaaacaattatttatgtttatttatta ttaaaaaaaaaacaaaactcaaaatttcttataaagtaacaaaacttt  
tatcgaataacctgcagccccgggggatgcagagggacagccccccccaaa  
gccccagggatgtaattacgtccctcccccgctagggggcagcagcgag  
ccgccccggggctccgctccggctccggcgctcccccgcatccccgagccg  
gcagcgtgcggggacagccccgggcacggggaaggtggcacgggatcgctt  
tcctctgaacgcttctcgtgctctttgagcctgcagacacctggggggga  
tacggggaaaagtgtactgtgcctttgatcgaacctggacagttagct ttgcaaagatggataaagtttaacagagaggaatctttgcagctaag  
gaccttctaggtcttgaaaggagtgggaattggctccgggtgccggtcagt  
gggcagagcgcacatcgccacagtcctccgagaagtgggggggaggggtc  
ggcaattgaaccggtgcctagagaaggtggcgcggggtaaactgggaaag  
tgatgtcgtgtactggctccgccttttcccaggggtgggggagaaccgt atataagtgcagtagtcgccgtgaacgttcttttcgcaacgggttgcc  
gccagaacacaggtaagtgcgtgtgtgggtcccgcgggcctggcctctt  
tacgggttatggccttgcgtgccttgaaacttccacctggctgcagt acgtgattcttgatcccgagcttcgggttggaagtgggtgggagagttcg  
aggccttgcgcttaaggagcccccttcgctcgtgcttgagttgaggcctg  
gcctgggcgctggggccgcgcgtgcgaatctgggtggcaccttcgcgcct gtctcgtgctttcgataagtctctagccatttaaaattttgatgacct  
gctgcgacgctttttctggcaagatagcttgtaaatgcgggccaaga  
tctgcacactggtatttcggttttggggccgcgggcggcgacggggccc  
gtgcgtcccagcgcacatgttcggcgaggcggggcctgcgagcgcggcca  
ccgagaatcggacgggggtagtctcaagctggccggcctgctctgggtgcc  
tggcctcgcgcccgctgtatcgccccgccctgggcggcaaggctggccc  
ggtcggcaccagttgcgtgagcggaaagatggccgcttcccgccctgct  
gcagggagctcaaaatggaggacgcggcgctcgggagagcgggcgggtga  
gtcaccacacaaaaggaaaaggccttccgtcctcagccgtcgctcat  
gtgactccacggagtagcgggcgccgtccaggcacctcgattagttctcg  
agcttttgagtagctcgtcttttaggttgggggggaggggtttatgcgat  
ggagtttccccacactgagtggttgagactgaagttaggccagcttggc acttgatgaattctccttggaatttgcctttttgagtttgatcttg  
ttcattctcaagcctcagacagtgggtcaaagtttttctccatttca ggtgtcgtgagaattctaatacgactcactataggggtgtgctgtctcatc  
at ttggcaaagattggccaccaagcttgcctgcaggaggggtcgacgcc  
tctagacggggcgccgctccgatccaggggtaccgatcacatatgcctt taattaaacactagttctatagtgacctaattcccttagtgagggt  
taatggccgtaggcccagaaattgggtccagacatgataagatacattg  
atgagtttgacaaaccacaactagaatgcagtgaaaaaatgctttatt tgtgaaatttgtgatgctattgctttattgtaaccattataagctgcaa  
taaacaagttaacaacaattgcattcattttatgtttcagggttcagg gggaggtgtgggaggtttttcggactctaggacctgcgcatgcgcttgg  
cgtaatcatggtcatagctgtttcctgttttccccgtatccccccaggtg  
tctgcaggctcaaaagagcagcgagaagcgttcagaggaaaagcgatcccgt  
gccaccttccccgtgcccgggctgtccccgcacgctgcgggctcggggat  
gcgggggggagcgcgggaccggagcggagccccgggcgggctcgtgctgcc  
ccctagcgggggagggacgtaattacatccctgggggctttggggggggg  
ctgtccctctcaccgcggtggagctccagcttttgtcgaattggggccc cccctcaggggtatcgatgatatctataacaagaaaatatatatataa  
agttatcacgtaagtagaacatgaaataacaataattatcgtatgagt taaatctaaaagtcacgtaaaaagataatcatgcgtcattttgactcacg  
cggtcgttatagttcaaaatcagtgcacttaccgcattgacaagcacgc  
ctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacg  
gattcgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtc aatttacgcagactatctttctaggggtaatctagctagccctaagggc

gcctattgcgttgcgctcactgcccgcctttccagtcgggaaacctgctgt  
gccagctgcattaatgaatcgccaacgcgcggggagaggcggttgcgt  
attgggcgctcttccgcttctcgcctcactgactcgtcgcctcggtcgt  
tcggctgcggcgagcggatcagctcactcaaaggcggtataacggttat  
ccacagaatcaggggataacgcaggaaagaacatgacaaaaatcccttaa  
cgtgagttttcgttccactgagcgtcagaccccgtagaaaagatcaaagg atcttcttgagatcctttttctgcgcgtaactctgctgcttgcaaaaa  
aaaaaccaccgctaccagcgggtggtttgttgcggatcaagagctacca  
actcttttccgaaggtaactggcttcagcagagcgcagataccaaatac tgttctttagttagccgtagtaggaccacttcaagaactctgtag  
caccgcctacatacctcgtctgctaactcgttaccagtggctgctgcc agtggcgataagtcgtgtcttaccgggttgactcaagacgatagttacc  
ggataaggcgcagcggctcgggctgaacgggggggttcgtgcacacagccca  
gcttgagcgaacgacctacaccgaactgagatacctacagcgtgagcta  
tgagaaagcgccacgcttcccgaaaggagaaaaggcgggacaggtatccggt  
aagcggcagggctcggaacaggagagcgcacgaggagctccagggggaa  
acgcctgggtatctttatagtcctgtcgggttccacctctgacttgag cgtcgattttgtgatgctcgtcagggggggcggagcctatggaaaaacgc  
cagcaacgcggccttttaccggttctggccttttctggtgcttctgctc acatgagattatcaaaaaggatcttcacctagatccttttaataaaaa  
tgaagttttaaatcaatcaaaagtatatatgagtaaacttggctgacag  
tcagaagaactcgtcaagaaggcgatagaaggcgatgcgctgcgaatcgg  
gagcggcgataccgtaaaagcacgaggaagcggctcagcccattcgccgcca  
agctcttcagcaatatcacgggtagccaacgctatgtcctgatagcggtc  
cgccacacccagccggccacagtcgatgaatccagaaaagcggccatttt  
ccaccatgatattcggcaagcaggcatcgccatgggtcacgacgagatcc  
tcgccgtcgggcatgctcgccttgagcctggcgaacagttcggctggcgc  
gagcccctgatgctcttcgtccagatcctcctgatcgacaagaccggctt ccacccgagtagctgctcgtcgtcgatgcgatgttccgcttggtggtcgaat  
gggcaggttagccggatcaagcgtatgcagccgccgattgcatcagccat  
gatggatactttctcggcaggagcaaggtgagatgacaggagatcctgcc  
ccggcacttcgccaatagcagccagtccttcccgttcagtgcacaacg  
tcgagcacagctgcgcaagggaacggcgctggtggccagccacgatagccg  
cgctgcctcgtcttgagttcattcagggcaccggacaggtcggcttga  
caaaaagaaccgggcgcccctgcgctgacagccggaacacggcggcacatca  
gagcagccgattgtctgttgcgccagtcatagccgaatagcctctccac  
ccaagcggccggagaacctgcgtgcaatccatcttgcctcaatcataatat tattgaagcatttatcagggttcgtctcgtccccggtctcctcccaatgca  
tgtcaatattggccattagccatattattcattggttatatagcataaat caatattggctattggccattgcatacgttgatctatatcataata

(623) The polynucleotides can optionally be joined to a vector containing a selectable marker for propagation in a host. Generally, a plasmid vector is introduced in a precipitate, such as a calcium phosphate precipitate, or in a complex with a charged lipid. If the vector is a virus, it can be packaged in vitro using an appropriate packaging cell line and then transduced into host cells.

(624) The DNA insert should be operatively linked to an appropriate promoter. The expression constructs will further contain sites for transcription initiation, termination and, in the transcribed region, a ribosome binding site for translation. The coding portion of the mature transcripts expressed by the constructs will preferably include a translation initiating at the beginning and a termination codon (e.g., UAA, UGA or UAG) appropriately positioned at the end of the mRNA to be translated, with UAA and UAG preferred for mammalian or eukaryotic cell expression.

(625) Expression vectors will preferably but optionally include at least one selectable marker. Such markers include, e.g., but are not limited to, ampicillin, zeocin (Sh bla gene), puromycin (pac gene), hygromycin B (hygB gene), G418/Geneticin (neo gene), mycophenolic acid, or glutamine synthetase (GS, U.S. Pat. Nos. 5,122,464; 5,770,359; 5,827,739), blasticidin (bsd gene), resistance genes for eukaryotic cell culture as well as ampicillin, zeocin (Sh bla gene), puromycin (pac gene), hygromycin B (hygB gene), G418/Geneticin (neo gene), kanamycin, spectinomycin, streptomycin, carbenicillin, bleomycin, erythromycin, polymyxin B, or tetracycline resistance genes for culturing in *E. coli* and other bacteria or prokaryotics (the above patents are entirely incorporated hereby by reference). Appropriate culture mediums and conditions for the above-described host cells are known in the art. Suitable vectors will be readily apparent to the skilled artisan. Introduction of a vector construct into a



host cell can be effected by calcium phosphate transfection, DEAE-dextran mediated transfection, cationic lipid-mediated transfection, electroporation, transduction, infection or other known methods. Such methods are described in the art, such as Sambrook, supra, Chapters 1-4 and 16-18; Ausubel, supra, Chapters 1, 9, 13, 15, 16.

(626) Expression vectors will preferably but optionally include at least one selectable cell surface marker for isolation of cells modified by the compositions and methods of the disclosure. Selectable cell surface markers of the disclosure comprise surface proteins, glycoproteins, or group of proteins that distinguish a cell or subset of cells from another defined subset of cells. Preferably the selectable cell surface marker distinguishes those cells modified by a composition or method of the disclosure from those cells that are not modified by a composition or method of the disclosure. Such cell surface markers include, e.g., but are not limited to, “cluster of designation” or “classification determinant” proteins (often abbreviated as “CD”) such as a truncated or full length form of CD19, CD271, CD34, CD22, CD20, CD33, CD52, or any combination thereof. Cell surface markers further include the suicide gene marker RQR8 (Philip B et al. Blood. 2014 Aug. 21; 124(8):1277-87).

(627) Expression vectors will preferably but optionally include at least one selectable drug resistance marker for isolation of cells modified by the compositions and methods of the disclosure. Selectable drug resistance markers of the disclosure may comprise wild-type or mutant Neo, DHFR, TYMS, FRANCF, RAD51C, GCS, MDR1, ALDH1, NKX2.2, or any combination thereof.

(628) At least one VHH or VCAR of the disclosure can be expressed in a modified form, such as a fusion protein, and can include not only secretion signals, but also additional heterologous functional regions. For instance, a region of additional amino acids, particularly charged amino acids, can be added to the N-terminus of a VHH or VCAR to improve stability and persistence in the host cell, during purification, or during subsequent handling and storage. Also, peptide moieties can be added to a VHH or VCAR of the disclosure to facilitate purification. Such regions can be removed prior to final preparation of a VHH or VCAR or at least one fragment thereof. Such methods are described in many standard laboratory manuals, such as Sambrook, supra, Chapters 17.29-17.42 and 18.1-18.74; Ausubel, supra, Chapters 16, 17 and 18.

(629) Those of ordinary skill in the art are knowledgeable in the numerous expression systems available for expression of a nucleic acid encoding a protein of the disclosure. Alternatively, nucleic acids of the disclosure can be expressed in a host cell by turning on (by manipulation) in a host cell that contains endogenous DNA encoding a VHH or VCAR of the disclosure. Such methods are well known in the art, e.g., as described in U.S. Pat. Nos. 5,580,734, 5,641,670, 5,733,746, and 5,733,761, entirely incorporated herein by reference.

(630) Illustrative of cell cultures useful for the production of the VHH or VCARS, specified portions or variants thereof, are bacterial, yeast, and mammalian cells as known in the art. Mammalian cell systems often will be in the form of monolayers of cells although mammalian cell suspensions or bioreactors can also be used. A number of suitable host cell lines capable of expressing intact glycosylated proteins have been developed in the art, and include the COS-1 (e.g., ATCC CRL 1650), COS-7 (e.g., ATCC CRL-1651), HEK293, BHK21 (e.g., ATCC CRL-10), CHO (e.g., ATCC CRL 1610) and BSC-1 (e.g., ATCC CRL-26) cell lines, Cos-7 cells, CHO cells, hep G2 cells, P3X63  $\mu$ g8.653, SP2/0-Ag14, 293 cells, HeLa cells and the like, which are readily available from, for example, American Type Culture Collection, Manassas, Va. (www.atcc.org). Preferred host cells include cells of lymphoid origin, such as myeloma and lymphoma cells. Particularly preferred host cells are P3X63Ag8.653 cells (ATCC Accession Number CRL-1580) and SP2/0-Ag14 cells (ATCC Accession Number CRL-1851). In a particularly preferred embodiment, the recombinant cell is a P3X63Ab8.653 or an SP2/0-Ag14 cell.

(631) Expression vectors for these cells can include one or more of the following expression control sequences, such as, but not limited to, an origin of replication; a promoter (e.g., late or early SV40 promoters, the CMV promoter (U.S. Pat. Nos. 5,168,062; 5,385,839), an HSV tk promoter, a pgk (phosphoglycerate kinase) promoter, an EF-1 alpha promoter (U.S. Pat. No. 5,266,491), at least one human promoter; an enhancer, and/or processing information sites, such as ribosome binding sites, RNA splice sites, polyadenylation sites (e.g., an SV40 large T Ag poly A addition site), and transcriptional terminator sequences. See, e.g., Ausubel et al., supra; Sambrook, et al., supra. Other cells



useful for production of nucleic acids or proteins of the present invention are known and/or available, for instance, from the American Type Culture Collection Catalogue of Cell Lines and Hybridomas ([www.atcc.org](http://www.atcc.org)) or other known or commercial sources.

(632) When eukaryotic host cells are employed, polyadenylation or transcription terminator sequences are typically incorporated into the vector. An example of a terminator sequence is the polyadenylation sequence from the bovine growth hormone gene. Sequences for accurate splicing of the transcript can also be included. An example of a splicing sequence is the VP1 intron from SV40 (Sprague, et al., J. Virol. 45:773-781 (1983)). Additionally, gene sequences to control replication in the host cell can be incorporated into the vector, as known in the art.

(633) Purification of a VH, VHH or VCAR

(634) A VHH protein or VCAR can be recovered and purified from recombinant cell cultures by well-known methods including, but not limited to, protein A purification, ammonium sulfate or ethanol precipitation, acid extraction, anion or cation exchange chromatography, phosphocellulose chromatography, hydrophobic interaction chromatography, affinity chromatography, hydroxylapatite chromatography and lectin chromatography. High performance liquid chromatography ("HPLC") can also be employed for purification. See, e.g., Colligan, Current Protocols in Immunology, or Current Protocols in Protein Science, John Wiley & Sons, NY, N.Y., (1997-2001), e.g., Chapters 1, 4, 6, 8, 9, 10, each entirely incorporated herein by reference.

(635) VHs, VHHs and/or VCARs of the disclosure include purified products, products of chemical synthetic procedures, and products produced by recombinant techniques from a prokaryotic or eukaryotic host, including, for example, *E. coli*, yeast, higher plant, insect and mammalian cells. Depending upon the host employed in a recombinant production procedure, the VCAR of the disclosure can be glycosylated or can be non-glycosylated. Such methods are described in many standard laboratory manuals, such as Sambrook, supra, Sections 17.37-17.42; Ausubel, supra, Chapters 10, 12, 13, 16, 18 and 20, Colligan, Protein Science, supra, Chapters 12-14, all entirely incorporated herein by reference.

(636) Amino Acid Codes

(637) The amino acids that make up VCARs of the disclosure are often abbreviated. The amino acid designations can be indicated by designating the amino acid by its single letter code, its three letter code, name, or three nucleotide codon(s) as is well understood in the art (see Alberts, B., et al., Molecular Biology of The Cell, Third Ed., Garland Publishing, Inc., New York, 1994). A VCAR of the disclosure can include one or more amino acid substitutions, deletions or additions, from spontaneous or mutations and/or human manipulation, as specified herein. Amino acids in a VCAR of the disclosure that are essential for function can be identified by methods known in the art, such as site-directed mutagenesis or alanine-scanning mutagenesis (e.g., Ausubel, supra, Chapters 8, 15; Cunningham and Wells, Science 244:1081-1085 (1989)). The latter procedure introduces single alanine mutations at every residue in the molecule. The resulting mutant molecules are then tested for biological activity, such as, but not limited to, at least one neutralizing activity. Sites that are critical for VCAR binding can also be identified by structural analysis, such as crystallization, nuclear magnetic resonance or photoaffinity labeling (Smith, et al., J. Mol. Biol. 224:899-904 (1992) and de Vos, et al., Science 255:306-312 (1992)).

(638) As those of skill will appreciate, the invention includes at least one biologically active VCAR of the disclosure. Biologically active VCARs have a specific activity at least 20%, 30%, or 40%, and, preferably, at least 50%, 60%, or 70%, and, most preferably, at least 80%, 90%, or 95%-99% or more of the specific activity of the native (non-synthetic), endogenous or related and known VCAR. Methods of assaying and quantifying measures of enzymatic activity and substrate specificity are well known to those of skill in the art.

(639) In another aspect, the disclosure relates to VHH proteins and fragments, as described herein, which are modified by the covalent attachment of an organic moiety. Such modification can produce a VCAR fragment with improved pharmacokinetic properties (e.g., increased in vivo serum half-life). The organic moiety can be a linear or branched hydrophilic polymeric group, fatty acid group, or fatty acid ester group. In particular embodiments, the hydrophilic polymeric group can have a molecular

weight of about 800 to about 120,000 Daltons and can be a polyalkane glycol (e.g., polyethylene glycol (PEG), polypropylene glycol (PPG)), carbohydrate polymer, amino acid polymer or polyvinyl pyrrolidone, and the fatty acid or fatty acid ester group can comprise from about eight to about forty carbon atoms.

(640) The modified VHH proteins and fragments of the disclosure can comprise one or more organic moieties that are covalently bonded, directly or indirectly, to the antibody. Each organic moiety that is bonded to a VHH protein or fragment of the disclosure can independently be a hydrophilic polymeric group, a fatty acid group or a fatty acid ester group. As used herein, the term “fatty acid” encompasses mono-carboxylic acids and dicarboxylic acids. A “hydrophilic polymeric group,” as the term is used herein, refers to an organic polymer that is more soluble in water than in octane. For example, polylysine is more soluble in water than in octane. Thus, a VHH protein modified by the covalent attachment of polylysine is encompassed by the disclosure. Hydrophilic polymers suitable for modifying VCARs of the disclosure can be linear or branched and include, for example, polyalkane glycols (e.g., PEG, monomethoxy-polyethylene glycol (mPEG), PPG and the like), carbohydrates (e.g., dextran, cellulose, oligosaccharides, polysaccharides and the like), polymers of hydrophilic amino acids (e.g., polylysine, polyarginine, polyaspartate and the like), polyalkane oxides (e.g., polyethylene oxide, polypropylene oxide and the like) and polyvinyl pyrrolidone. Preferably, the hydrophilic polymer that modifies the VCAR of the disclosure has a molecular weight of about 800 to about 150,000 Daltons as a separate molecular entity. For example, PEG5000 and PEG 20,000, wherein the subscript is the average molecular weight of the polymer in Daltons, can be used. The hydrophilic polymeric group can be substituted with one to about six alkyl, fatty acid or fatty acid ester groups. Hydrophilic polymers that are substituted with a fatty acid or fatty acid ester group can be prepared by employing suitable methods. For example, a polymer comprising an amine group can be coupled to a carboxylate of the fatty acid or fatty acid ester, and an activated carboxylate (e.g., activated with N,N-carbonyl diimidazole) on a fatty acid or fatty acid ester can be coupled to a hydroxyl group on a polymer.

(641) Fatty acids and fatty acid esters suitable for modifying VHH proteins of the disclosure can be saturated or can contain one or more units of unsaturation. Fatty acids that are suitable for modifying VCARs of the disclosure include, for example, n-dodecanoate (C12, laurate), n-tetradecanoate (C14, myristate), n-octadecanoate (C18, stearate), n-eicosanoate (C20, arachidate), n-docosanoate (C22, behenate), n-triacontanoate (C30), n-tetracontanoate (C40), cis- $\Delta^9$ -octadecanoate (C18, oleate), all cis- $\Delta^{5,8,11,14}$ -eicosatetraenoate (C20, arachidonate), octanedioic acid, tetradecanedioic acid, octadecanedioic acid, docosanedioic acid, and the like. Suitable fatty acid esters include mono-esters of dicarboxylic acids that comprise a linear or branched lower alkyl group. The lower alkyl group can comprise from one to about twelve, preferably, one to about six, carbon atoms.

(642) The modified VHH proteins and fragments can be prepared using suitable methods, such as by reaction with one or more modifying agents. A “modifying agent” as the term is used herein, refers to a suitable organic group (e.g., hydrophilic polymer, a fatty acid, a fatty acid ester) that comprises an activating group. An “activating group” is a chemical moiety or functional group that can, under appropriate conditions, react with a second chemical group thereby forming a covalent bond between the modifying agent and the second chemical group. For example, amine-reactive activating groups include electrophilic groups, such as tosylate, mesylate, halo (chloro, bromo, fluoro, iodo), N-hydroxysuccinimidyl esters (NHS), and the like. Activating groups that can react with thiols include, for example, maleimide, iodoacetyl, acryloyl, pyridyl disulfides, 5-thiol-2-nitrobenzoic acid thiol (TNB-thiol), and the like. An aldehyde functional group can be coupled to amine- or hydrazide-containing molecules, and an azide group can react with a trivalent phosphorous group to form phosphoramidate or phosphorimide linkages. Suitable methods to introduce activating groups into molecules are known in the art (see for example, Hermanson, G. T., *Bioconjugate Techniques*, Academic Press: San Diego, Calif. (1996)). An activating group can be bonded directly to the organic group (e.g., hydrophilic polymer, fatty acid, fatty acid ester), or through a linker moiety, for example, a divalent C1-C12 group wherein one or more carbon atoms can be replaced by a heteroatom, such as oxygen, nitrogen or sulfur. Suitable linker moieties include, for example, tetraethylene glycol,  $-(CH_2)_3-$ ,  $-NH-(CH_2)_6-NH-$ ,  $-(CH_2)_2-NH-$  and  $-CH_2-O-CH_2-CH_2-O-CH_2-CH_2-O-CH_2-NH-$ . Modifying agents that

comprise a linker moiety can be produced, for example, by reacting a mono-Boc-alkyldiamine (e.g., mono-Boc-ethylenediamine, mono-Boc-diaminohexane) with a fatty acid in the presence of 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide (EDC) to form an amide bond between the free amine and the fatty acid carboxylate. The Boc protecting group can be removed from the product by treatment with trifluoroacetic acid (TFA) to expose a primary amine that can be coupled to another carboxylate, as described, or can be reacted with maleic anhydride and the resulting product cyclized to produce an activated maleimido derivative of the fatty acid. (See, for example, Thompson, et al., WO 92/16221, the entire teachings of which are incorporated herein by reference.)

(643) The modified VHH proteins and fragments of the disclosure can be produced by reacting VHH protein or fragment with a modifying agent. For example, the organic moieties can be bonded to the VHH protein in a non-site specific manner by employing an amine-reactive modifying agent, for example, an NHS ester of PEG. Modified VHH proteins and fragments comprising an organic moiety that is bonded to specific sites of a VHH protein of the disclosure can be prepared using suitable methods, such as reverse proteolysis (Fisch et al., *Bioconjugate Chem.*, 3:147-153 (1992); Werlen et al., *Bioconjugate Chem.*, 5:411-417 (1994); Kumaran et al., *Protein Sci.* 6(10):2233-2241 (1997); Itoh et al., *Bioorg. Chem.*, 24(1): 59-68 (1996); Capellas et al., *Biotechnol. Bioeng.*, 56(4):456-463 (1997)), and the methods described in Hermanson, G. T., *Bioconjugate Techniques*, Academic Press: San Diego, Calif. (1996).

(644) VH or VHH Protein Compositions Comprising Further Therapeutically Active Ingredients

(645) VHH protein or VCAR compounds, compositions or combinations of the present disclosure can further comprise at least one of any suitable auxiliary, such as, but not limited to, diluent, binder, stabilizer, buffers, salts, lipophilic solvents, preservative, adjuvant or the like. Pharmaceutically acceptable auxiliaries are preferred. Non-limiting examples of, and methods of preparing such sterile solutions are well known in the art, such as, but limited to, Gennaro, Ed., *Remington's Pharmaceutical Sciences*, 18th Edition, Mack Publishing Co. (Easton, Pa.) 1990. Pharmaceutically acceptable carriers can be routinely selected that are suitable for the mode of administration, solubility and/or stability of the VHH protein or VCAR, fragment or variant composition as well known in the art or as described herein.

(646) Pharmaceutical excipients and additives useful in the present composition include, but are not limited to, proteins, peptides, amino acids, lipids, and carbohydrates (e.g., sugars, including monosaccharides, di-, tri-, tetra-, and oligosaccharides; derivatized sugars, such as alditols, aldonic acids, esterified sugars and the like; and polysaccharides or sugar polymers), which can be present singly or in combination, comprising alone or in combination 1-99.99% by weight or volume. Exemplary protein excipients include serum albumin, such as human serum albumin (HSA), recombinant human albumin (rHA), gelatin, casein, and the like. Representative amino acid/protein components, which can also function in a buffering capacity, include alanine, glycine, arginine, betaine, histidine, glutamic acid, aspartic acid, cysteine, lysine, leucine, isoleucine, valine, methionine, phenylalanine, aspartame, and the like. One preferred amino acid is glycine.

(647) Carbohydrate excipients suitable for use in the invention include, for example, monosaccharides, such as fructose, maltose, galactose, glucose, D-mannose, sorbose, and the like; disaccharides, such as lactose, sucrose, trehalose, cellobiose, and the like; polysaccharides, such as raffinose, melezitose, maltodextrins, dextrans, starches, and the like; and alditols, such as mannitol, xylitol, maltitol, lactitol, xylitol sorbitol (glucitol), myoinositol and the like. Preferred carbohydrate excipients for use in the present invention are mannitol, trehalose, and raffinose.

(648) VH protein, VHH protein or VCAR compositions can also include a buffer or a pH-adjusting agent; typically, the buffer is a salt prepared from an organic acid or base. Representative buffers include organic acid salts, such as salts of citric acid, ascorbic acid, gluconic acid, carbonic acid, tartaric acid, succinic acid, acetic acid, or phthalic acid; Tris, tromethamine hydrochloride, or phosphate buffers. Preferred buffers for use in the present compositions are organic acid salts, such as citrate.

(649) Additionally, VH protein, VHH protein or VCAR compositions of the disclosure can include polymeric excipients/additives, such as polyvinylpyrrolidones, ficolls (a polymeric sugar), dextrans (e.g., cyclodextrins, such as 2-hydroxypropyl- $\beta$ -cyclodextrin), polyethylene glycols, flavoring agents,

antimicrobial agents, sweeteners, antioxidants, antistatic agents, surfactants (e.g., polysorbates, such as “TWEEN 20” and “TWEEN 80”), lipids (e.g., phospholipids, fatty acids), steroids (e.g., cholesterol), and chelating agents (e.g., EDTA).

(650) These and additional known pharmaceutical excipients and/or additives suitable for use in the VH protein, VHH protein, VCAR, portion or variant compositions according to the invention are known in the art, e.g., as listed in “Remington: The Science & Practice of Pharmacy”, 19th ed., Williams & Williams, (1995), and in the “Physician's Desk Reference”, 52nd ed., Medical Economics, Montvale, N.J. (1998), the disclosures of which are entirely incorporated herein by reference. Preferred carrier or excipient materials are carbohydrates (e.g., saccharides and alditols) and buffers (e.g., citrate) or polymeric agents. An exemplary carrier molecule is the mucopolysaccharide, hyaluronic acid, which may be useful for intraarticular delivery.

(651) T Cell Isolation from a Leukapheresis Product

(652) A leukapheresis product or blood may be collected from a subject at clinical site using a closed system and standard methods (e.g., a COBE Spectra Apheresis System). Preferably, the product is collected according to standard hospital or institutional Leukapheresis procedures in standard Leukapheresis collection bags. For example, in preferred embodiments of the methods of the disclosure, no additional anticoagulants or blood additives (heparin, etc.) are included beyond those normally used during leukapheresis.

(653) Alternatively, white blood cells (WBC)/Peripheral Blood Mononuclear Cells (PBMC) (using Biosafe Sepax 2 (Closed/Automated)) or T cells (using CliniMACS® Prodigy (Closed/Automated)) may be isolated directly from whole blood. However, in certain subjects (e.g. those diagnosed and/or treated for cancer), the WBC/PBMC yield may be significantly lower when isolated from whole blood than when isolated by leukapheresis.

(654) Either the leukapheresis procedure and/or the direct cell isolation procedure may be used for any subject of the disclosure.

(655) The leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should be packed in insulated containers and should be kept at controlled room temperature (+19° C. to +25° C.) according to standard hospital or institutional blood collection procedures approved for use with the clinical protocol. The leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should not be refrigerated.

(656) The cell concentration leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should not exceed  $0.2 \times 10^9$  cells per mL during transportation. Intense mixing of the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should be avoided.

(657) If the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition has to be stored, e.g., overnight, it should be kept at controlled room temperature (same as above). During storage, the concentration of the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should never exceed  $0.2 \times 10^9$  cell per mL.

(658) Preferably, cells of the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should be stored in autologous plasma. In certain embodiments, if the cell concentration of the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition is higher than  $0.2 \times 10^9$  cell per mL, the product should be diluted with autologous plasma.

(659) Preferably, the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should not be older than 24 hours when starting the labeling and separation procedure. The leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition may be processed and/or prepared for cell labeling using a closed and/or automated system (e.g., CliniMACS Prodigy).

(660) An automated system may perform additional buffy coat isolation, possibly by ficolation, and/or washing of the cellular product (e.g., the leukapheresis product, blood, WBC/PBMC composition and/or T cell composition).

(661) A closed and/or automated system may be used to prepare and label cells for T-Cell isolation (from, for example, the leukapheresis product, blood, WBC/PBMC composition and/or T cell composition).

(662) Although WBC/PBMCs may be nucleofected directly (which is easier and saves additional steps),

the methods of the disclosure may include isolating T cells prior to nucleofection. The easier strategy of directly nucleofecting PBMC requires selective expansion of VCAR+ cells that is mediated via VCAR signaling, which by itself is proving to be an inferior expansion method that directly reduces the in vivo efficiency of the product by rendering T cells functionally exhausted. The product may be a heterogeneous composition of VCAR+ cells including T cells, NK cells, NKT cells, monocytes, or any combination thereof, which increases the variability in product from patient to patient and makes dosing and CRS management more difficult. Since T cells are thought to be the primary effectors in tumor suppression and killing, T cell isolation for the manufacture of an autologous product may result in significant benefits over the other more heterogeneous composition.

(663) T cells may be isolated directly, by enrichment of labeled cells or depletion of labeled cells in a one-way labeling procedure or, indirectly, in a two-step labeling procedure. According to certain enrichment strategies of the disclosure, T cells may be collected in a Cell Collection Bag and the non-labeled cells (non-target cells) in a Negative Fraction Bag. In contrast to an enrichment strategy of the disclosure, the non-labeled cells (target cells) are collected in a Cell Collection Bag and the labeled cells (non-target cells) are collected in a Negative Fraction Bag or in the Non-Target Cell Bag, respectively. Selection reagents may include, but are not limited to, antibody-coated beads. Antibody-coated beads may either be removed prior to a modification and/or an expansion step, or, retained on the cells prior to a modification and/or an expansion step. One or more of the following non-limiting examples of cellular markers may be used to isolate T-cells: CD3, CD4, CD8, CD25, anti-biotin, CD1c, CD3/CD19, CD3/CD56, CD14, CD19, CD34, CD45RA, CD56, CD62L, CD133, CD137, CD271, CD304, IFN-gamma, TCR alpha/beta, and/or any combination thereof. Methods for the isolation of T-cells may include one or more reagents that specifically bind and/or detectably-label one or more of the following non-limiting examples of cellular markers may be used to isolate T-cells: CD3, CD4, CD8, CD25, anti-biotin, CD1c, CD3/CD19, CD3/CD56, CD14, CD19, CD34, CD45RA, CD56, CD62L, CD133, CD137, CD271, CD304, IFN-gamma, TCR alpha/beta, and/or any combination thereof. These reagents may or may not be “Good Manufacturing Practices” (“GMP”) grade. Reagents may include, but are not limited to, Thermo DynaBeads and Miltenyi CliniMACS products. Methods of isolating T-cells of the disclosure may include multiple iterations of labeling and/or isolation steps. At any point in the methods of isolating T-cells of the disclosure, unwanted cells and/or unwanted cell types may be depleted from a T cell product composition of the disclosure by positively or negatively selecting for the unwanted cells and/or unwanted cell types. A T cell product composition of the disclosure may contain additional cell types that may express CD4, CD8, and/or another T cell marker(s).

(664) Methods of the disclosure for nucleofection of T cells may eliminate the step of T cell isolation by, for example, a process for nucleofection of T cells in a population or composition of WBC/PBMCs that, following nucleofection, includes an isolation step or a selective expansion step via TCR signaling.

(665) Certain cell populations may be depleted by positive or negative selection before or after T cell enrichment and/or sorting. Examples of cell compositions that may be depleted from a cell product composition may include myeloid cells, CD25+ regulatory T cells (T Regs), dendritic cells, macrophages, red blood cells, mast cells, gamma-delta T cells, natural killer (NK) cells, a Natural Killer (NK)-like cell (e.g., a Cytokine Induced Killer (CIK) cell), induced natural killer (iNK) T cells, NK T cells, B cells, or any combination thereof.

(666) T cell product compositions of the disclosure may include CD4+ and CD8+ T-Cells. CD4+ and CD8+ T-Cells may be isolated into separate collection bags during an isolation or selection procedure. CD4+ T cells and CD8+ T cells may be further treated separately, or treated after reconstitution (combination into the same composition) at a particular ratio.

(667) The particular ratio at which CD4+ T cells and CD8+ T cells may be reconstituted may depend upon the type and efficacy of expansion technology used, cell medium, and/or growth conditions utilized for expansion of T-cell product compositions. Examples of possible CD4+: CD8+ ratios include, but are not limited to, 50%:50%, 60%:40%, 40%:60% 75%:25% and 25%:75%.

(668) CD8+ T cells exhibit a potent capacity for tumor cell killing, while CD4+ T cells provide many of the cytokines required to support CD8+ T cell proliferative capacity and function. Because T cells isolated from normal donors are predominantly CD4+, the T-cell product compositions are artificially

adjusted *in vitro* with respect to the CD4+:CD8+ ratio to improve upon the ratio of CD4+ T cells to CD8+ T cells that would otherwise be present *in vivo*. An optimized ratio may also be used for the *ex vivo* expansion of the autologous T-cell product composition. In view of the artificially adjusted CD4+:CD8+ ratio of the T-cell product composition, it is important to note that the product compositions of the disclosure may be significantly different and provide significantly greater advantage than any endogenous-occurring population of T-cells.

(669) Preferred methods for T cell isolation may include a negative selection strategy for yielding untouched pan T cell, meaning that the resultant T-cell composition includes T-cells that have not been manipulated and that contain an endogenously-occurring variety/ratio of T-cells.

(670) Reagents that may be used for positive or negative selection include, but are not limited to, magnetic cell separation beads. Magnetic cell separation beads may or may not be removed or depleted from selected populations of CD4+ T cells, CD8+ T cells, or a mixed population of both CD4+ and CD8+ T cells before performing the next step in a T-cell isolation method of the disclosure.

(671) T cell compositions and T cell product compositions may be prepared for cryopreservation, storage in standard T Cell Culture Medium, and/or genetic modification.

(672) T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be cryopreserved using a standard cryopreservation method optimized for storing and recovering human cells with high recovery, viability, phenotype, and/or functional capacity. Commercially-available cryopreservation media and/or protocols may be used. Cryopreservation methods of the disclosure may include a DMSO free cryopreservant (e.g., CryoSOfree™ DMSO-free Cryopreservation Medium) reduce freezing-related toxicity.

(673) T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be stored in a culture medium. T cell culture media of the disclosure may be optimized for cell storage, cell genetic modification, cell phenotype and/or cell expansion. T cell culture media of the disclosure may include one or more antibiotics. Because the inclusion of an antibiotic within a cell culture media may decrease transfection efficiency and/or cell yield following genetic modification via nucleofection, the specific antibiotics (or combinations thereof) and their respective concentration(s) may be altered for optimal transfection efficiency and/or cell yield following genetic modification via nucleofection.

(674) T cell culture media of the disclosure may include serum, and, moreover, the serum composition and concentration may be altered for optimal cell outcomes. Human AB serum is preferred over FBS/FCS for culture of T cells because, although contemplated for use in T cell culture media of the disclosure, FBS/FCS may introduce xeno-proteins. Serum may be isolated from the blood of the subject for whom the T-cell composition in culture is intended for administration, thus, a T cell culture medium of the disclosure may comprise autologous serum. Serum-free media or serum-substitute may also be used in T-cell culture media of the disclosure. In certain embodiments of the T-cell culture media and methods of the disclosure, serum-free media or serum-substitute may provide advantages over supplementing the medium with xeno-serum, including, but not limited to, healthier cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies.

(675) T cell culture media may include a commercially-available cell growth media. Exemplary commercially-available cell growth media include, but are not limited to, PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium, or any combination thereof.

(676) T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be prepared for genetic modification. Preparation of T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof for genetic modification may include cell washing and/or resuspension in a desired nucleofection buffer. Cryopreserved T-cell compositions may be thawed and prepared for genetic modification by nucleofection. Cryopreserved cells may be thawed according to standard or known protocols. Thawing and preparation of cryopreserved cells may be optimized to yield

cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. For example, Grifols Albutein (25% human albumin) may be used in the thawing and/or preparation process.

(677) Genetic Modification of an Autologous T Cell Product Composition

(678) T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be genetically modified using, for example, a nucleofection strategy such as electroporation. The total number of cells to be nucleofected, the total volume of the nucleofection reaction, and the precise timing of the preparation of the sample may be optimized to yield cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies.

(679) Nucleofection and/or electroporation may be accomplished using, for example, Lonza Amaxa, MaxCyte PulseAgile, Harvard Apparatus BTX, and/or Invitrogen Neon. Non-metal electrode systems, including, but not limited to, plastic polymer electrodes, may be preferred for nucleofection.

(680) Prior to genetic modification by nucleofection, T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be resuspended in a nucleofection buffer. Nucleofection buffers of the disclosure include commercially available nucleofection buffers. Nucleofection buffers of the disclosure may be optimized to yield cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. Nucleofection buffers of the disclosure may include, but are not limited to, PBS, HBSS, OptiMEM, BTXpress, Amaxa Nucleofector, Human T cell nucleofection buffer and any combination thereof. Nucleofection buffers of the disclosure may comprise one or more supplemental factors to yield cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. Exemplary supplemental factors include, but are not limited to, recombinant human cytokines, chemokines, interleukins and any combination thereof. Exemplary cytokines, chemokines, and interleukins include, but are not limited to, IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN-gamma, IL-1 alpha/IL-1F1, IL-1 beta/IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 beta, IL-32 gamma, IL-33, LAP (TGF-beta 1), Lymphotoxin-alpha/TNF-beta, TGF-beta, TNF-alpha, TRANCE/TNFSF11/RANK L and any combination thereof. Exemplary supplemental factors include, but are not limited to, salts, minerals, metabolites or any combination thereof. Exemplary salts, minerals, and metabolites include, but are not limited to, HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl<sub>2</sub>, Na<sub>2</sub>HPO<sub>4</sub>, NaH<sub>2</sub>PO<sub>4</sub>, Sodium lactobionate, Mannitol, Sodium succinate, Sodium Chloride, CINA, Glucose, Ca(NO<sub>3</sub>)<sub>2</sub>, Tris/HCl, K<sub>2</sub>HPO<sub>4</sub>, KH<sub>2</sub>PO<sub>4</sub>, Polyethylenimine, Poly-ethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5, and any combination thereof. Exemplary supplemental factors include, but are not limited to, media such as PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium and any combination thereof. Exemplary supplemental factors include, but are not limited to, inhibitors of cellular DNA sensing, metabolism, differentiation, signal transduction, the apoptotic pathway and combinations thereof. Exemplary inhibitors include, but are not limited to, inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-1, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1B, PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3 $\beta$  (GSK-3 $\beta$ ) (e.g., TWS119),

Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK and any combination thereof. Exemplary supplemental factors include, but are not limited to, reagents that modify or stabilize one or more nucleic acids in a way to enhance cellular delivery, enhance nuclear delivery or transport, enhance the facilitated transport of nucleic acid into the nucleus, enhance degradation of epi-chromosomal nucleic acid, and/or decrease DNA-mediated toxicity. Exemplary reagents that modify or stabilize one or more nucleic acids include, but are not limited to, pH modifiers, DNA-binding proteins, lipids, phospholipids, CaPO<sub>4</sub>, net neutral charge DNA binding peptides with or without NLS sequences, TREX1 enzyme, and any combination thereof.

(681) Transposition reagents, including a transposon and a transposase, may be added to a nucleofection reaction of the disclosure prior to, simultaneously with, or after an addition of cells to a nucleofection buffer (optionally, contained within a nucleofection reaction vial or cuvette). Transposons of the disclosure may comprise plasmid DNA, linearized plasmid DNA, a PCR product, DOGGYBONE™ DNA, an mRNA template, a single or double-stranded DNA, a protein-nucleic acid combination or any combination thereof. Transposons of the disclosure may comprise one or more sequences that encode one or more TTAA site(s), one or more inverted terminal repeat(s) (ITRs), one or more long terminal repeat(s) (LTRs), one or more insulator(s), one or more promoter(s), one or more full-length or truncated gene(s), one or more polyA signal(s), one or more self-cleaving 2A peptide cleavage site(s), one or more internal ribosome entry site(s) (IRES), one or more enhancer(s), one or more regulator(s), one or more replication origin(s), and any combination thereof.

(682) Transposons of the disclosure may comprise one or more sequences that encode one or more full-length or truncated gene(s). Full-length and/or truncated gene(s) introduced by transposons of the disclosure may encode one or more of a signal peptide, a VCAR, a single chain variable fragment (scFv), a hinge, a transmembrane domain, a costimulatory domain, a chimeric antigen receptor (CAR), a VCAR, a chimeric T-cell receptor (CAR-T or VCAR-T), a receptor, a ligand, a cytokine, a drug resistance gene, a tumor antigen, an allo or auto antigen, an enzyme, a protein, a peptide, a polypeptide, a fluorescent protein, a mutein or any combination thereof.

(683) Transposons of the disclosure may be prepared in water, TAE, TBE, PBS, HBSS, media, a supplemental factor of the disclosure or any combination thereof.

(684) Transposons of the disclosure may be designed to optimize clinical safety and/or improve manufacturability. As a non-limiting example, transposons of the disclosure may be designed to optimize clinical safety and/or improve manufacturability by eliminating unnecessary sequences or regions and/or including a non-antibiotic selection marker. Transposons of the disclosure may or may not be GMP grade.

(685) Transposase enzymes of the disclosure may be encoded by one or more sequences of plasmid DNA, mRNA, protein, protein-nucleic acid combination or any combination thereof.

(686) Transposase enzymes of the disclosure may be prepared in water, TAE, TBE, PBS, HBSS, media, a supplemental factor of the disclosure or any combination thereof. Transposase enzymes of the disclosure or the sequences/constructs encoding or delivering them may or may not be GMP grade.

(687) Transposons and transposase enzymes of the disclosure may be delivered to a cell by any means.

(688) Although compositions and methods of the disclosure include delivery of a transposon and/or transposase of the disclosure to a cell by plasmid DNA (pDNA), the use of a plasmid for delivery may allow the transposon and/or transposase to be integrated into the chromosomal DNA of the cell, which may lead to continued transposase expression. Accordingly, transposon and/or transposase enzymes of the disclosure may be delivered to a cell as either mRNA or protein to remove any possibility for chromosomal integration.

(689) Transposons and transposases of the disclosure may be pre-incubated alone or in combination with one another prior to the introduction of the transposon and/or transposase into a nucleofection reaction. The absolute amounts of each of the transposon and the transposase, as well as the relative amounts, e.g., a ratio of transposon to transposase may be optimized.

(690) Following preparation of nucleofection reaction, optionally, in a vial or cuvette, the reaction may be loaded into a nucleofector apparatus and activated for delivery of an electric pulse according to the manufacturer's protocol. Electric pulse conditions used for delivery of a transposon and/or a transposase



of the disclosure (or a sequence encoding a transposon and/or a transposase of the disclosure) to a cell may be optimized for yielding cells with enhanced viability, higher nucleofection efficiency, greater viability post-nucleofection, desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. When using Amaxa nucleofector technology, each of the various nucleofection programs for the Amaxa 2B or 4D nucleofector are contemplated.

(691) Following a nucleofection reaction of the disclosure, cells may be gently added to a cell medium. For example, when T cells undergo the nucleofection reaction, the T cells may be added to a T cell medium. Post-nucleofection cell media of the disclosure may comprise any one or more commercially-available media. Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may be optimized to yield cells with greater viability, higher nucleofection efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may comprise PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium and any combination thereof. Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may comprise one or more supplemental factors of the disclosure to enhance viability, nucleofection efficiency, viability post-nucleofection, cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. Exemplary supplemental factors include, but are not limited to, recombinant human cytokines, chemokines, interleukins and any combination thereof. Exemplary cytokines, chemokines, and interleukins include, but are not limited to, IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN-gamma, IL-1 alpha/IL-1F1, IL-1 beta/IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 beta, IL-32 gamma, IL-33, LAP (TGF-beta 1), Lymphotoxin-alpha/TNF-beta, TGF-beta, TNF-alpha, TRANCE/TNFSF11/RANK L and any combination thereof. Exemplary supplemental factors include, but are not limited to, salts, minerals, metabolites or any combination thereof. Exemplary salts, minerals, and metabolites include, but are not limited to, HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl<sub>2</sub>, Na<sub>2</sub>HPO<sub>4</sub>, NaH<sub>2</sub>PO<sub>4</sub>, Sodium lactobionate, Mannitol, Sodium succinate, Sodium Chloride, CINa, Glucose, Ca(NO<sub>3</sub>)<sub>2</sub>, Tris/HCl, K<sub>2</sub>HPO<sub>4</sub>, KH<sub>2</sub>PO<sub>4</sub>, Polyethylenimine, Polyethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5, and any combination thereof. Exemplary supplemental factors include, but are not limited to, media such as PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium and any combination thereof. Exemplary supplemental factors include, but are not limited to, inhibitors of cellular DNA sensing, metabolism, differentiation, signal transduction, the apoptotic pathway and combinations thereof. Exemplary inhibitors include, but are not limited to, inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-1, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1B, PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3 $\beta$  (GSK-3 $\beta$ ) (e.g., TWS119), Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK and any combination thereof. Exemplary supplemental factors include, but are not limited to, reagents that modify or stabilize one or more nucleic acids in a way to enhance cellular delivery, enhance nuclear delivery or transport, enhance the facilitated transport of nucleic acid into the nucleus, enhance degradation of epi-chromosomal nucleic acid, and/or decrease DNA-mediated toxicity. Exemplary reagents that modify or stabilize one or more nucleic acids include, but are not limited to, pH modifiers, DNA-binding proteins, lipids, phospholipids, CaPO<sub>4</sub>, net neutral charge DNA binding peptides with or without NLS sequences, TREX1 enzyme, and

any combination thereof.

(692) Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may be used at room temperature or pre-warmed to, for example to between 32° C. to 37° C., inclusive of the endpoints. Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may be pre-warmed to any temperature that maintains or enhances cell viability and/or expression of a transposon or portion thereof of the disclosure.

(693) Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may be contained in tissue culture flasks or dishes, G-Rex flasks, Bioreactor or cell culture bags, or any other standard receptacle. Post-nucleofection cell cultures of the disclosure (including post-nucleofection T cell cultures of the disclosure) may be kept still, or, alternatively, they may be perturbed (e.g., rocked, swirled, or shaken).

(694) Post-nucleofection cell cultures may comprise genetically-modified cells. Post-nucleofection T cell cultures may comprise genetically-modified T cells. Genetically modified cells of the disclosure may be either rested for a defined period of time or stimulated for expansion by, for example, the addition of a T Cell Expander technology. In certain embodiments, genetically modified cells of the disclosure may be either rested for a defined period of time or immediately stimulated for expansion by, for example, the addition of a T Cell Expander technology. Genetically modified cells of the disclosure may be rested to allow them sufficient time to acclimate, time for transposition to occur, and/or time for positive or negative selection, resulting in cells with enhanced viability, higher nucleofection efficiency, greater viability post-nucleofection, desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. Genetically modified cells of the disclosure may be rested, for example, for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or more hours. In certain embodiments, genetically modified cells of the disclosure may be rested, for example, for an overnight. In certain aspects, an overnight is about 12 hours. Genetically modified cells of the disclosure may be rested, for example, for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more days.

(695) Genetically modified cells of the disclosure may be selected following a nucleofection reaction and prior to addition of an expander technology. For optimal selection of genetically-modified cells, the cells may be allowed to rest in a post-nucleofection cell medium for at least 2-14 days to facilitate identification of modified cells (e.g., differentiation of modified from non-modified cells).

(696) As early as 24-hours post-nucleofection, expression of a CAR/VCAR and selection marker of the disclosure may be detectable in modified T cells upon successful nucleofection of a transposon of the disclosure. Due to epi-chromosomal expression of the transposon, expression of a selection marker alone may not differentiate modified T cells (those cells in which the transposon has been successfully integrated) from unmodified T cells (those cells in which the transposon was not successfully integrated). When epi-chromosomal expression of the transposon obscures the detection of modified cells by the selection marker, the nucleofected cells (both modified and unmodified cells) may be rested for a period of time (e.g., 2-14 days) to allow the cells to cease expression or lose all epi-chromosomal transposon expression. Following this extended resting period, only modified T cells should remain positive for expression of selection marker. The length of this extended resting period may be optimized for each nucleofection reaction and selection process. When epi-chromosomal expression of the transposon obscures the detection of modified cells by the selection marker, selection may be performed without this extended resting period, however, an additional selection step may be included at a later time point (e.g., either during or after the expansion stage).

(697) Selection of genetically modified cells of the disclosure may be performed by any means. In certain embodiments of the methods of the disclosure, selection of genetically modified cells of the disclosure may be performed by isolating cells expressing a specific selection marker. Selection markers of the disclosure may be encoded by one or more sequences in the transposon. Selection markers of the disclosure may be expressed by the modified cell as a result of successful transposition (i.e., not encoded by one or more sequences in the transposon). In certain embodiments, genetically modified cells of the disclosure contain a selection marker that confers resistance to a deleterious compound of the post-nucleofection cell medium. The deleterious compound may comprise, for example, an antibiotic or a drug that, absent the resistance conferred by the selection marker to the

modified cells, would result in cell death. Exemplary selection markers include, but are not limited to, wild type (WT) or mutant forms of one or more of the following genes: neo, DHFR, TYMS, ALDH, MDR1, MGMT, FANCF, RAD51C, GCS, and NKX2.2. Exemplary selection markers include, but are not limited to, a surface-expressed selection marker or surface-expressed tag may be targeted by Ab-coated magnetic bead technology or column selection, respectively. A cleavable tag such as those used in protein purification may be added to a selection marker of the disclosure for efficient column selection, washing, and elution. In certain embodiments, selection markers of the disclosure are not expressed by the modified cells (including modified T cells) endogenously and, therefore, may be useful in the physical isolation of modified cells (by, for example, cell sorting techniques). Exemplary selection markers of the disclosure are not expressed by the modified cells (including modified T cells) endogenously include, but are not limited to, full-length, mutated, or truncated forms of CD271, CD19 CD52, CD34, RQR8, CD22, CD20, CD33 and any combination thereof.

(698) Genetically modified cells of the disclosure may be selective expanded following a nucleofection reaction. In certain embodiments, modified T cells comprising a VCAR may be selectively expanded by VCAR stimulation. Modified T cells comprising a VCAR may be stimulated by contact with a target-covered reagent (e.g., a tumor line or a normal cell line expressing a target or expander beads covered in a target). Alternatively, modified T cells comprising a VCAR may be stimulated by contact with an irradiated tumor cell, an irradiated allogeneic normal cell, an irradiated autologous PBMC. To minimize contamination of cell product compositions of the disclosure with a target-expressing cell used for stimulation, for example, when the cell product composition may be administered directly to a subject, the stimulation may be performed using expander beads coated with VCAR target protein. Selective expansion of modified T cells comprising a VCAR by VCAR stimulation may be optimized to avoid functionally-exhausting the modified T-cells.

(699) Selected genetically-modified cells of the disclosure may be cryopreserved, rested for a defined period of time, or stimulated for expansion by the addition of a Cell Expander technology. Selected genetically-modified cells of the disclosure may be cryopreserved, rested for a defined period of time, or immediately stimulated for expansion by the addition of a Cell Expander technology. When the selected genetically-modified cells are T cells, the T cells may be stimulated for expansion by the addition of a T-Cell Expander technology. Selected genetically modified cells of the disclosure may be rested, for example, for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or more hours. In certain embodiments, selected genetically modified cells of the disclosure may be rested, for example, for an overnight. In certain aspects, an overnight is about 12 hours. Selected genetically modified cells of the disclosure may be rested, for example, for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more days. Selected genetically modified cells of the disclosure may be rested for any period of time resulting in cells with enhanced viability, higher nucleofection efficiency, greater viability post-nucleofection, desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies.

(700) Selected genetically-modified cells (including selected genetically-modified T cells of the disclosure) may be cryopreserved using any standard cryopreservation method, which may be optimized for storing and/or recovering human cells with high recovery, viability, phenotype, and/or functional capacity. Cryopreservation methods of the disclosure may include commercially-available cryopreservation media and/or protocols.

(701) A transposition efficiency of selected genetically-modified cells (including selected genetically-modified T cells of the disclosure) may be assessed by any means. For example, prior to the application of an expander technology, expression of the transposon by selected genetically-modified cells (including selected genetically-modified T cells of the disclosure) may be measured by fluorescence-activated cell sorting (FACS). Determination of a transposition efficiency of selected genetically-modified cells (including selected genetically-modified T cells of the disclosure) may include determining a percentage of selected cells expressing the transposon (e.g., a VCAR). Alternatively, or in addition, a purity of T cells, a Mean Fluorescence Intensity (MFI) of the transposon expression (e.g., CAR expression), an ability of a VCAR (delivered in the transposon) to mediate degranulation and/or killing of a target cell expressing the VCAR ligand, and/or a phenotype of selected genetically-modified

cells (including selected genetically-modified T cells of the disclosure) may be assessed by any means. (702) Cell product compositions of the disclosure may be released for administration to a subject upon meeting certain release criteria. Exemplary release criteria may include, but are not limited to, a particular percentage of modified, selected and/or expanded T cells expressing detectable levels of a VCAR on the cell surface.

(703) Genetic Modification of an Autologous T Cell Product Composition

(704) Genetically-modified cells (including genetically-modified T cells) of the disclosure may be expanded using an expander technology. Expander technologies of the disclosure may comprise a commercially-available expander technology. Exemplary expander technologies of the disclosure include stimulation a genetically-modified T cell of the disclosure via the TCR. While all means for stimulation of a genetically-modified T cell of the disclosure are contemplated, stimulation a genetically-modified T cell of the disclosure via the TCR is a preferred method, yielding a product with a superior level of killing capacity.

(705) To stimulate a genetically-modified T cell of the disclosure via the TCR, Thermo Expander DynaBeads may be used at a 3:1 bead to T cell ratio. If the expander beads are not biodegradable, the beads may be removed from the expander composition. For example, the beads may be removed from the expander composition after about 5 days. To stimulate a genetically-modified T cell of the disclosure via the TCR, a Miltenyi T Cell Activation/Expansion Reagent may be used. To stimulate a genetically-modified T cell of the disclosure via the TCR, StemCell Technologies' ImmunoCult Human CD3/CD28 or CD3/CD28/CD2 T Cell Activator Reagent may be used. This technology may be preferred since the soluble tetrameric antibody complexes would degrade after a period and would not require removal from the process.

(706) Artificial antigen presenting cells (APCs) may be engineered to co-express the target antigen and may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure. Artificial APCs may comprise or may be derived from a tumor cell line (including, for example, the immortalized myelogenous leukemia line K562) and may be engineered to co-express multiple costimulatory molecules or technologies (such as CD28, 4-1BBL, CD64, mbIL-21, mbIL-15, CAR target molecule, etc.). When artificial APCs of the disclosure are combined with costimulatory molecules, conditions may be optimized to prevent the development or emergence of an undesirable phenotype and functional capacity, namely terminally-differentiated effector T cells.

(707) Irradiated PBMCs (auto or allo) may express some target antigens, such as CD19, and may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure. Alternatively, or in addition, irradiated tumor cells may express some target antigens and may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure.

(708) Plate-bound and/or soluble anti-CD3, anti-CD2 and/or anti-CD28 stimulate may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure.

(709) Antigen-coated beads may display target protein and may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure. Alternatively, or in addition, expander beads coated with a VCAR target protein may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure.

(710) Expansion methods drawn to stimulation of a cell or T-cell of the disclosure through the TCR or VCAR and via surface-expressed CD2, CD3, CD28, 4-1BB, and/or other markers on genetically-modified T cells.

(711) An expansion technology may be applied to a cell of the disclosure immediately post-nucleofection until approximately 24 hours post-nucleofection. While various cell media may be used during an expansion procedure, a desirable T Cell Expansion Media of the disclosure may yield cells with, for example, greater viability, cell phenotype, total expansion, or greater capacity for in vivo persistence, engraftment, and/or CAR-mediated killing. Cell media of the disclosure may be optimized to improve/enhance expansion, phenotype, and function of genetically-modified cells of the disclosure. A preferred phenotype of expanded T cells may include a mixture of T stem cell memory, T central, and T effector memory cells. Expander Dynabeads may yield mainly central memory T cells which may lead to superior performance in the clinic.

(712) Exemplary T cell expansion media of the disclosure may include, in part or in total, PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium, or any combination thereof. T cell expansion media of the disclosure may further include one or more supplemental factors. Supplemental factors that may be included in a T cell expansion media of the disclosure enhance viability, cell phenotype, total expansion, or increase capacity for in vivo persistence, engraftment, and/or VCAR-mediated killing. Supplemental factors that may be included in a T cell expansion media of the disclosure include, but are not limited to, recombinant human cytokines, chemokines, and/or interleukins such as IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN-gamma, IL-1 alpha/IL-1F1, IL-1 beta/IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 beta, IL-32 gamma, IL-33, LAP (TGF-beta 1), Lymphotoxin-alpha/TNF-beta, TGF-beta, TNF-alpha, TRANCE/TNFSF11/RANK L, or any combination thereof. Supplemental factors that may be included in a T cell expansion media of the disclosure include, but are not limited to, salts, minerals, and/or metabolites such as HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl<sub>2</sub>, Na<sub>2</sub>HPO<sub>4</sub>, NaH<sub>2</sub>PO<sub>4</sub>, Sodium lactobionate, Mannitol, Sodium succinate, Sodium Chloride, ClNa, Glucose, Ca(NO<sub>3</sub>)<sub>2</sub>, Tris/HCl, K<sub>2</sub>HPO<sub>4</sub>, KH<sub>2</sub>PO<sub>4</sub>, Polyethylenimine, Poly-ethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5 or any combination thereof. Supplemental factors that may be included in a T cell expansion media of the disclosure include, but are not limited to, inhibitors of cellular DNA sensing, metabolism, differentiation, signal transduction, and/or the apoptotic pathway such as inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-1, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1B, PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3 $\beta$  (GSK-3 $\beta$ ) (e.g., TWS119), Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK, or any combination thereof.

(713) Supplemental factors that may be included in a T cell expansion media of the disclosure include, but are not limited to, reagents that modify or stabilize nucleic acids in a way to enhance cellular delivery, enhance nuclear delivery or transport, enhance the facilitated transport of nucleic acid into the nucleus, enhance degradation of epi-chromosomal nucleic acid, and/or decrease DNA-mediated toxicity, such as pH modifiers, DNA-binding proteins, lipids, phospholipids, CaPO<sub>4</sub>, net neutral charge DNA binding peptides with or without NLS sequences, TREX1 enzyme, or any combination thereof.

(714) Genetically-modified cells of the disclosure may be selected during the expansion process by the use of selectable drugs or compounds. For example, in certain embodiments, when a transposon of the disclosure may encode a selection marker that confers to genetically-modified cells resistance to a drug added to the culture medium, selection may occur during the expansion process and may require approximately 1-14 days of culture for selection to occur. Examples of drug resistance genes that may be used as selection markers encoded by a transposon of the disclosure, include, but are not limited to, wild type (WT) or mutant forms of the genes neo, DHFR, TYMS, ALDH, MDR1, MGMT, FANCF, RAD51C, GCS, NKX2.2, or any combination thereof. Examples of corresponding drugs or compounds that may be added to the culture medium to which a selection marker may confer resistance include, but are not limited to, G418, Puromycin, Ampicillin, Kanamycin, Methotrexate, Mephalan, Temozolomide, Vincristine, Etoposide, Doxorubicin, Bendamustine, Fludarabine, Aredia (Pamidronate Disodium), Becenun (Carmustine), BiCNU (Carmustine), Bortezomib, Carfilzomib, Carmubris (Carmustine), Carmustine, Clafen (Cyclophosphamide), Cyclophosphamide, Cytosan (Cyclophosphamide), Daratumumab, Darzalex (Daratumumab), Doxil (Doxorubicin Hydrochloride Liposome), Doxorubicin Hydrochloride Liposome, Dox-SL (Doxorubicin Hydrochloride Liposome), Elotuzumab, Emticiti (Elotuzumab), Evacet (Doxorubicin Hydrochloride Liposome), Farydak (Panobinostat), Ixazomib

Citrate, Kyprolis (Carfilzomib), Lenalidomide, LipoDox (Doxorubicin Hydrochloride Liposome), Mozobil (Plerixafor), Neosar (Cyclophosphamide), Ninlaro (Ixazomib Citrate), Pamidronate Disodium, Panobinostat, Plerixafor, Pomalidomide, Pomalyst (Pomalidomide), Revlimid (Lenalidomide), Synovir (Thalidomide), Thalidomide, Thalomid (Thalidomide), Velcade (Bortezomib), Zoledronic Acid, Zometa (Zoledronic Acid), or any combination thereof.

(715) A T-Cell Expansion process of the disclosure may occur in a cell culture bag in a WAVE Bioreactor, a G-Rex flask, or in any other suitable container and/or reactor.

(716) A cell or T-cell culture of the disclosure may be kept steady, rocked, swirled, or shaken.

(717) A cell or T-cell expansion process of the disclosure may optimize certain conditions, including, but not limited to culture duration, cell concentration, schedule for T cell medium addition/removal, cell size, total cell number, cell phenotype, purity of cell population, percentage of genetically-modified cells in growing cell population, use and composition of supplements, the addition/removal of expander technologies, or any combination thereof.

(718) A cell or T-cell expansion process of the disclosure may continue until a predefined endpoint prior to formulation of the resultant expanded cell population. For example, a cell or T-cell expansion process of the disclosure may continue for a predetermined amount of time: at least, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24 hours; at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 days; at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 weeks; at least 1, 2, 3, 4, 5, 6, months, or at least 1 year. A cell or T-cell expansion process of the disclosure may continue until the resultant culture reaches a predetermined overall cell density: 1, 10, 100, 1000, 104, 105, 106, 107, 108, 109, 1010 cells per volume ( $\mu$ l, ml, L) or any density in between. A cell or T-cell expansion process of the disclosure may continue until the genetically-modified cells of a resultant culture demonstrate a predetermined level of expression of a transposon of the disclosure: 1%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 100% or any percentage in between of a threshold level of expression (a minimum, maximum or mean level of expression indicating the resultant genetically-modified cells are clinically-efficacious). A cell or T-cell expansion process of the disclosure may continue until the proportion of genetically-modified cells of a resultant culture to the proportion of unmodified cells reaches a predetermined threshold: at least 1:10, 1:9, 1:8, 1:7, 1:6, 1:5, 1:4, 1:3, 1:2, 1:1, 2:1, 2:1, 4:1, 5:1, 6:1, 7:1, 8:1, 9:1 10:1 or any ratio in between.

(719) Analysis of Genetically-Modified Autologous T Cells for Release

(720) A percentage of genetically-modified cells may be assessed during or after an expansion process of the disclosure. Cellular expression of a transposon by a genetically-modified cell of the disclosure may be measured by fluorescence-activated cell sorting (FACS). For example, FACS may be used to determine a percentage of cells or T cells expressing a VCAR of the disclosure. Alternatively, or in addition, a purity of genetically-modified cells or T cells, the Mean Fluorescence Intensity (MFI) of a VCAR expressed by a genetically-modified cell or T cell of the disclosure, an ability of the VCAR to mediate degranulation and/or killing of a target cell expressing the VCAR ligand, and/or a phenotype of VCAR+ T cells may be assessed.

(721) Compositions of the disclosure intended for administration to a subject may be required to meet one or more “release criteria” that indicate that the composition is safe and efficacious for formulation as a pharmaceutical product and/or administration to a subject. Release criteria may include a requirement that a composition of the disclosure (e.g., a T-cell product of the disclosure) comprises a particular percentage of T cells expressing detectable levels of a VCAR of the disclosure on their cell surface.

(722) The expansion process should be continued until a specific criterion has been met (e.g., achieving a certain total number of cells, achieving a particular population of memory cells, achieving a population of a specific size).

(723) Certain criterion signal a point at which the expansion process should end. For example, cells should be formulated, reactivated, or cryopreserved once they reach a cell size of 300 fL (otherwise, cells reaching a size above this threshold may start to die). Cryopreservation immediately once a population of cells reaches an average cell size of less than 300 fL may yield better cell recovery upon thawing and culture because the cells haven't yet reached a fully quiescent state prior to

cryopreservation (a fully quiescent size is approximately 180 fL). Prior to expansion, T cells of the disclosure may have a cell size of about 180 fL, but may more than quadruple their cell size to approximately 900 fL at 3 days post-expansion. Over the next 6-12 days, the population of T-cells will slowly decrease cell size to full quiescence at 180 fL.

(724) A process for preparing a cell population for formulation may include, but is not limited to the steps of, concentrating the cells of the cell population, washing the cells, and/or further selection of the cells via drug resistance or magnetic bead sorting against a particular surface-expressed marker. A process for preparing a cell population for formulation may further include a sorting step to ensure the safety and purity of the final product. For example, if a tumor cell from a patient has been used to stimulate a genetically-modified T-cell of the disclosure or that have been genetically-modified in order to stimulate a genetically-modified T-cell of the disclosure that is being prepared for formulation, it is critical that no tumor cells from the patient are included in the final product.

(725) Cell Product Infusion and/or Cryopreservation for Infusion

(726) A pharmaceutical formulation of the disclosure may be distributed into bags for infusion, cryopreservation, and/or storage.

(727) A pharmaceutical formulation of the disclosure may be cryopreserved using a standard protocol and, optionally, an infusible cryopreservation medium. For example, a DMSO free cryopreservant (e.g., CryoSOfree™ DMSO-free Cryopreservation Medium) may be used to reduce freezing-related toxicity. A cryopreserved pharmaceutical formulation of the disclosure may be stored for infusion to a patient at a later date. An effective treatment may require multiple administrations of a pharmaceutical formulation of the disclosure and, therefore, pharmaceutical formulations may be packaged in pre-aliquoted “doses” that may be stored frozen but separated for thawing of individual doses.

(728) A pharmaceutical formulation of the disclosure may be stored at room temperature. An effective treatment may require multiple administrations of a pharmaceutical formulation of the disclosure and, therefore, pharmaceutical formulations may be packaged in pre-aliquoted “doses” that may be stored together but separated for administration of individual doses.

(729) A pharmaceutical formulation of the disclosure may be archived for subsequent re-expansion and/or selection for generation of additional doses to the same patient in the case of an allogenic therapy who may need an administration at a future date following, for example, a remission and relapse of a condition.

(730) Formulations

(731) As noted above, the disclosure provides for stable formulations, which preferably comprise a phosphate buffer with saline or a chosen salt, as well as preserved solutions and formulations containing a preservative as well as multi-use preserved formulations suitable for pharmaceutical or veterinary use, comprising at least one VCAR in a pharmaceutically acceptable formulation. Preserved formulations contain at least one known preservative or optionally selected from the group consisting of at least one phenol, m-cresol, p-cresol, o-cresol, chlorocresol, benzyl alcohol, phenylmercuric nitrite, phenoxyethanol, formaldehyde, chlorobutanol, magnesium chloride (e.g., hexahydrate), alkylparaben (methyl, ethyl, propyl, butyl and the like), benzalkonium chloride, benzethonium chloride, sodium dehydroacetate and thimerosal, polymers, or mixtures thereof in an aqueous diluent. Any suitable concentration or mixture can be used as known in the art, such as about 0.0015%, or any range, value, or fraction therein. Non-limiting examples include, no preservative, about 0.1-2% m-cresol (e.g., 0.2, 0.3, 0.4, 0.5, 0.9, 1.0%), about 0.1-3% benzyl alcohol (e.g., 0.5, 0.9, 1.1, 1.5, 1.9, 2.0, 2.5%), about 0.001-0.5% thimerosal (e.g., 0.005, 0.01), about 0.001-2.0% phenol (e.g., 0.05, 0.25, 0.28, 0.5, 0.9, 1.0%), 0.0005-1.0% alkylparaben(s) (e.g., 0.00075, 0.0009, 0.001, 0.002, 0.005, 0.0075, 0.009, 0.01, 0.02, 0.05, 0.075, 0.09, 0.1, 0.2, 0.3, 0.5, 0.75, 0.9, 1.0%), and the like.

(732) As noted above, the invention provides an article of manufacture, comprising packaging material and at least one vial comprising a solution of at least one VCAR with the prescribed buffers and/or preservatives, optionally in an aqueous diluent, wherein said packaging material comprises a label that indicates that such solution can be held over a period of 1, 2, 3, 4, 5, 6, 9, 12, 18, 20, 24, 30, 36, 40, 48, 54, 60, 66, 72 hours or greater. The invention further comprises an article of manufacture, comprising packaging material, a first vial comprising lyophilized at least one VCAR, and a second vial comprising

an aqueous diluent of preservative, wherein said packaging material comprises a label that instructs a patient to reconstitute the at least one VCAR in the aqueous diluent to form a solution that can be held over a period of twenty-four hours or greater.

(733) The at least one VCAR used in accordance with the present invention can be produced by recombinant means, including from mammalian cell or transgenic preparations, or can be purified from other biological sources, as described herein or as known in the art.

(734) The range of at least one VCAR in the product of the present invention includes amounts yielding upon reconstitution, if in a wet/dry system, concentrations from about 1.0 pg/ml to about 1000 mg/ml, although lower and higher concentrations are operable and are dependent on the intended delivery vehicle, e.g., solution formulations will differ from transdermal patch, pulmonary, transmucosal, or osmotic or micro pump methods.

(735) Preferably, the aqueous diluent optionally further comprises a pharmaceutically acceptable preservative. Preferred preservatives include those selected from the group consisting of phenol, m-cresol, p-cresol, o-cresol, chlorocresol, benzyl alcohol, alkylparaben (methyl, ethyl, propyl, butyl and the like), benzalkonium chloride, benzethonium chloride, sodium dehydroacetate and thimerosal, or mixtures thereof. The concentration of preservative used in the formulation is a concentration sufficient to yield an anti-microbial effect. Such concentrations are dependent on the preservative selected and are readily determined by the skilled artisan.

(736) Other excipients, e.g., isotonicity agents, buffers, antioxidants, and preservative enhancers, can be optionally and preferably added to the diluent. An isotonicity agent, such as glycerin, is commonly used at known concentrations. A physiologically tolerated buffer is preferably added to provide improved pH control. The formulations can cover a wide range of pHs, such as from about pH 4 to about pH 10, and preferred ranges from about pH 5 to about pH 9, and a most preferred range of about 6.0 to about 8.0. Preferably, the formulations of the present invention have a pH between about 6.8 and about 7.8. Preferred buffers include phosphate buffers, most preferably, sodium phosphate, particularly, phosphate buffered saline (PBS).

(737) Other additives, such as a pharmaceutically acceptable solubilizers like Tween 20 (polyoxyethylene (20) sorbitan monolaurate), Tween 40 (polyoxyethylene (20) sorbitan monopalmitate), Tween 80 (polyoxyethylene (20) sorbitan monooleate), Pluronic F68 (polyoxyethylene polyoxypropylene block copolymers), and PEG (polyethylene glycol) or non-ionic surfactants, such as polysorbate 20 or 80 or poloxamer 184 or 188, Pluronic® polyols, other block co-polymers, and chelators, such as EDTA and EGTA, can optionally be added to the formulations or compositions to reduce aggregation. These additives are particularly useful if a pump or plastic container is used to administer the formulation. The presence of pharmaceutically acceptable surfactant mitigates the propensity for the protein to aggregate.

(738) The formulations of the present invention can be prepared by a process which comprises mixing at least one VCAR and a preservative selected from the group consisting of phenol, m-cresol, p-cresol, o-cresol, chlorocresol, benzyl alcohol, alkylparaben, (methyl, ethyl, propyl, butyl and the like), benzalkonium chloride, benzethonium chloride, sodium dehydroacetate and thimerosal or mixtures thereof in an aqueous diluent. Mixing the at least one VCAR and preservative in an aqueous diluent is carried out using conventional dissolution and mixing procedures. To prepare a suitable formulation, for example, a measured amount of at least one VCAR in buffered solution is combined with the desired preservative in a buffered solution in quantities sufficient to provide the protein and preservative at the desired concentrations. Variations of this process would be recognized by one of ordinary skill in the art. For example, the order the components are added, whether additional additives are used, the temperature and pH at which the formulation is prepared, are all factors that can be optimized for the concentration and means of administration used.

(739) The claimed formulations can be provided to patients as clear solutions or as dual vials comprising a vial of lyophilized at least one VCAR that is reconstituted with a second vial containing water, a preservative and/or excipients, preferably, a phosphate buffer and/or saline and a chosen salt, in an aqueous diluent. Either a single solution vial or dual vial requiring reconstitution can be reused multiple times and can suffice for a single or multiple cycles of patient treatment and thus can provide a



more convenient treatment regimen than currently available.

(740) The present claimed articles of manufacture are useful for administration over a period ranging from immediate to twenty-four hours or greater. Accordingly, the presently claimed articles of manufacture offer significant advantages to the patient. Formulations of the invention can optionally be safely stored at temperatures of from about 2° C. to about 40° C. and retain the biological activity of the protein for extended periods of time, thus allowing a package label indicating that the solution can be held and/or used over a period of 6, 12, 18, 24, 36, 48, 72, or 96 hours or greater. If preserved diluent is used, such label can include use up to 1-12 months, one-half, one and a half, and/or two years.

(741) The solutions of at least one VCAR of the invention can be prepared by a process that comprises mixing at least one VCAR in an aqueous diluent. Mixing is carried out using conventional dissolution and mixing procedures. To prepare a suitable diluent, for example, a measured amount of at least one VCAR in water or buffer is combined in quantities sufficient to provide the protein and, optionally, a preservative or buffer at the desired concentrations. Variations of this process would be recognized by one of ordinary skill in the art. For example, the order the components are added, whether additional additives are used, the temperature and pH at which the formulation is prepared, are all factors that can be optimized for the concentration and means of administration used.

(742) The claimed products can be provided to patients as clear solutions or as dual vials comprising a vial of lyophilized at least one VCAR that is reconstituted with a second vial containing the aqueous diluent. Either a single solution vial or dual vial requiring reconstitution can be reused multiple times and can suffice for a single or multiple cycles of patient treatment and thus provides a more convenient treatment regimen than currently available.

(743) The claimed products can be provided indirectly to patients by providing to pharmacies, clinics, or other such institutions and facilities, clear solutions or dual vials comprising a vial of lyophilized at least one VCAR that is reconstituted with a second vial containing the aqueous diluent. The clear solution in this case can be up to one liter or even larger in size, providing a large reservoir from which smaller portions of the at least one VCAR solution can be retrieved one or multiple times for transfer into smaller vials and provided by the pharmacy or clinic to their customers and/or patients.

(744) Recognized devices comprising single vial systems include pen-injector devices for delivery of a solution, such as BD Pens, BD Autojector®, Humaject®, NovoPen®, B-D® Pen, AutoPen®, and OptiPen®, GenotropinPen®, Genotronorm Pen®, Humatro Pen®, Reco-Pen®, Roferon Pen®, Biojector®, Iject®, J-tip Needle-Free Injector®, Intraject®, Medi-Ject®, e.g., as made or developed by Becton Dickinson (Franklin Lakes, N.J., [www.bectondickenson.com](http://www.bectondickenson.com)), Disetronic (Burgdorf, Switzerland, [www.disetronic.com](http://www.disetronic.com)); Bioject, Portland, Oreg. ([www.bioject.com](http://www.bioject.com)); National Medical Products, Weston Medical (Peterborough, UK, [www.weston-medical.com](http://www.weston-medical.com)), Medi-Ject Corp (Minneapolis, Minn., [www.mediject.com](http://www.mediject.com)), and similarly suitable devices. Recognized devices comprising a dual vial system include those pen-injector systems for reconstituting a lyophilized drug in a cartridge for delivery of the reconstituted solution, such as the HumatroPen®. Examples of other devices suitable include pre-filled syringes, auto-injectors, needle free injectors and needle free IV infusion sets.

(745) The products presently claimed include packaging material. The packaging material provides, in addition to the information required by the regulatory agencies, the conditions under which the product can be used. The packaging material of the present invention provides instructions to the patient to reconstitute at least one VCAR in the aqueous diluent to form a solution and to use the solution over a period of 2-24 hours or greater for the two vial, wet/dry, product. For the single vial, solution product, the label indicates that such solution can be used over a period of 2-24 hours or greater. The presently claimed products are useful for human pharmaceutical product use.

(746) The formulations of the present invention can be prepared by a process that comprises mixing at least one VCAR and a selected buffer, preferably, a phosphate buffer containing saline or a chosen salt. Mixing at least one VCAR and buffer in an aqueous diluent is carried out using conventional dissolution and mixing procedures. To prepare a suitable formulation, for example, a measured amount of at least one VCAR in water or buffer is combined with the desired buffering agent in water in quantities sufficient to provide the protein and buffer at the desired concentrations. Variations of this

process would be recognized by one of ordinary skill in the art. For example, the order the components are added, whether additional additives are used, the temperature and pH at which the formulation is prepared, are all factors that can be optimized for the concentration and means of administration used. (747) The claimed stable or preserved formulations can be provided to patients as clear solutions or as dual vials comprising a vial of lyophilized VCAR that is reconstituted with a second vial containing a preservative or buffer and excipients in an aqueous diluent. Either a single solution vial or dual vial requiring reconstitution can be reused multiple times and can suffice for a single or multiple cycles of patient treatment and thus provides a more convenient treatment regimen than currently available. (748) Other formulations or methods of stabilizing the VCAR may result in other than a clear solution of lyophilized powder comprising the VCAR. Among non-clear solutions are formulations comprising particulate suspensions, said particulates being a composition containing the VCAR in a structure of variable dimension and known variously as a microsphere, microparticle, nanoparticle, nanosphere, or liposome. Such relatively homogenous, essentially spherical, particulate formulations containing an active agent can be formed by contacting an aqueous phase containing the active agent and a polymer and a nonaqueous phase followed by evaporation of the nonaqueous phase to cause the coalescence of particles from the aqueous phase as taught in U.S. Pat. No. 4,589,330. Porous microparticles can be prepared using a first phase containing active agent and a polymer dispersed in a continuous solvent and removing said solvent from the suspension by freeze-drying or dilution-extraction-precipitation as taught in U.S. Pat. No. 4,818,542. Preferred polymers for such preparations are natural or synthetic copolymers or polymers selected from the group consisting of gelatin agar, starch, arabinogalactan, albumin, collagen, polyglycolic acid, polylactic acid, glycolide-L(-) lactide poly(epsilon-caprolactone, poly(epsilon-caprolactone-CO-lactic acid), poly(epsilon-caprolactone-CO-glycolic acid), poly(beta-hydroxy butyric acid), polyethylene oxide, polyethylene, poly(alkyl-2-cyanoacrylate), poly(hydroxyethyl methacrylate), polyamides, poly(amino acids), poly(2-hydroxyethyl DL-aspartamide), poly(ester urea), poly(L-phenylalanine/ethylene glycol/1,6-diisocyanatohexane) and poly(methyl methacrylate). Particularly preferred polymers are polyesters, such as polyglycolic acid, polylactic acid, glycolide-L(-) lactide poly(epsilon-caprolactone, poly(epsilon-caprolactone-CO-lactic acid), and poly(epsilon-caprolactone-CO-glycolic acid. Solvents useful for dissolving the polymer and/or the active include: water, hexafluoroisopropanol, methylenechloride, tetrahydrofuran, hexane, benzene, or hexafluoroacetone sesquihydrate. The process of dispersing the active containing phase with a second phase may include pressure forcing said first phase through an orifice in a nozzle to affect droplet formation.

(749) Dry powder formulations may result from processes other than lyophilization, such as by spray drying or solvent extraction by evaporation or by precipitation of a crystalline composition followed by one or more steps to remove aqueous or nonaqueous solvent. Preparation of a spray-dried VCAR preparation is taught in U.S. Pat. No. 6,019,968. The VCAR-based dry powder compositions may be produced by spray drying solutions or slurries of the VCAR and, optionally, excipients, in a solvent under conditions to provide a respirable dry powder. Solvents may include polar compounds, such as water and ethanol, which may be readily dried. VCAR stability may be enhanced by performing the spray drying procedures in the absence of oxygen, such as under a nitrogen blanket or by using nitrogen as the drying gas. Another relatively dry formulation is a dispersion of a plurality of perforated microstructures dispersed in a suspension medium that typically comprises a hydrofluoroalkane propellant as taught in WO 9916419. The stabilized dispersions may be administered to the lung of a patient using a metered dose inhaler. Equipment useful in the commercial manufacture of spray dried medicaments are manufactured by Buchi Ltd. or Niro Corp.

(750) At least one VCAR in either the stable or preserved formulations or solutions described herein, can be administered to a patient in accordance with the present invention via a variety of delivery methods including SC or IM injection; transdermal, pulmonary, transmucosal, implant, osmotic pump, cartridge, micro pump, or other means appreciated by the skilled artisan, as well-known in the art.

(751) Therapeutic Applications

(752) The present invention also provides a method for modulating or treating a disease, in a cell, tissue, organ, animal, or patient, as known in the art or as described herein, using at least one VCAR of

the present invention, e.g., administering or contacting the cell, tissue, organ, animal, or patient with a therapeutic effective amount of VCAR. The present invention also provides a method for modulating or treating a disease, in a cell, tissue, organ, animal, or patient including, but not limited to, a malignant disease.

(753) The present invention also provides a method for modulating or treating at least one malignant disease in a cell, tissue, organ, animal or patient, including, but not limited to, at least one of leukemia, acute leukemia, acute lymphoblastic leukemia (ALL), acute lymphocytic leukemia, B-cell, T-cell or FAB ALL, acute myeloid leukemia (AML), acute myelogenous leukemia, chronic myelocytic leukemia (CML), chronic lymphocytic leukemia (CLL), hairy cell leukemia, myelodysplastic syndrome (MDS), a lymphoma, Hodgkin's disease, a malignant lymphoma, non-Hodgkin's lymphoma, Burkitt's lymphoma, multiple myeloma, Kaposi's sarcoma, colorectal carcinoma, pancreatic carcinoma, nasopharyngeal carcinoma, malignant histiocytosis, paraneoplastic syndrome/hypercalcemia of malignancy, solid tumors, bladder cancer, breast cancer, colorectal cancer, endometrial cancer, head cancer, neck cancer, hereditary nonpolyposis cancer, Hodgkin's lymphoma, liver cancer, lung cancer, non-small cell lung cancer, ovarian cancer, pancreatic cancer, prostate cancer, renal cell carcinoma, testicular cancer, adenocarcinomas, sarcomas, malignant melanoma, hemangioma, metastatic disease, cancer related bone resorption, cancer related bone pain, and the like.

(754) Any method of the present invention can comprise administering an effective amount of a composition or pharmaceutical composition comprising at least one VCAR to a cell, tissue, organ, animal or patient in need of such modulation, treatment or therapy. Such a method can optionally further comprise co-administration or combination therapy for treating such diseases or disorders, wherein the administering of said at least one VCAR, specified portion or variant thereof, further comprises administering, before concurrently, and/or after, at least one selected from at least one of an alkylating agent, an a mitotic inhibitor, and a radiopharmaceutical. Suitable dosages are well known in the art. See, e.g., Wells et al., eds., *Pharmacotherapy Handbook*, 2nd Edition, Appleton and Lange, Stamford, Conn. (2000); *PDR Pharmacopoeia*, Tarascon Pocket Pharmacopoeia 2000, Deluxe Edition, Tarascon Publishing, Loma Linda, Calif. (2000); *Nursing 2001 Handbook of Drugs*, 21st edition, Springhouse Corp., Springhouse, Pa., 2001; *Health Professional's Drug Guide 2001*, ed., Shannon, Wilson, Stang, Prentice-Hall, Inc, Upper Saddle River, N.J. each of which references are entirely incorporated herein by reference.

(755) Preferred doses can optionally include about 0.1-99 and/or 100-500 mg/kg/administration, or any range, value or fraction thereof, or to achieve a serum concentration of about 0.1-5000 µg/ml serum concentration per single or multiple administration, or any range, value or fraction thereof. A preferred dosage range for the VCAR of the present invention is from about 1 mg/kg, up to about 3, about 6 or about 12 mg/kg of body weight of the patient.

(756) Alternatively, the dosage administered can vary depending upon known factors, such as the pharmacodynamic characteristics of the particular agent, and its mode and route of administration; age, health, and weight of the recipient; nature and extent of symptoms, kind of concurrent treatment, frequency of treatment, and the effect desired. Usually a dosage of active ingredient can be about 0.1 to 100 milligrams per kilogram of body weight. Ordinarily 0.1 to 50, and preferably, 0.1 to 10 milligrams per kilogram per administration or in sustained release form is effective to obtain desired results.

(757) As a non-limiting example, treatment of humans or animals can be provided as a one-time or periodic dosage of at least one VCAR of the present invention about 0.1 to 100 mg/kg or any range, value or fraction thereof per day, on at least one of day 1-40, or, alternatively or additionally, at least one of week 1-52, or, alternatively or additionally, at least one of 1-20 years, or any combination thereof, using single, infusion or repeated doses.

(758) Dosage forms (composition) suitable for internal administration generally contain from about 0.001 milligram to about 500 milligrams of active ingredient per unit or container. In these pharmaceutical compositions the active ingredient will ordinarily be present in an amount of about 0.5-99.999% by weight based on the total weight of the composition.

(759) For parenteral administration, the VCAR can be formulated as a solution, suspension, emulsion, particle, powder, or lyophilized powder in association, or separately provided, with a pharmaceutically

acceptable parenteral vehicle. Examples of such vehicles are water, saline, Ringer's solution, dextrose solution, and about 1-10% human serum albumin. Liposomes and nonaqueous vehicles, such as fixed oils, can also be used. The vehicle or lyophilized powder can contain additives that maintain isotonicity (e.g., sodium chloride, mannitol) and chemical stability (e.g., buffers and preservatives). The formulation is sterilized by known or suitable techniques.

(760) Suitable pharmaceutical carriers are described in the most recent edition of Remington's Pharmaceutical Sciences, A. Osol, a standard reference text in this field. Alternative Administration

(761) Many known and developed modes can be used according to the present invention for administering pharmaceutically effective amounts of at least one VCAR according to the present invention. While pulmonary administration is used in the following description, other modes of administration can be used according to the present invention with suitable results. VCARS of the present invention can be delivered in a carrier, as a solution, emulsion, colloid, or suspension, or as a dry powder, using any of a variety of devices and methods suitable for administration by inhalation or other modes described here within or known in the art.

(762) Parenteral Formulations and Administration

(763) Formulations for parenteral administration can contain as common excipients sterile water or saline, polyalkylene glycols, such as polyethylene glycol, oils of vegetable origin, hydrogenated naphthalenes and the like. Aqueous or oily suspensions for injection can be prepared by using an appropriate emulsifier or humidifier and a suspending agent, according to known methods. Agents for injection can be a non-toxic, non-orally administrable diluting agent, such as aqueous solution, a sterile injectable solution or suspension in a solvent. As the usable vehicle or solvent, water, Ringer's solution, isotonic saline, etc. are allowed; as an ordinary solvent or suspending solvent, sterile involatile oil can be used. For these purposes, any kind of involatile oil and fatty acid can be used, including natural or synthetic or semisynthetic fatty oils or fatty acids; natural or synthetic or semisynthetic mono- or di- or tri-glycerides. Parental administration is known in the art and includes, but is not limited to, conventional means of injections, a gas pressured needle-less injection device as described in U.S. Pat. No. 5,851,198, and a laser perforator device as described in U.S. Pat. No. 5,839,446 entirely incorporated herein by reference.

(764) Alternative Delivery

(765) The invention further relates to the administration of at least one VCAR by parenteral, subcutaneous, intramuscular, intravenous, intraarticular, intrabronchial, intraabdominal, intracapsular, intracartilaginous, intracavitary, intracelial, intracerebellar, intracerebroventricular, intracolic, intracervical, intragastric, intrahepatic, intramyocardial, intraosteal, intrapelvic, intrapericardiac, intraperitoneal, intrapleural, intraprostatic, intrapulmonary, intrarectal, intrarenal, intraretinal, intraspinal, intrasynovial, intrathoracic, intrauterine, intravesical, intralesional, bolus, vaginal, rectal, buccal, sublingual, intranasal, or transdermal means. At least one VCAR composition can be prepared for use for parenteral (subcutaneous, intramuscular or intravenous) or any other administration particularly in the form of liquid solutions or suspensions; for use in vaginal or rectal administration particularly in semisolid forms, such as, but not limited to, creams and suppositories; for buccal, or sublingual administration, such as, but not limited to, in the form of tablets or capsules; or intranasally, such as, but not limited to, the form of powders, nasal drops or aerosols or certain agents; or transdermally, such as not limited to a gel, ointment, lotion, suspension or patch delivery system with chemical enhancers such as dimethyl sulfoxide to either modify the skin structure or to increase the drug concentration in the transdermal patch (Junginger, et al. In "Drug Permeation Enhancement;" Hsieh, D. S., Eds., pp. 59-90 (Marcel Dekker, Inc. New York 1994, entirely incorporated herein by reference), or with oxidizing agents that enable the application of formulations containing proteins and peptides onto the skin (WO 98/53847), or applications of electric fields to create transient transport pathways, such as electroporation, or to increase the mobility of charged drugs through the skin, such as iontophoresis, or application of ultrasound, such as sonophoresis (U.S. Pat. Nos. 4,309,989 and 4,767,402) (the above publications and patents being entirely incorporated herein by reference).

(766) Pulmonary/Nasal Administration

(767) For pulmonary administration, preferably, at least one VCAR composition is delivered in a

particle size effective for reaching the lower airways of the lung or sinuses. According to the invention, at least one VCAR can be delivered by any of a variety of inhalation or nasal devices known in the art for administration of a therapeutic agent by inhalation. These devices capable of depositing aerosolized formulations in the sinus cavity or alveoli of a patient include metered dose inhalers, nebulizers, dry powder generators, sprayers, and the like. Other devices suitable for directing the pulmonary or nasal administration of VCARs are also known in the art. All such devices can use formulations suitable for the administration for the dispensing of VCAR in an aerosol. Such aerosols can be comprised of either solutions (both aqueous and nonaqueous) or solid particles.

(768) Metered dose inhalers like the Ventolin metered dose inhaler, typically use a propellant gas and require actuation during inspiration (See, e.g., WO 94/16970, WO 98/35888). Dry powder inhalers like Turbuhaler™ (Astra), Rotahaler® (Glaxo), Diskus® (Glaxo), Spiros™ inhaler (Dura), devices marketed by Inhale Therapeutics, and the Spinhaler® powder inhaler (Fisons), use breath-actuation of a mixed powder (U.S. Pat. No. 4,668,218 Astra, EP 237507 Astra, WO 97/25086 Glaxo, WO 94/08552 Dura, U.S. Pat. No. 5,458,135 Inhale, WO 94/06498 Fisons, entirely incorporated herein by reference). Nebulizers like AERx™ Aradigm, the Ultravent® nebulizer (Mallinckrodt), and the Acorn II® nebulizer (Marquest Medical Products) (U.S. Pat. No. 5,404,871 Aradigm, WO 97/22376), the above references entirely incorporated herein by reference, produce aerosols from solutions, while metered dose inhalers, dry powder inhalers, etc. generate small particle aerosols. These specific examples of commercially available inhalation devices are intended to be a representative of specific devices suitable for the practice of this invention, and are not intended as limiting the scope of the invention.

(769) Preferably, a composition comprising at least one VCAR is delivered by a dry powder inhaler or a sprayer. There are several desirable features of an inhalation device for administering at least one VCAR of the present invention. For example, delivery by the inhalation device is advantageously reliable, reproducible, and accurate. The inhalation device can optionally deliver small dry particles, e.g., less than about 10 µm, preferably about 1-5 µm, for good respirability.

(770) Administration of VCAR Compositions as a Spray

(771) A spray including VCAR composition can be produced by forcing a suspension or solution of at least one VCAR through a nozzle under pressure. The nozzle size and configuration, the applied pressure, and the liquid feed rate can be chosen to achieve the desired output and particle size. An electrospray can be produced, for example, by an electric field in connection with a capillary or nozzle feed. Advantageously, particles of at least one VCAR composition delivered by a sprayer have a particle size less than about 10 µm, preferably, in the range of about 1 µm to about 5 µm, and, most preferably, about 2 µm to about 3 µm.

(772) Formulations of at least one VCAR composition suitable for use with a sprayer typically include VCAR composition in an aqueous solution at a concentration of about 0.1 mg to about 100 mg of at least one VCAR composition per ml of solution or mg/gm, or any range, value, or fraction therein. The formulation can include agents, such as an excipient, a buffer, an isotonicity agent, a preservative, a surfactant, and, preferably, zinc. The formulation can also include an excipient or agent for stabilization of the VCAR composition, such as a buffer, a reducing agent, a bulk protein, or a carbohydrate. Bulk proteins useful in formulating VCAR compositions include albumin, protamine, or the like. Typical carbohydrates useful in formulating VCAR compositions include sucrose, mannitol, lactose, trehalose, glucose, or the like. The VCAR composition formulation can also include a surfactant, which can reduce or prevent surface-induced aggregation of the VCAR composition caused by atomization of the solution in forming an aerosol. Various conventional surfactants can be employed, such as polyoxyethylene fatty acid esters and alcohols, and polyoxyethylene sorbitol fatty acid esters. Amounts will generally range between 0.001 and 14% by weight of the formulation. Especially preferred surfactants for purposes of this invention are polyoxyethylene sorbitan monooleate, polysorbate 80, polysorbate 20, or the like. Additional agents known in the art for formulation of a protein, such as VCARs, or specified portions or variants, can also be included in the formulation.

(773) Administration of VCAR Compositions by a Nebulizer

(774) VCAR compositions of the invention can be administered by a nebulizer, such as jet nebulizer or an ultrasonic nebulizer. Typically, in a jet nebulizer, a compressed air source is used to create a high-

velocity air jet through an orifice. As the gas expands beyond the nozzle, a low-pressure region is created, which draws a solution of VCAR composition through a capillary tube connected to a liquid reservoir. The liquid stream from the capillary tube is sheared into unstable filaments and droplets as it exits the tube, creating the aerosol. A range of configurations, flow rates, and baffle types can be employed to achieve the desired performance characteristics from a given jet nebulizer. In an ultrasonic nebulizer, high-frequency electrical energy is used to create vibrational, mechanical energy, typically employing a piezoelectric transducer. This energy is transmitted to the formulation of VCAR composition either directly or through a coupling fluid, creating an aerosol including the VCAR composition. Advantageously, particles of VCAR composition delivered by a nebulizer have a particle size less than about 10  $\mu\text{m}$ , preferably, in the range of about 1  $\mu\text{m}$  to about 5  $\mu\text{m}$ , and, most preferably, about 2  $\mu\text{m}$  to about 3  $\mu\text{m}$ .

(775) Formulations of at least one VCAR suitable for use with a nebulizer, either jet or ultrasonic, typically include a concentration of about 0.1 mg to about 100 mg of at least one VCAR per ml of solution. The formulation can include agents, such as an excipient, a buffer, an isotonicity agent, a preservative, a surfactant, and, preferably, zinc. The formulation can also include an excipient or agent for stabilization of the at least one VCAR composition, such as a buffer, a reducing agent, a bulk protein, or a carbohydrate. Bulk proteins useful in formulating at least one VCAR compositions include albumin, protamine, or the like. Typical carbohydrates useful in formulating at least one VCAR include sucrose, mannitol, lactose, trehalose, glucose, or the like. The at least one VCAR formulation can also include a surfactant, which can reduce or prevent surface-induced aggregation of the at least one VCAR caused by atomization of the solution in forming an aerosol. Various conventional surfactants can be employed, such as polyoxyethylene fatty acid esters and alcohols, and polyoxyethylene sorbitol fatty acid esters. Amounts will generally range between about 0.001 and 4% by weight of the formulation. Especially preferred surfactants for purposes of this invention are polyoxyethylene sorbitan mono-oleate, polysorbate 80, polysorbate 20, or the like. Additional agents known in the art for formulation of a protein, such as VCAR, can also be included in the formulation.

(776) Administration of VCAR Compositions by a Metered Dose Inhaler

(777) In a metered dose inhaler (MDI), a propellant, at least one VCAR, and any excipients or other additives are contained in a canister as a mixture including a liquefied compressed gas. Actuation of the metering valve releases the mixture as an aerosol, preferably containing particles in the size range of less than about 10  $\mu\text{m}$ , preferably, about 1  $\mu\text{m}$  to about 5  $\mu\text{m}$ , and, most preferably, about 2  $\mu\text{m}$  to about 3  $\mu\text{m}$ . The desired aerosol particle size can be obtained by employing a formulation of VCAR composition produced by various methods known to those of skill in the art, including jet-milling, spray drying, critical point condensation, or the like. Preferred metered dose inhalers include those manufactured by 3M or Glaxo and employing a hydrofluorocarbon propellant. Formulations of at least one VCAR for use with a metered-dose inhaler device will generally include a finely divided powder containing at least one VCAR as a suspension in a non-aqueous medium, for example, suspended in a propellant with the aid of a surfactant. The propellant can be any conventional material employed for this purpose, such as chlorofluorocarbon, a hydrochlorofluorocarbon, a hydrofluorocarbon, or a hydrocarbon, including trichlorofluoromethane, dichlorodifluoromethane, dichlorotetrafluoroethanol and 1,1,1,2-tetrafluoroethane, HFA-134a (hydrofluoroalkane-134a), HFA-227 (hydrofluoroalkane-227), or the like. Preferably, the propellant is a hydrofluorocarbon. The surfactant can be chosen to stabilize the at least one VCAR as a suspension in the propellant, to protect the active agent against chemical degradation, and the like. Suitable surfactants include sorbitan trioleate, soya lecithin, oleic acid, or the like. In some cases, solution aerosols are preferred using solvents, such as ethanol. Additional agents known in the art for formulation of a protein can also be included in the formulation. One of ordinary skill in the art will recognize that the methods of the current invention can be achieved by pulmonary administration of at least one VCAR composition via devices not described herein.

(778) Oral Formulations and Administration

(779) Formulations for oral administration rely on the co-administration of adjuvants (e.g., resorcinols and nonionic surfactants, such as polyoxyethylene oleyl ether and n-hexadecylpolyethylene ether) to increase artificially the permeability of the intestinal walls, as well as the co-administration of

enzymatic inhibitors (e.g., pancreatic trypsin inhibitors, diisopropylfluorophosphate (DFF) and trasylol) to inhibit enzymatic degradation. Formulations for delivery of hydrophilic agents including proteins and VCARs and a combination of at least two surfactants intended for oral, buccal, mucosal, nasal, pulmonary, vaginal transmembrane, or rectal administration are taught in U.S. Pat. No. 6,309,663. The active constituent compound of the solid-type dosage form for oral administration can be mixed with at least one additive, including sucrose, lactose, cellulose, mannitol, trehalose, raffinose, maltitol, dextran, starches, agar, arginates, chitins, chitosans, pectins, gum tragacanth, gum arabic, gelatin, collagen, casein, albumin, synthetic or semisynthetic polymer, and glyceride. These dosage forms can also contain other type(s) of additives, e.g., inactive diluting agent, lubricant, such as magnesium stearate, paraben, preserving agent, such as sorbic acid, ascorbic acid, .alpha.-tocopherol, antioxidant such as cysteine, disintegrator, binder, thickener, buffering agent, sweetening agent, flavoring agent, perfuming agent, etc.

(780) Tablets and pills can be further processed into enteric-coated preparations. The liquid preparations for oral administration include emulsion, syrup, elixir, suspension and solution preparations allowable for medical use. These preparations can contain inactive diluting agents ordinarily used in said field, e.g., water. Liposomes have also been described as drug delivery systems for insulin and heparin (U.S. Pat. No. 4,239,754). More recently, microspheres of artificial polymers of mixed amino acids (proteinoids) have been used to deliver pharmaceuticals (U.S. Pat. No. 4,925,673). Furthermore, carrier compounds described in U.S. Pat. Nos. 5,879,681 and 5,871,753 and used to deliver biologically active agents orally are known in the art.

#### (781) Mucosal Formulations and Administration

(782) A formulation for orally administering a bioactive agent encapsulated in one or more biocompatible polymer or copolymer excipients, preferably, a biodegradable polymer or copolymer, affording microcapsules which due to the proper size of the resultant microcapsules results in the agent reaching and being taken up by the folliculi lymphatic aggregati, otherwise known as the "Peyer's patch," or "GALT" of the animal without loss of effectiveness due to the agent having passed through the gastrointestinal tract. Similar folliculi lymphatic aggregati can be found in the bronchei tubes (BALT) and the large intestine. The above-described tissues are referred to in general as mucosally associated lymphoreticular tissues (MALT). For absorption through mucosal surfaces, compositions and methods of administering at least one VCAR include an emulsion comprising a plurality of submicron particles, a mucoadhesive macromolecule, a bioactive peptide, and an aqueous continuous phase, which promotes absorption through mucosal surfaces by achieving mucoadhesion of the emulsion particles (U.S. Pat. No. 5,514,670). Mucous surfaces suitable for application of the emulsions of the present invention can include corneal, conjunctival, buccal, sublingual, nasal, vaginal, pulmonary, stomachic, intestinal, and rectal routes of administration. Formulations for vaginal or rectal administration, e.g., suppositories, can contain as excipients, for example, polyalkyleneglycols, vaseline, cocoa butter, and the like. Formulations for intranasal administration can be solid and contain as excipients, for example, lactose or can be aqueous or oily solutions of nasal drops. For buccal administration, excipients include sugars, calcium stearate, magnesium stearate, pregelatinized starch, and the like (U.S. Pat. No. 5,849,695).

#### (783) Transdermal Formulations and Administration

(784) For transdermal administration, the at least one VCAR is encapsulated in a delivery device, such as a liposome or polymeric nanoparticles, microparticle, microcapsule, or microspheres (referred to collectively as microparticles unless otherwise stated). A number of suitable devices are known, including microparticles made of synthetic polymers, such as polyhydroxy acids, such as polylactic acid, polyglycolic acid and copolymers thereof, polyorthoesters, polyanhydrides, and polyphosphazenes, and natural polymers, such as collagen, polyamino acids, albumin and other proteins, alginate and other polysaccharides, and combinations thereof (U.S. Pat. No. 5,814,599).

#### (785) Prolonged Administration and Formulations

(786) It can be desirable to deliver the compounds of the present invention to the subject over prolonged periods of time, for example, for periods of one week to one year from a single administration. Various slow release, depot or implant dosage forms can be utilized. For example, a dosage form can contain a

pharmaceutically acceptable non-toxic salt of the compounds that has a low degree of solubility in body fluids, for example, (a) an acid addition salt with a polybasic acid, such as phosphoric acid, sulfuric acid, citric acid, tartaric acid, tannic acid, pamoic acid, alginic acid, polyglutamic acid, naphthalene mono- or di-sulfonic acids, polygalacturonic acid, and the like; (b) a salt with a polyvalent metal cation, such as zinc, calcium, bismuth, barium, magnesium, aluminum, copper, cobalt, nickel, cadmium and the like, or with an organic cation formed from e.g., N,N'-dibenzyl-ethylenediamine or ethylenediamine; or (c) combinations of (a) and (b), e.g., a zinc tannate salt. Additionally, the compounds of the present invention or, preferably, a relatively insoluble salt, such as those just described, can be formulated in a gel, for example, an aluminum monostearate gel with, e.g., sesame oil, suitable for injection. Particularly preferred salts are zinc salts, zinc tannate salts, pamoate salts, and the like. Another type of slow release depot formulation for injection would contain the compound or salt dispersed for encapsulation in a slow degrading, non-toxic, non-antigenic polymer, such as a polylactic acid/polyglycolic acid polymer for example as described in U.S. Pat. No. 3,773,919. The compounds or, preferably, relatively insoluble salts, such as those described above, can also be formulated in cholesterol matrix silastic pellets, particularly for use in animals. Additional slow release, depot or implant formulations, e.g., gas or liquid liposomes, are known in the literature (U.S. Pat. No. 5,770,222 and "Sustained and Controlled Release Drug Delivery Systems", J. R. Robinson ed., Marcel Dekker, Inc., N.Y., 1978).

(787) Infusion of Modified Cells as Adoptive Cell Therapy

(788) The disclosure provides modified cells that express one or more CARs and/or VCARs of the disclosure that have been selected and/or expanded for administration to a subject in need thereof. Modified cells of the disclosure may be formulated for storage at any temperature including room temperature and body temperature. Modified cells of the disclosure may be formulated for cryopreservation and subsequent thawing. Modified cells of the disclosure may be formulated in a pharmaceutically acceptable carrier for direct administration to a subject from sterile packaging. Modified cells of the disclosure may be formulated in a pharmaceutically acceptable carrier with an indicator of cell viability and/or CAR/VCAR expression level to ensure a minimal level of cell function and CAR/VCAR expression. Modified cells of the disclosure may be formulated in a pharmaceutically acceptable carrier at a prescribed density with one or more reagents to inhibit further expansion and/or prevent cell death.

(789) Inducible Proapoptotic Polypeptides

(790) Inducible proapoptotic polypeptides of the disclosure are superior to existing inducible polypeptides because the inducible proapoptotic polypeptides of the disclosure are far less immunogenic. While inducible proapoptotic polypeptides of the disclosure are recombinant polypeptides, and, therefore, non-naturally occurring, the sequences that are recombined to produce the inducible proapoptotic polypeptides of the disclosure do not comprise non-human sequences that the host human immune system could recognize as "non-self" and, consequently, induce an immune response in the subject receiving an inducible proapoptotic polypeptide of the disclosure, a cell comprising the inducible proapoptotic polypeptide or a composition comprising the inducible proapoptotic polypeptide or the cell comprising the inducible proapoptotic polypeptide.

(791) The disclosure provides inducible proapoptotic polypeptides comprising a ligand binding region, a linker, and a proapoptotic peptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. In certain embodiments, the proapoptotic peptide is a caspase polypeptide. In certain embodiments, the caspase polypeptide is a caspase 9 polypeptide. In certain embodiments, the caspase 9 polypeptide is a truncated caspase 9 polypeptide. Inducible proapoptotic polypeptides of the disclosure may be non-naturally occurring.

(792) Caspase polypeptides of the disclosure include, but are not limited to, caspase 1, caspase 2, caspase 3, caspase 4, caspase 5, caspase 6, caspase 7, caspase 8, caspase 9, caspase 10, caspase 11, caspase 12, and caspase 14. Caspase polypeptides of the disclosure include, but are not limited to, those caspase polypeptides associated with apoptosis including caspase 2, caspase 3, caspase 6, caspase 7, caspase 8, caspase 9, and caspase 10. Caspase polypeptides of the disclosure include, but are not limited



to, those caspase polypeptides that initiate apoptosis, including caspase 2, caspase 8, caspase 9, and caspase 10. Caspase polypeptides of the disclosure include, but are not limited to, those caspase polypeptides that execute apoptosis, including caspase 3, caspase 6, and caspase 7.

(793) Caspase polypeptides of the disclosure may be encoded by an amino acid or a nucleic acid sequence having one or more modifications compared to a wild type amino acid or a nucleic acid sequence. The nucleic acid sequence encoding a caspase polypeptide of the disclosure may be codon optimized. The one or more modifications to an amino acid and/or nucleic acid sequence of a caspase polypeptide of the disclosure may increase an interaction, a cross-linking, a cross-activation, or an activation of the caspase polypeptide of the disclosure compared to a wild type amino acid or a nucleic acid sequence. Alternatively, or in addition, the one or more modifications to an amino acid and/or nucleic acid sequence of a caspase polypeptide of the disclosure may decrease the immunogenicity of the caspase polypeptide of the disclosure compared to a wild type amino acid or a nucleic acid sequence.

(794) Caspase polypeptides of the disclosure may be truncated compared to a wild type caspase polypeptide. For example, a caspase polypeptide may be truncated to eliminate a sequence encoding a Caspase Activation and Recruitment Domain (CARD) to eliminate or minimize the possibility of activating a local inflammatory response in addition to initiating apoptosis in the cell comprising an inducible caspase polypeptide of the disclosure. The nucleic acid sequence encoding a caspase polypeptide of the disclosure may be spliced to form a variant amino acid sequence of the caspase polypeptide of the disclosure compared to a wild type caspase polypeptide. Caspase polypeptides of the disclosure may be encoded by recombinant and/or chimeric sequences. Recombinant and/or chimeric caspase polypeptides of the disclosure may include sequences from one or more different caspase polypeptides. Alternatively, or in addition, recombinant and/or chimeric caspase polypeptides of the disclosure may include sequences from one or more species (e.g. a human sequence and a non-human sequence). Caspase polypeptides of the disclosure may be non-naturally occurring.

(795) The ligand binding region of an inducible proapoptotic polypeptide of the disclosure may include any polypeptide sequence that facilitates or promotes the dimerization of a first inducible proapoptotic polypeptide of the disclosure with a second inducible proapoptotic polypeptide of the disclosure, the dimerization of which activates or induces cross-linking of the proapoptotic polypeptides and initiation of apoptosis in the cell.

(796) The ligand-binding (“dimerization”) region may comprise any polypeptide or functional domain thereof that will allow for induction using an endogenous or non-naturally-occurring ligand (i.e. and induction agent), for example, a non-naturally-occurring synthetic ligand. The ligand-binding region may be internal or external to the cellular membrane, depending upon the nature of the inducible proapoptotic polypeptide and the choice of ligand (i.e. induction agent). A wide variety of ligand-binding polypeptides and functional domains thereof, including receptors, are known. Ligand-binding regions of the disclosure may include one or more sequences from a receptor. Of particular interest are ligand-binding regions for which ligands (for example, small organic ligands) are known or may be readily produced. These ligand-binding regions or receptors may include, but are not limited to, the FKBP and cyclophilin receptors, the steroid receptors, the tetracycline receptor, and the like, as well as “non-naturally-occurring” receptors, which can be obtained from antibodies, particularly the heavy or light chain subunit, mutated sequences thereof, random amino acid sequences obtained by stochastic procedures, combinatorial syntheses, and the like. In certain embodiments, the ligand-binding region is selected from the group consisting of a FKBP ligand-binding region, a cyclophilin receptor ligand-binding region, a steroid receptor ligand-binding region, a cyclophilin receptors ligand-binding region, and a tetracycline receptor ligand-binding region.

(797) The ligand-binding regions comprising one or more receptor domain(s) may be at least about 50 amino acids, and fewer than about 350 amino acids, usually fewer than 200 amino acids, either as the endogenous domain or truncated active portion thereof. The binding region may, for example, be small (<25 kDa, to allow efficient transfection in viral vectors), monomeric, nonimmunogenic, have synthetically accessible, cell permeable, nontoxic ligands that can be configured for dimerization.

(798) The ligand-binding regions comprising one or more receptor domain(s) may be intracellular or

extracellular depending upon the design of the inducible proapoptotic polypeptide and the availability of an appropriate ligand (i.e. induction agent). For hydrophobic ligands, the binding region can be on either side of the membrane, but for hydrophilic ligands, particularly protein ligands, the binding region will usually be external to the cell membrane, unless there is a transport system for internalizing the ligand in a form in which it is available for binding. For an intracellular receptor, the inducible proapoptotic polypeptide or a transposon or vector comprising the inducible proapoptotic polypeptide may encode a signal peptide and transmembrane domain 5' or 3' of the receptor domain sequence or may have a lipid attachment signal sequence 5' of the receptor domain sequence. Where the receptor domain is between the signal peptide and the transmembrane domain, the receptor domain will be extracellular.

(799) Antibodies and antibody subunits, e.g., heavy or light chain, particularly fragments, more particularly all or part of the variable region, or fusions of heavy and light chain to create high-affinity binding, can be used as a ligand binding region of the disclosure. Antibodies that are contemplated include ones that are an ectopically expressed human product, such as an extracellular domain that would not trigger an immune response and generally not expressed in the periphery (i.e., outside the CNS/brain area). Such examples, include, but are not limited to low affinity nerve growth factor receptor (LNGFR), and embryonic surface proteins (i.e., carcinoembryonic antigen). Yet further, antibodies can be prepared against haptenic molecules, which are physiologically acceptable, and the individual antibody subunits screened for binding affinity. The cDNA encoding the subunits can be isolated and modified by deletion of the constant region, portions of the variable region, mutagenesis of the variable region, or the like, to obtain a binding protein domain that has the appropriate affinity for the ligand. In this way, almost any physiologically acceptable haptenic compound can be employed as the ligand or to provide an epitope for the ligand. Instead of antibody units, endogenous receptors can be employed, where the binding region or domain is known and there is a useful or known ligand for binding.

(800) For multimerizing the receptor, the ligand for the ligand-binding region/receptor domains of the inducible proapoptotic polypeptides may be multimeric in the sense that the ligand can have at least two binding sites, with each of the binding sites capable of binding to a ligand receptor region (i.e. a ligand having a first binding site capable of binding the ligand-binding region of a first inducible proapoptotic polypeptide and a second binding site capable of binding the ligand-binding region of a second inducible proapoptotic polypeptide, wherein the ligand-binding regions of the first and the second inducible proapoptotic polypeptides are either identical or distinct). Thus, as used herein, the term "multimeric ligand binding region" refers to a ligand-binding region of an inducible proapoptotic polypeptide of the disclosure that binds to a multimeric ligand. Multimeric ligands of the disclosure include dimeric ligands. A dimeric ligand of the disclosure may have two binding sites capable of binding to the ligand receptor domain. In certain embodiments, multimeric ligands of the disclosure are a dimer or higher order oligomer, usually not greater than about tetrameric, of small synthetic organic molecules, the individual molecules typically being at least about 150 Da and less than about 5 kDa, usually less than about 3 kDa. A variety of pairs of synthetic ligands and receptors can be employed. For example, in embodiments involving endogenous receptors, dimeric FK506 can be used with an FKBP12 receptor, dimerized cyclosporin A can be used with the cyclophilin receptor, dimerized estrogen with an estrogen receptor, dimerized glucocorticoids with a glucocorticoid receptor, dimerized tetracycline with the tetracycline receptor, dimerized vitamin D with the vitamin D receptor, and the like. Alternatively higher orders of the ligands, e.g., trimeric can be used. For embodiments involving non-naturally occurring receptors, e.g., antibody subunits, modified antibody subunits, single chain antibodies comprised of heavy and light chain variable regions in tandem, separated by a flexible linker, or modified receptors, and mutated sequences thereof, and the like, any of a large variety of compounds can be used. A significant characteristic of the units comprising a multimeric ligand of the disclosure is that each binding site is able to bind the receptor with high affinity, and preferably, that they are able to be dimerized chemically. Also, methods are available to balance the hydrophobicity/hydrophilicity of the ligands so that they are able to dissolve in serum at functional levels, yet diffuse across plasma membranes for most applications.

(801) Activation of proapoptotic polypeptides of the disclosure may be accomplished through, for example, chemically induced dimerization (CID) mediated by an induction agent to produce a conditionally controlled protein or polypeptide. Proapoptotic polypeptides of the disclosure not only inducible, but the induction of these polypeptides is also reversible, due to the degradation of the labile dimerizing agent or administration of a monomeric competitive inhibitor.

(802) In certain embodiments, the ligand-binding region comprises a FKBP12 binding protein 12 (FKBP12) polypeptide. In certain embodiments, the ligand-binding region comprises a FKBP12 polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In certain embodiments, in which the ligand binding region comprises a FKBP12 polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V), the induction agent may comprise AP1903, a synthetic drug (CAS Index Name: 2-Piperidinecarboxylic acid, 1-[(2S)-1-oxo-2-(3,4,5-trimethoxyphenyl)butyl]-, 1,2-ethanediylbis[imino(2-oxo-2,1-ethanediyl)oxy-3,1-phenylene[(1R)-3-(3,4-dimethoxyphenyl)propylidene]]ester, [2S-[1(R\*),2R\*[S\*[S\*[1(R\*),2R\*]]]]-(9C1) CAS Registry Number: 195514-63-7; Molecular Formula: C<sub>78</sub>H<sub>98</sub>N<sub>4</sub>O<sub>20</sub>; Molecular Weight: 1411.65)). In certain embodiments, in which the ligand-binding region comprises a FKBP12 polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V), the induction agent may comprise AP20187 (CAS Registry Number: 195514-80-8 and Molecular Formula: C<sub>82</sub>H<sub>107</sub>N<sub>5</sub>O<sub>20</sub>). In certain embodiments, the induction agent is an AP20187 analog, such as, for example, AP1510. As used herein, the induction agents AP20187, AP1903 and AP1510 may be used interchangeably.

(803) AP1903 API is manufactured by Alphora Research Inc. and AP1903 Drug Product for Injection is made by Formatech Inc. It is formulated as a 5 mg/mL solution of AP1903 in a 25% solution of the non-ionic solubilizer Solutol HS 15 (250 mg/mL, BASF). At room temperature, this formulation is a clear, slightly yellow solution. Upon refrigeration, this formulation undergoes a reversible phase transition, resulting in a milky solution. This phase transition is reversed upon re-warming to room temperature. The fill is 2.33 mL in a 3 mL glass vial (approximately 10 mg AP1903 for Injection total per vial). Upon determining a need to administer AP1903, patients may be, for example, administered a single fixed dose of AP1903 for Injection (0.4 mg/kg) via IV infusion over 2 hours, using a non-DEHP, non-ethylene oxide sterilized infusion set. The dose of AP1903 is calculated individually for all patients, and is not be recalculated unless body weight fluctuates by  $\geq 10\%$ . The calculated dose is diluted in 100 mL in 0.9% normal saline before infusion. In a previous Phase I study of AP1903, 24 healthy volunteers were treated with single doses of AP1903 for Injection at dose levels of 0.01, 0.05, 0.1, 0.5 and 1.0 mg/kg infused IV over 2 hours. AP1903 plasma levels were directly proportional to dose, with mean C<sub>max</sub> values ranging from approximately 10-1275 ng/mL over the 0.01-1.0 mg/kg dose range. Following the initial infusion period, blood concentrations demonstrated a rapid distribution phase, with plasma levels reduced to approximately 18, 7, and 1% of maximal concentration at 0.5, 2 and 10 hours post-dose, respectively. AP1903 for Injection was shown to be safe and well tolerated at all dose levels and demonstrated a favorable pharmacokinetic profile. Iuliucci J D, et al., J Clin Pharmacol. 41: 870-9, 2001.

(804) The fixed dose of AP1903 for injection used, for example, may be 0.4 mg/kg intravenously infused over 2 hours. The amount of AP1903 needed in vitro for effective signaling of cells is 10-100 nM (1600 Da MW). This equates to 16-160 pg/L or 0.016-1.6 pg/kg (1.6-160 pg/kg). Doses up to 1 mg/kg were well tolerated in the Phase I study of AP1903 described above. Therefore, 0.4 mg/kg may be a safe and effective dose of AP1903 for this Phase I study in combination with the therapeutic cells.

(805) The amino acid and/or nucleic acid sequence encoding ligand binding of the disclosure may contain sequence one or more modifications compared to a wild type amino acid or nucleic acid sequence. For example, the amino acid and/or nucleic acid sequence encoding ligand-binding region of the disclosure may be a codon-optimized sequence. The one or more modifications may increase the binding affinity of a ligand (e.g., an induction agent) for the ligand-binding region of the disclosure compared to a wild type polypeptide. Alternatively, or in addition, the one or more modifications may decrease the immunogenicity of the ligand-binding region of the disclosure compared to a wild type polypeptide. Ligand binding regions of the disclosure and/or induction agents of the disclosure may be

non-naturally occurring.

(806) Inducible proapoptotic polypeptides of the disclosure comprise a ligand-binding region, a linker and a proapoptotic peptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. The linker may comprise any organic or inorganic material that permits, upon dimerization of the ligand-binding region, interaction, cross-linking, cross-activation, or activation of the proapoptotic polypeptides such that the interaction or activation of the proapoptotic polypeptides initiates apoptosis in the cell. In certain embodiments, the linker is a polypeptide. In certain embodiments, the linker is a polypeptide comprising a G/S rich amino acid sequence (a “GS” linker). In certain embodiments, the linker is a polypeptide comprising the amino acid sequence GGGGS (SEQ ID NO: 18024). In preferred embodiments, the linker is a polypeptide and the nucleic acid encoding the polypeptide does not contain a restriction site for a restriction endonuclease. Linkers of the disclosure may be non-naturally occurring.

(807) Inducible proapoptotic polypeptides of the disclosure may be expressed in a cell under the transcriptional regulation of any promoter capable of initiating and/or regulating the expression of an inducible proapoptotic polypeptide of the disclosure in that cell. The term “promoter” as used herein refers to a promoter that acts as the initial binding site for RNA polymerase to transcribe a gene. For example, inducible proapoptotic polypeptides of the disclosure may be expressed in a mammalian cell under the transcriptional regulation of any promoter capable of initiating and/or regulating the expression of an inducible proapoptotic polypeptide of the disclosure in a mammalian cell, including, but not limited to native, endogenous, exogenous, and heterologous promoters. Preferred mammalian cells include human cells. Thus, inducible proapoptotic polypeptides of the disclosure may be expressed in a human cell under the transcriptional regulation of any promoter capable of initiating and/or regulating the expression of an inducible proapoptotic polypeptide of the disclosure in a human cell, including, but not limited to, a human promoter or a viral promoter. Exemplary promoters for expression in human cells include, but are not limited to, a human cytomegalovirus (CMV) immediate early gene promoter, a SV40 early promoter, a Rous sarcoma virus long terminal repeat, j-actin promoter, a rat insulin promoter and a glyceraldehyde-3-phosphate dehydrogenase promoter, each of which may be used to obtain high-level expression of an inducible proapoptotic polypeptide of the disclosure. The use of other viral or mammalian cellular or bacterial phage promoters that are well known in the art to achieve expression of an inducible proapoptotic polypeptide of the disclosure is contemplated as well, provided that the levels of expression are sufficient for initiating apoptosis in a cell. By employing a promoter with well-known properties, the level and pattern of expression of the protein of interest following transfection or transformation can be optimized.

(808) Selection of a promoter that is regulated in response to specific physiologic or synthetic signals can permit inducible expression of the inducible proapoptotic polypeptide of the disclosure. The ecdysone system (Invitrogen, Carlsbad, Calif.) is one such system. This system is designed to allow regulated expression of a gene of interest in mammalian cells. It consists of a tightly regulated expression mechanism that allows virtually no basal level expression of a transgene, but over 200-fold inducibility. The system is based on the heterodimeric ecdysone receptor of *Drosophila*, and when ecdysone or an analog such as muristerone A binds to the receptor, the receptor activates a promoter to turn on expression of the downstream transgene high levels of mRNA transcripts are attained. In this system, both monomers of the heterodimeric receptor are constitutively expressed from one vector, whereas the ecdysone-responsive promoter, which drives expression of the gene of interest, is on another plasmid. Engineering of this type of system into a vector of interest may therefore be useful. Another inducible system that may be useful is the Tet-Off™ or Tet-On™ system (Clontech, Palo Alto, Calif.) originally developed by Gossen and Bujard (Gossen and Bujard, Proc. Natl. Acad. Sci. USA, 89:5547-5551, 1992; Gossen et al., Science, 268:1766-1769, 1995). This system also allows high levels of gene expression to be regulated in response to tetracycline or tetracycline derivatives such as doxycycline. In the Tet-On™ system, gene expression is turned on in the presence of doxycycline, whereas in the Tet-Off™ system, gene expression is turned on in the absence of doxycycline. These systems are based on two regulatory elements derived from the tetracycline resistance operon of *E. coli*:

the tetracycline operator sequence (to which the tetracycline repressor binds) and the tetracycline repressor protein. The gene of interest is cloned into a plasmid behind a promoter that has tetracycline-responsive elements present in it. A second plasmid contains a regulatory element called the tetracycline-controlled transactivator, which is composed, in the Tet-Off™ system, of the VP16 domain from the herpes simplex virus and the wild-type tetracycline repressor. Thus in the absence of doxycycline, transcription is constitutively on. In the Tet-On™ system, the tetracycline repressor is not wild type and in the presence of doxycycline activates transcription. For gene therapy vector production, the Tet-Off™ system may be used so that the producer cells could be grown in the presence of tetracycline or doxycycline and prevent expression of a potentially toxic transgene, but when the vector is introduced to the patient, the gene expression would be constitutively on.

(809) In some circumstances, it is desirable to regulate expression of a transgene in a gene therapy vector. For example, different viral promoters with varying strengths of activity are utilized depending on the level of expression desired. In mammalian cells, the CMV immediate early promoter is often used to provide strong transcriptional activation. The CMV promoter is reviewed in Donnelly, J. J., et al., 1997. *Annu. Rev. Immunol.* 15:617-48. Modified versions of the CMV promoter that are less potent have also been used when reduced levels of expression of the transgene are desired. When expression of a transgene in hematopoietic cells is desired, retroviral promoters such as the LTRs from MLV or MMTV are often used. Other viral promoters that are used depending on the desired effect include SV40, RSV LTR, HIV-1 and HIV-2 LTR, adenovirus promoters such as from the E1A, E2A, or MLP region, AAV LTR, HSV-TK, and avian sarcoma virus.

(810) In other examples, promoters may be selected that are developmentally regulated and are active in particular differentiated cells. Thus, for example, a promoter may not be active in a pluripotent stem cell, but, for example, where the pluripotent stem cell differentiates into a more mature cell, the promoter may then be activated.

(811) Similarly, tissue specific promoters are used to effect transcription in specific tissues or cells so as to reduce potential toxicity or undesirable effects to non-targeted tissues. These promoters may result in reduced expression compared to a stronger promoter such as the CMV promoter, but may also result in more limited expression, and immunogenicity (Bojak, A., et al., 2002. *Vaccine.* 20:1975-79; Cazeaux, N., et al., 2002. *Vaccine* 20:3322-31). For example, tissue specific promoters such as the PSA associated promoter or prostate-specific glandular kallikrein, or the muscle creatine kinase gene may be used where appropriate.

(812) Examples of tissue specific or differentiation specific promoters include, but are not limited to, the following: B29 (B cells); CD14 (monocytic cells); CD43 (leukocytes and platelets); CD45 (hematopoietic cells); CD68 (macrophages); desmin (muscle); elastase-1 (pancreatic acinar cells); endoglin (endothelial cells); fibronectin (differentiating cells, healing tissues); and Flt-1 (endothelial cells); GFAP (astrocytes).

(813) In certain indications, it is desirable to activate transcription at specific times after administration of the gene therapy vector. This is done with such promoters as those that are hormone or cytokine regulatable. Cytokine and inflammatory protein responsive promoters that can be used include K and T kininogen (Kageyama et al., (1987) *J. Biol. Chem.*, 262, 2345-2351), c-fos, TNF-alpha, C-reactive protein (Arcone, et al., (1988) *Nucl. Acids Res.*, 16(8), 3195-3207), haptoglobin (Oliviero et al., (1987) *EMBO J.*, 6, 1905-1912), serum amyloid A2, C/EBP alpha, IL-1, IL-6 (Poli and Cortese, (1989) *Proc. Nat'l Acad. Sci. USA*, 86, 8202-8206), Complement C3 (Wilson et al., (1990) *Mol. Cell. Biol.*, 6:6181-6191), IL-8, alpha-1 acid glycoprotein (Prowse and Baumann, (1988) *Mol Cell Biol*, 8, 42-51), alpha-1 antitrypsin, lipoprotein lipase (Zechner et al., *Mol. Cell. Biol.*, 2394-2401, 1988), angiotensinogen (Ron, et al., (1991) *Mol. Cell. Biol.*, 2887-2895), fibrinogen, c-jun (inducible by phorbol esters, TNF-alpha, UV radiation, retinoic acid, and hydrogen peroxide), collagenase (induced by phorbol esters and retinoic acid), metallothionein (heavy metal and glucocorticoid inducible), Stromelysin (inducible by phorbol ester, interleukin-1 and EGF), alpha-2 macroglobulin and alpha-1 anti-chymotrypsin. Other promoters include, for example, SV40, MMTV, Human Immunodeficiency Virus (MV), Moloney virus, ALV, Epstein Barr virus, Rous Sarcoma virus, human actin, myosin, hemoglobin, and creatine.

(814) It is envisioned that any of the above promoters alone or in combination with another can be

useful depending on the action desired. Promoters, and other regulatory elements, are selected such that they are functional in the desired cells or tissue. In addition, this list of promoters should not be construed to be exhaustive or limiting; other promoters that are used in conjunction with the promoters and methods disclosed herein.

#### (815) Armored T-Cells “Knock Down” Strategy

(816) T-cells of the disclosure may be genetically modified to enhance their therapeutic potential. Alternatively, or in addition, T-cells of the disclosure may be modified to render them less sensitive to immunologic and/or metabolic checkpoints. Modifications of this type “armor” the T cells of the disclosure, which, following the modification, may be referred to here as “armored” T cells. Armored T cells of the disclosure may be produced by, for example, blocking and/or diluting specific checkpoint signals delivered to the T-cells (i.e. checkpoint inhibition) naturally, within the tumor immunosuppressive microenvironment, for example.

(817) In some embodiments, an armored T-cell of the disclosure is derived from a T cell, a NK cell, a hematopoietic progenitor cell, a peripheral blood (PB) derived T cell (including a T cell isolated or derived from G-CSF-mobilized peripheral blood), or an umbilical cord blood (UCB) derived T cell. In some embodiments, an armored T-cell of the disclosure comprises one or more of a chimeric ligand receptor (CLR comprising a single domain antibody,)/chimeric antigen receptor (CAR comprising a protein scaffold, an antibody, an ScFv, or an antibody mimetic), a CARTyrin (a CAR comprising a Centyrin), and/or a VCAR (a CAR comprising a camelid VHH or a single domain VH) of the disclosure. In some embodiments, an armored T-cell of the disclosure comprises an inducible proapoptotic polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In some embodiments, the non-human sequence is a restriction site. In some embodiments, the ligand binding region inducible caspase polypeptide comprises a FK506 binding protein 12 (FKBP12) polypeptide. In some embodiments, the amino acid sequence of the FK506 binding protein 12 (FKBP12) polypeptide comprises a modification at position 36 of the sequence. In some embodiments, the modification is a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In some embodiments, an armored T-cell of the disclosure comprises an exogenous sequence. In some embodiments, the exogenous sequence comprises a sequence encoding a therapeutic protein. Exemplary therapeutic proteins may be nuclear, cytoplasmic, intracellular, transmembrane, cell-surface bound, or secreted proteins. Exemplary therapeutic proteins expressed by the armored T cell may modify an activity of the armored T cell or may modify an activity of a second cell. In some embodiments, an armored T-cell of the disclosure comprises a selection gene or a selection marker. In some embodiments, an armored T-cell of the disclosure comprises a synthetic gene expression cassette (also referred to herein as an inducible transgene construct).

(818) In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression one or more gene(s) encoding receptor(s) of inhibitory checkpoint signals to produce an armored T-cell of the disclosure. Examples of inhibitory checkpoint signals include, but are not limited to, a PD-L1 ligand binding to a PD-1 receptor on a CAR-T cell of the disclosure or a TGFβ cytokine binding to a TGFβRII receptor on a CAR-T cell. Receptors of inhibitory checkpoint signals are expressed on the cell surface or within the cytoplasm of a T-cell. Silencing or reducing expressing of the gene encoding the receptor of the inhibitory checkpoint signal results a loss of protein expression of the inhibitory checkpoint receptors on the surface or within the cytoplasm of an armored T-cell of the disclosure. Thus, armored T cells of the disclosure having silenced or reduced expression of one or more genes encoding an inhibitory checkpoint receptor is resistant, non-receptive or insensitive to checkpoint signals. The armored T cell's resistance or decreased sensitivity to inhibitory checkpoint signals enhances the armored T cell's therapeutic potential in the presence of these inhibitory checkpoint signals. Inhibitory checkpoint signals include but are not limited to the examples listed in Table 1. Exemplary inhibitory checkpoint signals that may be silenced in an armored T cell of the disclosure include, but are not limited to, PD-1 and TGFβRII.

(819) TABLE-US-00222 TABLE 1 Exemplary Inhibitory Checkpoint Signals (and proteins that induce immunosuppression). A CSR of the disclosure may comprise an endodomain of any one of the proteins

of this table. Full Name Abbreviation SEQ ID NO: Programmed cell death protein 1 PD1 14643-14644 transforming growth factor  $\beta$  Receptor 1 TGF $\beta$ R1 14645 transforming growth factor  $\beta$  Receptor 2 TGF $\beta$ R2 14646 T-cell immunoglobulin and mucin-domain TIM3 14647 containing-3 Lymphocyte-activation gene 3 LAG3 14648 Cytotoxic T-lymphocyte protein 4 CTLA4 14649 B- and T-lymphocyte attenuator BTLA 14650 Killer cell immunoglobulin-like receptor KIR 14651 Alpha-2A adrenergic receptor A2aR 14652 V-type immunoglobulin domain-containing VISTA 14653 suppressor of T-cell activation T-cell immunoreceptor with Ig and ITIM TIGIT 14654 domains Programmed cell death 1 ligand 1 B7H1 or PD-L1 14655 Programmed cell death 1 ligand 2 B7DC or PD-L2 14656 T-lymphocyte activation antigen CD80 B7-1 or CD80 14657 T-lymphocyte activation antigen CD86 B7-2 or CD86 14658 CD160 antigen CD160 14659 Leukocyte-associated immunoglobulin-like LAIR1 14660 receptor 1 T-cell immunoglobulin and mucin domain- TIM4 or TIMD4 14661 containing protein 4 Natural killer cell receptor 2B4 2B4 or CD244 14662 Major Histocompatibility Complex type I MHC I 14663 Major Histocompatibility Complex type II MHC II Putative 2-methylcitrate dehydratase receptor PDH1R T-cell immunoglobulin and mucin domain 1 TIM1R receptor T-cell immunoglobulin and mucin domain 4 TIM4R receptor B7-H3 receptor B7H3R or CD176 Receptor B7-H4 receptor B7H4R Immunoglobulin-like transcript (ILT) 3 receptor ILT3R phosphoinositide 3-kinase, subunit alpha PI3K alpha 14664 phosphoinositide 3-kinase, subunit gamma PI3K gamma 14665 Tyrosine-protein phosphatase non-receptor type SHP2 or PTPN11 14666 11 Protein phosphatase 2, subunit gamma PP2A gamma 14667 Protein phosphatase 2, subunit beta PP2A beta 14668 Protein phosphatase 2, subunit delta PP2A delta 14669 Protein phosphatase 2, subunit epsilon PP2A epsilon 14670 Protein phosphatase 2, subunit alpha PP2A alpha 14671 T-cell Receptor, subunit alpha TCR alpha 14672 T-cell Receptor, subunit beta TCR beta 14673 T-cell Receptor, subunit zeta TCR zeta 14674 T-cell Receptor, subunit CD3 epsilon TCR CD3 epsilon 14675 T-cell Receptor, subunit CD3 gamma TCR CD3 gamma 14676 T-cell Receptor, subunit CD3 delta TCR CD3 delta 14677 Cluster of Differentiation 28 CD28 14678 Galectins Galectins Galectin 9 Galectin 9 14679 High Mobility Group Box 1 HMGB1 14680 Arginase 1 ARG1 14681 Prostaglandin-Endoperoxide Synthase 1 PTGS1 14682 Prostaglandin-Endoperoxide Synthase 2 PTGS2 14683 Mucin 1, Cell Surface Associated MUC1 14684 Mucin 2, Oligomeric Mucus/Gel-Forming MUC2 14685 Mucin 3A, Cell Surface Associated MUC3A 14686 Mucin 3B, Cell Surface Associated MUC3B 14687 Mucin 4, Cell Surface Associated MUC4 14688 Mucin 5AC, Oligomeric Mucus/Gel-Forming MUC5AC 14689 Mucin 5B, Oligomeric Mucus/Gel-Forming MUC5B 14690 Mucin 6, Oligomeric Mucus/Gel-Forming MUC6 14691 Mucin 7, Secreted MUC7 14692 Mucin 8 MUC8 Mucin 12, Cell Surface Associated MUC12 14693 Mucin 13, Cell Surface Associated MUC13 14694 Mucin 15, Cell Surface Associated MUC15 14695 Mucin 16, Cell Surface Associated MUC16 14696 Mucin 17, Cell Surface Associated MUC17 14697 Mucin 19, Oligomeric MUC19 14698 Mucin 20, Cell Surface Associated MUC20 14699 Mucin 21, Cell Surface Associated MUC21 14700 Mucin 22 MUC22 14701 Indoleamine 2,3-Dioxygenase 1 IDO1 14702 Indoleamine 2,3-Dioxygenase 2 IDO2 14703 Inducible T Cell Costimulator Ligand ICOSLG 14704 ROS Proto-Oncogene 1, Receptor Tyrosine ROS1 14705 Kinase Tumor Necrosis Factor Receptor Superfamily 4-1BB, CD137, ILA or 14706 Member 9 TNFRSF9 4-1BB Ligand 4-1BB-L 14707 Glucocorticoid-induced TNFR family related GITR 14708 gene Glucocorticoid-induced TNFR family related GITRL 14709 gene ligand

(820) In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding intracellular proteins involved in checkpoint signaling to produce an armored T-cell of the disclosure. The activity of a T-cell of the disclosure may be enhanced by targeting any intracellular signaling protein involved in a checkpoint-signaling pathway, thereby achieving checkpoint inhibition or interference to one or more checkpoint pathways. Intracellular signaling proteins involved in checkpoint signaling include, but are not limited to, exemplary intracellular signaling proteins listed in Table 2.

(821) TABLE-US-00223 TABLE 2 Exemplary Intracellular Signaling Proteins. Full Name Abbreviation SEQ ID NO: phosphoinositide 3-kinase, subunit alpha PI3K alpha 14710 phosphoinositide 3-kinase, subunit gamma PI3K gamma 14711 Tyrosine-protein phosphatase non-receptor type SHP2 or PTPN11 14712 11 Protein phosphatase 2, subunit gamma PP2A gamma 14713 Protein phosphatase 2, subunit

beta PP2A beta 14714 Protein phosphatase 2, subunit delta PP2A delta 14715 Protein phosphatase 2, subunit epsilon PP2A epsilon 14716 Protein phosphatase 2, subunit alpha PP2A alpha 14717 RAC-alpha serine/threonine-protein kinase AKT or PKB 14718 Tyrosine-protein kinase ZAP-70 ZAP70 14719 Amino acid sequence (KIEELE)-containing KIEELE-domain domain protein containing proteins BCL2 associated athanogene 6 Bat3, Bag6 or Scythe 14720 B-cell lymphoma-extra large Bcl-xL 14721 Bcl-2-related protein A1 Bfl-1 or BCL2A1 14722

(822) In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding a transcription factor that hinders the efficacy of a therapy to produce an armored T-cell of the disclosure. The activity of armored T-cells may be enhanced or modulated by silencing or reducing expression (or repressing a function) of a transcription factor that hinders the efficacy of a therapy. Exemplary transcription factors that may be modified to silence or reduce expression or to repress a function thereof include, but are not limited to, the exemplary transcription factors listed in Table 3. For example, expression of a FOXP3 gene may be silenced or reduced in an armored T cell of the disclosure to prevent or reduce the formation of T regulatory CAR-T-cells (CAR-Treg cells), the expression or activity of which may reduce efficacy of a therapy.

(823) TABLE-US-00224 TABLE 3 Exemplary Transcription Factors. Full Name Abbreviation SEQ ID NO: activity-dependent neuroprotector homeobox ADNP 14723 ADNP homeobox 2 ADNP2 14724 AE binding protein 1 AEBP1 14725 AE binding protein 2 AEBP2 14726 AF4/FMR2 family member 1 AFF1 14727 AF4/FMR2 family member 2 AFF2 14728 AF4/FMR2 family member 3 AFF3 14729 AF4/FMR2 family member 4 AFF4 14730 AT-hook containing transcription factor 1 AHCTF1 14731 aryl hydrocarbon receptor AHR 14732 aryl-hydrocarbon receptor repressor AHRR 14733 autoimmune regulator AIRE 14734 AT-hook transcription factor AKNA 14735 ALX homeobox 1 ALX1 14736 ALX homeobox 3 ALX3 14737 ALX homeobox 4 ALX4 14738 ankyrin repeat and zinc finger domain containing 1 ANKZF1 14739 adaptor related protein complex 5 zeta 1 subunit AP5Z1 14740 androgen receptor AR 14741 arginine-fifty homeobox ARGFX 14742 Rho GTPase activating protein 35 ARHGAP35 14743 AT-rich interaction domain 1A ARID1A 14744 AT-rich interaction domain 1B ARID1B 14745 AT-rich interaction domain 2 ARID2 14746 AT-rich interaction domain 3A ARID3A 14747 AT-rich interaction domain 3B ARID3B 14748 AT-rich interaction domain 3C ARID3C 14749 AT-rich interaction domain 4A ARID4A 14750 AT-rich interaction domain 4B ARID4B 14751 AT-rich interaction domain 5A ARID5A 14752 AT-rich interaction domain 5B ARID5B 14753 aryl hydrocarbon receptor nuclear translocator ARNT 14754 aryl hydrocarbon receptor nuclear translocator 2 ARNT2 14755 aryl hydrocarbon receptor nuclear translocator like ARNTL 14756 aryl hydrocarbon receptor nuclear translocator like 2 ARNTL2 14757 aristaless related homeobox ARX 14758 achaete-scute family bHLH transcription factor 1 ASCL1 14759 achaete-scute family bHLH transcription factor 2 ASCL2 14760 achaete-scute family bHLH transcription factor 3 ASCL3 14761 achaete-scute family bHLH transcription factor 4 ASCL4 14762 achaete-scute family bHLH transcription factor 5 ASCL5 14763 ash1 (absent, small, or homeotic)-like (*Drosophila*) ASH1L 14764 ash2 (absent, small, or homeotic)-like (*Drosophila*) ASH2L 14765 activating transcription factor 1 ATF1 14766 activating transcription factor 2 ATF2 14767 activating transcription factor 3 ATF3 14768 activating transcription factor 4 ATF4 14769 activating transcription factor 5 ATF5 14770 activating transcription factor 6 ATF6 14771 activating transcription factor 6 beta ATF6B 14772 activating transcription factor 7 ATF7 14773 atonal bHLH transcription factor 1 ATOH1 14774 atonal bHLH transcription factor 7 ATOH7 14775 atonal bHLH transcription factor 8 ATOH8 14776 alpha thalassemia/mental retardation syndrome X-ATRX 14777 linked ataxin 7 ATXN7 14778 BTB and CNC homology 1, basic leucine zipper BACH1 14779-14780 transcription factor1 BTB domain and CNC homolog 2 BACH2 14781 BarH like homeobox 1 BARHL1 14782 BarH like homeobox 2 BARHL2 14783 BARX homeobox 1 BARX1 14784 BARX homeobox 2 BARX2 14785 Basic Leucine Zipper ATF-Like Transcription Factor, Batf 14786 basic leucine zipper transcription factor, ATF-like BATF 14786 basic leucine zipper transcription factor, ATF-like 2 BATF2 14787 basic leucine zipper transcription factor, ATF-like 3 BATF3 14788 bobby sox homolog (*Drosophila*) BBX 14789 B-cell CLL/lymphoma 11A BCL11A 14790 B-cell CLL/lymphoma 11B BCL11B 14791 B-cell CLL/lymphoma 3 BCL3 14792 B-cell CLL/lymphoma 6 BCL6 14793 B-cell CLL/lymphoma 6, member B BCL6B 14794 BCL2 associated transcription factor



1 BCLAF1 14795 basic helix-loop-helix family member a15 BHLHA15 14796 basic helix-loop-helix family member a9 BHLHA9 14797 basic helix-loop-helix domain containing, class B, 9 BHLHB9 14798 basic helix-loop-helix family member e22 BHLHE22 14799 basic helix-loop-helix family member e23 BHLHE23 14800 basic helix-loop-helix family member e40 BHLHE40 14801 basic helix-loop-helix family member e41 BHLHE41 14802 Beta-Interferon Gene Positive-Regulatory Domain I Blimp-1 14803 Binding Factor bone morphogenetic protein 2 BMP2 14804 basonuclein 1 BNC1 14805 basonuclein 2 BNC2 14806 bola family member 1 BOLA1 14807 bola family member 2 BOLA2 14808 bola family member 3 BOLA3 14809 bromodomain PHD finger transcription factor BPTF 14810 breast cancer 1 BRCA1 14811 brain specific homeobox BSX 14812 chromosome 20 open reading frame 194 C20orf194 14813 calmodulin binding transcription activator 1 CAMTA1 14814 calmodulin binding transcription activator 2 CAMTA2 14815 calcium regulated heat stable protein 1 CARHSP1 14816 castor zinc finger 1 CASZ1 14817 core-binding factor, beta subunit CBFB 14818 coiled-coil domain containing 79 CCDC79 14819 cell division cycle 5 like CDC5L 14820 caudal type homeobox 1 CDX1 14821 caudal type homeobox 2 CDX2 14822 caudal type homeobox 4 CDX4 14823 CCAAT/enhancer binding protein alpha CEBPA 14824 CCAAT/enhancer binding protein beta CEBPB 14825 CCAAT/enhancer binding protein delta CEBPD 14826 CCAAT/enhancer binding protein epsilon CEBPE 14827 CCAAT/enhancer binding protein gamma CEBPG 14828 CCAAT/enhancer binding protein zeta CEBPZ 14829 centromere protein T CENPT 14830 ceramide synthase 3 CERS3 14831 ceramide synthase 6 CERS6 14832 chromosome alignment maintaining phosphoprotein 1 CHAMP1 14833 capicua transcriptional repressor CIC 14834 CDKN1A interacting zinc finger protein 1 CIZ1 14835 clock circadian regulator CLOCK 14836 CCR4-NOT transcription complex subunit 4 CNOT4 14837 CPX chromosome region, candidate 1 CPXCR1 14838 cramped chromatin regulator homolog 1 CRAMP1 14839 cAMP responsive element binding protein 1 CREB1 14840 cAMP responsive element binding protein 3 CREB3 14841 cAMP responsive element binding protein 3-like 1 CREB3L1 14842 cAMP responsive element binding protein 3-like 2 CREB3L2 14843 cAMP responsive element binding protein 3-like 3 CREB3L3 14844 cAMP responsive element binding protein 3-like 4 CREB3L4 14845 cAMP responsive element binding protein 5 CREB5 14846 CREB binding protein CREBBP 14847 cAMP responsive element binding protein-like 2 CREBL2 14848 CREB3 regulatory factor CREBRF 14849 CREB/ATF bZIP transcription factor CREBZF 14850 cAMP responsive element modulator CREM 14851 cone-rod homeobox CRX 14852 cysteine-serine-rich nuclear protein 1 CSRNP1 14853 cysteine-serine-rich nuclear protein 2 CSRNP2 14854 cysteine-serine-rich nuclear protein 3 CSRNP3 14855 CCCTC-binding factor (zinc finger protein) CTCF 14856 CCCTC-binding factor like CTCFL 14857 cut-like homeobox 1 CUX1 14858-14859 cut-like homeobox 2 CUX2 14860 CXXC finger protein 1 CXXC1 14861 dachshund family transcription factor 1 DACH1 14862 dachshund family transcription factor 2 DACH2 14863 D site of albumin promoter (albumin D-box) binding DBP 14864 protein developing brain homeobox 1 DBX1 14865 developing brain homeobox 2 DBX2 14866 damage specific DNA binding protein 2 DDB2 14867 DNA damage inducible transcript 3 DDIT3 14868 DEAF1, transcription factor DEAF1 14869 distal-less homeobox 1 DLX1 14870 distal-less homeobox 2 DLX2 14871 distal-less homeobox 3 DLX3 14872 distal-less homeobox 4 DLX4 14873 distal-less homeobox 5 DLX5 14874 distal-less homeobox 6 DLX6 14875 DNA methyltransferase 1 associated protein 1 DMAP1 14876 diencephalon/mesencephalon homeobox 1 DMBX1 14877 doublesex and mab-3 related transcription factor 1 DMRT1 14878 doublesex and mab-3 related transcription factor 2 DMRT2 14879 doublesex and mab-3 related transcription factor 3 DMRT3 14880 DMRT like family A1 DMRTA1 14881 DMRT like family A2 DMRTA2 14882 DMRT like family B with proline rich C-terminal 1 DMRTB1 14883 DMRT like family C1 DMRTC1 14884 DMRT like family C1B DMRTC1B 14884 DMRT like family C2 DMRTC2 14885 cyclin D binding myb like transcription factor 1 DMTF1 14886 DnaJ heat shock protein family (Hsp40) member C1 DNAJC1 14887 DnaJ heat shock protein family (Hsp40) member C2 DNAJC2 14888 DnaJ heat shock protein family (Hsp40) member C21 DNAJC21 14889 DNA (cytosine-5-)-methyltransferase 1 DNMT1 14890 DNA (cytosine-5-)-methyltransferase 3 alpha DNMT3A 14891 DNA (cytosine-5-)-methyltransferase 3 beta DNMT3B 14892 DNA (cytosine-5-)-methyltransferase 3-like DNMT3L 14893 double PHD fingers 1 DPF1 14894 double PHD fingers 2 DPF2 14895 double PHD fingers 3 DPF3

14896 divergent-paired-related protein 1 DRP1 14897 down-regulator of transcription 1 DR1 14898 DR1 associated protein 1 DRAP1 14899 dorsal root ganglia homeobox DRGX 14900 double homeobox 4 DUX4 14901 double homeobox 4 like 9 DUX4L9 14902 double homeobox A DUXA 14903 E2F transcription factor 1 E2F1 14904 E2F transcription factor 2 E2F2 14905 E2F transcription factor 3 E2F3 14906 E2F transcription factor 4 E2F4 14907 E2F transcription factor 5 E2F5 14908 E2F transcription factor 6 E2F6 14909 E2F transcription factor 7 E2F7 14910 E2F transcription factor 8 E2F8 14911 E4F transcription factor 1 E4F1 14912 early B-cell factor 1 EBF1 14913 early B-cell factor 2 EBF2 14914 early B-cell factor 3 EBF3 14915 early B-cell factor 4 EBF4 14916 early growth response 1 EGR1 14917 early growth response 2 EGR2 14918 early growth response 3 EGR3 14919 early growth response 4 EGR4 14920 ets homologous factor EHF 14921 E74-like factor 1 (ets domain transcription factor) ELF1 14922 E74-like factor 2 (ets domain transcription factor) ELF2 14923 E74-like factor 3 (ets domain transcription factor, ELF3 14924 epithelial-specific) E74-like factor 4 (ets domain transcription factor) ELF4 14925 E74-like factor 5 (ets domain transcription factor) ELF5 14926 ELK1, member of ETS oncogene family ELK1 14927 ELK3, ETS-domain protein (SRF accessory protein 2) ELK3 14928 ELK4, ETS-domain protein (SRF accessory protein 1) ELKA 14929 ELM2 and Myb/SANT-like domain containing 1 ELMSAN1 14930 empty spiracles homeobox 1 EMX1 14931 empty spiracles homeobox 2 EMX2 14932 engrailed homeobox 1 EN1 14933 engrailed homeobox 2 EN2 14934 enolase 1, (alpha) ENO1 14935 eomesodermin EOMES 14936 endothelial PAS domain protein 1 EPAS1 14937 Ets2 repressor factor ERF 14938 v-ets avian erythroblastosis virus E26 oncogene ERG 14939-14940 homolog estrogen receptor 1 ESR1 14941 estrogen receptor 2 (ER beta) ESR2 14942 estrogen related receptor alpha ESRA 14943 estrogen related receptor beta ESRRB 14944 estrogen related receptor gamma ESRRG 14945 ESX homeobox 1 ESX1 14946 v-ets avian erythroblastosis virus E26 oncogene ETS1 14947 homolog 1 v-ets avian erythroblastosis virus E26 oncogene ETS2 14948 homolog 2 ets variant 1 ETV1 14949 ets variant 2 ETV2 14950 ets variant 3 ETV3 14951 ets variant 3-like ETV3L 14952 ets variant 4 ETV4 14953 ets variant 5 ETV5 14954 ets variant 6 ETV6 14955 ets variant 7 ETV7 14956 even-skipped homeobox 1 EVX1 14957 even-skipped homeobox 2 EVX2 14958 enhancer of zeste 1 polycomb repressive complex 2 EZH1 14959 subunit enhancer of zeste 2 polycomb repressive complex 2 EZH2 14960 subunit family with sequence similarity 170 member A FAM170A 14961 Fer3-like bHLH transcription factor FERD3L 14962 FEV (ETS oncogene family) FEV 14963 FEZ family zinc finger 1 FEZF1 14964 FEZ family zinc finger 2 FEZF2 14965 folliculogenesis specific bHLH transcription factor FIGLA 14966 FLT3-interacting zinc finger 1 FIZ1 14967 Fli-1 proto-oncogene, ETS transcription factor FLI1 14968 FBJ murine osteosarcoma viral oncogene homolog FOS 14969 FBJ murine osteosarcoma viral oncogene homolog B FOSB 14970 FOS like antigen 1 FOSL1 14971 FOS like antigen 2 FOSL2 14972 forkhead box A1 FOXA1 14973 forkhead box A2 FOXA2 14974 forkhead box A3 FOXA3 14975 forkhead box B1 FOXB1 14976 forkhead box B2 FOXB2 14977 forkhead box C1 FOXC1 14978 forkhead box C2 FOXC2 14979 forkhead box D1 FOXD1 14980 forkhead box D2 FOXD2 14981 forkhead box D3 FOXD3 14982 forkhead box D4 FOXD4 14983 forkhead box D4-like 1 FOXD4L1 14984 forkhead box D4-like 3 FOXD4L3 14985 forkhead box D4-like 4 FOXD4L4 14986 forkhead box D4-like 5 FOXD4L5 14987 forkhead box D4-like 6 FOXD4L6 14988 forkhead box E1 FOXE1 14989 forkhead box E3 FOXE3 14990 forkhead box F1 FOXF1 14991 forkhead box F2 FOXF2 14992 forkhead box G1 FOXG1 14993 forkhead box H1 FOXH1 14994 forkhead box I1 FOXI1 14995 forkhead box I2 FOXI2 14996 forkhead box I3 FOXI3 14997 forkhead box J1 FOXJ1 14998 forkhead box J2 FOXJ2 14999 forkhead box J3 FOXJ3 15000 forkhead box K1 FOXK1 15001 forkhead box K2 FOXK2 15002 forkhead box L1 FOXL1 15003 forkhead box L2 FOXL2 15004 forkhead box M1 FOXM1 15005 forkhead box N1 FOXN1 15006 forkhead box N2 FOXN2 15007 forkhead box N3 FOXN3 15008 forkhead box N4 FOXN4 15009 forkhead box O1 FOXO1 15010 forkhead box O3 FOXO3 15011 forkhead box O4 FOXO4 15012 forkhead box O6 FOXO6 15013 forkhead box P1 FOXP1 15014 forkhead box P2 FOXP3 15015 forkhead box P3 FOXP4 15016 forkhead box P4 FOXQ1 15017 forkhead box Q1 FOXR1 15018 forkhead box R1 FOXR2 15019 forkhead box R2 FOXS1 15020 forkhead box S1 FOXP3 15021 far upstream element binding protein 1 FUBP1 15022 far upstream element (FUSE) binding protein 3 FUBP3 15023 GA binding protein transcription factor alpha subunit

GABPA 15024 GABA binding protein transcription factor, beta subunit 1 GABPB1 15025 GA binding protein transcription factor, beta subunit 2 GABPB2 15026 GATA binding protein 1 (globin transcription factor 1) GATA1 15027 GATA binding protein 2 GATA2 15028 GATA binding protein 3 GATA3 15029 GATA binding protein 4 GATA4 15030 GATA binding protein 5 GATA5 15031 GATA binding protein 6 GATA6 15032 GATA zinc finger domain containing 1 GATAD1 15033 GATA zinc finger domain containing 2A GATAD2A 15034 GATA zinc finger domain containing 2B GATAD2B 15035 gastrulation brain homeobox 1 GBX1 15036 gastrulation brain homeobox 2 GBX2 15037 GC-rich sequence DNA-binding factor 2 GCFC2 15038 glial cells missing homolog 1 GCM1 15039 glial cells missing homolog 2 GCM2 15040 growth factor independent 1 transcription repressor GFI1 15041 growth factor independent 1B transcription repressor GFI1B 15042 GLI family zinc finger 1 GLI1 15043 GLI family zinc finger 2 GLI2 15044 GLI family zinc finger 3 GLI3 15045 GLI family zinc finger 4 GLI4 15046 GLIS family zinc finger 1 GLIS1 15047 GLIS family zinc finger 2 GLIS2 15048 GLIS family zinc finger 3 GLIS3 15049 glucocorticoid modulatory element binding protein 1 GMEB1 15050 glucocorticoid modulatory element binding protein 2 GMEB2 15051 gon-4-like (*C. elegans*) GON4L 15052 grainyhead like transcription factor 1 GRHL1 15053 grainyhead like transcription factor 2 GRHL2 15054 grainyhead like transcription factor 3 GRHL3 15055 goosecoid homeobox GSC 15056 goosecoid homeobox 2 GSC2 15057 GS homeobox 1 GSX1 15058 GS homeobox 2 GSX2 15059 general transcription factor IIIi GTF2I 15060 general transcription factor IIIA GTF3A 15061 GDNF inducible zinc finger protein 1 GZF1 15062 heart and neural crest derivatives expressed 1 HAND1 15063 heart and neural crest derivatives expressed 2 HAND2 15064 HMG-box transcription factor 1 HBP1 15065-15066 highly divergent homeobox HDX 15067 helt bHLH transcription factor HELT 15068 hes family bHLH transcription factor 1 HES1 15069-15070 hes family bHLH transcription factor 2 HES2 15071 hes family bHLH transcription factor 3 HES3 15072 hes family bHLH transcription factor 4 HES4 15073 hes family bHLH transcription factor 5 HES5 15074 hes family bHLH transcription factor 6 HES6 15075 hes family bHLH transcription factor 7 HES7 15076 HESX homeobox 1 HESX1 15077 hes-related family bHLH transcription factor with HEY1 15078 YRPW motif 1 hes-related family bHLH transcription factor with HEY2 15079 YRPW motif 2 hes-related family bHLH transcription factor with HEYL 15080 YRPW motif-like hematopoietically expressed homeobox HHEX 15081 hypermethylated in cancer 1 HIC1 15082 hypermethylated in cancer 2 HIC2 15083 hypoxia inducible factor 1, alpha subunit (basic helix- HIF1A 15084 loop-helix transcription factor) hypoxia inducible factor 3, alpha subunit HIF3A 15085 histone H4 transcription factor HINFP 15086 human immunodeficiency virus type I enhancer HIVEP1 15087 binding protein 1 human immunodeficiency virus type I enhancer HIVEP2 15088 binding protein 2 human immunodeficiency virus type I enhancer HIVEP3 15089 binding protein 3 HKR1, GLI-Kruppel zinc finger family member HKR1 15090 hepatic leukemia factor HLF 15091 helicase-like transcription factor HLTF 15092 H2.0-like homeobox HLX 15093 homeobox containing 1 HMBOX1 15094 high mobility group 20A HMG20A 15095 high mobility group 20B HMG20B 15096 high mobility group AT-hook 1 HMGA1 15097 high mobility group AT-hook 2 HMGA2 15098 HMG-box containing 3 HMGXB3 15099 HMG-box containing 4 HMGXB4 15100 H6 family homeobox 1 HMX1 15101 H6 family homeobox 2 HMX2 15102 H6 family homeobox 3 HMX3 15103-15104 HNF1 homeobox A HNF1A 15105 HNF1 homeobox B HNF1B 15106 hepatocyte nuclear factor 4 alpha HNF4A 15107 hepatocyte nuclear factor 4 gamma HNF4G 15108 heterogeneous nuclear ribonucleoprotein K HNRNPK 15109 homeobox and leucine zipper encoding HOMEZ 15110 HOP homeobox HOPX 15111 homeobox A1 HOXA1 15112 homeobox A10 HOXA10 15113 homeobox A11 HOXA11 15114 homeobox A13 HOXA13 15115 homeobox A2 HOXA2 15116 homeobox A3 HOXA3 15117 homeobox A4 HOXA4 15118 homeobox A5 HOXA5 15119 homeobox A6 HOXA6 15120 homeobox A7 HOXA7 15121 homeobox A9 HOXA9 15122 homeobox B1 HOXB1 15123 homeobox B13 HOXB13 15124 homeobox B2 HOXB2 15125 homeobox B3 HOXB3 15126 homeobox B4 HOXB4 15127 homeobox B5 HOXB5 15128 homeobox B6 HOXB6 15129 homeobox B7 HOXB7 15130 homeobox B8 HOXB8 15131 homeobox B9 HOXB9 15132 homeobox C10 HOXC10 15133 homeobox C11 HOXC11 15134 homeobox C12 HOXC12 15135 homeobox C13 HOXC13 15136 homeobox C4 HOXC4 15137 homeobox C5 HOXC5 15138 homeobox C6 HOXC6 15139 homeobox C8 HOXC8 15140 homeobox C9 HOXC9 15141 homeobox

D11 HOXD1 15142 homeobox D10 HOXD10 15143 homeobox D11 HOXD11 15144 homeobox D12 HOXD12 15145 homeobox D13 HOXD13 15146 homeobox D3 HOXD3 15147 homeobox D4 HOXD4 15148 homeobox D8 HOXD8 15149 homeobox D9 HOXD9 15150 heat shock transcription factor 1 HSF1 15151 heat shock transcription factor 2 HSF2 15152 heat shock transcription factor 4 HSF4 15153 heat shock transcription factor family member 5 HSF5 15154 heat shock transcription factor family, X-linked 1 HSFY1 15155 heat shock transcription factor, Y-linked 1 HSFY1 15156 heat shock transcription factor, Y-linked 2 HSFY2 15156 inhibitor of DNA binding 1, dominant negative helix- ID1 15157 loop-helix protein inhibitor of DNA binding 2, dominant negative helix- ID2 15158 loop-helix protein inhibitor of DNA binding 3, dominant negative helix- ID3 15159 loop-helix protein inhibitor of DNA binding 4, dominant negative helix- ID4 15160 loop-helix protein interferon, gamma-inducible protein 16 IFI16 15161 IKAROS family zinc finger 1 IKZF1 15162 IKAROS family zinc finger 2 IKZF2 15163 IKAROS family zinc finger 3 IKZF3 15164 IKAROS family zinc finger 4 IKZF4 15165 IKAROS family zinc finger 5 IKZF5 15166 insulinoma associated 1 INSM1 15167 insulinoma-associated 2 INSM2 15168 interferon regulatory factor 1 IRF1 15169 interferon regulatory factor 2 IRF2 15170 interferon regulatory factor 3 IRF3 15171 interferon regulatory factor 4 IRF4 15172 interferon regulatory factor 5 IRF5 15173 interferon regulatory factor 6 IRF6 15174 interferon regulatory factor 7 IRF7 15175 interferon regulatory factor 8 IRF8 15176 interferon regulatory factor 9 IRF9 15177 iroquois homeobox 1 IRX1 15178 iroquois homeobox 2 IRX2 15179 iroquois homeobox 3 IRX3 15180 iroquois homeobox 4 IRX4 15181 iroquois homeobox 5 IRX5 15182 iroquois homeobox 6 IRX6 15183 ISL LIM homeobox 1 ISL1 15184 ISL LIM homeobox 2 ISL2 15185 intestine specific homeobox ISX 15186 jumonji and AT-rich interaction domain containing 2 JARID2 15187 JAZF zinc finger 1 JAZF1 15188 Jun dimerization protein 2 JDP2 15189 jun proto-oncogene JUN 15190 jun B proto-oncogene JUNB 15191 jun D proto-oncogene JUND 15192 K(lysine) acetyltransferase 5 KAT5 15193 lysine acetyltransferase 6A KAT6A 15194 lysine acetyltransferase 6B KAT6B 15195 lysine acetyltransferase 7 KAT7 15196 lysine acetyltransferase 8 KAT8 15197 potassium channel modulatory factor 1 KCMF1 15198 potassium voltage-gated channel interacting protein 3 KCNIP3 15199 lysine demethylase 2A KDM2A 15200 lysine demethylase 5A KDM5A 15201 lysine demethylase 5B KDM5B 15202 lysine demethylase 5C KDM5C 15203 lysine demethylase 5D KDM5D 15204 KH-type splicing regulatory protein KHSRP 15205 KIAA1549 KIAA1549 15206 Kruppel-like factor 1 (erythroid) KLF1 15207 Kruppel-like factor 10 KLF10 15208 Kruppel-like factor 11 KLF11 15209 Kruppel-like factor 12 KLF12 15210 Kruppel-like factor 13 KLF13 15211 Kruppel-like factor 14 KLF14 15212 Kruppel-like factor 15 KLF15 15213 Kruppel-like factor 16 KLF16 15214 Kruppel-like factor 17 KLF17 15215 Kruppel-like factor 2 KLF2 15216 Kruppel-like factor 3 (basic) KLF3 15217 Kruppel-like factor 4 (gut) KLF4 15218 Kruppel-like factor 5 (intestinal) KLF5 15219 Kruppel-like factor 6 KLF6 15220 Kruppel-like factor 7 (ubiquitous) KLF7 15221 Kruppel-like factor 8 KLF8 15222 Kruppel-like factor 9 KLF9 15223 lysine methyltransferase 2A KMT2A 15224 lysine methyltransferase 2B KMT2B 15225 lysine methyltransferase 2C KMT2C 15226 lysine methyltransferase 2E KMT2E 15227 l(3)mbt-like 1 (*Drosophila*) L3MBTL1 15228 l(3)mbt-like 2 (*Drosophila*) L3MBTL2 15229 l(3)mbt-like 3 (*Drosophila*) L3MBTL3 15230 l(3)mbt-like 4 (*Drosophila*) L3MBTL4 15231 ladybird homeobox 1 LBX1 15232 ladybird homeobox 2 LBX2 15233 ligand dependent nuclear receptor corepressor LCOR 15234 ligand dependent nuclear receptor corepressor like LCORL 15235 lymphoid enhancer binding factor 1 LEF1 15236 leucine twenty homeobox LEUTX 15237 LIM homeobox 1 LHX1 15238 LIM homeobox 2 LHX2 15239 LIM homeobox 3 LHX3 15240 LIM homeobox 4 LHX4 15241 LIM homeobox 5 LHX5 15242 LIM homeobox 6 LHX6 15243 LIM homeobox 8 LHX8 15244 LIM homeobox 9 LHX9 15245 LIM homeobox transcription factor 1, alpha LMX1A 15246 LIM homeobox transcription factor 1, beta LMX1B 15247 LOC730110 LOC730110 leucine rich repeat (in FLII) interacting protein 1 LRRFIP1 15248 leucine rich repeat (in FLII) interacting protein 2 LRRFIP2 15249 Ly1 antibody reactive LYAR 15250 lymphoblastic leukemia associated hematopoiesis LYL1 15251 regulator 1 maelstrom spermatogenic transposon silencer MAEL 15252 v-maf avian musculoaponeurotic fibrosarcoma MAF 15253 oncogene homolog MAF1 homolog, negative regulator of RNA MAF1 15254 polymerase III v-maf avian musculoaponeurotic fibrosarcoma MAFA 15255-15256 oncogene homolog A v-maf avian

musculoaponeurotic fibrosarcoma MAFF 15257 oncogene homolog B v-maf avian musculoaponeurotic fibrosarcoma MAFF 15258 oncogene homolog F v-maf avian musculoaponeurotic fibrosarcoma MAFG 15259 oncogene homolog G v-maf avian musculoaponeurotic fibrosarcoma MAFK 15260 oncogene homolog K matrin 3 MATR3 15261 MYC associated factor X MAX 15262 MYC associated zinc finger protein MAZ 15263 methyl-CpG binding domain protein 1 MBD1 15264 methyl-CpG binding domain protein 2 MBD2 15265 methyl-CpG binding domain protein 3 MBD3 15266 methyl-CpG binding domain protein 3-like 1 MBD3L1 15267 methyl-CpG binding domain protein 3-like 2 MBD3L2 15268 methyl-CpG binding domain 4 DNA glycosylase MBD4 15269 methyl-CpG binding domain protein 5 MBD5 15270 methyl-CpG binding domain protein 6 MBD6 15271 muscleblind like splicing regulator 3 MBNL3 15272 MDS1 and EVI1 complex locus MECOM 15273 methyl-CpG binding protein 2 MECP2 15274 myocyte enhancer factor 2A MEF2A 15275 myocyte enhancer factor 2B MEF2B 15276 myocyte enhancer factor 2C MEF2C 15277 myocyte enhancer factor 2D MEF2D 15278 Meis homeobox 1 MEIS1 15279 Meis homeobox 2 MEIS2 15280 Meis homeobox 3 MEIS3 15281 Meis homeobox 3 pseudogene 1 MEIS3P1 15282 Meis homeobox 3 pseudogene 2 MEIS3P2 15283 mesenchyme homeobox 1 MEOX1 15284 mesenchyme homeobox 2 MEOX2 15285 mesoderm posterior bHLH transcription factor 1 MESP1 15286 mesoderm posterior bHLH transcription factor 2 MESP2 15287 MGA, MAX dimerization protein MGA 15288-15289 MIER1 transcriptional regulator MIER1 15290 MIER family member 2 MIER2 15291 MIER family member 3 MIER3 15292 MIS18 binding protein 1 MIS18BP1 15293 microphthalmia-associated transcription factor MITF 15294 Mix paired-like homeobox MIXL1 15295 mohawk homeobox MKX 15296 myeloid/lymphoid or mixed-lineage leukemia; MLLT1 15297 translocated to, 1 myeloid/lymphoid or mixed-lineage leukemia; MLLT10 15298 translocated to, 10 myeloid/lymphoid or mixed-lineage leukemia; MLLT11 15299 translocated to, 11 myeloid/lymphoid or mixed-lineage leukemia; MLLT3 15300 translocated to, 3 myeloid/lymphoid or mixed-lineage leukemia; MLLT4 15301 translocated to, 4 myeloid/lymphoid or mixed-lineage leukemia; MLLT6 15302 translocated to, 6 MLX, MAX dimerization protein MLX 15303 MLX interacting protein MLXIP 15304 MLX interacting protein-like MLXIPL 15305 MAX network transcriptional repressor MNT 15306 motor neuron and pancreas homeobox 1 MNX1 15307 musculin MSC 15308 mesogenin 1 MSGN1 15309 msh homeobox 1 MSX1 15310 msh homeobox 2 MSX2 15311 metastasis associated 1 MTA1 15312 metastasis associated 1 family member 2 MTA2 15313 metastasis associated 1 family member 3 MTA3 15314 metal-regulatory transcription factor 1 MTF1 15315 metal response element binding transcription factor 2 MTF2 15316 MAX dimerization protein 1 MXD1 15317 MAX dimerization protein 3 MXD3 15318 MAX dimerization protein 4 MXD4 15319 MAX interactor 1, dimerization protein MXI1 15320 v-myb avian myeloblastosis viral oncogene homolog MYB 15321 v-myb avian myeloblastosis viral oncogene homolog- MYBL1 15322 like 1 v-myb avian myeloblastosis viral oncogene homolog- MYBL2 15323 like 2 v-myc avian myelocytomatosis viral oncogene MYC 15324 homolog v-myc avian myelocytomatosis viral oncogene lung MYCL 15325 carcinoma derived homolog MYCL pseudogene 1 MYCLP1 15326 v-myc avian myelocytomatosis viral oncogene MYCN 15327 neuroblastoma derived homolog myogenic factor 5 MYF5 15328 myogenic factor 6 MYF6 15329 myoneurin MYNN 15330 myogenic differentiation 1 MYOD1 15331 myogenin (myogenic factor 4) MYOG 15332 myelin regulatory factor MYRF 15333 Myb-like, SWIRM and MPN domains 1 MYSM1 15334 myelin transcription factor 1 MYT1 15335-15336 myelin transcription factor 1 like MYT1L 15337 myeloid zinc finger 1 MZF1 15338 Nanog homeobox NANOG 15339 NANOG neighbor homeobox NANOGNB 15340 Nanog homeobox pseudogene 1 NANOGP1 15341 Nanog homeobox pseudogene 8 NANOGP8 15342 nuclear receptor coactivator 1 NCOA1 15343 nuclear receptor coactivator 2 NCOA2 15344 nuclear receptor coactivator 3 NCOA3 15345 nuclear receptor coactivator 4 NCOA4 15346 nuclear receptor coactivator 5 NCOA5 15347 nuclear receptor coactivator 6 NCOA6 15348 nuclear receptor coactivator 7 NCOA7 15349 nuclear receptor corepressor 1 NCOR1 15350 nuclear receptor corepressor 2 NCOR2 15351 neuronal differentiation 1 NEUROD1 15352 neuronal differentiation 2 NEUROD2 15353 neuronal differentiation 4 NEUROD4 15354 neuronal differentiation 6 NEUROD6 15355 neurogenin 1 NEUROG1 15356 neurogenin 2 NEUROG2 15357 neurogenin 3 NEUROG3 15358 nuclear factor of activated T-cells 5, tonicity- NFAT5 15359 responsive nuclear factor of activated T-cells, cytoplasmic,

NFATC1 15360 calcineurin-dependent 1 nuclear factor of activated T-cells, cytoplasmic, NFATC2 15361 calcineurin-dependent 2 nuclear factor of activated T-cells, cytoplasmic, NFATC3 15362 calcineurin-dependent 3 nuclear factor of activated T-cells, cytoplasmic, NFATC4 15363 calcineurin-dependent 4 nuclear factor, erythroid 2 NFE2 15364 nuclear factor, erythroid 2 like 1 NFE2L1 15365 nuclear factor, erythroid 2 like 2 NFE2L2 15366 nuclear factor, erythroid 2 like 3 NFE2L3 15367 nuclear factor I/A NFIA 15368 nuclear factor I/B NFIB 15369 nuclear factor I/C (CCAAT-binding transcription NFIC 15370 factor) nuclear factor, interleukin 3 regulated NFIL3 15371 nuclear factor I/X (CCAAT-binding transcription NFIX 15372 factor) nuclear factor of kappa light polypeptide gene NFkB1 15373 enhancer in B-cells 1 nuclear factor of kappa light polypeptide gene NFkB2 15374 enhancer in B-cells 2 (p49/p100) nuclear factor of kappa light polypeptide gene NFkBIA 15375 enhancer in B-cells inhibitor, alpha nuclear factor of kappa light polypeptide gene NFkBIB 15376 enhancer in B-cells inhibitor, beta nuclear factor of kappa light polypeptide gene NFkBID 15377 enhancer in B-cells inhibitor, delta nuclear factor of kappa light polypeptide gene NFkBIE 15378 enhancer in B-cells inhibitor, epsilon nuclear factor of kappa light polypeptide gene NFkBIL1 15379 enhancer in B-cells inhibitor-like 1 nuclear factor of kappa light polypeptide gene NFkBIZ 15380 enhancer in B-cells inhibitor, zeta nuclear factor related to kappaB binding protein NFRKB 15381 nuclear transcription factor, X-box binding 1 NFX1 15382 nuclear transcription factor, X-box binding-like 1 NFXL1 15383 nuclear transcription factor Y subunit alpha NFYA 15384 nuclear transcription factor Y subunit beta NFYB 15385 nuclear transcription factor Y subunit gamma NFYC 15386 nescient helix-loop-helix 1 NHLH1 15387 nescient helix-loop-helix 2 NHLH2 15388 NFkB repressing factor NKRF 15389 NK1 homeobox 1 NKX1-1 15390 NK1 homeobox 2 NKX1-2 15391 NK2 homeobox 1 NKX2-1 15392 NK2 homeobox 2 NKX2-2 15393 NK2 homeobox 3 NKX2-3 15394 NK2 homeobox 4 NKX2-4 15395 NK2 homeobox 5 NKX2-5 15396 NK2 homeobox 6 NKX2-6 15397 NK2 homeobox 8 NKX2-8 15398 NK3 homeobox 1 NKX3-1 15399 NK3 homeobox 2 NKX3-2 15400 NK6 homeobox 1 NKX6-1 15401 NK6 homeobox 2 NKX6-2 15402 NK6 homeobox 3 NKX6-3 15403 NOBOX oogenesis homeobox NOBOX 15404 NOC3 like DNA replication regulator NOC3L 15405 nucleolar complex associated 4 homolog NOC4L 15406 non-POU domain containing, octamer-binding NONO 15407 notochord homeobox NOTO 15408 neuronal PAS domain protein 1 NPAS1 15409 neuronal PAS domain protein 2 NPAS2 15410 neuronal PAS domain protein 3 NPAS3 15411 neuronal PAS domain protein 4 NPAS4 15412 nuclear receptor subfamily 0 group B member 1 NR0B1 15413 nuclear receptor subfamily 0 group B member 2 NR0B2 15414 nuclear receptor subfamily 1 group D member 1 NR1D1 15415 nuclear receptor subfamily 1 group D member 2 NR1D2 15416 nuclear receptor subfamily 1 group H member 2 NR1H2 15417 nuclear receptor subfamily 1 group H member 3 NR1H3 15418 nuclear receptor subfamily 1 group H member 4 NR1H4 15419 nuclear receptor subfamily 1 group I member 2 NR1I2 15420 nuclear receptor subfamily 1 group I member 3 NR1I3 15421 nuclear receptor subfamily 2 group C member 1 NR2C1 15422 nuclear receptor subfamily 2 group C member 2 NR2C2 15423 nuclear receptor subfamily 2 group E member 1 NR2E1 15424 nuclear receptor subfamily 2 group E member 3 NR2E3 15425 nuclear receptor subfamily 2 group F member 1 NR2F1 15426 nuclear receptor subfamily 2 group F member 2 NR2F2 15427 nuclear receptor subfamily 2 group F member 6 NR2F6 15428 nuclear receptor subfamily 3 group C member 1 NR3C1 15429 nuclear receptor subfamily 3 group C member 2 NR3C2 15430 nuclear receptor subfamily 4 group A member 1 NR4A1 15431 nuclear receptor subfamily 4 group A member 2 NR4A2 15432 nuclear receptor subfamily 4 group A member 3 NR4A3 15433 nuclear receptor subfamily 5 group A member 1 NR5A1 15434 nuclear receptor subfamily 5 group A member 2 NR5A2 15435 nuclear receptor subfamily 6 group A member 1 NR6A1 15436 nuclear respiratory factor 1 NRF1 15437-15438 neural retina leucine zipper NRL 15439 oligodendrocyte transcription factor 1 OLIG1 15440 oligodendrocyte lineage transcription factor 2 OLIG2 15441 oligodendrocyte transcription factor 3 OLIG3 15442 one cut homeobox 1 ONECUT1 15443 one cut homeobox 2 ONECUT2 15444 one cut homeobox 3 ONECUT3 15445 odd-skipped related transcription factor 1 OSR1 15446 odd-skipped related transcription factor 2 OSR2 15447 orthopedia homeobox OTP 15448 orthodenticle homeobox 1 OTX1 15449 orthodenticle homeobox 2 OTX2 15450 ovo like zinc finger 1 OVOL1 15451 ovo like zinc finger 2 OVOL2 15452 ovo like zinc finger 3 OVOL3 15453 poly(ADP-ribose) polymerase 1 PARP1

15454 poly(ADP-ribose) polymerase family member 12 PARP12 15455 POZ/BTB and AT hook containing zinc finger 1 PATZ1 15456 PRKC, apoptosis, WT1, regulator PAWR 15457 paired box 1 PAX1 15458 paired box 2 PAX2 15459 paired box 3 PAX3 15460 paired box 4 PAX4 15461 paired box 5 PAX5 15462 paired box 6 PAX6 15463 paired box 7 PAX7 15464 paired box 8 PAX8 15465 paired box 9 PAX9 15466 PAX3 and PAX7 binding protein 1 PAXBP1 15467 polybromo 1 PBRM1 15468 pre-B-cell leukemia homeobox 1 PBX1 15469 pre-B-cell leukemia homeobox 2 PBX2 15470 pre-B-cell leukemia homeobox 3 PBX3 15471 pre-B-cell leukemia homeobox 4 PBX4 15472 poly(rC) binding protein 1 PCBP1 15473 poly(rC) binding protein 2 PCBP2 15474 poly(rC) binding protein 3 PCBP3 15475 poly(rC) binding protein 4 PCBP4 15476 poly comb group ring finger 6 PCGF6 15477 pancreatic and duodenal homeobox 1 PDX1 15478-15479 paternally expressed 3 PEG3 15480 progesterone receptor PGR 15481 prohibitin PHB 15482 prohibitin 2 PHB2 15483 PHD finger protein 20 PHF20 15484 PHD finger protein 5A PHF5A 15485 paired like homeobox 2a PHOX2A 15486 paired like homeobox 2b PHOX2B 15487 putative homeodomain transcription factor 1 PHTF1 15488 putative homeodomain transcription factor 2 PHTF2 15489 paired like homeodomain 1 PITX1 15490 paired like homeodomain 2 PITX2 15491 paired like homeodomain 3 PITX3 15492 PBX/knotted 1 homeobox 1 PKNOX1 15493 PBX/knotted 1 homeobox 2 PKNOX2 15494 PLAG1 zinc finger PLAG1 15495 PLAG1 like zinc finger 1 PLAGL1 15496 PLAG1 like zinc finger 2 PLAGL2 15497 pleckstrin PLEK 15498 promyelocytic leukaemia zinc finger PLZF 15499 pogo transposable element with ZNF domain POGZ 15500 POU class 1 homeobox 1 POU1F1 15501 POU class 2 associating factor 1 POU2AF1 15502 POU class 2 homeobox 1 POU2F1 15503 POU class 2 homeobox 2 POU2F2 15504 POU class 2 homeobox 3 POU2F3 15505 POU class 3 homeobox 1 POU3F1 15506 POU class 3 homeobox 2 POU3F2 15507 POU class 3 homeobox 3 POU3F3 15508 POU class 3 homeobox 4 POU3F4 15509 POU class 4 homeobox 1 POU4F1 15510 POU class 4 homeobox 2 POU4F2 15511 POU class 4 homeobox 3 POU4F3 15512 POU class 5 homeobox 1 POU5F1 15513 POU class 5 homeobox 1B POU5F1B 15514 POU domain class 5, transcription factor 2 POU5F2 15515 POU class 6 homeobox 1 POU6F1 15516 POU class 6 homeobox 2 POU6F2 15517 peroxisome proliferator activated receptor alpha PPARA 15518 peroxisome proliferator activated receptor delta PPARD 15519 peroxisome proliferator activated receptor gamma PPARG 15520 protein phosphatase 1 regulatory subunit 13 like PPP1R13L 15521 PR domain 1 PRDM1 15522 PR domain 10 PRDM10 15523 PR domain 11 PRDM11 15524 PR domain 12 PRDM12 15525 PR domain 13 PRDM13 15526 PR domain 14 PRDM14 15527 PR domain 15 PRDM15 15528 PR domain 16 PRDM16 15529 PR domain 2 PRDM2 15530 PR domain 4 PRDM4 15531 PR domain 5 PRDM5 15532 PR domain 6 PRDM6 15533 PR domain 7 PRDM7 15534 PR domain 8 PRDM8 15535 PR domain 9 PRDM9 15536 prolactin regulatory element binding PREB 15537 PROP paired-like homeobox 1 PROP1 15538 prospero homeobox 1 PROX1 15539 prospero homeobox 2 PROX2 15540 paired related homeobox 1 PRRX1 15541 paired related homeobox 2 PRRX2 15542 paraspeckle component 1 PSPC1 15543 pancreas specific transcription factor, 1a PTF1A 15544 purine-rich element binding protein A PURA 15545 purine-rich element binding protein B PURB 15546 purine-rich element binding protein G PURG 15547 retinoic acid receptor alpha RARA 15548 retinoic acid receptor beta RARB 15549 retinoic acid receptor gamma RARG 15550 retina and anterior neural fold homeobox RAX 15551-15552 retina and anterior neural fold homeobox 2 RAX2 15553 RB associated KRAB zinc finger RBAK 15554 RNA binding motif protein 22 RBM22 15555 recombination signal binding protein for RBPJ 15556 immunoglobulin kappa J region recombination signal binding protein for RBPJL 15557 immunoglobulin kappa J region-like ring finger and CCCH-type domains 1 RC3H1 15558 ring finger and CCCH-type domains 2 RC3H2 15559 REST corepressor 1 RCOR1 15560 REST corepressor 2 RCOR2 15561 REST corepressor 3 RCOR3 15562 v-rel avian reticuloendotheliosis viral oncogene REL 15563 homolog v-rel avian reticuloendotheliosis viral oncogene RELA 15564 homolog A v-rel avian reticuloendotheliosis viral oncogene RELB 15565 homolog B arginine-glutamic acid dipeptide (RE) repeats RERE 15566 RE1-silencing transcription factor REST 15567 regulatory factor X1 RFX1 15568 regulatory factor X2 RFX2 15569 regulatory factor X3 RFX3 15570 regulatory factor X4 RFX4 15571 regulatory factor X5 RFX5 15572 regulatory factor X6 RFX6 15573 regulatory factor X7 RFX7 15574 RFX family member 8, lacking RFX DNA binding RFX8 15575 domain regulatory factor X



associated ankyrin containing RFXANK 15576 protein regulatory factor X associated protein RFXAP 15577 RhoX homeobox family member 1 RHOF1 15578 RhoX homeobox family member 2 RHOF2 15579 RhoX homeobox family member 2B RHOF2B 15580 rearranged L-myc fusion RLF 15581-15582 RAR related orphan receptor A RORA 15583 RAR related orphan receptor B RORB 15584 RAR related orphan receptor C RORC 15585 retinoic acid receptor-related orphan nuclear receptor RORgT 15586 gamma ras responsive element binding protein 1 RREB1 15587 runt related transcription factor 1 RUNX1 15588 runt related transcription factor 1; translocated to, 1 RUNX1T1 15589 (cyclin D related) runt related transcription factor 2 RUNX2 15590 runt related transcription factor 3 RUNX3 15591 retinoid X receptor alpha RXRA 15592 retinoid X receptor beta RXRB 15593 retinoid X receptor gamma RXRG 15594 spalt-like transcription factor 1 SALL1 15595 spalt-like transcription factor 2 SALL2 15596 spalt-like transcription factor 3 SALL3 15597 spalt-like transcription factor 4 SALL4 15598 SATB homeobox 1 SATB1 15599 SATB homeobox 2 SATB2 15600 S-phase cyclin A-associated protein in the ER SCAPER 15601 scratch family zinc finger 1 SCRT1 15602 scratch family zinc finger 2 SCRT2 15603 scleraxis bHLH transcription factor SCX 15604 SEBOX homeobox SEBOX 15605 SET binding protein 1 SETBP1 15606 splicing factor proline/glutamine-rich SFPQ 15607 short stature homeobox SHOX 15608 short stature homeobox 2 SHOX2 15609 single-minded family bHLH transcription factor 1 SIM1 15610 single-minded family bHLH transcription factor 2 SIM2 15611 SIX homeobox 1 SIX1 15612 SIX homeobox 2 SIX2 15613 SIX homeobox 3 SIX3 15614 SIX homeobox 4 SIX4 15615 SIX homeobox 5 SIX5 15616 SIX homeobox 6 SIX6 15617 SKI proto-oncogene SKI 15618 SKI-like proto-oncogene SKIL 15619 SKI family transcriptional corepressor 1 SKOR1 15620 SKI family transcriptional corepressor 2 SKOR2 15621 solute carrier family 30 (zinc transporter), member 9 SLC30A9 15622 SMAD family member 1 SMAD1 15623 SMAD family member 2 SMAD2 15624 SMAD family member 3 SMAD3 15625 SMAD family member 4 SMAD4 15626 SMAD family member 5 SMAD5 15627 SMAD family member 6 SMAD6 15628 SMAD family member 7 SMAD7 15629 SMAD family member 9 SMAD9 15630 SWI/SNF related, matrix associated, actin dependent SMARCA1 15631 regulator of chromatin, subfamily a, member 1 SWI/SNF related, matrix associated, actin dependent SMARCA2 15632 regulator of chromatin, subfamily a, member 2 SWI/SNF related, matrix associated, actin dependent SMARCA4 15633 regulator of chromatin, subfamily a, member 4 SWI/SNF related, matrix associated, actin dependent SMARCA5 15634 regulator of chromatin, subfamily a, member 5 SWI/SNF-related, matrix-associated actin-dependent SMARCD1 15635 regulator of chromatin, subfamily a, containing DEAD/H box 1 SWI/SNF related, matrix associated, actin dependent SMARCAL1 15636 regulator of chromatin, subfamily a-like 1 SWI/SNF related, matrix associated, actin dependent SMARCB1 15637 regulator of chromatin, subfamily b, member 1 SWI/SNF related, matrix associated, actin dependent SMARCC1 15638 regulator of chromatin, subfamily c, member 1 SWI/SNF related, matrix associated, actin dependent SMARCC2 15639 regulator of chromatin, subfamily c, member 2 SWI/SNF related, matrix associated, actin dependent SMARCD1 15640 regulator of chromatin, subfamily d, member 1 SWI/SNF related, matrix associated, actin dependent SMARCD2 15641 regulator of chromatin, subfamily d, member 2 SWI/SNF related, matrix associated, actin dependent SMARCD3 15642 regulator of chromatin, subfamily d, member 3 SWI/SNF related, matrix associated, actin dependent SMARCE1 15643 regulator of chromatin, subfamily e, member 1 snail family zinc finger 1 SNAI1 15644 snail family zinc finger 2 SNAI2 15645 snail family zinc finger 3 SNAI3 15646 small nuclear RNA activating complex polypeptide 4 SNAPC4 15647 spermatogenesis and oogenesis specific basic helix- SOHLH1 15648 loop-helix 1 spermatogenesis and oogenesis specific basic helix- SOHLH2 15649 loop-helix 2 SRY-box 1 SOX1 15650 SRY-box 10 SOX10 15651 SRY-box 11 SOX11 15652 SRY-box 12 SOX12 15653 SRY-box 13 SOX13 15654 SRY-box 14 SOX14 15655 SRY-box 15 SOX15 15656 SRY-box 17 SOX17 15657 SRY-box 18 SOX18 15658 SRY-box 2 SOX2 15659 SRY-box 21 SOX21 15660 SRY-box 3 SOX3 15661 SRY-box 30 SOX30 15662 SRY-box 4 SOX4 15663 SRY-box 5 SOX5 15664 SRY-box 6 SOX6 15665 SRY-box 7 SOX7 15666 SRY-box 8 SOX8 15667 SRY-box 9 SOX9 15668 Sp1 transcription factor SP1 15669-15670 SP100 nuclear antigen SP100 15671 SP110 nuclear body protein SP110 15672 SP140 nuclear body protein SP140 15673 SP140 nuclear body protein like SP140L 15674 Sp2 transcription factor SP2 15675 Sp3 transcription factor SP3 15676 Sp4



transcription factor SP4 15677 Sp5 transcription factor SP5 15678 Sp6 transcription factor SP6 15679 Sp7 transcription factor SP7 15680 Sp8 transcription factor SP8 15681 Sp9 transcription factor SP9 15682 SAM pointed domain containing ETS transcription SPDEF 15683 factor Spi-1 proto-oncogene SPI1 15684 Spi-B transcription factor (Spi-1/PU.1 related) SPIB 15685 Spi-C transcription factor (Spi-1/PU.1 related) SPIC 15686 spermatogenic leucine zipper 1 SPZ1 15687 sterol regulatory element binding transcription factor 1 SREBF1 15688 sterol regulatory element binding transcription factor 2 SREBF2 15689 serum response factor SRF 15690 sex determining region Y SRY 15691 structure specific recognition protein 1 SSRP1 15692 suppression of tumorigenicity 18, zinc finger ST18 15693 signal transducer and activator of transcription 1 STAT1 15694 signal transducer and activator of transcription 2 STAT2 15695 signal transducer and activator of transcription 3 STAT3 15696 (acute-phase response factor) signal transducer and activator of transcription 4 STAT4 15697 signal transducer and activator of transcription 5 STAT5 15698 signal transducer and activator of transcription 5A STAT5A 15699 signal transducer and activator of transcription 5B STAT5B 15700 signal transducer and activator of transcription 6, STAT6 15701 interleukin-4 induced transcriptional adaptor 2A TADA2A 15702 transcriptional adaptor 2B TADA2B 15703 TATA-box binding protein associated factor 1 TAF1 15704 T-cell acute lymphocytic leukemia 1 TAL1 15705 T-cell acute lymphocytic leukemia 2 TAL2 15706 Tax1 (human T-cell leukemia virus type I) binding TAX1BP1 15707 protein 1 Tax1 (human T-cell leukemia virus type I) binding TAX1BP3 15708 protein 3 T-box transcription factor T-bet Tbet 15709 TATA-box binding protein TBP 15710 TATA-box binding protein like 1 TBPL1 15711 TATA-box binding protein like 2 TBPL2 15712 T-box, brain 1 TBR1 15713 T-box 1 TBX1 15714 T-box 10 TBX10 15715 T-box 15 TBX15 15716 T-box 18 TBX18 15717 T-box 19 TBX19 15718 T-box 2 TBX2 15719 T-box 20 TBX20 15720 T-box 21 TBX21 15721 T-box 22 TBX22 15722 T-box 3 TBX3 15723 T-box 4 TBX4 15724 T-box 5 TBX5 15725 T-box 6 TBX6 15726 transcription factor 12 TCF12 15727 transcription factor 15 (basic helix-loop-helix) TCF15 15728 transcription factor 19 TCF19 15729 transcription factor 20 (AR1) TCF20 15730 transcription factor 21 TCF21 15731 transcription factor 23 TCF23 15732 transcription factor 24 TCF24 15733 transcription factor 25 (basic helix-loop-helix) TCF25 15734 transcription factor 3 TCF3 15735 transcription factor 4 TCF4 15736 transcription factor 7 (T-cell specific, HMG-box, TCF7 15737 TCF1) transcription factor 7 like 1 TCF7L1 15738 transcription factor 7 like 2 TCF7L2 15739 transcription factor-like 5 (basic helix-loop-helix) TCFL5 15740 TEA domain transcription factor 1 TEAD1 15741 TEA domain transcription factor 2 TEAD2 15742 TEA domain transcription factor 3 TEAD3 15743 TEA domain transcription factor 4 TEAD4 15744 thyrotrophic embryonic factor TEF 15745 telomeric repeat binding factor (NIMA-interacting) 1 TERF1 15746 telomeric repeat binding factor 2 TERF2 15747 tet methylcytosine dioxygenase 1 TET1 15748 tet methylcytosine dioxygenase 2 TET2 15749 tet methylcytosine dioxygenase 3 TET3 15750 transcription factor A, mitochondrial TFAM 15751 transcription factor AP-2 alpha (activating enhancer TFAP2A 15752 binding protein 2 alpha) transcription factor AP-2 beta (activating enhancer TFAP2B 15753 binding protein 2 beta) transcription factor AP-2 gamma (activating enhancer TFAP2C 15754 binding protein 2 gamma) transcription factor AP-2 delta (activating enhancer TFAP2D 15755 binding protein 2 delta) transcription factor AP-2 epsilon (activating enhancer TFAP2E 15756 binding protein 2 epsilon) transcription factor AP-4 (activating enhancer binding TFAP4 15757 protein 4) transcription factor B1, mitochondrial TFB1M 15758 transcription factor B2, mitochondrial TFB2M 15759 transcription factor CP2 TFCP2 15760 transcription factor CP2-like 1 TFCP2L1 15761 transcription factor Dp-1 TFDP1 15762 transcription factor Dp-2 (E2F dimerization partner 2) TFDP2 15763 transcription factor Dp family member 3 TFDP3 15764 transcription factor binding to IGHM enhancer 3 TFE3 15765 transcription factor EB TFEB 15766 transcription factor EC TFEC 15767 TGFB induced factor homeobox 1 TGIF1 15768 TGFB induced factor homeobox 2 TGIF2 15769 TGFB induced factor homeobox 2 like, X-linked TGIF2LX 15770 TGFB induced factor homeobox 2 like, Y-linked TGIF2LY 15771 THAP domain containing, apoptosis associated protein THAP1 15772 1 THAP domain containing 10 THAP10 15773 THAP domain containing 11 THAP11 15774 THAP domain containing 12 THAP12 15775 THAP domain containing, apoptosis associated protein THAP2 15776 2 THAP domain containing, apoptosis associated protein THAP3 15777 3 THAP domain containing 4 THAP4 15778 THAP domain

containing 5 THAP5 THAP domain containing 6 THAP6 15780 THAP domain containing 7 THAP7 15781 THAP domain containing 8 THAP8 15782 THAP domain containing 9 THAP9 15783 Th inducing POZ-Kruppel Factor ThPOK 15784 thyroid hormone receptor, alpha THRA 15785 thyroid hormone receptor, beta THRB 15786 T-cell leukemia homeobox 1 TLX1 15787 T-cell leukemia homeobox 2 TLX2 15788 T-cell leukemia homeobox 3 TLX3 15789 target of EGR1, member 1 (nuclear) TOE1 15790 tonsoku-like, DNA repair protein TONSL 15791 topoisomerase I binding, arginine/serine-rich, E3 TOPORS 15792 ubiquitin protein ligase thymocyte selection associated high mobility group TOX 15793 box TOX high mobility group box family member 2 TOX2 15794 TOX high mobility group box family member 3 TOX3 15795 TOX high mobility group box family member 4 TOX4 15796 tumor protein p53 TP53 15797 tumor protein p63 TP63 15798 tumor protein p73 TP73 15799 tetra-peptide repeat homeobox 1 TPRX1 15800 tetra-peptide repeat homeobox-like TPRXL 15801 transcriptional regulating factor 1 TRERF1 15802 trichorhinophalangeal syndrome I TRPS1 15803 TSC22 domain family member 1 TSC22D1 15804 TSC22 domain family member 2 TSC22D2 15805 TSC22 domain family member 3 TSC22D3 15806 TSC22 domain family member 4 TSC22D4 15807 teashirt zinc finger homeobox 1 TSHZ1 15808 teashirt zinc finger homeobox 2 TSHZ2 15809 teashirt zinc finger homeobox 3 TSHZ3 15810 transcription termination factor, RNA polymerase I TTF1 15811-15812 transcription termination factor, RNA polymerase II TTF2 15813-15814 tubby bipartite transcription factor TUB 15815 twist family bHLH transcription factor 1 TWIST1 15816 twist family bHLH transcription factor 2 TWIST2 15817 upstream binding protein 1 (LBP-1a) UBP1 15818 upstream binding transcription factor, RNA UBTF 15819 polymerase I upstream binding transcription factor, RNA UBTF1 15820 polymerase I-like 1 upstream binding transcription factor, RNA UBTF6 15821 polymerase I-like 6 (pseudogene) UNC homeobox UNCX 15822 unkempt family zinc finger UNK 15823 unkempt family like zinc finger UNKL 15824 upstream transcription factor 1 USF1 15825 upstream transcription factor 2, c-fos interacting USF2 15826 upstream transcription factor family member 3 USF3 15827 undifferentiated embryonic cell transcription factor 1 UTF1 15828 ventral anterior homeobox 1 VAX1 15829 ventral anterior homeobox 2 VAX2 15830 vitamin D (1,25-dihydroxyvitamin D3) receptor VDR 15831 VENT homeobox VENTX 15832 vascular endothelial zinc finger 1 VEZF1 15833 visual system homeobox 1 VSX1 15834 visual system homeobox 2 VSX2 15835 WD repeat and HMG-box DNA binding protein 1 WDHD1 15836 Wolf-Hirschhorn syndrome candidate 1 WHSC1 15837 widely interspaced zinc finger motifs WIZ 15838 Wilms tumor 1 WT1 15839 X-box binding protein 1 XBP1 15840 Y-box binding protein 1 YBX1 15841 Y-box binding protein 2 YBX2 15842 Y-box binding protein 3 YBX3 15843 YEATS domain containing 2 YEATS2 15844 YEATS domain containing 4 YEATS4 15845 YY1 transcription factor YY1 15846 YY2 transcription factor YY2 15847 zinc finger BED-type containing 1 ZBED1 15848 zinc finger BED-type containing 2 ZBED2 15849 zinc finger BED-type containing 3 ZBED3 15850 zinc finger BED-type containing 4 ZBED4 15851 zinc finger BED-type containing 5 ZBED5 15852 zinc finger, BED-type containing 6 ZBED6 15853 Z-DNA binding protein 1 ZBP1 15854-15855 zinc finger and BTB domain containing 1 ZBTB1 15856 zinc finger and BTB domain containing 10 ZBTB10 15857 zinc finger and BTB domain containing 11 ZBTB11 15858 zinc finger and BTB domain containing 12 ZBTB12 15859 zinc finger and BTB domain containing 14 ZBTB14 15860 zinc finger and BTB domain containing 16 ZBTB16 15861 zinc finger and BTB domain containing 17 ZBTB17 15862 zinc finger and BTB domain containing 18 ZBTB18 15863 zinc finger and BTB domain containing 2 ZBTB2 15864 zinc finger and BTB domain containing 20 ZBTB20 15865 zinc finger and BTB domain containing 21 ZBTB21 15866 zinc finger and BTB domain containing 22 ZBTB22 15867 zinc finger and BTB domain containing 24 ZBTB24 15868 zinc finger and BTB domain containing 25 ZBTB25 15869 zinc finger and BTB domain containing 26 ZBTB26 15870 zinc finger and BTB domain containing 3 ZBTB3 15871 zinc finger and BTB domain containing 32 ZBTB32 15872 zinc finger and BTB domain containing 33 ZBTB33 15873 zinc finger and BTB domain containing 34 ZBTB34 15874 zinc finger and BTB domain containing 37 ZBTB37 15875 zinc finger and BTB domain containing 38 ZBTB38 15876 zinc finger and BTB domain containing 39 ZBTB39 15877 zinc finger and BTB domain containing 4 ZBTB4 15878 zinc finger and BTB domain containing 40 ZBTB40 15879 zinc finger and BTB domain containing 41 ZBTB41 15880 zinc finger and BTB domain containing 42 ZBTB42 15881

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ZNF616 16300 zinc finger protein 618 ZNF618 16301 zinc finger protein 619 ZNF619 16302 zinc  
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ZNF622 16305 zinc finger protein 623 ZNF623 16306 zinc finger protein 624 ZNF624 16307 zinc  
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ZNF627 16310 zinc finger protein 628 ZNF628 16311 zinc finger protein 629 ZNF629 16312 zinc  
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ZNF644 16315 zinc finger protein 645 ZNF645 16316 zinc finger protein 646 ZNF646 16317 zinc  
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ZNF652 16320 zinc finger protein 653 ZNF653 16321 zinc finger protein 654 ZNF654 16322 zinc  
finger protein 655 ZNF655 16323 zinc finger protein 658 ZNF658 16324 zinc finger protein 658B  
(pseudogene) ZNF658B 16325 zinc finger protein 66 ZNF66 16326 zinc finger protein 660 ZNF660  
16327 zinc finger protein 662 ZNF662 16328 zinc finger protein 664 ZNF664 16329 zinc finger protein  
665 ZNF665 16330 zinc finger protein 667 ZNF667 16331 zinc finger protein 668 ZNF668 16332 zinc  
finger protein 669 ZNF669 16333 zinc finger protein 670 ZNF670 16334 zinc finger protein 671  
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ZNF677 16340 zinc finger protein 678 ZNF678 16341 zinc finger protein 679 ZNF679 16342 zinc  
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ZNF682 16345 zinc finger protein 683 ZNF683 16346 zinc finger protein 684 ZNF684 16347 zinc  
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16355 zinc finger protein 697 ZNF697 16356 zinc finger protein 699 ZNF699 16357 zinc finger protein  
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protein 702, pseudogene ZNF702P 16361 zinc finger protein 703 ZNF703 16362 zinc finger protein  
704 ZNF704 16363 zinc finger protein 705A ZNF705A 16364 zinc finger protein 705D ZNF705D  
16365 zinc finger protein 705E ZNF705E 16366 zinc finger protein 705G ZNF705G 16367 zinc finger  
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finger protein 721 ZNF721 16381 zinc finger protein 724, pseudogene ZNF724P 16382 zinc finger  
protein 726 ZNF726 16383 zinc finger protein 727 ZNF727 16384 zinc finger protein 729 ZNF729  
16385 zinc finger protein 730 ZNF730 16386 zinc finger protein 732 ZNF732 16387 zinc finger protein  
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ZNF747 16393 zinc finger protein 749 ZNF749 16394 zinc finger protein 750 ZNF750 16395 zinc

finger protein 75a ZNF75A 16396 zinc finger protein 75D ZNF75D 16397 zinc finger protein 76  
 ZNF76 16398 zinc finger protein 761 ZNF761 16399 zinc finger protein 763 ZNF763 16400 zinc finger  
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 16403 zinc finger protein 768 ZNF768 16404 zinc finger protein 77 ZNF77 16405 zinc finger protein  
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 finger family member 788 ZNF788 16424 zinc finger protein 789 ZNF789 16425 zinc finger protein 79  
 ZNF79 16426 zinc finger protein 790 ZNF790 16427 zinc finger protein 791 ZNF791 16428 zinc finger  
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 16431 zinc finger protein 8 ZNF8 16432 zinc finger protein 80 ZNF80 16433 zinc finger protein 800  
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 ZNF808 16439 zinc finger protein 81 ZNF81 16440 zinc finger protein 813 ZNF813 16441 zinc finger  
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 16444 zinc finger protein 823 ZNF823 16445 zinc finger protein 827 ZNF827 16446 zinc finger protein  
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 16454 zinc finger protein 839 ZNF839 16455 zinc finger protein 84 ZNF84 16456 zinc finger protein  
 840, pseudogene ZNF840P 16457 zinc finger protein 841 ZNF841 16458 zinc finger protein 843  
 ZNF843 16459 zinc finger protein 844 ZNF844 16460 zinc finger protein 845 ZNF845 16461 zinc  
 finger protein 846 ZNF846 16462 zinc finger protein 85 ZNF85 16463 zinc finger protein 853 ZNF853  
 16464 zinc finger protein 860 ZNF860 16465 zinc finger protein 876, pseudogene ZNF876P 16466 zinc  
 finger protein 878 ZNF878 16467 zinc finger protein 879 ZNF879 16468 zinc finger protein 880  
 ZNF880 16469 zinc finger protein 891 ZNF891 16470 zinc finger protein 90 ZNF90 16471 zinc finger  
 protein 91 ZNF91 16472 zinc finger protein 92 ZNF92 16473 zinc finger protein 93 ZNF93 16474 zinc  
 finger protein 98 ZNF98 16475 zinc finger protein 99 ZNF99 16476 zinc finger, NFX1-type containing  
 1 ZNFX1 16477 zinc finger and SCAN domain containing 1 ZSCAN1 16478 zinc finger and SCAN  
 domain containing 10 ZSCAN10 16479 zinc finger and SCAN domain containing 12 ZSCAN12 16480  
 zinc finger and SCAN domain containing 16 ZSCAN16 16481 zinc finger and SCAN domain  
 containing 18 ZSCAN18 16482 zinc finger and SCAN domain containing 2 ZSCAN2 16483 zinc finger  
 and SCAN domain containing 20 ZSCAN20 16484 zinc finger and SCAN domain containing 21  
 ZSCAN21 16485 zinc finger and SCAN domain containing 22 ZSCAN22 16486 zinc finger and SCAN  
 domain containing 23 ZSCAN23 16487 zinc finger and SCAN domain containing 25 ZSCAN25 16488  
 zinc finger and SCAN domain containing 26 ZSCAN26 16489 zinc finger and SCAN domain  
 containing 29 ZSCAN29 16490 zinc finger and SCAN domain containing 30 ZSCAN30 16491 zinc  
 finger and SCAN domain containing 31 ZSCAN31 16492 zinc finger and SCAN domain containing 32  
 ZSCAN32 16493 zinc finger and SCAN domain containing 4 ZSCAN4 16494 zinc finger and SCAN  
 domain containing 5A ZSCAN5A 16495 zinc finger and SCAN domain containing 5B ZSCAN5B  
 16496 zinc finger and SCAN domain containing 5C, ZSCAN5CP 16497 pseudogene zinc finger and  
 SCAN domain containing 9 ZSCAN9 16498 zinc finger with UFM1-specific peptidase domain ZUFSP  
 16499 zinc finger, X-linked, duplicated A ZXDA 16500 zinc finger, X-linked, duplicated B ZXDB  
 16501 ZXD family zinc finger C ZXDC 16502 zinc finger ZZ-type containing 3 ZZZ3 16503

(824) In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one  
 or more gene(s) encoding a cell death or cell apoptosis receptor to produce an armored T-cell of the  
 disclosure. Interaction of a death receptor and its endogenous ligand results in the initiation of  
 apoptosis. Disruption of an expression, an activity, or an interaction of a cell death and/or cell apoptosis



receptor and/or ligand render an armored T-cell of the disclosure less receptive to death signals, consequently, making the armored T cell of the disclosure more efficacious in a tumor environment. An exemplary cell death receptor which may be modified in an armored T cell of the disclosure is Fas (CD95). Exemplary cell death and/or cell apoptosis receptors and ligands of the disclosure include, but are not limited to, the exemplary receptors and ligands provided in Table 4.

(825) TABLE-US-00225 TABLE 4 Exemplary Cell Death and/or Cell Apoptosis Receptors and Ligands. Full Name Abbreviation SEQ ID NO: Cluster of Differentiation 120 CD120a 16504-16505 Death receptor 3 DR3 16506 Death receptor 6 DR6 16507 first apoptosis signal (Fas) receptor Fas 16508-16509 (CD95/APO-1) Fas Ligand FasL 16510 cellular tumor antigen p53 p53 16511 Tumor necrosis factor receptor 1 TNF-R1 16512 Tumor necrosis factor receptor 2 TNF-R2 16513 Tumor necrosis factor-related apoptosis- TRAIL-R1 16514 inducing ligand receptor 1 (DR4) Tumor necrosis factor-related apoptosis- TRAIL-R2 16515 inducing ligand receptor 2 (DR5) Fas-associated protein with death domain FADD 16516 Tumor necrosis factor receptor type 1- TRADD 16517 associated DEATH domain protein Bcl-2-associated X protein Bax 16518 Bcl-2 homologous killer BAK 16519 14-3-3 protein 14-3-3 16520 B-cell lymphoma 2 Bcl-2 16521 Cytochrome C CytC 16522 Second mitochondria-derived activator of Smac/Diablo 16523 caspase High temperature requirement protein A2 HTRA2/Omi 16524 Apoptosis inducing factor AIF 16525 Endonuclease G EXOG 16526 Caspase 9 Cas9 16527 Caspase 2 Cas2 16528 Caspase 8 Cas8 16529 Caspase 10 Cas10 16530 Caspase 3 Cas3 16531 Caspase 6 Cas6 16532 Caspase 7 Cas7 16533 Tumor Necrosis Factor alpha TNF-alpha 16534 TNF-related weak inducer of apoptosis TWEAK 16535 TNF-related weak inducer of apoptosis TWEAK -R 16536 receptor Tumor necrosis factor-related apoptosis- TRAIL 16537 inducing ligand TNF ligand-related molecule 1 TL1A 16538 Receptor-interacting serine/threonine- RIP1 16539 protein kinase 1 Cellular inhibitor of apoptosis 1 cIAP-1 16540 TNF receptor-associated factor 2 TRAF-2 16541

(826) In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding a metabolic sensing protein to produce an armored T-cell of the disclosure. Disruption to the metabolic sensing of the immunosuppressive tumor microenvironment (characterized by low levels of oxygen, pH, glucose and other molecules) by an armored T-cell of the disclosure leads to extended retention of T-cell function and, consequently, more tumor cell skilled per armored T cell. For example, HIF1a and VHL play a role in T-cell function while in a hypoxic environment. An armored T-cell of the disclosure may have silenced or reduced expression of one or more genes encoding HIF1a or VHL. Genes and proteins involved in metabolic sensing include, but are not limited to, the exemplary genes and proteins provided in Table 5.

(827) TABLE-US-00226 TABLE 5 Exemplary Metabolic Sensing Genes (and encoded Proteins). Full Name Metabolite Abbreviation SEQ ID NO: hypoxia-inducible factor 1 $\alpha$  Low oxygen HIF-1 $\alpha$  16542 von Hippel-Lindau tumor suppressor Low oxygen VHL 16543 Prolyl-hydroxylase domain proteins High oxygen PHD proteins Glucose transporter 1 glucose GLUT1 16544 Linker of Activated T cells Amino acid (leucine) LAT 16545 CD98 glycoprotein Amino acid (leucine) CD98 16546 Alanine, serine, cysteine-preferring Cationic Amino acid ASCT2/Slc1a5 16547 transporter 2 (glutamine) Solute carrier family 7 member 1 Cationic Amino acids Slc7a1 16548 Solute carrier family 7 member 2 Cationic Amino acids Slc7a2 16549 Solute carrier family 7 member 3 Cationic Amino acids Slc7a3 16550 Solute carrier family 7 member 4 Cationic Amino acids Slc7a4 16551 Solute carrier family 7 member 5 Glycoprotein Slc7a5 16552 associated Amino acids Solute carrier family 7 member 6 Glycoprotein Slc7a6 16553 associated Amino acids Solute carrier family 7 member 7 Glycoprotein Slc7a7 16554 associated Amino acids Solute carrier family 7 member 8 Glycoprotein Slc7a8 16555 associated Amino acids Solute carrier family 7 member 9 Glycoprotein Slc7a9 16556 associated Amino acids Solute carrier family 7 member 10 Glycoprotein Slc7a10 16557 associated Amino acids Solute carrier family 7 member 11 Glycoprotein Slc7a11 16558 associated Amino acids Solute carrier family 7 member 13 Glycoprotein Slc7a13 16559 associated Amino acids Solute carrier family 7 member 14 Cationic Amino acids Slc7a14 16560 Solute carrier family 3 member 2 Amino acid Slc3a2 16561 Calcium transport protein 2 Cationic Amino acid CAT2 16562 (arginine) Calcium transport protein 3 Cationic Amino acid CAT3 16563 (arginine) Calcium transport protein 4 Cationic Amino acid CAT4



16564 (arginine) Bromodomain adjacent to zinc finger Amino acid (arginine) BAZ1B 16565 domain protein 1B PC4 and SFRS1-interacting protein Amino acid (arginine) PSIP1 16566 Translin Amino acid (arginine) TSN 16567 G-protein-coupled receptors Fatty Acid and GPCRs Cholesterol T-cell Receptor, subunit alpha Fatty Acid and TCR alpha 16568 Cholesterol T-cell Receptor, subunit beta Fatty Acid and TCR beta 16569 Cholesterol T-cell Receptor, subunit zeta Fatty Acid and TCR zeta 16570 Cholesterol T-cell Receptor, subunit CD3 epsilon Fatty Acid and TCR CD3 epsilon 16571 Cholesterol T-cell Receptor, subunit CD3 Fatty Acid and TCR CD3 gamma 16572 gamma Cholesterol T-cell Receptor, subunit CD3 delta Fatty Acid and TCR CD3 delta 16573 Cholesterol peroxisome proliferator-activated Fatty Acid and PPARs receptors Cholesterol AMP-activated protein kinase Energy homeostasis AMPK 16574-16575 (intracellular AMP to ATP ratio) P2X purinoceptor 7 Redox homeostasis P2X7 16576 (828) In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding proteins that confer sensitivity to a cancer therapy, including a monoclonal antibody, to produce an armored T-cell of the disclosure. Thus, an armored T-cell of the disclosure can function and may demonstrate superior function or efficacy whilst in the presence of a cancer therapy (e.g., a chemotherapy, a monoclonal antibody therapy, or another anti-tumor treatment). Proteins involved in conferring sensitivity to a cancer therapy include, but are not limited to, the exemplary proteins provided in Table 6.

(829) TABLE-US-00227 TABLE 6 Exemplary Proteins that Confer Sensitivity to a Cancer Therapeutic. Full Name Abbreviation SEQ ID NO: Copper-transporting ATPase 2 ATP7B 16577 Breakpoint cluster region protein BCR 16578 Abelson tyrosine-protein kinase 1 ABL 16579 Breast cancer resistance protein BCRP 16580 Breast cancer type 1 susceptibility protein BRCA1 16581 Breast cancer type 2 susceptibility protein BRCA2 16582 CAMPATH-1 antigen CD52 16583 Cytochrome P450 2D6 CYP2D6 16584 Deoxycytidine kinase dCK 16585 Dihydrofolate reductase DHFR 16586 Dihydropyrimidine dehydrogenase [NADP(+)] DPYD 16587 Epidermal growth factor receptor EGFR 16588 DNA excision repair protein ERCC-1 ERCC1 16589 Estrogen Receptor ESR 16590 Low affinity immunoglobulin gamma Fc region FCGR3A 16591 receptor III-A Receptor tyrosine-protein kinase erbB-2 HER2 or ERBB2 16592 Insulin-like growth factor 1 receptor IGF1R 16593 GTPase KRas KRAS 16594 Multidrug resistance protein 1 MDR1 or ABCB1 16595 Methylated-DNA--protein-cysteine methyltransferase MGMT 16596 Multidrug resistance-associated protein 1 MRP1 or ABCC1 16597 Progesterone Receptor PGR 16598 Regulator of G-protein signaling 10 RGS10 16599 Suppressor of cytokine signaling 3 SOCS-3 16600 Thymidylate synthase TYMS 16601 UDP-glucuronosyltransferase 1-1 UGT1A1 16602

(830) In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding a growth advantage factor to produce an armored T-cell. Silencing or reducing expression of an oncogene can confer a growth advantage for an armored T-cell of the disclosure. For example, silencing or reducing expression (e.g., disrupting expression) of a TET2 gene during a CAR-T manufacturing process results in the generation of an armored CAR-T with a significant capacity for expansion and subsequent eradication of a tumor when compared to a non-armored CAR-T lacking this capacity for expansion. This strategy may be coupled to a safety switch (e.g., an iC9 safety switch of the disclosure), which allows for the targeted disruption of an armored CAR-T-cell in the event of an adverse reaction from a subject or uncontrolled growth of the armored CAR-T. Exemplary growth advantage factors include, but are not limited to, the factors provided in Table 7.

(831) TABLE-US-00228 TABLE 7 Exemplary Growth Advantage Factors. Full Name Abbreviation SEQ ID NO: Ten Eleven Translocation 2 TET2 16603 DNA (cytosine-5)-methyltransferase 3A DNMT3A 16604 Transforming protein RhoA RHOA 16605 Proto-oncogene vav VAV1 16606 Rhombotin-2 LMO2 16607 T-cell acute lymphocytic leukemia TAL1 16608 protein 1 Suppressor of cytokine signaling 1 SOCS1 16609 herpes virus entry mediator HVEM 16610 T cell death-associated gene 8 TDAG8 16611 BCL6 corepressor BCOR 16612 B and T cell attenuator BTLA 16613 SPARC-like protein 1 SPARCL1 16614 Msh homeobox 1-like protein MSX1 16615

Armored T-Cells “Null or Switch Receptor” Strategy

(832) In some embodiments, a T-cell of the disclosure is modified to express a modified/chimeric

checkpoint receptor to produce an armored T-cell of the disclosure.

(833) In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor. A null receptor, decoy receptor or dominant negative receptor of the disclosure may be modified/chimeric receptor/protein. A null receptor, decoy receptor or dominant negative receptor of the disclosure may be truncated for expression of the intracellular signaling domain. Alternatively, or in addition, a null receptor, decoy receptor or dominant negative receptor of the disclosure may be mutated within an intracellular signaling domain at one or more amino acid positions that are determinative or required for effective signaling. Truncation or mutation of null receptor, decoy receptor or dominant negative receptor of the disclosure may result in loss of the receptor's capacity to convey or transduce a checkpoint signal to the cell or within the cell.

(834) For example, a dilution or a blockage of an immunosuppressive checkpoint signal from a PD-L1 receptor expressed on the surface of a tumor cell may be achieved by expressing a modified/chimeric PD-1 null receptor on the surface of an armored T-cell of the disclosure, which effectively competes with the endogenous (non-modified) PD-1 receptors also expressed on the surface of the armored T-cell to reduce or inhibit the transduction of the immunosuppressive checkpoint signal through endogenous PD-1 receptors of the armored T cell. In this exemplary embodiment, competition between the two different receptors for binding to PD-L1 expressed on the tumor cell reduces or diminishes a level of effective checkpoint signaling, thereby enhancing a therapeutic potential of the armored T-cell expressing the PD-1 null receptor.

(835) In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor that is a transmembrane receptor.

(836) In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor that is a membrane-associated or membrane-linked receptor/protein.

(837) In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor that is an intracellular receptor/protein.

(838) In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor that is an intracellular receptor/protein. Exemplary null, decoy, or dominant negative intracellular receptors/proteins of the disclosure include, but are not limited to, signaling components downstream of an inhibitory checkpoint signal (as provided, for example, in Tables 1 and 2), a transcription factor (as provided, for example, in Table 3), a cytokine or a cytokine receptor, a chemokine or a chemokine receptor, a cell death or apoptosis receptor/ligand (as provided, for example, in Table 4), a metabolic sensing molecule (as provided, for example, in Table 5), a protein conferring sensitivity to a cancer therapy (as provided, for example, in Table 6), and an oncogene or a tumor suppressor gene (as provided, for example, in Table 7). Exemplary cytokines, cytokine receptors, chemokines and chemokine receptors of the disclosure include, but are not limited to, the cytokines and cytokine receptors as well as chemokines and chemokine receptors provided in Table 8.

(839) TABLE-US-00229 TABLE 8 Exemplary Cytokines, Cytokine receptors, Chemokines and Chemokine Receptors. Full Name Abbreviation SEQ ID NO: 4-1BB Ligand 4-1BBL 16616 Tumor necrosis factor receptor Apo3 or TNFRSF25 16617 superfamily member 25 Tumor necrosis factor receptor APRIL or TNFRSF13 16618 superfamily member 13 Bcl2-associated agonist of cell death Bcl-xL or BAD 16619 Tumor necrosis factor receptor BCMA or TNFRSF17 16620 superfamily member 17 C-C motif chemokine 1 CCL1 16621 C-C motif chemokine 11 CCL11 16622 C-C motif chemokine 13 CCL13 16623 C-C motif chemokine 14 CCL14 16624 C-C motif chemokine 15 CCL15 16625 C-C motif chemokine 16 CCL16 16626 C-C motif chemokine 17 CCL17 16627 C-C motif chemokine 18 CCL18 16628 C-C motif chemokine 19 CCL19 16629 C-C motif chemokine 2 CCL2 16630 C-C motif chemokine 20 CCL20 16631 C-C motif chemokine 21 CCL21 16632 C-C motif chemokine 22 CCL22 16633 C-C motif chemokine 23 CCL23 16634 C-C motif chemokine 24 CCL24 16635 C-C motif chemokine 25 CCL25 16636 C-C motif chemokine 26 CCL26 16637 C-C motif chemokine 27 CCL27 16638 C-C motif chemokine 28 CCL28 16639 C-C motif chemokine 3 CCL3 16640 C-C motif chemokine 4 CCL4 16641 C-C motif chemokine 5 CCL5 16642 C-C motif chemokine 7 CCL7 16643 C-C motif chemokine 8 CCL8 16644 C-C chemokine receptor type 1 CCR1 16645 C-C chemokine

receptor type 10 CCR11 16646 C-C chemokine receptor type 11 CCR11 16647 C-C chemokine receptor type 2 CCR2 16648 C-C chemokine receptor type 3 CCR3 16649 C-C chemokine receptor type 4 CCR4 16650 C-C chemokine receptor type 5 CCR5 16651 C-C chemokine receptor type 6 CCR6 16652 C-C chemokine receptor type 7 CCR7 16653 C-C chemokine receptor type 8 CCR8 16654 C-C chemokine receptor type 9 CCR9 16655 Granulocyte colony-stimulating factor CD114 or CSF3R 16656 receptor Macrophage colony-stimulating factor 1 CD115 or CSF1R 16657 receptor Granulocyte-macrophage colony- CD116 or CSF2RA 16658 stimulating factor receptor subunit alpha Mast/stem cell growth factor receptor CD117 or KIT 16659 Kit Leukemia inhibitory factor receptor CD118 or LIFR 16660 Tumor necrosis factor receptor CD120a or TNFRSF1A 16661 superfamily member 1A Tumor necrosis factor receptor CD120b or TNFRSF1B 16662 superfamily member 1B Interleukin-1 receptor type 1 CD121a or IL1R1 16663 Interleukin-2 receptor subunit beta CD122 or IL2RB 16664 Interleukin-3 receptor subunit alpha CD123 or IL3RA 16665 Interleukin-4 receptor subunit alpha CD124 or IL4R 16666 Interleukin-6 receptor subunit alpha CD126 or IL6R 16667 Interleukin-7 receptor subunit alpha CD127 or IL7R 16668 Interleukin-6 receptor subunit beta CD130 or IL6ST 16669 Cytokine receptor common subunit CD132 or IL2RG 16670 gamma Tumor necrosis factor ligand CD153 or TNFSF8 16671 superfamily member 8 CD40 ligand CD154 or CD40L 16672 Tumor necrosis factor ligand CD178 or FASLG 16673 superfamily member 6 Interleukin-12 receptor subunit beta-1 CD212 or IL12RB1 16674 Interleukin-13 receptor subunit alpha-1 CD213a1 or IL13RA1 16675 Interleukin-13 receptor subunit alpha-2 CD213a2 or IL13RA2 16676 Interleukin-2 receptor subunit alpha CD25 or IL2RA 16677 CD27 antigen CD27 16678 Tumor necrosis factor receptor CD30 or TNFRSF8 16679 superfamily member 8 T-cell surface glycoprotein CD4 CD4 16680 Tumor necrosis factor receptor CD40 or TNFRSF5 16681 superfamily member 5 CD70 antigen CD70 16682 Tumor necrosis factor receptor CD95 or FAS or 16683 superfamily member 6 TNFRSF6 Granulocyte-macrophage colony- CDw116 or CSF2RA 16684 stimulating factor receptor subunit alpha Interferon gamma receptor 1 CDw119 or IFNGR1 16685 Interleukin-1 receptor type 2 CDw121b or IL1R2 16686 Interleukin-5 receptor subunit alpha CDw125 or IL5RA 16687 Cytokine receptor common subunit beta CDw131 or CSF2RB 16688 Tumor necrosis factor receptor CDw137 or TNFRSF9 16689 superfamily member 9 Interleukin-10 receptor CDw210 or IL10R 16690 Interleukin-17 receptor A CDw217 or IL17RA 16691 C-X3-C motif chemokine 1 CX3CL1 16692 CX3C chemokine receptor 1 CX3CR1 16693 C-X-C motif chemokine 1 CXCL1 16694 C-X-C motif chemokine 10 CXCL10 16695 C-X-C motif chemokine 11 CXCL11 16696 C-X-C motif chemokine 12 CXCL12 16697 C-X-C motif chemokine 13 CXCL13 16698 C-X-C motif chemokine 14 CXCL14 16699 C-X-C motif chemokine 16 CXCL16 16700 C-X-C motif chemokine 2 CXCL2 16701 C-X-C motif chemokine 3 CXCL3 16702 C-X-C motif chemokine 4 CXCL4 16703 C-X-C motif chemokine 5 CXCL5 16704 C-X-C motif chemokine 6 CXCL6 16705 C-X-C motif chemokine 7 CXCL7 16706 C-X-C motif chemokine 8 CXCL8 16707 C-X-C motif chemokine 9 CXCL9 16708 C-X-C chemokine receptor type 1 CXCR1 16709 C-X-C chemokine receptor type 2 CXCR2 16710 C-X-C chemokine receptor type 3 CXCR3 16711 C-X-C chemokine receptor type 4 CXCR4 16712 C-X-C chemokine receptor type 5 CXCR5 16713 C-X-C chemokine receptor type 6 CXCR6 16714 C-X-C chemokine receptor type 7 CXCR7 16715 Atypical chemokine receptor 1 DARC or ACKR1 16716 Erythropoietin Epo 16717 Erythropoietin receptor EpoR 16718 Receptor-type tyrosine-protein kinase Flt-3 16719 FLT3 FLT3 Ligand Flt-3L 16720 Granulocyte colony-stimulating factor G-CSF or GSF3R 16721 receptor Tumor necrosis factor receptor GITR or TNFRSF18 16722 superfamily member 18 GITR Ligand GITRL 16723 Cytokine receptor common subunit beta GM-CSF or CSF2RB 16724 Interleukin-6 receptor subunit beta gp130 or IL6ST 16725 Tumor necrosis factor receptor HVEM or TNFRSF14 16726 superfamily member 14 Interferon gamma IENy 16727 Interferon gamma receptor 2 IFNGR2 16728 Interferon-alpha IFN- $\alpha$  16729 Interferon-beta IFN- $\beta$  16730 Interleukin-1 alpha IL1 16731 Interleukin-10 IL10 16732 Interleukin-10 receptor IL10R 16733 Interleukin-11 IL-11 16734 Interleukin-11 receptor alpha IL-11Ra 16735 Interleukin-12 IL12 16736 Interleukin-13 IL13 16737 Interleukin-13 receptor IL13R 16738 Interleukin-14 IL-14 16739 Interleukin-15 IL15 16740 Interleukin-15 receptor alpha IL-15Ra 16741 Interleukin-16 IL-16 16742 Interleukin-17 IL17 16743 Interleukin-17 receptor IL17R 16744 Interleukin-18 IL18 16745 Interleukin-1 receptor alpha IL-1RA

16746 Interleukin-1 alpha IL-1 $\alpha$  16747 Interleukin-1beta IL-1 $\beta$  16748 Interleukin-2 IL2 16749 Interleukin-20 IL-20 16750 Interleukin-20 receptor alpha IL-20R $\alpha$  16751 Interleukin-20 receptor beta IL-20R $\beta$  16752 Interleukin-21 IL21 16753 Interleukin-3 IL-3 16754 Interleukin-35 IL35 16755 Interleukin-4 IL4 16756 Interleukin-4 receptor IL4R 16757 Interleukin-5 IL5 16758 Interleukin-5 receptor IL5R 16759 Interleukin-6 IL6 16760 Interleukin-6 receptor IL6R 16761 Interleukin-7 IL7 16762 Interleukin-9 receptor IL-9R 16763 Leukemia inhibitory factor LIF 16764 Leukemia inhibitory factor receptor LIFR 16765 tumor necrosis factor superfamily LIGHT or TNFSF14 16766 member 14 Tumor necrosis factor receptor LT $\beta$ R or TNFRSF3 16767 superfamily member 3 Lymphotoxin-beta LT- $\beta$  16768 Macrophage colony-stimulating factor 1 M-CSF 16769 Tumor necrosis factor receptor OPG or TNFRSF11B 16770 superfamily member 11B Oncostatin-M OSM 16771 Oncostatin-M receptor OSMR 16772 Tumor necrosis factor receptor OX40 or TNFRSF4 16773 superfamily member 4 Tumor necrosis factor ligand OX40L or TNFSF4 16774 superfamily member 4 Tumor necrosis factor receptor RANK or TNFRSF11A 16775 superfamily member 11A Kit Ligand SCF or KITLG 16776 Tumor necrosis factor receptor TACI or TNFRSF13B 16777 superfamily member 13B Tumor necrosis factor ligand TALL-1 or TNFSF13B 16778 superfamily member 13B TGF-beta receptor type-1 TGF- $\beta$ R1 16779 TGF-beta receptor type-2 TGF- $\beta$ R2 16780 TGF-beta receptor type-3 TGF- $\beta$ R3 16781 Transforming growth factor beta-1 TGF- $\beta$ 1 16782 Transforming growth factor beta-2 TGF- $\beta$ 2 16783 Transforming growth factor beta-3 TGF- $\beta$ 3 16784 Tumor necrosis factor alpha TNF or TNF- $\alpha$  16785 Tumor necrosis factor beta TNF- $\beta$  16786 Thyroid peroxidase Tpo 16787 Thyroid peroxidase receptor TpoR 16788 Tumor necrosis factor ligand TRAIL or TNFSF10 16789 superfamily member 10 Tumor necrosis factor receptor TRAILR1 or 16790 superfamily member 10A TNFRSF10A Tumor necrosis factor receptor TRAILR2 or 16791 superfamily member 10B TNFRSF10B Tumor necrosis factor ligand TRANCE or TNFSF11 16792 superfamily member 11 Tumor necrosis factor ligand TWEAK or TNFSF11 16793 superfamily member 12 Lymphotoxin XCL1 16794 Cytokine SCM-1 beta XCL2 16795

(840) In some embodiments, the modified/chimeric checkpoint receptor comprises a switch receptor. Exemplary switch receptors may comprise a modified/chimeric receptor/protein of the disclosure wherein a native or wild type intracellular signaling domain is switched or replaced with a different intracellular signaling domain that is either non-native to the protein and/or not a wild-type domain. For example, replacement of an inhibitory signaling domain with a stimulatory signaling domain would switch an immunosuppressive signal into an immunostimulatory signal. Alternatively, replacement of an inhibitory signaling domain with a different inhibitory domain can reduce or enhance the level of inhibitory signaling. Expression or overexpression, of a switch receptor can result in the dilution and/or blockage of a cognate checkpoint signal via competition with an endogenous wildtype checkpoint receptor (not a switch receptor) for binding to the cognate checkpoint receptor expressed within the immunosuppressive tumor microenvironment. Armored T cells of the disclosure may comprise a sequence encoding switch receptors of the disclosure, leading to the expression of one or more switch receptors of the disclosure, and consequently, altering an activity of an armored T-cell of the disclosure. Armored T cells of the disclosure may express a switch receptor of the disclosure that targets an intracellularly expressed protein downstream of a checkpoint receptor, a transcription factor, a cytokine receptor, a death receptor, a metabolic sensing molecule, a cancer therapy, an oncogene, and/or a tumor suppressor protein or gene of the disclosure.

(841) Exemplary switch receptors of the disclosure may comprise or may be derived from a protein including, but are not limited to, the signaling components downstream of an inhibitory checkpoint signal (as provided, for example, in Tables 1 and 2), a transcription factor (as provided, for example, in Table 3), a cytokine or a cytokine receptor, a chemokine or a chemokine receptor, a cell death or apoptosis receptor/ligand (as provided, for example, in Table 4), a metabolic sensing molecule (as provided, for example, in Table 5), a protein conferring sensitivity to a cancer therapy (as provided, for example, in Table 6), and an oncogene or a tumor suppressor gene (as provided, for example, in Table 7). Exemplary cytokines, cytokine receptors, chemokines and chemokine receptors of the disclosure include, but are not limited to, the cytokines and cytokine receptors as well as chemokines and chemokine receptors provided in Table 8.

(842) Armored T-Cells—Conditional Gene Expression System

(843) In some embodiments, a T-cell of the disclosure is modified to express chimeric ligand receptor (CLR) or a chimeric antigen receptor (CAR) that mediates conditional gene expression to produce an armored T-cell of the disclosure. The combination of the CLR/CAR and the conditional gene expression system in the nucleus of the armored T cell constitutes a synthetic gene expression system that is conditionally activated upon binding of cognate ligand(s) with CLR or cognate antigen(s) with CAR. This system may help to ‘armor’ or enhance therapeutic potential of modified T cells by reducing or limiting synthetic gene expression at the site of ligand or antigen binding, at or within the tumor environment for example.

(844) Exogenous Receptors

(845) In some embodiments, the armored T-cell comprises a composition comprising (a) an inducible transgene construct, comprising a sequence encoding an inducible promoter and a sequence encoding a transgene, and (b) a receptor construct, comprising a sequence encoding a constitutive promoter and a sequence encoding an exogenous receptor, such as a CLR or CAR, wherein, upon integration of the construct of (a) and the construct of (b) into a genomic sequence of a cell, the exogenous receptor is expressed, and wherein the exogenous receptor, upon binding a ligand or antigen, transduces an intracellular signal that targets directly or indirectly the inducible promoter regulating the expression of the inducible transgene (a) to modify gene expression.

(846) In some embodiments of a synthetic gene expression system of the disclosure, the composition modifies gene expression by decreasing gene expression. In some embodiments, the composition modifies gene expression by transiently modifying gene expression (e.g., for the duration of binding of the ligand to the exogenous receptor). In some embodiments, the composition modifies gene expression acutely (e.g., the ligand reversibly binds to the exogenous receptor). In some embodiments, the composition modifies gene expression chronically (e.g., the ligand irreversibly binds to the exogenous receptor).

(847) In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises an endogenous receptor with respect to the genomic sequence of the cell. Exemplary receptors include, but are not limited to, intracellular receptors, cell-surface receptors, transmembrane receptors, ligand-gated ion channels, and G-protein coupled receptors.

(848) In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the non-naturally occurring receptor is a synthetic, modified, recombinant, mutant or chimeric receptor. In some embodiments, the non-naturally occurring receptor comprises one or more sequences isolated or derived from a T-cell receptor (TCR). In some embodiments, the non-naturally occurring receptor comprises one or more sequences isolated or derived from a scaffold protein. In some embodiments, including those wherein the non-naturally occurring receptor does not comprise a transmembrane domain, the non-naturally occurring receptor interacts with a second transmembrane, membrane-bound and/or an intracellular receptor that, following contact with the non-naturally occurring receptor, transduces an intracellular signal.

(849) In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the non-naturally occurring receptor is a synthetic, modified, recombinant, mutant or chimeric receptor. In some embodiments, the non-naturally occurring receptor comprises one or more sequences isolated or derived from a T-cell receptor (TCR). In some embodiments, the non-naturally occurring receptor comprises one or more sequences isolated or derived from a scaffold protein. In some embodiments, the non-naturally occurring receptor comprises a transmembrane domain. In some embodiments, the non-naturally occurring receptor interacts with an intracellular receptor that transduces an intracellular signal. In some embodiments, the non-naturally occurring receptor comprises an intracellular signaling domain. In some embodiments, the non-naturally occurring receptor is a chimeric ligand receptor (CLR). In some embodiments, the CLR is a chimeric antigen receptor (CAR).

(850) In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the CLR is a chimeric antigen

receptor (CAR). In some embodiments, the chimeric ligand receptor comprises (a) an ectodomain comprising a ligand recognition region, wherein the ligand recognition region comprises at least scaffold protein; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the ectodomain of (a) further comprises a signal peptide. In some embodiments, the ectodomain of (a) further comprises a hinge between the ligand recognition region and the transmembrane domain.

(851) In some embodiments of the CLR/CARs of the disclosure, the signal peptide comprises a sequence encoding a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR signal peptide. In some embodiments, the signal peptide comprises a sequence encoding a human CD8α signal peptide. In some embodiments, the signal peptide comprises an amino acid sequence comprising MALPVTALLPLALLHAARP (SEQ ID NO: 17503). In some embodiments, the signal peptide is encoded by a nucleic acid sequence comprising atggcactgccagtcaccgcctgtgctgcctctggctgtgctgcacgcagctagacca (SEQ ID NO: 17504).

(852) In some embodiments of the CLR/CARs of the disclosure, the transmembrane domain comprises a sequence encoding a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR transmembrane domain. In some embodiments, the transmembrane domain comprises a sequence encoding a human CD8α transmembrane domain. In some embodiments, the transmembrane domain comprises an amino acid sequence comprising IYIWAPLAGTCGVLLSLVITLYC (SEQ ID NO: 17505). In some embodiments, the transmembrane domain is encoded by a nucleic acid sequence comprising

(853) TABLE-US-00230 (SEQ ID NO: 17506)

atctacattggggcaccactggccgggacctgtggagtgtgctgtgag cctggatcacactgtactgc.

(854) In some embodiments of the CLR/CARs of the disclosure, the endodomain comprises a human CD3ζ endodomain. In some embodiments, the at least one costimulatory domain comprises a human 4-1BB, CD28, CD3ζ, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof. In some embodiments, the at least one costimulatory domain comprises a human CD3ζ and/or a 4-1BB costimulatory domain. In some embodiments, the CD3ζ costimulatory domain comprises an amino acid sequence comprising

RVKFSRSADAPAYKQGQNQLYNELNLGRREEYDVLDKRRGRDPGEMGGKPRRKNPQ  
EGLYNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQALP PR (SEQ ID NO: 17507). In some embodiments, the CD3ζ costimulatory domain is encoded by a nucleic acid sequence comprising

cgcgtgaagttagtcgatcagcagatgccccagcttacaacagggacagaaccagctgtataacgagctgaatctgggcccgcga  
gaggaatatgacgtgctggataagcggagaggacgcgacccccgaaatgggaggcaagcccaggcgcaaaaccctcaggaagg  
cctgtataacgagctgcagaaggacaaaatggcagaagcctattctgagatcgcatgaagggggagcgacggagaggcaaagg  
gcacgatgggctgtaccaggactgagcaccgccacaaaggacacctatgatgctctgcatatgcaggcactgcctccaagg (SEQ ID NO: 17508). In some embodiments, the 4-1BB costimulatory domain comprises an amino acid sequence comprising KRGRKKLLYIFKQPFMRPVQTTQEEDGCSRFPEEEGGCEL (SEQ ID NO: 17509).

In some embodiments, the 4-1BB costimulatory domain is encoded by a nucleic acid sequence comprising

aagagaggcaggaagaaactgctgtatatttcaaacagccctcatgcgccccgtgcagactaccagagggaagacgggtgctcc  
tgtcgattccctgaggaagaggaaggcgggtgtgagctg (SEQ ID NO: 17510). In some embodiments, the 4-1BB costimulatory domain is located between the transmembrane domain and the CD3ζ costimulatory domain.

(855) In some embodiments of the CLR/CARs of the disclosure, the hinge comprises a sequence derived from a human CD8α, IgG4, and/or CD4 sequence. In some embodiments, the hinge comprises a sequence derived from a human CD8α sequence. In some embodiments, the hinge comprises an amino acid sequence comprising

(856) TABLE-US-00231 (SEQ ID NO: 17511)

TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACD.

In some embodiments, the hinge is encoded by a nucleic acid sequence comprising

actaccacaccagcacctagaccaccaactccagctccaaccatcgcgagtcagcccctgagctgagacctgaggcctgcaggcc

agctgcaggaggagctgtgcacaccaggggctgccttgacttcgcctgcgac (SEQ ID NO: 17512). In some embodiments, the hinge is encoded by

(857) TABLE-US-00232 (SEQ ID NO: 17513)

ACCACAACCCCTGCCCCCAGACCTCCCACACCCGCCCTACCATCGCGAG  
TCAGCCCCTGAGTCTGAGACCTGAGGCCTGCAGGCCAGCTGCAGGAGGAG  
CTGTGCACACCAGGGGCCTGGACTTCGCCTGCGAC.

(858) In some embodiments, the at least one protein scaffold specifically binds the ligand.

(859) In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the CLR is a chimeric antigen receptor (CAR). In some embodiments, the chimeric ligand receptor comprises (a) an ectodomain comprising a ligand recognition region, wherein the ligand recognition region comprises at least scaffold protein; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the at least one protein scaffold comprises an antibody, an antibody fragment, a single domain antibody, a single chain antibody, an antibody mimetic, or a Centyrin (referred to herein as a CARTyrin). In some embodiments, the ligand recognition region comprises one or more of an antibody, an antibody fragment, a single domain antibody, a single chain antibody, an antibody mimetic, and a Centyrin. In some embodiments, the single domain antibody comprises or consists of a VHH or a VH (referred to herein as a VCAR). In some embodiments, the single domain antibody comprises or consists of a VHH or a VH comprising human complementarity determining regions (CDRs). In some embodiments, the VH is a recombinant or chimeric protein. In some embodiments, the VH is a recombinant or chimeric human protein. In some embodiments, the antibody mimetic comprises or consists of an affibody, an affililn, an affimer, an affitin, an alphabody, an anticalin, an avimer, a DARPin, a Fynomer, a Kunitz domain peptide or a monobody. In some embodiments, the Centyrin comprises or consists of a consensus sequence of at least one fibronectin type III (FN3) domain.

(860) In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the CLR is a chimeric antigen receptor (CAR). In some embodiments, the chimeric ligand receptor comprises (a) an ectodomain comprising a ligand recognition region, wherein the ligand recognition region comprises at least scaffold protein; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the Centyrin comprises or consists of a consensus sequence of at least one fibronectin type III (FN3) domain. In some embodiments, the at least one fibronectin type III (FN3) domain is derived from a human protein. In some embodiments, the human protein is Tenascin-C. In some embodiments, the consensus sequence comprises

LPAPKNLVVSEVTEDSLRLSWTAPDAAAFDSFLIQYQESEKVGGEAINLTVPGSERSYDL

TGLKPGTEYTVSIYGVKGGHRSNPLSAEFTT (SEQ ID NO: 17514). In some embodiments, the consensus sequence comprises

MLPAPKNLVVSEVTEDSLRLSWTAPDAAAFDSFLIQYQESEKVGGEAINLTVPGSERSYD

LTGLKPGTEYTVSIYGVKGGHRSNPLSAEFTT (SEQ ID NO: 17515). In some embodiments, the consensus sequence is modified at one or more positions within (a) a A-B loop comprising or consisting of the amino acid residues TEDS (SEQ ID NO: 17516) at positions 13-16 of the consensus sequence; (b) a B-C loop comprising or consisting of the amino acid residues TAPDAAF (SEQ ID NO: 17517) at positions 22-28 of the consensus sequence; (c) a C-D loop comprising or consisting of the amino acid residues SEKVGGE (SEQ ID NO: 17518) at positions 38-43 of the consensus sequence; (d) a D-E loop comprising or consisting of the amino acid residues GSER (SEQ ID NO: 17519) at positions 51-54 of the consensus sequence; (e) a E-F loop comprising or consisting of the amino acid residues GLKPG (SEQ ID NO: 17520) at positions 60-64 of the consensus sequence; (f) a F-G loop comprising or consisting of the amino acid residues KGGHRSN (SEQ ID NO: 17521) at positions 75-81 of the consensus sequence; or (g) any combination of (a)-(f). In some embodiments, the Centyrin comprises a consensus sequence of at least 5 fibronectin type III (FN3) domains. In some embodiments, the Centyrin comprises a consensus sequence of at least 10 fibronectin type III (FN3) domains. In some embodiments, the Centyrin comprises a consensus sequence of at least 15 fibronectin type III (FN3)

domains. In some embodiments, the scaffold binds an antigen with at least one affinity selected from a K<sub>sub</sub>.D of less than or equal to 10<sup>sup</sup>.-9M, less than or equal to 10<sup>sup</sup>.-10M, less than or equal to 10<sup>sup</sup>.-11M, less than or equal to 10<sup>sup</sup>.-12M, less than or equal to 10<sup>sup</sup>.-13M, less than or equal to 10<sup>sup</sup>.-14M, and less than or equal to 10<sup>sup</sup>.-15M. In some embodiments, the K<sub>sub</sub>.D is determined by surface plasmon resonance.

#### (861) Inducible Promoters

(862) In some embodiments of the compositions of the disclosure, the sequence encoding the inducible promoter of (a) comprises a sequence encoding an NFκB promoter. In some embodiments of the compositions of the disclosure, the sequence encoding the inducible promoter of (a) comprises a sequence encoding an interferon (IFN) promoter or a sequence encoding an interleukin-2 promoter. In some embodiments, the interferon (IFN) promoter is an IFNγ promoter. In some embodiments of the compositions of the disclosure, the inducible promoter is isolated or derived from the promoter of a cytokine or a chemokine. In some embodiments, the cytokine or chemokine comprises IL2, IL3, IL4, IL5, IL6, IL10, IL12, IL13, IL17A/F, IL21, IL22, IL23, transforming growth factor beta (TGFβ), colony stimulating factor 2 (GM-CSF), interferon gamma (IFNγ), Tumor necrosis factor (TNFα), LTα, perform, Granzyme C (Gzmc), Granzyme B (Gzmb), C—C motif chemokine ligand 5 (CCL5), C—C motif chemokine ligand 4 (Ccl4), C—C motif chemokine ligand 3 (Ccl3), X-C motif chemokine ligand 1 (Xcl1) and LIF interleukin 6 family cytokine (Lif).

(863) In some embodiments of the compositions of the disclosure, the inducible promoter is isolated or derived from the promoter of a gene comprising a surface protein involved in cell differentiation, activation, exhaustion and function. In some embodiments, the gene comprises CD69, CD71, CTLA4, PD-1, TIGIT, LAG3, TIM-3, GITR, MHCII, COX-2, FASL and 4-1BB.

(864) In some embodiments of the compositions of the disclosure, the inducible promoter is isolated or derived from the promoter of a gene involved in CD metabolism and differentiation. In some embodiments of the compositions of the disclosure, the inducible promoter is isolated or derived from the promoter of Nr4a1, Nr4a3, Tnfrsf9 (4-1BB), Sema7a, Zfp3612, Gadd45b, Dusp5, Dusp6 and Neto2.

#### (865) Inducible Transgene

(866) In some embodiments, the inducible transgene construct comprises or drives expression of a signaling component downstream of an inhibitory checkpoint signal (as provided, for example, in Tables 1 and 2), a transcription factor (as provided, for example, in Table 3), a cytokine or a cytokine receptor, a chemokine or a chemokine receptor, a cell death or apoptosis receptor/ligand (as provided, for example, in Table 4), a metabolic sensing molecule (as provided, for example, in Table 5), a protein conferring sensitivity to a cancer therapy (as provided, for example, in Table 6 or 9), and an oncogene or a tumor suppressor gene (as provided, for example, in Table 7). Exemplary cytokines, cytokine receptors, chemokines and chemokine receptors of the disclosure include, but are not limited to, the cytokines and cytokine receptors as well as chemokines and chemokine receptors provided in Table 8.

(867) TABLE-US-00233 TABLE 9 Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).  
Gene Name Gene Description Protein SEQ ID NO A1BG Alpha-1-B glycoprotein SEQ ID NOS: 1-2 A2M Alpha-2-macroglobulin SEQ ID NOS: 3-6 A2ML1 Alpha-2-macroglobulin-like 1 SEQ ID NOS: 7-12 A4GNT Alpha-1,4-N-acetylglucosaminyltransferase SEQ ID NO: 13 AADACL2 Arylacetamide deacetylase-like 2 SEQ ID NOS: 14-15 AANAT Aralkylamine N-acetyltransferase SEQ ID NOS: 16-19 ABCG1 ATP-binding cassette, sub-family G SEQ ID NOS: 20-26 (WHITE), member 1 ABHD1 Abhydrolase domain containing 1 SEQ ID NOS: 27-31 ABHD10 Abhydrolase domain containing 10 SEQ ID NOS: 32-35 ABHD14A Abhydrolase domain containing 14A SEQ ID NOS: 36-40 ABHD15 Abhydrolase domain containing 15 SEQ ID NO: 41 ABI3BP ABI family, member 3 (NESH) binding SEQ ID NOS: 42-63 protein AC008641.1 SEQ ID NO: 73 AC009133.22 SEQ ID NO: 76 AC009491.2 SEQ ID NO: 77 AC011513.3 SEQ ID NOS: 92-93 AC136352.5 SEQ ID NO: 88 AC145212.4 MaFF-interacting protein SEQ ID NO: 90 AC233755.1 SEQ ID NO: 91 ACACB Acetyl-CoA carboxylase beta SEQ ID NOS: 94-100 ACAN Aggrecan SEQ ID NOS: 101-108 ACE Angiotensin I converting enzyme SEQ ID NOS: 109-121 ACHE Acetylcholinesterase (Yt blood group) SEQ ID NOS: 122-134 ACP2 Acid phosphatase 2, lysosomal SEQ ID NOS: 135-142 ACP5 Acid phosphatase 5, tartrate resistant SEQ ID NOS: 143-151 ACP6 Acid phosphatase 6, lysophosphatidic



SEQ ID NOS: 152-158 ACPA phosphatase, prostate SEQ ID NOS: 163-167 ACR Acrosin SEQ ID NOS: 168-169 ACRBP Acrosin binding protein SEQ ID NOS: 170-174 ACRV1 Acrosomal vesicle protein 1 SEQ ID NOS: 175-178 ACSF2 Acyl-CoA synthetase family member 2 SEQ ID NOS: 179-187 ACTL10 Actin-like 10 SEQ ID NO: 188 ACVR1 Activin A receptor, type I SEQ ID NOS: 189-197 ACVR1C Activin A receptor, type IC SEQ ID NOS: 198-201 ACVRL1 Activin A receptor type II-like 1 SEQ ID NOS: 202-207 ACYP1 Acylphosphatase 1, erythrocyte (common) SEQ ID NOS: 208-213 type ACYP2 Acylphosphatase 2, muscle type SEQ ID NOS: 214-221 ADAM10 ADAM metalloproteinase domain 10 SEQ ID NOS: 230-237 ADAM12 ADAM metalloproteinase domain 12 SEQ ID NOS: 238-240 ADAM15 ADAM metalloproteinase domain 15 SEQ ID NOS: 241-252 ADAM17 ADAM metalloproteinase domain 17 SEQ ID NOS: 253-255 ADAM18 ADAM metalloproteinase domain 18 SEQ ID NOS: 256-260 ADAM22 ADAM metalloproteinase domain 22 SEQ ID NOS: 261-269 ADAM28 ADAM metalloproteinase domain 28 SEQ ID NOS: 270-275 ADAM29 ADAM metalloproteinase domain 29 SEQ ID NOS: 276-284 ADAM32 ADAM metalloproteinase domain 32 SEQ ID NOS: 285-291 ADAM33 ADAM metalloproteinase domain 33 SEQ ID NOS: 292-296 ADAM7 ADAM metalloproteinase domain 7 SEQ ID NOS: 297-300 ADAM8 ADAM metalloproteinase domain 8 SEQ ID NOS: 301-305 ADAM9 ADAM metalloproteinase domain 9 SEQ ID NOS: 306-311 ADAMDEC1 ADAM-like, decysin 1 SEQ ID NOS: 312-314 ADAMTS1 ADAM metalloproteinase with SEQ ID NOS: 315-318 thrombospondin type 1 motif, 1 ADAMTS10 ADAM metalloproteinase with SEQ ID NOS: 319-324 thrombospondin type 1 motif, 10 ADAMTS12 ADAM metalloproteinase with SEQ ID NOS: 325-327 thrombospondin type 1 motif, 12 ADAMTS13 ADAM metalloproteinase with SEQ ID NOS: 328-335 thrombospondin type 1 motif, 13 ADAMTS14 ADAM metalloproteinase with SEQ ID NOS: 336-337 thrombospondin type 1 motif, 14 ADAMTS15 ADAM metalloproteinase with SEQ ID NO: 338 thrombospondin type 1 motif, 15 ADAMTS16 ADAM metalloproteinase with SEQ ID NOS: 339-340 thrombospondin type 1 motif, 16 ADAMTS17 ADAM metalloproteinase with SEQ ID NOS: 341-344 thrombospondin type 1 motif, 17 ADAMTS18 ADAM metalloproteinase with SEQ ID NOS: 345-348 thrombospondin type 1 motif, 18 ADAMTS19 ADAM metalloproteinase with SEQ ID NOS: 349-352 thrombospondin type 1 motif, 19 ADAMTS2 ADAM metalloproteinase with SEQ ID NOS: 353-355 thrombospondin type 1 motif, 2 ADAMTS20 ADAM metalloproteinase with SEQ ID NOS: 356-359 thrombospondin type 1 motif, 20 ADAMTS3 ADAM metalloproteinase with SEQ ID NOS: 360-361 thrombospondin type 1 motif, 3 ADAMTS5 ADAM metalloproteinase with SEQ ID NO: 362 thrombospondin type 1 motif, 5 ADAMTS6 ADAM metalloproteinase with SEQ ID NOS: 363-364 thrombospondin type 1 motif, 6 ADAMTS7 ADAM metalloproteinase with SEQ ID NO: 365 thrombospondin type 1 motif, 7 ADAMTS8 ADAM metalloproteinase with SEQ ID NO: 366 thrombospondin type 1 motif, 8 ADAMTS9 ADAM metalloproteinase with SEQ ID NOS: 367-371 thrombospondin type 1 motif, 9 ADAMTSL1 ADAMTS-like 1 SEQ ID NOS: 372-382 ADAMTSL2 ADAMTS-like 2 SEQ ID NOS: 383-385 ADAMTSL3 ADAMTS-like 3 SEQ ID NOS: 386-387 ADAMTSL4 ADAMTS-like 4 SEQ ID NOS: 388-391 ADAMTSL5 ADAMTS-like 5 SEQ ID NOS: 392-397 ADCK1 AarF domain containing kinase 1 SEQ ID NOS: 398-402 ADCYAP1 Adenylate cyclase activating polypeptide 1 SEQ ID NOS: 403-404 (pituitary) ADCYAP1R1 Adenylate cyclase activating polypeptide 1 SEQ ID NOS: 405-411 (pituitary) receptor type I ADGRA3 Adhesion G protein-coupled receptor A3 SEQ ID NOS: 412-416 ADGRB2 Adhesion G protein-coupled receptor B2 SEQ ID NOS: 417-425 ADGRD1 Adhesion G protein-coupled receptor D1 SEQ ID NOS: 426-431 ADGRE3 Adhesion G protein-coupled receptor E3 SEQ ID NOS: 432-436 ADGRE5 Adhesion G protein-coupled receptor E5 SEQ ID NOS: 437-442 ADGRF1 Adhesion G protein-coupled receptor F1 SEQ ID NOS: 443-447 ADGRG1 Adhesion G protein-coupled receptor G1 SEQ ID NOS: 448-512 ADGRG5 Adhesion G protein-coupled receptor G5 SEQ ID NOS: 513-515 ADGRG6 Adhesion G protein-coupled receptor G6 SEQ ID NOS: 516-523 ADGRV1 Adhesion G protein-coupled receptor V1 SEQ ID NOS: 524-540 ADI1 Acireductone dioxygenase 1 SEQ ID NOS: 541-543 ADIG Adipogenin SEQ ID NOS: 544-547 ADIPOQ Adiponectin, C1Q and collagen domain SEQ ID NOS: 548-549 containing ADM Adrenomedullin SEQ ID NOS: 550-557 ADM2 Adrenomedullin 2 SEQ ID NOS: 558-559 ADM5 Adrenomedullin 5 (putative) SEQ ID NO: 560 ADPGK ADP-dependent glucokinase SEQ ID NOS: 561-570 ADPRHL2 ADP-ribosylhydrolase like 2 SEQ ID NO: 571 AEBP1 AE binding

protein 1 SEQ ID NOS: 572-579 AFM Afamin SEQ ID NOS: 584 AFP Alpha-fetoprotein SEQ ID NOS: 585-586 AGA Aspartylglucosaminidase SEQ ID NOS: 587-589 AGER Advanced glycosylation end product- SEQ ID NOS: 590-600 specific receptor AGK Acylglycerol kinase SEQ ID NOS: 601-606 AGPS Alkylglycerone phosphate synthase SEQ ID NOS: 607-610 AGR2 Anterior gradient 2, protein disulphide SEQ ID NOS: 611-614 isomerase family member AGR3 Anterior gradient 3, protein disulphide SEQ ID NOS: 615-617 isomerase family member AGRN Agrin SEQ ID NOS: 618-621 AGRP Agouti related neuropeptide SEQ ID NO: 622 AGT Angiotensinogen (serpin peptidase inhibitor, SEQ ID NO: 623 clade A, member 8) AGTPBP1 ATP/GTP binding protein 1 SEQ ID NOS: 624-627 AGTRAP Angiotensin II receptor-associated protein SEQ ID NOS: 628-635 AHCYL2 Adenosylhomocysteinase-like 2 SEQ ID NOS: 636-642 AHSG Alpha-2-HS-glycoprotein SEQ ID NOS: 643-644 AIG1 Androgen-induced 1 SEQ ID NOS: 645-653 AK4 Adenylate kinase 4 SEQ ID NOS: 654-657 AKAP10 A kinase (PRKA) anchor protein 10 SEQ ID NOS: 658-666 AKR1C1 Aldo-keto reductase family 1, member C1 SEQ ID NOS: 667-669 AL356289.1 SEQ ID NO: 677 AL589743.1 SEQ ID NO: 678 ALAS2 5'-aminolevulinate synthase 2 SEQ ID NOS: 684-691 ALB Albumin SEQ ID NOS: 692-701 ALDH9A1 Aldehyde dehydrogenase 9 family, member SEQ ID NO: 702 A1 ALDOA Aldolase A, fructose-bisphosphate SEQ ID NOS: 703-717 ALG1 ALG1, chitobiosyldiphosphodolichol beta- SEQ ID NOS: 718-723 mannosyltransferase ALG5 ALG5, dolichyl-phosphate beta- SEQ ID NOS: 724-725 glucosyltransferase ALG9 ALG9, alpha-1,2-mannosyltransferase SEQ ID NOS: 726-736 ALKBH1 AlkB homolog 1, histone H2A dioxygenase SEQ ID NOS: 746-748 ALKBH5 AlkB homolog 5, RNA demethylase SEQ ID NOS: 749-750 ALPI Alkaline phosphatase, intestinal SEQ ID NOS: 751-752 ALPL Alkaline phosphatase, liver/bone/kidney SEQ ID NOS: 753-757 ALPP Alkaline phosphatase, placental SEQ ID NO: 758 ALPPL2 Alkaline phosphatase, placental-like 2 SEQ ID NO: 759 AMBN Ameloblastin (enamel matrix protein) SEQ ID NOS: 760-762 AMBP Alpha-1-microglobulin/bikunin precursor SEQ ID NOS: 763-765 AMELX Amelogenin, X-linked SEQ ID NOS: 766-768 AMELY Amelogenin, Y-linked SEQ ID NOS: 769-770 AMH Anti-Mullerian hormone SEQ ID NO: 771 AMICA1 Adhesion molecule, interacts with CXADR SEQ ID NOS: 7348- antigen 1 7356 AMPD1 Adenosine monophosphate deaminase 1 SEQ ID NOS: 772-774 AMTN Amelotin SEQ ID NOS: 775-776 AMY1A Amylase, alpha 1A (salivary) SEQ ID NOS: 777-779 AMY1B Amylase, alpha 1B (salivary) SEQ ID NOS: 780-783 AMY1C Amylase, alpha 1C (salivary) SEQ ID NO: 784 AMY2A Amylase, alpha 2A (pancreatic) SEQ ID NOS: 785-787 AMY2B Amylase, alpha 2B (pancreatic) SEQ ID NOS: 788-792 ANG Angiogenin, ribonuclease, RNase A family, SEQ ID NOS: 793-794 5 ANGEL1 Angel homolog 1 (*Drosophila*) SEQ ID NOS: 795-798 ANGPT1 Angiopoietin 1 SEQ ID NOS: 799-803 ANGPT2 Angiopoietin 2 SEQ ID NOS: 804-807 ANGPT4 Angiopoietin 4 SEQ ID NO: 808 ANGPTL1 Angiopoietin-like 1 SEQ ID NOS: 809-811 ANGPTL2 Angiopoietin-like 2 SEQ ID NOS: 812-813 ANGPTL3 Angiopoietin-like 3 SEQ ID NO: 814 ANGPTL4 Angiopoietin-like 4 SEQ ID NOS: 815-822 ANGPTL5 Angiopoietin-like 5 SEQ ID NOS: 823-824 ANGPTL6 Angiopoietin-like 6 SEQ ID NOS: 825-827 ANGPTL7 Angiopoietin-like 7 SEQ ID NO: 828 ANK1 Ankyrin 1, erythrocytic SEQ ID NOS: 833-843 ANKDD1A Ankyrin repeat and death domain containing SEQ ID NOS: 844-850 1A ANKRD54 Ankyrin repeat domain 54 SEQ ID NOS: 851-859 ANKRD60 Ankyrin repeat domain 60 SEQ ID NO: 860 ANO7 Anoctamin 7 SEQ ID NOS: 861-864 ANO1 #N/A SEQ ID NO: 865 ANTXR1 Anthrax toxin receptor 1 SEQ ID NOS: 866-869 AOA AH Acyloxyacyl hydrolase (neutrophil) SEQ ID NOS: 870-874 AOC1 Amine oxidase, copper containing 1 SEQ ID NOS: 875-880 AOC2 Amine oxidase, copper containing 2 (retina- SEQ ID NOS: 881-882 specific) AOC3 Amine oxidase, copper containing 3 SEQ ID NOS: 883-889 AP000721.4 SEQ ID NO: 890 APBB1 Amyloid beta (A4) precursor protein- SEQ ID NOS: 891-907 binding, family B, member 1 (Fe65) APCDD1 Adenomatosis polyposis coli down- SEQ ID NOS: 908-913 regulated 1 APCS Amyloid P component, serum SEQ ID NO: 914 APELA Apelin receptor early endogenous ligand SEQ ID NOS: 915-917 APLN Apelin SEQ ID NO: 918 APLP2 Amyloid beta (A4) precursor-like protein 2 SEQ ID NOS: 919-928 APOA1 Apolipoprotein A-I SEQ ID NOS: 929-933 APOA1BP Apolipoprotein A-I binding protein SEQ ID NOS: 9177- 9179 APOA2 Apolipoprotein A-II SEQ ID NOS: 934-942 APOA4 Apolipoprotein A-IV SEQ ID NO: 943 APOA5 Apolipoprotein A-V SEQ ID NOS: 944-946 APOB Apolipoprotein B SEQ ID NOS: 947-948 APOC1 Apolipoprotein C-I SEQ ID NOS: 949-957 APOC2 Apolipoprotein C-II SEQ ID

NOS: 958-962 APOC3 Apolipoprotein C-III SEQ ID NOS: 959-966 APOC4 Apolipoprotein C-IV SEQ ID NOS: 967-968 APOC4- APOC4-APOC2 readthrough (NMD SEQ ID NOS: 969-970 APOC2 candidate) APOD Apolipoprotein D SEQ ID NOS: 971-974 APOE Apolipoprotein E SEQ ID NOS: 975-978 APOF Apolipoprotein F SEQ ID NO: 979 APOH Apolipoprotein H (beta-2-glycoprotein I) SEQ ID NOS: 980-983 APOL1 Apolipoprotein L, 1 SEQ ID NOS: 984-994 APOL3 Apolipoprotein L, 3 SEQ ID NOS: 995-1009 APOM Apolipoprotein M SEQ ID NOS: 1010- 1012 APOOL Apolipoprotein O-like SEQ ID NOS: 1013- 1015 ARCN1 Archain 1 SEQ ID NOS: 1016- 1020 ARFIP2 ADP-ribosylation factor interacting protein SEQ ID NOS: 1021- 2 1027 ARHGAP36 Rho GTPase activating protein 36 SEQ ID NOS: 1028- 1033 ARHGAP6 Rho GTPase activating protein 6 SEQ ID NOS: 1043- 1048 ARHGEF4 Rho guanine nucleotide exchange factor SEQ ID NOS: 1049- (GEF) 4 1059 ARL16 ADP-ribosylation factor-like 16 SEQ ID NOS: 1060- 1068 ARMC5 Armadillo repeat containing 5 SEQ ID NOS: 1069- 1075 ARNTL Aryl hydrocarbon receptor nuclear SEQ ID NOS: 1076- translocator-like 1090 ARSA Arylsulfatase A SEQ ID NOS: 1091- 1096 ARSB Arylsulfatase B SEQ ID NOS: 1097- 1100 ARSE Arylsulfatase E (chondrodysplasia punctata SEQ ID NOS: 1101- 1) 1104 ARSG Arylsulfatase G SEQ ID NOS: 1105- 1108 ARSI Arylsulfatase family, member I SEQ ID NOS: 1109- 1111 ARSK Arylsulfatase family, member K SEQ ID NOS: 1112- 1116 ART3 ADP-ribosyltransferase 3 SEQ ID NOS: 1117- 1124 ART4 ADP-ribosyltransferase 4 (Dombrock blood SEQ ID NOS: 1125- group) 1128 ART5 ADP-ribosyltransferase 5 SEQ ID NOS: 1129- 1133 ARTN Artemin SEQ ID NOS: 1134- 1144 ASAH1 N-acylsphingosine amidohydrolase (acid SEQ ID NOS: 1145- ceramidase) 1 1195 ASAH2 N-acylsphingosine amidohydrolase (non- SEQ ID NOS: 1196- lysosomal ceramidase) 2 1201 ASCL1 Achaete-scute family bHLH transcription SEQ ID NO: 1202 factor 1 ASIP Agouti signaling protein SEQ ID NOS: 1203- 1204 ASPN Asporin SEQ ID NOS: 1205- 1206 ASTL Astacin-like metallo-endopeptidase (M12 SEQ ID NO: 1207 family) ATAD5 ATPase family, AAA domain containing 5 SEQ ID NOS: 1208- 1209 ATAT1 Alpha tubulin acetyltransferase 1 SEQ ID NOS: 1210- 1215 ATG2A Autophagy related 2A SEQ ID NOS: 1216- 1218 ATG5 Autophagy related 5 SEQ ID NOS: 1219- 1227 ATMIN ATM interactor SEQ ID NOS: 1228- 1231 ATP13A1 ATPase type 13A1 SEQ ID NOS: 1232- 1234 ATP5F1 ATP synthase, H<sup>+</sup> transporting, SEQ ID NOS: 1235- mitochondrial Fo complex, subunit B1 1236 ATP6AP1 ATPase, H<sup>+</sup> transporting, lysosomal SEQ ID NOS: 1237- accessory protein 1 1244 ATP6AP2 ATPase, H<sup>+</sup> transporting, lysosomal SEQ ID NOS: 1245- accessory protein 2 1267 ATPAF1 ATP synthase mitochondrial F1 complex SEQ ID NOS: 1268- assembly factor 1 1278 AUH AU RNA binding protein/enoyl-CoA SEQ ID NOS: 1279- hydratase 1280 AVP Arginine vasopressin SEQ ID NO: 1281 AXIN2 Axin 2 SEQ ID NOS: 1282- 1289 AZGP1 Alpha-2-glycoprotein 1, zinc-binding SEQ ID NOS: 1290- 1292 AZU1 Azurocidin 1 SEQ ID NOS: 1293- 1294 B2M Beta-2-microglobulin SEQ ID NOS: 1295- 1301 B3GALNT1 Beta-1,3-N-acetylgalactosaminyltransferase SEQ ID NOS: 1302- 1 (globoside blood group) 1314 B3GALNT2 Beta-1,3-N-acetylgalactosaminyltransferase SEQ ID NOS: 1315- 2 1317 B3GALT1 UDP-Gal:betaGlcNAc beta 1,3- SEQ ID NO: 1318 galactosyltransferase, polypeptide 1 B3GALT4 UDP-Gal:betaGlcNAc beta 1,3- SEQ ID NO: 1319 galactosyltransferase, polypeptide 4 B3GALT5 UDP-Gal:betaGlcNAc beta 1,3- SEQ ID NOS: 1320- galactosyltransferase, polypeptide 5 1324 B3GALT6 UDP-Gal:betaGal beta 1,3- SEQ ID NO: 1325 galactosyltransferase polypeptide 6 B3GAT3 Beta-1,3-glucuronyltransferase 3 SEQ ID NOS: 1326- 1330 B3GLCT Beta 3-glucosyltransferase SEQ ID NO: 1331 B3GNT3 UDP-GlcNAc:betaGal beta-1,3-N- SEQ ID NOS: 1332- acetylglucosaminyltransferase 3 1335 B3GNT4 UDP-GlcNAc:betaGal beta-1,3-N- SEQ ID NOS: 1336- acetylglucosaminyltransferase 4 1339 B3GNT6 UDP-GlcNAc:betaGal beta-1,3-N- SEQ ID NOS: 1340- acetylglucosaminyltransferase 6 1341 B3GNT7 UDP-GlcNAc:betaGal beta-1,3-N- SEQ ID NO: 1342 acetylglucosaminyltransferase 7 B3GNT8 UDP-GlcNAc:betaGal beta-1,3-N- SEQ ID NO: 1343 acetylglucosaminyltransferase 8 B3GNT9 UDP-GlcNAc:betaGal beta-1,3-N- SEQ ID NO: 1344 acetylglucosaminyltransferase 9 B4GALNT1 Beta-1,4-N-acetyl-galactosaminyl SEQ ID NOS: 1345- transferase 1 1356 B4GALNT3 Beta-1,4-N-acetyl-galactosaminyl SEQ ID NOS: 1357- transferase 3 1358 B4GALNT4 Beta-1,4-N-acetyl-galactosaminyl SEQ ID NOS: 1359- transferase 4 1361 B4GALT4 UDP-Gal:betaGlcNAc beta 1,4- SEQ ID NOS: 1362- galactosyltransferase, polypeptide 4 1374 B4GALT5 UDP-Gal:betaGlcNAc beta 1,4- SEQ ID NO: 1375 galactosyltransferase, polypeptide 5 B4GALT6 UDP-Gal:betaGlcNAc beta

1,4- SEQ ID NOS: 1376- galactosyltransferase, polypeptide 6 1379 B4GAT1 Beta-1,4-glucuronyltransferase 1 SEQ ID NO: 1380 B9D1 B9 protein domain 1 SEQ ID NOS: 1381- 1397 BACE2 Beta-site APP-cleaving enzyme 2 SEQ ID NOS: 1398- 1400 BAGE5 B melanoma antigen family, member 5 SEQ ID NO: 1401 BCAM Basal cell adhesion molecule (Lutheran SEQ ID NOS: 1402- blood group) 1405 BCAN Brevican SEQ ID NOS: 1406- 1412 BCAP29 B-cell receptor-associated protein 29 SEQ ID NOS: 1413- 1425 BCAR1 Breast cancer anti-estrogen resistance 1 SEQ ID NOS: 1426- 1443 BCHE Butyrylcholinesterase SEQ ID NOS: 1444- 1448 BCKDHB Branched chain keto acid dehydrogenase SEQ ID NOS: 1449- E1, beta polypeptide 1451 BDNF Brain-derived neurotrophic factor SEQ ID NOS: 1452- 1469 BGLAP Bone gamma-carboxyglutamate (gla) SEQ ID NO: 1470 protein BGN Biglycan SEQ ID NOS: 1471- 1472 BLVRB Biliverdin reductase B SEQ ID NOS: 1473- 1477 BMP1 Bone morphogenetic protein 1 SEQ ID NOS: 1478- 1489 BMP10 Bone morphogenetic protein 10 SEQ ID NO: 1490 BMP15 Bone morphogenetic protein 15 SEQ ID NO: 1491 BMP2 Bone morphogenetic protein 2 SEQ ID NO: 1492 BMP3 Bone morphogenetic protein 3 SEQ ID NO: 1493 BMP4 Bone morphogenetic protein 4 SEQ ID NOS: 1494- 1501 BMP6 Bone morphogenetic protein 6 SEQ ID NO: 1502 BMP7 Bone morphogenetic protein 7 SEQ ID NOS: 1503- 1506 BMP8A Bone morphogenetic protein 8a SEQ ID NO: 1507 BMP8B Bone morphogenetic protein 8b SEQ ID NO: 1508 BMPER BMP binding endothelial regulator SEQ ID NOS: 1509- 1512 BNC1 Basonudin 1 SEQ ID NOS: 1513- 1514 BOC BOC cell adhesion associated, oncogene SEQ ID NOS: 1515- regulated 1525 BOD1 Biorientation of chromosomes in cell SEQ ID NOS: 1526- division 1 1530 BOLA1 Bola family member 1 SEQ ID NOS: 1531- 1533 BPI Bactericidal/permeability-increasing protein SEQ ID NOS: 1534- 1537 BPIFA1 BPI fold containing family A, member 1 SEQ ID NOS: 1538- 1541 BPIFA2 BPI fold containing family A, member 2 SEQ ID NOS: 1542- 1543 BPIFA3 BPI fold containing family A, member 3 SEQ ID NOS: 1544- 1545 BPIFB1 BPI fold containing family B, member 1 SEQ ID NOS: 1546- 1547 BPIFB2 BPI fold containing family B, member 2 SEQ ID NO: 1548 BPIFB3 BPI fold containing family B, member 3 SEQ ID NO: 1549 BPIFB4 BPI fold containing family B, member 4 SEQ ID NOS: 1550- 1551 BPIFB6 BPI fold containing family B, member 6 SEQ ID NOS: 1552- 1553 BPIFC BPI fold containing family C SEQ ID NOS: 1554- 1557 BRF1 BRF1, RNA polymerase III transcription SEQ ID NOS: 1558- initiation factor 90 kDa subunit 1573 BRINP1 Bone morphogenetic protein/retinoic acid SEQ ID NOS: 1574- inducible neural-specific 1 1575 BRINP2 Bone morphogenetic protein/retinoic acid SEQ ID NO: 1576 inducible neural-specific 2 BRINP3 Bone morphogenetic protein/retinoic acid SEQ ID NOS: 1577- inducible neural-specific 3 1579 BSG Basigin (Ok blood group) SEQ ID NOS: 1580- 1590 BSPH1 Binder of sperm protein homolog 1 SEQ ID NO: 1591 BST1 Bone marrow stromal cell antigen 1 SEQ ID NOS: 1592- 1596 BTBD17 BTB (POZ) domain containing 17 SEQ ID NO: 1597 BTD Biotinidase SEQ ID NOS: 1598- 1607 BTN2A2 Butyrophilin, subfamily 2, member A2 SEQ ID NOS: 1608- 1621 BTN3A1 Butyrophilin, subfamily 3, member A1 SEQ ID NOS: 1622- 1628 BTN3A2 Butyrophilin, subfamily 3, member A2 SEQ ID NOS: 1629- 1639 BTN3A3 Butyrophilin, subfamily 3, member A3 SEQ ID NOS: 1640- 1648 C10orf10 Chromosome 10 open reading frame 10 SEQ ID NOS: 4169- 4170 C10orf99 Chromosome 10 open reading frame 99 SEQ ID NO: 1650 C11orf1 Chromosome 11 open reading frame 1 SEQ ID NOS: 1651- 1655 C11orf24 Chromosome 11 open reading frame 24 SEQ ID NOS: 1656- 1658 C11orf45 Chromosome 11 open reading frame 45 SEQ ID NOS: 1659- 1660 C11orf94 Chromosome 11 open reading frame 94 SEQ ID NO: 1661 C12orf10 Chromosome 12 open reading frame 10 SEQ ID NOS: 1662- 1665 C12orf49 Chromosome 12 open reading frame 49 SEQ ID NOS: 1666- 1669 C12orf73 Chromosome 12 open reading frame 73 SEQ ID NOS: 1670- 1679 C12orf76 Chromosome 12 open reading frame 76 SEQ ID NOS: 1680- 1687 C14orf80 Chromosome 14 open reading frame 80 SEQ ID NOS: 13083- 13096 C14orf93 Chromosome 14 open reading frame 93 SEQ ID NOS: 1688- 1703 C16orf89 Chromosome 16 open reading frame 89 SEQ ID NOS: 1704- 1706 C16orf90 Chromosome 16 open reading frame 90 SEQ ID NOS: 1707- 1708 C17orf67 Chromosome 17 open reading frame 67 SEQ ID NO: 1709 C17orf75 Chromosome 17 open reading frame 75 SEQ ID NOS: 1710- 1718 C17orf99 Chromosome 17 open reading frame 99 SEQ ID NOS: 1719- 1721 C18orf54 Chromosome 18 open reading frame 54 SEQ ID NOS: 1722- 1726 C19orf47 Chromosome 19 open reading frame 47 SEQ ID NOS: 1727- 1734 C19orf70 Chromosome 19 open reading frame 70

SEQ ID NOS: 1735- 1738 C19orf80 Chromosome 19 open reading frame 80 SEQ ID NOS: 829-832  
 C1GALT1 Core 1 synthase, glycoprotein-N- SEQ ID NOS: 1739- acetylgalactosamine 3-beta- 1743  
 galactosyltransferase 1 C1orf127 Chromosome 1 open reading frame 127 SEQ ID NOS: 1744- 1747  
 C1orf159 Chromosome 1 open reading frame 159 SEQ ID NOS: 1748- 1760 C1orf198 Chromosome 1  
 open reading frame 198 SEQ ID NOS: 1761- 1765 C1orf234 Chromosome 1 open reading frame 234  
 SEQ ID NOS: 13118- 13120 C1orf54 Chromosome 1 open reading frame 54 SEQ ID NOS: 1766- 1768  
 C1orf56 Chromosome 1 open reading frame 56 SEQ ID NO: 1769 C1QA Complement component 1, q  
 SEQ ID NOS: 1770- subcomponent, A chain 1772 C1QB Complement component 1, q SEQ ID NOS:  
 1773- subcomponent, B chain 1776 C1QC Complement component 1, q SEQ ID NOS: 1777-  
 subcomponent, C chain 1779 C1QL1 Complement component 1, q SEQ ID NO: 1780 subcomponent-  
 like 1 C1QL2 Complement component 1, q SEQ ID NO: 1781 subcomponent-like 2 C1QL3  
 Complement component 1, q SEQ ID NOS: 1782- subcomponent-like 3 1783 C1QL4 Complement  
 component 1, q SEQ ID NO: 1784 subcomponent-like 4 C1QTNF1 C1q and tumor necrosis factor  
 related SEQ ID NOS: 1785- protein 1 1794 C1QTNF2 C1q and tumor necrosis factor related SEQ ID  
 NO: 1796 protein 2 C1QTNF3 C1q and tumor necrosis factor related SEQ ID NOS: 1797- protein 3  
 1798 C1QTNF4 C1q and tumor necrosis factor related SEQ ID NOS: 1799- protein 4 1800 C1QTNF5  
 C1q and tumor necrosis factor related SEQ ID NOS: 1801- protein 5 1803 C1QTNF7 C1q and tumor  
 necrosis factor related SEQ ID NOS: 1804- protein 7 1808 C1QTNF8 C1q and tumor necrosis factor  
 related SEQ ID NOS: 1809- protein 8 1810 C1QTNF9 C1q and tumor necrosis factor related SEQ ID  
 NOS: 1811- protein 9 1812 C1QTNF9B C1q and tumor necrosis factor related SEQ ID NOS: 1813-  
 protein 9B 1815 C1R Complement component 1, r subcomponent SEQ ID NOS: 1816- 1824 C1RL  
 Complement component 1, r subcomponent- SEQ ID NOS: 1825- like 1833 C1S Complement  
 component 1, s subcomponent SEQ ID NOS: 1834- 1843 C2 Complement component 2 SEQ ID NOS:  
 1844- 1858 C21orf33 Chromosome 21 open reading frame 33 SEQ ID NOS: 1859- 1867 C21orf62  
 Chromosome 21 open reading frame 62 SEQ ID NOS: 1868- 1871 C22orf15 Chromosome 22 open  
 reading frame 15 SEQ ID NOS: 1872- 1874 C22orf46 Chromosome 22 open reading frame 46 SEQ ID  
 NO: 1875 C2CD2 C2 calcium-dependent domain containing 2 SEQ ID NOS: 1876- 1878 C2orf40  
 Chromosome 2 open reading frame 40 SEQ ID NOS: 1879- 1881 C2orf66 Chromosome 2 open reading  
 frame 66 SEQ ID NO: 1882 C2orf69 Chromosome 2 open reading frame 69 SEQ ID NO: 1883 C2orf78  
 Chromosome 2 open reading frame 78 SEQ ID NO: 1884 C3 Complement component 3 SEQ ID NOS:  
 1885- 1889 C3orf33 Chromosome 3 open reading frame 33 SEQ ID NOS: 1890- 1894 C3orf58  
 Chromosome 3 open reading frame 58 SEQ ID NOS: 1895- 1898 C4A Complement component 4A  
 (Rodgers blood SEQ ID NOS: 1899- group) 1900 C4B Complement component 4B (Chido blood SEQ  
 ID NOS: 1901- group) 1902 C4BPA Complement component 4 binding protein, SEQ ID NOS: 1903-  
 alpha 1905 C4BPB Complement component 4 binding protein, SEQ ID NOS: 1906- beta 1910 C4orf26  
 Chromosome 4 open reading frame 26 SEQ ID NOS: 9751- 9754 C4orf48 Chromosome 4 open reading  
 frame 48 SEQ ID NOS: 1911- 1912 C5 Complement component 5 SEQ ID NO: 1913 C5orf46  
 Chromosome 5 open reading frame 46 SEQ ID NOS: 1914- 1915 C6 Complement component 6 SEQ  
 ID NOS: 1916- 1919 C6orf120 Chromosome 6 open reading frame 120 SEQ ID NO: 1920 C6orf15  
 Chromosome 6 open reading frame 15 SEQ ID NO: 1921 C6orf25 Chromosome 6 open reading frame  
 25 SEQ ID NOS: 8832- 8839 C6orf58 Chromosome 6 open reading frame 58 SEQ ID NO: 1922 C7  
 Complement component 7 SEQ ID NO: 1923 C7orf57 Chromosome 7 open reading frame 57 SEQ ID  
 NOS: 1924- 1928 C7orf73 Chromosome 7 open reading frame 73 SEQ ID NOS: 12924- 12925 C8A  
 Complement component 8, alpha SEQ ID NO: 1929 polypeptide C8B Complement component 8, beta  
 polypeptide SEQ ID NOS: 1930- 1932 C8G Complement component 8, gamma SEQ ID NOS: 1933-  
 polypeptide 1934 C9 Complement component 9 SEQ ID NO: 1935 C9orf47 Chromosome 9 open  
 reading frame 47 SEQ ID NOS: 1936- 1938 CA10 Carbonic anhydrase X SEQ ID NOS: 1939- 1945  
 CA11 Carbonic anhydrase XI SEQ ID NOS: 1946- 1947 CA6 Carbonic anhydrase VI SEQ ID NOS:  
 1948- 1952 CA9 Carbonic anhydrase IX SEQ ID NOS: 1953- 1954 CABLES1 Cdk5 and Abl enzyme  
 substrate 1 SEQ ID NOS: 1955- 1960 CABP1 Calcium binding protein 1 SEQ ID NOS: 1961- 1964  
 CACNA2D1 Calcium channel, voltage-dependent, alpha SEQ ID NOS: 1965- 2/delta subunit 1 1968  
 CACNA2D4 Calcium channel, voltage-dependent, alpha SEQ ID NOS: 1969- 2/delta subunit 4 1982

CADM3 Cadherin 3 SEQ ID NOS: 1983- 1985 CALCA Calcitonin-related polypeptide alpha SEQ ID NOS: 1986- 1990 CALCB Calcitonin-related polypeptide beta SEQ ID NOS: 1991- 1993 CALCR Calcitonin receptor SEQ ID NOS: 1994- 2000 CALCRL Calcitonin receptor-like SEQ ID NOS: 2001- 2005 CALR Calreticulin SEQ ID NOS: 2011- 2014 CCDC88B Coiled-coil domain containing 88B SEQ ID NOS: 2112- 2114 CCER2 Coiled-coil glutamate-rich protein 2 SEQ ID NOS: 2115- 2116 CCK Cholecystokinin SEQ ID NOS: 2117- 2119 CCL1 Chemokine (C-C motif) ligand 1 SEQ ID NO: 2120 CCL11 Chemokine (C-C motif) ligand 11 SEQ ID NO: 2121 CCL13 Chemokine (C-C motif) ligand 13 SEQ ID NOS: 2122- 2123 CCL14 Chemokine (C-C motif) ligand 14 SEQ ID NOS: 2124- 2127 CCL15 Chemokine (C-C motif) ligand 15 SEQ ID NOS: 2128- 2129 CCL16 Chemokine (C-C motif) ligand 16 SEQ ID NOS: 2130- 2132 CCL17 Chemokine (C-C motif) ligand 17 SEQ ID NOS: 2133- 2134 CCL18 Chemokine (C-C motif) ligand 18 SEQ ID NO: 2135 (pulmonary and activation-regulated) CCL19 Chemokine (C-C motif) ligand 19 SEQ ID NOS: 2136- 2137 CCL2 Chemokine (C-C motif) ligand 2 SEQ ID NOS: 2138- 2139 CCL20 Chemokine (C-C motif) ligand 20 SEQ ID NOS: 2140- 2142 CCL21 Chemokine (C-C motif) ligand 21 SEQ ID NOS: 2143- 2144 CCL22 Chemokine (C-C motif) ligand 22 SEQ ID NO: 2145 CCL23 Chemokine (C-C motif) ligand 23 SEQ ID NOS: 2146- 2148 CCL24 Chemokine (C-C motif) ligand 24 SEQ ID NOS: 2149- 2150 CCL25 Chemokine (C-C motif) ligand 25 SEQ ID NOS: 2151- 2154 CCL26 Chemokine (C-C motif) ligand 26 SEQ ID NOS: 2155- 2156 CCL27 Chemokine (C-C motif) ligand 27 SEQ ID NO: 2157 CCL28 Chemokine (C-C motif) ligand 28 SEQ ID NOS: 2158- 2160 CCL3 Chemokine (C-C motif) ligand 3 SEQ ID NO: 2161 CCL3L3 Chemokine (C-C motif) ligand 3-like 3 SEQ ID NO: 2162 CCL4 Chemokine (C-C motif) ligand 4 SEQ ID NOS: 2163- 2164 CCL4L2 Chemokine (C-C motif) ligand 4-like 2 SEQ ID NOS: 2165- 2174 CCL5 Chemokine (C-C motif) ligand 5 SEQ ID NOS: 2175- 2177 CCL7 Chemokine (C-C motif) ligand 7 SEQ ID NOS: 2178- 2180 CCL8 Chemokine (C-C motif) ligand 8 SEQ ID NO: 2181 CCNB1IP1 Cyclin B1 interacting protein 1, E3 SEQ ID NOS: 2182- ubiquitin protein ligase 2193 CCNL1 Cyclin L1 SEQ ID NOS: 2194- 2202 CCNL2 Cyclin L2 SEQ ID NOS: 2203- 2210 CD14 CD14 molecule SEQ ID NOS: 2211- 2215 CD160 CD160 molecule SEQ ID NOS: 2216- 2220 CD164 CD164 molecule, sialomucin SEQ ID NOS: 2221- 2226 CD177 CD177 molecule SEQ ID NOS: 2227- 2229 CD1E CD1e molecule SEQ ID NOS: 2230- 2243 CD2 CD2 molecule SEQ ID NOS: 2244- 2245 CD200 CD200 molecule SEQ ID NOS: 2246- 2252 CD200R1 CD200 receptor 1 SEQ ID NOS: 2253- 2257 CD22 CD22 molecule SEQ ID NOS: 2258- 2275 CD226 CD226 molecule SEQ ID NOS: 2276- 2283 CD24 CD24 molecule SEQ ID NOS: 2284- 2290 CD276 CD276 molecule SEQ ID NOS: 2291- 2306 CD300A CD300a molecule SEQ ID NOS: 2307- 2311 CD300LB CD300 molecule-like family member b SEQ ID NOS: 2312- 2313 CD300LF CD300 molecule-like family member f SEQ ID NOS: 2314- 2322 CD300LG CD300 molecule-like family member g SEQ ID NOS: 2323- 2328 CD3D CD3d molecule, delta (CD3-TCR complex) SEQ ID NOS: 2329- 2332 CD4 CD4 molecule SEQ ID NOS: 2333- 2335 CD40 CD40 molecule, TNF receptor superfamily SEQ ID NOS: 2336- member 5 2339 CD44 CD44 molecule (Indian blood group) SEQ ID NOS: 2340- 2366 CD48 CD48 molecule SEQ ID NOS: 2367- 2369 CD5 CD5 molecule SEQ ID NOS: 2370- 2371 CD55 CD55 molecule, decay accelerating factor SEQ ID NOS: 2372- for complement (Cromer blood group) 2382 CD59 CD59 molecule, complement regulatory SEQ ID NOS: 2383- protein 2393 CD5L CD5 molecule-like SEQ ID NO: 2394 CD6 CD6 molecule SEQ ID NOS: 2395- 2402 CD68 CD68 molecule SEQ ID NOS: 2403- 2406 CD7 CD7 molecule SEQ ID NOS: 2407- 2412 CD79A CD79a molecule, immunoglobulin- SEQ ID NOS: 2413- associated alpha 2415 CD80 CD80 molecule SEQ ID NOS: 2416- 2418 CD86 CD86 molecule SEQ ID NOS: 2419- 2425 CD8A CD8a molecule SEQ ID NOS: 2426- 2429 CD8B CD8b molecule SEQ ID NOS: 2430- 2435 CD99 CD99 molecule SEQ ID NOS: 2436- 2444 CDC23 Cell division cycle 23 SEQ ID NOS: 2445- 2449 CDC40 Cell division cycle 40 SEQ ID NOS: 2450- 2452 CDC45 Cell division cycle 45 SEQ ID NOS: 2453- 2459 CDCP1 CUB domain containing protein 1 SEQ ID NOS: 2460- 2461 CDCP2 CUB domain containing protein 2 SEQ ID NOS: 2462- 2463 CDH1 Cadherin 1, type 1 SEQ ID NOS: 2464- 2471 CDH11 Cadherin 11, type 2, OB-cadherin SEQ ID NOS: 2472- (osteoblast) 2481 CDH13 Cadherin 13 SEQ ID NOS: 2482- 2491 CDH17 Cadherin 17, LI cadherin (liver-intestine) SEQ ID NOS: 2492- 2496 CDH18 Cadherin 18, type 2 SEQ ID NOS: 2497- 2503 CDH19 Cadherin 19, type 2 SEQ ID NOS:

2504- 2508 CDH23 Cadherin-related 23 SEQ ID NOS: 2509- 2524 CDH5 Cadherin 5, type 2 (vascular endothelium) SEQ ID NOS: 2525- 2532 CDHR1 Cadherin-related family member 1 SEQ ID NOS: 2533- 2538 CDHR4 Cadherin-related family member 4 SEQ ID NOS: 2539- 2543 CDHR5 Cadherin-related family member 5 SEQ ID NOS: 2544- 2550 CDKN2A Cyclin-dependent kinase inhibitor 2A SEQ ID NOS: 2551- 2561 CDNF Cerebral dopamine neurotrophic factor SEQ ID NOS: 2562- 2563 CDON Cell adhesion associated, oncogene SEQ ID NOS: 2564- regulated 2571 CDSN Corneodesmosin SEQ ID NO: 2572 CEACAM16 Carcinoembryonic antigen-related cell SEQ ID NOS: 2573- adhesion molecule 16 2574 CEACAM18 Carcinoembryonic antigen-related cell SEQ ID NO: 2575 adhesion molecule 18 CEACAM19 Carcinoembryonic antigen-related cell SEQ ID NOS: 2576- adhesion molecule 19 2582 CEACAM5 Carcinoembryonic antigen-related cell SEQ ID NOS: 2583- adhesion molecule 5 2590 CEACAM7 Carcinoembryonic antigen-related cell SEQ ID NOS: 2591- adhesion molecule 7 2593 CEACAM8 Carcinoembryonic antigen-related cell SEQ ID NOS: 2594- adhesion molecule 8 2595 CECR1 Cat eye syndrome chromosome region, SEQ ID NOS: 222-229 candidate 1 CECR5 Cat eye syndrome chromosome region, SEQ ID NOS: 6411- candidate 5 6413 CEL Carboxyl ester lipase SEQ ID NO: 2596 CELA2A Chymotrypsin-like elastase family, member SEQ ID NO: 2597 2A CELA2B Chymotrypsin-like elastase family, member SEQ ID NOS: 2598- 2B 2599 CELA3A Chymotrypsin-like elastase family, member SEQ ID NOS: 2600- 3A 2602 CELA3B Chymotrypsin-like elastase family, member SEQ ID NOS: 2603- 3B 2605 CEMIP Cell migration inducing protein, hyaluronan SEQ ID NOS: 2606- binding 2610 CEP89 Centrosomal protein 89 kDa SEQ ID NOS: 2611- 2616 CER1 Cerberus 1, DAN family BMP antagonist SEQ ID NO: 2617 CERCAM Cerebral endothelial cell adhesion molecule SEQ ID NOS: 2618- 2625 CERS1 Ceramide synthase 1 SEQ ID NOS: 2626- 2630 CES1 Carboxylesterase 1 SEQ ID NOS: 2631- 2636 CES3 Carboxylesterase 3 SEQ ID NOS: 2637- 2641 CES4A Carboxylesterase 4A SEQ ID NOS: 2642- 2647 CES5A Carboxylesterase 5A SEQ ID NOS: 2648- 2655 CETP Cholesteryl ester transfer protein, plasma SEQ ID NOS: 2656- 2658 CFB Complement factor B SEQ ID NOS: 2669- 2673 CFC1 Cripto, FRL-1, cryptic family 1 SEQ ID NOS: 2674- 2676 CFC1B Cripto, FRL-1, cryptic family 1B SEQ ID NOS: 2677- 2679 CFD Complement factor D (adipsin) SEQ ID NOS: 2680- 2681 CFDP1 Craniofacial development protein 1 SEQ ID NOS: 2682- 2685 CFH Complement factor H SEQ ID NOS: 2686- 2688 CFHR1 Complement factor H-related 1 SEQ ID NOS: 2689- 2690 CFHR2 Complement factor H-related 2 SEQ ID NOS: 2691- 2692 CFHR3 Complement factor H-related 3 SEQ ID NOS: 2693- 2697 CFHR4 Complement factor H-related 4 SEQ ID NOS: 2698- 2701 CFHR5 Complement factor H-related 5 SEQ ID NO: 2702 CFI Complement factor I SEQ ID NOS: 2703- 2707 CFP Complement factor properdin SEQ ID NOS: 2708- 2711 CGA Glycoprotein hormones, alpha polypeptide SEQ ID NOS: 2712- 2716 CGB Chorionic gonadotropin, beta polypeptide SEQ ID NO: 2721 CGB1 Chorionic gonadotropin, beta polypeptide 1 SEQ ID NOS: 2717- 2718 CGB2 Chorionic gonadotropin, beta polypeptide 2 SEQ ID NOS: 2719- 2720 CGB5 Chorionic gonadotropin, beta polypeptide 5 SEQ ID NO: 2722 CGB7 Chorionic gonadotropin, beta polypeptide 7 SEQ ID NOS: 2723- 2725 CGB8 Chorionic gonadotropin, beta polypeptide 8 SEQ ID NO: 2726 CGREF1 Cell growth regulator with EF-hand domain SEQ ID NOS: 2727- 1 2734 CH507-9B2.3 SEQ ID NOS: 5532- 5538 CHAD Chondroadherin SEQ ID NOS: 2735- 2737 CHADL Chondroadherin-like SEQ ID NOS: 2738- 2740 CHEK2 Checkpoint kinase 2 SEQ ID NOS: 2741- 2762 CHGA Chromogranin A SEQ ID NOS: 2763- 2765 CHGB Chromogranin B SEQ ID NOS: 2766- 2767 CHI3L1 Chitinase 3-like 1 (cartilage glycoprotein- SEQ ID NOS: 2768- 39) 2769 CHI3L2 Chitinase 3-like 2 SEQ ID NOS: 2770- 2783 CHIA Chitinase, acidic SEQ ID NOS: 2784- 2792 CHID1 Chitinase domain containing 1 SEQ ID NOS: 2793- 2811 CHIT1 Chitinase 1 (chitotriosidase) SEQ ID NOS: 2812- 2815 CHL1 Cell adhesion molecule L1-like SEQ ID NOS: 2816- 2824 CHN1 Chimerin 1 SEQ ID NOS: 2825- 2835 CHPF Chondroitin polymerizing factor SEQ ID NOS: 2836- 2838 CHPF2 Chondroitin polymerizing factor 2 SEQ ID NOS: 2839- 2842 CHRDL1 Chordin-like 1 SEQ ID NOS: 2843- 2848 CHRDL2 Chordin-like 2 SEQ ID NOS: 2849- 2853 CHRDL2 Chordin-like 2 SEQ ID NOS: 2854- 2862 CHRNA2 Cholinergic receptor, nicotinic, alpha 2 SEQ ID NOS: 2863- (neuronal) 2871 CHRNA5 Cholinergic receptor, nicotinic, alpha 5 SEQ ID NOS: 2872- (neuronal) 2875 CHRNB1 Cholinergic receptor, nicotinic, beta 1 SEQ ID NOS: 2876- (muscle) 2881 CHRND Cholinergic receptor, nicotinic, delta SEQ ID NOS: 2882- (muscle) 2887



CHST1 Carbohydrate (sulfate Gal-6) SEQ ID NO: 2888 sulfotransferase 1 CHST10  
 Carbohydrate sulfotransferase 10 SEQ ID NOS: 2889- 2896 CHST11 Carbohydrate (chondroitin 4)  
 SEQ ID NOS: 2897- sulfotransferase 11 2901 CHST13 Carbohydrate (chondroitin 4) SEQ ID NOS:  
 2902- sulfotransferase 13 2903 CHST4 Carbohydrate (N-acetylglucosamine 6-O) SEQ ID NOS: 2904-  
 sulfotransferase 4 2905 CHST5 Carbohydrate (N-acetylglucosamine 6-O) SEQ ID NOS: 2906-  
 sulfotransferase 5 2907 CHST6 Carbohydrate (N-acetylglucosamine 6-O) SEQ ID NOS: 2908-  
 sulfotransferase 6 2909 CHST7 Carbohydrate (N-acetylglucosamine 6-O) SEQ ID NO: 2910  
 sulfotransferase 7 CHST8 Carbohydrate (N-acetylgalactosamine 4-O) SEQ ID NOS: 2911-  
 sulfotransferase 8 2914 CHSY1 Chondroitin sulfate synthase 1 SEQ ID NOS: 2915- 2916 CHSY3  
 Chondroitin sulfate synthase 3 SEQ ID NO: 2917 CHTF8 Chromosome transmission fidelity factor 8  
 SEQ ID NOS: 2918- 2928 CILP Cartilage intermediate layer protein, SEQ ID NO: 2929 nucleotide  
 pyrophosphohydrolase CILP2 Cartilage intermediate layer protein 2 SEQ ID NOS: 2930- 2931  
 CIRH1A Cirrhosis, autosomal recessive 1A (cirhin) SEQ ID NOS: 13974- 13983 CKLF Chemokine-  
 like factor SEQ ID NOS: 2932- 2937 CKMT1A Creatine kinase, mitochondrial 1A SEQ ID NOS: 2938-  
 2943 CKMT1B Creatine kinase, mitochondrial 1B SEQ ID NOS: 2944- 2953 CLCA1 Chloride channel  
 accessory 1 SEQ ID NOS: 2954- 2955 CLCF1 Cardiotrophin-like cytokine factor 1 SEQ ID NOS:  
 2956- 2957 CLDN15 Claudin 15 SEQ ID NOS: 2958- 2963 CLDN7 Claudin 7 SEQ ID NOS: 2964-  
 2970 CLDND1 Claudin domain containing 1 SEQ ID NOS: 2971- 2996 CLEC11A C-type lectin  
 domain family 11, member A SEQ ID NOS: 2997- 2999 CLEC16A C-type lectin domain family 16,  
 member A SEQ ID NOS: 3000- 3005 CLEC18A C-type lectin domain family 18, member A SEQ ID  
 NOS: 3006- 3011 CLEC18B C-type lectin domain family 18, member B SEQ ID NOS: 3012- 3015  
 CLEC18C C-type lectin domain family 18, member C SEQ ID NOS: 3016- 3022 CLEC19A C-type  
 lectin domain family 19, member A SEQ ID NOS: 3023- 3026 CLEC2B C-type lectin domain family 2,  
 member B SEQ ID NOS: 3027- 3028 CLEC3A C-type lectin domain family 3, member A SEQ ID  
 NOS: 3029- 3030 CLEC3B C-type lectin domain family 3, member B SEQ ID NOS: 3031- 3032  
 CLGN Calmegin SEQ ID NOS: 3033- 3035 CLN5 Ceroid-lipofuscinosis, neuronal 5 SEQ ID NOS:  
 3036- 3047 CLPS Colipase, pancreatic SEQ ID NOS: 3048- 3050 CLPSL1 Colipase-like 1 SEQ ID  
 NOS: 3051- 3052 CLPSL2 Colipase-like 2 SEQ ID NOS: 3053- 3054 CLPX Caseinolytic  
 mitochondrial matrix peptidase SEQ ID NOS: 3055- chaperone subunit 3057 CLSTN3 Calsynenin 3  
 SEQ ID NOS: 3058- 3064 CLU Clusterin SEQ ID NOS: 3065- 3078 CLUL1 Clusterin-like 1 (retinal)  
 SEQ ID NOS: 3079- 3086 CMA1 Chymase 1, mast cell SEQ ID NOS: 3087- 3088 CMPK1 Cytidine  
 monophosphate (UMP-CMP) SEQ ID NOS: 3089- kinase 1, cytosolic 3092 CNBD1 Cyclic nucleotide  
 binding domain SEQ ID NOS: 3093- containing 1 3096 CNDP1 Carnosine dipeptidase 1  
 (metallopeptidase SEQ ID NOS: 3097- M20 family) 3099 CNPY2 Canopy FGF signaling regulator 2  
 SEQ ID NOS: 3107- 3111 CNPY3 Canopy FGF signaling regulator 3 SEQ ID NOS: 3112- 3113  
 CNPY4 Canopy FGF signaling regulator 4 SEQ ID NOS: 3114- 3116 CNTFR Ciliary neurotrophic  
 factor receptor SEQ ID NOS: 3117- 3120 CNTN1 Contactin 1 SEQ ID NOS: 3121- 3130 CNTN2  
 Contactin 2 (axonal) SEQ ID NOS: 3131- 3142 CNTN3 Contactin 3 (plasmacytoma associated) SEQ  
 ID NO: 3143 CNTN4 Contactin 4 SEQ ID NOS: 3144- 3152 CNTN5 Contactin 5 SEQ ID NOS: 3153-  
 3158 CNTNAP2 Contactin associated protein-like 2 SEQ ID NOS: 3159- 3162 CNTNAP3 Contactin  
 associated protein-like 3 SEQ ID NOS: 3163- 3167 CNTNAP3B Contactin associated protein-like 3B  
 SEQ ID NOS: 3168- 3176 COASY CoA synthase SEQ ID NOS: 3177- 3186 COCH Cochlin SEQ ID  
 NOS: 3187- 3198 COG3 Component of oligomeric golgi complex 3 SEQ ID NOS: 3199- 3202  
 COL10A1 Collagen, type X, alpha 1 SEQ ID NOS: 3203- 3206 COL11A1 Collagen, type XI, alpha 1  
 SEQ ID NOS: 3207- 3217 COL11A2 Collagen, type XI, alpha 2 SEQ ID NOS: 3218- 3222 COL12A1  
 Collagen, type XII, alpha 1 SEQ ID NOS: 3223- 3230 COL14A1 Collagen, type XIV, alpha 1 SEQ ID  
 NOS: 3231- 3238 COL15A1 Collagen, type XV, alpha 1 SEQ ID NOS: 3239- 3240 COL16A1  
 Collagen, type XVI, alpha 1 SEQ ID NOS: 3241- 3245 COL18A1 Collagen, type XVIII, alpha 1 SEQ  
 ID NOS: 3246- 3250 COL19A1 Collagen, type XIX, alpha 1 SEQ ID NOS: 3251- 3253 COL1A1  
 Collagen, type I, alpha 1 SEQ ID NOS: 3254- 3255 COL1A2 Collagen, type I, alpha 2 SEQ ID NOS:  
 3256- 3257 COL20A1 Collagen, type XX, alpha 1 SEQ ID NOS: 3258- 3261 COL21A1 Collagen, type  
 XXI, alpha 1 SEQ ID NOS: 3262- 3267 COL22A1 Collagen, type XXII, alpha 1 SEQ ID NOS: 3268-



3270 COL24A1 Collagen, type XXIV, alpha 1 SEQ ID NOS: 3271- 3274 COL26A1 Collagen, type XXVI, alpha 1 SEQ ID NOS: 3275- 3276 COL27A1 Collagen, type XXVII, alpha 1 SEQ ID NOS: 3277- 3279 COL28A1 Collagen, type XXVIII, alpha 1 SEQ ID NOS: 3280- 3284 COL2A1 Collagen, type II, alpha 1 SEQ ID NOS: 3285- 3286 COL3A1 Collagen, type III, alpha 1 SEQ ID NOS: 3287- 3289 COL4A1 Collagen, type IV, alpha 1 SEQ ID NOS: 3290- 3292 COL4A2 Collagen, type IV, alpha 2 SEQ ID NOS: 3293- 3295 COL4A3 Collagen, type IV, alpha 3 (Goodpasture SEQ ID NOS: 3296- antigen) 3299 COL4A4 Collagen, type IV, alpha 4 SEQ ID NOS: 3300- 3301 COL4A5 Collagen, type IV, alpha 5 SEQ ID NOS: 3302- 3308 COL4A6 Collagen, type IV, alpha 6 SEQ ID NOS: 3309- 3314 COL5A1 Collagen, type V, alpha 1 SEQ ID NOS: 3315- 3317 COL5A2 Collagen, type V, alpha 2 SEQ ID NOS: 3318- 3319 COL5A3 Collagen, type V, alpha 3 SEQ ID NO: 3320 COL6A1 Collagen, type VI, alpha 1 SEQ ID NOS: 3321- 3322 COL6A2 Collagen, type VI, alpha 2 SEQ ID NOS: 3323- 3328 COL6A3 Collagen, type VI, alpha 3 SEQ ID NOS: 3329- 3337 COL6A5 Collagen, type VI, alpha 5 SEQ ID NOS: 3338- 3342 COL6A6 Collagen, type VI, alpha 6 SEQ ID NOS: 3343- 3345 COL7A1 Collagen, type VII, alpha 1 SEQ ID NOS: 3346- 3347 COL8A1 Collagen, type VIII, alpha 1 SEQ ID NOS: 3348- 3351 COL8A2 Collagen, type VIII, alpha 2 SEQ ID NOS: 3352- 3354 COL9A1 Collagen, type IX, alpha 1 SEQ ID NOS: 3355- 3358 COL9A2 Collagen, type IX, alpha 2 SEQ ID NOS: 3359- 3362 COL9A3 Collagen, type IX, alpha 3 SEQ ID NOS: 3363- 3364 COLEC10 Collectin sub-family member 10 (C-type SEQ ID NO: 3365 lectin) COLEC11 Collectin sub-family member 11 SEQ ID NOS: 3366- 3375 COLGALT1 Collagen beta(1-O)galactosyltransferase 1 SEQ ID NOS: 3376- 3378 COLGALT2 Collagen beta(1-O)galactosyltransferase 2 SEQ ID NOS: 3379- 3381 COLQ Collagen-like tail subunit (single strand of SEQ ID NOS: 3382- homotrimer) of asymmetric 3386 acetylcholinesterase COMP Cartilage oligomeric matrix protein SEQ ID NOS: 3387- 3389 COPS6 COP9 signalosome subunit 6 SEQ ID NOS: 3390- 3393 COQ6 Coenzyme Q6 monooxygenase SEQ ID NOS: 3394- 3401 CORT Cortistatin SEQ ID NO: 3402 CP Ceruloplasmin (ferroxidase) SEQ ID NOS: 3403- 3407 CPA1 Carboxypeptidase A1 (pancreatic) SEQ ID NOS: 3408- 3412 CPA2 Carboxypeptidase A2 (pancreatic) SEQ ID NOS: 3413- 3414 CPA3 Carboxypeptidase A3 (mast cell) SEQ ID NO: 3415 CPA4 Carboxypeptidase A4 SEQ ID NOS: 3416- 3421 CPA6 Carboxypeptidase A6 SEQ ID NOS: 3422- 3424 CPAMD8 C3 and PZP-like, alpha-2-macroglobulin SEQ ID NOS: 3425- domain containing 8 3430 CPB1 Carboxypeptidase B1 (tissue) SEQ ID NOS: 3431- 3435 CPB2 Carboxypeptidase B2 (plasma) SEQ ID NOS: 3436- 3438 CPE Carboxypeptidase E SEQ ID NOS: 3439- 3443 CPM Carboxypeptidase M SEQ ID NOS: 3444- 3453 CPN1 Carboxypeptidase N, polypeptide 1 SEQ ID NOS: 3454- 3455 CPN2 Carboxypeptidase N, polypeptide 2 SEQ ID NOS: 3456- 3457 CPO Carboxypeptidase O SEQ ID NO: 3458 CPQ Carboxypeptidase Q SEQ ID NOS: 3459- 3464 CPVL Carboxypeptidase, vitellogenic-like SEQ ID NOS: 3465- 3475 CPXM1 Carboxypeptidase X (M14 family), member SEQ ID NO: 3476 1 CPXM2 Carboxypeptidase X (M14 family), member SEQ ID NOS: 3477- 2 3478 CPZ Carboxypeptidase Z SEQ ID NOS: 3479- 3482 CR1L Complement component (3b/4b) receptor 1- SEQ ID NOS: 3483- like 3484 CRB2 Crumbs family member 2 SEQ ID NOS: 3485- 3487 CREG1 Cellular repressor of E1A-stimulated genes SEQ ID NO: 3488 1 CREG2 Cellular repressor of E1A-stimulated genes SEQ ID NO: 3489 2 CRELD1 Cysteine-rich with EGF-like domains 1 SEQ ID NOS: 3490- 3495 CRELD2 Cysteine-rich with EGF-like domains 2 SEQ ID NOS: 3496- 3500 CRH Corticotropin releasing hormone SEQ ID NO: 3501 CRHBP Corticotropin releasing hormone binding SEQ ID NOS: 3502- protein 3503 CRHR1 Corticotropin releasing hormone receptor 1 SEQ ID NOS: 3504- 3515 CRHR2 Corticotropin releasing hormone receptor 2 SEQ ID NOS: 3516- 3522 CRISP1 Cysteine-rich secretory protein 1 SEQ ID NOS: 3523- 3526 CRISP2 Cysteine-rich secretory protein 2 SEQ ID NOS: 3527- 3529 CRISP3 Cysteine-rich secretory protein 3 SEQ ID NOS: 3530- 3533 CRISPLD2 Cysteine-rich secretory protein LCCL SEQ ID NOS: 3534- domain containing 2 3541 CRLF1 Cytokine receptor-like factor 1 SEQ ID NOS: 3542- 3543 CRP C-reactive protein, pentraxin-related SEQ ID NOS: 3544- 3548 CRTAC1 Cartilage acidic protein 1 SEQ ID NOS: 3549- 3553 CRTAP Cartilage associated protein SEQ ID NOS: 3554- 3555 CRY2 Cryptochrome circadian clock 2 SEQ ID NOS: 3556- 3559 CSAD Cysteine sulfinic acid decarboxylase SEQ ID NOS: 3560- 3572 CSF1 Colony stimulating factor 1 (macrophage) SEQ ID NOS: 3573- 3580 CSF1R Colony stimulating factor 1 receptor SEQ ID NOS: 3581- 3585 CSF2 Colony stimulating factor 2 (granulocyte- SEQ ID NO: 3586 macrophage) CSF2RA

Colony stimulating factor 2 receptor, alpha, SEQ ID NOS: 3587- low-affinity (granulocyte-macrophage) 3598 CSF3 Colony stimulating factor 3 (granulocyte) SEQ ID NOS: 3599- 3605  
 CSGALNACT1 Chondroitin sulfate N- SEQ ID NOS: 3606- acetylgalactosaminyltransferase 1 3614  
 CSH1 Chorionic somatomammotropin hormone 1 SEQ ID NOS: 3615- (placental lactogen) 3618 CSH2  
 Chorionic somatomammotropin hormone 2 SEQ ID NOS: 3619- 3623 CSHL1 Chorionic  
 somatomammotropin hormone- SEQ ID NOS: 3624- like 1 3630 CSN1S1 Casein alpha s1 SEQ ID  
 NOS: 3631- 3636 CSN2 Casein beta SEQ ID NO: 3637 CSN3 Casein kappa SEQ ID NO: 3638 CST1  
 Cystatin SN SEQ ID NOS: 3639- 3640 CST11 Cystatin 11 SEQ ID NOS: 3641- 3642 CST2 Cystatin  
 SA SEQ ID NO: 3643 CST3 Cystatin C SEQ ID NOS: 3644- 3646 CST4 Cystatin S SEQ ID NO: 3647  
 CST5 Cystatin D SEQ ID NO: 3648 CST6 Cystatin E/M SEQ ID NO: 3649 CST7 Cystatin F  
 (leukocystatin) SEQ ID NO: 3650 CST8 Cystatin 8 (cystatin-related epididymal SEQ ID NOS: 3651-  
 specific) 3652 CST9 Cystatin 9 (testatin) SEQ ID NO: 3653 CST9L Cystatin 9-like SEQ ID NO: 3654  
 CSTL1 Cy statin-like 1 SEQ ID NOS: 3655- 3657 CT55 Cancer/testis antigen 55 SEQ ID NOS: 3658-  
 3659 CTB-60B18.6 SEQ ID NOS: 74-75 CTBS Chitobiase, di-N-acetyl- SEQ ID NOS: 3660- 3662  
 CTD- SEQ ID NO: 4160 2313N18.7 CTD- SEQ ID NOS: 81-84 2370N5.3 CTGF Connective tissue  
 growth factor SEQ ID NO: 3663 CTHRC1 Collagen triple helix repeat containing 1 SEQ ID NOS:  
 3664- 3667 CTLA4 Cytotoxic T-lymphocyte-associated protein SEQ ID NOS: 3668- 4 3671 CTNS  
 Cystinosin, lysosomal cystine transporter SEQ ID NOS: 3672- 3679 CTRB1 Chymotrypsinogen B1  
 SEQ ID NOS: 3680- 3682 CTRB2 Chymotrypsinogen B2 SEQ ID NOS: 3683- 3686 CTRC  
 Chymotrypsin C (caldecrin) SEQ ID NOS: 3687- 3688 CTRL Chymotrypsin-like SEQ ID NOS: 3689-  
 3691 CTSA Cathepsin A SEQ ID NOS: 3692- 3700 CTSB Cathepsin B SEQ ID NOS: 3701- 3725  
 CTSC Cathepsin C SEQ ID NOS: 3726- 3730 CTSD Cathepsin D SEQ ID NOS: 3731- 3741 CTSE  
 Cathepsin E SEQ ID NOS: 3742- 3743 CTSF Cathepsin F SEQ ID NOS: 3744- 3747 CTSG Cathepsin  
 G SEQ ID NO: 3748 CTSH Cathepsin H SEQ ID NOS: 3749- 3754 CTSK Cathepsin K SEQ ID NOS:  
 3755- 3756 CTSL Cathepsin L SEQ ID NOS: 3757- 3759 CTSO Cathepsin O SEQ ID NO: 3760 CTSS  
 Cathepsin S SEQ ID NOS: 3761- 3765 CTSV Cathepsin V SEQ ID NOS: 3766- 3767 CTSW Cathepsin  
 W SEQ ID NOS: 3768- 3770 CTSZ Cathepsin Z SEQ ID NO: 3771 CUBN Cubilin (intrinsic factor-  
 cobalamin receptor) SEQ ID NOS: 3772- 3775 CUTA CutA divalent cation tolerance homolog SEQ ID  
 NOS: 3776- (*E. coli*) 3785 CX3CL1 Chemokine (C-X3-C motif) ligand 1 SEQ ID NOS: 3786- 3789  
 CXADR Coxsackie virus and adenovirus receptor SEQ ID NOS: 3790- 3794 CXCL1 Chemokine (C-X-  
 C motif) ligand 1 SEQ ID NO: 3795 (melanoma growth stimulating activity, alpha) CXCL10  
 Chemokine (C-X-C motif) ligand 10 SEQ ID NO: 3796 CXCL11 Chemokine (C-X-C motif) ligand 11  
 SEQ ID NOS: 3797- 3798 CXCL12 Chemokine (C-X-C motif) ligand 12 SEQ ID NOS: 3799- 3804  
 CXCL13 Chemokine (C-X-C motif) ligand 13 SEQ ID NO: 3805 CXCL14 Chemokine (C-X-C motif)  
 ligand 14 SEQ ID NOS: 3806- 3807 CXCL17 Chemokine (C-X-C motif) ligand 17 SEQ ID NOS:  
 3808- 3809 CXCL2 Chemokine (C-X-C motif) ligand 2 SEQ ID NO: 3810 CXCL3 Chemokine (C-X-C  
 motif) ligand 3 SEQ ID NO: 3811 CXCL5 Chemokine (C-X-C motif) ligand 5 SEQ ID NO: 3812  
 CXCL6 Chemokine (C-X-C motif) ligand 6 SEQ ID NOS: 3813- 3814 CXCL8 Chemokine (C-X-C  
 motif) ligand 8 SEQ ID NOS: 3815- 3816 CXCL9 Chemokine (C-X-C motif) ligand 9 SEQ ID NO:  
 3817 CXorf36 Chromosome X open reading frame 36 SEQ ID NOS: 3818- 3819 CYB5D2 Cytochrome  
 b5 domain containing 2 SEQ ID NOS: 3820- 3823 CYHR1 Cysteine/histidine-rich 1 SEQ ID NOS:  
 3824- 3831 CYP17A1 Cytochrome P450, family 17, subfamily A, SEQ ID NOS: 3832- polypeptide 1  
 3836 CYP20A1 Cytochrome P450, family 20, subfamily A, SEQ ID NOS: 3837- polypeptide 1 3843  
 CYP21A2 Cytochrome P450, family 21, subfamily A, SEQ ID NOS: 3844- polypeptide 2 3851  
 CYP26B1 Cytochrome P450, family 26, subfamily B, SEQ ID NOS: 3852- polypeptide 1 3856  
 CYP2A6 Cytochrome P450, family 2, subfamily A, SEQ ID NOS: 3857- polypeptide 6 3858 CYP2A7  
 Cytochrome P450, family 2, subfamily A, SEQ ID NOS: 3859- polypeptide 7 3861 CYP2B6  
 Cytochrome P450, family 2, subfamily B, SEQ ID NOS: 3862- polypeptide 6 3865 CYP2C18  
 Cytochrome P450, family 2, subfamily C, SEQ ID NOS: 3866- polypeptide 18 3867 CYP2C19  
 Cytochrome P450, family 2, subfamily C, SEQ ID NOS: 3868- polypeptide 19 3869 CYP2C8  
 Cytochrome P450, family 2, subfamily C, SEQ ID NOS: 3870- polypeptide 8 3877 CYP2C9  
 Cytochrome P450, family 2, subfamily C, SEQ ID NOS: 3878- polypeptide 9 3880 CYP2E1

Cytochrome P450, family 2, subfamily E, SEQ ID NOS: 3881- polypeptide 1 3886 CYP2F1  
Cytochrome P450, family 2, subfamily F, SEQ ID NOS: 3887- polypeptide 1 3890 CYP2J2  
Cytochrome P450, family 2, subfamily J, SEQ ID NO: 3891 polypeptide 2 CYP2R1 Cytochrome P450,  
family 2, subfamily R, SEQ ID NOS: 3892- polypeptide 1 3897 CYP2S1 Cytochrome P450, family 2,  
subfamily S, SEQ ID NOS: 3898- polypeptide 1 3903 CYP2W1 Cytochrome P450, family 2, subfamily  
W, SEQ ID NOS: 3904- polypeptide 1 3906 CYP46A1 Cytochrome P450, family 46, subfamily A, SEQ  
ID NOS: 3907- polypeptide 1 3911 CYP4F11 Cytochrome P450, family 4, subfamily F, SEQ ID NOS:  
3912- polypeptide 11 3916 CYP4F2 Cytochrome P450, family 4, subfamily F, SEQ ID NOS: 3917-  
polypeptide 2 3921 CYR61 Cysteine-rich, angiogenic inducer, 61 SEQ ID NO: 3922 CYTL1 Cytokine-  
like 1 SEQ ID NOS: 3923- 3925 D2HGDH D-2-hydroxyglutarate dehydrogenase SEQ ID NOS: 3926-  
3934 DAG1 Dystroglycan 1 (dystrophin-associated SEQ ID NOS: 3935- glycoprotein 1) 3949 DAND5  
DAN domain family member 5, BMP SEQ ID NOS: 3950- antagonist 3951 DAO D-amino-acid  
oxidase SEQ ID NOS: 3952- 3957 DAZAP2 DAZ associated protein 2 SEQ ID NOS: 3958- 3966 DBH  
Dopamine beta-hydroxylase (dopamine SEQ ID NOS: 3967- beta-monooxygenase) 3968 DBNL  
Drebrin-like SEQ ID NOS: 3969- 3986 DCD Dermcidin SEQ ID NOS: 3987- 3989 DCN Decorin SEQ  
ID NOS: 3990- 4008 DDIAS DNA damage-induced apoptosis suppressor SEQ ID NOS: 4009- 4018  
DDOST Dolichyl-diphosphooligosaccharide--protein SEQ ID NOS: 4019- glycosyltransferase subunit  
(non-catalytic) 4022 DDR1 Discoidin domain receptor tyrosine kinase 1 SEQ ID NOS: 4023- 4068  
DDR2 Discoidin domain receptor tyrosine kinase 2 SEQ ID NOS: 4069- 4074 DDT D-dopachrome  
tautomerase SEQ ID NOS: 4075- 4080 DDX17 DEAD (Asp-Glu-Ala-Asp) box helicase 17 SEQ ID  
NOS: 4081- 4085 DDX20 DEAD (Asp-Glu-Ala-Asp) box polypeptide SEQ ID NOS: 4086- 20 4088  
DDX25 DEAD (Asp-Glu-Ala-Asp) box helicase 25 SEQ ID NOS: 4089- 4095 DDX28 DEAD (Asp-  
Glu-Ala-Asp) box polypeptide SEQ ID NO: 4096 28 DEAF1 DEAF1 transcription factor SEQ ID  
NOS: 4097- 4099 DEF8 Differentially expressed in FDCP 8 SEQ ID NOS: 4100- homolog (mouse)  
4119 DEFA1 Defensin, alpha 1 SEQ ID NOS: 4120- 4121 DEFA1B Defensin, alpha 1B SEQ ID NO:  
4122 DEFA3 Defensin, alpha 3, neutrophil-specific SEQ ID NO: 4123 DEFA4 Defensin, alpha 4,  
corticostatin SEQ ID NO: 4124 DEFA5 Defensin, alpha 5, Paneth cell-specific SEQ ID NO: 4125  
DEFA6 Defensin, alpha 6, Paneth cell-specific SEQ ID NO: 4126 DEFB1 Defensin, beta 1 SEQ ID  
NO: 4127 DEFB103A Defensin, beta 103A SEQ ID NO: 4128 DEFB103B Defensin, beta 103B SEQ  
ID NO: 4129 DEFB104A Defensin, beta 104A SEQ ID NO: 4130 DEFB104B Defensin, beta 104B  
SEQ ID NO: 4131 DEFB105A Defensin, beta 105A SEQ ID NO: 4132 DEFB105B Defensin, beta  
105B SEQ ID NO: 4133 DEFB106A Defensin, beta 106A SEQ ID NO: 4134 DEFB106B Defensin,  
beta 106B SEQ ID NO: 4135 DEFB107A Defensin, beta 107A SEQ ID NO: 4136 DEFB107B  
Defensin, beta 107B SEQ ID NO: 4137 DEFB108B Defensin, beta 108B SEQ ID NO: 4138 DEFB110  
Defensin, beta 110 SEQ ID NOS: 4139- 4140 DEFB113 Defensin, beta 113 SEQ ID NO: 4141  
DEFB114 Defensin, beta 114 SEQ ID NO: 4142 DEFB115 Defensin, beta 115 SEQ ID NO: 4143  
DEFB116 Defensin, beta 116 SEQ ID NO: 4144 DEFB118 Defensin, beta 118 SEQ ID NO: 4145  
DEFB119 Defensin, beta 119 SEQ ID NOS: 4146- 4148 DEFB121 Defensin, beta 121 SEQ ID NO:  
4149 DEFB123 Defensin, beta 123 SEQ ID NO: 4150 DEFB124 Defensin, beta 124 SEQ ID NO: 4151  
DEFB125 Defensin, beta 125 SEQ ID NO: 4152 DEFB126 Defensin, beta 126 SEQ ID NO: 4153  
DEFB127 Defensin, beta 127 SEQ ID NO: 4154 DEFB128 Defensin, beta 128 SEQ ID NO: 4155  
DEFB129 Defensin, beta 129 SEQ ID NO: 4156 DEFB130 Defensin, beta 130 SEQ ID NO: 4157  
DEFB131 Defensin, beta 131 SEQ ID NO: 4159 DEFB132 Defensin, beta 132 SEQ ID NO: 4161  
DEFB133 Defensin, beta 133 SEQ ID NO: 4162 DEFB134 Defensin, beta 134 SEQ ID NOS: 4163-  
4164 DEFB135 Defensin, beta 135 SEQ ID NO: 4165 DEFB136 Defensin, beta 136 SEQ ID NO: 4166  
DEFB4A Defensin, beta 4A SEQ ID NO: 4167 DEFB4B Defensin, beta 4B SEQ ID NO: 4168 DFNA5  
Deafness, autosomal dominant 5 SEQ ID NOS: 6271- 6279 DFNB31 Deafness, autosomal recessive 31  
SEQ ID NOS: 14251- 14254 DGCR2 DiGeorge syndrome critical region gene 2 SEQ ID NOS: 4171-  
4174 DHH Desert hedgehog SEQ ID NO: 4175 DHRS4 Dehydrogenase/reductase (SDR family) SEQ  
ID NOS: 4176- member 4 4183 DHRS4L2 Dehydrogenase/reductase (SDR family) SEQ ID NOS:  
4184- member 4 like 2 4193 DHRS7 Dehydrogenase/reductase (SDR family) SEQ ID NOS: 4194-  
member 7 4201 DHRS7C Dehydrogenase/reductase (SDR family) SEQ ID NOS: 4202- member 7C

4204 DHRS9 Dehydrogenase/reductase (SDR family) SEQ ID NOS: 4205- member 9 4212 DHRSX  
 Dehydrogenase/reductase (SDR family) X- SEQ ID NOS: 4213- linked 4217 DHX29 DEAH (Asp-Glu-  
 Ala-His) box polypeptide SEQ ID NOS: 4218- 29 4220 DHX30 DEAH (Asp-Glu-Ala-His) box  
 helicase 30 SEQ ID NOS: 4221- 4228 DHX8 DEAH (Asp-Glu-Ala-His) box polypeptide SEQ ID  
 NOS: 4229- 8 4233 DIO2 Deiodinase, iodothyronine, type II SEQ ID NOS: 4234- 4243 DIXDC1 DIX  
 domain containing 1 SEQ ID NOS: 4244- 4247 DKK1 Dickkopf WNT signaling pathway inhibitor  
 SEQ ID NO: 4248 1 DKK2 Dickkopf WNT signaling pathway inhibitor SEQ ID NOS: 4249- 2 4251  
 DKK3 Dickkopf WNT signaling pathway inhibitor SEQ ID NOS: 4252- 3 4257 DKK4 Dickkopf WNT  
 signaling pathway inhibitor SEQ ID NO: 4258 4 DKKL1 Dickkopf-like 1 SEQ ID NOS: 4259- 4264  
 DLG4 Discs, large homolog 4 (*Drosophila*) SEQ ID NOS: 4265- 4273 DLK1 Delta-like 1 homolog  
 (*Drosophila*) SEQ ID NOS: 4274- 4277 DLL1 Delta-like 1 (*Drosophila*) SEQ ID NOS: 4278- 4279  
 DLL3 Delta-like 3 (*Drosophila*) SEQ ID NOS: 4280- 4282 DMBT1 Deleted in malignant brain tumors  
 1 SEQ ID NOS: 4283- 4289 DMKN Dermokine SEQ ID NOS: 4290- 4336 DMP1 Dentin matrix acidic  
 phosphoprotein 1 SEQ ID NOS: 4337- 4338 DMRTA2 DMRT-like family A2 SEQ ID NOS: 4339-  
 4340 DNAAF5 Dynein, axonemal, assembly factor 5 SEQ ID NOS: 4341- 4344 DNAH14 Dynein,  
 axonemal, heavy chain 14 SEQ ID NOS: 4345- 4359 DNAJB11 DnaJ (Hsp40) homolog, subfamily B,  
 SEQ ID NOS: 4360- member 11 4361 DNAJB9 DnaJ (Hsp40) homolog, subfamily B, SEQ ID NO:  
 4362 member 9 DNAJC25- DNAJC25-GNG10 readthrough SEQ ID NO: 4363 GNG10 DNAJC3 DnaJ  
 (Hsp40) homolog, subfamily C, SEQ ID NOS: 4364- member 3 4365 DNASE1 Deoxyribonuclease I  
 SEQ ID NOS: 4366- 4376 DNASE1L1 Deoxyribonuclease I-like 1 SEQ ID NOS: 4377- 4387  
 DNASE1L2 Deoxyribonuclease I-like 2 SEQ ID NOS: 4388- 4393 DNASE1L3 Deoxyribonuclease I-  
 like 3 SEQ ID NOS: 4394- 4399 DNASE2 Deoxyribonuclease II, lysosomal SEQ ID NOS: 4400- 4401  
 DNASE2B Deoxyribonuclease II beta SEQ ID NOS: 4402- 4403 DPEP1 Dipeptidase 1 (renal) SEQ ID  
 NOS: 4404- 4408 DPEP2 Dipeptidase 2 SEQ ID NOS: 4409- 4415 DPEP3 Dipeptidase 3 SEQ ID NO:  
 4416 DPF3 D4, zinc and double PHD fingers, family 3 SEQ ID NOS: 4417- 4423 DPP4 Dipeptidyl-  
 peptidase 4 SEQ ID NOS: 4424- 4428 DPP7 Dipeptidyl-peptidase 7 SEQ ID NOS: 4429- 4434 DPT  
 Dermatopontin SEQ ID NO: 4435 DRAXIN Dorsal inhibitory axon guidance protein SEQ ID NO:  
 4436 DSE Dermatan sulfate epimerase SEQ ID NOS: 4437- 4445 DSG2 Desmoglein 2 SEQ ID NOS:  
 4446- 4447 DSPP Dentin sialophosphoprotein SEQ ID NOS: 4448- 4449 DST Dystonin SEQ ID NOS:  
 4450- 4468 DUOX1 Dual oxidase 1 SEQ ID NOS: 4469- 4473 DYNLT3 Dynein, light chain, Tctex-  
 type 3 SEQ ID NOS: 4474- 4476 E2F5 E2F transcription factor 5, p130-binding SEQ ID NOS: 4477-  
 4483 EBAG9 Estrogen receptor binding site associated, SEQ ID NOS: 4484- antigen, 9 4492 EB13  
 Epstein-Barr virus induced 3 SEQ ID NO: 4493 ECHDC1 Ethylmalonyl-CoA decarboxylase 1 SEQ ID  
 NOS: 4494- 4512 ECM1 Extracellular matrix protein 1 SEQ ID NOS: 4513- 4515 ECM2 Extracellular  
 matrix protein 2, female organ SEQ ID NOS: 4516- and adipocyte specific 4519 ECSIT ECSIT  
 signalling integrator SEQ ID NOS: 4520- 4531 EDDM3A Epididymal protein 3A SEQ ID NO: 4532  
 EDDM3B Epididymal protein 3B SEQ ID NO: 4533 EDEM2 ER degradation enhancer, mannosidase  
 SEQ ID NOS: 4534- alpha-like 2 4535 EDEM3 ER degradation enhancer, mannosidase SEQ ID NOS:  
 4536- alpha-like 3 4538 EDIL3 EGF-like repeats and discoidin I-like SEQ ID NOS: 4539- domains 3  
 4540 EDN1 Endothelin 1 SEQ ID NO: 4541 EDN2 Endothelin 2 SEQ ID NO: 4542 EDN3 Endothelin  
 3 SEQ ID NOS: 4543- 4548 EDNRB Endothelin receptor type B SEQ ID NOS: 4549- 4557 EFEMP1  
 EGF containing fibulin-like extracellular SEQ ID NOS: 4558- matrix protein 1 4568 EFEMP2 EGF  
 containing fibulin-like extracellular SEQ ID NOS: 4569- matrix protein 2 4580 EFNA1 Ephrin-A1 SEQ  
 ID NOS: 4581- 4582 EFNA2 Ephrin-A2 SEQ ID NO: 4583 EFNA4 Ephrin-A4 SEQ ID NOS: 4584-  
 4586 EGFL6 EGF-like-domain, multiple 6 SEQ ID NOS: 4587- 4588 EGFL7 EGF-like-domain,  
 multiple 7 SEQ ID NOS: 4589- 4593 EGFL8 EGF-like-domain, multiple 8 SEQ ID NOS: 4594- 4596  
 EGFLAM EGF-like, fibronectin type III and laminin G SEQ ID NOS: 4597- domains 4605 EGFR  
 Epidermal growth factor receptor SEQ ID NOS: 4606- 4613 EHBP1 EH domain binding protein 1 SEQ  
 ID NOS: 4614- 4625 EHF Ets homologous factor SEQ ID NOS: 4626- 4635 EHMT1 Euchromatic  
 histone-lysine N- SEQ ID NOS: 4636- methyltransferase 1 4661 EHMT2 Euchromatic histone-lysine  
 N- SEQ ID NOS: 4662- methyltransferase 2 4666 EIF2AK1 Eukaryotic translation initiation factor 2-  
 SEQ ID NOS: 4667- alpha kinase 1 4670 ELANE Elastase, neutrophil expressed SEQ ID NOS: 4671-

4672 ELN Elastin SEQ ID NOS: 4673- 4692 ELP2 Elongator acetyltransferase complex subunit SEQ ID NOS: 4696- 2 4708 ELSPBP1 Epididymal sperm binding protein 1 SEQ ID NOS: 4709- 4714 EMC1 ER membrane protein complex subunit 1 SEQ ID NOS: 4715- 4721 EMC10 ER membrane protein complex subunit 10 SEQ ID NOS: 4722- 4728 EMC9 ER membrane protein complex subunit 9 SEQ ID NOS: 4729- 4732 EMCN Endomucin SEQ ID NOS: 4733- 4737 EMID1 EMI domain containing 1 SEQ ID NOS: 4738- 4744 EMILIN1 Elastin microfibril interfacier 1 SEQ ID NOS: 4745- 4746 EMILIN2 Elastin microfibril interfacier 2 SEQ ID NO: 4747 EMILIN3 Elastin microfibril interfacier 3 SEQ ID NO: 4748 ENAM Enamelin SEQ ID NO: 4749 ENDOG Endonuclease G SEQ ID NO: 4750 ENDOU Endonuclease, polyU-specific SEQ ID NOS: 4751- 4753 ENHO Energy homeostasis associated SEQ ID NO: 4754 ENO4 Enolase family member 4 SEQ ID NOS: 4755- 4759 ENPP6 Ectonucleotide SEQ ID NOS: 4760- pyrophosphatase/phosphodiesterase 6 4761 ENPP7 Ectonucleotide SEQ ID NOS: 4762- pyrophosphatase/phosphodiesterase 7 4763 ENTPD5 Ectonucleoside triphosphate SEQ ID NOS: 4764- diphosphohydrolase 5 4768 ENTPD8 Ectonucleoside triphosphate SEQ ID NOS: 4769- diphosphohydrolase 8 4772 EOGT EGF domain-specific O-linked N- SEQ ID NOS: 4773- acetylglucosamine (GlcNAc) transferase 4780 EPCAM Epithelial cell adhesion molecule SEQ ID NOS: 4781- 4784 EPDR1 Ependymin related 1 SEQ ID NOS: 4785- 4788 EPGN Epithelial mitogen SEQ ID NOS: 4789- 4797 EPHA10 EPH receptor A10 SEQ ID NOS: 4798- 4805 EPHA3 EPH receptor A3 SEQ ID NOS: 4806- 4808 EPHA4 EPH receptor A4 SEQ ID NOS: 4809- 4818 EPHA7 EPH receptor A7 SEQ ID NOS: 4819- 4820 EPHA8 EPH receptor A8 SEQ ID NOS: 4821- 4822 EPHB2 EPH receptor B2 SEQ ID NOS: 4823- 4827 EPHB4 EPH receptor B4 SEQ ID NOS: 4828- 4830 EPHX3 Epoxide hydrolase 3 SEQ ID NOS: 4831- 4834 EPO Erythropoietin SEQ ID NO: 4835 EPPIN Epididymal peptidase inhibitor SEQ ID NOS: 4836- 4838 EPPIN- EPPIN-WFDC6 readthrough SEQ ID NO: 4839 WFDC6 EPS15 Epidermal growth factor receptor pathway SEQ ID NOS: 4840- substrate 15 4842 EPS8L1 EPS8-like 1 SEQ ID NOS: 4843- 4848 EPX Eosinophil peroxidase SEQ ID NO: 4849 EPYC Epiphycan SEQ ID NOS: 4850- 4851 EQTN Equatorin, sperm acrosome associated SEQ ID NOS: 4852- 4854 ERAP1 Endoplasmic reticulum aminopeptidase 1 SEQ ID NOS: 4855- 4859 ERAP2 Endoplasmic reticulum aminopeptidase 2 SEQ ID NOS: 4860- 4867 ERBB3 Erb-b2 receptor tyrosine kinase 3 SEQ ID NOS: 4868- 4881 ERLIN1 ER lipid raft associated 1 SEQ ID NOS: 4885- 4887 ERLIN2 ER lipid raft associated 2 SEQ ID NOS: 4888- 4896 ERN1 Endoplasmic reticulum to nucleus signaling SEQ ID NOS: 4897- 1 4898 ERN2 Endoplasmic reticulum to nucleus signaling SEQ ID NOS: 4899- 2 4903 ERO1A Endoplasmic reticulum oxidoreductase SEQ ID NOS: 4904- alpha 4910 ERO1B Endoplasmic reticulum oxidoreductase beta SEQ ID NOS: 4911- 4913 ERP27 Endoplasmic reticulum protein 27 SEQ ID NOS: 4914- 4915 ERP29 Endoplasmic reticulum protein 29 SEQ ID NOS: 4916- 4919 ERP44 Endoplasmic reticulum protein 44 SEQ ID NO: 4920 ERV3-1 Endogenous retrovirus group 3, member 1 SEQ ID NO: 4921 ESM1 Endothelial cell-specific molecule 1 SEQ ID NOS: 4922- 4924 ESRP1 Epithelial splicing regulatory protein 1 SEQ ID NOS: 4925- 4933 EXOG Endo/exonuclease (5'-3'), endonuclease G- SEQ ID NOS: 4934- like 4947 EXTL1 Exostosin-like glycosyltransferase 1 SEQ ID NO: 4948 EXTL2 Exostosin-like glycosyltransferase 2 SEQ ID NOS: 4949- 4953 F10 Coagulation factor X SEQ ID NOS: 4954- 4957 F11 Coagulation factor XI SEQ ID NOS: 4958- 4962 F12 Coagulation factor XII (Hageman factor) SEQ ID NO: 4963 F13B Coagulation factor XIII, B polypeptide SEQ ID NO: 4964 F2 Coagulation factor II (thrombin) SEQ ID NOS: 4965- 4967 F2R Coagulation factor II (thrombin) receptor SEQ ID NOS: 4968- 4969 F2RL3 Coagulation factor II (thrombin) receptor- SEQ ID NOS: 4970- like 3 4971 F5 Coagulation factor V (proaccelerin, labile SEQ ID NOS: 4972- factor) 4973 F7 Coagulation factor VII (serum prothrombin SEQ ID NOS: 4974- conversion accelerator) 4977 F8 Coagulation factor VIII, procoagulant SEQ ID NOS: 4978- component 4983 F9 Coagulation factor IX SEQ ID NOS: 4984- 4985 FABP6 Fatty acid binding protein 6, ileal SEQ ID NOS: 4986- 4988 FAM107B Family with sequence similarity 107, SEQ ID NOS: 4989- member B 5010 FAM131A Family with sequence similarity 131, SEQ ID NOS: 5011- member A 5019 FAM132A Family with sequence similarity 132, SEQ ID NO: 1795 member A FAM132B Family with sequence similarity 132, SEQ ID NOS: 4882- member B 4884 FAM150A Family with sequence similarity 150, SEQ ID NOS: 737-738 member A FAM150B Family with sequence similarity 150, SEQ ID NOS: 739-745 member B FAM171A1 Family

with sequence similarity 171, SEQ ID NOS: 5020- member A1 5021 FAM171B Family with sequence similarity 171, SEQ ID NOS: 5022- member B 5023 FAM172A Family with sequence similarity 172, SEQ ID NOS: 5024- member A 5028 FAM175A Family with sequence similarity 175, SEQ ID NOS: 64-71 member A FAM177A1 Family with sequence similarity 177, SEQ ID NOS: 5029- member A1 5038 FAM179B Family with sequence similarity 179, SEQ ID NOS: 13628- member B 13633 FAM180A Family with sequence similarity 180, SEQ ID NOS: 5039- member A 5041 FAM189A1 Family with sequence similarity 189, SEQ ID NOS: 5042- member A1 5043 FAM198A Family with sequence similarity 198, SEQ ID NOS: 5044- member A 5046 FAM19A1 Family with sequence similarity 19 SEQ ID NOS: 5047- (chemokine (C-C motif)-like), member A1 5049 FAM19A2 Family with sequence similarity 19 SEQ ID NOS: 5050- (chemokine (C-C motif)-like), member A2 5057 FAM19A3 Family with sequence similarity 19 SEQ ID NOS: 5058- (chemokine (C-C motif)-like), member A3 5059 FAM19A4 Family with sequence similarity 19 SEQ ID NOS: 5060- (chemokine (C-C motif)-like), member A4 5062 FAM19A5 Family with sequence similarity 19 SEQ ID NOS: 5063- (chemokine (C-C motif)-like), member A5 5066 FAM20A Family with sequence similarity 20, SEQ ID NOS: 5067- member A 5070 FAM20C Family with sequence similarity 20, SEQ ID NO: 5071 member C FAM213A Family with sequence similarity 213, SEQ ID NOS: 5072- member A 5077 FAM26D Family with sequence similarity 26, SEQ ID NOS: 2006- member D 2010 FAM46B Family with sequence similarity 46, SEQ ID NO: 5078 member B FAM57A Family with sequence similarity 57, SEQ ID NOS: 5079- member A 5084 FAM78A Family with sequence similarity 78, SEQ ID NOS: 5085- member A 5087 FAM96A Family with sequence similarity 96, SEQ ID NOS: 5088- member A 5092 FAM9B Family with sequence similarity 9, member SEQ ID NOS: 5093- B 5096 FAP Fibroblast activation protein, alpha SEQ ID NOS: 5097- 5103 FAS Fas cell surface death receptor SEQ ID NOS: 5104- 5113 FAT1 FAT atypical cadherin 1 SEQ ID NOS: 5114- 5120 FBLN1 Fibulin 1 SEQ ID NOS: 5121- 5133 FBLN2 Fibulin 2 SEQ ID NOS: 5134- 5139 FBLN5 Fibulin 5 SEQ ID NOS: 5140- 5145 FBLN7 Fibulin 7 SEQ ID NOS: 5146- 5151 FBN1 Fibrillin 1 SEQ ID NOS: 5152- 5155 FBN2 Fibrillin 2 SEQ ID NOS: 5156- 5161 FBN3 Fibrillin 3 SEQ ID NOS: 5162- 5166 FBXW7 F-box and WD repeat domain containing 7, SEQ ID NOS: 5167- E3 ubiquitin protein ligase 5177 FCAR Fc fragment of IgA receptor SEQ ID NOS: 5178- 5187 FCGBP Fc fragment of IgG binding protein SEQ ID NOS: 5188- 5190 FCGR1B Fc fragment of IgG, high affinity Ib, SEQ ID NOS: 5191- receptor (CD64) 5196 FCGR3A Fc fragment of IgG, low affinity IIIa, SEQ ID NOS: 5197- receptor (CD16a) 5203 FCGR4 Fc fragment of IgG, receptor, transporter, SEQ ID NOS: 5204- alpha 5214 FCMR Fc fragment of IgM receptor SEQ ID NOS: 5215- 5221 FCN1 Ficolin (collagen/fibrinogen domain SEQ ID NOS: 5222- containing) 1 5223 FCN2 Ficolin (collagen/fibrinogen domain SEQ ID NOS: 5224- containing lectin) 2 5225 FCN3 Ficolin (collagen/fibrinogen domain SEQ ID NOS: 5226- containing) 3 5227 FCRL1 Fc receptor-like 1 SEQ ID NOS: 5228- 5230 FCRL3 Fc receptor-like 3 SEQ ID NOS: 5231- 5236 FCRL5 Fc receptor-like 5 SEQ ID NOS: 5237- 5239 FCRLA Fc receptor-like A SEQ ID NOS: 5240- 5251 FCRLB Fc receptor-like B SEQ ID NOS: 5252- 5256 FDCSP Follicular dendritic cell secreted protein SEQ ID NO: 5257 FETUB Fetuin B SEQ ID NOS: 5258- 5264 FGA Fibrinogen alpha chain SEQ ID NOS: 5265- 5267 FGB Fibrinogen beta chain SEQ ID NOS: 5268- 5270 FGF10 Fibroblast growth factor 10 SEQ ID NOS: 5271- 5272 FGF17 Fibroblast growth factor 17 SEQ ID NOS: 5273- 5274 FGF18 Fibroblast growth factor 18 SEQ ID NO: 5275 FGF19 Fibroblast growth factor 19 SEQ ID NO: 5276 FGF21 Fibroblast growth factor 21 SEQ ID NOS: 5277- 5278 FGF22 Fibroblast growth factor 22 SEQ ID NOS: 5279- 5280 FGF23 Fibroblast growth factor 23 SEQ ID NO: 5281 FGF3 Fibroblast growth factor 3 SEQ ID NO: 5282 FGF4 Fibroblast growth factor 4 SEQ ID NO: 5283 FGF5 Fibroblast growth factor 5 SEQ ID NOS: 5284- 5286 FGF7 Fibroblast growth factor 7 SEQ ID NOS: 5287- 5291 FGF8 Fibroblast growth factor 8 (androgen- SEQ ID NOS: 5292- induced) 5297 FGFBP1 Fibroblast growth factor binding protein 1 SEQ ID NO: 5298 FGFBP2 Fibroblast growth factor binding protein 2 SEQ ID NO: 5299 FGFBP3 Fibroblast growth factor binding protein 3 SEQ ID NO: 5300 FGFR1 Fibroblast growth factor receptor 1 SEQ ID NOS: 5301- 5322 FGFR2 Fibroblast growth factor receptor 2 SEQ ID NOS: 5323- 5344 FGFR3 Fibroblast growth factor receptor 3 SEQ ID NOS: 5345- 5352 FGFR4 Fibroblast growth factor receptor 4 SEQ ID NOS: 5353- 5362 FGFR11 Fibroblast growth factor receptor-like 1 SEQ ID NOS: 5363- 5368 FGG Fibrinogen gamma chain SEQ



ID NOS: 5369- 5379 FGL1 Fibrogen-like 1 SEQ ID NOS: 5379- 5381 FGL2 Fibrogen-like 2 SEQ ID NOS: 5382- 5383 FHL1 Four and a half LIM domains 1 SEQ ID NOS: 5384- 5411 FHOD3 Formin homology 2 domain containing 3 SEQ ID NOS: 5412- 5418 FIBIN Fin bud initiation factor homolog SEQ ID NO: 5419 (zebrafish) FICD FIC domain containing SEQ ID NOS: 5420- 5423 FIGF C-fos induced growth factor (vascular SEQ ID NO: 14054 endothelial growth factor D) FJX1 Four jointed box 1 SEQ ID NO: 5424 FKBP10 FK506 binding protein 10, 65 kDa SEQ ID NOS: 5425- 5430 FKBP11 FK506 binding protein 11, 19 kDa SEQ ID NOS: 5431- 5437 FKBP14 FK506 binding protein 14, 22 kDa SEQ ID NOS: 5438- 5440 FKBP2 FK506 binding protein 2, 13 kDa SEQ ID NOS: 5441- 5444 FKBP7 FK506 binding protein 7 SEQ ID NOS: 5445- 5450 FKBP9 FK506 binding protein 9, 63 kDa SEQ ID NOS: 5451- 5454 FLT1 Fms-related tyrosine kinase 1 SEQ ID NOS: 5455- 5463 FLT4 Fms-related tyrosine kinase 4 SEQ ID NOS: 5464- 5468 FMO1 Flavin containing monooxygenase 1 SEQ ID NOS: 5469- 5473 FMO2 Flavin containing monooxygenase 2 (non- SEQ ID NOS: 5474- functional) 5476 FMO3 Flavin containing monooxygenase 3 SEQ ID NOS: 5477- 5479 FMO5 Flavin containing monooxygenase 5 SEQ ID NOS: 5480- 5486 FMOD Fibromodulin SEQ ID NO: 5487 FN1 Fibronectin 1 SEQ ID NOS: 5488- 5500 FNDC1 Fibronectin type III domain containing 1 SEQ ID NOS: 5501- 5502 FNDC7 Fibronectin type III domain containing 7 SEQ ID NOS: 5503- 5504 FOCAD Focadhesin SEQ ID NOS: 5505- 5511 FOLR2 Folate receptor 2 (fetal) SEQ ID NOS: 5512- 5521 FOLR3 Folate receptor 3 (gamma) SEQ ID NOS: 5522- 5526 FOXRED2 FAD-dependent oxidoreductase domain SEQ ID NOS: 5527- containing 2 5530 FP325331.1 Uncharacterized protein SEQ ID NO: 5531 UNQ6126/PRO20091 FPGS Folylpolyglutamate synthase SEQ ID NOS: 5539- 5545 FRAS1 Fraser extracellular matrix complex subunit SEQ ID NOS: 5546- 1 5551 FREM1 FRAS1 related extracellular matrix 1 SEQ ID NOS: 5552- 5556 FREM3 FRAS1 related extracellular matrix 3 SEQ ID NO: 5557 FRMPD2 FERM and PDZ domain containing 2 SEQ ID NOS: 5558- 5561 FRZB Frizzled-related protein SEQ ID NO: 5562 FSHB Follicle stimulating hormone, beta SEQ ID NOS: 5563- polypeptide 5565 FSHR Follicle stimulating hormone receptor SEQ ID NOS: 5566- 5569 FST Follistatin SEQ ID NOS: 5570- 5573 FSTL1 Follistatin-like 1 SEQ ID NOS: 5574- 5577 FSTL3 Follistatin-like 3 (secreted glycoprotein) SEQ ID NOS: 5578- 5583 FSTL4 Follistatin-like 4 SEQ ID NOS: 5584- 5586 FSTL5 Follistatin-like 5 SEQ ID NOS: 5587- 5589 FTCDNL1 Formiminotransferase cyclodeaminase N- SEQ ID NOS: 5590- terminal like 5593 FUCA1 Fucosidase, alpha-L- 1, tissue SEQ ID NO: 5594 FUCA2 Fucosidase, alpha-L- 2, plasma SEQ ID NOS: 5595- 5596 FURIN Furin (paired basic amino acid cleaving SEQ ID NOS: 5597- enzyme) 5603 FUT10 Fucosyltransferase 10 (alpha (1,3) SEQ ID NOS: 5604- fucosyltransferase) 5606 FUT11 Fucosyltransferase 11 (alpha (1,3) SEQ ID NOS: 5607- fucosyltransferase) 5608 FXN Frataxin SEQ ID NOS: 5609- 5616 FXR1 Fragile X mental retardation, autosomal SEQ ID NOS: 5617- homolog 1 5629 FXYD3 FXYD domain containing ion transport SEQ ID NOS: 5630- regulator 3 5642 GABBR1 Gamma-aminobutyric acid (GABA) B SEQ ID NOS: 5643- receptor, 1 5654 GABRA1 Gamma-aminobutyric acid (GABA) A SEQ ID NOS: 5655- receptor, alpha 1 5670 GABRA2 Gamma-aminobutyric acid (GABA) A SEQ ID NOS: 5671- receptor, alpha 2 5685 GABRA5 Gamma-aminobutyric acid (GABA) A SEQ ID NOS: 5686- receptor, alpha 5 5694 GABRG3 Gamma-aminobutyric acid (GABA) A SEQ ID NOS: 5695- receptor, gamma 3 5700 GABRP Gamma-aminobutyric acid (GABA) A SEQ ID NOS: 5701- receptor, pi 5709 GAL Galanin/GMAP prepropeptide SEQ ID NO: 5710 GAL3ST1 Galactose-3-O-sulfotransferase 1 SEQ ID NOS: 5711- 5732 GAL3ST2 Galactose-3-O-sulfotransferase 2 SEQ ID NO: 5733 GAL3ST3 Galactose-3-O-sulfotransferase 3 SEQ ID NOS: 5734- 5735 GALC Galactosylceramidase SEQ ID NOS: 5736- 5745 GALNS Galactosamine (N-acetyl)-6-sulfatase SEQ ID NOS: 5746- 5751 GALNT10 Polypeptide N- SEQ ID NOS: 5752- acetylgalactosaminyltransferase 10 5755 GALNT12 Polypeptide N- SEQ ID NOS: 5756- acetylgalactosaminyltransferase 12 5757 GALNT15 Polypeptide N- SEQ ID NOS: 5758- acetylgalactosaminyltransferase 15 5761 GALNT2 Polypeptide N- SEQ ID NO: 5762 acetylgalactosaminyltransferase 2 GALNT6 Polypeptide N- SEQ ID NOS: 5763- acetylgalactosaminyltransferase 6 5774 GALNT8 Polypeptide N- SEQ ID NOS: 5775- acetylgalactosaminyltransferase 8 5778 GALNTL6 Polypeptide N- SEQ ID NOS: 5779- acetylgalactosaminyltransferase-like 6 5782 GALP Galanin-like peptide SEQ ID NOS: 5783- 5785 GANAB Glucosidase, alpha; neutral AB SEQ ID NOS: 5786- 5794 GARS Glycyl-tRNA synthetase

SEQ ID NOS: 5795- 5798 GAS1 Growth arrest-specific 1 SEQ ID NO: 5799 GAS6 Growth arrest-specific 6 SEQ ID NO: 5800 GAST Gastrin SEQ ID NO: 5801 GBA Glucosidase, beta, acid SEQ ID NOS: 5811- 5814 GBGT1 Globoside alpha-1,3-N- SEQ ID NOS: 5815- acetylglactosaminyltransferase 1 5823 GC Group-specific component (vitamin D SEQ ID NOS: 5824- binding protein) 5828 GCG Glucagon SEQ ID NOS: 5829- 5830 GCGR Glucagon receptor SEQ ID NOS: 5831- 5833 GCNT7 Glucosaminyl (N-acetyl) transferase family SEQ ID NOS: 5834- member 7 5835 GCSH Glycine cleavage system protein H SEQ ID NOS: 5836- (aminomethyl carrier) 5844 GDF1 Growth differentiation factor 1 SEQ ID NO: 5845 GDF10 Growth differentiation factor 10 SEQ ID NO: 5846 GDF11 Growth differentiation factor 11 SEQ ID NOS: 5847- 5848 GDF15 Growth differentiation factor 15 SEQ ID NOS: 5849- 5851 GDF2 Growth differentiation factor 2 SEQ ID NO: 5852 GDF3 Growth differentiation factor 3 SEQ ID NO: 5853 GDF5 Growth differentiation factor 5 SEQ ID NOS: 5854- 5855 GDF6 Growth differentiation factor 6 SEQ ID NOS: 5856- 5858 GDF7 Growth differentiation factor 7 SEQ ID NO: 5859 GDF9 Growth differentiation factor 9 SEQ ID NOS: 5860- 5864 GDNF Glial cell derived neurotrophic factor SEQ ID NOS: 5865- 5872 GFOD2 Glucose-fructose oxidoreductase domain SEQ ID NOS: 5873- containing 2 5878 GFPT2 Glutamine-fructose-6-phosphate SEQ ID NOS: 5879- transaminase 2 5881 GFRA2 GDNF family receptor alpha 2 SEQ ID NOS: 5882- 5888 GFRA4 GDNF family receptor alpha 4 SEQ ID NOS: 5889- 5891 GGA2 Golgi-associated, gamma adaptin ear SEQ ID NOS: 5892- containing, ARF binding protein 2 5900 GGH Gamma-glutamyl hydrolase (conjugase, SEQ ID NO: 5901 folylpolyglutamyld hydrolase) GGT1 Gamma-glutamyltransferase 1 SEQ ID NOS: 5902- 5924 GGT5 Gamma-glutamyltransferase 5 SEQ ID NOS: 5925- 5929 GH1 Growth hormone 1 SEQ ID NOS: 5930- 5934 GH2 Growth hormone 2 SEQ ID NOS: 5935- 5939 GHDC GH3 domain containing SEQ ID NOS: 5940- 5947 GHRH Growth hormone releasing hormone SEQ ID NOS: 5948- 5950 GHRHR Growth hormone releasing hormone SEQ ID NOS: 5951- receptor 5956 GHRL Ghrelin/obestatin prepropeptide SEQ ID NOS: 5957- 5967 GIF Gastric intrinsic factor (vitamin B synthesis) SEQ ID NOS: 5968- 5969 GIP Gastric inhibitory polypeptide SEQ ID NO: 5970 GKN1 Gastroke 1 SEQ ID NO: 5971 GKN2 Gastroke 2 SEQ ID NOS: 5972- 5973 GLA Galactosidase, alpha SEQ ID NOS: 5974- 5975 GLB1 Galactosidase, beta 1 SEQ ID NOS: 5976- 5984 GLB1L Galactosidase, beta 1-like SEQ ID NOS: 5985- 5992 GLB1L2 Galactosidase, beta 1-like 2 SEQ ID NOS: 5993- 5994 GLCE Glucuronic acid epimerase SEQ ID NOS: 5995- 5996 GLG1 Golgi glycoprotein 1 SEQ ID NOS: 5997- 6004 GLIPR1 GLI pathogenesis-related 1 SEQ ID NOS: 6005- 6008 GLIPR1L1 GLI pathogenesis-related 1 like 1 SEQ ID NOS: 6009- 6012 GLIS3 GLIS family zinc finger 3 SEQ ID NOS: 6013- 6021 GLMP Glycosylated lysosomal membrane protein SEQ ID NOS: 6022- 6030 GLRB Glycine receptor, beta SEQ ID NOS: 6031- 6036 GLS Glutaminase SEQ ID NOS: 6037- 6044 GLT6D1 Glycosyltransferase 6 domain containing 1 SEQ ID NOS: 6045- 6046 GLTPD2 Glycolipid transfer protein domain SEQ ID NO: 6047 containing 2 GLUD1 Glutamate dehydrogenase 1 SEQ ID NO: 6048 GM2A GM2 ganglioside activator SEQ ID NOS: 6049- 6051 GML Glycosylphosphatidylinositol anchored SEQ ID NOS: 6052- molecule like 6053 GNAS GNAS complex locus SEQ ID NOS: 6054- 6075 GNLY Granulysin SEQ ID NOS: 6076- 6079 GNPTG N-acetylglucosamine-1-phosphate SEQ ID NOS: 6080- transferase, gamma subunit 6084 GNRH1 Gonadotropin-releasing hormone 1 SEQ ID NOS: 6085- (luteinizing-releasing hormone) 6086 GNRH2 Gonadotropin-releasing hormone 2 SEQ ID NOS: 6087- 6090 GNS Glucosamine (N-acetyl)-6-sulfatase SEQ ID NOS: 6091- 6096 GOLM1 Golgi membrane protein 1 SEQ ID NOS: 6097- 6101 GORAB Golgin, RAB6-interacting SEQ ID NOS: 6102- 6104 GOT2 Glutamic-oxaloacetic transaminase 2, SEQ ID NOS: 6105- mitochondrial 6107 GP2 Glycoprotein 2 (zymogen granule SEQ ID NOS: 6108- membrane) 6116 GP6 Glycoprotein VI (platelet) SEQ ID NOS: 6117- 6120 GPC2 Glypican 2 SEQ ID NOS: 6121- 6122 GPC5 Glypican 5 SEQ ID NOS: 6123- 6125 GPC6 Glypican 6 SEQ ID NOS: 6126- 6127 GPD2 Glycerol-3-phosphate dehydrogenase 2 SEQ ID NOS: 6128- (mitochondrial) 6136 GPER1 G protein-coupled estrogen receptor 1 SEQ ID NOS: 6137- 6143 GPHA2 Glycoprotein hormone alpha 2 SEQ ID NOS: 6144- 6146 GPHB5 Glycoprotein hormone beta 5 SEQ ID NOS: 6147- 6148 GPIHBP1 Glycosylphosphatidylinositol anchored high SEQ ID NO: 6149 density lipoprotein binding protein 1 GPLD1 Glycosylphosphatidylinositol specific SEQ ID NO: 6150 phospholipase D1 GPNMB Glycoprotein (transmembrane) nmb SEQ ID NOS: 6151- 6153 GPR162 G protein-coupled receptor 162



SEQ ID NOS: 6154- 6157 GPX3 Glutathione peroxidase 3 SEQ ID NOS: 6158- 6165 GPX4  
Glutathione peroxidase 4 SEQ ID NOS: 6166- 6176 GPX5 Glutathione peroxidase 5 SEQ ID NOS:  
6177- 6178 GPX6 Glutathione peroxidase 6 SEQ ID NOS: 6179- 6181 GPX7 Glutathione peroxidase 7  
SEQ ID NO: 6182 GREM1 Gremlin 1, DAN family BMP antagonist SEQ ID NOS: 6183- 6185  
GREM2 Gremlin 2, DAN family BMP antagonist SEQ ID NO: 6186 GRHL3 Grainyhead-like  
transcription factor 3 SEQ ID NOS: 6187- 6192 GRIA2 Glutamate receptor, ionotropic, AMPA 2 SEQ  
ID NOS: 6193- 6204 GRIA3 Glutamate receptor, ionotropic, AMPA 3 SEQ ID NOS: 6205- 6210  
GRIA4 Glutamate receptor, ionotropic, AMPA 4 SEQ ID NOS: 6211- 6222 GRIK2 Glutamate receptor,  
ionotropic, kainate 2 SEQ ID NOS: 6223- 6231 GRIN2B Glutamate receptor, ionotropic, N-methyl  
SEQ ID NOS: 6232- D-aspartate 2B 6235 GRM2 Glutamate receptor, metabotropic 2 SEQ ID NOS:  
6236- 6239 GRM3 Glutamate receptor, metabotropic 3 SEQ ID NOS: 6240- 6244 GRM5 Glutamate  
receptor, metabotropic 5 SEQ ID NOS: 6245- 6249 GRN Granulin SEQ ID NOS: 6250- 6265 GRP  
Gastrin-releasing peptide SEQ ID NOS: 6266- 6270 GSG1 Germ cell associated 1 SEQ ID NOS: 6280-  
6288 GSN Gelsolin SEQ ID NOS: 6289- 6297 GTDC1 Glycosyltransferase-like domain containing  
SEQ ID NOS: 6298- 1 6311 GTPBP10 GTP-binding protein 10 (putative) SEQ ID NOS: 6312- 6320  
GUCA2A Guanylate cyclase activator 2A (guanylin) SEQ ID NO: 6321 GUCA2B Guanylate cyclase  
activator 2B SEQ ID NO: 6322 (uroguanylin) GUSB Glucuronidase, beta SEQ ID NOS: 6323- 6327  
GVQW1 GVQW motif containing 1 SEQ ID NO: 6328 GXYLT1 Glucoside xylosyltransferase 1 SEQ  
ID NOS: 6329- 6330 GXYLT2 Glucoside xylosyltransferase 2 SEQ ID NOS: 6331- 6333 GYLTL1B  
Glycosyltransferase-like 1B SEQ ID NOS: 7702- 7707 GYPB Glycophorin B (MNS blood group) SEQ  
ID NOS: 6334- 6342 GZMA Granzyme A (granzyme 1, cytotoxic T- SEQ ID NO: 6343 lymphocyte-  
associated serine esterase 3) GZMB Granzyme B (granzyme 2, cytotoxic T- SEQ ID NOS: 6344-  
lymphocyte-associated serine esterase 1) 6352 GZMH Granzyme H (cathepsin G-like 2, protein h- SEQ  
ID NOS: 6353- CCPX) 6355 GZMK Granzyme K (granzyme 3; tryptase II) SEQ ID NO: 6356 GZMM  
Granzyme M (lymphocyte met-ase 1) SEQ ID NOS: 6357- 6358 H6PD Hexose-6-phosphate  
dehydrogenase SEQ ID NOS: 6359- (glucose 1-dehydrogenase) 6360 HABP2 Hyaluronan binding  
protein 2 SEQ ID NOS: 6361- 6362 HADHB Hydroxyacyl-CoA dehydrogenase/3- SEQ ID NOS: 6363-  
ketoacyl-CoA thiolase/enoyl-CoA hydratase 6369 (trifunctional protein), beta subunit HAMP Hepcidin  
antimicrobial peptide SEQ ID NOS: 6370- 6371 HAPLN1 Hyaluronan and proteoglycan link protein 1  
SEQ ID NOS: 6372- 6378 HAPLN2 Hyaluronan and proteoglycan link protein 2 SEQ ID NOS: 6379-  
6380 HAPLN3 Hyaluronan and proteoglycan link protein 3 SEQ ID NOS: 6381- 6384 HAPLN4  
Hyaluronan and proteoglycan link protein 4 SEQ ID NO: 6385 HARS2 Histidyl-tRNA synthetase 2,  
mitochondrial SEQ ID NOS: 6386- 6401 HAVCR1 Hepatitis A virus cellular receptor 1 SEQ ID NOS:  
6402- 6406 HCCS Holocytochrome c synthase SEQ ID NOS: 6407- 6409 HCRT Hypocretin (orexin)  
neuropeptide precursor SEQ ID NO: 6410 HEATR5A HEAT repeat containing 5A SEQ ID NOS: 6414-  
6420 HEPH Hephaestin SEQ ID NOS: 6421- 6428 HEXA Hexosaminidase A (alpha polypeptide) SEQ  
ID NOS: 6429- 6438 HEXB Hexosaminidase B (beta polypeptide) SEQ ID NOS: 6439- 6444 HFE2  
Hemochromatosis type 2 (juvenile) SEQ ID NOS: 6445- 6451 HGF Hepatocyte growth factor  
(hepapoietin A; SEQ ID NOS: 6452- scatter factor) 6462 HGFAC HGF activator SEQ ID NOS: 6463-  
6464 HHIP Hedgehog interacting protein SEQ ID NOS: 6465- 6466 HHIPL1 HHIP-like 1 SEQ ID  
NOS: 6467- 6468 HHIPL2 HHIP-like 2 SEQ ID NO: 6469 HHLA1 HERV-H LTR-associating 1 SEQ  
ID NOS: 6470- 6471 HHLA2 HERV-H LTR-associating 2 SEQ ID NOS: 6472- 6482 HIBADH 3-  
hydroxyisobutyrate dehydrogenase SEQ ID NOS: 6483- 6485 HINT2 Histidine triad nucleotide binding  
protein 2 SEQ ID NO: 6486 HLA-A Major histocompatibility complex, class I, SEQ ID NOS: 6487- A  
6491 HLA-C Major histocompatibility complex, class I, C SEQ ID NOS: 6492- 6496 HLA-DOA Major  
histocompatibility complex, class II, SEQ ID NOS: 6497- DO alpha 6498 HLA-DPA1 Major  
histocompatibility complex, class II, SEQ ID NOS: 6499- DP alpha 1 6502 HLA-DQA1 Major  
histocompatibility complex, class II, SEQ ID NOS: 6503- DQ alpha 1 6508 HLA-DQB1 Major  
histocompatibility complex, class II, SEQ ID NOS: 6509- DQ beta 1 6514 HLA-DQB2 Major  
histocompatibility complex, class II, SEQ ID NOS: 6515- DQ beta 2 6518 HMCN1 Hemicentin 1 SEQ  
ID NOS: 6519- 6520 HMCN2 Hemicentin 2 SEQ ID NOS: 6521- 6524 HMGCL 3-hydroxymethyl-3-  
methylglutaryl-CoA SEQ ID NOS: 6525- lyase 6528 HMHA1 Histocompatibility (minor) HA-1 SEQ

ID NOS: 1034- 1042 H1MSD Histocompatibility (minor) serpin domain SEQ ID NOS: 6529- containing  
6530 HP Haptoglobin SEQ ID NOS: 6531- 6544 HPR Haptoglobin-related protein SEQ ID NOS: 6545-  
6547 HPSE Heparanase SEQ ID NOS: 6548- 6554 HPSE2 Heparanase 2 (inactive) SEQ ID NOS:  
6555- 6560 HPX Hemopexin SEQ ID NOS: 6561- 6562 HRC Histidine rich calcium binding protein  
SEQ ID NOS: 6563- 6565 HRG Histidine-rich glycoprotein SEQ ID NO: 6566 HRSP12 Heat-  
responsive protein 12 SEQ ID NOS: 11389- 11392 HS2ST1 Heparan sulfate 2-O-sulfotransferase 1  
SEQ ID NOS: 6567- 6569 HS3ST1 Heparan sulfate (glucosamine) 3-O- SEQ ID NOS: 6570-  
sulfotransferase 1 6572 HS6ST1 Heparan sulfate 6-O-sulfotransferase 1 SEQ ID NO: 6573 HS6ST3  
Heparan sulfate 6-O-sulfotransferase 3 SEQ ID NOS: 6574- 6575 HSD11B1L Hydroxysteroid (11-beta)  
dehydrogenase 1- SEQ ID NOS: 6576- like 6594 HSD17B11 Hydroxysteroid (17-beta) dehydrogenase  
11 SEQ ID NOS: 6595- 6596 HSD17B7 Hydroxysteroid (17-beta) dehydrogenase 7 SEQ ID NOS:  
6597- 6601 HSP90B1 Heat shock protein 90 kDa beta (Grp94), SEQ ID NOS: 6602- member 1 6607  
HSPA13 Heat shock protein 70 kDa family, member SEQ ID NO: 6608 13 HSPA5 Heat shock 70 kDa  
protein 5 (glucose- SEQ ID NO: 6609 regulated protein, 78 kDa) HSPG2 Heparan sulfate proteoglycan  
2 SEQ ID NOS: 6610- 6614 HTATIP2 HIV-1 Tat interactive protein 2, 30 kDa SEQ ID NOS: 6615-  
6622 HTN1 Histatin 1 SEQ ID NOS: 6623- 6625 HTN3 Histatin 3 SEQ ID NOS: 6626- 6628 HTRA1  
HtrA serine peptidase 1 SEQ ID NOS: 6629- 6630 HTRA3 HtrA serine peptidase 3 SEQ ID NOS:  
6631- 6632 HTRA4 HtrA serine peptidase 4 SEQ ID NO: 6633 HYAL1 Hyaluronoglucosaminidase 1  
SEQ ID NOS: 6634- 6642 HYAL2 Hyaluronoglucosaminidase 2 SEQ ID NOS: 6643- 6651 HYAL3  
Hyaluronoglucosaminidase 3 SEQ ID NOS: 6652- 6658 HYOU1 Hypoxia up-regulated 1 SEQ ID  
NOS: 6659- 6673 IAPP Islet amyloid polypeptide SEQ ID NOS: 6674- 6678 IBSP Integrin-binding  
sialoprotein SEQ ID NO: 6679 ICAM1 Intercellular adhesion molecule 1 SEQ ID NOS: 6680- 6682  
ICAM2 Intercellular adhesion molecule 2 SEQ ID NOS: 6683- 6693 ICAM4 Intercellular adhesion  
molecule 4 SEQ ID NOS: 6694- (Landsteiner-Wiener blood group) 6696 ID1 Inhibitor of DNA binding  
1, dominant SEQ ID NOS: 6697- negative helix-loop-helix protein 6698 IDE Insulin-degrading enzyme  
SEQ ID NOS: 6699- 6702 IDNK IdnK, gluconokinase homolog (*E. coli*) SEQ ID NOS: 6703- 6708  
IDS Iduronate 2-sulfatase SEQ ID NOS: 6709- 6714 IDUA Iduronidase, alpha-L- SEQ ID NOS: 6715-  
6720 IFI27L2 Interferon, alpha-inducible protein 27-like 2 SEQ ID NOS: 6721- 6722 IFI30 Interferon,  
gamma-inducible protein 30 SEQ ID NOS: 6723- 6724 IFNA1 Interferon, alpha 1 SEQ ID NO: 6725  
IFNA10 Interferon, alpha 10 SEQ ID NO: 6726 IFNA13 Interferon, alpha 13 SEQ ID NOS: 6727- 6728  
IFNA14 Interferon, alpha 14 SEQ ID NO: 6729 IFNA16 Interferon, alpha 16 SEQ ID NO: 6730  
IFNA17 Interferon, alpha 17 SEQ ID NO: 6731 IFNA2 Interferon, alpha 2 SEQ ID NO: 6732 IFNA21  
Interferon, alpha 21 SEQ ID NO: 6733 IFNA4 Interferon, alpha 4 SEQ ID NO: 6734 IFNA5 Interferon,  
alpha 5 SEQ ID NO: 6735 IFNA6 Interferon, alpha 6 SEQ ID NOS: 6736- 6737 IFNA7 Interferon,  
alpha 7 SEQ ID NO: 6738 IFNA8 Interferon, alpha 8 SEQ ID NO: 6739 IFNAR1 Interferon (alpha,  
beta and omega) receptor SEQ ID NOS: 6740- 1 6741 IFNB1 Interferon, beta 1, fibroblast SEQ ID NO:  
6742 IFNE Interferon, epsilon SEQ ID NO: 6743 IFNG Interferon, gamma SEQ ID NO: 6744 IFNGR1  
Interferon gamma receptor 1 SEQ ID NOS: 6745- 6755 IFNL1 Interferon, lambda 1 SEQ ID NO: 6756  
IFNL2 Interferon, lambda 2 SEQ ID NO: 6757 IFNL3 Interferon, lambda 3 SEQ ID NOS: 6758- 6759  
IFNLR1 Interferon, lambda receptor 1 SEQ ID NOS: 6760- 6764 IFNW1 Interferon, omega 1 SEQ ID  
NO: 6765 IGF1 Insulin-like growth factor 1 (somatomedin SEQ ID NOS: 6766- C) 6771 IGF2 Insulin-  
like growth factor 2 SEQ ID NOS: 6772- 6779 IGFBP1 Insulin-like growth factor binding protein, SEQ  
ID NOS: 6780- acid labile subunit 6782 IGFBP1 Insulin-like growth factor binding protein 1 SEQ ID  
NOS: 6783- 6785 IGFBP2 Insulin-like growth factor binding protein 2, SEQ ID NOS: 6786- 36 kDa  
6789 IGFBP3 Insulin-like growth factor binding protein 3 SEQ ID NOS: 6790- 6797 IGFBP4 Insulin-  
like growth factor binding protein 4 SEQ ID NO: 6798 IGFBP5 Insulin-like growth factor binding  
protein 5 SEQ ID NOS: 6799- 6800 IGFBP6 Insulin-like growth factor binding protein 6 SEQ ID NOS:  
6801- 6803 IGFBP7 Insulin-like growth factor binding protein 7 SEQ ID NOS: 6804- 6805 IGFBPL1  
Insulin-like growth factor binding protein- SEQ ID NO: 6806 like 1 IGFL1 IGF-like family member 1  
SEQ ID NO: 6807 IGFL2 IGF-like family member 2 SEQ ID NOS: 6808- 6810 IGFL3 IGF-like family  
member 3 SEQ ID NO: 6811 IGFLR1 IGF-like family receptor 1 SEQ ID NOS: 6812- 6820 IGIP IgA-  
inducing protein SEQ ID NO: 6821 IGLON5 IgLON family member 5 SEQ ID NO: 6822 IGSF1

Immunoglobulin superfamily, member 1 SEQ ID NOS: 6823- 6828 IGSF10 Immunoglobulin superfamily, member 10 SEQ ID NOS: 6829- 6830 IGSF11 Immunoglobulin superfamily, member 11 SEQ ID NOS: 6831- 6838 IGSF21 Immunoglobulin superfamily, member 21 SEQ ID NO: 6839 IGSF8 Immunoglobulin superfamily, member 8 SEQ ID NOS: 6840- 6843 IGSF9 Immunoglobulin superfamily, member 9 SEQ ID NOS: 6844- 6846 IHH Indian hedgehog SEQ ID NO: 6847 IL10 Interleukin 10 SEQ ID NOS: 6848- 6849 IL11 Interleukin 11 SEQ ID NOS: 6850- 6853 IL11RA Interleukin 11 receptor, alpha SEQ ID NOS: 6854- 6864 IL12B Interleukin 12B SEQ ID NO: 6865 IL12RB1 Interleukin 12 receptor, beta 1 SEQ ID NOS: 6866- 6871 IL12RB2 Interleukin 12 receptor, beta 2 SEQ ID NOS: 6872- 6876 IL13 Interleukin 13 SEQ ID NOS: 6877- 6878 IL13RA1 Interleukin 13 receptor, alpha 1 SEQ ID NOS: 6879- 6880 IL15RA Interleukin 15 receptor, alpha SEQ ID NOS: 6881- 6898 IL17A Interleukin 17A SEQ ID NO: 6899 IL17B Interleukin 17B SEQ ID NO: 6900 IL17C Interleukin 17C SEQ ID NO: 6901 IL17D Interleukin 17D SEQ ID NOS: 6902- 6904 IL17F Interleukin 17F SEQ ID NO: 6905 IL17RA Interleukin 17 receptor A SEQ ID NOS: 6906- 6907 IL17RC Interleukin 17 receptor C SEQ ID NOS: 6908- 6923 IL17RE Interleukin 17 receptor E SEQ ID NOS: 6924- 6930 IL18BP Interleukin 18 binding protein SEQ ID NOS: 6931- 6941 IL18R1 Interleukin 18 receptor 1 SEQ ID NOS: 6942- 6945 IL18RAP Interleukin 18 receptor accessory protein SEQ ID NOS: 6946- 6948 IL19 Interleukin 19 SEQ ID NOS: 6949- 6951 IL1R1 Interleukin 1 receptor, type I SEQ ID NOS: 6952- 6964 IL1R2 Interleukin 1 receptor, type II SEQ ID NOS: 6965- 6968 IL1RAP Interleukin 1 receptor accessory protein SEQ ID NOS: 6969- 6982 IL1RL1 Interleukin 1 receptor-like 1 SEQ ID NOS: 6983- 6988 IL1RL2 Interleukin 1 receptor-like 2 SEQ ID NOS: 6989- 6991 IL1RN Interleukin 1 receptor antagonist SEQ ID NOS: 6992- 6996 IL2 Interleukin 2 SEQ ID NO: 6997 IL20 Interleukin 20 SEQ ID NOS: 6998- 7000 IL20RA Interleukin 20 receptor, alpha SEQ ID NOS: 7001- 7007 IL21 Interleukin 21 SEQ ID NOS: 7008- 7009 IL22 Interleukin 22 SEQ ID NOS: 7010- 7011 IL22RA2 Interleukin 22 receptor, alpha 2 SEQ ID NOS: 7012- 7014 IL23A Interleukin 23, alpha subunit p19 SEQ ID NO: 7015 IL24 Interleukin 24 SEQ ID NOS: 7016- 7021 IL25 Interleukin 25 SEQ ID NOS: 7022- 7023 IL26 Interleukin 26 SEQ ID NO: 7024 IL27 Interleukin 27 SEQ ID NOS: 7025- 7026 IL2RB Interleukin 2 receptor, beta SEQ ID NOS: 7027- 7031 IL3 Interleukin 3 SEQ ID NO: 7032 IL31 Interleukin 31 SEQ ID NO: 7033 IL31RA Interleukin 31 receptor A SEQ ID NOS: 7034- 7041 IL32 Interleukin 32 SEQ ID NOS: 7042- 7071 IL34 Interleukin 34 SEQ ID NOS: 7072- 7075 IL3RA Interleukin 3 receptor, alpha (low affinity) SEQ ID NOS: 7076- 7078 IL4 Interleukin 4 SEQ ID NOS: 7079- 7081 IL4I1 Interleukin 4 induced 1 SEQ ID NOS: 7082- 7089 IL4R Interleukin 4 receptor SEQ ID NOS: 7090- 7103 IL5 Interleukin 5 SEQ ID NOS: 7104- 7105 IL5RA Interleukin 5 receptor, alpha SEQ ID NOS: 7106- 7115 IL6 Interleukin 6 SEQ ID NOS: 7116- 7122 IL6R Interleukin 6 receptor SEQ ID NOS: 7123- 7128 IL6ST Interleukin 6 signal transducer SEQ ID NOS: 7129- 7138 IL7 Interleukin 7 SEQ ID NOS: 7139- 7146 IL7R Interleukin 7 receptor SEQ ID NOS: 7147- 7153 IL9 Interleukin 9 SEQ ID NO: 7154 ILDR1 Immunoglobulin-like domain containing SEQ ID NOS: 7155- receptor 1 7159 ILDR2 Immunoglobulin-like domain containing SEQ ID NOS: 7160- receptor 2 7166 IMP4 IMP4, U3 small nucleolar ribonucleoprotein SEQ ID NOS: 7167- 7172 IMPG1 Interphotoreceptor matrix proteoglycan 1 SEQ ID NOS: 7173- 7176 INHA Inhibin, alpha SEQ ID NO: 7177 INHBA Inhibin, beta A SEQ ID NOS: 7178- 7180 INHBB Inhibin, beta B SEQ ID NO: 7181 INHBC Inhibin, beta C SEQ ID NO: 7182 INHBE Inhibin, beta E SEQ ID NOS: 7183- 7184 INPP5A Inositol polyphosphate-5-phosphatase A SEQ ID NOS: 7185- 7189 INS Insulin SEQ ID NOS: 7190- 7194 INS-IGF2 INS-IGF2 readthrough SEQ ID NOS: 7195- 7196 INSL3 Insulin-like 3 (Leydig cell) SEQ ID NOS: 7197- 7199 INSL4 Insulin-like 4 (placenta) SEQ ID NO: 7200 INSL5 Insulin-like 5 SEQ ID NO: 7201 INSL6 Insulin-like 6 SEQ ID NO: 7202 INTS3 Integrator complex subunit 3 SEQ ID NOS: 7203- 7208 IPO11 Importin 11 SEQ ID NOS: 7209- 7217 IPO9 Importin 9 SEQ ID NOS: 7218- 7219 IQCF6 IQ motif containing F6 SEQ ID NOS: 7220- 7221 IRAK3 Interleukin-1 receptor-associated kinase 3 SEQ ID NOS: 7222- 7224 IRS4 Insulin receptor substrate 4 SEQ ID NO: 7225 ISLR Immunoglobulin superfamily containing SEQ ID NOS: 7226- leucine-rich repeat 7229 ISLR2 Immunoglobulin superfamily containing SEQ ID NOS: 7230- leucine-rich repeat 2 7239 ISM1 Isthmin 1, angiogenesis inhibitor SEQ ID NO: 7240 ISM2 Isthmin 2 SEQ ID NOS: 7241- 7246 ITGA4 Integrin, alpha 4 (antigen CD49D, alpha 4 SEQ ID NOS: 7247- subunit of VLA-4 receptor) 7249 ITGA9

Integrin, alpha 9 SEQ ID NOS: 7250- 7252 ITGAL Integrin, alpha L (antigen CD11A (p180), SEQ ID NOS: 7253- lymphocyte function-associated antigen 1; 7262 alpha polypeptide) ITGAX Integrin, alpha X (complement component 3 SEQ ID NOS: 7263- receptor 4 subunit) 7265 ITGB1 Integrin, beta 1 (fibronectin receptor, beta SEQ ID NOS: 7266- polypeptide, antigen CD29 includes MDF2, 7281 MSK12) ITGB2 Integrin, beta 2 (complement component 3 SEQ ID NOS: 7282- receptor 3 and 4 subunit) 7298 ITGB3 Integrin, beta 3 (platelet glycoprotein IIIa, SEQ ID NOS: 7299- antigen CD61) 7301 ITGB7 Integrin, beta 7 SEQ ID NOS: 7302- 7309 ITGBL1 Integrin, beta-like 1 (with EGF-like repeat SEQ ID NOS: 7310- domains) 7315 ITIH1 Inter-alpha-trypsin inhibitor heavy chain 1 SEQ ID NOS: 7316- 7321 ITIH2 Inter-alpha-trypsin inhibitor heavy chain 2 SEQ ID NOS: 7322- 7324 ITIH3 Inter-alpha-trypsin inhibitor heavy chain 3 SEQ ID NOS: 7325- 7327 ITIH4 Inter-alpha-trypsin inhibitor heavy chain SEQ ID NOS: 7328- family, member 4 7331 ITIH5 Inter-alpha-trypsin inhibitor heavy chain SEQ ID NOS: 7332- family, member 5 7335 ITIH6 Inter-alpha-trypsin inhibitor heavy chain SEQ ID NO: 7336 family, member 6 ITLN1 Intelectin 1 (galactofuranose binding) SEQ ID NO: 7337 ITLN2 Intelectin 2 SEQ ID NO: 7338 IZUMO1R IZUMO1 receptor, JUNO SEQ ID NOS: 7339- 7340 IZUMO4 IZUMO family member 4 SEQ ID NOS: 7341- 7347 JCHAIN Joining chain of multimeric IgA and IgM SEQ ID NOS: 7357- 7362 JMJD8 Jumonji domain containing 8 SEQ ID NOS: 7363- 7367 JSRP1 Junctional sarcoplasmic reticulum protein 1 SEQ ID NO: 7368 KANSL2 KAT8 regulatory NSL complex subunit 2 SEQ ID NOS: 7369- 7379 KAZALD1 Kazal-type serine peptidase inhibitor SEQ ID NO: 7380 domain 1 KCNIP3 Kv channel interacting protein 3, calsenilin SEQ ID NOS: 7381- 7383 KCNK7 Potassium channel, two pore domain SEQ ID NOS: 7384- subfamily K, member 7 7389 KCNN4 Potassium channel, calcium activated SEQ ID NOS: 7390- intermediate/small conductance subfamily 7395 N alpha, member 4 KCNU1 Potassium channel, subfamily U, member 1 SEQ ID NOS: 7396- 7400 KCP Kielin/chordin-like protein SEQ ID NOS: 7401- 7404 KDEL1 KDEL (Lys-Asp-Glu-Leu) containing 1 SEQ ID NO: 7405 KDEL2 KDEL (Lys-Asp-Glu-Leu) containing 2 SEQ ID NOS: 7406- 7409 KDM1A Lysine (K)-specific demethylase 1A SEQ ID NOS: 7410- 7413 KDM3B Lysine (K)-specific demethylase 3B SEQ ID NOS: 7414- 7417 KDM6A Lysine (K)-specific demethylase 6A SEQ ID NOS: 7418- 7427 KDM7A Lysine (K)-specific demethylase 7A SEQ ID NOS: 7428- 7429 KDSR 3-ketodihydrosphingosine reductase SEQ ID NOS: 7430- 7436 KERA Keratocan SEQ ID NO: 7437 KIAA0100 KIAA0100 SEQ ID NOS: 7438- 7443 KIAA0319 KIAA0319 SEQ ID NOS: 7444- 7449 KIAA1324 KIAA1324 SEQ ID NOS: 7450- 7458 KIFC2 Kinesin family member C2 SEQ ID NOS: 7459- 7461 KIR2DL4 Killer cell immunoglobulin-like receptor, SEQ ID NOS: 7462- two domains, long cytoplasmic tail, 4 7468 KIR3DX1 Killer cell immunoglobulin-like receptor, SEQ ID NOS: 7469- three domains, X1 7473 KIRREL2 Kin of IRRE like 2 (*Drosophila*) SEQ ID NOS: 7474- 7478 KISS1 KiSS-1 metastasis-suppressor SEQ ID NOS: 7479- 7480 KLHL11 Kelch-like family member 11 SEQ ID NO: 7481 KLHL22 Kelch-like family member 22 SEQ ID NOS: 7482- 7488 KLK1 Kallikrein 1 SEQ ID NOS: 7489- 7490 KLK10 Kallikrein-related peptidase 10 SEQ ID NOS: 7491- 7495 KLK11 Kallikrein-related peptidase 11 SEQ ID NOS: 7496- 7504 KLK12 Kallikrein-related peptidase 12 SEQ ID NOS: 7505- 7511 KLK13 Kallikrein-related peptidase 13 SEQ ID NOS: 7512- 7520 KLK14 Kallikrein-related peptidase 14 SEQ ID NOS: 7521- 7522 KLK15 Kallikrein-related peptidase 15 SEQ ID NOS: 7523- 7527 KLK2 Kallikrein-related peptidase 2 SEQ ID NOS: 7528- 7540 KLK3 Kallikrein-related peptidase 3 SEQ ID NOS: 7541- 7552 KLK4 Kallikrein-related peptidase 4 SEQ ID NOS: 7553- 7557 KLK5 Kallikrein-related peptidase 5 SEQ ID NOS: 7558- 7561 KLK6 Kallikrein-related peptidase 6 SEQ ID NOS: 7562- 7568 KLK7 Kallikrein-related peptidase 7 SEQ ID NOS: 7569- 7573 KLK8 Kallikrein-related peptidase 8 SEQ ID NOS: 7574- 7581 KLK9 Kallikrein-related peptidase 9 SEQ ID NOS: 7582- 7583 KLKB1 Kallikrein B, plasma (Fletcher factor) 1 SEQ ID NOS: 7584- 7588 KNDC1 Kinase non-catalytic C-lobe domain SEQ ID NOS: 7593- (KIND) containing 1 7594 KNG1 Kininogen 1 SEQ ID NOS: 7595- 7599 KRBA2 KRAB-A domain containing 2 SEQ ID NOS: 7600- 7603 KREMEN2 Kringle containing transmembrane protein 2 SEQ ID NOS: 7604- 7609 KRTDAP Keratinocyte differentiation-associated SEQ ID NOS: 7610- protein 7611 L1CAM L1 cell adhesion molecule SEQ ID NOS: 7612- 7621 L3MBTL2 L(3)mbt-like 2 (*Drosophila*) SEQ ID NOS: 7622- 7626 LA16c- SEQ ID NO: 72 380H5.3 LACE1 Lactation elevated 1 SEQ ID NOS: 580-583 LACRT Lacritin SEQ ID NOS: 7627- 7629 LACTB Lactamase, beta SEQ ID NOS:

7630- 7632 LAG3 Lymphocyte-activation gene 3 SEQ ID NOS: 7633- 7634 LAIR2 Leukocyte-associated immunoglobulin-like SEQ ID NOS: 7635- receptor 2 7638 LALBA Lactalbumin, alpha-SEQ ID NOS: 7639- 7640 LAMA1 Laminin, alpha 1 SEQ ID NOS: 7641- 7642 LAMA2 Laminin, alpha 2 SEQ ID NOS: 7643- 7646 LAMA3 Laminin, alpha 3 SEQ ID NOS: 7647- 7656 LAMA4 Laminin, alpha 4 SEQ ID NOS: 7657- 7671 LAMA5 Laminin, alpha 5 SEQ ID NOS: 7672- 7674 LAMB1 Laminin, beta 1 SEQ ID NOS: 7675- 7679 LAMB2 Laminin, beta 2 (laminin S) SEQ ID NOS: 7680- 7682 LAMB3 Laminin, beta 3 SEQ ID NOS: 7683- 7687 LAMB4 Laminin, beta 4 SEQ ID NOS: 7688- 7691 LAMC1 Laminin, gamma 1 (formerly LAMB2) SEQ ID NOS: 7692- 7693 LAMC2 Laminin, gamma 2 SEQ ID NOS: 7694- 7695 LAMC3 Laminin, gamma 3 SEQ ID NOS: 7696- 7697 LAMP3 Lysosomal-associated membrane protein 3 SEQ ID NOS: 7698- 7701 LAT Linker for activation of T cells SEQ ID NOS: 7708- 7717 LAT2 Linker for activation of T cells family, SEQ ID NOS: 7718- member 2 7726 LBP Lipopolysaccharide binding protein SEQ ID NO: 7727 LCAT Lecithin-cholesterol acyltransferase SEQ ID NOS: 7728- 7734 LCN1 Lipocalin 1 SEQ ID NOS: 7735- 7736 LCN10 Lipocalin 10 SEQ ID NOS: 7737- 7742 LCN12 Lipocalin 12 SEQ ID NOS: 7743- 7745 LCN15 Lipocalin 15 SEQ ID NO: 7746 LCN2 Lipocalin 2 SEQ ID NOS: 7747- 7749 LCN6 Lipocalin 6 SEQ ID NOS: 7750- 7751 LCN8 Lipocalin 8 SEQ ID NOS: 7752- 7753 LCN9 Lipocalin 9 SEQ ID NOS: 7754- 7755 LCORL Ligand dependent nuclear receptor SEQ ID NOS: 7756- corepressor-like 7761 LDLR Low density lipoprotein receptor SEQ ID NOS: 7762- 7770 LDLRAD2 Low density lipoprotein receptor class A SEQ ID NOS: 7771- domain containing 2 7772 LEAP2 Liver expressed antimicrobial peptide 2 SEQ ID NO: 7773 LECT2 Leukocyte cell-derived chemotaxin 2 SEQ ID NOS: 7774- 7777 LEFTY1 Left-right determination factor 1 SEQ ID NOS: 7778- 7779 LEFTY2 Left-right determination factor 2 SEQ ID NOS: 7780- 7781 LEP Leptin SEQ ID NO: 7782 LFNG LFNG O-fucosylpeptide 3-beta-N- SEQ ID NOS: 7783- acetylglucosaminyltransferase 7788 LGALS3BP Lectin, galactoside-binding, soluble, 3 SEQ ID NOS: 7789- binding protein 7803 LGI1 Leucine-rich, glioma inactivated 1 SEQ ID NOS: 7804- 7822 LGI2 Leucine-rich repeat LGI family, member 2 SEQ ID NOS: 7823- 7824 LGI3 Leucine-rich repeat LGI family, member 3 SEQ ID NOS: 7825- 7828 LGI4 Leucine-rich repeat LGI family, member 4 SEQ ID NOS: 7829- 7832 LGMN Legumain SEQ ID NOS: 7833- 7846 LGR4 Leucine-rich repeat containing G protein- SEQ ID NOS: 7847- coupled receptor 4 7849 LHB Luteinizing hormone beta polypeptide SEQ ID NO: 7850 LHCGR Luteinizing hormone/choriogonadotropin SEQ ID NOS: 7851- receptor 7855 LIF Leukemia inhibitory factor SEQ ID NOS: 7856- 7857 LIFR Leukemia inhibitory factor receptor alpha SEQ ID NOS: 7858- 7862 LILRA1 Leukocyte immunoglobulin-like receptor, SEQ ID NOS: 7863- subfamily A (with TM domain), member 1 7864 LILRA2 Leukocyte immunoglobulin-like receptor, SEQ ID NOS: 7865- subfamily A (with TM domain), member 2 7871 LILRB3 Leukocyte immunoglobulin-like receptor, SEQ ID NOS: 7872- subfamily B (with TM and ITIM domains), 7876 member 3 LIME1 Lck interacting transmembrane adaptor 1 SEQ ID NOS: 7877- 7882 LINGO1 Leucine rich repeat and Ig domain SEQ ID NOS: 7883- containing 1 7893 LIPA Lipase A, lysosomal acid, cholesterol SEQ ID NOS: 7894- esterase 7898 LIPC Lipase, hepatic SEQ ID NOS: 7899- 7902 LIPF Lipase, gastric SEQ ID NOS: 7903- 7906 LIPG Lipase, endothelial SEQ ID NOS: 7907- 7912 LIPH Lipase, member H SEQ ID NOS: 7913- 7917 LIPK Lipase, family member K SEQ ID NO: 7918 LIPM Lipase, family member M SEQ ID NOS: 7919- 7920 LIPN Lipase, family member N SEQ ID NO: 7921 LMAN2 Lectin, mannose-binding 2 SEQ ID NOS: 7922- 7926 LMNTD1 Lamin tail domain containing 1 SEQ ID NOS: 7927- 7937 LNX1 Ligand of numb-protein X 1, E3 ubiquitin SEQ ID NOS: 7938- protein ligase 7944 LOX Lysyl oxidase SEQ ID NOS: 7945- 7947 LOXL1 Lysyl oxidase-like 1 SEQ ID NOS: 7948- 7949 LOXL2 Lysyl oxidase-like 2 SEQ ID NOS: 7950- 7958 LOXL3 Lysyl oxidase-like 3 SEQ ID NOS: 7959- 7965 LOXL4 Lysyl oxidase-like 4 SEQ ID NO: 7966 LPA Lipoprotein, Lp(a) SEQ ID NOS: 7967- 7969 LPL Lipoprotein lipase SEQ ID NOS: 7970- 7974 LPO Lactoperoxidase SEQ ID NOS: 7975- 7981 LRAT Lecithin retinol acyltransferase SEQ ID NOS: 7982- (phosphatidylcholine--retinol O- 7984 acyltransferase) LRCH3 Leucine-rich repeats and calponin SEQ ID NOS: 7985- homology (CH) domain containing 3 7993 LRCOL1 Leucine rich colipase-like 1 SEQ ID NOS: 7994- 7997 LRFN4 Leucine rich repeat and fibronectin type III SEQ ID NOS: 7998- domain containing 4 7999 LRFN5 Leucine rich repeat and fibronectin type III SEQ ID NOS: 8000- domain containing 5

LRG1 Leucine-rich alpha-2-glycoprotein 1 SEQ ID NO: 8003 LRP1 Low density lipoprotein receptor-related SEQ ID NOS: 8004- protein 1 8009 LRP11 Low density lipoprotein receptor-related SEQ ID NOS: 8010- protein 11 8011 LRP1B Low density lipoprotein receptor-related SEQ ID NOS: 8012- protein 1B 8015 LRP2 Low density lipoprotein receptor-related SEQ ID NOS: 8016- protein 2 8017 LRP4 Low density lipoprotein receptor-related SEQ ID NOS: 8018- protein 4 8019 LRPAP1 Low density lipoprotein receptor-related SEQ ID NOS: 8020- protein associated protein 1 8021 LRRC17 Leucine rich repeat containing 17 SEQ ID NOS: 8022- 8024 LRRC32 Leucine rich repeat containing 32 SEQ ID NOS: 8025- 8028 LRRC3B Leucine rich repeat containing 3B SEQ ID NOS: 8029- 8033 LRRC4B Leucine rich repeat containing 4B SEQ ID NOS: 8034- 8036 LRRC70 Leucine rich repeat containing 70 SEQ ID NOS: 8037- 8038 LRRN3 Leucine rich repeat neuronal 3 SEQ ID NOS: 8039- 8042 LRRTM1 Leucine rich repeat transmembrane SEQ ID NOS: 8043- neuronal 1 8049 LRRTM2 Leucine rich repeat transmembrane SEQ ID NOS: 8050- neuronal 2 8052 LRRTM4 Leucine rich repeat transmembrane SEQ ID NOS: 8053- neuronal 4 8058 LRTM2 Leucine-rich repeats and transmembrane SEQ ID NOS: 8059- domains 2 8063 LSR Lipolysis stimulated lipoprotein receptor SEQ ID NOS: 8064- 8074 LST1 Leukocyte specific transcript 1 SEQ ID NOS: 8075- 8092 LTA Lymphotoxin alpha SEQ ID NOS: 8093- 8094 LTBP1 Latent transforming growth factor beta SEQ ID NOS: 8095- binding protein 1 8104 LTBP2 Latent transforming growth factor beta SEQ ID NOS: 8105- binding protein 2 8108 LTBP3 Latent transforming growth factor beta SEQ ID NOS: 8109- binding protein 3 8121 LTBP4 Latent transforming growth factor beta SEQ ID NOS: 8122- binding protein 4 8137 LTBR Lymphotoxin beta receptor (TNFR SEQ ID NOS: 8138- superfamily, member 3) 8143 LTF Lactotransferrin SEQ ID NOS: 8144- 8148 LTK Leukocyte receptor tyrosine kinase SEQ ID NOS: 8149- 8152 LUM Lumican SEQ ID NO: 8153 LUZP2 Leucine zipper protein 2 SEQ ID NOS: 8154- 8157 LVRN Laeverin SEQ ID NOS: 8158- 8163 LY6E Lymphocyte antigen 6 complex, locus E SEQ ID NOS: 8164- 8177 LY6G5B Lymphocyte antigen 6 complex, locus G5B SEQ ID NOS: 8178- 8179 LY6G6D Lymphocyte antigen 6 complex, locus G6D SEQ ID NOS: 8180- 8181 LY6G6E Lymphocyte antigen 6 complex, locus G6E SEQ ID NOS: 8182- (pseudogene) 8185 LY6H Lymphocyte antigen 6 complex, locus H SEQ ID NOS: 8186- 8189 LY6K Lymphocyte antigen 6 complex, locus K SEQ ID NOS: 8190- 8193 LY86 Lymphocyte antigen 86 SEQ ID NOS: 8195- 8196 LY96 Lymphocyte antigen 96 SEQ ID NOS: 8197- 8198 LYG1 Lysozyme G-like 1 SEQ ID NOS: 8199- 8200 LYG2 Lysozyme G-like 2 SEQ ID NOS: 8201- 8206 LYNX1 Ly6/neurotoxin 1 SEQ ID NOS: 8207- 8211 LYPD1 LY6/PLAUR domain containing 1 SEQ ID NOS: 8212- 8214 LYPD2 LY6/PLAUR domain containing 2 SEQ ID NO: 8215 LYPD4 LY6/PLAUR domain containing 4 SEQ ID NOS: 8216- 8218 LYPD6 LY6/PLAUR domain containing 6 SEQ ID NOS: 8219- 8223 LYPD6B LY6/PLAUR domain containing 6B SEQ ID NOS: 8224- 8230 LYPD8 LY6/PLAUR domain containing 8 SEQ ID NOS: 8231- 8232 LYZ Lysozyme SEQ ID NOS: 8233- 8235 LYZL4 Lysozyme-like 4 SEQ ID NOS: 8236- 8237 LYZL6 Lysozyme-like 6 SEQ ID NOS: 8238- 8240 M6PR Mannose-6-phosphate receptor (cation SEQ ID NOS: 8241- dependent) 8251 MAD1L1 MAD1 mitotic arrest deficient-like 1 (yeast) SEQ ID NOS: 8252- 8264 MAG Myelin associated glycoprotein SEQ ID NOS: 8265- 8270 MAGT1 Magnesium transporter 1 SEQ ID NOS: 8271- 8274 MALSU1 Mitochondrial assembly of ribosomal large SEQ ID NO: 8275 subunit 1 MAMDC2 MAM domain containing 2 SEQ ID NO: 8276 MAN2B1 Mannosidase, alpha, class 2B, member 1 SEQ ID NOS: 8277- 8282 MAN2B2 Mannosidase, alpha, class 2B, member 2 SEQ ID NOS: 8283- 8285 MANBA Mannosidase, beta A, lysosomal SEQ ID NOS: 8286- 8299 MANEAL Mannosidase, endo-alpha-like SEQ ID NOS: 8300- 8304 MANF Mesencephalic astrocyte-derived SEQ ID NOS: 8305- neurotrophic factor 8306 MANSC1 MANSC domain containing 1 SEQ ID NOS: 8307- 8310 MAP3K9 Mitogen-activated protein kinase 9 SEQ ID NOS: 8311- 8316 MASP1 Mannan-binding lectin serine peptidase 1 SEQ ID NOS: 8317- (C4/C2 activating component of Ra-reactive 8324 factor) MASP2 Mannan-binding lectin serine peptidase 2 SEQ ID NOS: 8325- 8326 MATN1 Matrilin 1, cartilage matrix protein SEQ ID NO: 8327 MATN2 Matrilin 2 SEQ ID NOS: 8328- 8340 MATN3 Matrilin 3 SEQ ID NOS: 8341- 8342 MATN4 Matrilin 4 SEQ ID NOS: 8343- 8347 MATR3 Matrin 3 SEQ ID NOS: 8348- 8375 MAU2 MAU2 sister chromatid cohesion factor SEQ ID NOS: 8376- 8378 MAZ MYC-associated zinc finger protein (purine- SEQ ID NOS: 8379- binding transcription factor) 8393 MBD6 Methyl-CpG binding domain protein 6 SEQ ID

NOS: 8394- 8405 MBL2 Mannose-binding lectin (protein C) 2, SEQ ID NO: 8406 soluble MBNL1 Muscleblind-like splicing regulator 1 SEQ ID NOS: 8407- 8425 MCCC1 Methylcrotonoyl-CoA carboxylase 1 (alpha) SEQ ID NOS: 8426- 8437 MCCD1 Mitochondrial coiled-coil domain 1 SEQ ID NO: 8438 MCEE Methylmalonyl CoA epimerase SEQ ID NOS: 8439- 8442 MCF2L MCF.2 cell line derived transforming SEQ ID NOS: 8443- sequence-like 8464 MCFD2 Multiple coagulation factor deficiency 2 SEQ ID NOS: 8465- 8476 MDFIC MyoD family inhibitor domain containing SEQ ID NOS: 8477- 8484 MDGA1 MAM domain containing SEQ ID NOS: 8485- glycosylphosphatidylinositol anchor 1 8490 MDK Midkine (neurite growth-promoting factor SEQ ID NOS: 8491- 2) 8500 MED20 Mediator complex subunit 20 SEQ ID NOS: 8501- 8505 MEGF10 Multiple EGF-like-domains 10 SEQ ID NOS: 8506- 8509 MEGF6 Multiple EGF-like-domains 6 SEQ ID NOS: 8510- 8513 MEI1 Meiotic double-stranded break formation SEQ ID NOS: 8514- protein 1 8517 MEI4 Meiotic double-stranded break formation SEQ ID NO: 8518 protein 4 MEIS1 Meis homeobox 1 SEQ ID NOS: 8519- 8524 MEIS3 Meis homeobox 3 SEQ ID NOS: 8525- 8534 MEPE Matrix extracellular phosphoglycoprotein SEQ ID NOS: 8538- 8544 MESDC2 Mesoderm development candidate 2 SEQ ID NOS: 8545- 8549 MEST Mesoderm specific transcript SEQ ID NOS: 8550- 8563 MET MET proto-oncogene, receptor tyrosine SEQ ID NOS: 8564- kinase 8569 METRN Meteorin, glial cell differentiation regulator SEQ ID NOS: 8570- 8574 METRNL Meteorin, glial cell differentiation regulator- SEQ ID NOS: 8575- like 8578 METTL17 Methyltransferase like 17 SEQ ID NOS: 8579- 8589 METTL24 Methyltransferase like 24 SEQ ID NO: 8590 METTL7B Methyltransferase like 7B SEQ ID NOS: 8591- 8592 METTL9 Methyltransferase like 9 SEQ ID NOS: 8593- 8601 MEX3C Mex-3 RNA binding family member C SEQ ID NOS: 8602- 8604 MFAP2 Microfibrillar-associated protein 2 SEQ ID NOS: 8605- 8606 MFAP3 Microfibrillar-associated protein 3 SEQ ID NOS: 8607- 8611 MFAP3L Microfibrillar-associated protein 3-like SEQ ID NOS: 8612- 8621 MFAP4 Microfibrillar-associated protein 4 SEQ ID NOS: 8622- 8624 MFAP5 Microfibrillar associated protein 5 SEQ ID NOS: 8625- 8635 MFGE8 Milk fat globule-EGF factor 8 protein SEQ ID NOS: 8636- 8642 MFI2 Antigen p97 (melanoma associated) SEQ ID NOS: 8535- identified by monoclonal antibodies 133.2 8537 and 96.5 MFNG MFNG O-fucosylpeptide 3-beta-N- SEQ ID NOS: 8643- acetylglucosaminyltransferase 8650 MGA MGA, MAX dimerization protein SEQ ID NOS: 8651- 8659 MGAT2 Mannosyl (alpha-1,6-)-glycoprotein beta- SEQ ID NO: 8660 1,2-N-acetylglucosaminyltransferase MGAT3 Mannosyl (beta-1,4-)-glycoprotein beta-1,4- SEQ ID NOS: 8661- N-acetylglucosaminyltransferase 8663 MGAT4A Mannosyl (alpha-1,3-)-glycoprotein beta- SEQ ID NOS: 8664- 1,4-N-acetylglucosaminyltransferase, 8668 isozyme A MGAT4B Mannosyl (alpha-1,3-)-glycoprotein beta- SEQ ID NOS: 8669- 1,4-N-acetylglucosaminyltransferase, 8679 isozyme B MGAT4D MGAT4 family, member D SEQ ID NOS: 8680- 8685 MGLL Monoglyceride lipase SEQ ID NOS: 8686- 8695 MGP Matrix Gla protein SEQ ID NOS: 8696- 8698 MGST2 Microsomal glutathione S-transferase 2 SEQ ID NOS: 8699- 8702 MIA Melanoma inhibitory activity SEQ ID NOS: 8703- 8708 MIA2 Melanoma inhibitory activity 2 SEQ ID NO: 8709 MIA3 Melanoma inhibitory activity family, SEQ ID NOS: 8710- member 3 8714 MICU1 Mitochondrial calcium uptake 1 SEQ ID NOS: 8715- 8724 MIER1 Mesoderm induction early response 1, SEQ ID NOS: 8725- transcriptional regulator 8733 MINOS1- MINOS1-NBL1 readthrough SEQ ID NOS: 8734- NBL1 8736 MINPP1 Multiple inositol-polyphosphate SEQ ID NOS: 8737- phosphatase 1 8739 MLEC Malectin SEQ ID NOS: 8740- 8743 MLN Motilin SEQ ID NOS: 8744- 8746 MLXIP MLX interacting protein SEQ ID NOS: 8747- 8752 MLXIPL MLX interacting protein-like SEQ ID NOS: 8753- 8760 MMP1 Matrix metallopeptidase 1 SEQ ID NO: 8761 MMP10 Matrix metallopeptidase 10 SEQ ID NOS: 8762- 8763 MMP11 Matrix metallopeptidase 11 SEQ ID NOS: 8764- 8767 MMP12 Matrix metallopeptidase 12 SEQ ID NO: 8768 MMP13 Matrix metallopeptidase 13 SEQ ID NOS: 8769- 8771 MMP14 Matrix metallopeptidase 14 (membrane- SEQ ID NOS: 8772- inserted) 8774 MMP17 Matrix metallopeptidase 17 (membrane- SEQ ID NOS: 8775- inserted) 8782 MMP19 Matrix metallopeptidase 19 SEQ ID NOS: 8783- 8788 MMP2 Matrix metallopeptidase 2 SEQ ID NOS: 8789- 8796 MMP20 Matrix metallopeptidase 20 SEQ ID NO: 8797 MMP21 Matrix metallopeptidase 21 SEQ ID NO: 8798 MMP25 Matrix metallopeptidase 25 SEQ ID NOS: 8799- 8800 MMP26 Matrix metallopeptidase 26 SEQ ID NOS: 8801- 8802 MMP27 Matrix metallopeptidase 27 SEQ ID NO: 8803 MMP28 Matrix metallopeptidase 28 SEQ ID NOS: 8804- 8809



MMP3 Matrix metallopeptidase 3 SEQ ID NOS: 8810- 8812 MMP7 Matrix metallopeptidase 7 SEQ ID NO: 8813 MMP8 Matrix metallopeptidase 8 SEQ ID NOS: 8814- 8819 MMP9 Matrix metallopeptidase 9 SEQ ID NO: 8820 MMRN1 Multimerin 1 SEQ ID NOS: 8821- 8823 MMRN2 Multimerin 2 SEQ ID NOS: 8824- 8828 MOXD1 Monooxygenase, DBH-like 1 SEQ ID NOS: 8829- 8831 MPO Myeloperoxidase SEQ ID NOS: 8840- 8841 MPPED1 Metallophosphoesterase domain containing SEQ ID NOS: 8842- 1 8845 MPZL1 Myelin protein zero-like 1 SEQ ID NOS: 8846- 8850 MR1 Major histocompatibility complex, class I- SEQ ID NOS: 8851- related 8856 MRPL2 Mitochondrial ribosomal protein L2 SEQ ID NOS: 8857- 8861 MRPL21 Mitochondrial ribosomal protein L21 SEQ ID NOS: 8862- 8868 MRPL22 Mitochondrial ribosomal protein L22 SEQ ID NOS: 8869- 8873 MRPL24 Mitochondrial ribosomal protein L24 SEQ ID NOS: 8874- 8878 MRPL27 Mitochondrial ribosomal protein L27 SEQ ID NOS: 8879- 8884 MRPL32 Mitochondrial ribosomal protein L32 SEQ ID NOS: 8885- 8887 MRPL34 Mitochondrial ribosomal protein L34 SEQ ID NOS: 8888- 8892 MRPL35 Mitochondrial ribosomal protein L35 SEQ ID NOS: 8893- 8896 MRPL52 Mitochondrial ribosomal protein L52 SEQ ID NOS: 8897- 8907 MRPL55 Mitochondrial ribosomal protein L55 SEQ ID NOS: 8908- 8933 MRPS14 Mitochondrial ribosomal protein S14 SEQ ID NOS: 8934- 8935 MRPS22 Mitochondrial ribosomal protein S22 SEQ ID NOS: 8936- 8944 MRPS28 Mitochondrial ribosomal protein S28 SEQ ID NOS: 8945- 8952 MS4A14 Membrane-spanning 4-domains, subfamily SEQ ID NOS: 8953- A, member 14 8963 MS4A3 Membrane-spanning 4-domains, subfamily SEQ ID NOS: 8964- A, member 3 (hematopoietic cell-specific) 8968 MSH3 MutS homolog 3 SEQ ID NO: 8969 MSH5 MutS homolog 5 SEQ ID NOS: 8970- 8981 MSLN Mesothelin SEQ ID NOS: 8982- 8989 MSMB Microseminoprotein, beta- SEQ ID NOS: 8990- 8991 MSRA Methionine sulfoxide reductase A SEQ ID NOS: 8992- 8999 MSRB2 Methionine sulfoxide reductase B2 SEQ ID NOS: 9000- 9001 MSRB3 Methionine sulfoxide reductase B3 SEQ ID NOS: 9002- 9015 MST1 Macrophage stimulating 1 SEQ ID NOS: 9016- 9017 MSTN Myostatin SEQ ID NO: 9018 MT1G Metallothionein 1G SEQ ID NOS: 9019- 9022 MTHFD2 Methylenetetrahydrofolate dehydrogenase SEQ ID NOS: 9023- (NADP+ dependent) 2, 9027 methenyltetrahydrofolate cyclohydrolase MTMR14 Myotubularin related protein 14 SEQ ID NOS: 9028- 9038 MTRNR2L11 MT-RNR2-like 11 (pseudogene) SEQ ID NO: 9039 MTRR 5-methyltetrahydrofolate-homocysteine SEQ ID NOS: 9040- methyltransferase reductase 9052 MTTP Microsomal triglyceride transfer protein SEQ ID NOS: 9053- 9063 MTX2 Metaxin 2 SEQ ID NOS: 9064- 9068 MUC1 Mucin 1, cell surface associated SEQ ID NOS: 9069- 9094 MUC13 Mucin 13, cell surface associated SEQ ID NOS: 9095- 9096 MUC20 Mucin 20, cell surface associated SEQ ID NOS: 9097- 9101 MUC3A Mucin 3A, cell surface associated SEQ ID NOS: 9102- 9104 MUC5AC Mucin 5AC, oligomeric mucus/gel-forming SEQ ID NO: 9105 MUC5B Mucin 5B, oligomeric mucus/gel-forming SEQ ID NOS: 9106- 9107 MUC6 Mucin 6, oligomeric mucus/gel-forming SEQ ID NOS: 9108- 9111 MUC7 Mucin 7, secreted SEQ ID NOS: 9112- 9115 MUCL1 Mucin-like 1 SEQ ID NOS: 9116- 9118 MXRA5 Matrix-remodelling associated 5 SEQ ID NO: 9119 MXRA7 Matrix-remodelling associated 7 SEQ ID NOS: 9120- 9126 MYDGF Myeloid-derived growth factor SEQ ID NOS: 9127- 9129 MYL1 Myosin, light chain 1, alkali; skeletal, fast SEQ ID NOS: 9130- 9131 MYOC Myocilin, trabecular meshwork inducible SEQ ID NOS: 9132- glucocorticoid response 9133 MYRFL Myelin regulatory factor-like SEQ ID NOS: 9134- 9138 MZB1 Marginal zone B and B1 cell-specific SEQ ID NOS: 9139- protein 9143 N4BP2L2 NEDD4 binding protein 2-like 2 SEQ ID NOS: 9144- 9149 NAA38 N(alpha)-acetyltransferase 38, NatC SEQ ID NOS: 9150- auxiliary subunit 9155 NAAA N-acyl ethanolamine acid amidase SEQ ID NOS: 9156- 9161 NAGA N-acetylgalactosaminidase, alpha- SEQ ID NOS: 9162- 9164 NAGLU N-acetylglucosaminidase, alpha SEQ ID NOS: 9165- 9169 NAGS N-acetylglutamate synthase SEQ ID NOS: 9170- 9171 NAPSA Napsin A aspartic peptidase SEQ ID NOS: 9172- 9174 NBL1 Neuroblastoma 1, DAN family BMP SEQ ID NOS: 9180- antagonist 9193 NCAM1 Neural cell adhesion molecule 1 SEQ ID NOS: 9194- 9213 NCAN Neurocan SEQ ID NOS: 9214- 9215 NCBP2-AS2 NCBP2 antisense RNA 2 (head to head) SEQ ID NO: 9216 NCSTN Nicastrin SEQ ID NOS: 9217- 9226 NDNF Neuron-derived neurotrophic factor SEQ ID NOS: 9227- 9229 NDP Norrie disease (pseudoglioma) SEQ ID NOS: 9230- 9232 NDUFA10 NADH dehydrogenase (ubiquinone) 1 alpha SEQ ID NOS: 9233- subcomplex, 10, 42 kDa 9242 NDUFB5 NADH dehydrogenase (ubiquinone) 1 beta SEQ ID NOS: 9243- subcomplex, 5, 16 kDa 9251 NDUFS8 NADH



dehydrogenase (ubiquinone) Fe—S SEQ ID NOS: 9252- protein 8, 23 kDa (NADH-coenzyme Q 9261 reductase) NDUFV1 NADH dehydrogenase (ubiquinone) SEQ ID NOS: 9262- flavoprotein 1, 51 kDa 9275 NECAB3 N-terminal EF-hand calcium binding SEQ ID NOS: 9276- protein 3 9285 NELL1 Neural EGFL like 1 SEQ ID NOS: 9289- 9292 NELL2 Neural EGFL like 2 SEQ ID NOS: 9293- 9307 NENF Neudesin neurotrophic factor SEQ ID NO: 9308 NETO1 Neuropilin (NRP) and tolloid (TLL)-like 1 SEQ ID NOS: 9309- 9312 NFASC Neurofascin SEQ ID NOS: 9313- 9327 NFE2L1 Nuclear factor, erythroid 2-like 1 SEQ ID NOS: 9328- 9346 NFE2L3 Nuclear factor, erythroid 2-like 3 SEQ ID NOS: 9347- 9348 NGEF Neuronal guanine nucleotide exchange SEQ ID NOS: 9349- factor 9354 NGF Nerve growth factor (beta polypeptide) SEQ ID NO: 9355 NGLY1 N-glycanase 1 SEQ ID NOS: 9356- 9362 NGRN Neugrin, neurite outgrowth associated SEQ ID NOS: 9363- 9364 NHLRC3 NHL repeat containing 3 SEQ ID NOS: 9365- 9367 NID1 Nidogen 1 SEQ ID NOS: 9368- 9369 NID2 Nidogen 2 (osteonidogen) SEQ ID NOS: 9370- 9372 NKG7 Natural killer cell granule protein 7 SEQ ID NOS: 9373- 9377 NLGN3 Neuroligin 3 SEQ ID NOS: 9378- 9382 NLGN4Y Neuroligin 4, Y-linked SEQ ID NOS: 9383- 9389 NLRP5 NLR family, pyrin domain containing 5 SEQ ID NOS: 9390- 9392 NMB Neuromedin B SEQ ID NOS: 9393- 9394 NME1 NME/NM23 nucleoside diphosphate kinase SEQ ID NOS: 9395- 1 9401 NME1-NME2 NME1-NME2 readthrough SEQ ID NOS: 9402- 9404 NME3 NME/NM23 nucleoside diphosphate kinase SEQ ID NOS: 9405- 3 9409 NMS Neuromedin S SEQ ID NO: 9410 NMU Neuromedin U SEQ ID NOS: 9411- 9414 NOA1 Nitric oxide associated 1 SEQ ID NO: 9415 NODAL Nodal growth differentiation factor SEQ ID NOS: 9416- 9417 NOG Noggin SEQ ID NO: 9418 NOMO3 NODAL modulator 3 SEQ ID NOS: 9419- 9425 NOS1AP Nitric oxide synthase 1 (neuronal) adaptor SEQ ID NOS: 9426- protein 9430 NOTCH3 Notch 3 SEQ ID NOS: 9431- 9434 NOTUM Notum pectinacetyltransferase homolog SEQ ID NOS: 9435- (*Drosophila*) 9437 NOV Nephroblastoma overexpressed SEQ ID NO: 9438 NPB Neuropeptide B SEQ ID NOS: 9439- 9440 NPC2 Niemann-Pick disease, type C2 SEQ ID NOS: 9441- 9449 NPFF Neuropeptide FF-amide peptide precursor SEQ ID NO: 9450 NPFFR2 Neuropeptide FF receptor 2 SEQ ID NOS: 9451- 9454 NPHS1 Nephrosis 1, congenital, Finnish type SEQ ID NOS: 9455- (nephrin) 9456 NPNT Nephronectin SEQ ID NOS: 9457- 9467 NPPA Natriuretic peptide A SEQ ID NOS: 9468- 9470 NPPB Natriuretic peptide B SEQ ID NO: 9471 NPPC Natriuretic peptide C SEQ ID NOS: 9472- 9473 NPS Neuropeptide S SEQ ID NO: 9474 NPTX1 Neuronal pentraxin I SEQ ID NO: 9475 NPTX2 Neuronal pentraxin II SEQ ID NO: 9476 NPTXR Neuronal pentraxin receptor SEQ ID NOS: 9477- 9478 NPVF Neuropeptide VF precursor SEQ ID NO: 9479 NPW Neuropeptide W SEQ ID NOS: 9480- 9482 NPY Neuropeptide Y SEQ ID NOS: 9483- 9485 NQO2 NAD(P)H dehydrogenase, quinone 2 SEQ ID NOS: 9486- 9494 NRCAM Neuronal cell adhesion molecule SEQ ID NOS: 9495- 9507 NRG1 Neuregulin 1 SEQ ID NOS: 9508- 9525 NRN1L Neuritin 1-like SEQ ID NOS: 9526- 9528 NRP1 Neuropilin 1 SEQ ID NOS: 9529- 9542 NRP2 Neuropilin 2 SEQ ID NOS: 9543- 9549 NRTN Neurturin SEQ ID NO: 9550 NRXN1 Neurexin 1 SEQ ID NOS: 9551- 9581 NRXN2 Neurexin 2 SEQ ID NOS: 9582- 9590 NT5C3A 5'-nucleotidase, cytosolic IIIA SEQ ID NOS: 9591- 9601 NT5DC3 5'-nucleotidase domain containing 3 SEQ ID NOS: 9602- 9604 NT5E 5'-nucleotidase, ecto (CD73) SEQ ID NOS: 9605- 9609 NTF3 Neurotrophin 3 SEQ ID NOS: 9610- 9611 NTF4 Neurotrophin 4 SEQ ID NOS: 9612- 9613 NTM Neurotrimin SEQ ID NOS: 9614- 9623 NTN1 Netrin 1 SEQ ID NOS: 9624- 9625 NTN3 Netrin 3 SEQ ID NO: 9626 NTN4 Netrin 4 SEQ ID NOS: 9627- 9631 NTN5 Netrin 5 SEQ ID NOS: 9632- 9633 NTNG1 Netrin G1 SEQ ID NOS: 9634- 9640 NTNG2 Netrin G2 SEQ ID NOS: 9641- 9642 NTS Neurotensin SEQ ID NOS: 9643- 9644 NUBPL Nucleotide binding proteindike SEQ ID NOS: 9645- 9651 NUCB1 Nucleobindin 1 SEQ ID NOS: 9652- 9658 NUCB2 Nucleobindin 2 SEQ ID NOS: 9659- 9674 NUDT19 Nudix (nucleoside diphosphate linked SEQ ID NO: 9675 moiety X)-type motif 19 NUDT9 Nudix (nucleoside diphosphate linked SEQ ID NOS: 9676- moiety X)-type motif 9 9680 NUP155 Nucleoporin 155 kDa SEQ ID NOS: 9681- 9684 NUP214 Nucleoporin 214 kDa SEQ ID NOS: 9685- 9696 NUP85 Nucleoporin 85 kDa SEQ ID NOS: 9697- 9711 NXPE3 Neurexophilin and PC-esterase domain SEQ ID NOS: 9712- family, member 3 9716 NXPE4 Neurexophilin and PC-esterase domain SEQ ID NOS: 9717- family, member 4 9718 NXPH1 Neurexophilin 1 SEQ ID NOS: 9719- 9722 NXPH2 Neurexophilin 2 SEQ ID NO: 9723 NXPH3 Neurexophilin 3 SEQ ID NOS: 9724- 9725 NXPH4 Neurexophilin 4 SEQ ID NOS: 9726- 9727 NYX Nyctalopin SEQ ID NOS: 9728- 9729

OAF Out at first homolog SEQ ID NOS: 9730- 9731 OBP2A Odorant binding protein 2A SEQ ID NOS: 9732- 9738 OBP2B Odorant binding protein 2B SEQ ID NOS: 9739- 9742 OC90 Otoconin 90 SEQ ID NO: 9743 OCLN Occludin SEQ ID NOS: 9744- 9746 ODAM Odontogenic, ameloblast associated SEQ ID NOS: 9747- 9750 OGG1 8-oxoguanine DNA glycosylase SEQ ID NOS: 9755- 9768 OGN Osteoglycin SEQ ID NOS: 9769- 9771 OIT3 Oncoprotein induced transcript 3 SEQ ID NOS: 9772- 9773 OLFM1 Olfactomedin 1 SEQ ID NOS: 9774- 9784 OLFM2 Olfactomedin 2 SEQ ID NOS: 9785- 9788 OLFM3 Olfactomedin 3 SEQ ID NOS: 9789- 9791 OLFM4 Olfactomedin 4 SEQ ID NO: 9792 OLFML1 Olfactomedin-like 1 SEQ ID NOS: 9793- 9796 OLFML2A Olfactomedin-like 2A SEQ ID NOS: 9797- 9799 OLFML2B Olfactomedin-like 2B SEQ ID NOS: 9800- 9804 OLFML3 Olfactomedin-like 3 SEQ ID NOS: 9805- 9807 OMD Osteomodulin SEQ ID NO: 9808 OMG Oligodendrocyte myelin glycoprotein SEQ ID NO: 9809 OOSP2 Oocyte secreted protein 2 SEQ ID NOS: 9810- 9811 OPCML Opioid binding protein/cell adhesion SEQ ID NOS: 9812- molecule-like 9816 OPTC Opticin SEQ ID NOS: 9818- 9819 ORAI1 ORAI calcium release-activated calcium SEQ ID NO: 9820 modulator 1 ORM1 Orosomucoid 1 SEQ ID NO: 9821 ORM2 Orosomucoid 2 SEQ ID NO: 9822 ORMDL2 ORMDL sphingolipid biosynthesis SEQ ID NOS: 9823- regulator 2 9826 OS9 Osteosarcoma amplified 9, endoplasmic SEQ ID NOS: 9827- reticulum lectin 9841 OSCAR Osteoclast associated, immunoglobulin-like SEQ ID NOS: 9842- receptor 9852 OSM Oncostatin M SEQ ID NOS: 9853- 9855 OSMR Oncostatin M receptor SEQ ID NOS: 9856- 9860 OSTN Osteocrin SEQ ID NOS: 9861- 9862 OTOA Otoanchorin SEQ ID NOS: 9863- 9868 OTOG Otogelin SEQ ID NOS: 9869- 9871 OTOGL Otogelin-like SEQ ID NOS: 9872- 9878 OTOL1 Otolin 1 SEQ ID NO: 9879 OTOR Otoraplin SEQ ID NO: 9880 OTOS Otospiralin SEQ ID NOS: 9881- 9882 OVCH1 Ovochymase 1 SEQ ID NOS: 9883- 9885 OVCH2 Ovochymase 2 (gene/pseudogene) SEQ ID NOS: 9886- 9887 OVGP1 Oviductal glycoprotein 1, 120 kDa SEQ ID NO: 9888 OXCT1 3-oxoacid CoA transferase 1 SEQ ID NOS: 9889- 9892 OXCT2 3-oxoacid CoA transferase 2 SEQ ID NO: 9893 OXNAD1 Oxidoreductase NAD-binding domain SEQ ID NOS: 9894- containing 1 9900 OXT Oxytocin/neurophysin I prepropeptide SEQ ID NO: 9901 P3H1 Prolyl 3-hydroxylase 1 SEQ ID NOS: 9902- 9906 P3H2 Prolyl 3-hydroxylase 2 SEQ ID NOS: 9907- 9910 P3H3 Prolyl 3-hydroxylase 3 SEQ ID NO: 9911 P3H4 Prolyl 3-hydroxylase family member 4 SEQ ID NOS: 9912- (non-enzymatic) 9916 P4HA1 Prolyl 4-hydroxylase, alpha polypeptide I SEQ ID NOS: 9917- 9921 P4HA2 Prolyl 4-hydroxylase, alpha polypeptide II SEQ ID NOS: 9922- 9936 P4HA3 Prolyl 4-hydroxylase, alpha polypeptide III SEQ ID NOS: 9937- 9941 P4HB Prolyl 4-hydroxylase, beta polypeptide SEQ ID NOS: 9942- 9953 PAEP Progestagen-associated endometrial protein SEQ ID NOS: 9954- 9962 PAM Peptidylglycine alpha-amidating SEQ ID NOS: 9963- monooxygenase 9976 PAMR1 Peptidase domain containing associated SEQ ID NOS: 9977- with muscle regeneration 1 9983 PAPL Iron/zinc purple acid phosphatase-like SEQ ID NOS: 159-162 protein PAPLN Papilin, proteoglycan-like sulfated SEQ ID NOS: 9984- glycoprotein 9991 PAPPA Pregnancy-associated plasma protein A, SEQ ID NO: 9992 pappalysin 1 PAPPA2 Pappalysin 2 SEQ ID NOS: 9993- 9994 PARP15 Poly (ADP-ribose) polymerase family, SEQ ID NOS: 9995- member 15 9998 PARVB Parvin, beta SEQ ID NOS: 9999- 10003 PATE1 Prostate and testis expressed 1 SEQ ID NOS: 10004- 10005 PATE2 Prostate and testis expressed 2 SEQ ID NOS: 10006- 10007 PATE3 Prostate and testis expressed 3 SEQ ID NO: 10008 PATE4 Prostate and testis expressed 4 SEQ ID NOS: 10009- 10010 PATL2 Protein associated with topoisomerase II SEQ ID NOS: 10011- homolog 2 (yeast) 10016 PAX2 Paired box 2 SEQ ID NOS: 10017- 10022 PAX4 Paired box 4 SEQ ID NOS: 10023- 10029 PCCB Propionyl CoA carboxylase, beta SEQ ID NOS: 10030- polypeptide 10044 PCDH1 Protocadherin 1 SEQ ID NOS: 10045- 10050 PCDH12 Protocadherin 12 SEQ ID NOS: 10051- 10052 PCDH15 Protocadherin-related 15 SEQ ID NOS: 10053- 10086 PCDHA1 Protocadherin alpha 1 SEQ ID NOS: 10087- 10089 PCDHA10 Protocadherin alpha 10 SEQ ID NOS: 10090- 10092 PCDHA11 Protocadherin alpha 11 SEQ ID NOS: 10093- 10095 PCDHA6 Protocadherin alpha 6 SEQ ID NOS: 10096- 10098 PCDHB12 Protocadherin beta 12 SEQ ID NOS: 10099- 10101 PCDHGA11 Protocadherin gamma subfamily A, 11 SEQ ID NOS: 10102- 10104 PCF11 PCF11 cleavage and polyadenylation factor SEQ ID NOS: 10105- subunit 10109 PCOLCE Procollagen C-endopeptidase enhancer SEQ ID NO: 10110 PCOLCE2 Procollagen C-endopeptidase enhancer 2 SEQ ID NOS: 10111- 10114 PCSK1 Proprotein convertase subtilisin/kexin type SEQ ID NOS: 10115- 1 10117

PCSK1N Proprotein convertase subtilisin/kexin type SEQ ID NOS: 10118- 1 inhibitor PCSK2 Proprotein convertase subtilisin/kexin type SEQ ID NOS: 10119- 2 10121 PCSK4 Proprotein convertase subtilisin/kexin type SEQ ID NOS: 10122- 4 10124 PCSK5 Proprotein convertase subtilisin/kexin type SEQ ID NOS: 10125- 5 10129 PCSK9 Proprotein convertase subtilisin/kexin type SEQ ID NO: 10130 9 PCYOX1 Prenylcysteine oxidase 1 SEQ ID NOS: 10131- 10135 PCYOX1L Prenylcysteine oxidase 1 like SEQ ID NOS: 10136- 10140 PDDC1 Parkinson disease 7 domain containing 1 SEQ ID NOS: 5802- 5810 PDE11A Phosphodiesterase 11A SEQ ID NOS: 10141- 10146 PDE2A Phosphodiesterase 2A, cGMP-stimulated SEQ ID NOS: 10147- 10168 PDE7A Phosphodiesterase 7A SEQ ID NOS: 10169- 10172 PDF Peptide deformylase (mitochondrial) SEQ ID NO: 10173 PDGFA Platelet-derived growth factor alpha SEQ ID NOS: 10174- polypeptide 10177 PDGFB Platelet-derived growth factor beta SEQ ID NOS: 10178- polypeptide 10181 PDGFC Platelet derived growth factor C SEQ ID NOS: 10182- 10185 PDGFD Platelet derived growth factor D SEQ ID NOS: 10186- 10188 PDGFRA Platelet-derived growth factor receptor, SEQ ID NOS: 10189- alpha polypeptide 10195 PDGFRB Platelet-derived growth factor receptor, beta SEQ ID NOS: 10196- polypeptide 10199 PDGFRL Platelet-derived growth factor receptor-like SEQ ID NOS: 10200- 10201 PDHA1 Pyruvate dehydrogenase (lipoamide) alpha SEQ ID NOS: 10202- 1 10210 PDIA2 Protein disulfide isomerase family A, SEQ ID NOS: 10211- member 2 10214 PDIA3 Protein disulfide isomerase family A, SEQ ID NOS: 10215- member 3 10218 PDIA4 Protein disulfide isomerase family A, SEQ ID NOS: 10219- member 4 10220 PDIA5 Protein disulfide isomerase family A, SEQ ID NOS: 10221- member 5 10224 PDIA6 Protein disulfide isomerase family A, SEQ ID NOS: 10225- member 6 10231 PDILT Protein disulfide isomerase-like, testis SEQ ID NOS: 10232- expressed 10233 PDYN Prodynorphin SEQ ID NOS: 10234- 10236 PDZD8 PDZ domain containing 8 SEQ ID NO: 10237 PDZRN4 PDZ domain containing ring finger 4 SEQ ID NOS: 10238- 10240 PEAR1 Platelet endothelial aggregation receptor 1 SEQ ID NOS: 10241- 10244 PEBP4 Phosphatidylethanolamine-binding protein 4 SEQ ID NOS: 10245- 10246 PECAM1 Platelet/endothelial cell adhesion molecule SEQ ID NOS: 10247- 1 10250 PENK Proenkephalin SEQ ID NOS: 10251- 10256 PET117 PET117 homolog SEQ ID NO: 10257 PF4 Platelet factor 4 SEQ ID NO: 10258 PF4V1 Platelet factor 4 variant 1 SEQ ID NO: 10259 PFKP Phosphofructokinase, platelet SEQ ID NOS: 10260- 10268 PFN1 Profilin 1 SEQ ID NOS: 10269- 10271 PGA3 Pepsinogen 3, group I (pepsinogen A) SEQ ID NOS: 10272- 10275 PGA4 Pepsinogen 4, group I (pepsinogen A) SEQ ID NOS: 10276- 10278 PGA5 Pepsinogen 5, group I (pepsinogen A) SEQ ID NOS: 10279- 10281 PGAM5 PGAM family member 5, serine/threonine SEQ ID NOS: 10282- protein phosphatase, mitochondrial 10285 PGAP3 Post-GPI attachment to proteins 3 SEQ ID NOS: 10286- 10293 PGC Progastricsin (pepsinogen C) SEQ ID NOS: 10294- 10297 PGF Placental growth factor SEQ ID NOS: 10298- 10301 PGLYRP1 Peptidoglycan recognition protein 1 SEQ ID NO: 10302 PGLYRP2 Peptidoglycan recognition protein 2 SEQ ID NOS: 10303- 10306 PGLYRP3 Peptidoglycan recognition protein 3 SEQ ID NO: 10307 PGLYRP4 Peptidoglycan recognition protein 4 SEQ ID NOS: 10308- 10309 PHACTR1 Phosphatase and actin regulator 1 SEQ ID NOS: 10310- 10316 PHB Prohibitin SEQ ID NOS: 10317- 10325 PI15 Peptidase inhibitor 15 SEQ ID NOS: 10326- 10327 PI3 Peptidase inhibitor 3, skin-derived SEQ ID NO: 10328 PIANP PILR alpha associated neural protein SEQ ID NOS: 10329- 10334 PIGK Phosphatidylinositol glycan anchor SEQ ID NOS: 10335- biosynthesis, class K 10338 PIGL Phosphatidylinositol glycan anchor SEQ ID NOS: 10339- biosynthesis, class L 10346 PIGT Phosphatidylinositol glycan anchor SEQ ID NOS: 10347- biosynthesis, class T 10400 PIGZ Phosphatidylinositol glycan anchor SEQ ID NOS: 10401- biosynthesis, class Z 10403 PIK3AP1 Phosphoinositide-3-kinase adaptor protein 1 SEQ ID NOS: 10404- 10406 PIK3IP1 Phosphoinositide-3-kinase interacting SEQ ID NOS: 10407- protein 1 10410 PILRA Paired immunoglobulin-like type 2 receptor SEQ ID NOS: 10411- alpha 10415 PILRB Paired immunoglobulin-like type 2 receptor SEQ ID NOS: 10416- beta 10427 PINLYP Phospholipase A2 inhibitor and SEQ ID NOS: 10428- LY6/PLAUR domain containing 10432 PIP Prolactin-induced protein SEQ ID NO: 10433 PIWIL4 Piwi-like RNA-mediated gene silencing 4 SEQ ID NOS: 10434- 10438 PKDCC Protein kinase domain containing, SEQ ID NOS: 10439- cytoplasmic 10440 PKHD1 Polycystic kidney and hepatic disease 1 SEQ ID NOS: 10441- (autosomal recessive) 10442 PLA1A Phospholipase A1 member A SEQ ID NOS: 10443- 10447 PLA2G10 Phospholipase A2, group X SEQ

ID NOS: 10448- 10452 PLA2G12A Phospholipase A2, group XIIA SEQ ID NOS: 10450- 10451 PLA2G12B Phospholipase A2, group XIIB SEQ ID NOS: 10453 PLA2G15 Phospholipase A2, group XV SEQ ID NOS: 10454- 10461 PLA2G1B Phospholipase A2, group IB (pancreas) SEQ ID NOS: 10462- 10464 PLA2G2A Phospholipase A2, group IIA (platelets, SEQ ID NOS: 10465- synovial fluid) 10466 PLA2G2C Phospholipase A2, group IIC SEQ ID NOS: 10467- 10468 PLA2G2D Phospholipase A2, group IID SEQ ID NOS: 10469- 10470 PLA2G2E Phospholipase A2, group IIE SEQ ID NOS: 10471 PLA2G3 Phospholipase A2, group III SEQ ID NOS: 10472 PLA2G5 Phospholipase A2, group V SEQ ID NOS: 10473 PLA2G7 Phospholipase A2, group VII (platelet- SEQ ID NOS: 10474- activating factor acetylhydrolase, plasma) 10475 PLA2R1 Phospholipase A2 receptor 1, 180 kDa SEQ ID NOS: 10476- 10477 PLAC1 Placenta-specific 1 SEQ ID NOS: 10478 PLAC9 Placenta-specific 9 SEQ ID NOS: 10479- 10481 PLAT Plasminogen activator, tissue SEQ ID NOS: 10482- 10490 PLAU Plasminogen activator, urokinase SEQ ID NOS: 10491- 10493 PLAUR Plasminogen activator, urokinase receptor SEQ ID NOS: 10494- 10505 PLBD1 Phospholipase B domain containing 1 SEQ ID NOS: 10506- 10508 PLBD2 Phospholipase B domain containing 2 SEQ ID NOS: 10509- 10511 PLG Plasminogen SEQ ID NOS: 10512- 10514 PLGLB1 Plasminogen-like B1 SEQ ID NOS: 10515- 10518 PLGLB2 Plasminogen-like B2 SEQ ID NOS: 10519- 10520 PLOD1 Procollagen-lysine, 2-oxoglutarate 5- SEQ ID NOS: 10521- dioxygenase 1 10523 PLOD2 Procollagen-lysine, 2-oxoglutarate 5- SEQ ID NOS: 10524- dioxygenase 2 10529 PLOD3 Procollagen-lysine, 2-oxoglutarate 5- SEQ ID NOS: 10530- dioxygenase 3 10536 PLTP Phospholipid transfer protein SEQ ID NOS: 10537- 10541 PLXNA4 Plexin A4 SEQ ID NOS: 10542- 10545 PLXNB2 Plexin B2 SEQ ID NOS: 10546- 10554 PM20D1 Peptidase M20 domain containing 1 SEQ ID NOS: 10555 PMCH Pro-melanin-concentrating hormone SEQ ID NOS: 10556 PMEL Premelanosome protein SEQ ID NOS: 10557- 10568 PMEPA1 Prostate transmembrane protein, androgen SEQ ID NOS: 10569- induced 1 10575 PNLIP Pancreatic lipase SEQ ID NOS: 10576 PNLIPRP1 Pancreatic lipase-related protein 1 SEQ ID NOS: 10577- 10585 PNLIPRP3 Pancreatic lipase-related protein 3 SEQ ID NOS: 10586 PNOC Prepronociceptin SEQ ID NOS: 10587- 10589 PNP Purine nucleoside phosphorylase SEQ ID NOS: 10590- 10593 PNPLA4 Patatin-like phospholipase domain SEQ ID NOS: 10594- containing 4 10597 PODNL1 Podocan-like 1 SEQ ID NOS: 10598- 10609 POFUT1 Protein O-fucosyltransferase 1 SEQ ID NOS: 10610- 10611 POFUT2 Protein O-fucosyltransferase 2 SEQ ID NOS: 10612- 10617 POGLUT1 Protein O-glucosyltransferase 1 SEQ ID NOS: 10618- 10622 POLL Polymerase (DNA directed), lambda SEQ ID NOS: 10623- 10635 POMC Proopiomelanocortin SEQ ID NOS: 10636- 10640 POMGNT2 Protein O-linked mannose N- SEQ ID NOS: 10641- acetylglucosaminyltransferase 2 (beta 1,4-) 10642 PON1 Paraoxonase 1 SEQ ID NOS: 10643- 10644 PON2 Paraoxonase 2 SEQ ID NOS: 10645- 10657 PON3 Paraoxonase 3 SEQ ID NOS: 10658- 10663 POSTN Periostin, osteoblast specific factor SEQ ID NOS: 10664- 10669 PPBP Pro-platelet basic protein (chemokine (C-X- SEQ ID NOS: 10670 C motif) ligand 7) PPIB Peptidylprolyl isomerase B (cyclophilin B) SEQ ID NOS: 10671 PPIC Peptidylprolyl isomerase C (cyclophilin C) SEQ ID NOS: 10672 PPOX Protoporphyrinogen oxidase SEQ ID NOS: 10673- 10683 PPP1CA Protein phosphatase 1, catalytic subunit, SEQ ID NOS: 10684- alpha isozyme 10689 PPT1 Palmitoyl-protein thioesterase 1 SEQ ID NOS: 10690- 10706 PPT2 Palmitoyl-protein thioesterase 2 SEQ ID NOS: 10707- 10714 PPY Pancreatic polypeptide SEQ ID NOS: 10715- 10719 PRAC2 Prostate cancer susceptibility candidate 2 SEQ ID NOS: 10720- 10721 PRADC1 Protease-associated domain containing 1 SEQ ID NOS: 10722 PRAP1 Proline-rich acidic protein 1 SEQ ID NOS: 10723- 10724 PRB1 Proline-rich protein BstNI subfamily 1 SEQ ID NOS: 10725- 10728 PRB2 Proline-rich protein BstNI subfamily 2 SEQ ID NOS: 10729- 10730 PRB3 Proline-rich protein BstNI subfamily 3 SEQ ID NOS: 10731- 10732 PRB4 Proline-rich protein BstNI subfamily 4 SEQ ID NOS: 10733- 10736 PRCD Progressive rod-cone degeneration SEQ ID NOS: 10737- 10738 PRCP Prolylcarboxypeptidase (angiotensinase C) SEQ ID NOS: 10739- 10750 PRDM12 PR domain containing 12 SEQ ID NOS: 10751 PRDX4 Peroxiredoxin 4 SEQ ID NOS: 10752- 10755 PRELP Proline/arginine-rich end leucine-rich repeat SEQ ID NOS: 10756 protein PRF1 Perforin 1 (pore forming protein) SEQ ID NOS: 10757- 10759 PRG2 Proteoglycan 2, bone marrow (natural killer SEQ ID NOS: 10760- cell activator, eosinophil granule major 10762 basic protein) PRG3 Proteoglycan 3 SEQ ID NOS: 10763 PRG4 Proteoglycan 4 SEQ ID NOS: 10764- 10769 PRH1 Proline-rich protein HaeIII subfamily 1 SEQ ID

NOS: 10770- 10772 PRH2 Proline-rich protein HaellI subfamily 2 SEQ ID NOS: 10773- 10774 PRKAG1 Protein kinase, AMP-activated, gamma 1 SEQ ID NOS: 10775- non-catalytic subunit 10789 PRKCSH Protein kinase C substrate 80K-H SEQ ID NOS: 10790- 10799 PRKD1 Protein kinase D1 SEQ ID NOS: 10800- 10805 PRL Prolactin SEQ ID NOS: 10806- 10808 PRLH Prolactin releasing hormone SEQ ID NO: 10809 PRLR Prolactin receptor SEQ ID NOS: 10810- 10828 PRNP Prion protein SEQ ID NOS: 10829- 10832 PRNT Prion protein (testis specific) SEQ ID NO: 10833 PROC Protein C (inactivator of coagulation factors SEQ ID NOS: 10834- Va and VIIa) 10841 PROK1 Prokineticin 1 SEQ ID NO: 10842 PROK2 Prokineticin 2 SEQ ID NOS: 10843- 10844 PROL1 Proline rich, lacrimal 1 SEQ ID NO: 9817 PROM1 Prominin 1 SEQ ID NOS: 10845- 10856 PROS1 Protein S (alpha) SEQ ID NOS: 10857- 10860 PROZ Protein Z, vitamin K-dependent plasma SEQ ID NOS: 10861- glycoprotein 10862 PRR27 Proline rich 27 SEQ ID NOS: 10863- 10866 PRR4 Proline rich 4 (lacrimal) SEQ ID NOS: 10867- 10869 PRRG2 Proline rich Gla (G-carboxyglutamic acid) 2 SEQ ID NOS: 10870- 10872 PRRT3 Proline-rich transmembrane protein 3 SEQ ID NOS: 10873- 10875 PRRT4 Proline-rich transmembrane protein 4 SEQ ID NOS: 10876- 10882 PRSS1 Protease, serine, 1 (trypsin 1) SEQ ID NOS: 10883- 10886 PRSS12 Protease, serine, 12 (neurotrypsin, SEQ ID NO: 10887 motopsin) PRSS16 Protease, serine, 16 (thymus) SEQ ID NOS: 10888- 10895 PRSS2 Protease, serine, 2 (trypsin 2) SEQ ID NOS: 10896- 10899 PRSS21 Protease, serine, 21 (testisin) SEQ ID NOS: 10900- 10905 PRSS22 Protease, serine, 22 SEQ ID NOS: 10906- 10908 PRSS23 Protease, serine, 23 SEQ ID NOS: 10909- 10912 PRSS27 Protease, serine 27 SEQ ID NOS: 10913- 10915 PRSS3 Protease, serine, 3 SEQ ID NOS: 10916- 10920 PRSS33 Protease, serine, 33 SEQ ID NOS: 10921- 10924 PRSS35 Protease, serine, 35 SEQ ID NO: 10925 PRSS36 Protease, serine, 36 SEQ ID NOS: 10926- 10929 PRSS37 Protease, serine, 37 SEQ ID NOS: 10930- 10933 PRSS38 Protease, serine, 38 SEQ ID NO: 10934 PRSS42 Protease, serine, 42 SEQ ID NOS: 10935- 10936 PRSS48 Protease, serine, 48 SEQ ID NOS: 10937- 10938 PRSS50 Protease, serine, 50 SEQ ID NO: 10939 PRSS53 Protease, serine, 53 SEQ ID NO: 10940 PRSS54 Protease, serine, 54 SEQ ID NOS: 10941- 10945 PRSS55 Protease, serine, 55 SEQ ID NOS: 10946- 10948 PRSS56 Protease, serine, 56 SEQ ID NOS: 10949- 10950 PRSS57 Protease, serine, 57 SEQ ID NOS: 10951- 10952 PRSS58 Protease, serine, 58 SEQ ID NOS: 10953- 10954 PRSS8 Protease, serine, 8 SEQ ID NOS: 10955- 10958 PRTG Protogenin SEQ ID NOS: 10959- 10962 PRTN3 Proteinase 3 SEQ ID NOS: 10963- 10964 PSAP Prosaposin SEQ ID NOS: 10965- 10968 PSAPL1 Prosaposin-like 1 (gene/pseudogene) SEQ ID NO: 10969 PSG1 Pregnancy specific beta-1-glycoprotein 1 SEQ ID NOS: 10970- 10977 PSG11 Pregnancy specific beta-1-glycoprotein 11 SEQ ID NOS: 10978- 10982 PSG2 Pregnancy specific beta-1-glycoprotein 2 SEQ ID NOS: 10983- 10984 PSG3 Pregnancy specific beta-1-glycoprotein 3 SEQ ID NOS: 10985- 10988 PSG4 Pregnancy specific beta-1-glycoprotein 4 SEQ ID NOS: 10989- 11000 PSG5 Pregnancy specific beta-1-glycoprotein 5 SEQ ID NOS: 11001- 11006 PSG6 Pregnancy specific beta-1-glycoprotein 6 SEQ ID NOS: 11007- 11012 PSG7 Pregnancy specific beta-1-glycoprotein 7 SEQ ID NOS: 11013- (gene/pseudogene) 11015 PSG8 Pregnancy specific beta-1-glycoprotein 8 SEQ ID NOS: 11016- 11020 PSG9 Pregnancy specific beta-1-glycoprotein 9 SEQ ID NOS: 11021- 11028 PSMD1 Proteasome 26S subunit, non-ATPase 1 SEQ ID NOS: 11029- 11036 PSORS1C2 Psoriasis susceptibility 1 candidate 2 SEQ ID NO: 11037 PSPN Persephin SEQ ID NOS: 11038- 11039 PTGDS Prostaglandin D2 synthase 21 kDa (brain) SEQ ID NOS: 11040- 11044 PTGIR Prostaglandin I2 (prostacyclin) receptor (IP) SEQ ID NOS: 11045- 11049 PTGS1 Prostaglandin-endoperoxide synthase 1 SEQ ID NOS: 11050- (prostaglandin G/H synthase and 11058 cyclooxygenase) PTGS2 Prostaglandin-endoperoxide synthase 2 SEQ ID NOS: 11059- (prostaglandin G/H synthase and 11060 cyclooxygenase) PTH Parathyroid hormone SEQ ID NOS: 11061- 11062 PTH2 Parathyroid hormone 2 SEQ ID NO: 11063 PTHLH Parathyroid hormone-like hormone SEQ ID NOS: 11064- 11072 PTK7 Protein tyrosine kinase 7 (inactive) SEQ ID NOS: 11073- 11088 PTN Pleiotrophin SEQ ID NOS: 11089- 11090 PTPRA Protein tyrosine phosphatase, receptor type, SEQ ID NOS: 11091- A 11098 PTPRB Protein tyrosine phosphatase, receptor type, SEQ ID NOS: 11099- B 11106 PTPRC Protein tyrosine phosphatase, receptor type, SEQ ID NOS: 11107- C 11117 PTPRCAP Protein tyrosine phosphatase, receptor type, SEQ ID NO: 11118 C-associated protein PTPRD Protein tyrosine phosphatase, receptor type, SEQ ID NOS: 11119- D 11130 PTPRF Protein tyrosine phosphatase, receptor type, SEQ ID NOS: 11131- F

11138 PTPRJ Protein tyrosine phosphatase, receptor type, SEQ ID NOS: 11139- J 11144 PTPRO  
 Protein tyrosine phosphatase, receptor type, SEQ ID NOS: 11145- O 11153 PTPRS Protein tyrosine  
 phosphatase, receptor type, SEQ ID NOS: 11154- S 11161 PTTG1IP Pituitary tumor-transforming 1  
 interacting SEQ ID NOS: 11162- protein 11165 PTX3 Pentraxin 3, long SEQ ID NO: 11166 PTX4  
 Pentraxin 4, long SEQ ID NOS: 11167- 11169 PVR Poliovirus receptor SEQ ID NOS: 11170- 11175  
 PVRL1 Poliovirus receptor-related 1 (herpesvirus SEQ ID NOS: 9286- entry mediator C) 9288 PXDN  
 Peroxidasin SEQ ID NOS: 11176- 11180 PXDNL Peroxidasin-like SEQ ID NOS: 11181- 11183  
 PXYLP1 2-phosphoxylose phosphatase 1 SEQ ID NOS: 11184- 11196 PYY Peptide YY SEQ ID NOS:  
 11197- 11198 PZP Pregnancy-zone protein SEQ ID NOS: 11199- 11200 QPCT Glutaminyl-peptide  
 cyclotransferase SEQ ID NOS: 11201- 11203 QPRT Quinolate phosphoribosyltransferase SEQ ID  
 NOS: 11204- 11205 QRFP Pyroglutamylated RFamide peptide SEQ ID NOS: 11206- 11207 QSOX1  
 Quiescin Q6 sulfhydryl oxidase 1 SEQ ID NOS: 11208- 11211 R3HDM1 R3H domain containing-like  
 SEQ ID NO: 11212 RAB26 RAB26, member RAS oncogene family SEQ ID NOS: 11213- 11216  
 RAB36 RAB36, member RAS oncogene family SEQ ID NOS: 11217- 11219 RAB9B RAB9B, member  
 RAS oncogene family SEQ ID NO: 11220 RAET1E Retinoic acid early transcript 1E SEQ ID NOS:  
 11221- 11226 RAET1G Retinoic acid early transcript 1G SEQ ID NOS: 11227- 11229 RAMP2  
 Receptor (G protein-coupled) activity SEQ ID NOS: 11230- modifying protein 2 11234 RAPGEF5 Rap  
 guanine nucleotide exchange factor SEQ ID NOS: 11235- (GEF) 5 11241 RARRES1 Retinoic acid  
 receptor responder (tazarotene SEQ ID NOS: 11242- induced) 1 11243 RARRES2 Retinoic acid  
 receptor responder (tazarotene SEQ ID NOS: 11244- induced) 2 11247 RASA2 RAS p21 protein  
 activator 2 SEQ ID NOS: 11248- 11250 RBM3 RNA binding motif (RNP1, RRM) protein 3 SEQ ID  
 NOS: 11251- 11253 RBP3 Retinol binding protein 3, interstitial SEQ ID NO: 11254 RBP4 Retinol  
 binding protein 4, plasma SEQ ID NOS: 11255- 11258 RCN1 Reticulocalbin 1, EF-hand calcium  
 binding SEQ ID NOS: 11259- domain 11262 RCN2 Reticulocalbin 2, EF-hand calcium binding SEQ  
 ID NOS: 11263- domain 11266 RCN3 Reticulocalbin 3, EF-hand calcium binding SEQ ID NOS:  
 11267- domain 11270 RCOR1 REST corepressor 1 SEQ ID NOS: 11271- 11272 RDH11 Retinol  
 dehydrogenase 11 (all-trans/9- SEQ ID NOS: 11273- cis/11-cis) 11280 RDH12 Retinol dehydrogenase  
 12 (all-trans/9- SEQ ID NOS: 11281- cis/11-cis) 11282 RDH13 Retinol dehydrogenase 13 (all-trans/9-  
 cis) SEQ ID NOS: 11283- 11291 RDH5 Retinol dehydrogenase 5 (11-cis/9-cis) SEQ ID NOS: 11292-  
 11296 RDH8 Retinol dehydrogenase 8 (all-trans) SEQ ID NOS: 11297- 11298 REG1A Regenerating  
 islet-derived 1 alpha SEQ ID NO: 11299 REG1B Regenerating islet-derived 1 beta SEQ ID NOS:  
 11300- 11301 REG3A Regenerating islet-derived 3 alpha SEQ ID NOS: 11302- 11304 REG3G  
 Regenerating islet-derived 3 gamma SEQ ID NOS: 11305- 11307 REG4 Regenerating islet-derived  
 family, member SEQ ID NOS: 11308- 4 11311 RELN Reelin SEQ ID NOS: 11312- 11315 RELT RELT  
 tumor necrosis factor receptor SEQ ID NOS: 11316- 11319 REN Renin SEQ ID NOS: 11320- 11321  
 REPIN1 Replication initiator 1 SEQ ID NOS: 11322- 11335 REPS2 RALBP1 associated Eps domain  
 containing SEQ ID NOS: 11336- 2 11337 RET Ret proto-oncogene SEQ ID NOS: 11338- 11343 RETN  
 Resistin SEQ ID NOS: 11344- 11346 RETNLB Resistin like beta SEQ ID NO: 11347 RETSAT Retinol  
 saturase (all-trans-retinol 13,14- SEQ ID NOS: 11348- reductase) 11352 RFNG RFNG O-  
 fucosylpeptide 3-beta-N- SEQ ID NOS: 11353- acetylglucosaminyltransferase 11355 RGCC Regulator  
 of cell cycle SEQ ID NO: 11356 RGL4 Ral guanine nucleotide dissociation SEQ ID NOS: 11357-  
 stimulator-like 4 11363 RGMA Repulsive guidance molecule family SEQ ID NOS: 11364- member a  
 11373 RGMB Repulsive guidance molecule family SEQ ID NOS: 11374- member b 11375 RHOQ Ras  
 homolog family member Q SEQ ID NOS: 11376- 11380 RIC3 RIC3 acetylcholine receptor chaperone  
 SEQ ID NOS: 11381- 11388 RIMS1 Regulating synaptic membrane exocytosis 1 SEQ ID NOS: 11393-  
 11408 RIPPLY1 Ripply transcriptional repressor 1 SEQ ID NOS: 11409- 11410 RLN1 Relaxin 1 SEQ  
 ID NO: 11411 RLN2 Relaxin 2 SEQ ID NOS: 11412- 11413 RLN3 Relaxin 3 SEQ ID NOS: 11414-  
 11415 RMDN1 Regulator of microtubule dynamics 1 SEQ ID NOS: 11416- 11429 RNASE1  
 Ribonuclease, RNase A family, 1 SEQ ID NOS: 11430- (pancreatic) 11434 RNASE10 Ribonuclease,  
 RNase A family, 10 (non- SEQ ID NOS: 11435- active) 11436 RNASE11 Ribonuclease, RNase A  
 family, 11 (non- SEQ ID NOS: 11437- active) 11447 RNASE12 Ribonuclease, RNase A family, 12  
 (non- SEQ ID NO: 11448 active) RNASE13 Ribonuclease, RNase A family, 13 (non- SEQ ID NO:

11449 active) RNASE2 Ribonuclease, RNase A family, 2 (liver, SEQ ID NO: 11450 eosinophil-derived neurotoxin) RNASE3 Ribonuclease, RNase A family, 3 SEQ ID NO: 11451 RNASE4 Ribonuclease, RNase A family, 4 SEQ ID NOS: 11452- 11454 RNASE6 Ribonuclease, RNase A family, k6 SEQ ID NO: 11455 RNASE7 Ribonuclease, RNase A family, 7 SEQ ID NOS: 11456- 11457 RNASE8 Ribonuclease, RNase A family, 8 SEQ ID NO: 11458 RNASE9 Ribonuclease, RNase A family, 9 (non-SEQ ID NOS: 11459- active) 11469 RNASEH1 Ribonuclease H1 SEQ ID NOS: 11470- 11472 RNASET2 Ribonuclease T2 SEQ ID NOS: 11473- 11480 RNF146 Ring finger protein 146 SEQ ID NOS: 11481- 11492 RNF148 Ring finger protein 148 SEQ ID NOS: 11493- 11494 RNF150 Ring finger protein 150 SEQ ID NOS: 11495- 11499 RNF167 Ring finger protein 167 SEQ ID NOS: 11500- 11510 RNF220 Ring finger protein 220 SEQ ID NOS: 11511- 11517 RNF34 Ring finger protein 34, E3 ubiquitin protein SEQ ID NOS: 11518- ligase 11525 RNLS Renalase, FAD-dependent amine oxidase SEQ ID NOS: 11526- 11528 RNPEP Arginyl aminopeptidase (aminopeptidase B) SEQ ID NOS: 11529- 11534 ROR1 Receptor tyrosine kinase-like orphan SEQ ID NOS: 11535- receptor 1 11537 RP11- SEQ ID NO: 4158 1236K1.1 RP11-14J7.7 SEQ ID NOS: 674-675 RP11- SEQ ID NOS: 85-87 196G11.1 RP11- SEQ ID NO: 683 350O14.18 RP11- SEQ ID NO: 8194 520P18.5 RP11- SEQ ID NO: 89 812E19.9 RP11- SEQ ID NO: 676 903H12.5 RP11- SEQ ID NOS: 78-80 977G19.10 RP4-576H24.4 SEQ ID NOS: 670-672 RP4-608O15.3 Complement factor H-related protein 2 SEQ ID NO: 1649 RPL3 Ribosomal protein L3 SEQ ID NOS: 11538- 11543 RPLP2 Ribosomal protein, large, P2 SEQ ID NOS: 11544- 11546 RPN2 Ribophorin II SEQ ID NOS: 11547- 11553 RPS27L Ribosomal protein S27-like SEQ ID NOS: 11554- 11559 RQCD1 RCD1 required for cell differentiation1 SEQ ID NOS: 3100- homolog (*S. pombe*) 3106 RS1 Retinoschisin 1 SEQ ID NO: 11560 RSF1 Remodeling and spacing factor 1 SEQ ID NOS: 11561- 11567 RSPO1 R-spondin 1 SEQ ID NOS: 11568- 11571 RSPO2 R-spondin 2 SEQ ID NOS: 11572- 11579 RSPO3 R-spondin 3 SEQ ID NOS: 11580- 11581 RSPO4 R-spondin 4 SEQ ID NOS: 11582- 11583 RSPRY1 Ring finger and SPRY domain containing 1 SEQ ID NOS: 11584- 11590 RTBDN Retbindin SEQ ID NOS: 11591- 11603 RTN4RL1 Reticulon 4 receptor-like 1 SEQ ID NO: 11604 RTN4RL2 Reticulon 4 receptor-like 2 SEQ ID NOS: 11605- 11607 SAA1 Serum amyloid A1 SEQ ID NOS: 11608- 11610 SAA2 Serum amyloid A2 SEQ ID NOS: 11611- 11616 SAA4 Serum amyloid A4, constitutive SEQ ID NO: 11617 SAP30 Sin3A-associated protein, 30 kDa SEQ ID NO: 11618 SAR1A Secretion associated, Ras related GTPase SEQ ID NOS: 11619- 1A 11625 SARAF Store-operated calcium entry-associated SEQ ID NOS: 11626- regulatory factor 11636 SARM1 Sterile alpha and TIR motif containing 1 SEQ ID NOS: 11637- 11640 SATB1 SATB homeobox 1 SEQ ID NOS: 11641- 11653 SAXO2 Stabilizer of axonemal microtubules 2 SEQ ID NOS: 11654- 11658 SBSN Suprabasin SEQ ID NOS: 11659- 11661 SBSPON Somatomedin B and thrombospondin, type SEQ ID NO: 11662 1 domain containing SCARF1 Scavenger receptor class F, member 1 SEQ ID NOS: 11663- 11667 SCG2 Secretogranin II SEQ ID NOS: 11668- 11670 SCG3 Secretogranin III SEQ ID NOS: 11671- 11673 SCG5 Secretogranin V SEQ ID NOS: 11674- 11678 SCGB1A1 Secretoglobulin, family 1A, member 1 SEQ ID NOS: 11679- (uteroglobin) 11680 SCGB1C1 Secretoglobulin, family 1C, member 1 SEQ ID NO: 11681 SCGB1C2 Secretoglobulin, family 1C, member 2 SEQ ID NO: 11682 SCGB1D1 Secretoglobulin, family 1D, member 1 SEQ ID NO: 11683 SCGB1D2 Secretoglobulin, family 1D, member 2 SEQ ID NO: 11684 SCGB1D4 Secretoglobulin, family 1D, member 4 SEQ ID NO: 11685 SCGB2A1 Secretoglobulin, family 2A, member 1 SEQ ID NO: 11686 SCGB2A2 Secretoglobulin, family 2A, member 2 SEQ ID NOS: 11687- 11688 SCGB2B2 Secretoglobulin, family 2B, member 2 SEQ ID NOS: 11689- 11690 SCGB3A1 Secretoglobulin, family 3A, member 1 SEQ ID NO: 11691 SCGB3A2 Secretoglobulin, family 3A, member 2 SEQ ID NOS: 11692- 11693 SCN1B Sodium channel, voltage gated, type I beta SEQ ID NOS: 11694- subunit 11699 SCN3B Sodium channel, voltage gated, type III beta SEQ ID NOS: 11700- subunit 11704 SCPEP1 Serine carboxypeptidase 1 SEQ ID NOS: 11705- 11712 SCRG1 Stimulator of chondrogenesis 1 SEQ ID NOS: 11713- 11714 SCT Secretin SEQ ID NO: 11715 SCUBE1 Signal peptide, CUB domain, EGF-like 1 SEQ ID NOS: 11716- 11719 SCUBE2 Signal peptide, CUB domain, EGF-like 2 SEQ ID NOS: 11720- 11726 SCUBE3 Signal peptide, CUB domain, EGF-like 3 SEQ ID NO: 11727 SDC1 Syndecan 1 SEQ ID NOS: 11728- 11732 SDF2 Stromal cell-derived factor 2 SEQ ID NOS: 11733- 11735 SDF2L1 Stromal cell-derived factor 2-like 1 SEQ ID NO: 11736 SDF4 Stromal cell derived factor 4 SEQ ID NOS: 11737- 11740 SDHAF2



Succinate dehydrogenase complex assembly SEQ ID NOS: 11741- factor 2 11748 SDHAF4 Succinate dehydrogenase complex assembly SEQ ID NO: 11749 factor 4 SDHB Succinate dehydrogenase complex, subunit SEQ ID NOS: 11750- B, iron sulfur (Ip) 11752 SDHD Succinate dehydrogenase complex, subunit SEQ ID NOS: 11753- D, integral membrane protein 11762 SEC14L3 SEC14-like lipid binding 3 SEQ ID NOS: 11763- 11769 SEC16A SEC16 homolog A, endoplasmic reticulum SEQ ID NOS: 11770- export factor 11776 SEC16B SEC16 homolog B, endoplasmic reticulum SEQ ID NOS: 11777- export factor 11780 SEC22C SEC22 homolog C, vesicle trafficking SEQ ID NOS: 11781- protein 11793 SEC31A SEC31 homolog A, COPII coat complex SEQ ID NOS: 11794- component 11823 SECISBP2 SECIS binding protein 2 SEQ ID NOS: 11824- 11828 SECTM1 Secreted and transmembrane 1 SEQ ID NOS: 11829- 11836 SEL1L Sel-1 suppressor of lin-12-like (*C. elegans*) SEQ ID NOS: 11837- 11839 SELM Selenoprotein M SEQ ID NOS: 11847- 11849 SELO Selenoprotein O SEQ ID NOS: 11854- 11855 SEMA3A Sema domain, immunoglobulin domain SEQ ID NOS: 11862- (Ig), short basic domain, secreted, 11866 (semaphorin) 3A SEMA3B Sema domain, immunoglobulin domain SEQ ID NOS: 11867- (Ig), short basic domain, secreted, 11873 (semaphorin) 3B SEMA3C Sema domain, immunoglobulin domain SEQ ID NOS: 11874- (Ig), short basic domain, secreted, 11878 (semaphorin) 3C SEMA3E Sema domain, immunoglobulin domain SEQ ID NOS: 11879- (Ig), short basic domain, secreted, 11883 (semaphorin) 3E SEMA3F Sema domain, immunoglobulin domain SEQ ID NOS: 11884- (Ig), short basic domain, secreted, 11890 (semaphorin) 3F SEMA3G Sema domain, immunoglobulin domain SEQ ID NOS: 11891- (Ig), short basic domain, secreted, 11893 (semaphorin) 3G SEMA4A Sema domain, immunoglobulin domain SEQ ID NOS: 11894- (Ig), transmembrane domain (TM) and short 11902 cytoplasmic domain, (semaphorin) 4A SEMA4B Sema domain, immunoglobulin domain SEQ ID NOS: 11903- (Ig), transmembrane domain (TM) and short 11913 cytoplasmic domain, (semaphorin) 4B SEMA4C Sema domain, immunoglobulin domain SEQ ID NOS: 11914- (Ig), transmembrane domain (TM) and short 11916 cytoplasmic domain, (semaphorin) 4C SEMA4D Sema domain, immunoglobulin domain SEQ ID NOS: 11917- (Ig), transmembrane domain (TM) and short 11930 cytoplasmic domain, (semaphorin) 4D SEMA4F Sema domain, immunoglobulin domain SEQ ID NOS: 11931- (Ig), transmembrane domain (TM) and short 11939 cytoplasmic domain, (semaphorin) 4F SEMA4G Sema domain, immunoglobulin domain SEQ ID NOS: 11940- (Ig), transmembrane domain (TM) and short 11947 cytoplasmic domain, (semaphorin) 4G SEMA5A Sema domain, seven thrombospondin SEQ ID NOS: 11948- repeats (type 1 and type 1-like), 11949 transmembrane domain (TM) and short cytoplasmic domain, (semaphorin) 5A SEMA6A Sema domain, transmembrane domain SEQ ID NOS: 11950- (TM), and cytoplasmic domain, 11957 (semaphorin) 6A SEMA6C Sema domain, transmembrane domain SEQ ID NOS: 11958- (TM), and cytoplasmic domain, 11963 (semaphorin) 6C SEMA6D Sema domain, transmembrane domain SEQ ID NOS: 11964- (TM), and cytoplasmic domain, 11977 (semaphorin) 6D SEMG1 Semenogelin I SEQ ID NO: 11978 SEMG2 Semenogelin II SEQ ID NO: 11979 SEPN1 Selenoprotein N, 1 SEQ ID NOS: 11850- 11853 SEPP1 Selenoprotein P, plasma, 1 SEQ ID NOS: 11856- 11861 SEPT15 15 kDa selenoprotein SEQ ID NOS: 11840- 11846 SEPT9 Septin 9 SEQ ID NOS: 11980- 12016 SERPINA1 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NOS: 12017- antiproteinase, antitrypsin), member 1 12033 SERPINA10 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NOS: 12034- antiproteinase, antitrypsin), member 10 12037 SERPINA11 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NO: 12038 antiproteinase, antitrypsin), member 11 SERPINA12 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NOS: 12039- antiproteinase, antitrypsin), member 12 12040 SERPINA3 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NOS: 673- antiproteinase, antitrypsin), member 3 12047 SERPINA4 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NOS: 12048- antiproteinase, antitrypsin), member 4 12050 SERPINA5 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NOS: 12051- antiproteinase, antitrypsin), member 5 12062 SERPINA6 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NOS: 12063- antiproteinase, antitrypsin), member 6 12065 SERPINA7 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NOS: 12066- antiproteinase, antitrypsin), member 7 12067 SERPINA9 Serpin peptidase inhibitor, clade A (alpha-1 SEQ ID NOS: 12068- antiproteinase, antitrypsin), member 9 12074 SERPINB2 Serpin peptidase inhibitor, clade B SEQ ID NOS: 12075- (ovalbumin), member 2 12079 SERPINC1 Serpin peptidase inhibitor, clade C SEQ ID NOS: 12080- (antithrombin), member 1



12081 SERPIND1 Serpin peptidase inhibitor, clade D (heparin SEQ ID NOS: 12082- cofactor), member 1 12083 SERPINE1 Serpin peptidase inhibitor, clade E (nexin. SEQ ID NO: 12084 plasminogen activator inhibitor type 1), member 1 SERPINE2 Serpin peptidase inhibitor, clade E (nexin, SEQ ID NOS: 12085- plasminogen activator inhibitor type 1), 12091 member 2 SERPINE3 Serpin peptidase inhibitor, clade E (nexin, SEQ ID NOS: 12092- plasminogen activator inhibitor type 1), 12095 member 3 SERPINF1 Serpin peptidase inhibitor, clade F (alpha-2 SEQ ID NOS: 12096- antiplasmin, pigment epithelium derived 12104 factor), member 1 SERPINF2 Serpin peptidase inhibitor, clade F (alpha-2 SEQ ID NOS: 12105- antiplasmin, pigment epithelium derived 12109 factor), member 2 SERPING1 Serpin peptidase inhibitor, clade G (C1 SEQ ID NOS: 12110- inhibitor), member 1 12120 SERPINH1 Serpin peptidase inhibitor, clade H (heat SEQ ID NOS: 12121- shock protein 47), member 1, (collagen 12135 binding protein 1) SERPINI1 Serpin peptidase inhibitor, clade I SEQ ID NOS: 12136- (neuroserpin), member 1 12140 SERPINI2 Serpin peptidase inhibitor, clade I (pancpin), SEQ ID NOS: 12141- member 2 12147 SETD8 SET domain containing (lysine SEQ ID NOS: 7589- methyltransferase) 8 7592 SEZ6L2 Seizure related 6 homolog (mouse)-like 2 SEQ ID NOS: 12148- 12154 SFRP1 Secreted frizzled-related protein 1 SEQ ID NOS: 12155- 12156 SFRP2 Secreted frizzled-related protein 2 SEQ ID NO: 12157 SFRP4 Secreted frizzled-related protein 4 SEQ ID NOS: 12158- 12159 SFRP5 Secreted frizzled-related protein 5 SEQ ID NO: 12160 SFTA2 Surfactant associated 2 SEQ ID NOS: 12161- 12162 SFTPA1 Surfactant protein A1 SEQ ID NOS: 12163- 12167 SFTPA2 Surfactant protein A2 SEQ ID NOS: 12168- 12172 SFTPB Surfactant protein B SEQ ID NOS: 12173- 12177 SFTPD Surfactant protein D SEQ ID NOS: 12178- 12179 SFXN5 Sideroflexin 5 SEQ ID NOS: 12180- 12184 SGCA Sarcoglycan, alpha (50 kDa dystrophin- SEQ ID NOS: 12185- associated glycoprotein) 12192 SGSH N-sulfoglucosamine sulfohydrolase SEQ ID NOS: 12193- 12201 SH3RF3 SH3 domain containing ring finger 3 SEQ ID NO: 12202 SHBG Sex hormone-binding globulin SEQ ID NOS: 12203- 12221 SHE Src homology 2 domain containing E SEQ ID NOS: 12222- 12224 SHH Sonic hedgehog SEQ ID NOS: 12225- 12228 SHKBP1 SH3KBP1 binding protein 1 SEQ ID NOS: 12229- 12244 SIAE Sialic acid acetyltransferase SEQ ID NOS: 12245- 12247 SIDT2 SID1 transmembrane family, member 2 SEQ ID NOS: 12248- 12257 SIGLEC10 Sialic acid binding Ig-like lectin 10 SEQ ID NOS: 12258- 12266 SIGLEC6 Sialic acid binding Ig-like lectin 6 SEQ ID NOS: 12267- 12272 SIGLEC7 Sialic acid binding Ig-like lectin 7 SEQ ID NOS: 12273- 12277 SIGLECL1 SIGLEC family like 1 SEQ ID NOS: 12278- 12283 SIGMAR1 Sigma non-opioid intracellular receptor 1 SEQ ID NOS: 12284- 12287 SIL1 SIL1 nucleotide exchange factor SEQ ID NOS: 12288- 12296 SIRPB1 Signal-regulatory protein beta 1 SEQ ID NOS: 12297- 12309 SIRPD Signal-regulatory protein delta SEQ ID NOS: 12310- 12312 SLAMF1 Signaling lymphocytic activation molecule SEQ ID NOS: 12313- family member 1 12315 SLAMF7 SLAM family member 7 SEQ ID NOS: 12316- 12324 SLC10A3 Solute carrier family 10, member 3 SEQ ID NOS: 12325- 12329 SLC15A3 Solute carrier family 15 (oligopeptide SEQ ID NOS: 12330- transporter), member 3 12335 SLC25A14 Solute carrier family 25 (mitochondrial SEQ ID NOS: 12336- carrier, brain), member 14 12342 SLC25A25 Solute carrier family 25 (mitochondrial SEQ ID NOS: 12343- carrier; phosphate carrier), member 25 12349 SLC2A5 Solute carrier family 2 (facilitated SEQ ID NOS: 12350- glucose/fructose transporter), member 5 12358 SLC35E3 Solute carrier family 35, member E3 SEQ ID NOS: 12359- 12360 SLC39A10 Solute carrier family 39 (zinc transporter), SEQ ID NOS: 12361- member 10 12367 SLC39A14 Solute carrier family 39 (zinc transporter), SEQ ID NOS: 12368- member 14 12378 SLC39A4 Solute carrier family 39 (zinc transporter), SEQ ID NOS: 12379- member 4 12381 SLC39A5 Solute carrier family 39 (zinc transporter), SEQ ID NOS: 12382- member 5 12388 SLC3A1 Solute carrier family 3 (amino acid SEQ ID NOS: 12389- transporter heavy chain), member 1 12398 SLC51A Solute carrier family 51, alpha subunit SEQ ID NOS: 12399- 12403 SLC52A2 Solute carrier family 52 (riboflavin SEQ ID NOS: 12404- transporter), member 2 12414 SLC5A6 Solute carrier family 5 SEQ ID NOS: 12415- (sodium/multivitamin and iodide 12425 cotransporter), member 6 SLC6A9 Solute carrier family 6 (neurotransmitter SEQ ID NOS: 12426- transporter, glycine), member 9 12433 SLC8A1 Solute carrier family 8 (sodium/calcium SEQ ID NOS: 12434- exchanger), member 1 12445 SLC8B1 Solute carrier family 8 SEQ ID NOS: 12446- (sodium/lithium/calcium exchanger), 12456 member B1 SLC9A6 Solute carrier family 9, subfamily A SEQ ID NOS: 12457-

(NHE6, cation proton antiporter 6), member 12468 6 SLC01A2 Solute carrier organic anion transporter  
 SEQ ID NOS: 12469- family, member 1A2 12481 SLIT1 Slit guidance ligand 1 SEQ ID NOS: 12482-  
 12485 SLIT2 Slit guidance ligand 2 SEQ ID NOS: 12486- 12494 SLIT3 Slit guidance ligand 3 SEQ ID  
 NOS: 12495- 12497 SLITRK3 SLIT and NTRK-like family, member 3 SEQ ID NOS: 12498- 12500  
 SLPI Secretory leukocyte peptidase inhibitor SEQ ID NO: 12501 SLTM SAFB-like, transcription  
 modulator SEQ ID NOS: 12502- 12515 SLURP1 Secreted LY6/PLAUR domain containing 1 SEQ ID  
 NO: 12516 SMARCA2 SWI/SNF related, matrix associated, actin SEQ ID NOS: 12517- dependent  
 regulator of chromatin, subfamily 12562 a, member 2 SMG6 SMG6 nonsense mediated mRNA decay  
 SEQ ID NOS: 12563- factor 12574 SMIM7 Small integral membrane protein 7 SEQ ID NOS: 12575-  
 12591 SMOC1 SPARC related modular calcium binding 1 SEQ ID NOS: 12592- 12593 SMOC2  
 SPARC related modular calcium binding 2 SEQ ID NOS: 12594- 12598 SMPDL3A Sphingomyelin  
 phosphodiesterase, acid-like SEQ ID NOS: 12599- 3A 12600 SMPDL3B Sphingomyelin  
 phosphodiesterase, acid-like SEQ ID NOS: 12601- 3B 12605 SMR3A Submaxillary gland androgen  
 regulated SEQ ID NO: 12606 protein 3A SMR3B Submaxillary gland androgen regulated SEQ ID  
 NOS: 12607- protein 3B 12609 SNED1 Sushi, nidogen and EGF-like domains 1 SEQ ID NOS: 12610-  
 12616 SNTB1 Syntrophin, beta 1 (dystrophin-associated SEQ ID NOS: 12617- protein A1, 59 kDa,  
 basic component 1) 12619 SNTB2 Syntrophin, beta 2 (dystrophin-associated SEQ ID NOS: 12620-  
 protein A1, 59 kDa, basic component 2) 12624 SNX14 Sorting nexin 14 SEQ ID NOS: 12625- 12636  
 SOD3 Superoxide dismutase 3, extracellular SEQ ID NOS: 12637- 12638 SOST Sclerostin SEQ ID  
 NO: 12639 SOSTDC1 Sclerostin domain containing 1 SEQ ID NOS: 12640- 12641 SOWAHA  
 Sosondowah ankyrin repeat domain family SEQ ID NO: 12642 member A SPACA3 Sperm acrosome  
 associated 3 SEQ ID NOS: 12643- 12645 SPACA4 Sperm acrosome associated 4 SEQ ID NO: 12646  
 SPACA5 Sperm acrosome associated 5 SEQ ID NOS: 12647- 12648 SPACA5B Sperm acrosome  
 associated 5B SEQ ID NO: 12649 SPACA7 Sperm acrosome associated 7 SEQ ID NOS: 12650- 12653  
 SPAG11A Sperm associated antigen 11A SEQ ID NOS: 12654- 12662 SPAG11B Sperm associated  
 antigen 11B SEQ ID NOS: 12663- 12671 SPARC Secreted protein, acidic, cysteine-rich SEQ ID NOS:  
 12672- (osteonectin) 12676 SPARCL1 SPARC-like 1 (hevin) SEQ ID NOS: 12677- 12686 SPATA20  
 Spermatogenesis associated 20 SEQ ID NOS: 12687- 12700 SPESP1 Sperm equatorial segment protein  
 1 SEQ ID NO: 12701 SPINK1 Serine peptidase inhibitor, Kazal type 1 SEQ ID NOS: 12702- 12703  
 SPINK13 Serine peptidase inhibitor, Kazal type 13 SEQ ID NOS: 12704- (putative) 12706 SPINK14  
 Serine peptidase inhibitor, Kazal type 14 SEQ ID NOS: 12707- (putative) 12708 SPINK2 Serine  
 peptidase inhibitor, Kazal type 2 SEQ ID NOS: 12709- (acrosin-trypsin inhibitor) 12714 SPINK4  
 Serine peptidase inhibitor, Kazal type 4 SEQ ID NOS: 12715- 12716 SPINK5 Serine peptidase  
 inhibitor, Kazal type 5 SEQ ID NOS: 12717- 12722 SPINK6 Serine peptidase inhibitor, Kazal type 6  
 SEQ ID NOS: 12723- 12725 SPINK7 Serine peptidase inhibitor, Kazal type 7 SEQ ID NOS: 12726-  
 (putative) 12727 SPINK8 Serine peptidase inhibitor, Kazal type 8 SEQ ID NO: 12728 (putative)  
 SPINK9 Serine peptidase inhibitor, Kazal type 9 SEQ ID NOS: 12729- 12730 SPINT1 Serine peptidase  
 inhibitor, Kunitz type 1 SEQ ID NOS: 12731- 12738 SPINT2 Serine peptidase inhibitor, Kunitz type, 2  
 SEQ ID NOS: 12739- 12746 SPINT3 Serine peptidase inhibitor, Kunitz type, 3 SEQ ID NO: 12747  
 SPINT4 Serine peptidase inhibitor, Kunitz type 4 SEQ ID NO: 12748 SPOCK1 Sparc/osteonectin,  
 cwcv and kazal-like SEQ ID NOS: 12749- domains proteoglycan (testican) 1 12752 SPOCK2  
 Sparc/osteonectin, cwcv and kazal-like SEQ ID NOS: 12753- domains proteoglycan (testican) 2 12756  
 SPOCK3 Sparc/osteonectin, cwcv and kazal-like SEQ ID NOS: 12757- domains proteoglycan (testican)  
 3 12782 SPON1 Spondin 1, extracellular matrix protein SEQ ID NO: 12783 SPON2 Spondin 2,  
 extracellular matrix protein SEQ ID NOS: 12784- 12793 SPP1 Secreted phosphoprotein 1 SEQ ID  
 NOS: 12794- 12798 SPP2 Secreted phosphoprotein 2, 24 kDa SEQ ID NOS: 12799- 12801 SPRN  
 Shadow of prion protein homolog SEQ ID NO: 12802 (zebrafish) SPRYD3 SPRY domain containing 3  
 SEQ ID NOS: 12803- 12806 SPRYD4 SPRY domain containing 4 SEQ ID NO: 12807 SPTY2D1-AS1  
 SPTY2D1 antisense RNA 1 SEQ ID NOS: 12808- 12813 SPX Spexin hormone SEQ ID NOS: 12814-  
 12815 SRGN Serglycin SEQ ID NO: 12816 SRL Sarcalumenin SEQ ID NOS: 12817- 12819 SRP14  
 Signal recognition particle 14 kDa SEQ ID NOS: 12820- (homologous Alu RNA binding protein)  
 12823 SRPX Sushi-repeat containing protein, X-linked SEQ ID NOS: 12824- 12827 SRPX2 Sushi-

repeat contain protein X-linked 2 SEQ ID NOS: 12828- 12831 SSC4D Scavenger receptor cysteine rich family, 4 SEQ ID NO: 12832 domains SSC5D Scavenger receptor cysteine rich family, 5 SEQ ID NOS: 12833- domains 12836 SSPO SCO-spondin SEQ ID NO: 12837 SSR2 Signal sequence receptor, beta (translocon- SEQ ID NOS: 12838- associated protein beta) 12847 SST Somatostatin SEQ ID NO: 12848 ST3GAL1 ST3 beta-galactoside alpha-2,3- SEQ ID NOS: 12849- sialyltransferase 1 12856 ST3GAL4 ST3 beta-galactoside alpha-2,3- SEQ ID NOS: 12857- sialyltransferase 4 12872 ST6GAL1 ST6 beta-galactosamide alpha-2,6- SEQ ID NOS: 12873- sialyltransferase 1 12888 ST6GALNAC2 ST6 (alpha-N-acetyl-neuraminy-2,3-beta- SEQ ID NOS: 12889- galactosyl-1,3)-N-acetylgalactosaminide 12893 alpha-2,6-sialyltransferase 2 ST6GALNAC5 ST6 (alpha-N-acetyl-neuraminy-2,3-beta- SEQ ID NOS: 12894- galactosyl-1,3)-N-acetylgalactosaminide 12895 alpha-2,6-sialyltransferase 5 ST6GALNAC6 ST6 (alpha-N-acetyl-neuraminy-2,3-beta- SEQ ID NOS: 12896- galactosyl-1,3)-N-acetylgalactosaminide 12903 alpha-2,6-sialyltransferase 6 ST8SIA2 ST8 alpha-N-acetyl-neuraminide alpha-2,8- SEQ ID NOS: 12904- sialyltransferase 2 12906 ST8SIA4 ST8 alpha-N-acetyl-neuraminide alpha-2,8- SEQ ID NOS: 12907- sialyltransferase 4 12909 ST8SIA6 ST8 alpha-N-acetyl-neuraminide alpha-2,8- SEQ ID NOS: 12910- sialyltransferase 6 12911 STARD7 StAR-related lipid transfer (START) SEQ ID NOS: 12912- domain containing 7 12913 STATH Statherin SEQ ID NOS: 12914- 12916 STC1 Stanniocalcin 1 SEQ ID NOS: 12917- 12918 STC2 Stanniocalcin 2 SEQ ID NOS: 12919- 12921 STMND1 Stathmin domain containing 1 SEQ ID NOS: 12922- 12923 STOML2 Stomatin (EPB72)-like 2 SEQ ID NOS: 12926- 12929 STOX1 Storkhead box 1 SEQ ID NOS: 12930- 12934 STRC Stereocilin SEQ ID NOS: 12935- 12940 SUCLG1 Succinate-CoA ligase, alpha subunit SEQ ID NOS: 12941- 12942 SUDS3 SDS3 homolog, SIN3A corepressor SEQ ID NO: 12943 complex component SULF1 Sulfatase 1 SEQ ID NOS: 12944- 12954 SULF2 Sulfatase 2 SEQ ID NOS: 12955- 12959 SUMF1 Sulfatase modifying factor 1 SEQ ID NOS: 12960- 12964 SUMF2 Sulfatase modifying factor 2 SEQ ID NOS: 12965- 12978 SUSD1 Sushi domain containing 1 SEQ ID NOS: 12979- 12984 SUSD5 Sushi domain containing 5 SEQ ID NOS: 12985- 12986 SVEP1 Sushi, von Willebrand factor type A, EGF SEQ ID NOS: 12987- and pentraxin domain containing 1 12989 SWSAP1 SWIM-type zinc finger 7 associated protein SEQ ID NO: 12990 1 SYAP1 Synapse associated protein 1 SEQ ID NO: 12991 SYCN Syncollin SEQ ID NO: 12992 TAC1 Tachykinin, precursor 1 SEQ ID NOS: 12993- 12995 TAC3 Tachykinin 3 SEQ ID NOS: 12996- 13005 TAC4 Tachykinin 4 (hemokinin) SEQ ID NOS: 13006- 13011 TAGLN2 Transgelin 2 SEQ ID NOS: 13012- 13015 TAPBP TAP binding protein (tapasin) SEQ ID NOS: 13016- 13021 TAPBPL TAP binding protein-like SEQ ID NOS: 13022- 13023 TBL2 Transducin (beta)-like 2 SEQ ID NOS: 13024- 13036 TBX10 T-box 10 SEQ ID NO: 13037 TCF12 Transcription factor 12 SEQ ID NOS: 13038- 13051 TCN1 Transcobalamin I (vitamin B12 binding SEQ ID NO: 13052 protein, R binder family) TCN2 Transcobalamin II SEQ ID NOS: 13053- 13056 TCTN1 Tectonic family member 1 SEQ ID NOS: 13057- 13075 TCTN3 Tectonic family member 3 SEQ ID NOS: 13076- 13080 TDP2 Tyrosyl-DNA phosphodiesterase 2 SEQ ID NOS: 13081- 13082 TEK TEK tyrosine kinase, endothelial SEQ ID NOS: 13097- 13101 TEPP Testis, prostate and placenta expressed SEQ ID NOS: 13102- 13103 TEX101 Testis expressed 101 SEQ ID NOS: 13104- 13105 TEX264 Testis expressed 264 SEQ ID NOS: 13106- 13117 TF Transferrin SEQ ID NOS: 13121- 13127 TFAM Transcription factor A, mitochondrial SEQ ID NOS: 13128- 13130 TFF1 Trefoil factor 1 SEQ ID NO: 13131 TFF2 Trefoil factor 2 SEQ ID NO: 13132 TFF3 Trefoil factor 3 (intestinal) SEQ ID NOS: 13133- 13135 TFPI Tissue factor pathway inhibitor (lipoprotein- SEQ ID NOS: 13136- associated coagulation inhibitor) 13145 TFPI2 Tissue factor pathway inhibitor 2 SEQ ID NOS: 13146- 13147 TG Thyroglobulin SEQ ID NOS: 13148- 13157 TGFB1 Transforming growth factor, beta 1 SEQ ID NOS: 13158- 13159 TGFB2 Transforming growth factor, beta 2 SEQ ID NOS: 13160- 13161 TGFB3 Transforming growth factor, beta 3 SEQ ID NOS: 13162- 13163 TGFB1 Transforming growth factor, beta-induced, SEQ ID NOS: 13164- 68 kDa 13171 TGFB1 Transforming growth factor, beta receptor 1 SEQ ID NOS: 13172- 13181 TGFB3 Transforming growth factor, beta receptor SEQ ID NOS: 13182- III 13188 THBS1 Thrombospondin 1 SEQ ID NOS: 13189- 13190 THBS2 Thrombospondin 2 SEQ ID NOS: 13191- 13193 THBS3 Thrombospondin 3 SEQ ID NOS: 13194- 13198 THBS4 Thrombospondin 4 SEQ ID NOS: 13199- 13200 THOC3 THO complex 3 SEQ ID NOS: 13201- 13210 THPO Thrombopoietin SEQ ID NOS: 13211- 13216 THSD4 Thrombospondin, type I,

domain containing SEQ ID NOS: 13217- 4 13220 THY1 Thy-1 cell surface antigen SEQ ID NOS: 13221- 13226 TIE1 Tyrosine kinase with immunoglobulin-like SEQ ID NOS: 13227- and EGF-like domains 1 13228 TIMMDC1 Translocase of inner mitochondrial SEQ ID NOS: 13229- membrane domain containing 1 13236 TIMP1 TIMP metallopeptidase inhibitor 1 SEQ ID NOS: 13237- 13241 TIMP2 TIMP metallopeptidase inhibitor 2 SEQ ID NOS: 13242- 13246 TIMP3 TIMP metallopeptidase inhibitor 3 SEQ ID NO: 13247 TIMP4 TIMP metallopeptidase inhibitor 4 SEQ ID NO: 13248 TINAGL1 Tubulointerstitial nephritis antigen-like 1 SEQ ID NOS: 13249- 13251 TINF2 TERF1 (TRF1)-interacting nuclear factor 2 SEQ ID NOS: 13252- 13261 TLL2 Tolloid-like 2 SEQ ID NO: 13262 TLR1 Toll-like receptor 1 SEQ ID NOS: 13263- 13268 TLR3 Toll-like receptor 3 SEQ ID NOS: 13269- 13271 TM2D2 TM2 domain containing 2 SEQ ID NOS: 13272- 13277 TM2D3 TM2 domain containing 3 SEQ ID NOS: 13278- 13285 TM7SF3 Transmembrane 7 superfamily member 3 SEQ ID NOS: 13286- 13300 TM9SF1 Transmembrane 9 superfamily member 1 SEQ ID NOS: 13301- 13311 TMCO6 Transmembrane and coiled-coil domains 6 SEQ ID NOS: 13312- 13319 TMED1 Transmembrane p24 trafficking protein 1 SEQ ID NOS: 13320- 13326 TMED2 Transmembrane p24 trafficking protein 2 SEQ ID NOS: 13327- 13329 TMED3 Transmembrane p24 trafficking protein 3 SEQ ID NOS: 13330- 13333 TMED4 Transmembrane p24 trafficking protein 4 SEQ ID NOS: 13334- 13336 TMED5 Transmembrane p24 trafficking protein 5 SEQ ID NOS: 13337- 13340 TMED7 Transmembrane p24 trafficking protein 7 SEQ ID NOS: 13341- 13342 TMED7- TMED7-TICAM2 readthrough SEQ ID NOS: 13343- TICAM2 13344 TMEM108 Transmembrane protein 108 SEQ ID NOS: 13345- 13353 TMEM116 Transmembrane protein 116 SEQ ID NOS: 13354- 13365 TMEM119 Transmembrane protein 119 SEQ ID NOS: 13366- 13369 TMEM155 Transmembrane protein 155 SEQ ID NOS: 13370- 13373 TMEM168 Transmembrane protein 168 SEQ ID NOS: 13374- 13379 TMEM178A Transmembrane protein 178A SEQ ID NOS: 13380- 13381 TMEM179 Transmembrane protein 179 SEQ ID NOS: 13382- 13387 TMEM196 Transmembrane protein 196 SEQ ID NOS: 13388- 13392 TMEM199 Transmembrane protein 199 SEQ ID NOS: 13393- 13396 TMEM205 Transmembrane protein 205 SEQ ID NOS: 13397- 13410 TMEM213 Transmembrane protein 213 SEQ ID NOS: 13411- 13414 TMEM25 Transmembrane protein 25 SEQ ID NOS: 13415- 13431 TMEM30C Transmembrane protein 30C SEQ ID NO: 13432 TMEM38B Transmembrane protein 38B SEQ ID NOS: 13433- 13437 TMEM44 Transmembrane protein 44 SEQ ID NOS: 13438- 13447 TMEM52 Transmembrane protein 52 SEQ ID NOS: 13448- 13452 TMEM52B Transmembrane protein 52B SEQ ID NOS: 13453- 13455 TMEM59 Transmembrane protein 59 SEQ ID NOS: 13456- 13463 TMEM67 Transmembrane protein 67 SEQ ID NOS: 13464- 13475 TMEM70 Transmembrane protein 70 SEQ ID NOS: 13476- 13478 TMEM87A Transmembrane protein 87A SEQ ID NOS: 13479- 13488 TMEM94 Transmembrane protein 94 SEQ ID NOS: 13489- 13504 TMEM95 Transmembrane protein 95 SEQ ID NOS: 13505- 13507 TMIGD1 Transmembrane and immunoglobulin SEQ ID NOS: 13508- domain containing 1 13509 TMPRSS12 Transmembrane (C-terminal) protease, SEQ ID NOS: 13510- serine 12 13511 TMPRSS5 Transmembrane protease, serine 5 SEQ ID NOS: 13512- 13523 TMUB1 Transmembrane and ubiquitin-like domain SEQ ID NOS: 13524- containing 1 13530 TMX2 Thioredoxin-related transmembrane protein SEQ ID NOS: 13531- 2 13538 TMX3 Thioredoxin-related transmembrane protein SEQ ID NOS: 13539- 3 13546 TNC Tenascin C SEQ ID NOS: 13547- 13555 TNFAIP6 Tumor necrosis factor, alpha-induced SEQ ID NO: 13556 protein 6 TNFRSF11A Tumor necrosis factor receptor superfamily, SEQ ID NOS: 13557- member 11a, NFkB activator 13561 TNFRSF11B Tumor necrosis factor receptor superfamily, SEQ ID NOS: 13562- member 11b 13563 TNFRSF12A Tumor necrosis factor receptor superfamily, SEQ ID NOS: 13564- member 12A 13569 TNFRSF14 Tumor necrosis factor receptor superfamily, SEQ ID NOS: 13570- member 14 13576 TNFRSF18 Tumor necrosis factor receptor superfamily, SEQ ID NOS: 13577- member 18 13580 TNFRSF1A Tumor necrosis factor receptor superfamily, SEQ ID NOS: 13581- member 1A 13589 TNFRSF1B Tumor necrosis factor receptor superfamily, SEQ ID NOS: 13590- member 1B 13591 TNFRSF25 Tumor necrosis factor receptor superfamily, SEQ ID NOS: 13592- member 25 13603 TNFRSF6B Tumor necrosis factor receptor superfamily, SEQ ID NO: 13604 member 6b, decoy TNFSF11 Tumor necrosis factor (ligand) superfamily, SEQ ID NOS: 13605- member 11 13609 TNFSF12 Tumor necrosis factor (ligand) superfamily, SEQ ID NOS: 13610- member 12 13611

TNFSF12- TNFSF12-TNF through SEQ ID NO: 13612 TNFSF13 TNFSF13 Tumor necrosis factor (ligand) superfamily, SEQ ID NOS: 13613- member 15 13614 TNN Tenascin N SEQ ID NOS: 13615- 13617 TNR Tenascin R SEQ ID NOS: 13618- 13620 TNXB Tenascin XB SEQ ID NOS: 13621- 13627 TOMM7 Translocase of outer mitochondrial SEQ ID NOS: 13634- membrane 7 homolog (yeast) 13637 TOP1MT Topoisomerase (DNA) I, mitochondrial SEQ ID NOS: 13638- 13652 TOR1A Torsin family 1, member A (torsin A) SEQ ID NO: 13653 TOR1B Torsin family 1, member B (torsin B) SEQ ID NOS: 13654- 13655 TOR2A Torsin family 2, member A SEQ ID NOS: 13656- 13662 TOR3A Torsin family 3, member A SEQ ID NOS: 13663- 13667 TPD52 Tumor protein D52 SEQ ID NOS: 13668- 13680 TPO Thyroid peroxidase SEQ ID NOS: 13681- 13691 TPP1 Tripeptidyl peptidase I SEQ ID NOS: 13692- 13709 TPSAB1 Tryptase alpha/beta 1 SEQ ID NOS: 13710- 13712 TPSB2 Tryptase beta 2 (gene/pseudogene) SEQ ID NOS: 13713- 13715 TPSD1 Tryptase delta 1 SEQ ID NOS: 13716- 13717 TPST1 Tyrosylprotein sulfotransferase 1 SEQ ID NOS: 13718- 13720 TPST2 Tyrosylprotein sulfotransferase 2 SEQ ID NOS: 13721- 13729 TRABD2A TraB domain containing 2A SEQ ID NOS: 13730- 13732 TRABD2B TraB domain containing 2B SEQ ID NO: 13733 TREH Trehalase (brush-border membrane SEQ ID NOS: 13734- glycoprotein) 13736 TREM1 Triggering receptor expressed on myeloid SEQ ID NOS: 13737- cells 1 13740 TREM2 Triggering receptor expressed on myeloid SEQ ID NOS: 13741- cells 2 13743 TRH Thyrotropin-releasing hormone SEQ ID NOS: 13744- 13745 TRIM24 Tripartite motif containing 24 SEQ ID NOS: 13746- 13747 TRIM28 Tripartite motif containing 28 SEQ ID NOS: 13748- 13753 TRIO Trio Rho guanine nucleotide exchange SEQ ID NOS: 13754- factor 13760 TRNP1 TMF1-regulated nuclear protein 1 SEQ ID NOS: 13761- 13762 TSC22D4 TSC22 domain family, member 4 SEQ ID NOS: 13763- 13766 TSHB Thyroid stimulating hormone, beta SEQ ID NOS: 13767- 13768 TSHR Thyroid stimulating hormone receptor SEQ ID NOS: 13769- 13776 TSKU Tsukushi, small leucine rich proteoglycan SEQ ID NOS: 13777- 13781 TSLP Thymic stromal lymphopoietin SEQ ID NOS: 13782- 13784 TSPAN3 Tetraspanin 3 SEQ ID NOS: 13785- 13790 TSPAN31 Tetraspanin 31 SEQ ID NOS: 13791- 13797 TSPEAR Thrombospondin-type laminin G domain SEQ ID NOS: 13798- and EAR repeats 13801 TTC13 Tetratricopeptide repeat domain 13 SEQ ID NOS: 13802- 13808 TTC19 Tetratricopeptide repeat domain 19 SEQ ID NOS: 13809- 13814 TTC9B Tetratricopeptide repeat domain 9B SEQ ID NO: 13815 TTLL11 Tubulin tyrosine ligase-like family member SEQ ID NOS: 13816- 11 13820 TTR Transthyretin SEQ ID NOS: 13821- 13823 TWSG1 Twisted gastrulation BMP signaling SEQ ID NOS: 13824- modulator 1 13826 TXNDC12 Thioredoxin domain containing 12 SEQ ID NOS: 13827- (endoplasmic reticulum) 13829 TXNDC15 Thioredoxin domain containing 15 SEQ ID NOS: 13830- 13836 TXNDC5 Thioredoxin domain containing 5 SEQ ID NOS: 13837- (endoplasmic reticulum) 13838 TXNRD2 Thioredoxin reductase 2 SEQ ID NOS: 13839- 13851 TYRP1 Tyrosinase-related protein 1 SEQ ID NOS: 13852- 13854 UBAC2 UBA domain containing 2 SEQ ID NOS: 13855- 13859 UBALD1 UBA-like domain containing 1 SEQ ID NOS: 13860- 13868 UBAP2 Ubiquitin associated protein 2 SEQ ID NOS: 13869- 13875 UBXN8 UBX domain protein 8 SEQ ID NOS: 13876- 13882 UCMA Upper zone of growth plate and cartilage SEQ ID NOS: 13883- matrix associated 13884 UCN Urocortin SEQ ID NO: 13885 UCN2 Urocortin 2 SEQ ID NO: 13886 UCN3 Urocortin 3 SEQ ID NO: 13887 UGGT2 UDP-glucose glycoprotein SEQ ID NOS: 13888- glucosyltransferase 2 13893 UGT1A10 UDP glucuronosyltransferase 1 family, SEQ ID NOS: 13894- polypeptide A10 13895 UGT2A1 UDP glucuronosyltransferase 2 family, SEQ ID NOS: 13896- polypeptide A1, complex locus 13900 UGT2B11 UDP glucuronosyltransferase 2 family, SEQ ID NO: 13901 polypeptide B11 UGT2B28 UDP glucuronosyltransferase 2 family, SEQ ID NOS: 13902- polypeptide B28 13903 UGT2B4 UDP glucuronosyltransferase 2 family, SEQ ID NOS: 13904- polypeptide B4 13907 UGT2B7 UDP glucuronosyltransferase 2 family, SEQ ID NOS: 13908- polypeptide B7 13911 UGT3A1 UDP glycosyltransferase 3 family, SEQ ID NOS: 13912- polypeptide A1 13917 UGT3A2 UDP glycosyltransferase 3 family, SEQ ID NOS: 13918- polypeptide A2 13921 UGT8 UDP glycosyltransferase 8 SEQ ID NOS: 13922- 13924 ULBP3 UL16 binding protein 3 SEQ ID NOS: 13925- 13926 UMOD Uromodulin SEQ ID NOS: 13927- 13938 UNC5C Unc-5 netrin receptor C SEQ ID NOS: 13939- 13943 UPK3B Uroplakin 3B SEQ ID NOS: 13944- 13946 USP11 Ubiquitin specific peptidase 11 SEQ ID NOS: 13947- 13950 USP14 Ubiquitin specific peptidase 14 (tRNA- SEQ ID NOS: 13951- guanine transglycosylase) 13957 USP3 Ubiquitin specific peptidase 3

SEQ ID NOS: 13958- 13973 UTS2 SEQ ID NOS: 13984- 13986 UTS2B Urotensin 2B  
SEQ ID NOS: 13987- 13992 UTY Ubiquitously transcribed tetratricopeptide SEQ ID NOS: 13993-  
repeat containing, Y-linked 14005 UXS1 UDP-glucuronate decarboxylase 1 SEQ ID NOS: 14006-  
14013 VASH1 Vasohibin 1 SEQ ID NOS: 14014- 14016 VCAN Versican SEQ ID NOS: 14017- 14023  
VEGFA Vascular endothelial growth factor A SEQ ID NOS: 14024- 14049 VEGFB Vascular  
endothelial growth factor B SEQ ID NOS: 14050- 14052 VEGFC Vascular endothelial growth factor C  
SEQ ID NO: 14053 VGF VGF nerve growth factor inducible SEQ ID NOS: 14055- 14057 VIP  
Vasoactive intestinal peptide SEQ ID NOS: 14058- 14060 VIPR2 Vasoactive intestinal peptide receptor  
2 SEQ ID NOS: 14061- 14064 VIT Vitrin SEQ ID NOS: 14065- 14072 VKORC1 Vitamin K epoxide  
reductase complex, SEQ ID NOS: 14073- subunit 1 14080 VLDLR Very low density lipoprotein  
receptor SEQ ID NOS: 14081- 14083 VMO1 Vitelline membrane outer layer 1 homolog SEQ ID NOS:  
14084- (chicken) 14087 VNN1 Vanin 1 SEQ ID NO: 14088 VNN2 Vanin 2 SEQ ID NOS: 14089-  
14102 VNN3 Vanin 3 SEQ ID NOS: 14103- 14114 VOPP1 Vesicular, overexpressed in cancer, SEQ ID  
NOS: 14115- prosurvival protein 1 14127 VPREB1 Pre-B lymphocyte 1 SEQ ID NOS: 14128- 14129  
VPREB3 Pre-B lymphocyte 3 SEQ ID NOS: 14130- 14131 VPS37B Vacuolar protein sorting 37  
homolog B SEQ ID NOS: 14132- (*S. cerevisiae*) 14134 VPS51 Vacuolar protein sorting 51 homolog  
SEQ ID NOS: 14135- (*S. cerevisiae*) 14146 VSIG1 V-set and immunoglobulin domain SEQ ID NOS:  
14147- containing 1 14149 VSIG10 V-set and immunoglobulin domain SEQ ID NOS: 14150-  
containing 10 14151 VSTM1 V-set and transmembrane domain SEQ ID NOS: 14152- containing 1  
14158 VSTM2A V-set and transmembrane domain SEQ ID NOS: 14159- containing 2A 14162  
VSTM2B V-set and transmembrane domain SEQ ID NO: 14163 containing 2B VSTM2L V-set and  
transmembrane domain SEQ ID NOS: 14164- containing 2 like 14166 VSTM4 V-set and  
transmembrane domain SEQ ID NOS: 14167- containing 4 14168 VTN Vitronectin SEQ ID NOS:  
14169- 14170 VWA1 Von Willebrand factor A domain containing SEQ ID NOS: 14171- 1 14174  
VWA2 Von Willebrand factor A domain containing SEQ ID NOS: 14175- 2 14176 VWA5B2 Von  
Willebrand factor A domain containing SEQ ID NOS: 14177- 5B2 14178 VWA7 Von Willebrand factor  
A domain containing SEQ ID NO: 14179 7 VWC2 Von Willebrand factor C domain containing SEQ ID  
NO: 14180 2 VWC2L Von Willebrand factor C domain containing SEQ ID NOS: 14181- protein 2-like  
14182 VWCE Von Willebrand factor C and EGF domains SEQ ID NOS: 14183- 14187 VWDE Von  
Willebrand factor D and EGF domains SEQ ID NOS: 14188- 14193 VWF Von Willebrand factor SEQ  
ID NOS: 14194- 14196 WDR25 WD repeat domain 25 SEQ ID NOS: 14197- 14203 WDR81 WD  
repeat domain 81 SEQ ID NOS: 14204- 14213 WDR90 WD repeat domain 90 SEQ ID NOS: 14214-  
14221 WFDC1 WAP four-disulfide core domain 1 SEQ ID NOS: 14222- 14224 WFDC10A WAP four-  
disulfide core domain 10A SEQ ID NO: 14225 WFDC10B WAP four-disulfide core domain 10B SEQ  
ID NOS: 14226- 14227 WFDC11 WAP four-disulfide core domain 11 SEQ ID NOS: 14228- 14230  
WFDC12 WAP four-disulfide core domain 12 SEQ ID NO: 14231 WFDC13 WAP four-disulfide core  
domain 13 SEQ ID NO: 14232 WFDC2 WAP four-disulfide core domain 2 SEQ ID NOS: 14233-  
14237 WFDC3 WAP four-disulfide core domain 3 SEQ ID NOS: 14238- 14241 WFDC5 WAP four-  
disulfide core domain 5 SEQ ID NOS: 14242- 14243 WFDC6 WAP four-disulfide core domain 6 SEQ  
ID NOS: 14244- 14245 WFDC8 WAP four-disulfide core domain 8 SEQ ID NOS: 14246- 14247  
WFIKKN1 WAP, follistatin/kazal, immunoglobulin, SEQ ID NO: 14248 kunitz and netrin domain  
containing 1 WFIKKN2 WAP, follistatin/kazal, immunoglobulin, SEQ ID NOS: 14249- kunitz and  
netrin domain containing 2 14250 WIF1 WNT inhibitory factor 1 SEQ ID NOS: 14255- 14257 WISP1  
WNT1 inducible signaling pathway protein SEQ ID NOS: 14258- 1 14262 WISP2 WNT1 inducible  
signaling pathway protein SEQ ID NOS: 14263- 2 14265 WISP3 WNT1 inducible signaling pathway  
protein SEQ ID NOS: 14266- 3 14273 WNK1 WNK lysine deficient protein kinase 1 SEQ ID NOS:  
14274- 14287 WNT1 Wingless-type MMTV integration site SEQ ID NOS: 14288- family, member 1  
14289 WNT10B Wingless-type MMTV integration site SEQ ID NOS: 14290- family, member 10B  
14294 WNT11 Wingless-type MMTV integration site SEQ ID NOS: 14295- family, member 11 14297  
WNT16 Wingless-type MMTV integration site SEQ ID NOS: 14298- family, member 16 14299 WNT2  
Wingless-type MMTV integration site SEQ ID NOS: 14300- family member 2 14302 WNT3 Wingless-  
type MMTV integration site SEQ ID NO: 14303 family, member 3 WNT3A Wingless-type MMTV

integration site SEQ ID NO: 14304 family, member 3A WNT5A Wingless-type MMTV integration site  
SEQ ID NOS: 14305- family, member 5A 14308 WNT5B Wingless-type MMTV integration site SEQ  
ID NOS: 14309- family, member 5B 14315 WNT6 Wingless-type MMTV integration site SEQ ID NO:  
14316 family, member 6 WNT7A Wingless-type MMTV integration site SEQ ID NO: 14317 family,  
member 7A WNT7B Wingless-type MMTV integration site SEQ ID NOS: 14318- family, member 7B  
14322 WNT8A Wingless-type MMTV integration site SEQ ID NOS: 14323- family, member 8A 14326  
WNT8B Wingless-type MMTV integration site SEQ ID NO: 14327 family, member 8B WNT9A  
Wingless-type MMTV integration site SEQ ID NO: 14328 family, member 9A WNT9B Wingless-type  
MMTV integration site SEQ ID NOS: 14329- family, member 9B 14331 WSB1 WD repeat and SOCS  
box containing 1 SEQ ID NOS: 14332- 14341 WSCD1 WSC domain containing 1 SEQ ID NOS:  
14342- 14351 WSCD2 WSC domain containing 2 SEQ ID NOS: 14352- 14355 XCL1 Chemokine (C  
motif) ligand 1 SEQ ID NO: 14356 XCL2 Chemokine (C motif) ligand 2 SEQ ID NO: 14357  
XPNPEP2 X-prolyl aminopeptidase (aminopeptidase SEQ ID NOS: 14358- P) 2, membrane-bound  
14359 XXbac- SEQ ID NOS: 679-680 BPG116M5.17 XXbac- SEQ ID NO: 681 BPG181M17.5  
XXbac- SEQ ID NO: 682 BPG32J3.20 XXYL1 Xyloside xylosyltransferase 1 SEQ ID NOS: 14360-  
14365 XYLT1 Xylosyltransferase I SEQ ID NO: 14366 XYLT2 Xylosyltransferase II SEQ ID NOS:  
14367- 14372 ZFYVE21 Zinc finger, FYVE domain containing 21 SEQ ID NOS: 14373- 14377 ZG16  
Zymogen granule protein 16 SEQ ID NO: 14378 ZG16B Zymogen granule protein 16B SEQ ID NOS:  
14379- 14382 ZIC4 Zic family member 4 SEQ ID NOS: 14383- 14391 ZNF207 Zinc finger protein 207  
SEQ ID NOS: 14392- 14402 ZNF26 Zinc finger protein 26 SEQ ID NOS: 14403- 14406 ZNF34 Zinc  
finger protein 34 SEQ ID NOS: 14407- 14410 ZNF419 Zinc finger protein 419 SEQ ID NOS: 14411-  
14425 ZNF433 Zinc finger protein 433 SEQ ID NOS: 14426- 14435 ZNF449 Zinc finger protein 449  
SEQ ID NOS: 14436- 14437 ZNF488 Zinc finger protein 488 SEQ ID NOS: 14438- 14439 ZNF511  
Zinc finger protein 511 SEQ ID NOS: 14440- 14441 ZNF570 Zinc finger protein 570 SEQ ID NOS:  
14442- 14447 ZNF691 Zinc finger protein 691 SEQ ID NOS: 14448- 14455 ZNF98 Zinc finger protein  
98 SEQ ID NOS: 14456- 14459 ZPBP Zona pellucida binding protein SEQ ID NOS: 14460- 14463  
ZPBP2 Zona pellucida binding protein 2 SEQ ID NOS: 14464- 14467 ZSCAN29 Zinc finger and  
SCAN domain containing SEQ ID NOS: 14468- 29 14474

#### Cas-Clover

(868) The disclosure provides a composition comprising a guide RNA and a fusion protein or a  
sequence encoding the fusion protein wherein the fusion protein comprises a dCas9 and a Clo51  
endonuclease or a nuclease domain thereof.

(869) Small Cas9 (SaCas9)

(870) The disclosure provides compositions comprising a small, Cas9 (Cas9) operatively-linked to an  
effector. In certain embodiments, the disclosure provides a fusion protein comprising, consisting  
essentially of or consisting of a DNA localization component and an effector molecule, wherein the  
effector comprises a small, Cas9 (Cas9). In certain embodiments, a small Cas9 construct of the  
disclosure may comprise an effector comprising a type IIS endonuclease.

(871) Amino acid sequence of *Staphylococcus aureus* Cas9 with an active catalytic site.

(872) TABLE-US-00234 (SEQ ID NO: 18040) 1 mkrnyilgld igitsvgygi idyetrdivd  
agvrlfkean vennegrsk rgarrlkr 61 rhriqrkkl lfdynlltdh selsginpye arvkglsqkl seeefsaa  
hlakrrgvhn 121 vneveedtn elstkeqisr nskaleekyv aelqlerlkk dgevrgrsinr fktstdyvkea 181  
kqllkvqkay hqldqsfidt yidlletrrt yyegpgegsf fgwkdkewy emlmghctyf 241 peelrsvkya  
ynadlynaln dlennvitrd enekleyyek fqiienvfkq kkkptlkqia 301 keilvneedi kgyrvstgk  
peftnlkvyyh dikditarke iienaellldq iakiltiys 361 sediqeeltn lseltqeei eqisnlkgyt gthnlslkai  
nlildelwht ndnqiaifnr 421 lklvpkkvdl sqkqeiptl vddfilsppv krsfiqsikv inaiikkygl pndiiielar  
481 eknskdaqkm inemqkrnrq tnerieeir ttgenakyl iekiklhdmq egkclyslea 541 ipledllnnp  
fnyevdhiip rsysfdnsfn nkvlvkqeen skkgntpfq ylssdsksis 601 yetfkkhiln lakgkgrisk  
tkkeylleer dinrfsvqkd finrnldvtr yatrglmnll 661 rsyfrvnld vkvksinggf tsflrrkwkf  
kkernkgykh haedaliian adfifkewkk 721 ldkakkvmen qmfeekqaes mpeietegey keifitphqi  
khikdfkdyk yshrvdkkpn 781 relindtlys trkddkgntl ivnnlnglyd kdndklkkli nkspekllmy  
hhdpqtyqkl 841 klimeqygde knplykyeye tgnyltkysk kdngpvikki kyygnkl nah lditddypns 901



rnkvvklslk pyrfdvylnd gvykfvtvkn ldvikenyy evnscyeea kklkkisnqa 961 efiasfynnd  
likingelyr vigvnnldln rievnmidit yrelenmnd krpprikti 1021 asktsikky stdilnlye  
vkskhhpqii kkg.

Inactivated, Small Cas9 (dSaCas9)

(873) The disclosure provides compositions comprising an inactivated, small, Cas9 (dSaCas9) operatively-linked to an effector. In certain embodiments, the disclosure provides a fusion protein comprising, consisting essentially of or consisting of a DNA localization component and an effector molecule, wherein the effector comprises a small, inactivated Cas9 (dSaCas9). In certain embodiments, a small, inactivated Cas9 (dSaCas9) construct of the disclosure may comprise an effector comprising a type IIS endonuclease.

(874) dSaCas9 Sequence: D10A and N580A mutations (bold, capitalized, and underlined) inactivate the catalytic site.

(875) TABLE-US-00235 (SEQ ID NO: 18041) 1 mknyilgl**A** igitsvgygi idyetrdvid  
agvrlfkean vennegrsk rgarrlkr 61 rhriqrkkl lfdynlltdh selsginpye arvkglsqkl seeefsaall  
hlakrrgvhn 121 vneveedtn elstkeqisr nskaleekyv aelqlerlkk dgevrgsinr fktstdyvkea 181  
kqlkqvqkay hqldqsfiid yidlletrt yyegpgegsp fgwkdkewy emlmghctyf 241 peelrsvkya  
ynadlynaln dlennvitrd enekleyyek fqiienvfkq kkkptlkqia 301 keilvneedi kgyrvstgk  
peftnlkvyyh dikditarke iienaellq iakiltiys 361 sediqeeln lseltqeei eqisnlkgyt gthnlslkai  
nlildelwht ndnqiaifnr 421 lklvpkkvdl sqkkeipttl vddfilsppv krsfiqsikv inaiikkygl pndiielar  
481 eknskdaqkm inemqkrnrq tnerieeir ttgenakyl iekiklhdmq egkclyslea 541 ipledllnnp  
fnyevdhiip rsysfdnsfn nkvlvkqee**A** skkgnrtpfq ylssdsks 601 yetfkkhiln lakgkgrisk  
tkkeylleer dinrfsvqkd finrnldvtr yatrqlmnl 661 rsyfrvnnld vkvksinggf tsflrrkwkf  
kkernkgykh haedaliian adfifkewkk 721 ldkakkvmen qmfeekqaes mpeietevey keifitphqi  
khikdfkdyk yshrvdkkpn 781 relindtlys trkddkgntl ivnnlnglyd kdndklkkli nkspekllmy  
hhdpqtyqkl 841 klimeqygde knplykyee tgnyltkysk kdngpvikki kyygnklnah lditddypns 901  
rnkvvklslk pyrfdvylnd gvykfvtvkn ldvikenyy evnscyeea kklkkisnqa 961 efiasfynnd  
likingelyr vigvnnldln rievnmidit yrelenmnd krpprikti 1021 asktsikky stdilnlye  
vkskhhpqii kkg.

Inactivated Cas9 (dCas9)

(876) The disclosure provides compositions comprising an inactivated Cas9 (dCas9) operatively-linked to an effector. In certain embodiments, the disclosure provides a fusion protein comprising, consisting essentially of or consisting of a DNA localization component and an effector molecule, wherein the effector comprises an inactivated Cas9 (dCas9). In certain embodiments, an inactivated Cas9 (dCas9) construct of the disclosure may comprise an effector comprising a type IIS endonuclease.

(877) In certain embodiments, the dCas9 of the disclosure comprises a dCas9 isolated or derived from *Staphylococcus pyogenes*. In certain embodiments, the dCas9 comprises a dCas9 with substitutions at positions 10 and 840 of the amino acid sequence of the dCas9 that inactivate the catalytic site. In certain embodiments, these substitutions are D10A and H840A. In certain embodiments, the amino acid sequence of the dCas9 comprises the sequence of:

(878) TABLE-US-00236 (SEQ ID NO: 18042) 1 XDKKYSIGL**A** IGTNSVGWAV  
ITDEYKVPSK KFKVLGNTDR HSIKKNLIGA LLFDSGETAE 61 ATRLKRTARR  
RYTRRKNRIC YLQEIFSNEK AKVDDSFHR LEESFLVEED KKHHERHPIFG 121  
NIVDEVAYHE KYPTIYHLRK KLVDSTDKAD LRLIYLALAH MIKFRGHFLI  
EGDLNPDNSD 181 VDKLFIQLVQ TYNQLFEENP INASGVDAKA ILSARLSKSR  
RLENLIAQLP GEKKNGLFGN 241 LIALSLGLTP NFKSNEFDLAE DAKLQLSKDT  
YDDDLNLLA QIGDQYADLF LAAKNLSDAI 301 LLSDILRVNT EITKAPLSAS  
MIKRYDEHHQ DLTLKALVR QQLPEKYKEI FFDQSKNGYA 361 GYIDGGASQE  
EFYKFIKPIL EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 421  
AILRRQEDFY PFLKDNREKI EKILTFRIPI YVGPLARGNS RFAWMTRKSE  
ETITPWNFEE 481 VVDKGASASQ FIERMTNFDK NLPNEKVLPK HSLLYEYFTV  
YNELTKVKYV TEGMRKPAFL 541 SGEQKKAIVD LLFKTNRKVT VKQLKEDYFK  
KIECFDSVEI SGVEDRFNAS LGTYHDLLEKI 601 IKDKDFLDNE ENEDILEDIV



LTLTLFEDRE MIEERLKYTA HLFDDKVMKQ LKRRRYTGWG 661 RLSRKLINGI  
 RDKQSGKTIL DFLKSDGFAN RNFMQLIHDD SLTFKEDIQK AQVSGQGDSL 721  
 HEHIANLAGS PAIKKGILQT VKVDELVKV MGRHKPENIV IEMARENQTT  
 QKGQKNSRER 781 MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR  
 DMYVDQELDI NRLSDYDVDA 841 IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV  
 PSEEVVKKMK NYWRQLLNAK LITQRKFDNL 901 TKAERGGLSE LDKAGFIKRQ  
 LVETRQITKH VAQILDSRMN TKYDENDKLI REVKVITLKS 961 KLVSDFRKDF  
 QFYKVVREINN YHHAHDAYLN AVVGTAIIKK YPKLESEFVY GDYKVYDVVRK 1021  
 MIAKSEQEIG KATAKYFFYS NIMNFFKTEI TLANGEIRKR PLIETNGETG  
 EIVWDKGRDF 1081 ATVRKVLSP QVNIVKKTEV QTGGFSKESI LPKRNSDKLI  
 ARKKDWDPKK YGGFDSPTVA 1141 YSVLVVAKVE KGKSKKLKSV KELLGITIME  
 RSSFEKNPID FLEAKGYKEV KKDLIIKLPK 1201 YSLFELENGR KRMLASAGEL  
 QKGNELALPS KYVNFLYLAS HYEKLKGSPE DNEQKQLFVE 1261 QHKHYLDEII  
 EQISEFSKRV ILADANLDKV LSAYNKHHRDK PIREQAENII HLFTLTNLGA 1321  
 PAAFKYFDTT IDRKYRTSTK EVLDATLIHQ SITGLYETRI DLSQLGGD.

(879) In certain embodiments, the amino acid sequence of the dCas9 comprises the sequence of:

(880) TABLE-US-00237 (SEQ ID NO: 18043) 1 MDKKYSIGLA IGTVSVGWAV  
 ITDEYKVPSK KFKVLGNTDR HSIKKNLIGA LLFDSGETAE 61 ATRLKRTARR  
 RYTRRKNRIC YLQEIFSNEK AKVDDSFHR LEESFLVEED KKHHERHPIFG 121  
 NIVDEVAYHE KYPTIYHLRK KLVDSTDKAD LRLIYLALAH MIKFRGHFLI  
 EGDLPDNDSD 181 VDKLFIQLVQ TYNQLFEENP INASGVDAKA ILSARLSKSR  
 RLENLIAQLP GEKKNGLFGN 241 LIALSLGLTP NFKNFDLAE DAKLQLSKDT  
 YDDDLNLLA QIGDQYADLF LAAKNLSDAI 301 LLSLILRVNT EITKAPLSAS  
 MIKRYDEHHQ DLTLLKALVR QQLPEKYKEI FFDQSKNGYA 361 GYIDGGASQE  
 EFYKFIKPIL EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH 421  
 AILRRQEDFY PFLKDNREKI EKILTFRIPI YVGPLARGNS RFAWMTRKSE  
 ETITPWNFEE 481 VVDKGASASQ FIERMTNFDK NLPNEKVLPK HSLLYEYFTV  
 YNELTKVKYV TEGMRKPAFL 541 SGEQKKAIVD LLFKTNRKVT VKQLKEDYFK  
 KIECFDSVEI SGVEDRFNAS LGTYHDLLEKI 601 IKDKDFLDNE ENEDILEDIV  
 LTLTLFEDRE MIEERLKYTA HLFDDKVMKQ LKRRRYTGWG 661 RLSRKLINGI  
 RDKQSGKTIL DFLKSDGFAN RNFMQLIHDD SLTFKEDIQK AQVSGQGDSL 721  
 HEHIANLAGS PAIKKGILQT VKVDELVKV MGRHKPENIV IEMARENQTT  
 QKGQKNSRER 781 MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR  
 DMYVDQELDI NRLSDYDVDA 841 IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV  
 PSEEVVKKMK NYWRQLLNAK LITQRKFDNL 901 TKAERGGLSE LDKAGFIKRQ  
 LVETRQITKH VAQILDSRMN TKYDENDKLI REVKVITLKS 961 KLVSDFRKDF  
 QFYKVVREINN YHHAHDAYLN AVVGTAIIKK YPKLESEFVY GDYKVYDVVRK 1021  
 MIAKSEQEIG KATAKYFFYS NIMNFFKTEI TLANGEIRKR PLIETNGETG  
 EIVWDKGRDF 1081 ATVRKVLSP QVNIVKKTEV QTGGFSKESI LPKRNSDKLI  
 ARKKDWDPKK YGGFDSPTVA 1141 YSVLVVAKVE KGKSKKLKSV KELLGITIME  
 RSSFEKNPID FLEAKGYKEV KKDLIIKLPK 1201 YSLFELENGR KRMLASAGEL  
 QKGNELALPS KYVNFLYLAS HYEKLKGSPE DNEQKQLFVE 1261 QHKHYLDEII  
 EQISEFSKRV ILADANLDKV LSAYNKHHRDK PIREQAENII HLFTLTNLGA 1321  
 PAAFKYFDTT IDRKYRTSTK EVLDATLIHQ SITGLYETRI DLSQLGGD.

Clo051 Endonuclease

(881) An exemplary Clo051 nuclease domain may comprise, consist essentially of or consist of, the amino acid sequence of:

EGIKSNISLLKDELRGQISHISHEYLSDIDLAFDSKQNRLEFEMKVLELLVNEYGFKGRH  
 LGGSRKPDGIVYSTTLEDNFGIIVDTKAYSEGYSLPISQADEMERYVRENSNRDEEVN  
 PNKWWENFSEEVKKYFVFISGSFKGKFEEQLRRLSMTTGVNGSAVNVNLLLGA  
 KIRSGEMTIEELERAMFNSEFILKY (SEQ ID NO: 18044).

(882) Cas-Clover Fusion Protein

(883) In certain embodiments, an exemplary dCas9-Clo051 fusion protein (embodiment 1) may comprise, consist essentially of or consist of, the amino acid sequence of (Clo051 sequence underlined, linker bold italics, dCas9 sequence (*Streptococcus pyogenes*) in italics):

(884) TABLE-US-00238 (SEQ ID NO: 18045)

MAPKKKRKVEGIKSNISLLKDELRGQISHISHEYLSLIDLAFDSKQNRLF  
EMKVLELLVNEYGFKGRHLGGSRKPDGIVYSTTLEDNFGIIVDTKAYSEG  
YSLPISQADEMERYVRENSNRDEEVNPNKWWENFSEEVKKYYFVFISGSF  
KGKFEEQLRRLSMTTGVNGSAVNVNLLGAEKIRSGEMTIEELERAMFN NSEFILKY

*custom character* *DKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNT*  
*DRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFS*  
*N*  
*EMAKVDDSFHRLVESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHL*  
*RKKLVDSTDKADLRILIYALAHMIKFRGHFLIEGDLNPDNSDVKLFIQL*  
*VQTYNQLFEENPINASGVDAKAILSARLSKSRLENLIAQLPGEKKNGLF*  
*GNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLNLLAQIGDQYAD*  
*LFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKAL*  
*VRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEE*  
*LLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPFLKDNRE*  
*KIEKILTRIPYYVGPLARGNSRFAWMTRKSEETITPWNFEEVVDKGASA*  
*QSFIERMTNFDKNLPNEKVLPHKSLLEYFTVYNELTKVKYVTEGMRKPA*  
*FLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFN*  
*ASLGTYHDLKIIKDKDFLDNEENEDILEDIVLTLTLFEDREMIEERLKT*  
*YAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGF*  
*ANRNFMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIKKGIL*  
*QTVKVVDDELVKVMGRHKPENIVIAMARENQTTQKGQKNSRERMKRIEEGI*  
*KELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLSDYDV*  
*DAIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMKNYWRQLLN*  
*AKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSR*  
*MNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKVINNYHHAHDAY*  
*LNAVVGTAIIKKYPKLESEFVYGDYKVYDVRKMIKSEQEIGKATAKYFF*  
*YSNIMNFFKTEITLANGEIRKRLIETNGETGEIVWDKGRDFATVRKVLS*  
*MPQVNIVKKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPT*  
*VAYSVLVAKVEKGKSKKLKSVKELLGITIMERSSEKKNPIDFLEAKGYK*  
*EVKKDLIIKLPKYSLFELENGRKRMLASAGELQKGNELALPSKYVNFLYL*  
*ASHYEKLKGSPEDEQKQLFVEQHKHYLDEIIEQISEFSKRVILADANLD*  
*KVLSAYNKHDKPIREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTS*  
*TKEVLDTLIHQSIITGLYETRIDLSQLGGDGSPKKKRKVSS.*

(885) In certain embodiments, an exemplary dCas9-Clo051 fusion protein (embodiment 1) may comprise, consist essentially of or consist of, the nucleic acid sequence of (dCas9 sequence derived from *Streptococcus pyogenes*):

(886) TABLE-US-00239 (SEQ ID NO: 18046) 1 atggcaccaa agaagaaaag aaaagtggag  
ggcatcaagt caaacatcag cctgctgaaa 61 gacgaactgc ggggacagat tagtcacatc agtcacgagt  
acctgtcact gattgatctg 121 gccttcgaca gcaagcagaa tagactgttt gagatgaaaag tgctggaact gctggtcaac  
181 gagtatggct tcaagggcag acatctgggc ggggtctagga aacctgacgg catcgtgtac 241 agtaccacac  
tggaagacaa cttcggaatc attgtcgata ccaaggccta ttccgagggc 301 tactctctgc caattagtca  
ggcagatgag atggaaagggt acgtgcgcga aaactcaaat 361 agggacgagg aagtcaaccc caataagtgg  
tgggagaatt tcagcgagga agtgaagaaa 421 tactacttcg tctttatctc aggcagcttc aaaggggaagt  
ttgaggaaca gctgcggaga 481 ctgtccatga ctaccgggggt gaacggatct gctgtcaacg tggatcaatc  
gctgctgggc 541 gcagaaaaga tcaggtccgg ggagatgaca attgaggaac tggaacgcgc catgttcaac 601  
aattctgagt ttatctgaa gtatggaggc gggggaagcg ataagaaata ctccatcgga 661 ctggccattg  
gcaccaatc cgtgggctgg gctgtcatca cagacgagta caagtgcca 721 agcaagaagt tcaaggtcct  
ggggaacacc gatcgccaca gtatcaagaa aaatctgatt 781 ggagccctgc tgttcgactc aggcgagact  
gctgaagcaa cccgactgaa gcggactgct 841 aggcgccgat ataccggag aaaaaatcgg atctgtacc

tgaggagaaat tttcagcaac 901 gagatggcca aggtggacga tagtttcttt caccgccttg aggaatcctt cctggtggag  
961 gaagataaga aacacgagcg gcatccatc tttggcaaca ttgtggacga agtcgcttat 1021 caccagaaagt  
accctactat ctatcatctg aggaagaaac tgggtggactc caccgataag 1081 gcagacctgc gcctgatcta  
tctggccctg gctcacatga tcaagttccg ggggcatttt 1141 ctgatcgagg gagatctgaa ccctgacaat  
tctgatgtgg acaagctgtt catccagctg 1201 gtccagacat acaatcagct gtttgaggaa aaccaatta  
atgcctcagg cgtggacgca 1261 aaggccatcc tgagcgccag actgtccaaa ttagggccc tgaaaaacct  
gatcgctcag 1321 ctgccaggag agaagaaaaa cggcctgttt gggaatctga ttgactgtc cctgggcctg 1381  
acaccaact tcaagtctaa tttgatctg gccgaggacg ctaagctgca gctgtccaaa 1441 gacacttatg  
acgatgacct ggataacctg ctggctcaga tcggcgatca gtacgcagac 1501 ctgttcttg ccgctaagaa  
tctgagtac gccatcctgc tgcagatat tctgcgcgtg 1561 aacacagaga ttactaaggc cccactgagt  
gcttcaatga tcaaaagata tgacgagcac 1621 catcaggatc tgacctgtc gaaggctctg gtgaggcagc  
agctgcccga gaaatacaag 1681 gaaatcttct ttgatcagag caagaatgga tacgccggtc atattgacgg  
cggggcttcc 1741 caggaggagt tctacaagtt catcaagccc attctggaaa agatggacgg caccgaggaa 1801  
ctgctggtga agctgaatcg ggaggacctg ctgagaaaac agaggacatt tgataacgga 1861 agcatccctc  
accagattca tctgggcgaa ctgcacgcca tctgcgacg gcaggaggac 1921 ttctacccat ttctgaagga  
taaccgagag aaaatcgaaa agatcctgac cttcagaatc 1981 ccctactatg tggggcctct ggcacgggga  
aatagtagat ttgcctggat gacaagaaag 2041 tcagaggaaa ctatcaccct ctggaacttc gaggaagtgg  
tcgataaagg cgtagcgca 2101 cagtccttca ttgaaaggat gacaaatctt gacaagaacc tgccaaatga  
gaaggtgctg 2161 cccaaacaca gcctgctgta cgaatatctc acagtgtata acgagctgac taaagtgaag 2221  
tacgtcaccg aagggatgag caagcccga ttctgtccg gagagcagaa gaaagccatc 2281 gtggacctgc  
tgtttaagac aaatcgaaa gtgactgtca aacagctgaa ggaagactat 2341 ttcaagaaaa ttgagtgtt  
cgattcagtg gaaatcagcg gcgtcgagga caggtttaac 2401 gcctccctgg ggacctacca cgatctgctg  
aagatcatca aggataagga cttcctggac 2461 aacgaggaaa atgaggacat cctggaggac attgtgctga  
cactgactct gtttgaggat 2521 cgcgaaatga tcgaggaacg actgaagact tatgcccac tggtcgatga  
caaagtgatg 2581 aagcagctga aaagaaggcg ctacaccgga tggggacgcc tgagccgaaa actgatcaat 2641  
gggattagag acaagcagag cggaaaaact atctggact ttctgaagtc cgatggcttc 2701 gccaacagga  
acttcatgca gctgattcac gatgactctc tgacctcaa ggaggacatc 2761 cagaaagcac aggtgtctgg  
ccaggggggac agtctgcacg agcatatcg aaacctggcc 2821 ggcagccccg ccatcaagaa agggattctg  
cagaccgtga aggtggtgga cgaactggc 2881 aaggtcatgg gacgacaaa acctgagaac atcgtgattg  
agatggcccc cgaaaatcag 2941 acaactcaga agggccagaa aaacagtcga gaacggatga agagaatcga  
ggaaggcatc 3001 aaggagctgg ggtcacagat cctgaaggag catcctgtgg aaaacactca gctgcagaat 3061  
gagaaactgt atctgtacta tctgcagaat ggacgggata tgtactgga ccaggagctg 3121 gatattaaca  
gactgagtga ttatgactg gatgccatc tccctcagag cttctgaag 3181 gatgactcca ttgacaaca  
gggtgtgacc aggtccgaca agaaccgagg caaatcagat 3241 aatgtgcaa gcgaggaagt ggtcaagaaa  
atgaagaact actggaggca gctgctgaat 3301 gccaaactga tcacacagcg gaaattgat aacctgacta  
aggcagaaaag agggggcctg 3361 tctgagctgg acaaggccgg cttcatcaag cggcagctgg tggagacaag  
acagatcact 3421 aagcagctc ctgacttct ggatagcaga atgaacaaa agtacgatga aaacgacaag 3481  
ctgatcaggg aggtgaaagt cattactctg aaatccaagc tgggtgtctga ctttagaaag 3541 gatttccagt  
ttataaagt caggagatc aacaactacc accatgctca tgacgcatac 3601 ctgaacgcag tggtcgggac  
cgccctgatt aagaaatacc ccaagctgga gtccgagttc 3661 gtgtacggag actataaagt gtacgatgc  
cggaagatga tcgccaatc tgagcaggaa 3721 attggcaagg ccaccgtaa gtatttctt tacagtaaca  
tcatgaattt cttaagacc 3781 gaaatcacac tggcaaatgg ggagatcaga aaaaggcctc tgattgagac  
caacggggag 3841 acaggagaaa tcgtgtggga caagggaagg gatttctga ccgtgcgcaa agtctgtcc 3901  
atgccccaa gtaatatgt caagaaaact gaagtgcaga ccgggggatt cttaaggag 3961 agtattctgc  
ctaagcgaac ctctgataaa ctgatcccc ggaagaaaga ctgggacccc 4021 aagaagtatg gcgggttcga  
cttccaaca gtggcttaca gtgtcctggt ggtcgcaaag 4081 gtggaaaagg ggaagtccaa gaaactgaag  
tctgtcaaag agctgctggg aatcactatt 4141 atggaacgca gctcctcga gaagaatcct atcgatttc  
tggaagccaa gggctataaa 4201 gaggtgaaga aagacctgat cattaagctg ccaaaatact cactgtttga  
gctggaaaac 4261 ggacgaaagc gaatgtggc aagcgccgga gaactgcaga agggcaatga gctggccctg 4321  
ccctccaaat acgtgaactt cctgtatctg gctagccact acgagaaact gaaggggctc 4381 cctgaggata  
acgaacagaa gcagctgttt gtggagcagc acaaacatta tctggacgag 4441 atcattgaac agatttcaga  
gttcagcaag agagtgatcc tggctgacgc aaatctggat 4501 aaagtcctga gcgcataca caagcaccga

gacaaaccaa tccgggagca ggcgaaaat 4561 atcattcatc tgttcaccct gacaaacctg ggcgcccctg  
cagccttcaa gtattttgac 4621 accacaatcg atcggaagag atacacttct accaaagagg tgctggatgc  
taccctgatc 4681 caccagagta ttaccggcct gtatgagaca cgcacatgacc tgtcacagct gggaggcgat 4741  
gggagcccca agaaaaagcg gaaggtgtct agttaa.

(887) In certain embodiments, the nucleic acid sequence encoding a dCas9-Clo051 fusion protein (embodiment 1) of the disclosure may comprise a DNA. In certain embodiments, the nucleic acid sequence encoding a dCas9-Clo051 fusion protein (embodiment 1) of the disclosure may comprise an RNA.

(888) In certain embodiments, an exemplary dCas9-Clo051 fusion protein (embodiment 2) may comprise, consist essentially of or consist of, the amino acid sequence of (Clo051 sequence underlined, linker bold italics, dCas9 sequence (*Streptococcus pyogenes*) in italics):

(889) TABLE-US-00240 (SEQ ID NO: 18047) 1 MPKKRKVEG IKSNISLLKD  
ELRGQISHIS HEYLSLIDLA FDSKQNRLE MKVLELLVNE 61 YGFKGRHLGG  
SRKPDGIVYS TTLEDNFGII VDTKAYSEGY SLPISQADEM ERYVRENSNR 121  
DEEVNPNKWW ENFSEEVKKY YFVFISGSFK GKFEELRLRL SMTTGVNGSA  
VNVVNLLLGA 181 EKIRSGEMTI EELERAMFNN SEFILKY custom character  
custom character *DKKYSIGL AIGTNSVGWA VITDEYKVPS* 241 *KKFKVLGNTD*  
*RHSIKKNLIG ALLFDSGETA EATRLKRTAR RRYTRRKNRI CYLQEIFSNE* 301  
*MAKVDDSSFFH RLEESFLVEE DKKHERHPIF GNIVDEVAYH EKYPTIYHLR*  
*KKLVDSTDKA* 361 *DLRLIYLALA HMIKFRGHFL IEGDLNPDNS DVDKLFQVLV*  
*QTYNQLFEEN PINASGVDAK* 421 *AILSARLSKS RRLLENLIAQL PGEKKNGLFG*  
*NLIALSLGLT PNFKSNFDLA EDAKLQLSKD* 481 *TYDDDLNLL AQIGDQYADL*  
*FLAAKNLSDA ILLSDILRVN TEITKAPLSA SMIKRYDEHH* 541 *QDLTLLKALV*  
*RQQLPKEYKE IFFDQSKNGY AGYIDGGASQ EEFYKFIKPI LEKMDGTEEL* 601  
*LVKLNREDLL RKQRTFDNGS IPHQIHLGEL HAILRRQEDF YPFLKDNREK IEKILTRFIP*  
*661 YYVGPLARGN SRFWMTRKS EETITPWNFE EVVDKGASAQ SFIERMTNED*  
*KNLPNEKVLP* 721 *KHSLLEYEFT VYNELTKVKY VTEGMRKPAF LSGEQKKAIV*  
*DLLFKTNRKV TVKQLKEDYF* 781 *KKIECFDSVE ISGVEDRFNA SLGTYHDLK*  
*IIKDKDFLDN EENEDILEDI VLTTLTFEDR* 841 *EMIEERLKTY AHLFDDKVMK*  
*QLKRRRYTGW GRLSRKLING IRDKQSGKTI LDFLKSDGFA* 901 *NRNFMQLIHD*  
*DSLTFKEDIQ KAQVSGQGDS LHEHIANLAG SPAIKKGILQ TVKVVDDELVK* 961  
*VMGRHKPENI VIEMARENQT TQKGQKNSRE RMKRIEEGIK ELGSQILKEH*  
*PVENTQLQNE* 1021 *KLYLYYLQNG RDMYVDQELD INRLSDYDVD AIVPQSFLKD*  
*DSIDNKVLTR SDKNRGKSDN* 1081 *VPSEEVVKKM KNYWRQLLNA KLITQRKFDN*  
*LTKAERGGLS ELDKAGFIKR QLVETRQITK* 1141 *HVAQILDSRM NTKYDENDKL*  
*IREVKVITLK SKLVSDFRKD FQFYKVIN* 1201 *NAVVGITALIK*  
*KYPKLESEFV YGDYKVYDVR KMIKSEQEI GKATAKYFFY SNIMNFFKTE* 1261  
*ITLANGEIRK RPLIETNGET GEIVWDKGRD FATVRKVLSM PQVNIVKKTE VQTGGFSKES*  
*1321 ILPKRNSDKL IARKRDWDPK KYGGFDSPTV AYSVLVAKV EKGKSKKLKS*  
*VKELLGITIM* 1381 *ERSSFENPI DFLEAKGYKE VRKDLIIKLP KYSLFELENG*  
*RKRMLASAGE LQKGNELALP* 1441 *SKYVNFLYLA SHYEKLKGSP EDNEQKQLFV*  
*EQHKHYLDEI IEQISEFSKR VILADANLDK* 1501 *VLSAYNKHHRD KPIREQAENI*  
*IHLFTLTNLG APAAFKYFDT TIDRKRYTST KEVLDTLIH* 1561 *QSITGLYETR*  
*IDLSQLGGDG SPKKKRKV.*

(890) In certain embodiments, an exemplary dCas9-Clo051 fusion protein (embodiment 2) may comprise, consist essentially of or consist of, the nucleic acid sequence of (dCas9 sequence derived from *Streptococcus pyogenes*):

(891) TABLE-US-00241 (SEQ ID NO: 18048) 1 atgcctaaga agaagcggaa ggtggaaggc  
atcaaaagca acatctccct cctgaaagac 61 gaactccggg ggcagattag ccacattagt cacgaatacc  
tctccctcat cgacttggtc 121 ttcgatagca agcagaacag gctctttgag atgaaagtgc tggaactgct cgtaatgag  
181 tacgggttca agggtcgaca cctcggcgga tctaggaaac cagacggcat cgtgtatagt 241 accacactgg  
aagacaactt tgggatcatt gtggatacca aggcatactc tgagggttat 301 agtctgcca tttcacaggc

cgacgagatg gaacggtagc tgcgcgagaa 361 gatgaggaag tcaacctaa caagtggtagg  
gagaacttct ctgaggaagt gaagaaatac 421 tacttcgtct ttatcagcgg gtccttcaag ggtaaatttg  
aggaacagct caggagactg 481 agcatgacta ccggcgtgaa tggcagcgcc gtcaacgtgg tcaatctgct  
cctgggcgct 541 gaaaagattc ggagcggaga gatgaccatc gaagagctgg agagggcaat gtttaataat 601  
agcgagttta tcctgaaata cgggtggcggg ggatccgata aaaagtattc tattggttta 661 gccatcggca ctaattccgt  
tggatgggct gtcataaccg atgaatacaa agtaccttca 721 aagaaattta aggtgttggg gaacacagac  
cgtcattcga ttaaaaagaa tcttatcggg 781 gccctcctat tcgatagtgg cgaaacggca gaggcgactc  
gcctgaaacg aaccgctcgg 841 agaaggata cacgtcgaa gaaccgaata tgttacttac aagaaatttt tagcaatgag  
901 atggccaaag ttgacgattc tttcttcac cgtttggaag agtccttctt tgcgaagag 961 gacaagaaac  
atgaacggca ccccatcttt ggaaacatag tagatgaggt ggcataatcat 1021 gaaaagtacc caacgattta  
tcacctcaga aaaaagctag ttgactcaac tgataaagcg 1081 gacctgaggt taatctactt ggctcttgcc  
catatgataa agttccgtgg gcactttctc 1141 attgagggtg atctaaatcc ggacaactcg gatgtcgaca  
aactgttcat ccagttagta 1201 caaacctata atcagttgtt tgaagagaac cctataaatg caagtggcgt ggatgcgaag  
1261 gctattctta gcgcccgcct ctctaaatcc cgacggctag aaaacctgat cgcaacaatta 1321 cccggagaga  
agaaaaatgg gttgttcggg aaccttatag cgctctcact aggcctgaca 1381 ccaaatttta agtcgaactt  
cgacttagct gaagatgcca aattgcagct tagtaaggac 1441 acgtacgatg acgatctcga caatctactg  
gcacaaattg gagatcagta tgcggactta 1501 ttttggctg ccaaaaacct tagcgatgca atcctctat  
ctgacatact gagagttaat 1561 actgagatta ccaaggcgcc gttatccgct tcaatgatca aaaggtagca  
tgaacatcac 1621 caagacttga cacttctcaa ggccctagtc cgtcagcaac tgcctgagaa atataaggaa 1681  
atattctttg atcagtcgaa aaacgggtac gcaggttata ttgacggcgg agcgagtcaa 1741 gaggaattct  
acaagtttat caaacccata ttagagaaga tggatgggac ggaagagttg 1801 cttgtaaaac tcaatcgcgca  
agatctactg cgaaagcagc ggactttcga caacggtagc 1861 attccacatc aaatccactt aggcgaattg  
catgctatac ttgaaggca ggaggatttt 1921 tatccgttcc tcaaagacaa tcgtgaaaag attgagaaaa  
tcctaaccct tcgcatacct 1981 tactatgtgg gaccctggc ccgagggaac tctcggttcg catggatgac  
aagaaaagtc 2041 gaagaaacga ttactccatg gaattttgag gaagttgtcg ataaaggtgc gtcagctcaa 2101  
tcgttcatcg agaggatgac caactttgac aagaatttac cgaacgaaaa agtattgcct 2161 aagcacagtt  
tactttacga gtatttcaca gtgtacaatg aactcacgaa agttaagtat 2221 gtcactgagg gcatgcgtaa  
accgccttt ctaagcggag aacagaagaa agcaatagta 2281 gatctgttat tcaagaccaa ccgcaaagtg  
acagttaagc aattgaaaga ggactacttt 2341 aagaaaattg aatgcttcga ttctgtcgag atctccgggg  
tagaagatcg atttaatgcg 2401 tcaacttgta cgtatcatga cctcctaaag ataattaaag ataaggactt cctggataac  
2461 gaagagaatg aagatatctt agaagatata gtgtgactc ttaccctctt tgaagatcgg 2521 gaaatgattg  
aggaaagact aaaaacatac gtcacactgt tcgacgataa ggttatgaaa 2581 cagttaaaga ggcgtcgcta  
tacgggctgg ggacgattgt cgcggaact tatcaacggg 2641 ataagagaca agcaaagtgg taaaactatt  
ctgattttc taaagagcga cggttcgcc 2701 aataggaact ttatgcagct gatccatgat gactctttaa  
cctcaaaga ggatatacaa 2761 aaggcacagg ttccggaca aggggactca ttgcacgaac atattgcgaa  
tcttgctggg 2821 tcgccagcca tcaaaaaggg catactccag acagtcaaag tagtggatga gctagttaag 2881  
gtcatgggac gtcacaaacc ggaaaacatt gtaatcgaga tggcacgca aaatcaaacg 2941 actcagaagg  
ggcaaaaaaa cagtcgagag cggtatgaaga gaatagaaga gggattataa 3001 gaactgggca gccagatctt  
aaaggagcat cctgtggaaa ataccatt gcagaacgag 3061 aaactttacc tctattacct acaaaatgga  
agggacatgt atgttgatca ggaactggac 3121 ataaaccgtt tatctgatta cgacgtcgat gccattgtac  
cccaatcctt ttgaaggac 3181 gattcaatcg acaataaagt gcttacacgc tcggataaga accgagggaa  
aagtgacaat 3241 gtccaagcg aggaagtcgt aaagaaaatg aagaactatt ggccggcagct cctaatgcg 3301  
aaactgataa cgcaaagaaa gttcgataac ttaactaaag ctgagagggg tggcttgtct 3361 gaactgaca  
aggccggatt tattaacgt cagctcgtgg aaaccgcca aatcacaaag 3421 catgttgac agatactaga  
ttcccgaatg aatacgaaat acgacgagaa cgataagctg 3481 attcgggaag tcaaagtaat cactttaaag  
tcaaaattgg tgtcggactt cagaaaggat 3541 ttcaattct ataaagttag ggagataaat aactaccacc  
atgcgcacga cgcttatctt 3601 aatgccgtcg tagggaccgc actcattaag aaataccga agctagaaag  
tgagtttg 3661 tatggtgatt acaaagtta tgacgtccgt aagatgatcg cgaaaagcga acaggagata 3721  
ggcaaggcta cagccaaata cttctttat tctaacatta tgaattctt taagacggaa 3781 atcactctgg  
caaacggaga gatacgaaa cgaccttaa ttgaaaccaa tggggagaca 3841 ggtgaaatcg tatgggataa  
gggccgggac ttgcgcacgg tgagaaaagt ttgtccatg 3901 cccaagtca acatagtaaa gaaaactgag  
gtgcagaccg gagggtttc aaaggaatcg 3961 attcttcaa aaaggaatag tgataagctc atcgctcgta

aaaaggactg ggaccgaaa 4021 aagtcagggtg gcttcgatag ccctacagtt gcctattctg tcctagtagt  
 ggcaaaagtt 4081 gagaagggaa aatccaagaa actgaagtca gtcaaagaat tattggggat aacgattatg 4141  
 gagcgctcgt cttttgaaaa gaaccccatc gacttccttg aggcgaaagg ttacaaggaa 4201 gtaaaaaagg  
 attcataat taaactacca aagtatagtc tgtttgagtt agaaaatggc 4261 cgaaaacgga tgttggttag  
 cgccggagag cttcaaaagg ggaacgaact cgcactaccg 4321 tctaaatacg tgaatttctt gtatttagcg  
 tcccattacg agaagttgaa aggttcacct 4381 gaagataacg aacagaagca actttttgtt gagcagcaca  
 aacattatct cgacgaaatc 4441 atagagcaaa ttcggaatt cagtaagaga gtcatcctag ctgatgccaa tctggacaaa  
 4501 gtattaagcg catacaaaa gcacagggat aaaccctac gtgagcaggc ggaaaatatt 4561 atccatttgt  
 ttactcttac caacctcggc gctccagccg cattcaagta tttgacaca 4621 acgatagatc gcaaacgata  
 cacttctacc aaggagggtc tagacgcgac actgattcac 4681 caatccatca cgggattata tgaaactcgg  
 atagatttgt cacagcttgg ggggtgacgga 4741 tcccccaaga agaagaggaa agtctga.

(892) In certain embodiments, the nucleic acid sequence encoding a dCas9-Clo051 fusion protein (embodiment 2) of the disclosure may comprise a DNA. In certain embodiments, the nucleic acid sequence encoding a dCas9-Clo051 fusion protein (embodiment 2) of the disclosure may comprise an RNA.

## EXAMPLES

### Example 1: Knock Down Efficiency of Checkpoint Signaling Proteins on Armored T-Cells

(893) To create armored T-cells that have enhanced therapeutic potential, genetic modifications may be made in order to render the T-cells less sensitive to immunologic and/or metabolic checkpoints. One mechanism to produce armored T-cells is to inhibit checkpoint signaling is to knockout various checkpoint receptors. The Cas-CLOVER™ platform was used to target and knockout the checkpoint receptors PD-1, TGFβR2, LAG-3, Tim-3, and CTLA-4 in resting (or quiescent) primary pan T cells. As measured by flow cytometry, gene editing resulted in 30-70% loss of protein expression at the cell surface (FIG. 11). These results show that Cas-CLOVER™ is able to efficiently target the knockout of these genes resulting in loss of target protein expression on the T-cell surface. Knockout efficiency can significantly be increased by further optimization of guide RNA pairs, or by using additional guide RNA pairs targeting the same gene and/or regulators or promoters of the target gene.

### Example 2: Strategies for the Expression of Null or Switch Intracellular Signaling Proteins on Armored T-Cells

(894) Another strategy to produce armored T-cells is to reduce or inhibit endogenous checkpoint signaling by expressing various modified/chimeric checkpoint receptors that have an altered or absent intracellular signaling domain. Checkpoint signals that could be targeted using this strategy include PD-1 or TGFβRII of T-cells, which bind to the PD-L1 ligand and TGFβ cytokine, respectively. FIG. 12 shows a schematic diagram of various strategies for producing decoy/null/dominant negative receptor (Null receptors) for two different inhibitory receptors (PD-1 (top panel) and TGFβRII (bottom panel)). To design Null receptors, the intracellular domain (ICD) of PD1 or TGFβRII can be mutated (mutated null) or deleted (truncated null). As a result, binding of the cognate ligand(s) of the null receptor does not result in delivery of the checkpoint signal to the T-cells. Furthermore, since the Null receptor competes with wildtype receptors for binding of the endogenous ligand(s), any binding by the Null receptor sequesters endogenous ligand(s) from binding the wildtype receptor. This results in dilution of the overall level of checkpoint signaling effectively delivered to the T-cell, thus, reducing or blocking checkpoint inhibition. FIG. 12 also shows switch receptor design strategies for the inhibitory receptors PD-1 (top panel) and TGFβRII (bottom panel). In switch receptors, wildtype ICD is replaced with the ICD from either an immuno-stimulatory molecule (Co-stimulatory switch) or a different inhibitory molecule (Inhibitory switch). Immuno-stimulatory molecules include but are not limited to CD3z, CD28, 4-1BB and the examples listed in Table 1. Inhibitory molecules include but are not limited to CTLA4, PD1, Lag3 and the examples listed in Tables 1 and 9. In the former case, binding of the endogenous ligand by the modified switch receptor results in the delivery of a positive signal to the T-cells, thereby helping to enhance stimulation of the T-cell, facilitating continuation of tumor targeting and killing. In the latter case, binding of the endogenous ligand by the modified switch receptor results in the delivery of a negative signal to the T-cells, thereby helping to reduce stimulation and activity of the T-cell.

### Example 3: Surface Expression of PD1 and TGFβRII Null or Switch Intracellular Signaling Proteins on Armored T-Cells

(895) To create armored T-cells, a number of truncated null receptors expressing alternative signal peptides (SP) and transmembrane domains (TM) designed and tested for maximal expression on the surface of modified T-cells. FIG. 13 shows schematic diagrams of several null receptor constructs for PD-1 (top) and TGFβRII (bottom). Extracellular domains (ECD) of these proteins were modified such that the wildtype signal peptide (SP) and/or the transmembrane domains (TM) were replaced with that from the human T cell CD8α receptor (red arrows). Each of the six truncated null constructs shown in FIG. 13 were DNA synthesized and then subcloned into an mRNA IVT DNA vector (pRT). High quality mRNA was produced via IVT for each. Transfection of mRNA encoding each of the six molecules was performed using electroporation (EP) delivery into primary human T cells and FACS analysis was performed 24 hours post-EP to evaluate expression level of each construct on the cell surface (FIG. 14). By flow cytometry, replacement of the WT SP with the alternative CD8α (02.8aSP-PD-1 and 02.8aSP-TGFβRII) resulted in the highest level of expression at the T cell surface. 02.8aSP-PD-1 Null receptor exhibited an MFI of 43,680, which is 177-fold higher than endogenous T cell PD-1 expression and 2.8-fold higher than the WT PD-1 Null receptor. 02.8aSP-TGFβRII Null receptor exhibited an MFI of 13,809, which is 102-fold higher than endogenous T cell TGFβRII expression and 1.8-fold higher than the WT TGFβRII Null receptor. These results show that replacement of wildtype SP with the alternative CD8α SP for both PD1 and TGFβRII inhibitory proteins leads to enhanced surface expression of the Null or Switch receptor. This in turn will maximize checkpoint inhibition or co-stimulation, respectively, upon binding of the natural ligand(s).

### Example 4: Design of NF-KB Inducible Vectors for Expression in Modified T-Cells

(896) Two T cell activation NF-KB inducible vectors were developed (FIGS. 15A and 15B); one with the gene expression system (GES) in the forward orientation (A) and the other in the complementary direction (B), both preceding the constitutive EF1a promoter. These vectors also direct expression of a CAR molecule and a DHFR selection gene, separated by a T2A sequence. Both the conditional NF-KB inducible system and the EF1a directed genes are a part of a piggyBac transposon, which can be permanently integrated into T cells using EP. Once integrated into the genome, the T cells constitutively express the CAR on the membrane surface and the DHFR within the cell, while expression of the NF-KB inducible gene, GFP, will be expressed to the highest level only upon T cell activation.

### Example 5: NF-KB Inducible Vectors for GFP Expression in Modified T-Cells

(897) T cells were nucleofected with a piggyBac vector expressing an anti-BCMA CAR and a DHFR mutein gene under control of an EF1a promoter along with the absence (No gene expression system (GES) control) or presence of an NF-KB inducible expression system driving GFP expression in either the forward (pNFKB-GFP forward) or reverse orientation (pNFKB-GFP reverse). Cells were cultured in the presence of methotrexate selection until the cells were almost completely resting (Day 19) and GFP expression was assessed at Day 5 and Day 19. At Day 5, all T cells are proliferating and highly stimulated, with cells harboring the NF-KB inducible expression cassette producing high levels of GFP due to strong NFκB activity (see FIG. 16). The No GES control cells did not express detectable levels of GFP. By Day 19, the GES T cells were almost fully resting and GFP expression was significantly lower than Day 5 (~1/8 MFI), since NFκB activity is lower. GFP expression is still observed at Day 19, which may due to the long half-life of GFP protein (~30 hr), or, basal level of NFκB activity through, for example, a TCR, a CAR, a cytokine receptor, or a growth factor receptor signal.

### Example 6: NF-KB Inducible Vectors for Anti-BCMA CAR-Mediated GFP Expression in Modified T-Cells

(898) T cells were either unmodified (Mock T cells) or nucleofected with a piggyBac vector expressing an anti-BCMA CAR and a DHFR mutein gene under control of an EF1a promoter along with the absence (No GES control) or presence of an NF-KB inducible expression system driving GFP expression in either the forward (pNFKB-GFP forward) or reverse orientation (pNFKB-GFP reverse). All cells were cultured for 22 days, either with or without methotrexate selection (Mock T cells), until the cells were almost completely resting. Cells were then stimulated for 3 days in the absence (No stimulation) or presence of BCMA-(K562), BMCA+(RPMI 8226), or positive control anti-CD3 anti-



CD28 activation reagent (CD3/28 stimulation). GFP expression was undetectable under all conditions with the No GES control or Mock T cells. However, while pNFkB-GFP forward- and reverse-transposed cells exhibited little GFP expression over the No stimulation control when cultured with BCMA+ K562 cells, they both demonstrated dramatic upregulation of gene expression either in the presence of BCMA+ tumor cells or under positive control conditions (FIG. 17). Little difference in GFP expression was observed between the pNFkB-GFP forward- and reverse-transposed cells that were cocultured with BCMA+ tumor cells.

#### Example 7: Control of Anti-BCMA CAR-Mediated Expression in Modified T-Cells

(899) The expression level of inducible gene can be regulated by the number of response elements upstream or preceding the inducible promoter. T cells were nucleofected with a piggyBac vector encoding an anti-BCMA CARTyrin followed by a selection gene, both under control of a human EF1a promoter (FIG. 18). Further, vectors either additionally encoded the conditional NF-KB inducible gene expression system driving expression of a truncated CD19 protein (dCD19) and included a number of NFkB response elements (RE) varying from 0-5, no GES (No GES), or received an electroporation pulse but no piggyBac nucleic acid (Mock). Data are shown for only the GES in the reverse (opposite) direction/orientation. All cells were cultured for 18 days and included selection for piggyBac-modified T cells using methotrexate addition. Cells were then stimulated for 3 days using anti-CD3 anti-CD28 bead activation reagent and dCD19 surface expression was assessed by FACS at Days 0, 3 and 18, and data are shown as FACS histograms and MFI of target protein staining. Surface dCD19 expression was detected at low levels at Day 0 in all T cells transposed with vectors encoding the GES. At 3 days post-stimulation, dramatic upregulation of dCD19 expression was observed for all T cells expressing the GES, with a greater fold increase in surface expression in those with higher numbers of REs. Thus, surface dCD19 expression was directly proportional with the number of REs encoded in the GES. No dCD19 was detected on the surface of T cells that did not harbor the GES: No GES and Mock controls.

#### INCORPORATION BY REFERENCE

(900) Every document cited herein, including any cross referenced or related patent or application is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

#### Other Embodiments

(901) While particular embodiments of the disclosure have been illustrated and described, various other changes and modifications can be made without departing from the spirit and scope of the disclosure. The scope of the appended claims includes all such changes and modifications that are within the scope of this disclosure.

## Claims

1. A cell comprising a chimeric antigen receptor (CAR) comprising: (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH that specifically binds to B-Cell Maturation Antigen (BCMA); (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain; wherein the VH comprises: a complementarity determining region 1 (CDR1) comprising an amino acid sequence of SEQ ID NO: 18056; a complementarity determining region 2 (CDR2) comprising an amino acid sequence of SEQ ID NO: 18060; and a complementarity determining region 3 (CDR3) comprising an amino acid sequence of SEQ ID NO: 18064.
2. The cell of claim 1, wherein the VH comprises or consists of a recombinant or chimeric sequence.
3. The cell of claim 1, wherein the VH comprises or consists of a human or humanized sequence.
4. The cell of claim 1, wherein the ectodomain of (a) further comprises a signal peptide.



5. The cell of claim 1, wherein the ectodomain of (a) further comprises a hinge between the antigen recognition region and the transmembrane domain.
6. The cell of claim 4, wherein the signal peptide comprises an amino acid sequence of a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR signal peptide.
7. The cell of claim 6, wherein the signal peptide comprises an amino acid sequence of a human CD8a signal peptide.
8. The cell of claim 7, wherein the human CD8a signal peptide comprises an amino acid sequence of SEQ ID NO: 18012.
9. The cell of claim 1, wherein the transmembrane domain comprises an amino acid sequence of a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR transmembrane domain.
10. The cell of claim 9, wherein the transmembrane domain comprises an amino acid sequence of a human CD8a transmembrane domain.
11. The cell of claim 10, wherein the human CD8a transmembrane domain comprises an amino acid sequence of SEQ ID NO: 18014.
12. The cell of claim 1, wherein the endodomain comprises an amino acid sequence of a human CD3ζ endodomain.
13. The cell of claim 1, wherein the at least one costimulatory domain comprises an amino acid sequence of a human 4-1BB, CD28, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof.
14. The cell of claim 13, wherein the at least one costimulatory domain comprises an amino acid sequence of a human 4-1BB costimulatory domain.
15. The cell of claim 12, wherein the human CD35 endodomain comprises an amino acid sequence of SEQ ID NO: 18016.
16. The cell of claim 14, wherein the human 4-1BB costimulatory domain comprises an amino acid sequence of SEQ ID NO: 18018.
17. The cell of claim 5, wherein the hinge comprises an amino acid sequence of a human CD8α, IgG4, CD4 hinge or any combination thereof.
18. The cell of claim 17, wherein the hinge comprises an amino acid sequence of a human CD8α hinge.
19. The cell of claim 18, wherein the human CD8α comprises an amino acid sequence of SEQ ID NO: 18020.
20. A composition comprising the cell of claim 1 and at least one pharmaceutically acceptable carrier.
21. The cell of claim 1, wherein the cell is an immune cell.
22. The cell of claim 21, wherein the immune cell is a T-cell, a Natural Killer (NK) cell, a Natural Killer (NK)-like cell, a Cytokine Induced Killer (CIK) cell, a hematopoietic progenitor cell, a peripheral blood (PB) derived T cell or an umbilical cord blood (UCB) derived T-cell.
23. The cell of claim 1, wherein the cell is autologous.
24. The cell of claim 1, wherein the cell is allogeneic.
25. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 1.
26. The cell of claim 1, wherein the VH comprises an amino acid sequence of SEQ ID NO: 18051.
27. The cell of claim 1, wherein the CAR comprises amino acids 22-362 of the amino acid sequence of SEQ ID NO: 18006.
28. A cell comprising a chimeric antigen receptor (CAR) comprising: (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH that specifically binds to BCMA and wherein the VH comprises an amino acid sequence of SEQ ID NO: 18051; (b) a hinge domain comprising a human CD8α hinge domain; (c) a transmembrane domain comprising a human CD8α transmembrane domain; and (d) an endodomain comprising a human 4-1BB costimulatory domain and a human CD3ζ endodomain.
29. The cell of claim 28, wherein the human CD8α hinge domain comprises the amino acid sequence of SEQ ID NO: 18020, wherein the human CD8α transmembrane domain comprises the amino acid sequence of SEQ ID NO: 18014, wherein the human 4-1BB costimulatory domain comprises the amino

acid sequence of SEQ ID NO: 18018, and wherein the human CD3 $\zeta$  endodomain comprises the amino acid sequence of SEQ ID NO: 18016.

30. The cell of claim 28, wherein the cell is an immune cell.

31. The cell of claim 30, wherein the immune cell is a T-cell, a Natural Killer (NK) cell, a Natural Killer (NK)-like cell, a Cytokine Induced Killer (CIK) cell, a hematopoietic progenitor cell, a peripheral blood (PB) derived T cell or an umbilical cord blood (UCB) derived T-cell.

32. The cell of claim 31, wherein the immune cell is a T cell.

33. The cell of claim 32, wherein the immune cell is allogeneic.

34. The cell of claim 27, wherein the cell is a T cell.

35. The cell of claim 34, wherein the T cell is a memory stem T cell (T.sub.SCM).

36. The cell of claim 35, wherein the T.sub.SCM is allogeneic.

37. The cell of claim 27, wherein the cell further comprises an exogenous gene that confers resistance to a deleterious compound.

38. The cell of claim 37, wherein the gene is a Dihydrofolate Reductase (DHFR) gene.

39. The cell of claim 27, wherein the cell further comprises an exogenous inducible proapoptotic polypeptide.

40. The cell of claim 39, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

41. A composition comprising the cell of claim 29 and at least one pharmaceutically acceptable carrier.

42. A composition comprising the cell of claim 33 and at least one pharmaceutically acceptable carrier.

43. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 29.

44. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 33.

45. The cell of claim 1, wherein the CAR comprises an amino acid sequence of SEQ ID NO: 18006.

46. The cell of claim 28, wherein the ectodomain further comprises a human CD8a signal peptide.

47. The cell of claim 46, where in the human CD8a signal peptide comprises the amino acid sequence of SEQ ID NO: 18012.

48. The cell of claim 27, wherein the cell is an allogeneic T-cell.

49. The cell of claim 48, wherein the cell further comprises an exogenous gene that confers resistance to a deleterious compound.

50. The cell of claim 49, wherein the gene is a Dihydrofolate Reductase (DHFR) gene.

51. The cell of claim 48, wherein the cell further comprises an exogenous inducible proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

52. The cell of claim 50, wherein the cell further comprises an exogenous inducible proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

53. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 50.

54. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 52.

55. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 49.

56. The cell of claim 49, wherein the DHFR gene comprises the sequence of SEQ ID NO: 17012.

57. The cell of claim 48, wherein the truncated caspase 9 polypeptide comprises the sequence of SEQ ID NO: 18028.

58. The cell of **50**, wherein the truncated caspase 9 polypeptide comprises the sequence of SEQ ID NO: 18028.

59. A nucleic acid encoding a chimeric antigen receptor (CAR) comprising: (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH that specifically binds to B-Cell Maturation Antigen (BCMA); (b) a transmembrane domain, and (c)

an endodomain comprising at least one costimulatory domain; wherein the VH comprises: a complementarity determining region 1 (CDR1) comprising an amino acid sequence of SEQ ID NO: 18056; a complementarity determining region 2 (CDR2) comprising an amino acid sequence of SEQ ID NO: 18060; and a complementarity determining region 3 (CDR3) comprising an amino acid sequence of SEQ ID NO: 18064.

60. The nucleic acid of claim 59, wherein the ectodomain of (a) further comprises a signal peptide.

61. The nucleic acid of claim 59, wherein the ectodomain of (a) further comprises a hinge between the antigen recognition region and the transmembrane domain.

62. The nucleic acid of claim 60, wherein the signal peptide is a human CD8 $\alpha$  signal peptide.

63. The nucleic acid of claim 59, wherein the transmembrane domain comprises a human CD8 $\alpha$  transmembrane domain.

64. The nucleic acid of claim 59, wherein the endodomain comprises a human CD3 $\zeta$  endodomain.

65. The nucleic acid of claim 59, wherein the at least one costimulatory domain comprises an amino acid sequence of a human 4-1 BB costimulatory domain.

66. The nucleic acid of claim 59, wherein the hinge comprises an amino acid sequence of a human CD8 $\alpha$  hinge.

67. The nucleic acid of claim 59, wherein the VH comprises the amino acid sequence of SEQ ID NO: 18051.

68. The nucleic acid of claim 59, wherein the CAR comprises the amino acid sequence of SEQ ID NO: 18006.

69. The nucleic acid of claim 59 comprising the sequence of SEQ ID NO: 18007.

70. A transposon comprising the nucleic acid of claim 59.

71. The transposon of claim 70, wherein the transposon is a piggyBac transposon.

72. The nucleic acid of claim 68, wherein the polynucleotide further comprises an exogenous gene that confers resistance to a deleterious compound.

73. The nucleic acid of claim 72, wherein the gene is a Dihydrofolate Reductase (DHFR) gene.

74. The nucleic acid of claim 68, wherein the polynucleotide further encodes an exogenous inducible proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

75. The nucleic acid of claim 73, wherein the polynucleotide further encodes an exogenous inducible proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

76. A transposon comprising the nucleic acid of claim 73.

77. The transposon of claim 76, wherein the transposon is a piggyBac transposon.

78. A transposon comprising the nucleic acid of claim 75.

79. The transposon of claim 78, wherein the transposon is a piggyBac transposon.

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